

Compendio tecnico del sistema trading ibrido

Estratto, integrazione e traduzione operativa dei PDF sorgente

Questo documento integra le note del vault con i PDF sorgente: signal extraction, Fourier statistico, option pricing con Fourier, libreria frattale e documento Pythagoras. La parte finale contiene le fonti originali in appendice, così che la versione sintetica e quella integrale restino nello stesso file.

Criterio di costruzione: teoria matematica -> traduzione operativa -> validazione su dati reali -> rischio e logging.

1. Mappa delle fonti e dei concetti estratti

Questo nodo raccoglie la traccia di provenienza dei materiali analizzati e la loro traduzione operativa nel vault.

Fonte	Nucleo concettuale	Uso nel progetto
ECONSIGS.pdf	Wiener-Kolmogorov, filtri finiti, Butterworth digitale, processi band-limited	
FIVESTAT.pdf	Fourier statistico, DFT, matrice circolante, aliasing, sampling, Parseval	Base
Matsuda Intro FT Pricing.pdf	Fourier transform pricing, characteristic functions, DFT, Black	
fractals.pdf	Libreria di pattern frattali LONG, sequenze da 3 a 6 candele	Pattern matching
Pythagoras_Trading_Prediction_.pdf	Frattali, proiezione temporale, narrazione coscienza/mercato	

Regola di lettura

La parte teorica viene separata in tre livelli:

1. Descrizione matematica: formule, trasformate, densità, filtri, derivate. 2. Operatività di mercato: entry, timing, conferme, uscita, rischio. 3. Validazione: backtest, out-of-sample, metriche, robustezza.

Nota metodologica

Le componenti filosofiche o speculative presenti in alcuni materiali vengono mantenute come contesto storico o narrativo, ma nel progetto vengono convertite solo nelle parti osservabili e testabili: • ricorrenza di forme • trasformazioni invarianti • cicli dominanti • accelerazione/decelerazione del prezzo • metriche verificabili

2. Architettura del sistema ibrido

Obiettivo generale

Il sistema non nasce come singolo indicatore, ma come catena di trasformazione:

dati grezzi -> pulizia -> decomposizione -> pattern -> decisione -> esecuzione -> misurazione -> apprendimento

Il principio di progetto è conservare solo ciò che produce un vantaggio ripetibile. Il resto rimane ipotesi da testare.

Strati del motore

Strato	Funzione	Output
Data Layer	raccogliere OHLC, volumi, sessioni, spread, eventi	serie pulite e allineate
Transform Layer	applicare FFT, derivate, smoothing, filtri	segnali numerici e spettrali
Pattern Layer	riconoscere forme frattali, simmetrie, ricorrenze	pattern_id, confidenza, zone
Feature Layer	unire ampiezza, fase, pendenza, curvatura, volatilità	feature vector
Decision Engine	combinare regole e filtri	long / short / no trade
Risk Layer	size, SL, TP, RR, max exposure	ordine controllato
Backtest Layer	misurare edge, robustezza, degrado	report e ranking
Logging Layer	memorizzare risultati e anomalie	memoria strutturata

Regole di progetto estratte dal vault

- testare un concetto per volta;
- validare prima in isolamento;
- accettare solo combinazioni che migliorano il risultato netto;
- usare risk contenuto;
- impedire che l'ottimizzazione distrugga la generalizzazione;
- non confondere spiegazione teorica e vantaggio operativo.

Pipeline operativa

1. Normalizzazione dei dati. 2. Estrazione del segnale dominante. 3. Identificazione del contesto di mercato. 4. Verifica di timing con derivate. 5. Confluenza con pattern frattali. 6. Calcolo del rischio e della taglia. 7. Simulazione e logging. 8. Iterazione.

Criterio di utilita'

Un modulo entra nel sistema finale solo se risponde a tre domande:

- riduce rumore?
- migliora timing o selezione?
- mantiene robustezza su piu' periodi?

Se la risposta e' no, il modulo resta in osservazione.

3. Dati, pulizia e lettura del mercato

Dataset reale usato come riferimento

Nel corpus e' presente una serie minute su EURUSD (`eurusd_m1_2020_2025_combined.csv`). Un estratto iniziale mostra struttura OHLC standard con volume nullo su molte barre forex interbank.

Campo	Significato
datetime	timestamp della barra
open	apertura
high	massimo
low	minimo
close	chiusura
volume	volume registrato

Osservazioni preliminari sul campione analizzato

Su un campione iniziale di 100000 righe:

- media prezzo close circa 1.1016;
- deviazione standard del close circa 0.0157;
- deviazione standard dei rendimenti minuti circa 0.000178;
- skewness positiva;
- curtosi molto alta, quindi code grasse e outlier non trascurabili.

Questa combinazione indica che il mercato non si comporta come una gaussiana semplice e che il modello deve tollerare regime shift, spike e cluster di volatilita'.

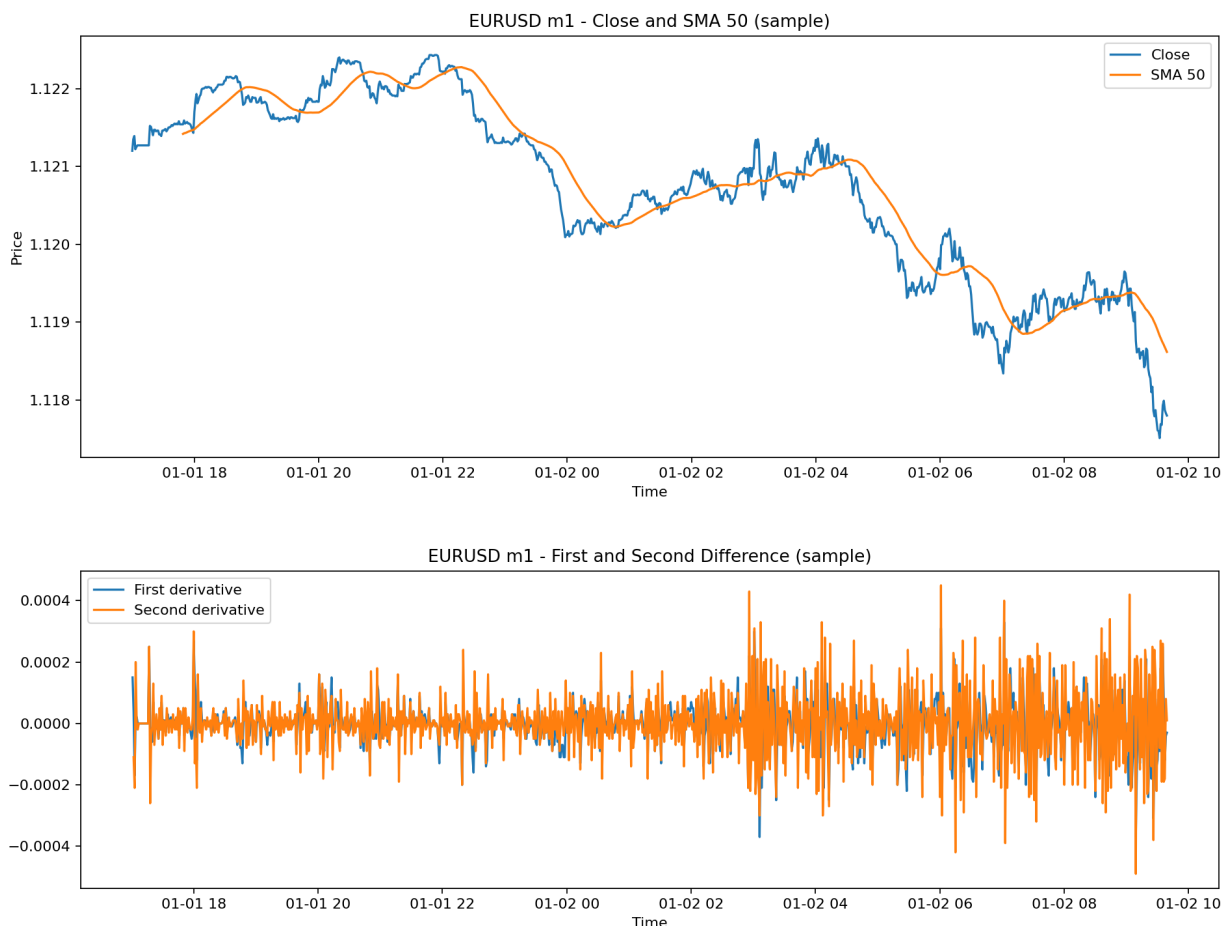
Pulizia minima necessaria

1. Ordinare per tempo. 2. Verificare duplicati. 3. Controllare buchi o barre anomale. 4. Uniformare timezone. 5. Separare training, validation e test. 6. Trattare i rendimenti come oggetto principale, non solo il prezzo.

Perché il prezzo da solo non basta

Il prezzo e' una traiettoria cumulata. Per il trading quantitativo interessano anche: • differenze discrete; • log-return; • volatilità realizzata; • range intrabar; • continuità o discontinuità del regime.

Chart di riferimento



Lettura operativa

Il prezzo grezzo e' utile per visualizzare il contesto, ma il motore deve ragionare su: • pendenza locale; • curvatura; • distanza da media; • ampiezza del movimento; • varianza recente; • posizione rispetto a ciclo e pattern.

Implicazione per il backtest

Un backtest corretto deve conservare il dato originale e, in parallelo, derivarne una vista feature-based. In pratica: • OHLC per la simulazione; • returns per il rischio; • FFT per il ciclo; • derivate per il timing; • pattern per la direzione condizionata.

4. Derivate discrete: direzione, accelerazione, inversione

Definizione pratica

Nel vault la derivata prima e la derivata seconda sono introdotte come versioni discrete e operative:

$$dP_t = P_t - P_{t-1}$$

$$d^2P_t = dP_t - dP_{t-1}$$

Nel linguaggio del progetto: • dP_t misura la direzione immediata; • d^2P_t misura la variazione della direzione, cioè accelerazione o rallentamento.

Interpretazione di mercato

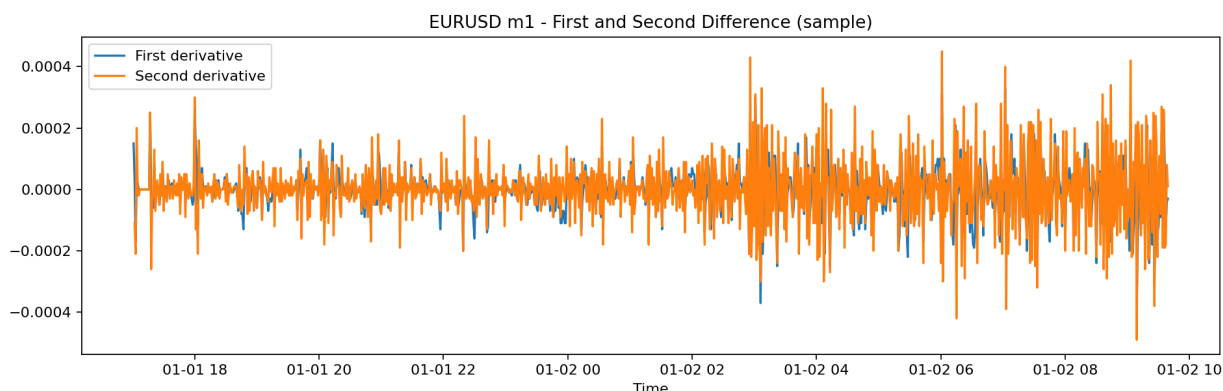
```
| Segno | Lettura |
|---|---|
|  $dP_t > 0$  | bias rialzista |
|  $dP_t < 0$  | bias ribassista |
|  $d^2P_t > 0$  | spinta crescente |
|  $d^2P_t < 0$  | spinta che si indebolisce |
```

Una derivata prima positiva non basta: un trend può essere positivo ma perdere energia. La seconda derivata serve a distinguere: • trend forte ma in estensione; • trend forte ma in esaurimento; • falsa partenza; • inversione probabile.

Uso nelle strategie

Le derivate sono un filtro di timing: 1. il contesto viene stimato da FFT o frattali; 2. la derivata prima conferma il verso; 3. la derivata seconda decide se entrare subito o attendere.

Chart



Collegamento con la fisica del prezzo

Il prezzo non viene trattato come segnale statico, ma come traiettoria: • la prima differenza corrisponde alla velocità discreta; • la seconda differenza corrisponde all'accelerazione discreta; • il cambio di segno della seconda differenza segnala un possibile punto di svolta.

Regola utile

In presenza di rumore elevato, la derivata va sempre letta dopo un minimo di smoothing. Senza questo passaggio si amplifica il rumore invece del segnale.

5. Fourier, FFT e lettura spettrale

Idea centrale

Il prezzo puo' essere visto come somma di componenti periodiche e non periodiche. La trasformata di Fourier sposta l'attenzione: • dal tempo alla frequenza; • dalla forma visibile alla struttura nascosta; • dal singolo movimento al ciclo dominante.

Formula continua

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-i\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{i\omega t} d\omega$$

DFT

Per una sequenza discreta di lunghezza N :

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}$$

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{i2\pi kn/N}$$

Proprieta' che contano nel trading

```
| Proprieta' | Impatto operativo |  
|---|---|  
| linearita' | sommare filtri e segnali |  
| simmetria | interpretazione delle frequenze positive/negative |  
| convoluzione | filtraggio nel dominio del tempo |  
| modulazione | traslazione di una banda di frequenza |  
| Parseval | energia uguale in tempo e frequenza |  
| differenziazione | enfasi sulle alte frequenze |  
| time shifting | ritardo/fase del segnale |
```

Parseval

$$\sum_{n=0}^{N-1} |x_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X_k|^2$$

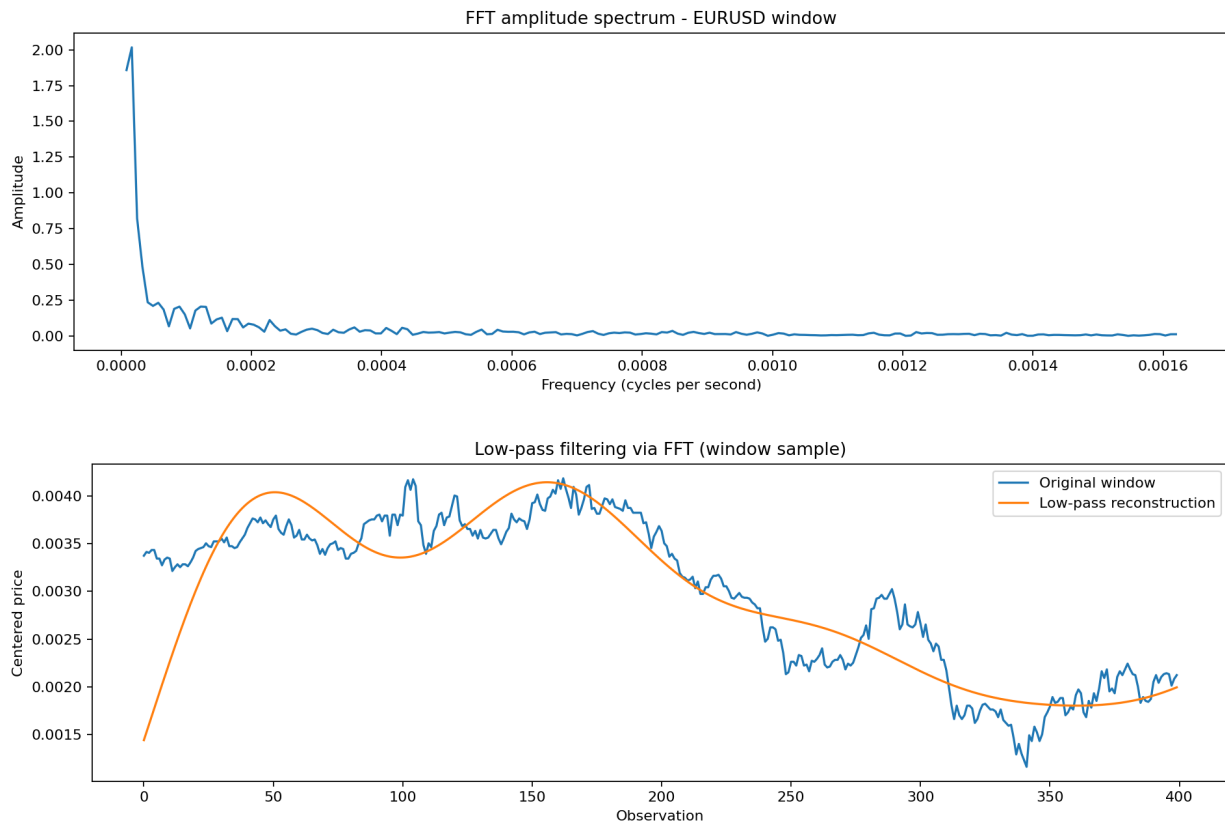
Questa identita' e' utile per verificare che il filtraggio non distrugga energia in modo incoerente.

Frequenze basse e alte

• frequenze basse: trend, cicli lenti, struttura di fondo; • frequenze alte: micro-rumore, micro-variazione, spike.

Il trading quantitativo ha senso solo se il filtro conserva la banda utile e sopprime il resto senza introdurre eccessivo lag.

Grafici dimostrativi



Problemi classici

Gibbs

Ai bordi di una discontinuità la ricostruzione può oscillare. Questo non implica fallimento del metodo, ma richiede consapevolezza della convergenza.

Aliasing

Se il campionamento è troppo lento rispetto alla frequenza del segnale, le frequenze alte si ripiegano sulle basse. Nel trading significa confondere rumore rapido con ciclo reale.

Scelta della finestra

La finestra è sempre un compromesso tra:

- risoluzione in frequenza;
- reattività nel tempo;
- stabilità del pattern.

Lettura operativa

La FFT non deve essere usata come oracolo, ma come:

- localizzatore di ciclo;
- filtro di rumore;
- strumento per stimare la banda principale;
- base per regole di conferma.

6. Estrazione del segnale: Wiener-Kolmogorov, Butterworth e processi band-limited

Nucleo concettuale

Il documento sulle signal extraction methods insiste su un'idea precisa: il segnale osservato e' una somma di componenti che si possono filtrare in modo ottimale se si conosce abbastanza della loro struttura statistica.

La forma classica e':

$$y(t) = \xi(t) + \eta(t)$$

dove: • $\xi(t)$ = segnale; • $\eta(t)$ = rumore.

Estimatori di Wiener-Kolmogorov

Per serie finite, gli stimatori minimi in senso quadratico sono:

$$\hat{\xi} = \Omega_{\xi}(\Omega_{\xi} + \Omega_{\eta})^{-1}y$$

$$\hat{\eta} = \Omega_{\eta}(\Omega_{\xi} + \Omega_{\eta})^{-1}y$$

Queste formule esprimono l'idea di separare il dato in due parti che si spiegano reciprocamente.

Filtri Butterworth

Il Butterworth digitale viene usato come prototipo di filtro con transizione regolare:

$$|H(\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}}$$

Dove: • ω_c = frequenza di taglio; • n = ordine del filtro.

Un ordine maggiore produce una transizione piu' netta ma aumenta il rischio di eccesso di fase e instabilita' pratica.

Band-limited processes

Il documento evidenzia che alcuni processi sono band-limited: la loro energia vive in bande separate da zone di silenzio spettrale. Per il progetto questo significa: • non tutte le frequenze sono uguali; • ci sono zone informative; • il filtro deve rispettare la separazione tra bande.

Forme operative nel trading

Oggetto Lettura
--- ---
banda bassa trend macro
banda media swing e cicli intermedi
banda alta microstruttura / rumore
dead space intervallo di scarsa informazione

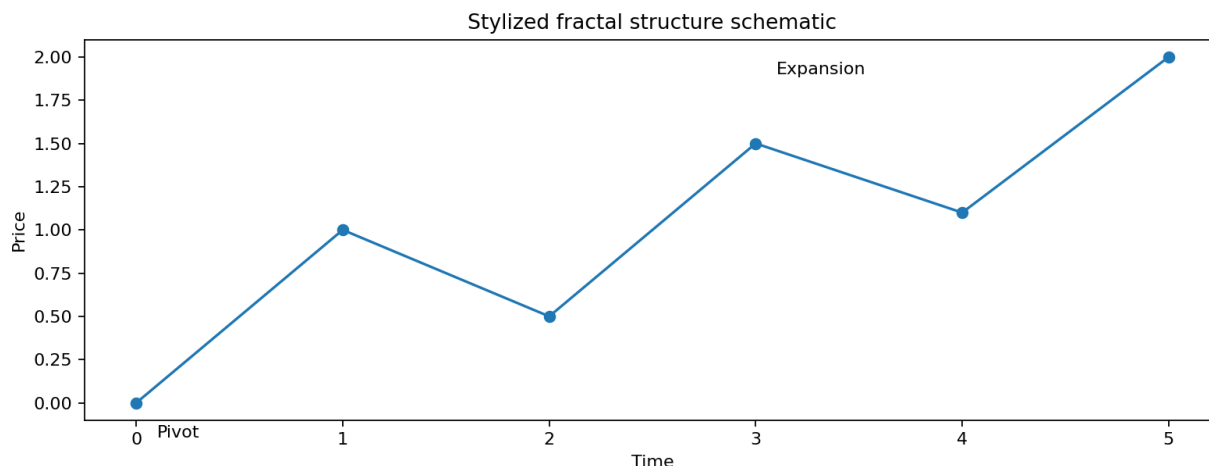
Confronto con Kalman

Il testo di Pollock sottolinea che il Kalman e' potente ma complesso. Nel progetto la lezione e': • usare il minimo modello sufficiente; • non complicare se il filtro locale basta; • preferire una struttura testabile e spiegabile.

Pipeline consigliata

1. filtro anti-rumore; 2. FFT per stimare la banda; 3. ricostruzione a banda limitata; 4. derivata del segnale filtrato; 5. decisione con regole di contesto; 6. backtest sul risultato.

Chart di supporto



7. Frattali: libreria di pattern e traslazioni

Cosa emerge dai materiali frattali

I PDF frattali mostrano una libreria di pattern LONG formata da sequenze di 3 fino a 6 candele, con estensioni successive fino a range piu' ampi nel vault. Il punto non e' solo riconoscere la forma, ma riconoscere: • struttura; • trasformazione; • ricorrenza; • posizione nel contesto.

Unita' minima del pattern

Ogni pattern candidato dovrebbe essere descritto da:

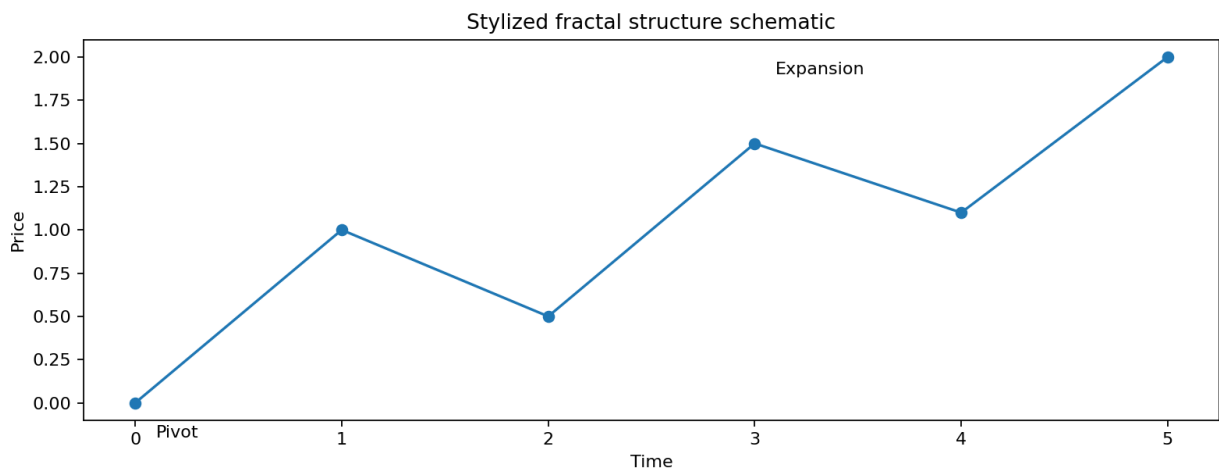
Campo	Significato
---	---
pattern_id	identificativo univoco
range_candele	ampiezza temporale della figura
pivot_count	numero di pivot principali
shape_vector	vettore descrittivo della forma
ampiezza_relativa	profondita' o ampiezza del movimento
durata_relativa	tempo impiegato
slope	pendenza complessiva
curvature	piegatura / variazione di pendenza
entry_zone	area di ingresso
target_zone	area target
invalidazione	livello che ne rompe la logica

Trasformazioni ammesse

Il materiale lascia intendere che la stessa identita' strutturale possa sopravvivere a trasformazioni diverse: • traslazione temporale; • traslazione verticale; • scaling in ampiezza; • scaling temporale; • inversione; • riflessione.

Questa idea e' utile per il matching dei pattern su asset diversi o su scale temporali diverse.

Pattern da 3 a 6 candele



Nel PDF originale le sequenze sono presentate come tavole di pattern. Nel vault si traduce la libreria in una struttura operativa che consente: • comparazione storica; • similarita' geometrica; • selezione di setup; • studio di deformazioni del pattern.

Fractal Circular Indicator

Il Circular Indicator viene interpretato come framework che unisce: • pivot frattali; • tempi ciclici; • geometria circolare o ellittica; • livelli armonici; • zone di espansione e ritorno.

La parte importante per il trading non e' la forma estetica, ma l'ipotesi che la relazione tempo-prezzo abbia cicli ricorrenti e punti di simmetria utili al timing.

Lettura operativa del frattale

1. individuare pivot; 2. misurare ampiezza e durata; 3. classificare la forma; 4. confrontarla con il catalogo storico; 5. attendere la conferma di volume, momentum o derivata; 6. entrare solo se il rischio e' definito.

Limite metodologico

Il frattale non va trattato come verita' assoluta. Va trattato come: • descrittore di forma; • filtro di contesto; • ipotesi da validare con backtest.

8. Analisi del documento Pythagoras e traduzione operativa

Contenuto concettuale del PDF

Il documento Pythagoras intreccia: • frattali; • Fourier; • derivate; • idea di proiezione del futuro; • narrazione su coscienza, osservatore e mercato.

La parte utile al progetto non e' la cornice filosofica in se, ma la catena di lavoro che suggerisce:

frattale -> Fourier -> estrazione del ciclo -> timing -> decisione

Traduzione in ipotesi testabili

Dal punto di vista operativo, questa catena si può riscrivere così:

1. il mercato mostra forme ricorrenti; 2. tali forme hanno una componente ciclica; 3. la componente ciclica si può scomporre; 4. la scomposizione aiuta il timing; 5. il timing diventa regola di ingresso o uscita.

Parti non trasferibili direttamente

Le sezioni su: • magia; • coscienza come causa fisica; • lettura non locale del tempo; • interpretazioni speculative dell'universo;

non vengono usate come ipotesi di trading. Rimangono come materiale narrativo del documento originario.

Parti trasferibili

```
| Concetto | Traduzione nel vault |  
|---|---|  
| ricorrenza | pattern matching |  
| simmetria | trasformazioni invarianti |  
| proiezione temporale | stima di turning window |  
| Fourier | decomposizione del prezzo |  
| derivata | direzione e accelerazione |  
| consapevolezza | in pratica = osservazione disciplinata e logging |
```

Lezione metodologica

Il trader non deve cercare conferme ideologiche, ma: • osservare; • misurare; • memorizzare; • verificare; • correggere.

Uso nel progetto

Il documento viene quindi letto come: • fonte di idee strutturali; • contenitore di pattern e metafore; • generatore di ipotesi di ricerca; • non come prova matematica del sistema.

9. Fourier transform pricing, Black-Scholes, Merton e Variance Gamma

Perché Fourier nel pricing

Il libro di Matsuda mostra che il pricing via Fourier diventa utile quando il modello ha una caratteristica funzione nota, anche se la densità di prezzo terminale non è scrivibile in forma semplice.

Il vantaggio è: • generalità; • velocità di calibrazione; • compatibilità con modelli a salto; • uso diretto della characteristic function.

Funzione caratteristica

$$\varphi_X(u) = E[e^{iuX}]$$

Da questa si ottengono: • momenti; • cumulanti; • pricing integrale.

Black-Scholes

Nel caso classico, il prezzo di una call europea può essere espresso in forma chiusa. Nel linguaggio del compendio, Black-Scholes resta il caso base: • solo diffusione continua; • no salti; • varianza lognormale.

Merton jump-diffusion

Il modello Merton aggiunge salti casuali: • numero di salti Poisson; • ampiezza dei salti gaussiana o comunque parametrica; • coda più realistica rispetto al solo Brownian motion.

Questo rende il modello più adatto ai mercati che presentano shock discreti.

Variance Gamma

Il VG usa una subordinazione: • tempo operativo casuale; • code più pesanti; • skewness e kurtosis più realistiche.

Formula di pricing via Fourier

Una forma standard del prezzo call damped è:

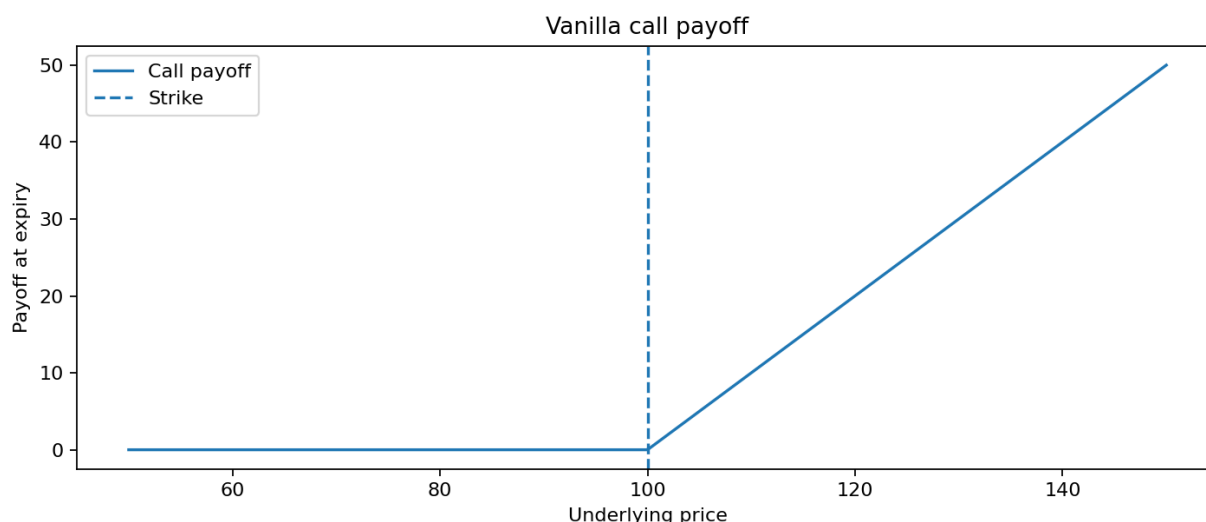
$$C(K) = \frac{e^{-rT - \alpha k}}{\pi} \int_0^\infty \operatorname{Re} \left[e^{-iuk} \frac{\phi(u - (\alpha + 1)i)}{\alpha^2 + \alpha - u^2 + i(2\alpha + 1)u} \right] du$$

dove: • $k = \ln K$; • α è il parametro di damping; • ϕ è la characteristic function del log-prezzo terminale.

DFT pricing

La DFT approssima l'integrale e permette calcolo rapido su una griglia di strike.

Payoff call



Uso nel progetto

Il blocco pricing non serve solo a prezzare opzioni, ma anche a insegnare come: • usare la characteristic function; • gestire distribuzioni non gaussiane; • lavorare con dati in frequenza; • mantenere il legame tra mercato e forma matematica del processo.

Messaggio chiave

Se il mercato ha salti, asimmetria e code pesanti, un modello puramente gaussiano e' spesso insufficiente. La coppia Fourier + Lévy e' piu' coerente con la struttura osservabile.

10. Metriche, rischio e validazione

Metriche di base

Metrics	Formula sintetica	Uso
Return	$\frac{V_t - V_{t-1}}{V_{t-1}}$	performance semplice
Log-return	$\ln(V_t/V_{t-1})$	additivita'
Win rate	trade vincenti / trade totali	frequenza
Sharpe	media eccedenza / deviazione standard	efficienza rischio
Max drawdown	picco meno valle successiva	peggior perdita
Profit factor	gross profit / gross loss	bilancio del sistema
Expectancy	$p \cdot \text{win} - (1-p) \cdot \text{loss}$	valore atteso

Drawdown

Il drawdown e' la metrica che piu' penalizza i sistemi fragili. Un sistema puo' avere buon win rate e fallire comunque per: • perdite rare ma enormi; • sequenze di trade avversi; • leva eccessiva; • assenza di stop coerenti.

Validazione corretta

1. In-sample: costruzione. 2. Out-of-sample: verifica su dati non visti. 3. Walk-forward: stabilita' nel tempo. 4. Stress test: slippage, spread, shock, dati mancanti. 5. Regime test: trend, range, volatilita' alta, volatilita' bassa.

Regola di sopravvivenza

Il motore non deve cercare solo il massimo profitto teorico, ma: • drawdown accettabile; • stabilita' temporale; • assenza di overfitting; • relazione chiara tra entry e rischio.

Lettura operativa

Un sistema puo' essere considerato utile solo se i risultati rimangono leggibili su: • periodi diversi; • asset diversi; • timeframe diversi; • condizioni di mercato differenti.

Collegamento con il dataset

Il dataset EURUSD minute e' adatto a: • test di trend; • test di micro-struttura; • ricerca di cicli; • confronto tra filtraggio e rumore.

11. Decision engine e logica del bot

Obiettivo del decisore

Il decisore non deve essere creativo in senso libero. Deve essere disciplinato, cioe' capace di combinare: • contesto; • timing; • rischio; • conferma; • invalidazione.

Sequenza logica

1. Il mercato e' in trend o in range? 2. Esiste un ciclo dominante? 3. La derivata conferma il verso? 4. Il pattern frattale e' valido? 5. Il rischio e' compatibile con la struttura? 6. L'entry ha un rapporto rischio/rendimento accettabile? 7. Il trade e' ripetibile in backtest?

Esempio di regola combinata

• se FFT indica banda bassa dominante, • e la pendenza locale e' positiva, • e il pattern frattale mostra compressione seguita da breakout, • allora la strategia puo' passare da osservazione a esecuzione.

Moduli attualmente presenti nel vault

• MA crossover testato; • derivata prima e seconda; • idee FFT trend following; • confluence frattale-circolare; • proiezione pattern frattale-FFT; • risk management; • broker API; • dashboard; • experiment protocol.

Principio di progetto

Il sistema finale deve rifiutare l'azione quando: • il segnale e' ambiguo; • il rumore e' troppo alto; • il pattern non e' confermato; • il rischio supera la soglia; • il regime non e' adatto.

Logica di memoria

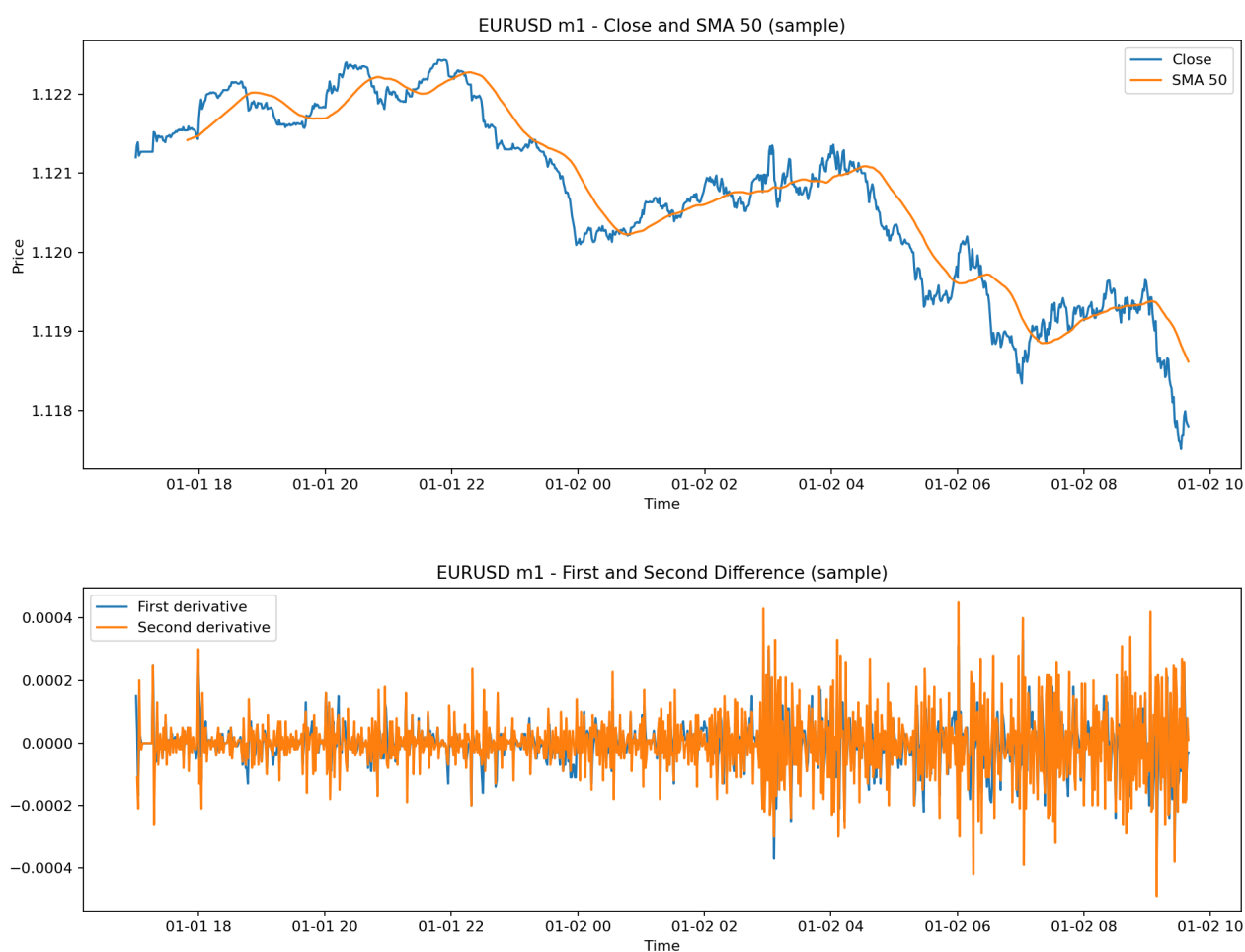
Ogni esperimento deve produrre: • input; • ipotesi; • parametri; • risultato; • motivo di successo o fallimento; • decisione successiva.

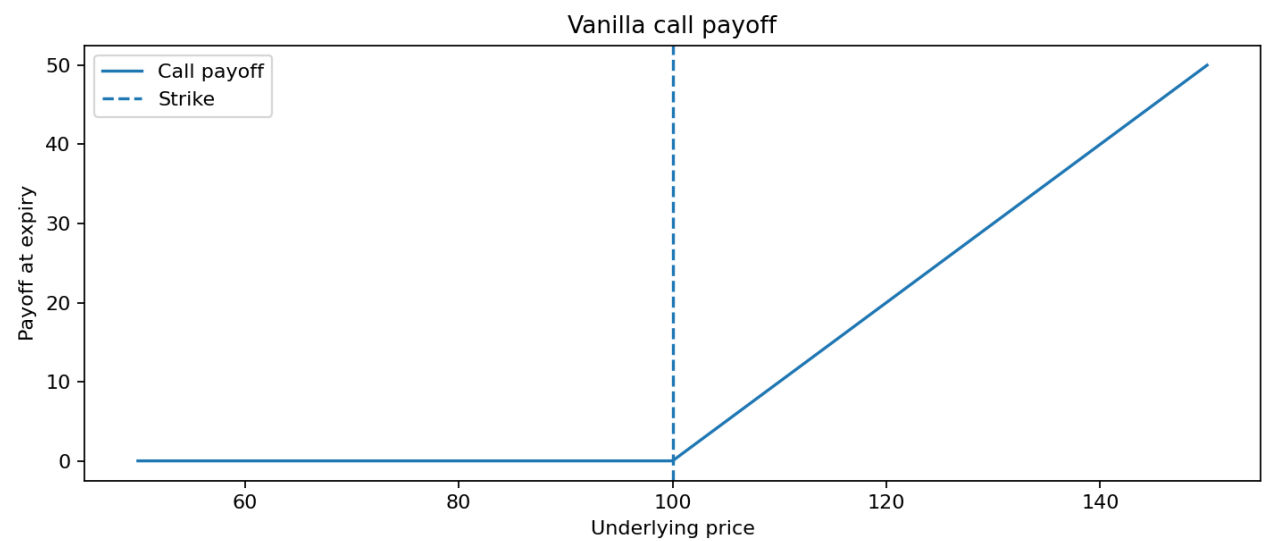
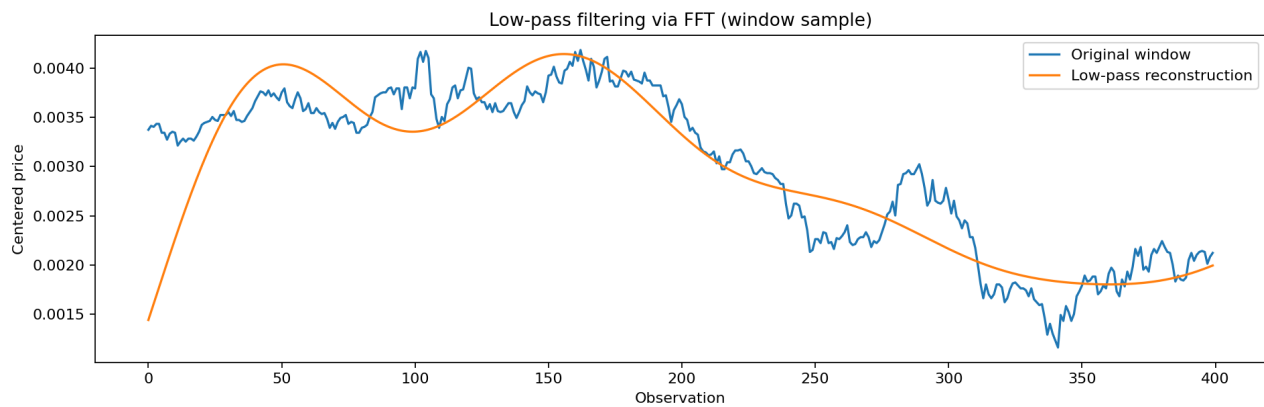
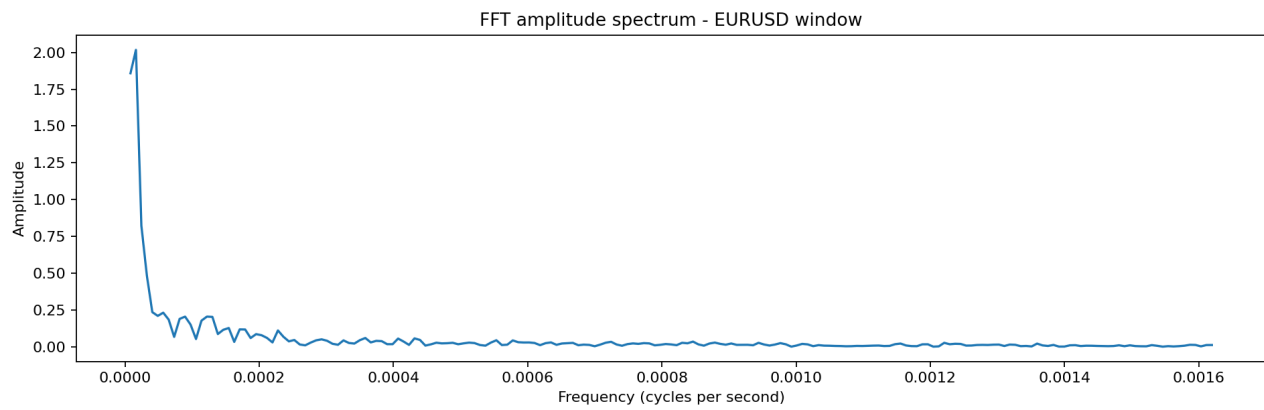
Questo e' il meccanismo che trasforma la ricerca in un sistema cumulativo invece che in una sequenza di test isolati.

12. Appendice numeriche e compendio dati

Elemento	Valore
Campione EURUSD	100000 righe iniziali
Start	2020-01-01 17:00:00
End	2020-04-08 07:19:00
Media close	1.1016
Std close	0.0157
Std return	0.0001777
Skew return	1.0871
Kurtosis return	49.7299

Chart di supporto su dati reali





13. Compendio formule essenziali

Differenze discrete:

$$dP_t = P_t - P_{(t-1)}$$

$$d^2P_t = dP_t - dP_{(t-1)}$$

DFT:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}$$

$$x_n = (1/N) \sum_{k=0}^{N-1} X_k e^{i2\pi kn/N}$$

Fourier continua:

$$X(\omega) = \int x(t) e^{-i\omega t} dt$$

$$x(t) = (1/2\pi) \int X(\omega) e^{i\omega t} d\omega$$

Function characteristic:

$$\phi_X(u) = E[e^{iuX}]$$

Wiener-Kolmogorov:

$$\hat{x} = \Omega_x (\Omega_x + \Omega_\eta)^{-1} y$$

Butterworth:

$$|H(\omega)|^2 = 1/(1 + (\omega/\omega_c)^{2n})$$

Drawdown:

$$DD_t = (\max_{s \leq t} V_s - V_t) / \max_{s \leq t} V_s$$

14. Glossario operativo esteso

Termine	Definizione
Aliasing	Piegatura delle alte frequenze su bande piu' basse quando il campionamento non
Band-limited	Processo concentrato in una banda ristretta del dominio della frequenza.
Butterworth	Filtro con risposta monotona e transizione regolare.
Characteristic function	Trasformata complessa della distribuzione di probabilita'.
Convolution	Operazione che combina due segnali tramite scorrimento e integrazione/somma.
Drawdown	Discesa dal massimo al valore corrente.
FFT	Algoritmo veloce per calcolare la DFT.
Frattale	Forma che conserva una relazione di scala o ricorrenza.
Gibbs	Oscillazione locale vicino a una discontinuita' nella ricostruzione di Fourier.
Lag	Ritardo introdotto da un filtro.
Lévy process	Processo con incrementi indipendenti e stazionari.
Low-pass	Filtro che conserva la banda bassa.
Nyquist	Vincolo di campionamento per evitare aliasing.
Parseval	Uguaglianza dell'energia tra tempo e frequenza.
Pivot	Punto locale di svolta nel prezzo.
Sharpe ratio	Rapporto rendimento eccedente / volatilita'.
Spectral density	Distribuzione dell'energia sulle frequenze.
Subordination	Tempo casuale che guida un altro processo.
Wiener-Kolmogorov	Soluzione ottimale per estrarre segnali da rumore.

15. Nota finale sulla fonte originale

Le fonti originali sono mantenute in appendice nel PDF finale per evitare perdita di dettaglio. La parte sintetica e' pensata per orientare il sistema, la parte integrale serve come archivio di riferimento e verifica.

ECONOMETRIC METHODS OF SIGNAL EXTRACTION

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The Wiener–Kolmogorov signal extraction filters, which are widely used in econometric analysis, are constructed on the basis of statistical models of the processes generating the data. In this paper, such models are used mainly as heuristic devices that are to be specified in whichever ways are appropriate to ensure that the filters have the desired characteristics. The digital Butterworth filters, which are described and illustrated in the paper, are specified in this way.

The components of an econometric time series often give rise to spectral structures that fall within well-defined frequency bands that are isolated from each other by spectral dead spaces. We find that the finite-sample Wiener–Kolmogorov formulation lends itself readily to a specialisation that is appropriate for dealing with band-limited components.

1. Introduction

The econometric methods of signal extraction that are based on linear filters have attained a high level of sophistication. They have now coalesced into two distinct categories.

In the first category are the methods that are favoured by the central statistical agencies of many of the OECD countries. These were developed in the North American agencies, notably in Statistics Canada (Dagum 1980) and in the U.S. Bureau of the Census (Findley *et al.* 1998), and they have been supported and refined in other agencies throughout Europe and in the Antipodes.

The methods are based upon a variety of moving-average smoothing filters, and they are used, principally, in trend extraction and in deseasonalising data series. Considerable efforts have been devoted to dealing with the end effects that can arise when such filters are applied to short non-stationary sequences.

The success of the methods of the central statistical agencies has entailed a widespread acceptance of a set of common conventions and definitions. Thus, it is commonly agreed that a deseasonalised data series can be fairly and simply defined as the product of the relevant methods of the central statistical office.

However, the acceptance of such conventions, whilst necessary to ensure comparability of statistics across nations, inhibits scientific innovation and discovery. Therefore, academic interest has been focused mainly on the so-called model-based procedures, which constitute the second category of econometric signal-extraction methodology.

The model-based approaches are derived from the idea that the components of an econometric times series, which are its trend, its secular cycles, its seasonal cycles and its irregular component, can all be modelled by autoregressive integrated moving-average (ARIMA) processes of low orders. There are several exponents of this genre who have taken slightly different approaches.

In the structural approach, which originated with Harrison and Stevens (1976) and which has been developed by Harvey (1989) and others, such models are fitted jointly to the data in a manner that renders their parameters readily accessible. (These methods have been implemented in the STAMP program of Koopman et al. 2000.) In the alternative canonical approach, which has been advocated by Hillmer and Tiao (1982) and by Maravall and Pierce (1987), amongst others, the models of the individual components must be disentangled from a fitted ARIMA model that represents their joint effects. (An accessible implementation is in the SEATS-TRAMO program of Gómez and Maravall, 1994, 1996.)

The model-based procedures have the seeming advantage that they subsist within a framework that facilitates conventional statistical inference. Thus, for example, confidence intervals are easily generated that can surround the estimated data components. However, the validity of such inferences depends crucially upon the cogency of the linear time-invariant ARIMA models that are applied to the components. The ability of such models to reflect the underlying data structures is limited. In particular, the models are liable to be subverted whenever the structures show any significant tendency to evolve through time.

In this paper, we also use models in deriving the filters that are used to isolate the data components, but the models will be treated mainly as heuristic devices that allow us to exploit the mathematical formalisms of the Wiener-Kolmogorov theory of signal extraction. This theory indicates that the optimal estimates of the data components are provided by their conditional expectations that are formed in the light of the observed data and of the models that are presumed to have generated them.

When the models themselves are to be fitted to the data, it is important that they should be realistic—otherwise they will provide an insecure basis for forming the supposedly optimal filters. However, in practice, they rarely achieve much realism. When the models are used merely as heuristic devices, they can be specified in whichever ways are appropriate to ensure that the filters have the desired characteristics. Moreover, the desirable characteristics will be unaffected, in the main, by the evolutions of the data components that the resulting filters are designed to isolate.

The original Wiener-Kolmogorov theory was developed under the fictional assumption that the data are generated by a stationary stochastic process and that they form a doubly infinite sequence. In econometrics, one has to contend with short non-stationary sequences; and, to cope with these, it is common to resort to the Kalman filter.

The Kalman filter and the associated smoothing algorithms are complicated and powerful devices, of which the workings can often seem obscure. The difficulty can be attributed to the all-encompassing nature of the algorithms.

In this paper, we shall pursue a simpler approach that fulfils the same objective of obtaining quasi minimum-mean-square-error estimates, but which deals directly with the specific features of the problem at hand. We shall use the finite-sample version of the bidirectional Wiener–Kolmogorov filter that has been expounded in previous papers of the present author (see Pollock, 1997, 2000, 2001, and 2002).

One of the contentions of this paper is that the components of an econometric time series often give rise to spectral structures that fall within well-defined frequency bands that are isolated from each other by spectral dead spaces. This leads us to consider the nature of band-limited stochastic processes, which are characterised by singular dispersion matrices. We find that the finite-sample Wiener–Kolmogorov formulation lends itself readily to a specialisation that is appropriate for dealing with band-limited components.

2. Filtering Short Stationary Sequences

We begin by considering the problem of estimating the signal component $\xi(t)$ and the noise component $\eta(t)$ of a data sequence

$$y(t) = \xi(t) + \eta(t), \quad (1)$$

where $t \in \{0, \pm 1, \pm 2, \dots\}$ is the index of the discrete-time observations. According to the classical assumptions, which we shall later amend in various ways, the signal and the noise are generated by stationary stochastic processes that are mutually independent. It follows that the autocovariance generating function of the data, defined by

$$\gamma(z) = \gamma_0 + \sum_{\tau=1}^{\infty} \gamma_{\tau}(z^{\tau} + z^{-\tau}), \quad (2)$$

is the sum of the autocovariance generating functions of its components. Thus

$$\gamma(z) = \gamma_{\xi}(z) + \gamma_{\eta}(z). \quad (3)$$

In practice, the available data will form a finite sequence which constitutes a vector $y = [y_0, y_1, \dots, y_{T-1}]'$ with a signal component ξ and a noise component η such that

$$y = \xi + \eta. \quad (4)$$

The data might owe their stationarity to a prior differencing operation or to an operation that has involved the extraction of a trend from the original data and the retention of the residue. For these vectors, the moment matrices are

$$\begin{aligned} E(\xi) &= 0, & D(\xi) &= \Omega_{\xi}, \\ E(\eta) &= 0, & D(\eta) &= \Omega_{\eta}, \\ \text{and } C(\xi, \eta) &= 0. \end{aligned} \quad (5)$$

The independence of ξ and η implies that $D(y) = \Omega = \Omega_\xi + \Omega_\eta$.

Here, the various variance–covariance or dispersion matrices, which have a Toeplitz structure, may be obtained by replacing the argument z within the relevant generating function by the matrix

$$L_T = [e_1, \dots, e_{T-1}, 0], \quad (6)$$

which is obtained from the identity matrix $I_T = [e_0, e_1, \dots, e_{T-1}]$ by deleting the leading column and appending a column of zeros to the end of the array. The matrix L_T , which has units on the first subdiagonal and zeros elsewhere, is the finite-sample version of the lag operator. Using it in place of z in $\gamma(z)$ gives

$$D(y) = \Omega = \gamma_0 I + \sum_{\tau=1}^{T-1} \gamma_\tau (L_T^\tau + F_T^\tau), \quad (7)$$

where $F_T = L_T'$ is in place of z^{-1} . Since L_T and F_T are nilpotent of degree T , such that $L_T^q, F_T^q = 0$ when $q \geq T$, the index of summation has an upper limit of $T - 1$.

The optimal predictors of the components ξ and η are their conditional expectations, denoted by x and h , respectively, in (8) and (9):

$$\begin{aligned} E(\xi|y) &= E(\xi) + C(\xi, y)D^{-1}(y)\{y - E(y)\} \\ &= \Omega_\xi(\Omega_\xi + \Omega_\eta)^{-1}y = x, \end{aligned} \quad (8)$$

$$\begin{aligned} E(\eta|y) &= E(\eta) + C(\eta, y)D^{-1}(y)\{y - E(y)\} \\ &= \Omega_\eta(\Omega_\xi + \Omega_\eta)^{-1}y = h. \end{aligned} \quad (9)$$

These are the so-called Wiener–Kolmogorov estimates. They constitute the minimum-mean-square-error estimates of the components, under the assumption that the specifications in (5) are correct. The assumptions provide a set of ordinary positive-definite variance–covariance matrices that pertain to conventional linear stochastic processes, which have spectra that extend across the frequency range.

We should observe that adding the estimates gives

$$y = x + h, \quad (10)$$

which is to say that the estimated components add up to the data vector y , as do the true components ξ and η in equation (4). To calculate the estimates, we may begin by solving the equation

$$(\Omega_\xi + \Omega_\eta)b = y \quad (11)$$

for the value of b . Thereafter, we can find

$$x = \Omega_\xi b \quad \text{and} \quad h = \Omega_\eta b. \quad (12)$$

The solution to equation (11) may be found via a Cholesky factorisation that sets $\Omega_\xi + \Omega_\eta = GG'$, where G is a lower-triangular matrix. The system $GG'b = y$ may be cast in the form of $Gp = y$ and solved for p . Then $G'b = p$ can be solved for b .

The solution via the Cholesky decomposition constitutes a recursive bidirectional filtering process that generates the vector b via two passes running in opposite directions through the data. The vector p is the product of a pass that runs forwards in time, and the vector b is generated from p in a reverse-time pass. Then b is subjected to further non-recursive filtering operations, described by (12), which produce x and h .

Exactly the same results would be obtained, albeit in a more complicated way, by using the Kalman filter, in a forwards pass, and an associated smoothing algorithm, in a backwards pass. (For an exposition of the latter procedures, see, for example, Pollock 2003c, where the method of Ansley Kohn 1985 for initiating the recursions is also analysed.)

It is notable that the recursive filter weights that are provided by the rows of the matrices G and G' vary as the filter progresses through the sample. As the sample size increases, the weights in the final rows of G will tend asymptotically to the set of constant weights that would be derived under the assumption of a doubly-infinite data sequence.

In some small-sample implementations of the Wiener–Kolmogorov filter, a set of constant weights has been applied to data samples that have been extended at either end by backcasting and forecasting. The additional extra-sample values have been used in a run-up to the filtering process wherein the filter is stabilised by providing it with a plausible presample history, if it is working in the direction of time, or with a plausible post-sample future, if it is working backward in time. The manner of treating the end-of sample problem that is implicit in constructions presented here has the theoretical sanction that it conforms with the theory of minimum-mean-square-error prediction in finite samples.

The Wiener–Kolmogorov principle of signal extraction is the foundation of the model-based methods of unobserved components analysis that are nowadays in widespread use. The parameters of the filters are determined in the process of fitting ARIMA models to the data components. However, the principle also supports a variety of heuristic filters of which the parameters are determined by rule of thumb or in view of the desired characteristics of their frequency response functions.

Amongst such heuristic filters is the digital version of the Butterworth filter, which has been advocated by Pollock (1997, 1999, 2000, 2001a, 2001b). The frequency response of the filter is maximally flat in the vicinity of the zero frequency and it has a transition band, centered on a chosen cut-off frequency, that can be narrowed by increasing the filter order. (Gómez 2001 has also advocated the Butterworth filter, but he has widened its definition to include filters, such as the filter of Hodrick and Prescott 1997, that do not share the property of maximal flatness.)

The Butterworth filter, which is of the lowpass Wiener–Kolmogorov va-

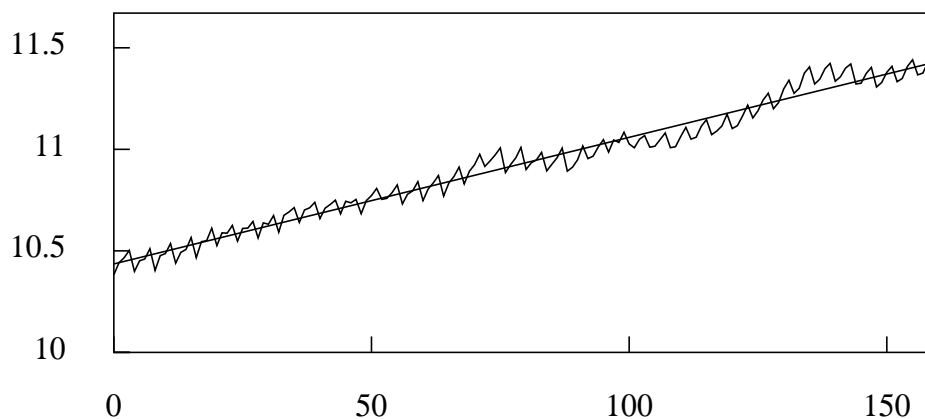


Figure 1. The quarterly series of the logarithms of consumption in the U.K., for the years 1955 to 1994, with a linear function interpolated by least-squares regression.

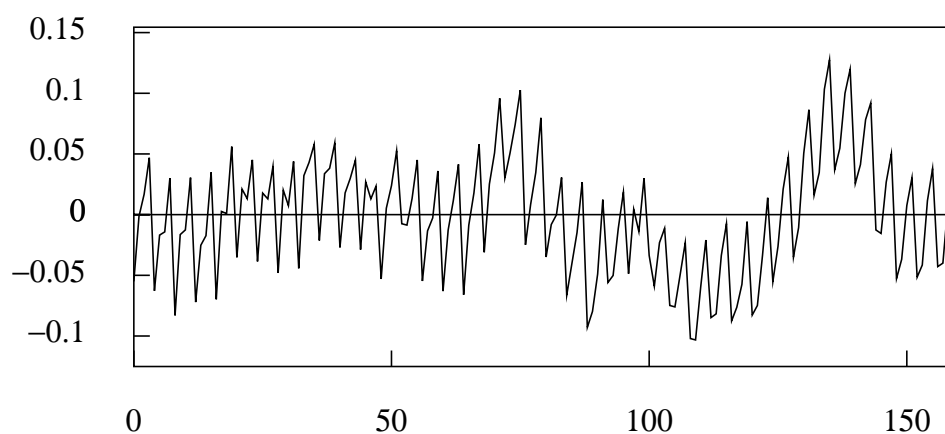


Figure 2. The residuals obtained by fitting a linear trend through the logarithmic consumption data of Figure 1.

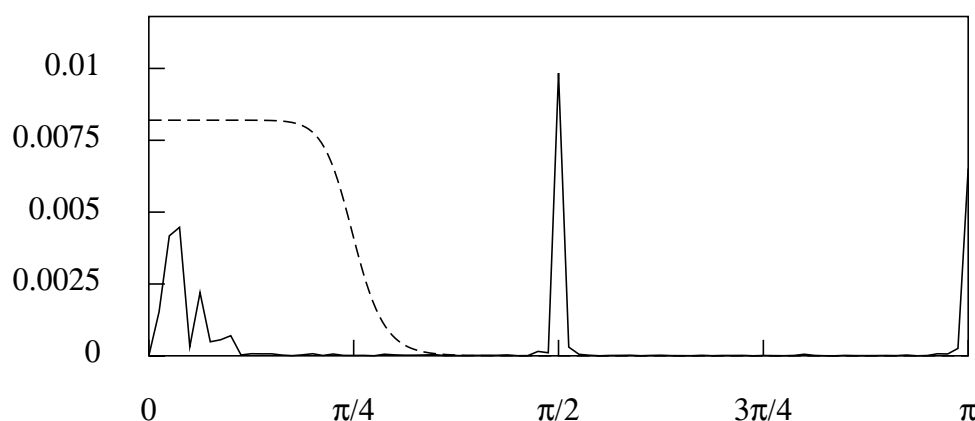


Figure 3. The periodogram of the residuals obtained by fitting a linear trend through the logarithmic consumption data of Figure 1. The gain of the lowpass Butterworth filter of order $n = 6$ and with a cut-off frequency of $\pi/4$ is represented by the dotted line. (The gain is unity at zero frequency.)

riety, originates in a function that is the ratio of two quasi autocovariance generating functions:

$$\begin{aligned}\psi(z) &= \frac{\sigma_\xi^2 \gamma_\xi(z)}{\sigma_\xi^2 \gamma_\xi(z) + \sigma_\eta^2 \gamma_\eta(z)} \\ &= \frac{(1+z)^n (1+z^{-1})^n}{(1+z)^n (1+z^{-1})^n + \lambda (1-z)^n (1-z^{-1})^n}.\end{aligned}\tag{13}$$

(Here, the normalised autocovariance functions $\gamma_\xi(z)$ and $\gamma_\eta(z)$, which are autocorrelation functions in other words, need to be scaled by the factors σ_ξ^2 and σ_η^2 respectively, which stand of the variances of the white-noise processes from which the signal and the noise components are supposedly derived by linear filtering.)

The autocovariance generating functions relate to an heuristic statistical model as opposed to a realistic one. The denominator function $\gamma(z) = \sigma_\xi^2 \gamma_\xi(z) + \sigma_\eta^2 \gamma_\eta(z)$ stands in place of that of the data process and the numerator function $\sigma_\xi^2 \gamma_\xi(z)$ corresponds to that of the signal. Here, $\lambda = \sigma_\eta^2 / \sigma_\xi^2 = \{1 / \tan(\omega_C / 2)\}^{2n}$ incorporates the nominal cut-off frequency of ω_C . By setting $z = L_T$ and $z^{-1} = F_T$ in the numerator and the denominator of $\psi(z)$, we derive the matrices Ω_ξ and $\Omega_\xi + \Omega_\eta$, respectively, which can be entered into equation (8).

To derive the two unidirectional filters, the rational function is factorised as $\psi(z) = \beta(z)\beta(z^{-1})$, where $\beta(z)$, which relates to the direct-time filter, contains the poles that lie outside the unit circle, and $\beta(z^{-1})$, which relates to the reverse-time filter, contains the poles that lie inside the circle. This factorisation is described as the Cramér–Wold decomposition. In the case of the Butterworth filter, analytic expressions for the roots of both the denominator and the numerator, i.e the poles and the zeros of the filter, are available. The roots of the denominator have been given by Pollock (2000).

For most other Wiener–Kolmogorov filters specified in the manner of the Butterworth filter, it is necessary to use an iterative procedure for finding the Cramér–Wold decomposition. (See for, example Pollock 2003b.) The algorithm of Wilson (1969), which is based on the Newton–Raphson procedure, is an effective way of achieving the factorisation; and versions which are coded in *C* and in *Pascal* have been provided by Pollock (1999). (See, also, Laurie 1980, 1982.)

There can be a reasonable objection to the assumption that the data components are generated by ordinary linear stochastic processes that comprise the full range of frequencies from zero up to the limiting Nyquist frequency of π radians per period. (In discretely sampled systems, the frequencies in excess of the Nyquist value will be aliased by frequencies within the interval $[0, \pi]$.) We shall illustrate the grounds for questioning the assumption via an analysis of a leading economic index.

Example 1. Figure 1 show the logarithms of the quarterly consumption data for the U.K. for the years 1955–1994, through which a linear trend has been interpolated by least-squares regression. When a quadratic polynomial trend was fitted, it was discovered that the coefficient associated with t^2 was not

significantly different from zero. This implies that, over the years in question, the underlying growth of the economy was at a constant exponential rate. The residual deviations from the trend, which are shown in Figure 2, represent a variable multiplicative factor by which underlying trend is modulated; and the residuals reveal both secular and seasonal variations in consumption.

The periodogram of the residuals is shown in Figure 3. This has a low-frequency spectral structure, which extends no further than the frequency value of $\pi/8$. The remainder of the periodogram shows a dead space that is punctuated by tall spikes in the vicinities of the frequencies of $\pi/2$ and π . The first of these spikes corresponds to the fundamental frequency of the seasonal fluctuations that play on the back of the more gentle variations that surround the ascending line in Figure 1. The spike at π corresponds to the first harmonic of the seasonal frequency.

The low-frequency structure of Figure 3, which occupies the frequency interval $[0, \pi/8]$, can be isolated successfully by any of a wide variety of filters. All that is required of such a filter is that its transition from pass band to stop band occurs within the spectral dead space that stretches from $\pi/8$ to the vicinity of $\pi/2$, where the spectral structure of the seasonal fluctuations is first encountered. The Butterworth filter of order $n = 6$ with a cut off frequency of $\pi/4$ fulfils this requirement. Its frequency response function is superimposed on Figure 3.

A more exacting task is the extraction of the low-frequency components from data that is observed at monthly intervals. In that case, the fundamental seasonal frequency is at $\pi/6$ and the transition of the filter must occur within a correspondingly reduced interval.

The sharpening of the transition can be achieved by raising the order n of the filter. However, a sharp transition in the low frequency range can be achieved with a recursive Wiener–Kolmogorov filter only at the cost of bringing the poles of the filter into close proximity with the perimeter of the unit circle. This can lead to problems of filter instability, which include the propagation of numerical rounding errors and the prolongation of the transient effects of ill-chosen start-up conditions.

These problems have been addressed within the context of the Wiener–Kolmogorov specification by Pollock (2003b). Alternative specifications for recursive filters have been investigated in Pollock (2003a). In the next section, we shall also deal with the problems of “sharp filtering” within the context of the Wiener–Kolmogorov theory; but we shall forsake the method of recursive filtering in favour of a method based on Fourier analysis.

3. Filtering via Circulant Matrices

A finite-sample analogue of a stationary stochastic process is a circular or periodic process $y(t) = \{y_t; t = 0, \pm 1, \pm 2, \dots\}$ that is completely specified by its values at T consecutive points such that $y_t = y_{t \bmod T}$. For such processes, the lag operator is replaced by the circulant matrix

$$K_T = [e_1, \dots, e_{T-1}, e_0], \quad (14)$$

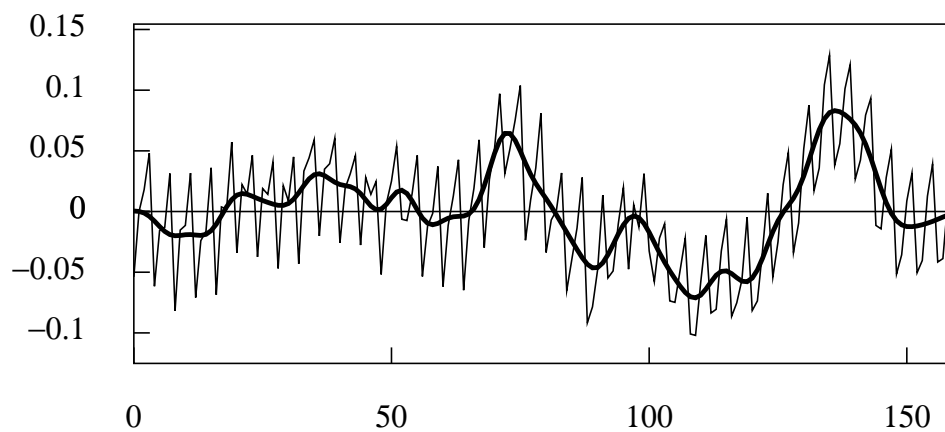


Figure 4. The low-frequency component of the consumption residuals of Figure 2. The component has been extracted by applying a lowpass Butterworth filter of order $n = 6$ with a cut off point at $\omega_c = \pi/4$.

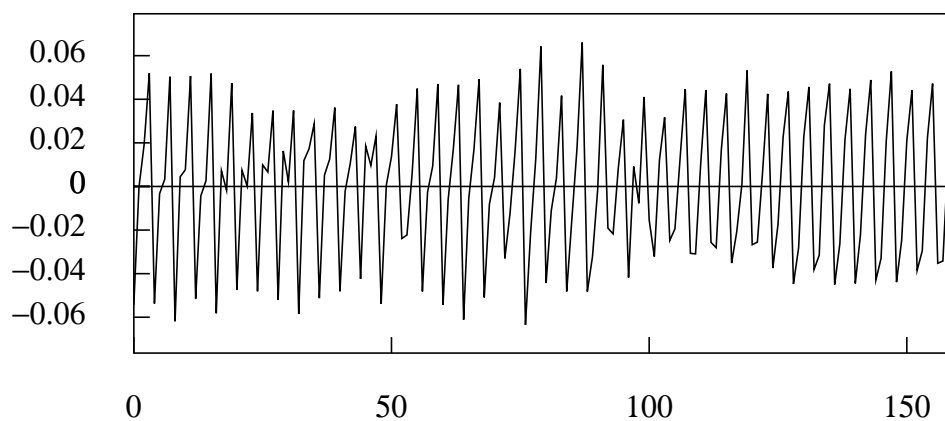


Figure 5. The component extracted from the consumption residuals by applying a highpass Butterworth filter of order 6 with a cut off point at $\omega_c = \pi/4$.

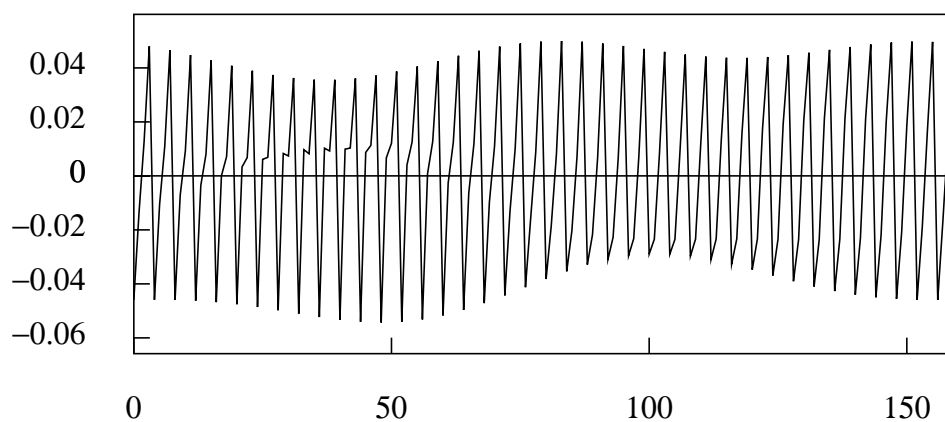


Figure 6. The seasonal component of the consumption residuals, synthesised from the Fourier ordinates in the vicinities of $\pi/2$ and π .

which is formed from the identity matrix I_T by moving the leading vector to the back of the array.

This operator effects the cyclic permutation of the elements of any (column) vector of order T . The matrix is T -periodic such that $K^{q+T} = K^q$. Whereas $L_T y = [0, y_0, \dots, y_{T-2}]$ is obtained from $y = [y_0, y_1, \dots, y_{T-1}]$ by deleting the final element and placing a zero in the leading position, the vector $K_T y = [y_{T-1}, y_0, \dots, y_{T-2}]$ is obtained from y by moving the final element to the leading position.

The powers of K form the basis for the set of circulant matrices. In particular, we may define a matrix of circular autocovariances via the formula

$$\begin{aligned} D^\circ(y) &= \Omega^\circ = \gamma(K) \\ &= \gamma_0 I + \sum_{\tau=1}^{\infty} \gamma_\tau (K^\tau + K^{-\tau}) \\ &= \gamma_0^\circ I + \sum_{\tau=1}^{T-1} \gamma_\tau^\circ (K^\tau + K^{-\tau}). \end{aligned} \tag{15}$$

Here, $\gamma_\tau^\circ; \tau = 0, \dots, T-1$ are the circular autocovariances defined by

$$\gamma_\tau^\circ = \sum_{j=0}^{\infty} \gamma_{(jT+\tau)}. \tag{16}$$

The matrix operator K has a spectral factorisation, which is particularly useful in analysing the properties of the discrete Fourier transform. The basis of this factorisation is the so-called Fourier matrix. This is a symmetric matrix

$$U = T^{-1/2} [W^{jt}; t, j = 0, \dots, T-1], \tag{17}$$

of which the generic element in the j th row and t th column is

$$\begin{aligned} W^{jt} &= \exp(-i2\pi tj/T) = \cos(\omega_j t) - i \sin(\omega_j t), \\ \text{where } \omega_j &= 2\pi j/T. \end{aligned} \tag{18}$$

The matrix U is a unitary, which is to say that it fulfils the condition

$$\bar{U}U = U\bar{U} = I, \tag{19}$$

where $\bar{U} = T^{-1/2} [W^{-jt}; t, j = 0, \dots, T-1]$ denotes the conjugate matrix.

The operator K can be factorised as

$$K = \bar{U} D U = U \bar{D} \bar{U}, \tag{20}$$

where

$$D = \text{diag}\{1, W, W^2, \dots, W^{T-1}\} \tag{21}$$

is a diagonal matrix whose elements are the T roots of unity, which are found on the circumference of the unit circle in the complex plane. Observe also that D is T -periodic, such that $D^{q+T} = D^q$, and that $K^q = \bar{U}D^qU = U\bar{D}^q\bar{U}$ for any integer q .

The spectral factorisation of the circulant autocovariance matrix gives

$$\Omega^\circ = \gamma(K) = \bar{U}\gamma(D)U. \quad (22)$$

Here, the j th element of the diagonal matrix $\gamma(D) = \Lambda$ is

$$\gamma(\exp\{i\omega_j\}) = \gamma_0 + 2 \sum_{\tau=1}^{\infty} \gamma_\tau \cos(\omega_j \tau). \quad (23)$$

This represents the cosine Fourier transform of the sequence of the ordinary autocovariances; and it corresponds to an ordinate (scaled by 2π) sampled at the point ω_j from the spectral density function of the linear (i.e. non-circular) stationary stochastic process. (An account of the algebra of circulant matrices has been provided by Pollock 2002. See, also, Gray 2002.)

The circulant autocovariance matrices that are the counterparts of the ordinary autocovariance matrices defined in (5) are

$$\begin{aligned} \Omega_\xi^\circ &= \bar{U}\Lambda_\xi U, & \Omega_\eta^\circ &= \bar{U}\Lambda_\eta U, \\ \Omega^\circ &= \bar{U}\Lambda U = \bar{U}(\Lambda_\xi + \Lambda_\eta)U, \end{aligned} \quad (24)$$

where Λ_η and Λ_ξ are diagonal matrices of spectral ordinates. Any of these autocovariance matrices may be singular in consequence of the presence of zero elements on the diagonals. Using the circulant matrices instead of the ordinary autocovariance matrices in the Wiener–Kolmogorov formulae of (8) and (9) gives

$$x = \bar{U}\Lambda_\xi\{\Lambda_\xi + \Lambda_\eta\}^+Uy = \bar{U}J_\xi Uy, \quad (25)$$

$$h = \bar{U}\Lambda_\eta\{\Lambda_\xi + \Lambda_\eta\}^+Uy = \bar{U}J_\eta Uy. \quad (26)$$

To accommodate the possibility that $\Lambda_\xi + \Lambda_\eta$ is singular, a generalised inverse has been applied to it instead of an ordinary inverse. A generalised inverse can be obtained by replacing the zero-value diagonal elements, which correspond to spectral ordinates falling within dead spaces, by nonzero values and, thereafter, by inverting the matrix that has acquired the full rank. Observe that, if Λ_ξ and Λ_η are disjoint such that $\Lambda_\xi\Lambda_\eta = 0$, then $J_\xi = \Lambda_\xi\{\Lambda_\xi + \Lambda_\eta\}^+$ is a matrix with units on the diagonal wherever Λ_ξ has nonzero elements and with zeros elsewhere. Analogous conditions apply to $J_\eta = \Lambda_\eta\{\Lambda_\xi + \Lambda_\eta\}^+$.

The formulae of (25) and (26) have a simple interpretation. First, the discrete Fourier transform is applied to the data vector y to translate it into the frequency domain. Then, a differential weighting, which might entail setting some values to zero, is applied to the spectral ordinates of the transformed

vector via the diagonal matrices J_ξ or J_η . Finally, to produce the estimate of the component, the inverse Fourier transform is applied.

Implicit in the use of the discrete Fourier transform is the assumption that the data sequence represents a single cycle of a periodic function. In the periodic extension of the data, the values from the interval $[0, T)$ are reproduced in successive segments of length T that precede and follow the data.

In one sense, there is no start-up problem affecting a Fourier-based filtering procedure, since the periodic extension constitutes a doubly-infinite sequence. However, there may be radical disjunctions at the points where one replication of the data ends and another begins.

Such features are liable to be reflected in the periodogram in a way that can obscure the underlying data structures. Thus, the ordinates of the Fourier transform may be affected by a slew of values which serve the purpose only of synthesising the end-of-sample disjunctions. One recourse is to taper both ends of the sample so that they arrive the same level. Another recourse is to join the sample to its mirror-image reflection and to use this combination in place of the original data.

The problems of an end-of-sample disjunction are particularly acute in the case of nonstationary data sequences that follow rising or falling trends; and the trends have to be eliminated before the filters are applied. So far, we have succeeded in eliminating the trend by fitting a polynomial function to the data. An alternative recourse, which we shall pursue in the next section, is to make use of differencing.

Example 2. Consider the task of extracting the seasonal component from the residuals that have been obtained by fitting a linear function to the logarithmic consumption data. The periodogram of Figure 3 suggests that the seasonal sequence should be synthesised from a small number of Fourier ordinates that are in the vicinity of the seasonal frequency and its harmonic. In addition to the ordinates at $\pi/2$, we may take two ordinates below and one above. Also, we may take the ordinate at π and the one immediately below.

The seasonal sequence, which is plotted in of Figure 6, is equally a component of the sequence of Figure 4, which represents the residuals from the linear detrending of the logarithmic consumption data, and a component of the sequence of Figure 5, which has been derived by applying a highpass Butterworth filter to remove a further low-frequency component—represented by the thick line in Figure 4—that is unrelated to the seasons. In terms of the variances, the seasonal sequence represents 47 percent of the Figure 4 sequence and 94 percent of the Figure 5 sequence.

4. Filtering Nonstationary Sequences

The problems of a trended data sequence may be overcome by differencing. The matrix that takes the d -th difference of a vector of order T is given by

$$\nabla_T^d = (I - L_T)^d. \quad (27)$$

We may partition the matrix so that $\nabla_T^d = [Q_*, Q']'$, where Q_* has d rows. The inverse matrix is partitioned conformably to give $\nabla_T^{-d} = [S_*, S]$. We may

observe that

$$\begin{bmatrix} S_* & S \end{bmatrix} \begin{bmatrix} Q'_* \\ Q' \end{bmatrix} = S_* Q'_* + S Q' = I_T, \quad (28)$$

and that

$$\begin{bmatrix} Q'_* \\ Q' \end{bmatrix} \begin{bmatrix} S_* & S \end{bmatrix} = \begin{bmatrix} Q'_* S_* & Q'_* S \\ Q' S_* & Q' S \end{bmatrix} = \begin{bmatrix} I_d & 0 \\ 0 & I_{T-d} \end{bmatrix}. \quad (29)$$

When the difference operator is applied to the data vector y , the first d elements of the product, which are in g_* , are not true differences and they are liable to be discarded:

$$\nabla_T^d y = \begin{bmatrix} Q'_* \\ Q' \end{bmatrix} y = \begin{bmatrix} g_* \\ g \end{bmatrix}. \quad (30)$$

However, if the elements of g_* are available, then the vector y can be recovered from $g = Q'y$ via the equation

$$y = S_* g_* + S g. \quad (31)$$

The columns of the matrix S_* provide a basis for the set of polynomials of degree $d - 1$ defined over the integer values $t = 0, 1, \dots, T - 1$. Therefore, $p = S_* g_*$ is a vector of polynomial ordinates whilst g_* can be regarded as a vector of d polynomial parameters.

We may approach the filtering of a trended data sequence in the following manner. First, we reduce the data to stationarity by differencing it an appropriate number of times. (We rarely need to difference the data more than twice.) From the differenced data, viewed in an appropriate manner, we may discern the nature and the frequency ranges of the various data structures that we wish to isolate.

Next, the components of the differenced data that correspond to these structures may be extracted, either by a recursive filtering process—using, for example, a Butterworth filter—or via the Fourier method described in the preceding section.

Finally, the components of the differenced data may be integrated, with an appropriate choice of initial conditions, to provide estimates of the components of the original trended sequence.

An apparent problem with this procedure is that the act of differencing is liable to attenuate the components of the low-frequency data structure to such an extent that they become invisible in the periodogram of the differenced data. The problem is illustrated in Figure 8, which shows the periodogram of $g = Q'y$ in the case of the once-differenced consumption data.

The problem vanishes when we recognise that we can discern the low-frequency structure via the periodogram of the residual sequence

$$\begin{aligned} y - p &= y - S_*(S'_* S_*)^{-1} S'_* y \\ &= Q(Q'Q)^{-1} Q' y, \end{aligned} \quad (32)$$

obtained by fitting to the data, by least-squares, a polynomial of degree $d - 1$. The identity $Q(Q'Q)^{-1} Q' = I - S_*(S'_* S_*)^{-1} S'_*$ follows from the fact that Q

and S_* are complementary matrices with $\text{Rank}[Q, S_*] = T$ and $Q'S_* = 0$. It will be recognised that the residuals contain the same information as does the differenced data $Q'y$. Their periodogram, in the case of the consumption data, has been displayed already in Figure 3.

To elucidate the procedures for extracting the components of a trended data sequence, let us consider the case of the data vector $y = \xi + \eta$, where η , which has $E(\eta) = 0$ and $D(\eta) = \Omega_\eta$, is from a stationary stochastic process and where ξ is from a process that requires a d -fold differencing in order to reduce it to a vector $\zeta = Q'\xi$ with a stationary distribution. Then we shall have

$$\begin{aligned} Q'y &= Q'\xi + Q'\eta, \\ &= \zeta + \kappa = g, \end{aligned} \quad (33)$$

and we may assume, by analogy with (5), that ζ and κ are characterised by their first and second moments, which are

$$\begin{aligned} E(\zeta) &= 0, & D(\zeta) &= \Omega_\zeta = Q'\Omega_\xi Q, \\ E(\kappa) &= 0, & D(\kappa) &= \Omega_\kappa = Q'\Omega_\eta Q, \\ \text{and } C(\zeta, \kappa) &= 0. \end{aligned} \quad (34)$$

Here, the derived dispersion matrices Ω_ζ and Ω_κ retain the Toeplitz structure that is a feature of Ω_ξ and Ω_η .

Let the estimates of ζ and κ be denoted by z and k . If x and h are the estimates of ξ and η respectively, then it is reasonable to require that $Q'x = z$ and $Q'h = k$ so that

$$\begin{aligned} Q'y &= Q'x + Q'h \\ &= z + k = g. \end{aligned} \quad (35)$$

The estimates z and k must be integrated to give

$$x = S_*z_* + Sz \quad \text{and} \quad h = S_*k_* + Sk. \quad (36)$$

The criterion for finding the starting value z_* is

$$\text{Minimise } (y - x)' \Omega_\eta^{-1} (y - x) = (y - S_*z_* - Sz)' \Omega_\eta^{-1} (y - S_*z_* - Sz). \quad (37)$$

This requires that the estimated trend x should adhere as closely as possible to the data. The minimising value is

$$z_* = (S_*' \Omega_\eta^{-1} S_*)^{-1} S_*' \Omega_\eta^{-1} (y - Sz) \quad (38)$$

Since $y - x = h$, an equivalent criterion is

$$\text{Minimise } h' \Omega_\eta^{-1} h = (S_*k_* + Sz)' \Omega_\eta^{-1} (S_*k_* + Sk). \quad (39)$$

for which the minimising value is

$$k_* = -(S_*' \Omega_\eta^{-1} S_*)^{-1} S_*' \Omega_\eta^{-1} Sk. \quad (40)$$

Using

$$P_* = S_*(S'_*\Omega_\eta^{-1}S_*)^{-1}S'_*\Omega_\eta^{-1}, \quad (41)$$

we get, from (36), the following values:

$$x = P_*y + (I - P_*)Sz, \quad \text{and} \quad h = (I - P_*)Sk. \quad (42)$$

The disadvantage in using these formulae directly is that the inverse matrix Ω_η^{-1} , which is of order T , is liable to have nonzero elements in every location. (This will be so whenever Ω_η has the form of an autocovariance matrix of a moving-average process, as it does in the case of the Butterworth filter, for example.)

The appropriate recourse is to use the identity

$$\begin{aligned} I - P_* &= I - S_*(S'_*\Omega_\eta^{-1}S_*)^{-1}S'_*\Omega_\eta^{-1} \\ &= \Omega_\eta Q(Q'\Omega_\eta Q)^{-1}Q' \end{aligned} \quad (43)$$

to provide an alternative expression for the projection matrix $I - P_*$ that incorporates the band-limited matrix Ω_η instead of its inverse. The equality follows from the fact that, if $\text{Rank}[R, S_*] = T$ and if $S'_*\Omega_\eta^{-1}R = 0$, then

$$I - S_*(S'_*\Omega_\eta^{-1}S_*)^{-1}S'_*\Omega_\eta^{-1} = R(R'\Omega_\eta^{-1}R)^{-1}R'\Omega_\eta^{-1}. \quad (44)$$

Setting $R = \Omega_\eta Q$ gives the result. Given $x = y - h$, it follows that we can write

$$\begin{aligned} x &= y - (I - P_*)Sk \\ &= y - \Omega_\eta Q(Q'\Omega_\eta Q)^{-1}k, \end{aligned} \quad (45)$$

where the second equality depends upon $Q'S = I$.

So far, we have not specified the precise method by which the estimates z and k of the differenced components have been obtained. They may be obtained equally via a recursive filtering method or via the Fourier that has been outlined in the preceding section. In case we have used the Fourier method, we might be inclined to use the circulant version of the dispersion matrix Ω_η within the foregoing formulae.

Let us consider, instead, the possibility of obtaining the estimate k via recursive filtering. Then, with reference to equation (9), we can see that the assumptions of (34) imply that the estimate should take the form of

$$k = Q'\Omega_\eta Q(\Omega_\zeta + Q'\Omega_\eta Q)^{-1}Q'y, \quad (46)$$

On substituting this in the equation of (45), we get

$$x = y - \Omega_\eta Q(\Omega_\zeta + Q'\Omega_\eta Q)^{-1}Q'y. \quad (47)$$

In the case of Butterworth filter, we take the quasi-autocorrelation functions of the nonstationary signal sequence $\xi(t)$ and of the stationary noise sequence $\eta(t)$ to be

$$\gamma_\xi(z) = \sigma_\xi^2 \frac{(1+z)^n(1+z^{-1})^n}{(1-z)^d(1-z^{-1})^d} \quad \text{and} \quad \gamma_\eta(z) = \sigma_\eta^2(1-z)^{n-d}(1-z^{-1})^{n-d} \quad (48)$$

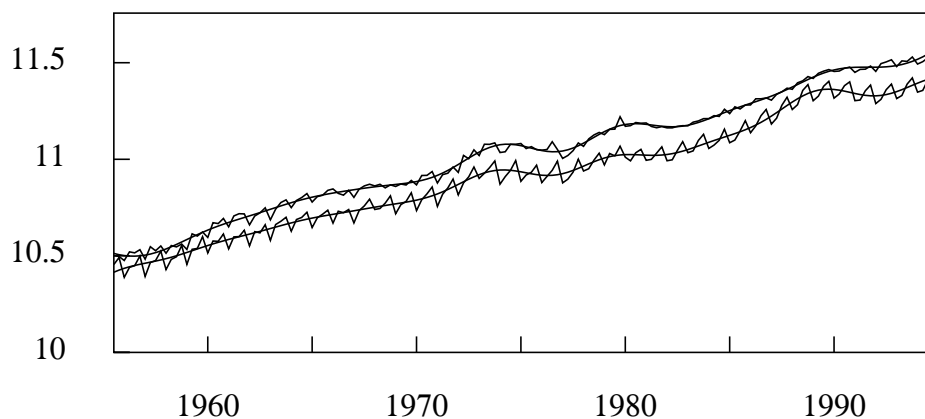


Figure 7. The quarterly series of the logarithms of income (upper) and consumption (lower) in the U.K for the years 1955 to 1994 together with their interpolated trends.

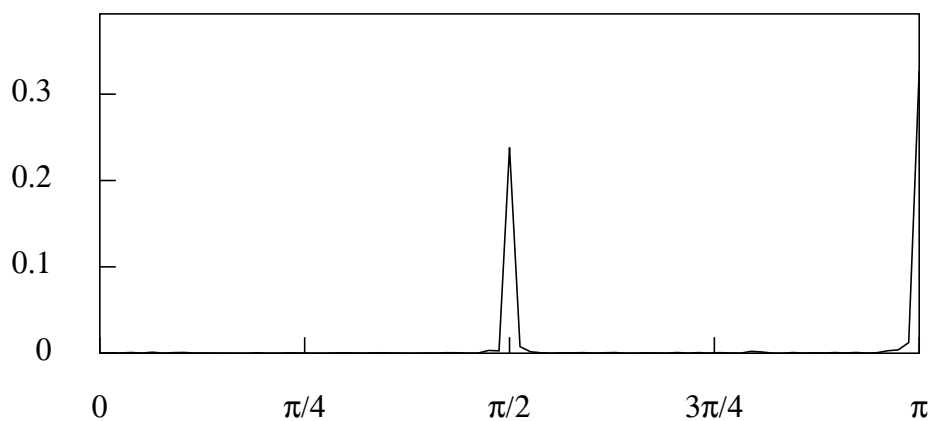


Figure 8. The periodogram of the first differences of the logarithmic consumption data.

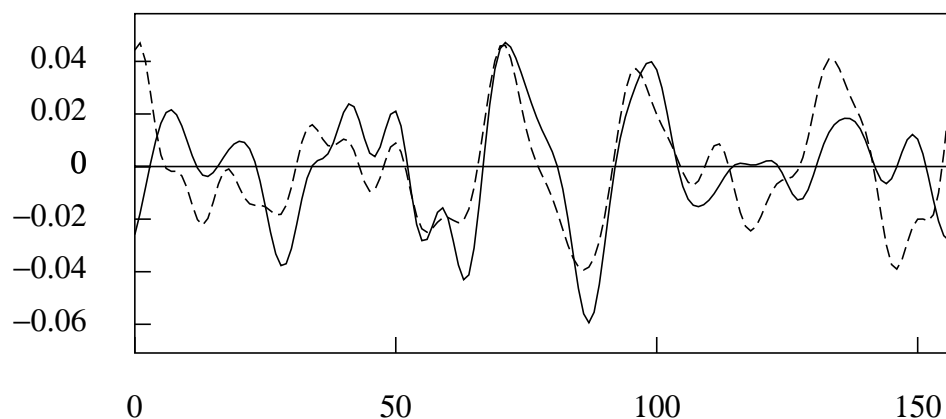


Figure 9. The bandpass estimates of the fluctuations, within the range of the business-cycle frequencies, of the logarithmic income series (solid line) and of the logarithmic consumption series (broken line).

respectively, which become the elements of (13) in the case where $d = 0$. We may also define

$$\begin{aligned}\gamma_\zeta(z) &= \sigma_\xi^2(1+z)^n(1+z^{-1})^n \quad \text{and} \quad \gamma_\kappa(z) = (1-z)^d\gamma_\eta(z)(1-z^{-1})^d \\ &= \sigma_\eta^2(1-z)^n(1-z^{-1})^n,\end{aligned}\tag{49}$$

which is a matter of renaming the elements of (13) when $d > 0$. The matrices Ω_ζ and $\Omega_\kappa = Q'\Omega_\eta Q$ are generated by setting $z = L_{T-d}$ and $z^{-1} = L'_{T-d} = F_{T-d}$ in $\gamma_\zeta(z)$ and $\gamma_\kappa(z)$ respectively and by scaling the resulting matrices by the appropriate variances. Observe that the generating functions of (49) are not affected by the order d of the differencing operator. Therefore, for the Butterworth filter, only the dimension of the matrix $\Omega_\zeta + Q'\Omega_\eta Q$ changes when d varies. Its essential structure remains the same.

The computational procedure that has been described in section 2 can also be applied when $d > 0$. That is to say, the solution of the equation $(\Omega_\zeta + Q'\Omega_\eta Q)b = g$, where $g = Q'y$, is found via the Cholesky factorisation of $\Omega_\zeta + Q'\Omega_\eta Q = GG'$. Thereafter, $h = \Omega_\eta b$ and $x = y - h$ are found.

Example 3. Figure 7 shows the quarterly sequences of the logarithms of income (upper) and consumption (lower) in the U.K. for the years 1955 to 1994 together with their interpolated trends. We can afford to treat the income sequence in the same manner as we treat the consumption sequence; and, in what follows, we shall concentrate on the latter.

The periodogram of Figure 3, suggests that both the trend component and the seasonal component of the consumption data are generated by band-limited processes. The trend component is confined to the frequency interval $[0, \pi/8]$ and the seasonal component comprises a handful of nonzero Fourier ordinates in the vicinities of $\pi/2$ and π . The remainder of the periodogram consists of virtual dead spaces. When equation (4) is applied to these circumstances, ξ , which is estimated by x , becomes the trend component and η , which is estimated by h , becomes the seasonal component.

The trend that interpolates the consumption data has been constructed by extracting from the untrended, twice-differenced data sequence $g = Q'y$ the Fourier elements that lie in the frequency interval $[0, \pi/8]$. The sequence z that is synthesised from these elements has then been integrated to create the trend $x = S_*z_* + Sz$. In seeking the starting value z_* with which to initiate the process of integration, we may consider minimising a criterion function in the form of $h'(\Omega_\eta^\circ)^+h = h'\bar{U}\Lambda_\eta^+Uh$, where $(\Omega_\eta^\circ)^+ = \bar{U}\Lambda_\eta^+U$ represents the generalised inverse of the singular circulant autocovariance matrix $D^\circ(\eta) = \Omega_\eta^\circ$.

The elements of Λ_η^+ that correspond to zero-valued elements of Uh , which lie in spectral dead spaces, can take arbitrary values. These values will have no effect upon the value of the criterion function. Therefore, the generalised inverse can be formed by replacing the nonzero elements of Λ_η by their inverses and by placing arbitrary values elsewhere on the diagonal.

For want of a better assumption, we may assume that the Fourier ordinates of the seasonal process are distributed uniformly within their designated bands.

In that case, the corresponding elements of Λ_η should all have same value, and so, likewise, should the corresponding elements of Λ_η^+ .

The remaining elements of Λ_η^+ , which correspond to zero-valued Fourier ordinates and which can take arbitrary values, may be set to the same values as the elements corresponding to the seasonal ordinates. Thus Λ_η^+ , which needs to be determined only up to a scalar factor, becomes an arbitrary multiple of the identity matrix—and it may as well become the identity matrix itself. In that case, we should have $\Omega_\eta^\circ = \bar{U}U = I$ and $(\Omega_\eta^\circ)^+ = I$.

This simplification allows us to specialise equation (38) to give

$$z_* = (S'_* S_*)^{-1} S'_* (y - Sz). \quad (50)$$

In the case where the data is differenced twice, there is

$$S'_* = \begin{bmatrix} 1 & 2 & \dots & T-1 & T \\ 0 & 1 & \dots & T-2 & T-1 \end{bmatrix} \quad (51)$$

The elements of the matrix $S'_* S_*$ can be found via the formulae

$$\begin{aligned} \sum_{t=1}^T t^2 &= \frac{1}{6} T(T+1)(2T+1) \quad \text{and} \\ \sum_{t=1}^T t(t-1) &= \frac{1}{6} T(T+1)(2T+1) - \frac{1}{2} T(T+1). \end{aligned} \quad (52)$$

(A compendium of such results has been provided by Jolly 1961, and proofs of the present results were given by Hall and Knight 1899.) The matrix is somewhat ill-conditioned. Moreover, when the order of differencing exceeds two or three, it is necessary, in calculating the polynomial ordinates of $p = S_* z_*$, to use to an orthogonal basis in place of the monomial basis that is provided by the columns of S_* . However, this case is rare.

5. Bandpass Filtering

Econometricians often characterise the business cycle in terms of a sinusoid that fluctuates around a slow-moving trend. According to the definitions of Burns and Mitchell (1946), the effects of the business cycle within an economic index correspond to the sinusoidal elements therein that have periods of no less than one-and-a-half years and of no more than eight years. A duration of one-and-a-half years seems too short, and we prefer to set the shortest duration at 2 years—and this seems to be a common preference (see, for example, Christiano and Fitzgerald 1998).

The business cycle, defined in this manner, is unlikely to correspond to any self-contained spectral structure that might be discerned by inspecting the relevant periodogram. In the case of quarterly data, the business cycle frequencies range from $\pi/16$ radians per period to $\pi/4$ radians per period (corresponding to a duration of 2 years.) Neither of these values corresponds to a natural break in the periodogram of the consumption residuals of Figure 3.

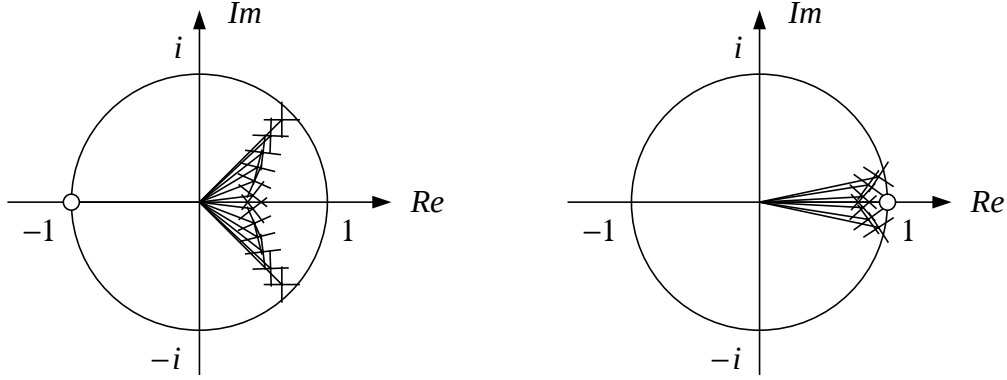


Figure 10. The pole-zero diagrams (left) of the lowpass Butterworth filter of order $n = 12$ with a cut-off frequency of $\omega_U = \pi/4$ and (right) of the highpass filter of order $n = 6$ with a cut-off frequency of $\omega_L = \pi/16$.

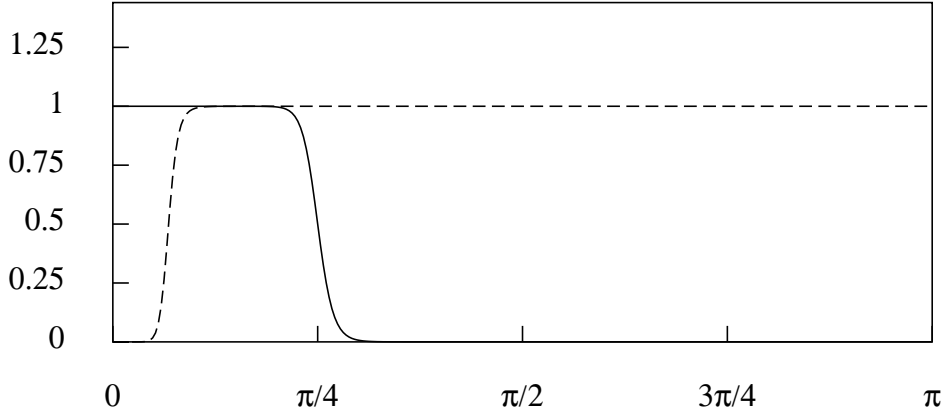


Figure 11. The gain of the 12th order digital Butterworth lowpass filter with cut-off frequency of $\omega_U = \pi/4$ and of 6th order highpass filter with cut-off frequency of $\omega_L = \pi/16$, superimposed on the same diagram.

The business-cycle frequencies may be extracted from the data using a bandpass filter with nominal cut-off points at the designated frequencies. For this purpose, economists have tended to use finite-impulse-response (FIR) or moving-average filters that are derived by truncating the doubly-infinite sequence of filter coefficients associated with the unrealisable ideal bandpass filter. (See, for example, Baxter and King, 1999.) The effect of the truncation is to create ripples in the stopbands of the frequency response function, which entail considerable spectral leakage.

A superior bandpass filter can be realised using the Butterworth formulation. One way of creating a bandpass filter is to apply the so-called Constantinides (1970) transformation to a prototype lowpass filter with a nominal cut-off point at $\pi/2$. The method is also described by Pollock (1999). In the current application of business cycle analysis, this transformation will result in a filter with a frequency response that has a far wider transition band at the upper cut-off frequency than at the lower cut-off frequency.

A better way of creating a bandpass filter for the current application is to apply two filters in succession. The first filter is a lowpass filter that is intended to remove the components of frequencies in excess of $\pi/4$. The second is a highpass filter that preserves the remaining components of frequencies in excess of $\pi/16$ and eliminates those of lesser frequencies. The order of the first filter should exceed that of the second filter so as to enhance the rate of transition at the upper cut-off frequency.

Figure 10 shows the pole-zero diagrams of the 12th order lowpass and the 6th order highpass filters; while Figure 11 shows the frequency response functions of the two filters superimposed on the same diagram. It can be seen that some of the poles of the highpass filter come very close to the circumference of the unit circle. This feature can lead to problems of numerical instability.

One way of overcoming the problems of numerical instability is to subsample the data that has resulted from applying the first filter. Since there is no information in this data remaining in the interval $[\pi/2, \pi]$, we can afford to omit alternate points so as to create a semi-annual sequence. The effect is that the contents of the original data that lie in the frequency interval $[0, \pi/2]$ are mapped into the wider interval $[0, \pi]$. In the process, the lower cut-off frequency moves from $\pi/16$ to $\pi/8$. The poles of the 6th-order Butterworth filter with this cut-off point are no longer so close to the perimeter of the unit circle, which implies a greater numerical stability. (More general methods of sample-rate conversion have been described by Vaidyanathan 1993, amongst others.)

An alternative recourse is to base the estimate of the business cycle component on the Fourier ordinates of the data that fall within the specified frequency range. In principal, the method entails no spectral leakage so long as it is applied to data that have been detrended in a manner that will ensure that there are no disjunctions in the periodic extension where the end of one data segment joins the beginning of another. This can be achieved by a process of differencing followed by a judicious tapering of the ends of the data segment.

Since the business cycle is an artificial construct, it is difficult to relate the method of extraction to an underlying statistical model. However, under certain assumptions, it becomes appropriate to treat this component in the same manner as the noise component η within the trended vector $y = \xi + \eta$, which has been the subject of the previous section.

Now the component vector ξ becomes the repository of the Fourier elements with frequencies that are less than the value of the lower cut-off frequency of the pass band. The components of frequencies in excess of the upper cut-off frequency can be assigned to a third component, which is eliminated via the first lowpass filtering operation. The vector y can be taken to represent the product of this operation.

Under these constructions, there are no spectral overlaps amongst the various components; and the appropriate statistical model is one that comprises separable band-limited processes. It follows that the appropriate method for extracting the business-cycle component is, indeed, the Fourier-based method of Section 3. This is well-adapted to dealing with band-limited processes. The

relevant Fourier components of the business cycle, contained in the vector k , must be extracted from a data vector $g = z + k$ that has been detrended by differencing. The estimate

$$h = S_* k_* + S k \quad (53)$$

of the business cycle component is obtained by a process of summation that reverses the differencing.

In the absence of prior knowledge of the distribution of the spectral ordinates, we may set $\Omega_\eta = I$. In that case, the starting values are provided by the simplified formula

$$k_* = (S'_* S_*)^{-1} S'_* S k. \quad (54)$$

which is derived from equation (40) by setting $\Omega_\eta^{-1} = I$. The simplification extends to the identity of (43), which becomes

$$\begin{aligned} P_* &= S_* (S'_* S_*)^{-1} S'_* \\ &= I - Q(Q'Q)^{-1}Q' = I - P_Q. \end{aligned} \quad (55)$$

Therefore, the estimate of the business cycle component is also provided by

$$h = (I - P_*) S k = Q(Q'Q)^{-1} k, \quad (56)$$

wherein the condition $Q'S = I$ has been effective in simplifying the final expression.

Example 4. Figure 9 shows the business cycle fluctuations that have been extracted from the quarterly logarithmic income and consumption data for the U.K. over the period 1955 to 1994. In both cases, a Fourier bandpass filter has been applied that has a lower cut-off point at $\pi/16$ radians per period (corresponding to a cycle of 8 years duration) and an upper cut-off point of $\pi/4$ radians per period (corresponding to a cycle of 2 years duration).

There is evidence here that the fluctuations in consumption precede those in income. This contradicts the common supposition that the business cycle is driven by variations in “autonomous expenditures”, which do not include consumption, and in the rate of investment.

One might be doubtful of the comparisons at the beginning and the end of the sample, where the interpolated functions are not tied down by preceding or succeeding data points and where they appear to be heading in opposite directions. The problem could be overcome by adding a few extrapolated points at either end of the sample that would serve to tie down the functions.

6. Multiple Components

The problems of econometric signal extraction have been handled, so far, within the context of a model, described by equation (4), that has only a signal component and a noise component. Allowance has been made for a non stationary signal component. However, it might be required to partition the data amongst more than two components. Thus, in a classical econometric time-series analysis, at least four components are identified. These are the trend, the business cycle, the seasonal cycle and an irregular component.

The two-component model can also serve the purpose of extracting several components, for the reason that its components are readily amenable, if necessary, to further decompositions. Thus, for example, an initial decomposition of the data sequence into a trend/cycle component and a residue can be followed by decomposition of the residue into a seasonal cycle and an irregular cycle. If the data are stationary, it is unnecessary to perform such a multiple decomposition sequentially—each component can be extracted separately.

If the data are nonstationary and if there are more than one nonstationary component, then a sequential decomposition might be called for. A typical model of an econometric time series, described by the equation $y = \xi + \eta = (\mu + \rho) + \eta$, comprises both a trend/cycle component μ and a seasonal component ρ that are described by ARIMA models with real and complex unit roots respectively.

To reduce the data to stationarity, an operator is used that is the product of the d -fold difference operator $\nabla_T^d = (I - L_T)^d$ and a deseasonalising operator $\Sigma_T = (I - L_T^s)(I - L_T)^{-1}$. (The operator Σ is used instead of $(I - L_T^s)$ because it can be assumed, without loss of generality, that the seasonal deviations from the trend have zero mean.) Let the product of the two operators be denoted by $M_T = \Sigma_T \nabla_T^d = [Q_*, Q']'$, where Q'_* contains the first $d + s - 1$ rows of the matrix, and let the inverse operator $M_T^{-1} = [S_*, S]$ be partitioned conformably such that S_* contains the first $d + s - 1$ columns. The factors of M_T^{-1} are further partitioned as $\Sigma_T^{-1} = [S_{\Sigma*}, S_{\Sigma}]$ and $\nabla_T^{-d} = [S_{\nabla*}, S_{\nabla}]$.

Let the components of the differenced data be denoted by $Q'\xi = \zeta$, $Q'\mu = \zeta_\mu$ and $Q'\rho = \zeta_\rho$. Then there is

$$\begin{aligned} Q'y &= Q'\xi + Q'\eta \\ &= Q'(\mu + \rho) + \kappa = (\zeta_\mu + \zeta_\rho) + \kappa. \end{aligned} \tag{57}$$

Also, let the estimates of μ and ρ be denoted by m and r and those of ζ_μ and ζ_ρ by z_m and z_r . Then, in parallel with equation (57), there is

$$\begin{aligned} Q'y &= Q'x + Q'h \\ &= Q'(m + r) + k = (z_m + z_r) + k. \end{aligned} \tag{58}$$

The estimates z_m , z_r and k may be obtained from the differenced data $g = Q'y$ by a process of linear filtering. It is then required to form m , r and h from these elements. First, consider

$$\begin{aligned} x &= (m + r) = S_* z_* + Sz \\ &= S_* z_* + S(z_m + z_r). \end{aligned} \tag{59}$$

Here, z_* is computed according to the formula of (38). Given x , an estimate $h = y - x$ of the irregular component can be formed. Next, there is an equation

$$S_* z_* = [S_{\nabla*} \quad S_{\Sigma*}] \begin{bmatrix} z_{*m} \\ z_{*r} \end{bmatrix}. \tag{60}$$

This may be solved uniquely for z_{*m} and z_{*r} ; and, for this purpose, only the first $s + d - 1$ rows of the system are required. Thereafter, the estimates of μ and ρ are given by

$$m = S_{\nabla*} z_{*m} + S z_m \quad \text{and} \quad r = S_{\Sigma*} z_{*r} + S z_r. \quad (61)$$

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STATISTICAL FOURIER ANALYSIS: CLARIFICATIONS AND INTERPRETATIONS

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This paper expounds some of the results of Fourier theory that are essential to the statistical analysis of time series. It employs the algebra of circulant matrices to expose the structure of the discrete Fourier transform and to elucidate the filtering operations that may be applied to finite data sequences.

An ideal filter with a gain of unity throughout the pass band and a gain of zero throughout the stop band is commonly regarded as incapable of being realised in finite samples. It is shown here that, to the contrary, such a filter can be realised both in the time domain and in the frequency domain.

The algebra of circulant matrices is also helpful in revealing the nature of statistical processes that are band limited in the frequency domain. In order to apply the conventional techniques of autoregressive moving-average modelling, the data generated by such processes must be subjected to anti-aliasing filtering and sub sampling. These techniques are also described.

It is argued that band-limited processes are more prevalent in statistical and econometric time series than is commonly recognised.

1. Introduction

Statistical Fourier analysis is an important part of modern time-series analysis, yet it frequently poses an impediment that prevents a full understanding of temporal stochastic processes and of the manipulations to which their data are amenable. This paper provides a survey of the theory that is not overburdened by inessential complications, and it addresses some enduring misapprehensions.

Amongst these is a misunderstanding of the effects in the frequency domain of linear filtering operations. It is commonly maintained, for example, that an ideal frequency-selective filter that preserves some of the elements of a time series whilst nullifying all others is incapable of being realised in finite samples.

This paper shows that such finite-sample filters are readily available and that they possess time-domain and frequency-domain representations that are both tractable. Their representations are directly related to the classical Wiener–Kolmogorov theory of filtering, which presupposes that the data can be treated as if they form a doubly-infinite or a semi-infinite sequence.

A related issue, which the paper aims to clarify, concerns the nature of continuous stochastic processes that are band-limited in frequency. The paper provides a model of such processes that challenges the common supposition that the natural *primum mobile* of any continuous-time stochastic process is Wiener process, which consists of a stream of infinitesimal impulses.

The sampling of a Wiener process is inevitably accompanied by an aliasing effect, whereby elements of high frequencies are confounded with elements of frequencies that fall within the range that is observable in sampling. Here, it is shown, to the contrary, that there exist simple models of continuous band-limited processes for which no aliasing need occur in sampling.

The spectral theory of time series is a case of a non-canonical Fourier theory. It has some peculiarities that originally caused considerable analytic difficulties, which were first overcome in a satisfactory manner by Norbert Wiener (1930) via his theory of a generalised harmonic analysis. The statistical filtering theory that is appropriate to such time series originated with Wiener (1941) and Kolmogorov (1941a).

Many of the difficulties of the spectral representation of a time series can be avoided by concentrating exclusively on its correlation structure. Then, one can exploit a simple canonical Fourier relationship that exists between the autocovariance function of a discrete stationary process and its spectral density function.

This relationship is by virtue of an inverted form of the classical Fourier series relationship that exists between a continuous, or piecewise continuous, periodic function and its transform, which is a sequence of Fourier coefficients. (In the inverted form of the relationship, which is described as the discrete-time Fourier transform, it is the sequence that is the primary function and the continuous periodic function that is its transform.) However, it is important to have a mathematical model of the process itself, and this is where some of the complications arise.

Figure 1 depicts what may be described as the canonical Fourier transforms. For conciseness, they have been expressed using complex exponentials,

whereas they might be more intelligible when expressed in terms of ordinary trigonometrical functions. When dealing with the latter, only positive frequencies are considered, whereas complex exponentials are defined for both positive and negative frequencies.

In the diagrams, the real-valued time-domain functions are symmetric, with the consequence that their frequency-domain transforms are also real-valued and symmetric. Under other conditions, the transforms would be complex-valued, giving rise to a pair of functions.

2. Canonical Fourier Analysis

The classical Fourier series, illustrated in Figure 1.(ii), expresses a continuous, or piecewise continuous, period function $x(t) = x(t+T)$ in terms of a sum of sine and cosine functions of harmonically increasing frequencies $\{\omega_j = 2\pi j/T; j = 1, 2, 3, \dots\}$:

$$\begin{aligned} x(t) &= \alpha_0 + \sum_{j=1}^{\infty} \alpha_j \cos(\omega_j t) + \sum_{j=1}^{\infty} \beta_j \sin(\omega_j t) \\ &= \alpha_0 + \sum_{j=1}^{\infty} \rho_j \cos(\omega_j t - \theta_j). \end{aligned} \quad (1)$$

Here, $\omega_1 = 2\pi/T$ is the fundamental frequency or angular velocity, which corresponds to a trigonometric function that completes a single cycle in the interval $[0, T]$, or in any other interval of length T such as $[-T/2, T/2]$. The second expression depends upon the definitions

$$\rho_j^2 = \alpha_j^2 + \beta_j^2 \quad \text{and} \quad \theta_j = \tan^{-1}(\beta_j/\alpha_j). \quad (2)$$

The equality follows from the trigonometrical identity

$$\cos(A - B) = \cos(A) \cos(B) + \sin(A) \sin(B). \quad (3)$$

In representing the periodic nature of $x(t)$, it is helpful to consider mapping the interval $[0, T]$, over which the function is defined, onto the circumference of a circle, such that the end points of the interval coincide. Then, successive laps or circuits of the circle will generate successive cycles of the function.

According to Euler's equations, there are

$$\cos(\omega_j t) = \frac{1}{2}(e^{i\omega_j t} + e^{-i\omega_j t}) \quad \text{and} \quad \sin(\omega_j t) = \frac{-i}{2}(e^{i\omega_j t} - e^{-i\omega_j t}). \quad (4)$$

Therefore, equation (1) can be expressed as

$$x(t) = \alpha_0 + \sum_{j=1}^{\infty} \frac{\alpha_j + i\beta_j}{2} e^{-i\omega_j t} + \sum_{j=1}^{\infty} \frac{\alpha_j - i\beta_j}{2} e^{i\omega_j t}, \quad (5)$$

which can be written concisely as

$$x(t) = \sum_{j=-\infty}^{\infty} \xi_j e^{i\omega_j t}, \quad (6)$$

(i) The Fourier integral:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \xi(\omega) e^{i\omega t} d\omega \quad \longleftrightarrow \quad \xi(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$$



(ii) The classical Fourier series:

$$x(t) = \sum_{j=-\infty}^{\infty} \xi_j e^{i\omega_j t} \quad \longleftrightarrow \quad \xi_j = \frac{1}{T} \int_0^T x(t) e^{-i\omega_j t} dt$$



(iii) The discrete-time Fourier transform:

$$x_t = \frac{1}{2\pi} \int_{-\pi}^{\pi} \xi(\omega) e^{i\omega t} d\omega \quad \longleftrightarrow \quad \xi(\omega) = \sum_{t=-\infty}^{\infty} x_t e^{-i\omega t}$$



(iv) The discrete Fourier transform:

$$x_t = \sum_{j=0}^{T-1} \xi_j e^{i\omega_j t} \quad \longleftrightarrow \quad \xi_j = \frac{1}{T} \sum_{t=0}^{T-1} x_t e^{-i\omega_j t}$$



Figure 1. The classes of the Fourier transforms.

where

$$\xi_0 = \alpha_0, \quad \xi_j = \frac{\alpha_j - i\beta_j}{2} \quad \text{and} \quad \xi_{-j} = \xi_j^* = \frac{\alpha_j + i\beta_j}{2}. \quad (7)$$

The inverse of the classical transform of (6) is demonstrated by writing

$$\begin{aligned} \xi_j &= \frac{1}{T} \int_0^T x(t) e^{-i\omega_j t} dt = \frac{1}{T} \int_0^T \left\{ \sum_{k=-\infty}^{\infty} \xi_k e^{i\omega_k t} \right\} e^{-i\omega_j t} dt \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} \xi_k \left\{ \int_0^T e^{i(\omega_k - \omega_j)t} dt \right\} = \xi_j, \end{aligned} \quad (8)$$

where the final equality follows from an orthogonality condition in the form of

$$\int_0^T e^{i(\omega_k - \omega_j)t} dt = \begin{cases} 0, & \text{if } j \neq k; \\ T, & \text{if } j = k. \end{cases} \quad (9)$$

The relationship between the continuous periodic function and its Fourier transform can be summarised by writing

$$x(t) = \sum_{j=-\infty}^{\infty} \xi_j e^{i\omega_j t} \quad \longleftrightarrow \quad \xi_j = \frac{1}{T} \int_0^T x(t) e^{-i\omega_j t} dt. \quad (10)$$

The question of the conditions that are sufficient for the existence of such a Fourier relationship is an essential one; and there are a variety of Fourier theories. The theory of Fourier series is concerned with establishing the conditions under which the partial sums converge to the function, in some specified sense, as the number of the terms increases.

At the simplest level, it is sufficient for convergence that $x(t)$ should be continuous and bounded in the interval $[0, T]$. However, the classical theory of Fourier series is concerned with the existence of the relationship in the case where $x(t)$ is bounded but is also permitted to have a finite number of maxima and minima and a finite number of jump discontinuities. It can be shown that, in that case, as successive terms are added, the Fourier series of (1) converges to

$$\frac{1}{2} \{x(t+0) + x(t-0)\}, \quad (11)$$

where $x(t+0)$ is the value as t is approached from the right and $x(t-0)$ is the value as it is approached from the left. If $x(t)$ is continuous at the point in question, then the Fourier series converges to $x(t)$.

Here, the relevant criterion is that of overall convergence in mean-square rather than of pointwise convergence. The question of the mode of converge was the issue at stake in the paradox known as Gibbs' phenomenon, which concerns the absence of pointwise convergence in the presence jump discontinuities.

This phenomenon is illustrated in Figure 2, where it is apparent that not all of the oscillations in the partial sums that approximate a periodic square wave (i.e. a time-domain rectangle) are decreasing at a uniform rate as the

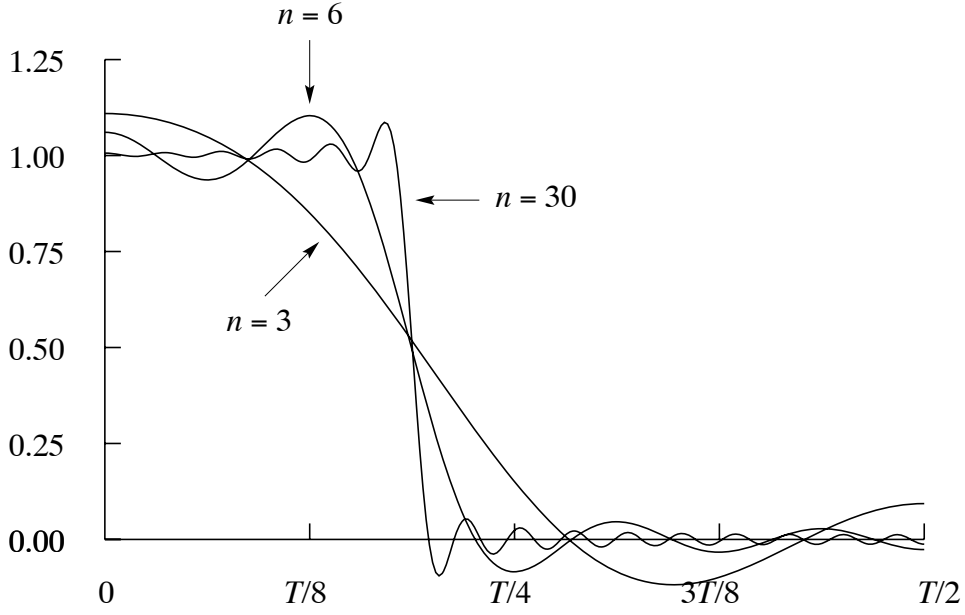


Figure 2. The Fourier approximation of a square wave. The diagram shows approximations to the positive half of a rectangle defined on the interval $[-T/2, T/2]$ and symmetric with respect to the vertical axis through zero.

number of terms n increases. Instead, the oscillations that are adjacent to the point of discontinuity are tending to a limiting amplitude, which is about 9% of the jump. However, as n increases, the width of these end-oscillations becomes vanishingly small; and thus the mean-square convergence of the Fourier series is assured. Gibbs' phenomenon is analysed in detail by Carslaw (1930); and Carslaw (1925) has also recounted the history of its discovery.

Parseval's theorem asserts that, under the stated conditions, which guarantee mean-square convergence, there is

$$\frac{1}{T} \int_0^T |x(t)|^2 dt = \alpha_0^2 + \frac{1}{4} \sum_{j=1}^{\infty} (\alpha_j^2 + \beta_j^2) = \sum_{j=-\infty}^{\infty} |\xi_j|^2, \quad (12)$$

where $|\xi_j|^2 = \xi_j \xi_j^* = \rho_j^2/4$. This indicates that the energy of the function $x(t)$, which becomes its average power if we divide by T , is equal to the sum of the energies of the sinusoidal components. Here, it is essential that the square integral converges, which is to say that the function possesses a finite energy. The fulfilment of such an energy or power condition characterises what we have chosen to describe as the canonical Fourier transforms.

There are some powerful symmetries between the two domains of the Fourier transforms. The discrete-time Fourier transform, which is fundamental to time-series analysis, is obtained by interchanging the two domains of the classical Fourier series transform. It effects the transformation of a sequence $x(t) = \{x_t; t = 0, \pm 1, \pm 2, \dots\}$ within the time domain, that is square summable,

into a continuous periodic function in the frequency domain via an integral expression that is mean-square convergent. It is illustrated in Figure 1.(iii). We may express the relationship in question by writing

$$x_t = \frac{1}{2\pi} \int_{-\pi}^{\pi} \xi(\omega) e^{i\omega t} d\omega \quad \longleftrightarrow \quad \xi(\omega) = \sum_{t=-\infty}^{\infty} x_t e^{-i\omega t}. \quad (13)$$

The periodicity is now in the frequency domain such that $\xi(\omega) = \xi(\omega + 2\pi)$ for all ω . The periodic function completes a single cycle in any interval of length 2π ; and, sometimes, it may be appropriate to define the function over the interval $[0, 2\pi]$ instead of the interval $[-\pi, \pi]$. In that case, the relationship of (13) will be unaffected; and its correspondence with (10), which defines the classical Fourier series, is clarified.

There remain two other Fourier transforms, which embody a complete symmetry between the two domains. The first of these is the Fourier integral transform of Figure 1.(i) which is denoted by

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \xi(\omega) e^{i\omega t} d\omega \quad \longleftrightarrow \quad \xi(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt. \quad (14)$$

Here, apart from the scaling, which can be revised to achieve symmetry, there is symmetry between the transform and its inverse. (The factor $1/2\pi$ can be eliminated from the first integral by changing the variable from angular velocity ω , measured in radians per period, to frequency $f = \omega/2\pi$, measured in cycles per period.)

One of the surprises on first encountering the Fourier integral is the aperiodic nature of the transform, which is in spite of the cyclical nature of the constituent sinusoidal elements. However, this is readily explained. Consider the sum of two elements within a Fourier synthesis that are at the frequencies ω_m and ω_n respectively. Then, their sum will constitute a periodic function of period $\tau = 2\pi/\omega_*$ if and only if $\omega_m = m\omega_*$ and $\omega_n = n\omega_*$ are integer multiples of a common frequency ω_* .

For a sum of many sinusoidal elements to constitute a periodic function, it is necessary and sufficient that all of the corresponding frequencies should be integer multiples of a fundamental frequency. Whereas this condition is fulfilled by the classical Fourier series of (10), it cannot be fulfilled by a Fourier integral comprising frequencies that are arbitrary irrational numbers.

A sequence that has been sampled at the integer time points from a continuous aperiodic function that is square-integrable will have a transform that is a periodic function. This outcome is manifest in the discrete-time Fourier transform, where it can be seen that the period in question has the length of 2π radians. Let $\xi(\omega)$ be the transform of the continuous aperiodic function, and let $\xi_S(\omega)$ be the transform of the sampled sequence $\{x_t; t = 0, \pm 1, \pm 2, \dots\}$. Then

$$x_t = \frac{1}{2\pi} \int_{-\infty}^{\infty} \xi(\omega) e^{i\omega t} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \xi_S(\omega) e^{i\omega t} d\omega. \quad (15)$$

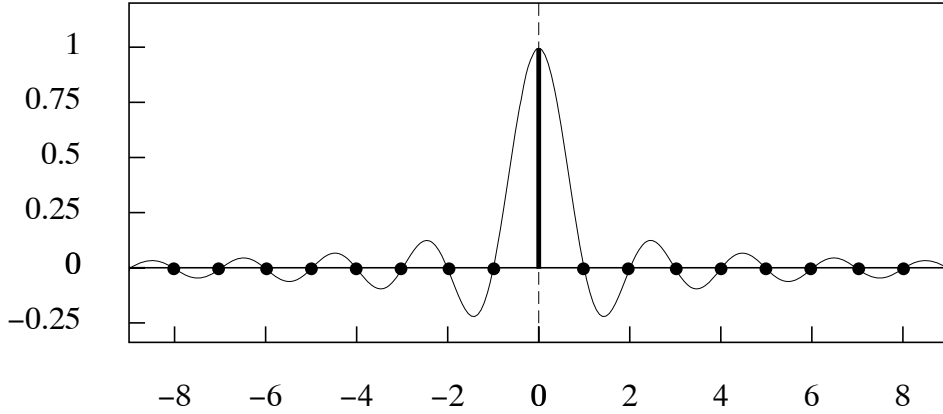


Figure 3. The sinc function wave-packet $\psi_0(t) = \sin(\pi t)/\pi t$ comprising frequencies in the interval $[0, \pi]$.

The equality of the two integrals implies that

$$\xi_S(\omega) = \sum_{j=-\infty}^{\infty} \xi(\omega + 2j\pi). \quad (16)$$

Thus, the periodic function $\xi_S(\omega)$ is obtained by wrapping $\xi(\omega)$ around a circle of circumference of 2π and adding the coincident ordinates.

The relationship between the Fourier series and the discrete-time Fourier transform enables us to infer that a similar effect will arise from the regular sampling of a continuous function of frequency. Thus, the corresponding time-domain function will be wrapped around a circle of circumferences $T = 2\pi/\omega_1$, where ω_1 is both the fundamental frequency and the sampling interval in the frequency domain.

The Fourier integral is employed extensively in mathematical physics. In quantum mechanics, for example, it is used in one domain to describe a localised wave train and, in the other domain, to describe its frequency composition, which is a resolution of its energy amongst a set of frequencies. There is an inverse relationship between the dispersion of the wave train in space and the dispersion of its frequency components. The product of the two dispersions is bounded below according to the famous Heisenberg uncertainty principle.

A simple and important example of the Fourier integral is afforded by the sinc function wave packet and its transform, which is a frequency-domain rectangle. Figure 3 depicts a continuous sinc function of which the Fourier transform is a rectangle on the frequency interval $[-\pi, \pi]$:

$$\psi_0(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i\omega t} d\omega(t) = \left[\frac{e^{i\omega t}}{i2\pi t} \right]_{-\pi}^{\pi} = \frac{\sin \pi t}{\pi t}. \quad (17)$$

Here, the final equality is by virtue of (4), which expresses a sine function as a combination of complex exponential functions.

The function $\psi_0(t)$ can also be construed as a wave packet centred on time $t = 0$. The figure also represents a sampled version the sinc function. This would be obtained, in the manner of a classical Fourier series, from the rectangle on the interval $[-\pi, \pi]$, if this were regarded as a single cycle of a periodic function.

The function $\psi_0(t)$ with $t \in \mathcal{I} = \{0, \pm 1, \pm 2, \dots\}$ is nothing but the unit impulse sequence. Therefore, the set of all sequences $\{\psi_0(t - k); t, k \in \mathcal{I}\}$, obtained by integer displacements k of $\psi_0(t)$, constitutes the ordinary orthogonal Cartesian basis in the time domain for the set of all real-valued time series.

When $t \in \mathcal{R}$ is a real-valued index of continuous time, the set of displaced sinc functions $\{\psi_0(t - k); t \in \mathcal{R}, k \in \mathcal{I}\}$ constitute a basis for the set of continuous functions of which the frequency content is bounded by the Nyquist value of π radians per unit time interval. In common with their discretely sampled counterparts, the sequence of continuous sinc functions at integer displacements constitutes an orthogonal basis.

To demonstrate the orthogonality, consider the fact that the corresponding frequency-domain rectangle is an idempotent function. When multiplied by itself it does not change. The time-domain operation corresponding to this frequency-domain multiplication is an autoconvolution. The time-domain functions are real and symmetric with $\psi_0(k - t) = \psi_0(t - k)$, so their autoconvolution is the same as their autocorrelation:

$$\gamma(k) = \int \psi_0(t)\psi_0(k - t)dt = \int \psi_0(t)\psi_0(t - k)dt. \quad (18)$$

Therefore, the sinc function is its own autocovariance function: $\gamma(k) = \psi_0(k)$. The zeros of the sinc function that are found at integer displacements from the centre correspond the orthogonality of sinc functions separated from each other by these distances.

A sinc function wave packet that is limited to the frequency band $[\alpha, \beta] \in [0, \pi]$, if we are talking only of positive frequencies, has the functional form of

$$\begin{aligned} \psi(t) &= \frac{1}{\pi t} \{\sin(\beta t) - \sin(\alpha t)\} \\ &= \frac{2}{\pi t} \cos\{(\alpha + \beta)t/2\} \sin\{(\beta - \alpha)t/2\} \\ &= \frac{2}{\pi t} \cos(\gamma t) \sin(\delta t), \end{aligned} \quad (19)$$

where $\gamma = (\alpha + \beta)/2$ is the centre of the band and $\delta = (\beta - \alpha)/2$ is half its width. The equality follows from the identity $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$.

Finally, we must consider the Fourier transform that is the most important one from the point of view of this paper. This is the discrete Fourier transform of Figure 1.(iv) that maps from a sequence of T data $\{x_t; t = 0, 1, \dots, T - 1\}$ in the time domain to a set of T ordinates $\{\xi_j; j = 0, 1, \dots, T - 1\}$ in the frequency domain at the points $\{\omega_j = 2\pi j/T; j = 0, 1, \dots, T - 1\}$, which are equally spaced within the interval $[0, 2\pi]$, and vice versa. The relationship can be expressed in terms of sines and cosines; but it is more conveniently expressed

Table 1. The classes of Fourier transforms*

	Periodic	Aperiodic
Continuous	Discrete aperiodic <i>Fourier series</i>	Continuous aperiodic <i>Fourier integral</i>
Discrete	Discrete periodic <i>Discrete FT</i>	Continuous periodic <i>Discrete-time FT</i>

* The class of the Fourier transform depends upon the nature of the function which is transformed. This function may be discrete or continuous and it may be periodic or aperiodic. The names and the natures of corresponding transforms are shown in the cells of the table.

in terms of complex exponentials, even in the case where the sequence $\{x_t\}$ is a set of real data values:

$$x_t = \sum_{j=0}^{T-1} \xi_j e^{i\omega_j t} \longleftrightarrow \xi_j = \frac{1}{T} \sum_{t=0}^{T-1} x_t e^{-i\omega_j t}. \quad (20)$$

Since all sequences in data analysis are finite, the discrete Fourier transform is used, in practice, to represent all manner of Fourier transforms.

Although the sequences in both domains are finite and of an equal number of elements, it is often convenient to regard both of them as representing single cycles of periodic functions defined over the entire set of positive and negative integers. These infinite sequences are described and the periodic extensions of their finite counterparts.

The various Fourier transforms may be summarised in a table that records the essential characteristics of the function that is to be transformed, namely whether it is discrete or continuous and whether it is periodic or aperiodic. This is table 1.

There are numerous accounts of Fourier analysis that can be accessed in pursuit of a more thorough treatment. Much of the text of Carslaw (1930) is devoted to the issues in real analysis that were raised in the course of the development of Fourier analysis. The book was first published in 1906, which is close to the dates of some of the essential discoveries that clarified such matters as Gibbs' phenomenon. It continues to be a serviceable and authoritative text. Titchmarsh (1937), from the same era, provides a classical account of the Fourier integral.

The more recent text of Kreider, Kuller, Ostberg and Perkins (1966) deals with some of the analytic complications of Fourier analysis in an accessible manner, as does the brief text of Churchill and Brown (1978). The recent text of Stade (2005) is an extensive resource. That of Nahin (2006) is both engaging and insightful.

Many of the texts that are intended primarily for electrical engineers are also useful to statisticians and econometricians. Baher (1990) gives much of the relevant analytic details, whereas Brigham (1988) provides a practical text that avoids analytic complications.

Following the discoveries of Wiener (1930) and Kolmogorov (1941b), the mathematical exposition of the spectral analysis of time series developed rapidly. Detailed treatments of the analytic issues are to be found in some of the classical texts of time series analysis, amongst which are those of Wold (1938), Doob (1953), Grenander and Rosenblatt (1957), Hannan (1960), Yaglom (1962) and Rozanov (1967). Yaglom and Rozanov were heirs to a Russian tradition that began with Kolmogorov (1941b)

It was not until the 1960's that spectral analysis began to find widespread application; and the book of Jenkins and Watts (1968) was symptomatic of the increasing practicality of the subject. Econometricians also began to take note of spectral analysis in the 1960's; and two influential books that brought it to their attention were those of Granger and Hatanaka (1964) and Fishman (1969). Nerlove, Grether and Carvalho (1979) continued the pursuit of the spectral analysis of economic data.

Latterly, Brillinger (1975), Priestly (1981), Rosenblatt (1985) and Brockwell and Davis (1987) have treated Fourier analysis with a view to the statistical analysis of time series, as has Pollock (1999).

3. Representations of the Discrete Fourier Transform

It is often helpful to express the discrete Fourier transform in terms of the notation of the z -transform. The z -transforms of the sequences $\{x_t; t = 0, \dots, T-1\}$ and $\{\xi_j; j = 0, \dots, T-1\}$ are the polynomials

$$\xi(z) = \sum_{j=0}^{T-1} \xi_j z^j \quad \text{and} \quad x(z) = \sum_{t=0}^{T-1} x_t z^t, \quad (21)$$

wherein z is an indeterminate algebraic symbol that is commonly taken to be a complex number, in accordance with the fact that the equations $\xi(z) = 0$ and $x(z) = 0$ have solutions within the complex plane.

The polynomial equation $z^T = 1$ is important in establishing the connection between the z -transform of a sequence of T elements and the corresponding discrete Fourier transform. The solutions of the equation are the complex numbers

$$W_T^j = \exp\left(\frac{-i2\pi j}{T}\right) = \cos\left(\frac{2\pi j}{T}\right) - i \sin\left(\frac{2\pi j}{T}\right); j = 0, 1, \dots, T-1. \quad (22)$$

These constitute a set of points, described as the T roots of unity, that are equally spaced around the circumference of the unit circle at angles of $\omega_j = 2\pi j/T$ radians. Multiplying any of these angles by T will generate a multiple of 2π radians, which coincides with an angle of zero radians, which is the angle, or argument, attributable to unity within the complex plane.

Using this conventional notation for the roots of unity, we can write the discrete Fourier transform of (20) and its inverse as

$$\xi_j = \frac{1}{T} \sum_{t=0}^{T-1} x_t W_T^{jt} \quad \text{and} \quad x_t = \sum_{j=0}^{T-1} \xi_j W_T^{-jt}. \quad (23)$$

The advantage of the z -transforms is that they enable us to deploy the ordinary algebra of polynomials and power series to manipulate the objects of Fourier analysis. In econometrics, it is common to replace z by the so-called lag operator L , which operates on doubly-infinite sequences and which has the effect that $Lx(t) = x(t - 1)$.

It is also useful to replace z by a variety of matrix operators. The matrix lag operator $L_T = [e_1, e_2, \dots, e_{T-1}, 0]$ is derived from the identity matrix $I_T = [e_0, e_1, \dots, e_{T-1}]$ of order T by deleting its leading column and appending a column of zeros to the end of the array.

Setting $z = L_T$ within the polynomial $x(z) = x_0 + x_1 z + \dots + x_{T-1} z^{T-1}$ gives rise to a banded lower-triangular Toeplitz matrix of order T . The matrices $I_T = L_T^0, L_T, L_T^2, \dots, L_T^{T-1}$ form a basis for the vector space comprising all such matrices. The ordinary polynomial algebra can be applied to these matrices with the provision that the argument L_T is nilpotent of degree T , which is to say that $L_T^q = 0$ for all $q \geq T$.

Circulant Matrices

A matrix argument that has properties that more closely resemble those of the complex exponentials is the circulant matrix $K_T = [e_1, \dots, e_{T-1}, e_0]$, which is formed by carrying the leading column of the identity matrix I_T to the back of the array. This is an orthonormal matrix of which the transpose is the inverse, such that $K_T' K_T = K_T K_T' = I_T$.

The powers of the matrix form a T -periodic sequence such that $K_T^{T+q} = K_T^q$ for all q . The periodicity of these powers is analogous to the periodicity of the powers of the argument $z = \exp\{-i2\pi/T\}$, which is to be found in the Fourier transform of a sequence of T elements.

The matrices $K_T^0 = I_T, K_T, \dots, K_T^{T-1}$ form a basis for the set of all circulant matrices of order T —a circulant matrix $X = [x_{ij}]$ of order T being defined as a matrix in which the value of the generic element x_{ij} is determined by the index $\{(i - j) \bmod T\}$. This implies that each column of X is equal to the previous column rotated downwards by one element. The generic circulant matrix has a form that may be illustrated follows:

$$X = \begin{bmatrix} x_0 & x_{T-1} & x_{T-2} & \dots & x_1 \\ x_1 & x_0 & x_{T-1} & \dots & x_2 \\ x_2 & x_1 & x_0 & \dots & x_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{T-1} & x_{T-2} & x_{T-3} & \dots & x_0 \end{bmatrix}. \quad (24)$$

There exists a one-to-one correspondence between the set of all polynomials of degree less than T and the set of all circulant matrices of order T . Thus, if $x(z)$ is a polynomial of degree less than T , then there exists a corresponding circulant matrix

$$X = x(K_T) = x_0 I_T + x_1 K_T + \dots + x_{T-1} K_T^{T-1}. \quad (25)$$

A convergent sequence of an indefinite length can also be mapped into a circulant matrix. If $\{c_i\}$ is an absolutely summable sequence obeying the

condition that $\sum |c_i| < \infty$, then the z -transform of the sequence, which is defined by $c(z) = \sum c_j z^j$, is an analytic function on the unit circle. In that case, replacing z by K_T gives rise to a circulant matrix $C^\circ = c(K_T)$ with finite-valued elements. In consequence of the periodicity of the powers of K_T , it follows that

$$\begin{aligned} C^\circ &= \left\{ \sum_{j=0}^{\infty} c_{jT} \right\} I_T + \left\{ \sum_{j=0}^{\infty} c_{(jT+1)} \right\} K_T + \cdots + \left\{ \sum_{j=0}^{\infty} c_{(jT+T-1)} \right\} K_T^{T-1} \\ &= c_0^\circ I_T + c_1^\circ K_T + \cdots + c_{T-1}^\circ K_T^{T-1}. \end{aligned} \quad (26)$$

Given that $\{c_i\}$ is a convergent sequence, it follows that the sequence of the matrix coefficients $\{c_0^\circ, c_1^\circ, \dots, c_{T-1}^\circ\}$ converges to $\{c_0, c_1, \dots, c_{T-1}\}$ as T increases. Notice that the matrix $c^\circ(K_T) = c_0^\circ I_T + c_1^\circ K_T + \cdots + c_{T-1}^\circ K_T^{T-1}$, which is derived from a polynomial $c^\circ(z)$ of degree $T-1$, is a synonym for the matrix $c(K_T)$, which is derived from the z -transform of an infinite convergent sequence.

The polynomial representation is enough to establish that circulant matrices commute in multiplication and that their product is also a polynomial in K . That is to say

$$\begin{aligned} \text{If } X = x(K_T) \text{ and } Y = y(K_T) \text{ are circulant matrices,} \\ \text{then } XY = YX \text{ is also a circulant matrix.} \end{aligned} \quad (27)$$

The matrix operator K_T has a spectral factorisation that is particularly useful in analysing the properties of the discrete Fourier transform. To demonstrate this factorisation, we must first define the so-called Fourier matrix. This is a symmetric matrix $U = T^{-1/2}[W^{jt}; t, j = 0, \dots, T-1]$, of which the generic element in the j th row and t th column is W^{jt} , where $W = \exp\{-i2\pi/T\}$ is the first root of unity, from which we are now omitting the subscripted T for ease of notation. On taking account of the T -periodicity of W^q , the matrix can be written explicitly as

$$U = \frac{1}{\sqrt{T}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W & W^2 & \cdots & W^{T-1} \\ 1 & W^2 & W^4 & \cdots & W^{T-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W^{T-1} & W^{T-2} & \cdots & W \end{bmatrix}. \quad (28)$$

The second row and the second column of this matrix contain the T roots of unity. The conjugate matrix is defined as $\bar{U} = T^{-1/2}[W^{-jt}; t, j = 0, \dots, T-1]$; and, by using $W^{-q} = W^{T-q}$, this can be written explicitly as

$$\bar{U} = \frac{1}{\sqrt{T}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W^{T-1} & W^{T-2} & \cdots & W \\ 1 & W^{T-2} & W^{T-4} & \cdots & W^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W & W^2 & \cdots & W^{T-1} \end{bmatrix}. \quad (29)$$

The matrix U is a unitary, which is to say that it fulfils the condition

$$\bar{U}U = U\bar{U} = I. \quad (30)$$

The operator K , from which we now omit the subscript, can be factorised as

$$K = \bar{U}DU = U\bar{D}\bar{U}, \quad (31)$$

where

$$D = \text{diag}\{1, W, W^2, \dots, W^{T-1}\} \quad (32)$$

is a diagonal matrix whose elements are the T roots of unity, which are found on the circumference of the unit circle in the complex plane. Observe also that D is T -periodic, such that $D^{q+T} = D^q$, and that $K^q = \bar{U}D^qU = U\bar{D}^q\bar{U}$ for any integer q . Since the powers of K form the basis for the set of circulant matrices, it follows that all circulant matrices are amenable to a spectral factorisation based on (31).

Circulant matrices have represented a mathematical curiosity ever since their first appearance in the literature in a paper by Catalan (1846). The literature on circulant matrices, from their introduction until 1920, was summarised in four papers by Muir (1911)–(1923). A recent treatise on the subject, which contains a useful bibliography, has been provided by Davis (1979); but his book does not deal with problems in time-series analysis. An up-to-date account, orientated towards statistical signal processing, has been provided by Gray (2002).

The Matrix Discrete Fourier Transform

The matrices U and $\bar{U}$ are entailed in the discrete Fourier transform. Thus, the equations of (20) or (23) can be written as

$$\xi = T^{-1/2}Ux = T^{-1}Wx \quad \text{and} \quad x = T^{1/2}\bar{U}\xi = \bar{W}\xi, \quad (33)$$

where $x = [x_0, x_1, \dots, x_{T-1}]'$ and $\xi = [\xi_0, \xi_1, \dots, \xi_{T-1}]'$, and where $W = [W^{jt}]$ and $\bar{W} = [W^{-jt}]$ are the matrices of (28) and (29) respectively, freed from their scaling factors. We may note, in particular, that

$$\iota = T^{1/2}Ue_0 = T^{1/2}\bar{U}e_0 \quad \text{and} \quad e_0 = T^{-1/2}U\iota = T^{-1/2}\bar{U}\iota, \quad (34)$$

where $e_0 = [1, 0, \dots, 0]'$ is the leading column of the identity matrix and $\iota = [1, 1, \dots, 1]'$ is the summation vector. Thus, a spike or an impulse located at $t = 0$ in the time domain or at $j = 0$ in the frequency domain is transformed into a constant sequence in the other domain and vice versa. These are extreme examples of the inverse relationship between the dispersion of a sequence and that of its transform.

Given the circulant the matrix $X = x(K)$ defined in (25), the discrete Fourier transform and its inverse may be represented by the following spectral factorisations:

$$X = \bar{U}x(D)U \quad \text{and} \quad x(D) = UX\bar{U}. \quad (35)$$

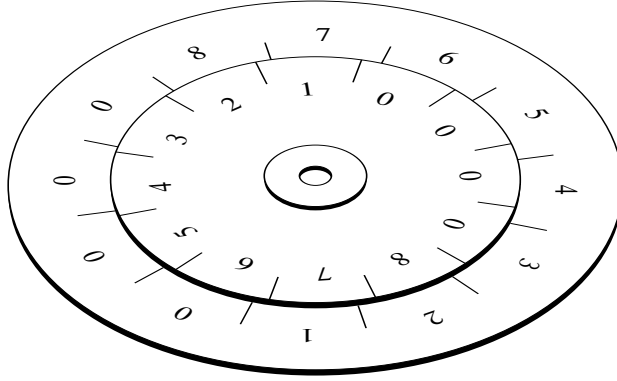


Figure 4. A device for finding the circular correlation of two sequences. The upper disc is rotated clockwise through successive angles of 30 degrees. Adjacent numbers on the two discs are multiplied and the products are summed and divided by the sample size T to obtain the circular covariances.

Here, $x(D) = T \text{diag}\{\xi_0, \xi_1, \dots, \xi_{T-1}\}$ is a diagonal matrix containing the spectral ordinates of the data. The equations of (33) are recovered by post multiplying the X by e_0 and $x(D)$ by ι and using the results of (34).

Assuming that the data are mean adjusted, i.e. that $\iota'X = 0$, the cross product of the matrix X gives rise to the matrix C° of the circular autocovariances of the data. Thus

$$\begin{aligned} T^{-1}X'X &= C^\circ = \bar{U}c^\circ(D)U \\ &= T^{-1}\bar{U}x(\bar{D})x(D)U, \end{aligned} \quad (36)$$

where we have used $X' = Ux(D)\bar{U} = \bar{U}x(\bar{D})U$.

The values of the elements of $[c_0^\circ, c_1^\circ, \dots, c_{T-1}^\circ]' = c^\circ = C^\circ e_0$ are given by the formula

$$\begin{aligned} c_\tau^\circ &= \frac{1}{T} \sum_{t=0}^{T-1} x_t x_{t+\tau}; \quad \text{where } x_t = x_{(t \bmod T)} \\ \text{or, equivalently,} \\ c_\tau^\circ &= \frac{1}{T} \sum_{t=0}^{T-1-\tau} x_t x_{t+\tau} + \frac{1}{T} \sum_{t=0}^{\tau-1} x_t x_{t+T-\tau} \\ &= c_\tau + c_{T-\tau}, \end{aligned} \quad (37)$$

where c_τ is the ordinary empirical autocovariance of lag τ .

The vector c° of circular autocovariances is obtained via a circular correlation of the data vector x . One can envisage the data disposed at T equally-spaced points around the circumferences of two discs on a common axis. The same numbers on the two discs are aligned and products are formed and added. This generates the sum of squares. Then, the upper disc is rotated by an angle of $2\pi/T$ radians and the displaced numbers are multiplied together and added. The process is repeated. The T values formed via a complete rotation are each

divided by T , and the result is the sequence of circular autocovariances. This device is illustrated in Figure 4, where $T = 12$ and where identical sequences on the two discs end with four zeros.

The core of the factorisation of the matrix of C° of (36) is the real-valued diagonal matrix

$$c^\circ(D) = T^{-1}x(\bar{D})x(D) = T\text{Diag}\{|\xi_0|^2, |\xi_1|^2, \dots, |\xi_{T-1}|^2\}, \quad (38)$$

of which the graph of the elements is described as the periodogram. (The frequencies indexed by $j = 0, \dots, T-1$ extend from 0 to $2\pi(T-1)/T$. However, for real-valued data, there is $\xi_j = \xi_{T-j}$, and so it is customary to plot the periodogram ordinates only over a frequency range from 0 to π , as in Figure 14.)

Since $e_0' C^\circ e_0 = c_0^\circ = c_0$ and $Ue_0 = \bar{U}e_0 = T^{-1/2}\iota$, we get the following expression for the variance:

$$\begin{aligned} c_0 &= T^{-1} \sum_{j=0}^{T-1} x_j^2 \\ &= T^{-2} \iota' x(\bar{D})x(D) \iota = \sum_{j=0}^{T-1} |\xi_j|^2. \end{aligned} \quad (39)$$

This is the discrete version of Parseval's theorem. In statistical terms, it represents a frequency-specific analysis of variance.

4. Fourier Analysis of Temporal Sequences

It is clear that the discrete-time Fourier transform is inappropriate to a doubly-infinite stationary stochastic sequence $y(t) = \{y_t; t = 0, \pm 1, \pm 2, \dots\}$ defined over the set of positive and negative integers. Such a sequence, which has infinite energy, is not summable.

In the appendix, it is demonstrated there there exists a spectral representation of $y(t)$ in the form of

$$y(t) = \int_{-\pi}^{\pi} e^{i\omega t} dZ(\omega), \quad (40)$$

where $Z(\omega)$ is continuous non-differentiable complex-valued stochastic process defined on the interval $[-\pi, \pi]$. The increments $dZ(\omega)$, $dZ(\lambda)$ of the process are uncorrelated for all $\lambda \neq \omega$. However, the statistical expectation

$$E\{dZ(\omega)dZ^*(\omega)\} = dF(\omega) = f(\omega)d\omega \quad (41)$$

comprises a function $F(\omega)$, known as the spectral distribution function, which is continuous and differentiable almost everywhere in the interval, provided that $y(t)$ contains no perfectly regular components of finite power. If there are regular components within $y(t)$, then $F(\omega)$ will manifest jump discontinuities at the corresponding frequencies.

The discrete-time transform comes into effect when it is applied not to the data sequence itself but, instead, to its autocovariance function

$$\gamma(\tau) = \{\gamma_t; \tau = 0, \pm 1, \pm 2, \dots\} \quad \text{where} \quad \gamma_\tau = \gamma_{-\tau} = E(y_t y_{t-\tau}). \quad (42)$$

To ensure that its transform is mean-square convergent, the autocovariance sequence must be square summable such that $\sum \gamma_\tau^2 < \infty$. This is a necessary and sufficient condition for the stationarity of the process $y(t)$.

The Fourier transform of the autocovariance function is the spectral density function or power spectrum $f(\omega)$, which, in view of the symmetry of $\gamma(\tau)$, can be expressed as

$$\begin{aligned} f(\omega) &= \frac{1}{2\pi} \left\{ \gamma_0 + \sum_{\tau=1}^{\infty} \gamma_\tau (e^{i\omega\tau} + e^{-i\omega\tau}) \right\} \\ &= \frac{1}{2\pi} \left\{ \gamma_0 + 2 \sum_{\tau=1}^{\infty} \gamma_\tau \cos(\omega\tau) \right\}. \end{aligned} \quad (43)$$

Notice, however, that compared with equation (14), the scaling factor $1/2\pi$ has migrated from the Fourier integral to the series. Now, the inverse transform takes the form of

$$\gamma_\tau = \int_{-\pi}^{\pi} e^{i\omega\tau} f(\omega) d\omega. \quad (44)$$

Setting $\tau = 0$ gives $\gamma_0 = \int f(\omega) d\omega$, which indicates that the power of the process, which is expressed by its variance, is equal to the integral of the spectral density function over the interval $[-\pi, \pi]$. This is a form of Parseval's theorem. The sequence $y(t)$ is not square summable and, therefore, its energy cannot be expressed in terms of a sum of squares. Instead, its power is expressed, in the time domain, via the statistical moment $\gamma_0 = E(y_t^2)$.

In the frequency domain, this power is attributed to an infinite set of sinusoidal components, of which the measure of power is provided by the continuous spectral density function $f(\omega)$. One can see, by a direct analogy with a continuous probability density function, that the power associated with a component at a particular frequency is vanishingly small, i.e. of measure zero.

Often, it is assumed that, in addition to square summability, the autocovariances obey the stronger condition that $\sum |\gamma_\tau| < \infty$, which is the condition of absolute summability. This condition is satisfied by processes generated by finite-order autoregressive moving-average (ARMA) models, which can be represented by setting $d = 0$ in the equation

$$y(z) = \frac{\theta(z)}{(1-z)^d \phi(z)} \varepsilon(z), \quad (45)$$

which describes an autoregressive integrated moving-average (ARIMA) model. Here $\varepsilon(z)$ and $y(z)$ are, respectively, the z -transform of a white-noise forcing function and of the output sequence and $\theta(z)$ and $\phi(z)$ are polynomials of finite degree. Provided that $d = 0$ and that the roots of the equation

$\phi(z) = \phi_0 + \phi_1 z + \cdots + \phi_p z^p = 0$ lie outside the unit circle, the spectral density function of the process will be bounded and everywhere continuous.

The absolute summability of the ARMA autocovariances is a consequence of the summability of the infinite sequence of moving-average coefficients generated by the series expansion of the rational transfer function $\theta(z)/\phi(z)$ that maps from the white-noise forcing function $\varepsilon(t)$ to the output $y(t)$. These coefficients constitute the impulse response of the transfer function; and the condition of their absolute summability is the bounded input–bounded output (BIBO) stability condition of linear systems analysis. (see Pollock 1999, for example.)

The autocovariance generating function of an ARMA process, which is the z -transform of the two-sided sequence $\{\gamma_\tau; \tau = 0, \pm 1, \pm 2, \dots\}$, is given by

$$\gamma(z) = \sigma^2 \frac{\theta(z)\theta(z^{-1})}{\phi(z)\phi(z^{-1})}, \quad (46)$$

where $\sigma^2 = V\{\varepsilon(t)\}$ is the variance of the white-noise forcing function. The spectral density function $f(\omega) = \gamma(\exp\{i\omega\})$ is produced when the locus of z is the unit circle in the complex plane.

The conditions of absolute and square summability are violated by the ARIMA models that can be used to describe trended processes. Unless there are finite starting values at a finite distance from the current value, the variance of such a process will be unbounded. (If equation (45) is to remain viable when $d = 1$, then it is necessary to define $y(z)$ and $\varepsilon(z)$ to be the z -transforms of finite sequences that incorporate the appropriate initial conditions—see Pollock (2008), for example.)

The spectral density function of an ARIMA process, which is defined in respect of the doubly infinite sequence $y(t)$ will not have a finite-valued integral. Therefore, it is liable to be described as a pseudo spectral density function. See, for example, Maravall and Piece (1987). The unit roots within the factor $(1 - z)^d$ of equation (45) contribute infinite power to the ARIMA process from within the neighbourhood of the zero frequency, where the pseudo spectrum is unbounded.

In a seasonal ARIMA model, which is used to model persistent seasonal fluctuations, the denominator of the transfer function incorporates the factor

$$S(z) = 1 + z + z^2 + \cdots + z^{s-1} = \prod_{j=1}^{s-1} (1 - e^{2\pi j/s} z). \quad (47)$$

where s is the number of seasons or months of the year. Here, the generic quadratic factor is

$$1 - 2 \cos(2\pi j/s) + z^2 = (1 - e^{2\pi j/s} z)(1 - e^{2\pi(s-j)/s} z), \quad (48)$$

where $j < s/2$. This factor, which will contribute infinite power to the pseudo spectrum, corresponds to an unbounded spike at the frequency of $\omega_j = 2\pi j/s$ which is the j th harmonic of the fundamental seasonal frequency. In the

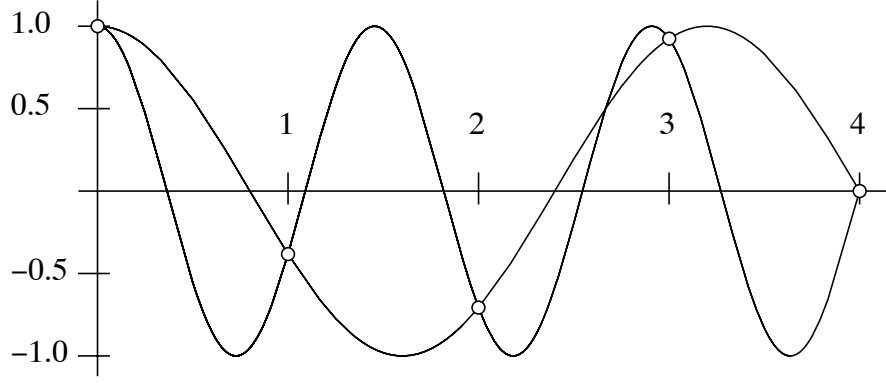


Figure 5. The values of the function $\cos\{(11/8)\pi t\}$ coincide with those of its alias $\cos\{(5/8)\pi t\}$ at the integer points $\{t = 0, \pm 1, \pm 2, \dots\}$.

numerator of the transfer function, there should be a polynomial $R(z) = 1 + \rho z + \rho^2 z^2 + \dots + \rho^{s-1} z^{s-1}$ with a value of $\rho < 1$ close to unity, which will serve to counteract the effects of $S(z)$ at the non-seasonal frequencies.

Latterly, econometricians and others have been focussing their attention on so-called long-memory processes that are characterised by autocovariance sequences that are square summable but not absolutely summable. Such processes are commonly modelled by replacing the difference operator of an ARIMA model by a corresponding fractional difference operator $(1 - z)^d$, where $d \in [0, 0.5)$. Such models have spectra that are unbounded at the zero frequency. When $|d| \geq 0.5$, the condition of the square summability of the autocovariances is violated and the process is no longer stationary.

Models incorporating the fractional difference operator were proposed by Granger and Joyeux (1980) and Hoskin (1981). They have been described in detail by Beran (1994) and by Palma (2007). Granger and Ding (1996) have considered the provenance of long-memory processes; and they have provided some alternative explanations.

Persistent seasonal fluctuations can also be described by long-memory models. Hoskin (1984) noted that, when $|d| < 0.5$ and $|\phi| < 1$, the process defined by $y(z) = \varepsilon(z)/(1 - 2\phi z + z^2)^d$ would generate a sequence displaying long-term persistence and quasi periodic behaviour. The transfer function of this model, which incorporates a fractional power of the seasonal polynomial of (48), is a so-called Gegenbauer polynomial. The lead of Hoskin has been followed by other authors, including Gray, Zhang and Woodward, (1989) Hidalgo (1997), Artech and Robinson (2000) and McCoy and Stephens (2004), who have defined models with spectra that are unbounded at multiple frequencies.

The Problem of Aliasing

Notice that the (positive) frequencies in the spectral analysis of a discrete-time process are limited to the interval $[0, \pi]$, bounded by the so-call Nyquist frequency π . The process of sampling imposes a limit of the observability of

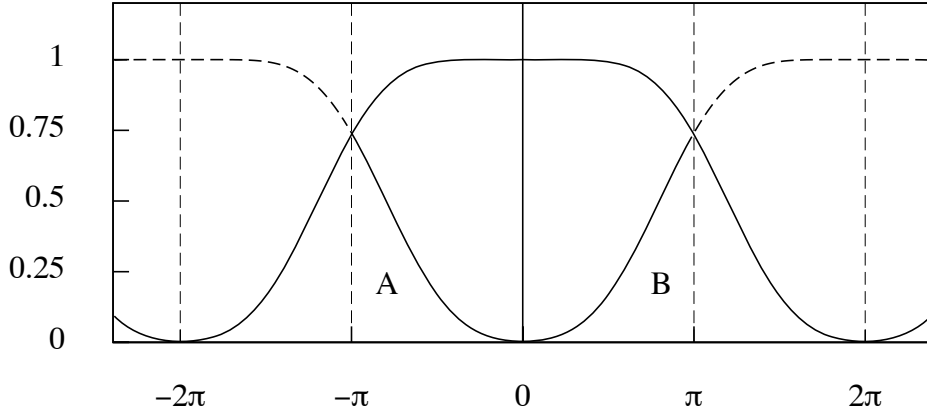


Figure 6. The function $f(\lambda/2 + \pi)$, represented by the semi-continuous line, superimposed upon the function $f(\lambda/2)$. The sum of these functions is a 2π -periodic function $f_H(\lambda)$, which represents the spectral density of a subsampled process. The segments A and B of $f(\lambda/2 + \pi)$ that fall within the interval $[-\pi, \pi]$ may be construed as the effects of wrapping $f(\lambda/2)$, defined over the interval $[-2\pi, 2\pi]$, around a circle of circumference 2π .

high-frequency components. To be observable in sampled data, a sinusoidal motion must take no less than two sample periods to complete its cycle, which limits its frequency to no more than π radians per period.

A motion of higher angular velocity will be mistaken for one of a velocity that lies within the observable interval, and this is described as the problem of aliasing. The problem is illustrated by Figure 5 which shows that, by sampling it at the integer points $\{t = 0, \pm 1, \pm 2, \dots\}$, a sinusoid with frequency of $(11/8)\pi$ radians per period, which is in excess of the limiting Nyquist value of π , will be mistaken for one with a frequency of $(5/8)\pi$.

The extent to which aliasing is a problem depends upon the structure of the particular time series under analysis. It will be suggested later that, for many econometric time series, the problem does not arise, for the reason that their (positive) frequencies are band-limited to a subinterval of the range $[0, \pi]$. However, this fact, which has its own problems, is not commonly recognised in econometric time-series analysis.

To understand the statistical aspects of aliasing, we may consider a stationary stochastic process $y(t) = \{y_t; t = 0, \pm 1, \pm 2, \dots\}$ with an autocovariance function $\gamma(\tau) = \{\gamma_\tau; \tau = 0, \pm 1, \pm 2, \dots\}$. The relationship between the spectral density function $f(\omega)$ and the autocovariance function is as follows:

$$\gamma(\tau) = \int_{-\pi}^{\pi} f(\omega) e^{i\omega\tau} d\omega \quad \longleftrightarrow \quad f(\omega) = \frac{1}{2\pi} \sum_{\tau=-\infty}^{\infty} \gamma_\tau e^{-i\omega\tau}. \quad (49)$$

When alternate values are selected from the data sequence, the autocovariance function is likewise subsampled, and there is

$$\begin{aligned}
 \gamma(2\tau) &= \int_{-\pi}^{\pi} f(\omega) e^{i\omega(2\tau)} d\omega \\
 &= \frac{1}{2} \int_{-2\pi}^{2\pi} f(\lambda/2) e^{i\lambda\tau} d\lambda \\
 &= \frac{1}{2} \left\{ \int_{-2\pi}^{-\pi} f(\lambda/2) e^{i\lambda\tau} d\lambda + \int_{-\pi}^{\pi} f(\lambda/2) e^{i\lambda\tau} d\lambda + \int_{\pi}^{2\pi} f(\lambda/2) e^{i\lambda\tau} d\lambda \right\}.
 \end{aligned} \tag{50}$$

Here, we have defined $\lambda = 2\omega$, and we have used the change of variable technique to obtain the second equality.

Within the integrand of (50), there is $\exp\{i\lambda\tau\} = \exp\{i(\lambda\tau \pm 2\pi)\}$. Also, $f(\{\lambda/2\} - \pi) = f(\{\lambda/2\} + \pi)$, by virtue of the 2π -periodicity of the function. Therefore,

$$\begin{aligned}
 \int_{-2\pi}^{-\pi} f(\lambda/2) e^{i\lambda\tau} d\lambda &= \int_0^{\pi} f(\{\lambda/2\} - \pi) e^{i(\lambda\tau - 2\pi)} d\lambda \\
 &= \int_0^{\pi} f(\{\lambda/2\} + \pi) e^{i\lambda\tau} d\lambda,
 \end{aligned} \tag{51}$$

where the first equality is an identity and the second exploits the results above. Likewise,

$$\begin{aligned}
 \int_{\pi}^{2\pi} f(\lambda/2) e^{i\lambda\tau} d\lambda &= \int_{-\pi}^0 f(\{\lambda/2\} + \pi) e^{i(\lambda\tau + 2\pi)} d\lambda \\
 &= \int_{-\pi}^0 f(\{\lambda/2\} + \pi) e^{i\lambda\tau} d\lambda.
 \end{aligned} \tag{52}$$

It follows that within the expression of (50), the first integral may be translated to the interval $[0, \pi]$, whereas the third integral may be translated to the interval $[-\pi, 0]$. After their translations, the first and the third integrands can be combined to form the segment of the function $f(\pi + \lambda/2)$ that falls in the interval $[-\pi, \pi]$. The consequence is that

$$\gamma(2\tau) = \frac{1}{2} \int_{-\pi}^{\pi} \{f(\lambda/2) + f(\pi + \lambda/2)\} e^{i\lambda\tau} d\lambda. \tag{53}$$

It follows that

$$\gamma(2\tau) \longleftrightarrow f_H(\lambda) = \frac{1}{2} \{f(\lambda/2) + f(\pi + \lambda/2)\}, \tag{54}$$

where $f_H(\lambda) = f_H(\lambda + 2\pi)$ is the spectral density function of the subsampled process. Figure 6 shows the relationship between $f(\lambda/2)$ and $f(\pi + \lambda/2)$.

The superimposition of the shifted frequency function $f(\pi + \lambda/2)$ upon the function $f(\lambda/2)$ corresponds to a process of aliasing. To visualise the process, one can imagine a single cycle of the original function $f(\omega)$ over the interval

$[-\pi, \pi]$. The dilated function $f(\lambda/2)$ manifests a single cycle over the interval $[-2\pi, 2\pi]$. By wrapping this segment twice around the circle defined by $\exp\{-i\lambda\}$ with $\lambda \in [-\pi, \pi]$, we obtain the aliased function $\{f(\lambda/2) + f(\pi + \lambda/2)\}$.

Observe that, if $f(\omega) = 0$ for all $|\omega| > \pi/2$, then the support of the dilated function would be the interval $[-\pi, \pi]$, and the first and third integrals would be missing from the final expression of (50). In that case, there would be no aliasing, and the function $f_H(\lambda)$ would represent the spectrum of the original process accurately.

In the case of a finite sample, the frequency-domain representation and the time-domain representation are connected via the discrete Fourier transform. We may assume that the size T of the original sample is an even number. Then, the elements of the sequence $\{x_0, x_2, x_4, \dots, x_{T-1}\}$, obtained by selecting alternate points, are described by

$$\begin{aligned} x_{2t} &= \sum_{j=0}^{T-1} \xi_j \exp\{i\omega_1 2tj\}, \quad \text{where } \omega_1 = \frac{2\pi}{T} \\ &= \sum_{j=0}^{(T/2)-1} \{\xi_j + \xi_{j+(T/2)}\} \exp\{i\omega_2 tj\}, \quad \text{where } \omega_2 = 2\omega_1 = \frac{4\pi}{T}. \end{aligned} \quad (55)$$

Here, the second equality follow from the fact that

$$\exp\{i\omega_2 j\} = \exp\{i\omega_2 [j \bmod (T/2)]\}.$$

To envisage the effect of equation (55), we can imagine wrapping the sequence $\xi_j; j = 0, 1, \dots, T-1$ twice around a circle of circumference $T/2$ so that its elements fall on $T/2$ points at angles of $\omega_2 j = 4\pi j/T; j = 1, 2, \dots, T/2$ from the horizontal.

Undoubtedly, the easiest way to recognise the effect of aliasing in the case of a finite sample is to examine the effect upon the matrix representation of the discrete Fourier transform. For conciseness, we adopt the expressions under (33), and we may take the case where $T = 8$. Subsampling by a factor of 2 is a matter of removing from the matrix $\bar{U}$ alternate rows corresponding to the odd-valued indices. This gives

$$\begin{aligned} \begin{bmatrix} x_0 \\ x_2 \\ x_4 \\ x_6 \end{bmatrix} &= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & W_8^6 & W_8^4 & W_8^2 & 1 & W_8^6 & W_8^4 & W_8^2 \\ 1 & W_8^4 & 1 & W_8^4 & 1 & W_8^4 & 1 & W_8^4 \\ 1 & W_8^2 & W_8^4 & W_8^6 & 1 & W_8^2 & W_8^4 & W_8^6 \end{bmatrix} \begin{bmatrix} \xi_0 \\ \xi_1 \\ \xi_2 \\ \xi_3 \\ \xi_4 \\ \xi_5 \\ \xi_6 \\ \xi_7 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4^3 & W_4^2 & W_4 \\ 1 & W_4^2 & 1 & W_4^2 \\ 1 & W_4 & W_4^2 & W_4^3 \end{bmatrix} \begin{bmatrix} \xi_0 + \xi_4 \\ \xi_1 + \xi_5 \\ \xi_2 + \xi_6 \\ \xi_3 + \xi_7 \end{bmatrix}. \end{aligned} \quad (56)$$

We see that, in the final vector on the RHS, the elements $\xi_0, \xi_1, \xi_2, \xi_3$ are conjoined with the elements $\xi_4, \xi_5, \xi_6, \xi_7$, which are associated with frequencies that exceed the Nyquist rate.

5. Linear Filters in Time and Frequency

In order to avoid the effects of aliasing in the process of subsampling the data, it is appropriate to apply an anti-aliasing filter to remove from the data those frequency components that are in excess of the Nyquist frequency of the subsampled data. In the case of downsampling by a factor of two, we should need to remove, via a preliminary operation, those components of frequencies in excess of $\pi/2$.

A linear filter is defined by a set of coefficients $\{\psi_k; k = -p, \dots, q\}$, which are applied to the data sequence $y(t)$ via a process of linear convolution to give a processed sequence

$$x(t) = \sum_{k=-p}^q \psi_k y(t-k). \quad (57)$$

On defining the z -transforms $\psi(z) = \sum_k \psi_k z^k$, $y(z) = \sum_t y_t z^t$ and $x(z) = \sum_t x_t z^t$, we may represent the filtering process by the following polynomial or series equation:

$$x(z) = \psi(z)y(z). \quad (58)$$

Here, $\psi(z)$ is described as the transfer function of the filter.

To ensure that the sequence $x(t)$ is bounded if $y(t)$ is bounded, it is necessary and sufficient for the coefficient sequence to be absolutely summable, such that $\sum_k |\psi_k| < \infty$. This is known by engineers as the bounded input–bounded output (BIBO) condition. The condition guarantees the existence of the discrete-time Fourier transform of the sequence. However, a meaning must be sought for this transform.

Consider, therefore, the mapping of a (doubly-infinite) cosine sequence $y(t) = \cos(\omega t)$ through a filter defined by the coefficients $\{\psi_k\}$. This produces the output

$$\begin{aligned} x(t) &= \sum_k \psi_k \cos(\omega[t-k]) \\ &= \sum_k \psi_k \cos(\omega k) \cos(\omega t) + \sum_k \psi_k \sin(\omega k) \sin(\omega t) \\ &= \alpha \cos(\omega t) + \beta \sin(\omega t) = \rho \cos(\omega t - \theta), \end{aligned} \quad (59)$$

where $\alpha = \sum_k \psi_k \cos(\omega k)$, $\beta = \sum_k \psi_k \sin(\omega k)$, $\rho = \sqrt{(\alpha^2 + \beta^2)}$ and $\theta = \tan^{-1}(\beta/\alpha)$. These results follow in view of the trigonometrical identity of (3).

The equation indicates that the filter serves to alter the amplitude of the cosine signal by a factor of ρ and to displace it in time by a delay of $\tau = \theta/\omega$ periods, corresponding to an angular displacement of θ radians. A (positive) phase delay is inevitable if the filter is operating in real time, with $j \geq 0$, which is to say that only the present and past values of $y(t)$ are comprised by the filter. However, if the filter is symmetric, with $\psi_{-k} = \psi_k$, which necessitates working off-line, then $\beta = 0$ and, therefore, $\theta = 0$.

Observe that the values of ρ and θ are specific to the frequency ω of the cosine signal. The discrete-time Fourier transform of the filter coefficients

allows the gain and the phase effects to be assessed over the entire range of frequencies in the interval $[0, \pi]$. The transform may be obtained by setting $z = \exp\{-i\omega\} = \cos(\omega) - i\sin(\omega)$ within $\psi(z)$, which constrains this argument to lie on the unit circle in the complex plane. The resulting function

$$\begin{aligned}\psi(\exp\{-i\omega\}) &= \sum_k \psi_k \cos(\omega k) - i \sum_k \psi_k \sin(\omega k) \\ &= \alpha(\omega) - i\beta(\omega)\end{aligned}\tag{60}$$

is the frequency response function, which is, in general, a periodic complex-valued function of ω with a period of 2π . In the case of a symmetric filter, it becomes a real-valued and even function, which is symmetric about $\omega = 0$.

For a more concise notation, we may write $\psi(\omega)$ in place of $\psi(\exp\{-i\omega\})$. In that case, the relationship between the coefficients and their transform can be conveyed by writing $\{\psi_k\} \longleftrightarrow \psi(\omega)$.

The effect of a linear filter on a stationary stochastic process, which is not summable, can be represented in terms of the autocovariance generating functions $f_y(z)$ and $f_x(z)$ of the sequence and its transform, respectively. Thus

$$f_x(z) = |\psi(z)|^2 f_y(z),\tag{61}$$

where $|\psi(z)|^2 = \psi(z)\psi(z^{-1})$. Setting $z = \exp\{-i\omega\}$ gives the relationship between the spectral density functions, which entails the squared gain $|\psi(\omega)|^2 = |\psi(e^{-i\omega})|^2$ of the filter.

The Ideal Filter

Imagine that it is required to remove from a data sequence the components of frequencies in excess of ω_d within the interval $[0, \pi]$. This requirement may arise from an anti-aliasing operation, preliminary to sub sampling, or it may arise from a desire to isolate a component that lies in the frequency range $[0, \omega_d]$. We shall discuss the motivation in more detail later.

The ideal filter for the purpose would be one that preserves all components with (positive) frequencies in the subinterval $[0, \omega_d]$ and nullifies all those with frequencies in the subinterval $(\omega_d, \pi]$. The specification of the frequency response function of such a filter over the interval $[-\pi, \pi]$ is

$$\psi(\omega) = \begin{cases} 1, & \text{if } |\omega| \in (0, \omega_d), \\ 1/2, & \text{if } \omega = \pm\omega_d, \\ 0, & \text{otherwise.} \end{cases}\tag{62}$$

The coefficients of the filter are given by the sinc function:

$$\psi_k = \frac{1}{2\pi} \int_{-\omega_d}^{\omega_d} e^{i\omega k} d\omega = \begin{cases} \frac{\omega_d}{\pi}, & \text{if } k = 0; \\ \frac{\sin(\omega_d k)}{\pi k}, & \text{if } k \neq 0. \end{cases}\tag{63}$$

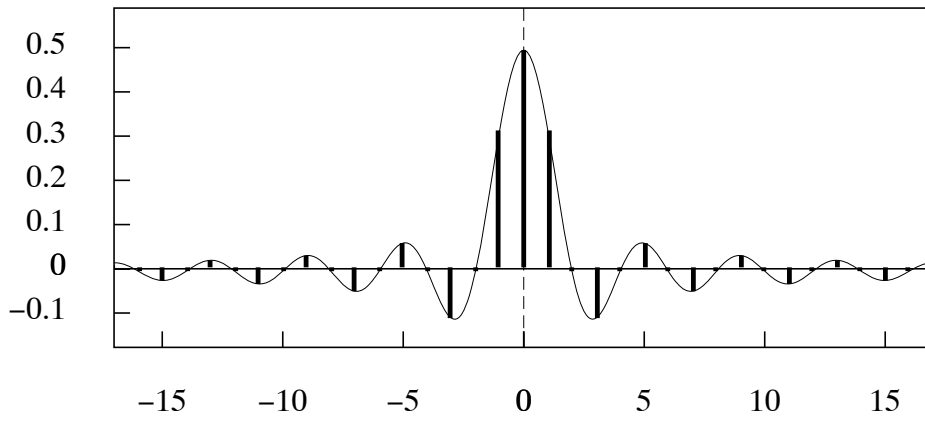


Figure 7. The central coefficients of the Fourier transform of the frequency response of an ideal low pass filter with a cut-off point at $\omega = \pi/2$. The sequence of coefficients extends indefinitely in both directions.

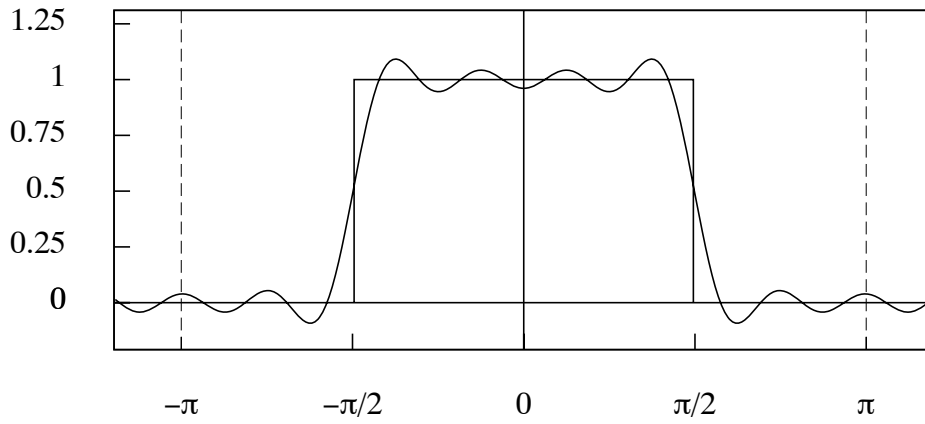


Figure 8. The result of applying a 17-point rectangular window to the coefficients of an ideal lowpass filter with a cut-off point at $\omega = \pi/2$.

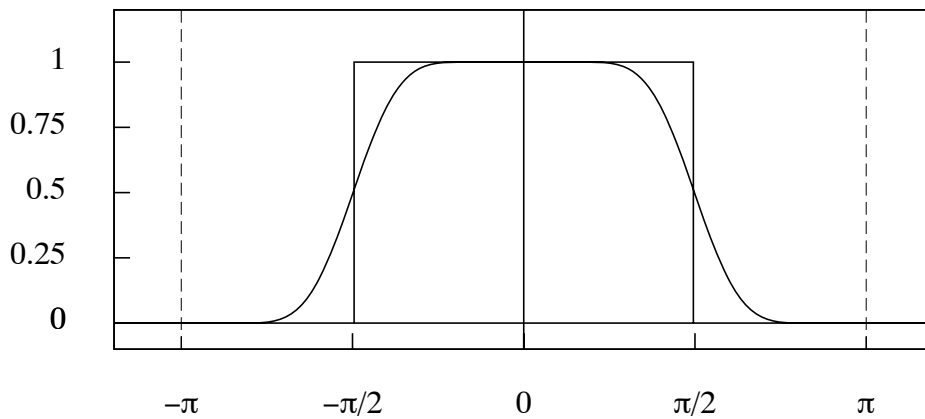


Figure 9. The result of applying a 17-point Blackman window to the coefficients of an ideal lowpass filter with a cut-off point at $\omega = \pi/2$.

These coefficients form a doubly-infinite sequence and, in order to apply such a filter to a data sample of size T , it is customary to truncate the sequence, retaining only a limited number of its central elements (Figure 7).

The truncation gives rise to a partial-sum approximation of the ideal frequency response that has undesirable characteristics. In particular, there is a ripple effect whereby the gain of the filter fluctuates within the pass band, where it should be constant with a unit value, and within the stop band, where it should be zero-valued. Within the stop band, there is a corresponding problem of leakage whereby the truncated filter transmits elements that ought to be blocked (Figure 8). This is nothing but Gibbs' effect, which has been discussed already in section 3 and illustrated in Figure 2.

The classical approach to these problems, which has been pursued by electrical engineers, has been to modulate the truncated filter sequence with a so-called window sequence, which applies a gradual taper to the higher-order filter coefficients. (A full account of this has been given by Pollock, 1999.)

The effect is to suppress the leakage that would otherwise occur in regions of the stop band that are remote from the regions where the transitions occur between stop band and pass band. The detriment of this approach is that it exacerbates the extent of the leakage within the transition regions (Figure 9).

The implication here seems to be that, if we wish to make sharp cuts in the frequency domain to separate closely adjacent spectral structures, then we require a filter of many coefficients and a very long data sequence to match it. However, Figures 8 and 9 are based upon the discrete-time Fourier transform, which entails an infinite sequence in the time domain and a continuous function in the frequency domain; and their appearances are liable be misleading.

In practice, one must work with a finite number of data points that transform into an equal number of frequency ordinates. If one is prepared to work in the frequency domain, then one can impose an ideal frequency selection upon these ordinates by setting to zero those that lie within the stop band of the ideal filter and conserving those that lie within the pass band. One can ignore whatever values a continuous frequency response function might take in the interstices between these frequency ordinates, because they have no implications for the finite data sequence.

The same effects can be achieved by operations performed within the time domain. These entail a circular filter, which is applied to the data by a process of circular convolution that is akin to the process of circular correlation described and illustrated in section 3. In the case of circular convolution, the data sequence and the sequence of filter coefficients are disposed around the circle in opposite directions—clockwise and anti clockwise. The set of T coefficients of a circular filter are derived by applying the (inverse) discrete Fourier transform to a sample of the ordinates of the ideal frequency response, taken at the T Fourier frequencies $\omega_j = 2\pi j/T$; $j = 0, 1, \dots, T - 1$.

To understand the nature of the filter, consider evaluating the frequency response of the ideal filter at one of the points ω_j . The response, expressed in

terms of the doubly-infinite coefficient sequence of the ideal filter, is

$$\psi(\omega_j) = \sum_{k=-\infty}^{\infty} \psi_k W^{jk}, \quad (64)$$

where $W^j = \exp\{-i\omega_j\} = \cos(\omega_j) - i\sin(\omega_j)$. However, since W^q is a T -periodic function, there is $W \uparrow q = W \uparrow (q \bmod T)$, where the upward arrow signifies exponentiation. Therefore,

$$\begin{aligned} \psi(\omega_j) &= \left\{ \sum_{q=-\infty}^{\infty} \psi_{qT} \right\} + \left\{ \sum_{q=-\infty}^{\infty} \psi_{qT+1} \right\} W^j + \cdots \\ &\quad + \left\{ \sum_{q=-\infty}^{\infty} \psi_{qT+T-1} \right\} W^{j(T-1)} \\ &= \psi_0^\circ + \psi_1^\circ W^j + \cdots + \psi_{T-1}^\circ W^{j(T-1)}, \quad \text{for } j = 0, 1, 2, \dots, T-1. \end{aligned} \quad (65)$$

Thus, the coefficients $\psi_0^\circ, \psi_1^\circ, \dots, \psi_{T-1}^\circ$ of the circular filter would be obtained by wrapping the infinite sequence of sinc function coefficients of the ideal filter around a circle of circumference T and adding the overlying coefficients. By applying a sampling process to the continuous frequency response function, we have created an aliasing effect in the time domain. However, summing the sinc function coefficients is not a practical way of obtaining the coefficients of the finite-sample filter. They should be obtained, instead, by transforming the frequency-response sequence or else from one of the available analytic formulae.

In the case where $\omega_d = d\omega_1 = 2\pi d/T$, the coefficients of the filter that fulfils the specification of (61) within a finite sample of T points are given by

$$\phi_d^\circ(k) = \begin{cases} \frac{2d}{T}, & \text{if } k = 0 \\ \frac{\cos(\omega_1 k/2) \sin(d\omega_1 k)}{T \sin(\omega_1 k/2)}, & \text{for } k = 1, \dots, [T/2], \end{cases} \quad (66)$$

where $\omega_1 = 2\pi/T$ and where $[T/2]$ is the integral part of $T/2$.

A more general specification for a bandpass filter is as follows:

$$\phi(\omega) = \begin{cases} 1, & \text{if } |\omega| \in (-\omega_a, \omega_b), \\ 1/2, & \text{if } \omega = \pm\omega_a, \pm\omega_b, \\ 0, & \text{for } \omega \text{ elsewhere in } [-\pi, \pi), \end{cases} \quad (67)$$

where $\omega_a = a\omega_1$ and $\omega_b = b\omega_1$. This can be fulfilled by subtracting one lowpass filter from another to create

$$\begin{aligned} \phi_{[a,b]}^\circ(k) &= \phi_b^\circ(k) - \phi_a^\circ(k) \\ &= \frac{\cos(\omega_1 k/2) \{\sin(b\omega_1 k) - \sin(a\omega_1 k)\}}{T \sin(\omega_1 k/2)} \\ &= 2 \cos(g\omega_1 k) \frac{\cos(\omega_1 k/2) \sin(d\omega_1 k)}{T \sin(\omega_1 k/2)}. \end{aligned} \quad (68)$$

Here, $2d = b - a$ is the width of the pass band (measured in terms of a number of sampled points) and $g = (a + b)/2$ is the index of its centre. These results have been demonstrated by Pollock (2009).

The relationship between the time-domain and the frequency-domain filtering can be elucidated by considering the matrix representation of the filter, which is obtained by replacing the generic argument $W^j = \exp\{-i\omega_j\}$ in (64) and (65) by the matrix operator K . This gives

$$\psi^\circ(K) = \Psi^\circ = \bar{U}\psi^\circ(D)U. \quad (69)$$

Replacing z in the z -transform $y(z)$ of the data sequence by K gives the circulant matrix $Y = \bar{U}y(D)U$. Multiplying the data matrix by the filter matrix gives

$$\begin{aligned} x(K) = X &= \Psi^\circ Y = \{\bar{U}\psi^\circ(D)U\}\{\bar{U}y(D)U\} \\ &= \bar{U}\{\psi^\circ(D)y(D)\}U = \bar{U}\psi^\circ(D)UY, \end{aligned} \quad (70)$$

where the final equality follows in view of the identity $UY = y(D)U$.

Here, there is a convolution in the time domain, represented by the product $X = \Psi^\circ Y$ of two circulant matrices, and a modulation in the frequency domain, represented by the product $x(D) = \psi^\circ(D)y(D)$ of two diagonal matrices. Of these matrices, $\psi^\circ(D)$ comprises the zeros and units of the ideal frequency response, whereas $y(D)$ comprises the spectral ordinates of the data.

It should be emphasised that $\psi(\omega)$ and $\psi^\circ(\omega_j)$ are distinct functions of which, in general, the values coincide only at the roots of unity. Strictly speaking, these are the only points for which the latter function is defined. Nevertheless, there may be some interest in discovering the values that $\psi^\circ(\omega_j)$ would deliver if its argument were free to travel around the unit circle.

Figure 10 is designed to satisfy such curiosity. Here, it can be seen that, within the stop band, the function $\psi^\circ(\omega)$ is zero-valued only at the Fourier frequencies. Since the data are compounded from sinusoidal functions with Fourier frequencies, this is enough to eliminate from the sample all elements that fall within the stop band and to preserve all that fall within the pass band.

In Figure 10, the transition between the pass band and the stop band occurs at a Fourier frequency. This feature is also appropriate to other filters that we might design, that have a more gradual transition with a mid point that also falls on a Fourier frequency. Figure 11 shows that it is possible, nevertheless, to make an abrupt transition within the space between two adjacent frequencies.

An alternative way of generating the filtered sequence within the time domain is to create a periodic extension of the data sequence via successive replications of the data. Then, the filtered values can be generated via an ordinary linear convolution of the wrapped filter coefficients of $\psi_0(z)$ with the data. It can be seen that these products are synonymous with the product of the infinite sequence of sinc function coefficients with an infinite periodic extension of the data.

A modulation in the frequency domain, which requires only T multiplications, is a far less laborious operation than the corresponding circular convolution in the time domain, which requires T^2 multiplications and $T(T - 1)$

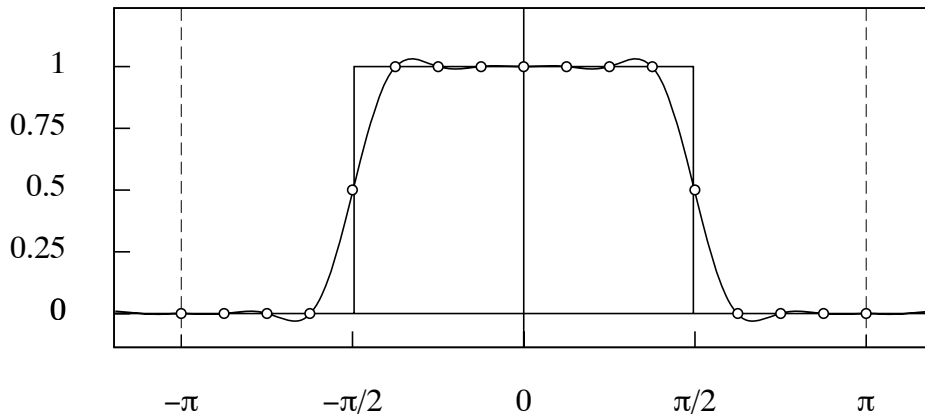


Figure 10. The frequency response of the 16-point wrapped filter defined over the interval $[-\pi, \pi)$. The values at the Fourier frequencies are marked by circles and dots.

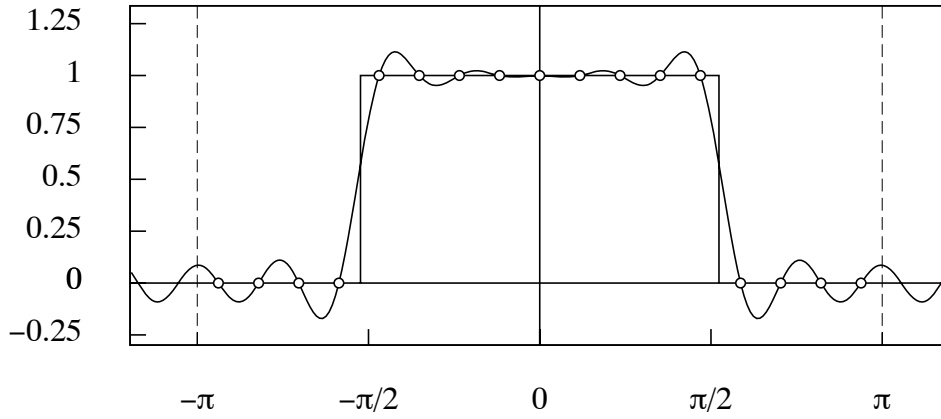


Figure 11. The frequency response of the 17-point wrapped filter defined over the interval $[-\pi, \pi)$. The values at the Fourier frequencies are marked by circles.

additions. To be set against the relative efficiency of the frequency-domain operations is the need to carry the data to that domain via a Fourier transform and to carry the products of the modulation back to the time domain via an inverse transformation. However, given the efficiency of the Fast Fourier transform, the advantage remains with the frequency-domain operations. Also, the time-domain filter may have to be formed by applying an inverse Fourier transform to a set of ordinates sampled from the frequency response of the filter.

The ideal lowpass filter has been implemented in a program that can be downloaded from the following web address:

<http://www.le.ac.uk/users/dsgp1/>

Various devices are available within the program for dealing with trended data sequences. A trend function can be interpolated through the data to deliver a sequence of residual deviations to which the filter can be applied. Thereafter, the filtered residuals and the trend can be recombined to produce

a trend-cycle trajectory.

Alternatively, the trend can be eliminated from the data by a twofold application of the difference operator. The filter can be applied to the differenced data and the result can be reinflated via the summation operator, which is the inverse of the difference operator.

The program also uses the method of frequency-domain filtering to create an ideal bandstop filter that is able to eliminate the seasonal component of a trended econometric data sequence.

Wiener–Kolmogorov Theory

The classical Wiener–Kolmogorov filtering theory concerns a doubly infinite data sequence $y(t) = \{y_t; t = 0, \pm 1, \pm 2, \dots\}$, which is the sum of a signal sequence $\xi(t)$ and a noise sequence $\eta(t)$:

$$y(t) = \xi(t) + \eta(t). \quad (71)$$

The signal and noise components are assumed to be generated by mutually independent stationary stochastic processes of zero mean. Therefore, the autocovariance generating function of the data, which may be represented by

$$\gamma_y(z) = \gamma_\xi(z) + \gamma_\eta(z), \quad (72)$$

is the sum of the autocovariance generating functions of the components.

On this basis, a filter may be formulated that produces a minimum mean-square-error estimate of the signal. Its transfer function is given by

$$\psi(z) = \frac{\gamma_\xi(z)}{\gamma_\xi(z) + \gamma_\eta(z)}. \quad (73)$$

The infinite-sample filter is a theoretical construct from which a practical implementation must be derived.

A straightforward way of deriving a filter that is applicable to the finite data sequence contained in a vector $y = [y_0, y_1, \dots, y_{T-1}]'$ is to replace the argument z within the various autocovariance generating functions by the matrix lag operator $L_T = [e_1, \dots, e_{T-1}, 0]$ and to replace z^{-1} by $F_T = L_T'$. Then, the generating function $\gamma_y(z) = \gamma_0 + \sum_{\tau=1}^{\infty} \gamma_\tau(z^{-\tau} + z^\tau)$ gives rise to matrix

$$\Omega_y = \gamma_0 I_T + \sum_{\tau=1}^{T-1} \gamma_\tau (F_T^\tau + L_T^\tau), \quad (74)$$

where the limit of the summation is when $\tau = T-1$, since $L_T^q = 0$ for all $q \geq T$. This is an ordinary autocovariance matrix. The autocovariance matrices of the components Ω_ξ and Ω_η are derived likewise, and then there is $\Omega_y = \Omega_\xi + \Omega_\eta$.

Using the matrices in place of the autocovariance generating functions in (72) gives rise to the following estimate x of the signal component ξ :

$$x = \Omega_\xi (\Omega_\xi + \Omega_\eta)^{-1} y \quad (75)$$

It is necessary to preserve the order of the factors of this product since, unlike the corresponding z -transforms, Ω_ξ and $\Omega_y^{-1} = (\Omega_\xi + \Omega_\eta)^{-1}$ do not commute in multiplication. Such filters have been discussed in greater detail by Pollock (2000), (2001), where the matter of trended data sequences is also dealt with.

For an alternative finite-sample representation of the filter, we may replace z in (72) and (73) by $K_T = \bar{U}DU$ and z^{-1} by $K'_T = K_T^{T-1} = UD\bar{U}$. Then, the generating functions are replaced by the following matrices of circular autocovariances:

$$\begin{aligned}\Omega_\xi^\circ &= \gamma_\xi(K) = \bar{U}\gamma_\xi(D)U, & \Omega_\eta^\circ &= \gamma_\eta(K) = \bar{U}\gamma_\eta(D)U \\ \text{and } \Omega_y^\circ &= \Omega_\xi^\circ + \Omega_\eta^\circ = \gamma_y(K) = \bar{U}\gamma_y(D)U.\end{aligned}\tag{76}$$

Putting these matrices in place of the autocovariance matrices of (75) gives rise to the following estimate of the signal:

$$x = \bar{U}\gamma_\xi(D)\{\gamma_\xi(D) + \gamma_\eta(D)\}^{-1}Uy = \bar{U}J_\xi Uy.\tag{77}$$

The estimate may be obtained in the following manner. First, a Fourier transform is applied to the data vector y to give Uy . Then, the elements of the transformed vector are multiplied by those of the diagonal weighting matrix $J_\xi = \gamma_\xi(D)\{\gamma_\xi(D) + \gamma_\eta(D)\}^{-1}$. Finally, the products are carried back to the time domain by the inverse Fourier transform.

The same method is indicated by equation (70), which represent the application of the ideal filter. When it is post multiplied by e_0 , the latter delivers the expression $x = \bar{U}\psi^\circ(D)Uy$, which can also be given the foregoing interpretation.

6. The Processes Underlying the Data

In this section, we shall discuss the relationship between the sampled data and the processes that generate it, which are presumed to operate in continuous time.

If the continuous process contains no sinusoidal elements with frequencies in excess of the Nyquist value of π radians, then a perfect reconstruction of the underlying function can be obtained from its sampled values by using the sinc function $\sin(\pi t)/\pi t$ of Figure 3 as an interpolator. To effect the interpolation, each of the data points is associated with a sinc function scaled by the sampled value. These functions, which are set at unit displacements along the real line, are added to form a continuous envelope.

The effect of a sinc function at any integer point t is to disperse the mass of the sampled value over the adjacent points of the real line. The function has a value of unity at t and a value of zero on all other integer points. Therefore, in the process of interpolation, it adds nothing to the values at the adjacent integer points. However, it does attribute values to the non-integer points that lie in the interstices, thereby producing a continuous function from discrete data. Moreover, the data can be recovered precisely by sampling the continuous function at the integers.

Only if the sampled data is free of aliasing, will this process of interpolation succeed in recovering the original continuous function. The question of whether aliasing has occurred is an empirical one that relates to the specific nature of the process underlying the data. However, the models of continuous-time processes that are commonly adopted by econometricians and other statisticians suggest that severe aliasing is bound to occur.

The discrete-time white-noise process $\varepsilon(t)$ that is the *primum mobile* of linear stochastic processes of the ARMA and ARIMA varieties is commonly thought to originate from sampling a Wiener process $W(t)$ defined by the following conditions:

- (a) $W(0) = 0$,
- (b) $E\{W(t)\} = 0$, for all t ,
- (c) $W(t)$ is normal,
- (d) $dW(s), dW(t)$ for all $t \neq s$ are independent stationary increments,
- (e) $\varepsilon(t) = \frac{1}{h}\{W(t+h) - W(t)\}$ for some $h > 0$.

The increments $dW(s), dW(t)$ are impulses that have a uniform power spectrum distributed over the entire real line, with infinitesimal power in any finite interval. Sampling $W(t)$ at regular intervals entails a process of aliasing whereby the spectral power of the cumulated increments gives rise to a uniform spectrum of finite power over the interval $[-\pi, \pi]$.

An alternative approach is to imagine that the discrete-time white noise is derived by sampling a continuous-time process that is inherently limited in frequency to the interval $[-\pi, \pi]$. The sinc function of Figure 3, which is limited in frequency to this interval, can also be construed as a wave packet centered on time $t = 0$. A continuous stochastic function that is band-limited by the Nyquist frequency of π will be generated by the superposition of such functions arriving at regular or irregular intervals of time and having random amplitudes. No aliasing would occur in the sampling of such a process.

(The wave packet is a concept from quantum mechanics—see, for example Dirac (1958) and Schiff (1981)—that has become familiar to statisticians via wavelet analysis—see, for example, Daubechies (1992) and Percival and Walden (2000).)

Band-Limited Processes

It is implausible to assume that a Wiener process could be the *primum mobile* of a stochastic process that is band limited within a sub interval of the (positive) frequency range $[0, \pi]$. Such processes appear to be quite common amongst the components of econometric time-series.

Figure 12 displays a sequence of the logarithms of the quarterly series of U.K. Gross Domestic Product (GDP) over the period from 1970 to 2005. Interpolated through this sequence is a quadratic trend, which represents the growth path of the economy.

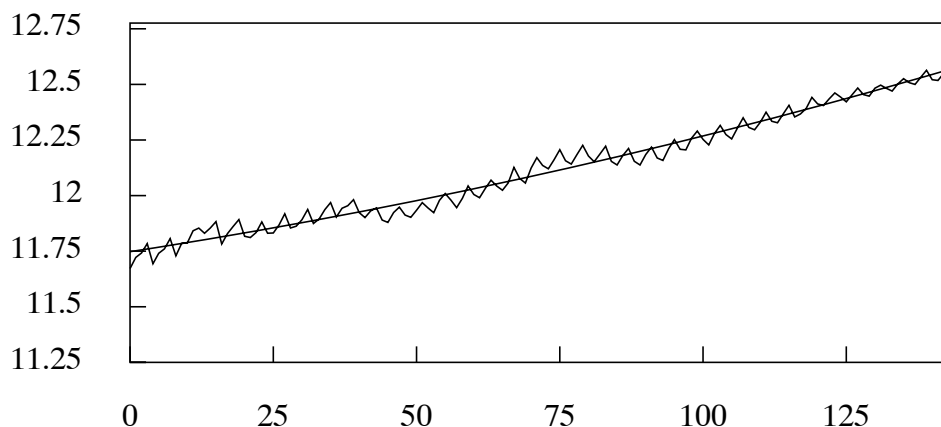


Figure 12. The quarterly sequence of the logarithms of the GDP in the U.K. for the years 1970 to 2005, inclusive, together with a quadratic trend interpolated by least squares regression.

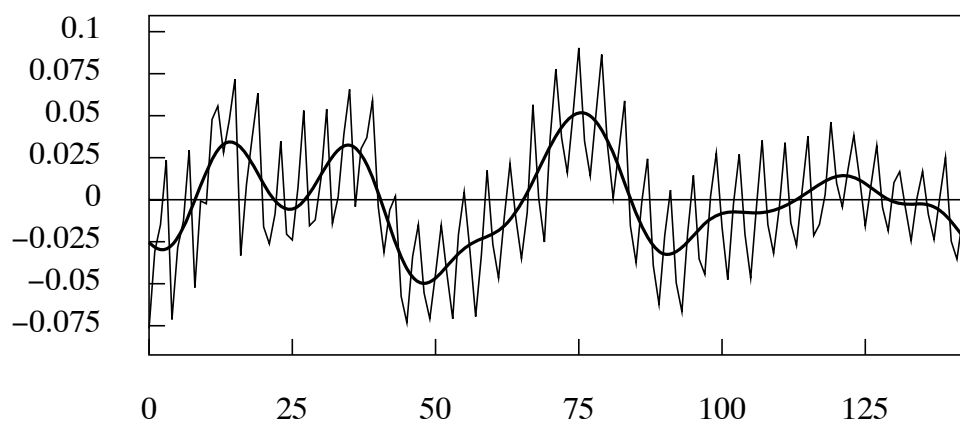


Figure 13. The residual sequence from fitting a quadratic trend to the income data of Figure 12. The interpolated line represents the business cycle.

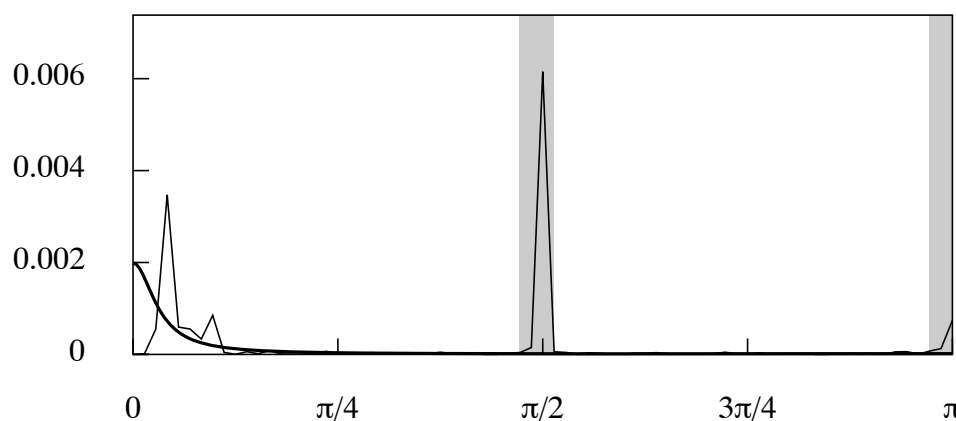


Figure 14. The periodogram of the residuals obtained by fitting a quadratic trend through the logarithmic sequence of U.K. GDP. The parametric spectrum of a fitted AR(2) model is superimposed upon the periodogram.

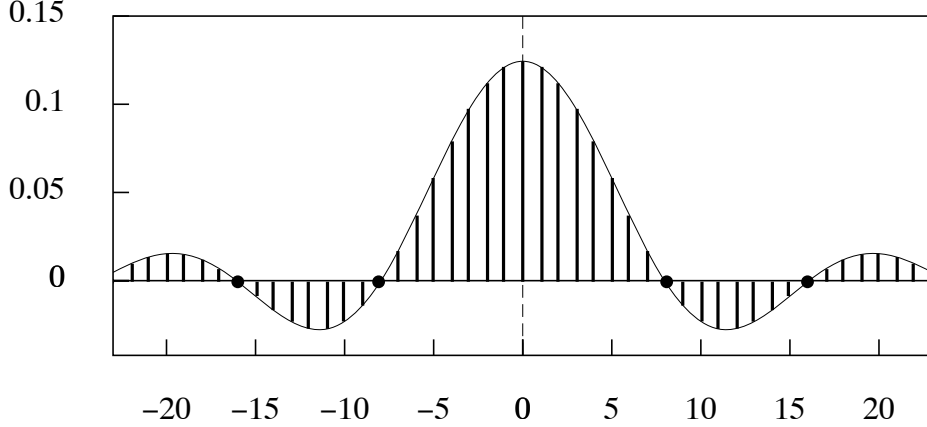


Figure 15. The sinc function wave-packet $\psi_3(t) = \sin(\pi t/8)/\pi t$ comprising frequencies in the interval $[0, \pi/8]$.

The deviations from this growth path are a combination of the low-frequency business cycle with the high-frequency fluctuations that are due to the seasonal nature of economic activity. These deviations are represented in Figure 13, which also shows an interpolated continuous function that is designed to represent the business cycle.

The periodogram of the deviations is shown in Figure 14. This gives a clear indication of the separability of the business cycle and the seasonal fluctuations. The spectral structure extending from zero frequency up to $\pi/8$ belongs to the business cycle. The prominent spikes located at the frequency $\pi/2$ and at the limiting Nyquist frequency of π are the property of the seasonal fluctuations. Elsewhere in the periodogram, there are wide dead spaces, which are punctuated by the spectral traces of minor elements of noise.

The slowly varying continuous function interpolated through the deviations of Figure 13 has been created by combining a set of sine and cosine functions of increasing frequencies in the manner of equation (1), but with the summation extending no further than the limiting frequency of the business cycle, which is $\pi/8$.

A process that is band limited to the frequency interval $[0, \pi/8]$ would be derived from the random arrival of wave packets of random amplitudes, likewise band limited in frequency. The sinc function is but one example such a packet; and it has the disadvantage of being widely dispersed in time. Its advantage is that it provides an orthogonal basis for the set of band-limited functions. The basis set is

$$\{\psi_3(t - 8k); k = 0, \pm 1, \pm 2, \dots\} \quad \text{with} \quad \psi_3(t) = \frac{\sin(\pi t/8)}{\pi t}, \quad (78)$$

wherein the functions are displaced, from one to the next, by 8 sample points. (Here the subscript on ψ_3 is in reference to the fact that $8 = 2^3$ and that the interval $[0, \pi/8]$ arises from the 3rd dyadic subdivision of $[0, \pi]$.) The function at $t = 0$ is represented in Figure 15. Whereas a continuous band-limited stochastic function would be constituted by adding the envelopes of the succession of

continuous sinc functions, its sampled version would be obtained by adding the sampled ordinates that are also represented in Figure 15.

A further point concerns the autocovariances of a band-limited process. The autocovariance function of an ordinary linear stochastic process of the ARMA variety is positive definite, which corresponds to the nonzero property of the spectral density function. Likewise, the matrix of autocovariances is positive definite, which implies that all of its eigenvalues are positive. Therefore, it is not possible to represent a band-limited process by such means.

However, it is straightforward to represent the autocovariances of a band-limited process via a circulant matrix of the form

$$\gamma(K) = \Gamma = \bar{U}\gamma(D)U. \quad (79)$$

Here, the diagonal matrix $\gamma(D)$ represents the spectral density of the process, sampled at T frequency points. The band-limited nature of the process will be reflected by the zero-valued elements of the diagonal of $\gamma(D)$ and by the singularity of Γ .

Fitting ARMA Models to the Business Cycle

One might wish to fit a parametric model to the detrended data of Figure 13. For this purpose, the multiplicative seasonal ARIMA model of Box and Jenkins (1976) seems, at first, to be a natural choice. Such a model embodies a seasonal difference operator of the form $(1 - z^4) = (1 - z)(1 + z + z^2 + z^3)$, of which the ordinary difference operator $(1 - z)$ is a factor. However, the data of Figure 13, have already been detrended. Therefore, the presence or the redundant difference operator renders the model inappropriate. A more appropriate recourse is to remove the seasonal component from the data before fitting an autoregressive model aimed at representing the business cycle component.

The seasonal component can be removed from the data of Figure 13 by nullifying the spectral ordinates at the seasonal frequencies and at some adjacent frequencies. This operation, which can be performed by the program mentioned in Section 5, is represented in Figure 14 by the highlighted bands that correspond to the stop bands of the deseasonalising filter. Also represented in Figure 14 is the parametric spectrum of a second-order autoregressive AR(2) model that has been fitted to the seasonally adjusted data,

One would expect to be able to characterise the low-frequency dynamics of the business cycle component via a pair of conjugate complex autoregressive roots to be found within the estimated model. The arguments of such roots should characterise the angular velocity of the cycle; and their moduli should represent the degree of its damping. However, it is found that the roots in question are real-valued. It has been reported by Pagan (1997), amongst others, that this is the almost invariable outcome of such exercises.

We might wish, nevertheless, to use an ARMA model to characterise the business cycle component that is supported on the interval $[0, \pi/8]$ and which is represented by the smooth continuous function that has been interpolated through the data of Figure 13. In general, when we wish to fit an ARMA model to a component of the data that is supported on a frequency interval $[0, \omega_d]$,

where $\omega_d < \pi$, it is necessary to map the component onto the full frequency interval $[0, \pi]$. For this purpose, it may be necessary to synthesise a continuous function from the Fourier ordinates within the subinterval $[0, \omega_d]$. The function can then be resampled at the lesser rate of one observation in every π/ω_d periods to produce data that are appropriate to the purpose.

In the case of the business cycle component of Figure 13, which has a spectral signature that falls within the interval $[0, \pi/8]$, it is sufficient to take one in eight of the data points of the anti-aliased sequence that has been obtained by subjecting the original data to an ideal lowpass filter with a cut-off at $\pi/8$. The periodogram of the resulting subsampled sequence, which has the frequency range of $[0, \pi]$, will have a profile identical to that of the spectral of structure that occupies the interval $[0, \pi/8]$ in Figure 14.

The AR(2) model that fits these data has a pair of conjugate complex roots with an argument of $\theta = 1.35$ radians (77°), which corresponds to a cyclical period of 37 quarters, and a modulus of $\rho = 0.695$. However, the evidence of the interpolated function of Figure 13 and of the periodogram of Figure 14 suggests that the estimated period is an average of the durations longer major cycle and a shorter minor cycle. However, with only 18 points available in the subsampled data, it seems that there is insufficient information to justify the presentation of the AR(4) model that reflects these putative cycles in two pairs of conjugate complex roots.

The Recognition of the Frequency Limitation

Some reasons should be given for why the prevalence of band limited processes has been overlooked by econometricians. One evident reason is that, in the past, econometricians have rarely examined the periodograms of their data sequences.

However, the band limited nature of a process is liable to be obscured whenever there are significant disjunctions in the periodic extension of the data at the points where the end of one replication of the sample joins the beginning of the next. Such disjunctions will give rise to a slew of spectral power that will serve to conceal the underlying structure of a band-limited process.

The presence of a barely perceptible trend in the data will be enough to cause such a disjunction. The periodic extension of a finite linearly trending sequence will have the appearance of the serrated edge of a carpenter's saw. The spectrum of a saw tooth function has the *one-over-f* profile of a rectangular hyperbola, which will mask the finer spectral details. (See Hamming 1989, for example.) Granger (1966) has described such spectra as typical of econometric time series. His characterisation has greatly influenced the perceptions of econometricians. It has suggested to many of them that the spectral analysis of economic data is liable to be difficult if not fruitless.

Several authors, including Granger and Hyung (2004), have observed that structural breaks within the processes generating the data can give rise to slowly decaying autocorrelations and to other properties of long-memory processes, including *one-over-f* periodograms that decay hyperbolically as the frequency rises. Such breaks are not dissimilar to the end-of-sample disjunctions which are

considered here. However, the latter seem to provide a more fruitful explanation for the occurrence of misleading long-memory properties.

A third reason for the failure to perceive the band-limited nature of some of these process may be that econometricians have been unduly influenced by the prevalence of the Wiener process as a model of the forcing functions of continuous-time stochastic processes. A Wiener process, which has a frequency band of infinite width, is the antithesis of a band-limited process.

Within the computer program that is an adjunct of this paper, there are several provisions for ensuring that the data can be wrapped seamlessly around the circumference of a circle in a manner that avoids disjunctions in the periodic extension.

If the data are to be detrended by the interpolation of a polynomial function, then extra weight can be given to the points at the beginning and the end of the sample to ensure that the polynomial passes through their midst. This may have the desired effect upon the residual sequence, which will be used thereafter in the analysis.

If the data have already been detrended, then it may be desirable to add tapered extrapolations at both ends which converge on a common asymptote. The extrapolations are liable to be discarded after the data have been filtered. Alternatively, if the detrended data display prominent seasonal fluctuations of a pattern that is changing over the course of the sample, then a segment may be interpolated into the circular data in which the pattern of the final year of the sample gradually morphs into the pattern of the first year.

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Appendix: The Spectral Representation of a Stationary Process

Consider a sample $\{y_0, y_1, \dots, y_{T-1}\}$ of T observations on a stationary stochastic process. These are amenable to a discrete Fourier transform, whereby they can be expressed as

$$y_t = \sum_{j=0}^n \left\{ \alpha_j \cos(\omega_j t) + \beta_j \sin(\omega_j t) \right\}, \quad (A.1)$$

where $\omega_j = 2\pi j/T$ and n is the integral part of $T/2$.

By allowing n to tend to infinity, it is possible to express a sequence of indefinite length in terms of a sum of sine and cosine functions. However, in the limit as $n \rightarrow \infty$, the coefficients α_j, β_j tend to vanish; and, therefore, an alternative representation in terms of differentials is called for.

By writing $\alpha_j = dA(\omega_j)$, $\beta_j = dB(\omega_j)$, where $A(\omega)$, $B(\omega)$ are step functions with discontinuities at the points $\{\omega_j; j = 0, \dots, n\}$, the expression (A.1) can be rendered as

$$y_t = \sum_j \left\{ \cos(\omega_j t) dA(\omega_j) + \sin(\omega_j t) dB(\omega_j) \right\}. \quad (\text{A.2})$$

In the limit, as $n \rightarrow \infty$, the summation is replaced by an integral to give the expression

$$y(t) = \int_0^\pi \left\{ \cos(\omega t) dA(\omega) + \sin(\omega t) dB(\omega) \right\}. \quad (\text{A.3})$$

Here, $\cos(\omega t)$ and $\sin(\omega t)$, and therefore $y(t)$, may be regarded as infinite sequences defined over the entire set of positive and negative integers.

Since $A(\omega)$ and $B(\omega)$ are discontinuous functions for which no derivatives exist, one must avoid using $\alpha(\omega)d\omega$ and $\beta(\omega)d\omega$ in place of $dA(\omega)$ and $dB(\omega)$. Moreover, the integral in (A.3) is a Fourier–Stieltjes integral.

In order to derive a statistical theory for the process that generates $y(t)$, one must make some assumptions concerning the functions $A(\omega)$ and $B(\omega)$. So far, the sequence $y(t)$ has been interpreted as a realisation of a stochastic process. If $y(t)$ is regarded as the stochastic process itself, then the functions $A(\omega)$, $B(\omega)$ must, likewise, be regarded as stochastic processes defined over the interval $[0, \pi]$. A single realisation of these processes now corresponds to a single realisation of the process $y(t)$.

The first assumption to be made is that the functions $dA(\omega)$ and $dB(\omega)$ represent a pair of stochastic processes of zero mean, which are indexed on the continuous parameter ω . Thus

$$E\{dA(\omega)\} = E\{dB(\omega)\} = 0. \quad (\text{A.4})$$

The second and third assumptions are that the two processes are mutually uncorrelated and that non-overlapping increments within each process are uncorrelated. Thus

$$\begin{aligned} E\{dA(\omega)dB(\lambda)\} &= 0 \quad \text{for all } \omega, \lambda, \\ E\{dA(\omega)dA(\lambda)\} &= 0 \quad \text{if } \omega \neq \lambda, \\ E\{dB(\omega)dB(\lambda)\} &= 0 \quad \text{if } \omega \neq \lambda. \end{aligned} \quad (\text{A.5})$$

The final assumption is that the variance of the increments is given by

$$\begin{aligned} V\{dA(\omega)\} &= V\{dB(\omega)\} = 2dF(\omega) \\ &= 2f(\omega)d\omega. \end{aligned} \quad (\text{A.6})$$

It can be seen that, unlike $A(\omega)$ and $B(\omega)$, $F(\omega)$ is a continuous function that is differentiable except, at most, on a set of measure zero. The function $F(\omega)$ and its derivative $f(\omega)$ are the spectral distribution function and the spectral density function, respectively.

In order to express (A.3) in terms of complex exponentials, we may define a pair of conjugate complex stochastic processes:

$$\begin{aligned} dZ(\omega) &= \frac{1}{2} \{dA(\omega) - i dB(\omega)\}, \\ dZ^*(\omega) &= \frac{1}{2} \{dA(\omega) + i dB(\omega)\}. \end{aligned} \quad (A.7)$$

Also, the domain of the functions $A(\omega)$, $B(\omega)$ may be extended from $[0, \pi]$ to $[-\pi, \pi]$ by regarding $A(\omega)$ as an even function such that $A(-\omega) = A(\omega)$ and by regarding $B(\omega)$ as an odd function such that $B(-\omega) = -B(\omega)$. Then there is

$$dZ^*(\omega) = dZ(-\omega). \quad (A.8)$$

From the conditions of (A.5), it follows that

$$\begin{aligned} E\{dZ(\omega)dZ^*(\lambda)\} &= 0 \quad \text{if } \omega \neq \lambda, \\ E\{dZ(\omega)dZ^*(\omega)\} &= f(\omega)d\omega. \end{aligned} \quad (A.9)$$

These results may be used to reexpress (A.3) as

$$\begin{aligned} y(t) &= \int_0^\pi \left\{ \frac{(e^{i\omega t} + e^{-i\omega t})}{2} dA(\omega) - i \frac{(e^{i\omega t} - e^{-i\omega t})}{2} dB(\omega) \right\} \\ &= \int_0^\pi \left\{ e^{i\omega t} \frac{\{dA(\omega) - i dB(\omega)\}}{2} + e^{-i\omega t} \frac{\{dA(\omega) + i dB(\omega)\}}{2} \right\} \\ &= \int_0^\pi \left\{ e^{i\omega t} dZ(\omega) + e^{-i\omega t} dZ^*(\omega) \right\}. \end{aligned} \quad (A.10)$$

When the integral is extended over the range $[-\pi, \pi]$, this becomes

$$y(t) = \int_{-\pi}^\pi e^{i\omega t} dZ(\omega), \quad (A.11)$$

which is commonly described as the spectral representation of the process $y(t)$.

Introduction to Option Pricing with Fourier Transform: Option Pricing with Exponential Lévy Models

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Abstract

This sequel is designed as an introduction to Fourier transform option pricing for readers who have zero previous knowledge of Fourier transform. First part of this sequel is devoted for the basic understanding of Fourier transform and discrete Fourier transform using numerous examples and providing important properties. Second part of this sequel applies FT and DFT option pricing approach for three exponential Lévy models: Classic Black-Scholes model which is the only continuous exponential Lévy model, Merton jump-diffusion model (1976) which is an exponential Lévy model with finite arrival rate of jumps, and variance gamma model by Madan, Carr, and Chang (1998) which is an exponential Lévy model with infinite arrival rate of jumps. Some readers may question that what the need for FT option pricing is since all three models above can price options with closed form formulae. The answer is that these three models are special cases of more general exponential Lévy models. Options cannot be priced with general exponential Lévy models using the traditional approach of the use of the risk-neutral density of the terminal stock price because it is not available. Therefore, Carr and Madan (1999) rewrite the option price in terms of a characteristic function of the log terminal stock price by the use of FT. The advantage of FT option pricing is its generality in the sense that the only thing necessary for FT option pricing is a characteristic function of the log terminal stock price. This generality of FT option pricing speeds up the calibration and Monte Carlo simulation with various exponential Lévy models. It is no doubt to us that FT option pricing will be a standard option pricing method from now on.

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[1] Introduction

Many of the option pricing models assume that a stock price process $\{S_t; 0 \leq t \leq T\}$ follows an exponential (geometric) Lévy process:

$$S_t = S_0 e^{L_t},$$

where $\{L_t; 0 \leq t \leq T\}$ is a Lévy process. The reason of the popularity of exponential (geometric) Lévy models is its mathematical tractability which comes from the independent and stationary increments of Lévy processes. Classic Black-Scholes (BS) model chooses a Brownian motion with drift process which is the only continuous Lévy process as their choice of a (risk-neutral) Lévy process:

$$L_t = \left(r - \frac{1}{2} \sigma^2 \right) t + \sigma B_t,$$

where $\{B_t; 0 \leq t \leq T\}$ is a standard Brownian motion process. This specification leads to a normally distributed conditional risk-neutral log return density:

$$\mathbb{Q}(\ln(S_T / S_0) | \mathcal{F}_0) = \frac{1}{\sqrt{2\pi\sigma^2 T}} \exp \left[-\frac{\left\{ \ln(S_T / S_0) - \left(r - \frac{1}{2} \sigma^2 \right) T \right\}^2}{2\sigma^2 T} \right].$$

BS call price can be simply calculated as the discounted value of the expected terminal payoff under risk-neutral measure $\mathbb{Q}$:

$$C(S_0, T) = e^{-rT} \int_K^\infty (S_T - K) \mathbb{Q}(S_T | \mathcal{F}_0) dS_T, \quad (1.1)$$

$$\text{where } \mathbb{Q}(S_T | \mathcal{F}_0) = \frac{1}{S_T \sqrt{2\pi\sigma^2 T}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_0 + \left(r - \frac{1}{2} \sigma^2 \right) T \right) \right\}^2}{2\sigma^2 T} \right] \text{ which is a}$$

lognormal density.

But even before BS model was developed, researchers knew that the empirical log return density is not normal, it shows excess kurtosis and skewness. Thus, all the option pricing models after BS (so called beyond BS) try to capture excess kurtosis and negative skewness of the risk-neutral log return density by the use of different techniques.

This sequel deals with Merton jump-diffusion model (we call it Merton JD model) and variance gamma model by Madan, Carr, and Chang (1998) (we call it VG model). These are both exponential Lévy models of different types. Merton's choice of Lévy process is a Brownian motion with drift process plus a compound Poisson jump process which has a continuous path with occasional jumps:

$$L_t = \left(r - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma B_t + \sum_{i=1}^{N_t} Y_i .$$

Merton JD model can be categorized as a finite activity exponential Lévy model because the expected number of jumps per unit of time (i.e. intensity λ) is finite and small. In other words, the Lévy measure $\ell(dx)$ of Merton JD model is finite:

$$\int \ell(dx) < \infty .$$

The only but important difference between the BS and the Merton JD model is the addition of a compound Poisson jump process $\sum_{i=1}^{N_t} Y_i$. Merton introduces three extra parameters λ (intensity of the Poisson process), μ (mean log stock price jump size), and δ (standard deviation of log stock price jump size) to the original BS framework and controls the (negative) skewness and excess kurtosis of the log return density.

Choice of Lévy process by Madan, Carr, and Chang (1998) is a VG process plus a drift:

$$L_t \equiv \left\{ r + \frac{1}{\kappa} \ln \left(1 - \theta \kappa - \frac{\sigma^2 \kappa}{2} \right) \right\} t + VG(x_t; \theta, \sigma, \kappa) .$$

A VG process $VG(x_t; \theta, \sigma, \kappa)$ is defined as a stochastic process $\{X_t; 0 \leq t \leq T\}$ created by random time changing (i.e. subordinating) a Brownian motion with drift process $\theta t + \sigma B_t$ by a tempered 0-stable subordinator (i.e. a gamma subordinator) $\{S_t; 0 \leq t \leq T\}$ with unit mean rate:

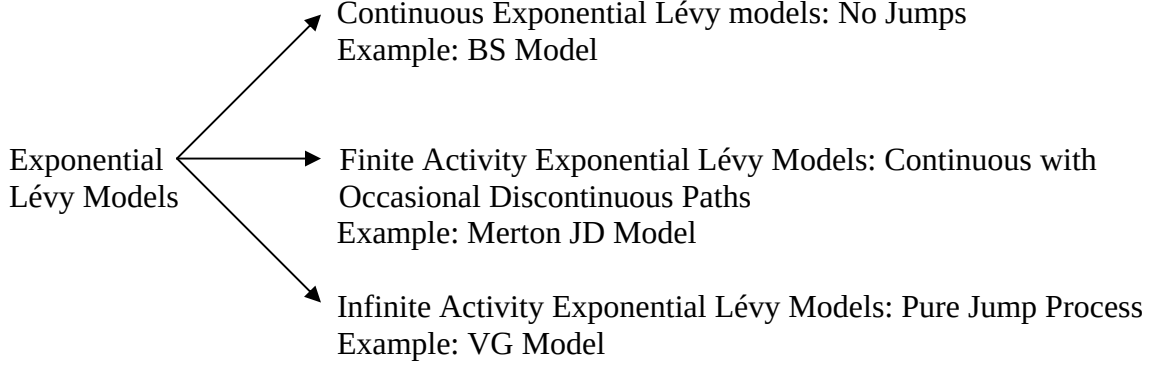
$$X_t \equiv \theta(S_t) + \sigma B_{S_t} .$$

A VG process $VG(x_t; \theta, \sigma, \kappa)$ is characterized as a pure jump Lévy process with infinite arrival rate of jumps. In other words, the Lévy measure of a VG process has an infinite integral:

$$\int_{-\infty}^{\infty} \ell(x) dx = \infty .$$

This means that a VG process has infinitely many small jumps but a finite number of large jumps. VG model introduces two extra parameters: One is variance rate parameter

κ which controls the degree of the randomness of the subordination and the larger κ implies the fatter tails of the log return density. The other is the drift parameter of the subordinated Brownian motion process θ which captures the skewness of the log return density.



Traditionally, both Merton JD call price and VG call price have been expressed as BS type closed form function using the conditional normality of both models. Merton JD call price can be expressed as the weighted average of the BS call price conditional on that the underlying stock price jumps i times to the expiry with weights being the probability that the underlying jumps i times to the expiry. Because of the subordination structure $X_t \equiv \theta(S_t) + \sigma B_{S_t}$ of the VG process $\{X_t; 0 \leq t \leq T\}$, the probability density of VG process can be expressed as the conditionally normal by conditioning on the fact that the realized value of the random time change by a gamma subordinator S_t with unit mean rate is $S_t = g$:

$$VG(x_t | t = S_t = g) = \frac{1}{\sqrt{2\pi\sigma^2 g}} \exp\left\{-\frac{(x_t - \theta g)^2}{2\sigma^2 g}\right\}.$$

Using this conditional normality of VG process, the conditional call price can be obtained as a BS type formula after lengthy and tedious process of numerous changes of variables.

The fact that a call price can be expressed as a BS type formula implies that the model has a closed form expression for the log return density $\mathbb{Q}(\ln(S_T / S_0) | \mathcal{F}_0)$. Merton JD model has a log return density of the form:

$$\mathbb{Q}_{Merton}(\ln(S_T / S_0) | \mathcal{F}_0) = \sum_{i=0}^{\infty} \frac{e^{-\lambda t} (\lambda t)^i}{i!} N\left(\ln(S_T / S_0); (r - \frac{\sigma^2}{2} - \lambda k)t + i\mu, \sigma^2 t + i\delta^2\right),$$

where $N(x; a, b) \equiv \frac{1}{\sqrt{2\pi b}} \exp\left\{-\frac{(x-a)^2}{2b}\right\}$ which is a normal density. VG model has a log return density of the form:

$$\mathbb{Q}_{VG}(\ln(S_T / S_0) | \mathcal{F}_0) = \frac{\sqrt{2} \exp\left(\frac{\theta}{\sigma^2} x_T\right)}{\kappa^{T/\kappa} \sigma \sqrt{\pi} \Gamma(T/\kappa)} \left(\frac{x_T^2}{\frac{2\sigma^2}{\kappa} + \theta^2}\right)^{\frac{T-1}{2\kappa-4}} K_{\frac{T-1}{\kappa-2}} \left(\frac{\sqrt{x_T^2 \left(\frac{2\sigma^2}{\kappa} + \theta^2\right)}}{\sigma^2}\right),$$

where $x_T \equiv \ln(S_T / S_0) - \left\{r + \frac{1}{\kappa} \ln\left(1 - \theta\kappa - \frac{\sigma^2 \kappa}{2}\right)\right\} t$. The existence of the closed form expression for the log return density $\mathbb{Q}(\ln(S_T / S_0) | \mathcal{F}_0)$ enables the use of the equation (1.1) to calculate a call price.

But Merton JD model and VG model are special cases of exponential Lévy models in the sense that more general exponential Lévy models do not have closed form log return densities $\mathbb{Q}(\ln(S_T / S_0) | \mathcal{F}_0)$ or they cannot be expressed using special functions of mathematics. Therefore, we cannot price plain vanilla options using the equation (1.1). How do we price options using general exponential Lévy models? The answer is to use a very interesting fact that characteristic functions (CF) of general exponential Lévy processes are always known in closed-forms or can be expressed in terms of special functions of mathematics although their probability densities are not. There is one-to-one relationship between a probability density and a CF (i.e. CF is just a Fourier transform of a probability density with FT parameters (1,1)) and both of which uniquely determine a probability distribution. If we can somehow rewrite (1.1) in terms of a characteristic function of the conditional terminal stock price $S_T | \mathcal{F}_0$ (i.e. log of $S_T | \mathcal{F}_0$ to be more precise) instead of its probability density $\mathbb{Q}(S_T | \mathcal{F}_0)$, we will be able to price options in general exponential Lévy models.

The purpose of this sequel is to introduce the basics of Fourier transform option pricing approach developed by Carr and Madan (1999) to the readers who have no previous knowledge of Fourier transforms. Carr and Madan (1999)'s contribution was rewriting the equation (1.1) in terms of a CF of the conditional log terminal stock price $\phi(\ln S_T | \mathcal{F}_0)$:

$$C(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} d\omega. \quad (1.2)$$

And approximate the equation (1.2) using DFT (i.e. simply by taking a sample of size N):

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k} \times \frac{1}{N} \sum_{j=0}^{N-1} w_j \{ \exp(i\pi j) \psi_T(\omega_j) \} \exp(-i2\pi jn/N), \quad (1.3)$$

which improves significant amount of computational time. This simple operation is critically important in the pricing with general exponential Lévy models. Why? Because without Fourier transform pricing approach, we cannot price options in general exponential Lévy models or we have to spend tremendous amount of energy just to come up with closed form solution like the VG model. The excellence of FT option pricing is its simplicity and generality and FT pricing works as long as a CF of the conditional log terminal stock price $S_T | \mathcal{F}_0$ is obtained in closed form.

The structure of this sequel is as follows. Chapter 2 provides readers with minimal necessary knowledge before learning Fourier transform. Chapter 3 defines FT and then, numerous examples are presented in order to intuitively understand the meaning of FT. We also present the important properties of FT. These are not necessary for the beginners, but we believe these will help readers as they proceed to more advanced integral transform pricing methods. In Chapter 4, a characteristic function is defined and its properties are discussed. We show, using several examples, how to obtain moments by using a CF (or characteristic exponent). Moment generating function is also dealt. Chapter 5 gives an introduction to the discrete Fourier transform which is just an approximation of FT. This approximation is done by sampling a finite number of points N of a continuous time domain function $g(t)$ with time domain sampling interval Δt (seconds) and sampling a finite number of points N of a continuous FT $\mathcal{G}(\omega)$ with angular frequency sampling interval $\Delta \omega$ Hz. In other words, both the original continuous time domain function $g(t)$ and the original continuous FT $\mathcal{G}(\omega)$ are approximated by a sample of N points. The use of DFT improves the computation time by a tremendous amount. Following Chapter 3, the examples of DFT and its properties are examined. In Chapter 6, a Lévy process is defined and its properties are discussed. We frequently use Lévy-Khinchin representation to obtain a CF of a Lévy process. Chapter 7 revisits the Black-Scholes model as an exponential Lévy model and its basic properties are reviewed. Chapter 8 gives Carr and Madan (1999)'s general FT call price and our version of general DFT call price. These general FT and DFT call prices are applied in the BS framework which shows that the original BS, BS-FT, and BS-DFT call prices are identical as expected. Chapter 9 illustrates the derivation of Merton JD model, factors which determine the skewness and the excess kurtosis of the log return density, Lévy measure of jump-diffusion process, and the traditional (i.e. PDE and martingale approach) option pricing with Merton JD model. Chapter 10 applies the general FT and DFT call price to the Merton JD model. This is simply done by substituting the CF of Merton JD log terminal stock price. Chapter 11 presents the derivation of VG model, factors which

determine the skewness and the excess kurtosis of the log return density, Lévy measure of VG process, and its closed form and numerical call price. Chapter 12 is an application of the general FT and DFT call price to the VG model which is simply done by substituting the CF of VG log terminal stock price. Chapter 13 gives concluding remarks.

[2] Prerequisite for Fourier Transform

In this section we present prerequisite knowledge before moving to Fourier transform.

[2.1] Radian

Radian is the unit of angle. A complete circle has $2\pi = 6.28319$ radians which is equal to 360° . This in turn means that one radian is equal to:

$$1 \text{ radian} = \frac{360^\circ}{2\pi} = 57.2958^\circ.$$

[2.2] Wavelength

Wavelength λ of a waveform is defined as a distance (d) between peaks or troughs. In other words, wavelength is the distance at which a waveform completes one cycle:

$$\lambda \equiv \frac{\text{distance}}{1 \text{ cycle}}. \quad (2.1)$$

Let v be the speed (distance/second), and f be the frequency (cycles/second, explained soon) of a waveform. These are related by:

$$\lambda(\text{distance / cycle}) \equiv \frac{v(\text{distance/second})}{f(\text{cycles/second})}. \quad (2.2)$$

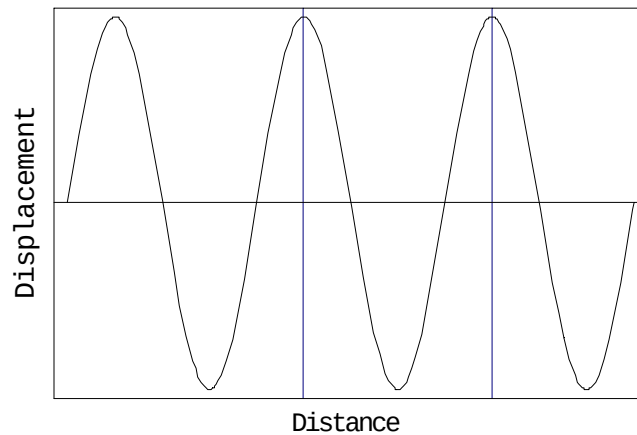


Figure 2.1: Illustration of Wavelength λ .

[2.3] Frequency, Angular Frequency, and Period of a Waveform

Period T of oscillation of a wave is the seconds (time) taken for a waveform to complete one wavelength:

$$T \equiv \frac{\text{seconds}}{1 \text{ wavelength (cycle)}} . \quad (2.3)$$

Period is by definition a reciprocal of a frequency. Let f be the frequency of a wave. Then:

$$T(\text{seconds/cycle}) \equiv \frac{1}{f(\text{cycles/second})} . \quad (2.4)$$

Frequency f of a wave measures the number of times for a wave to complete one wavelength (cycle) per second:

$$f \equiv \frac{\text{number of wavelengths (cycles)}}{1 \text{ second}} . \quad (2.5)$$

By definition, f is calculated as a reciprocal of the period of a wave:

$$f(\text{cycles/second}) \equiv \frac{1}{T(\text{seconds/cycle})} . \quad (2.6)$$

Frequency f is measured in Hertz (Hz). 1 Hz wave is a wave which completes one wavelength (cycle) per second. The frequency of the AC (alternating current) in U.S. is 60 Hz. Human beings can hear frequencies from about 20 to 20,000 Hz (called range of hearing). Alternatively, frequency can be calculated using the speed v and wavelength λ of a wave as:

$$f(\text{cycles/second}) \equiv \frac{v(\text{distance/second})}{\lambda(\text{distance/cycle})} . \quad (2.7)$$

Angular frequency (also called angular speed or radian frequency) ω is a measure of rotation rate (i.e. the speed at which an object rotates). The unit of measurement for ω is radians per 1 second. Since one cycle equals 2π radians, angular frequency ω is calculated as:

$$\omega(\text{radians/second}) = \frac{2\pi(\text{radians})}{T(\text{seconds/cycle})} = 2\pi f(\text{cycles/second}) . \quad (2.8)$$

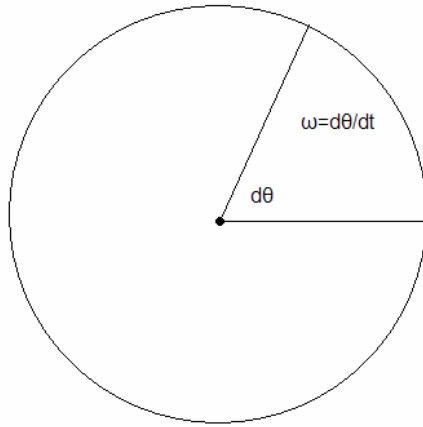


Figure 2.2: Definition of Angular Frequency.

Consider a sine wave $g(t) = \sin(2\pi(5)t)$ which is illustrated in Figure 2.3 for the time between 0 and 2 seconds. This sine wave has frequency $f = 5$ Hz (5 cycles per second) and angular frequency $\omega = 10\pi$ Hz (10π radians per second). Its period is

$$T \equiv \frac{1}{f} = \frac{1}{5} = 0.2 \text{ (seconds/cycle).}$$

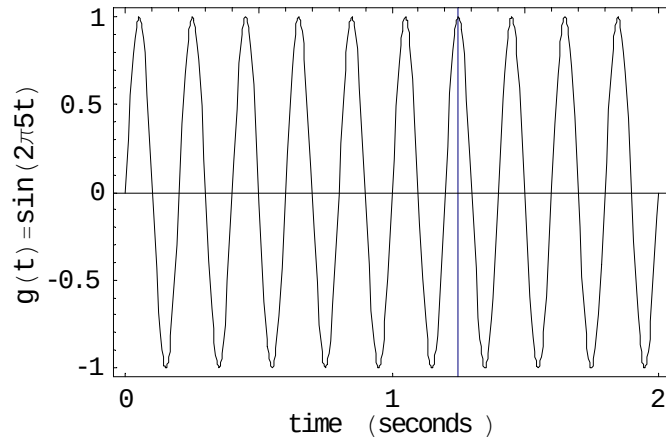


Figure 2.3: Plot of 5 Hz Sine Wave $g(t) = \sin(2\pi(5)t)$.

[2.4] Sine and Cosine

Let θ be an angle which is measured counterclockwise from the x -axis along an arc of a unit circle. Sine function ($\sin \theta$) is defines as a vertical coordinate of the arc endpoint. Cosine function ($\cos \theta$) is defined as a horizontal coordinate of the arc endpoint. Sine and cosine functions $\sin \theta$ and $\cos \theta$ are periodic functions with period 2π as illustrated in Figure 2.5:

$$\sin \theta = \sin(\theta + 2\pi h),$$

$$\cos \theta = \cos(\theta + 2\pi h),$$

where h is any integer.

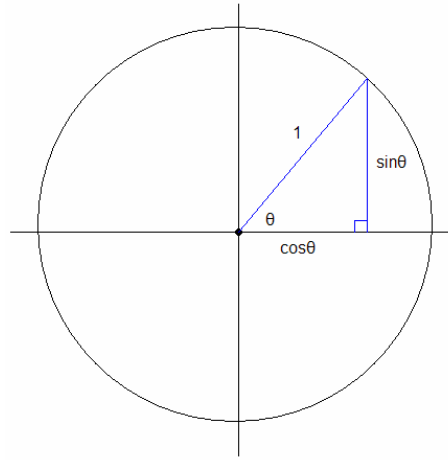


Figure 2.4: The Definition of Sine and Cosine Function with Unit Circle

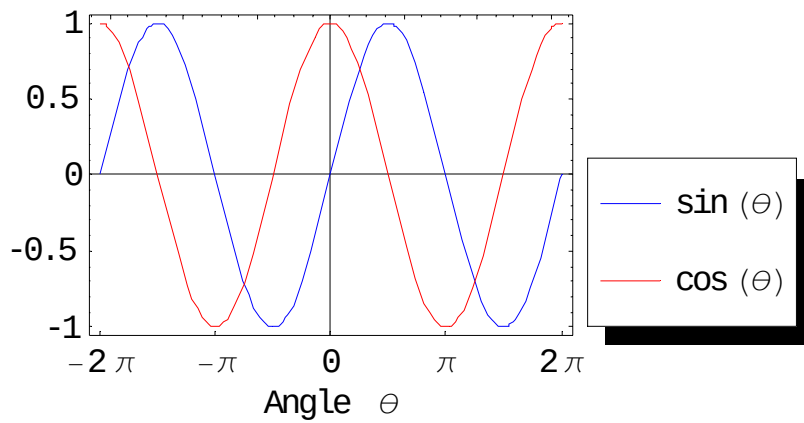


Figure 2.5: Sine and Cosine Function $\sin(\theta)$ and $\cos(\theta)$.

Following Pythagorean theorem, we have the identity:

$$\sin^2 \theta + \cos^2 \theta = 1. \quad (2.9)$$

[2.5] Derivative and Integral of Sine and Cosine Function

Let $\sin(x)$ and $\cos(x)$ be sine and cosine functions for $x \in \mathbb{R}$. The derivative of $\sin(x)$ can be expressed as:

$$\frac{d \sin(x)}{dx} = \cos(x). \quad (2.10)$$

The derivative of $\cos(x)$ can be expressed as:

$$\frac{d \cos(x)}{dx} = -\sin(x). \quad (2.11)$$

The integral of $\sin(x)$ can be expressed as:

$$\int_{-\infty}^{\infty} \sin(x) dx = -\cos(x). \quad (2.12)$$

Refer to any graduate school level trigonometry textbook for proofs.

[2.6] Series Definition of Sine Function and Cosine Function

For any $x \in \mathbb{R}$:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}, \quad (2.13)$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}. \quad (2.14)$$

Refer to any graduate school level trigonometry textbook for proofs.

[2.7] Euler's Formula

Euler's formula gives a very important relationship between the complex (imaginary) exponential function and the trigonometric functions. For any $x \in \mathbb{R}$:

$$e^{ix} = \cos(x) + i \sin(x). \quad (2.15)$$

From (2.15), variants of Euler's formula are derived:

$$e^{-ix} = \cos(x) - i \sin(x), \quad (2.16)$$

$$e^{ix} + e^{-ix} = 2 \cos(x), \quad (2.17)$$

$$e^{ix} - e^{-ix} = 2i \sin(x). \quad (2.18)$$

Consider sine and cosine functions with complex arguments z . Then, Euler's formula tells:

$$\sin z = \text{Im}(e^{iz}) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1} = \frac{e^{iz} - e^{-iz}}{2i}, \quad (2.19)$$

$$\cos z = \text{Re}(e^{iz}) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} = \frac{e^{iz} + e^{-iz}}{2}. \quad (2.20)$$

Refer to any graduate school level trigonometry textbook for proofs.

[2.8] Sine Wave: Sinusoid

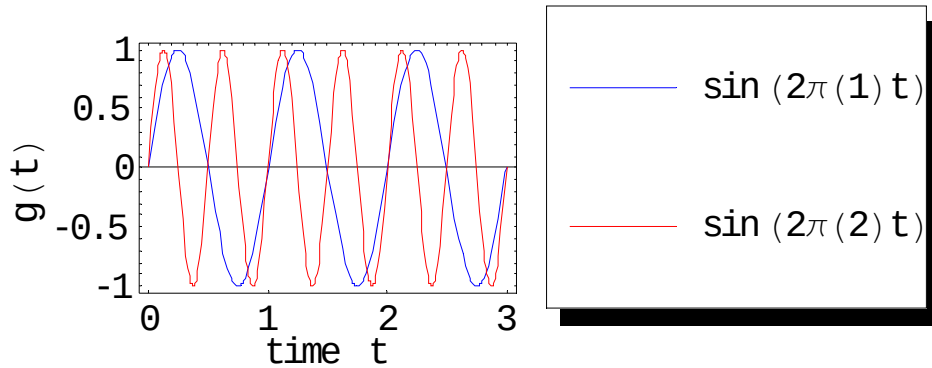
Sine wave is generally defined as a function of time t (seconds):

$$g(t) = a \sin(2\pi f_0 t + b), \quad (2.21)$$

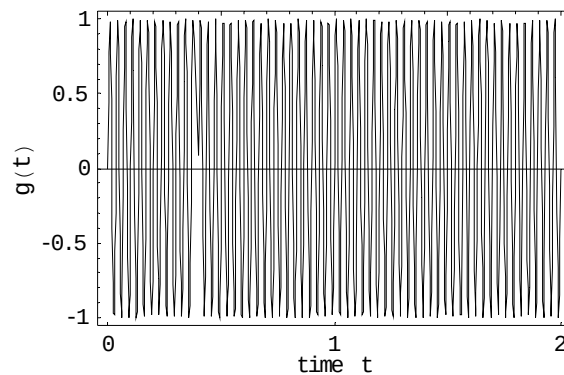
where a is the amplitude, f_0 is the fundamental frequency (cycles/second, Hz), and b changes the phase (angular position) of a sinusoid. In terms of a fundamental angular frequency ω_0 (radians/second), a sine wave is defined as (i.e. $\omega_0 \equiv 2\pi f_0$):

$$g(t) = a \sin(\omega_0 t + b). \quad (2.22)$$

Figure 2.6 illustrates the role of a fundamental frequency f_0 . When a fundamental frequency f_0 doubles from 1 (1 cycle /second) to 2 (cycles/second), its period becomes half from 1 to 1/2 seconds as illustrated in Panel A.



A) 1 Hz sine wave $\sin(2\pi(1)t)$ versus 2 Hz sine wave $\sin(2\pi(2)t)$.



B) 30 Hz sine wave $\sin(2\pi(30)t)$.

Figure 2.6: Plot of a Sine Wave $g(t) = \sin(2\pi ft)$ with Different Fundamental Frequency f_0 .

The role of amplitude a is to increase or decrease the magnitude of an oscillation. Figure 2.7 illustrates how magnitude of an oscillation changes for three different amplitudes $a = 1/2, 1$, and 2 . In audio study amplitude a determines how loud a sound is.

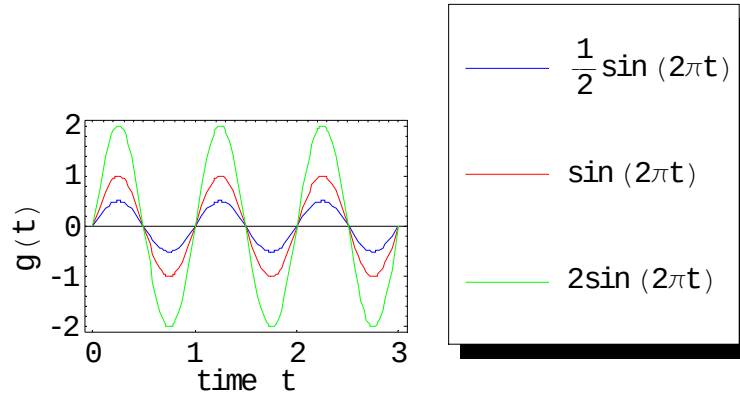


Figure 2.7: Plot of a 1 Hz Sine Wave $g(t) = a \sin(2\pi t)$ with Different Amplitudes $a = 1/2, 1$, and 2 .

Consider three 1 Hz sine waves $\sin(2\pi t)$, $\sin(2\pi t + \frac{\pi}{2})$, and $\sin(2\pi t - \frac{\pi}{2})$. A sine wave $\sin(2\pi t)$ has a phase 0, $\sin(2\pi t + \frac{\pi}{2}) = \cos(2\pi t)$ has a phase $\pi/2$, and $\sin(2\pi t - \frac{\pi}{2})$ has a phase $-\pi/2$. The role of a parameter b is to change the position of a waveform by an amount b as illustrated in Figure 2.8.

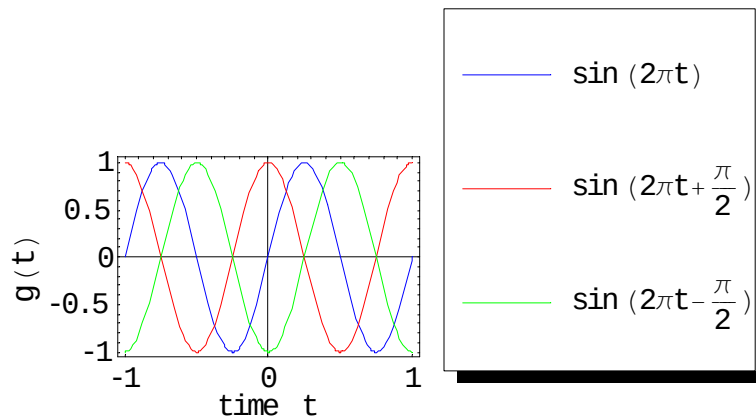


Figure 2.8: Plot of a 1 Hz Sine Wave $g(t) = \sin(2\pi t + b)$ with Different Phase $b = 0, \pi/2$, and $-\pi/2$.

This means that $g(t) = a \sin(2\pi f_0 t)$ is a sinusoid at phase zero and $g(t) = a \cos(2\pi f_0 t)$ is a sinusoid at phase $\pi/2$. For the purpose of defining a sinusoid, it really does not matter whether $\sin(\)$ or $\cos(\)$ is used.

[3] Fourier Transform (FT)

[3.1] Definition of Fourier Transform

We consider Fourier transform of a function $g(t)$ from a time domain t into an angular frequency domain ω (radians/second). This follows the convention in physics. In the field of signal processing which is a major application of FT, frequency f (cycles/second) is used instead of ω . But this difference is not important because ω and f are measuring the same thing (rotation speed) in different units and related by (2.8):

$$\omega = 2\pi f . \quad (3.1)$$

Table 3.1 gives a clear-cut relationship between frequency f and angular frequency ω .

Table 3.1: Relationship between Frequency f and Angular Frequency ω	
Frequency f (cycles/second)	Angular Frequency ω (radians/second)
1 Hz	$2\pi = 360^\circ$ Hz
10 Hz	$20\pi = 360^\circ \times 10$ Hz
100 Hz	$200\pi = 360^\circ \times 100$ Hz

We start from the most general definition of FT. FT from a function $g(t)$ to a function $\mathcal{G}(\omega)$ (thus, switching domains from t to ω) is defined using two arbitrary constants a and b called FT parameters as:

$$\mathcal{G}(\omega) \equiv \mathcal{F}[g(t)](\omega) \equiv \sqrt{\frac{|b|}{(2\pi)^{1-a}}} \int_{-\infty}^{\infty} e^{ib\omega t} g(t) dt . \quad (3.2)$$

Inverse Fourier transform from a function $\mathcal{G}(\omega)$ to a function $g(t)$ (thus, switching domains from ω to t) is defined as (i.e. the reverse procedure of (3.2)):

$$g(t) \equiv \mathcal{F}^{-1}[\mathcal{G}(\omega)](t) \equiv \sqrt{\frac{|b|}{(2\pi)^{1+a}}} \int_{-\infty}^{\infty} e^{-ib\omega t} \mathcal{G}(\omega) d\omega . \quad (3.3)$$

For our purpose which is to calculate characteristic functions, FT parameters are set as $(a, b) = (1, 1)$. Thus, (3.2) and (3.3) become:

$$\mathcal{G}(\omega) \equiv \mathcal{F}[g(t)](\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt , \quad (3.4)$$

$$g(t) \equiv \mathcal{F}^{-1}[\mathcal{G}(\omega)](t) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{G}(\omega) d\omega . \quad (3.5)$$

Euler's formula (2.15) is for $t \in \mathbb{R}$:

$$e^{it} = \cos t + i \sin t .$$

Thus, the FT of (3.4) can be rewritten as:

$$\mathcal{G}(\omega) \equiv \mathcal{F}[g(t)](\omega) \equiv \int_{-\infty}^{\infty} \cos(\omega t) g(t) dt + i \int_{-\infty}^{\infty} \sin(\omega t) g(t) dt .$$

Intuitively speaking, FT is a decomposition of a waveform $g(t)$ (i.e. in time domain t) into a sum of sinusoids (i.e. sine and cosine functions) of different frequencies (Hz) which sum to the original waveform. In other words, FT enables any function in time domain to be represented by an infinite number of sinusoidal functions. Therefore, FT is an angular frequency representation (i.e. different look) of a function $g(t)$ and $\mathcal{G}(\omega)$ contains the exact same information as the original function $g(t)$.

We can check if the inverse Fourier transform (3.5) is true:

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{G}(\omega) d\omega &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} e^{i\omega \tau} g(\tau) d\tau \right) e^{-i\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} g(\tau) \left(\int_{-\infty}^{\infty} e^{i\omega(\tau-t)} d\omega \right) d\tau \\ &= \int_{-\infty}^{\infty} g(\tau) \delta(\tau-t) d\tau \\ &= g(t) , \end{aligned}$$

where we used the identity of Dirac's delta function (3.12) which is proven soon.

Although FT parameters $(a, b) = (1, 1)$ are used for calculating characteristic functions, different pairs of (a, b) are used for other purposes. For example, $(a, b) = (1, -1)$ in pure mathematics:

$$\begin{aligned} \mathcal{G}(\omega) &\equiv \mathcal{F}_t[g(t)](\omega) \equiv \int_{-\infty}^{\infty} e^{-i\omega t} g(t) dt , \\ g(t) &\equiv \mathcal{F}_\omega^{-1}[\mathcal{G}(\omega)](t) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} \mathcal{G}(\omega) d\omega . \end{aligned}$$

Modern physics uses $(a, b) = (0, 1)$:

$$\begin{aligned} \mathcal{G}(\omega) &\equiv \mathcal{F}_t[g(t)](\omega) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt , \\ g(t) &\equiv \mathcal{F}_\omega^{-1}[\mathcal{G}(\omega)](t) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{G}(\omega) d\omega . \end{aligned}$$

In the field of signal processing and most of the standard textbooks of FT, FT parameters $(a, b) = (0, -2\pi)$ are used (i.e. they use frequency f instead of angular frequency ω):

$$\mathcal{G}(f) \equiv \mathcal{F}_t[g(t)](f) \equiv \int_{-\infty}^{\infty} e^{-2\pi i f t} g(t) dt, \quad (3.6)$$

$$g(t) \equiv \mathcal{F}_f^{-1}[\mathcal{G}(f)](t) \equiv \int_{-\infty}^{\infty} e^{2\pi i f t} \mathcal{G}(f) df. \quad (3.7)$$

We use (3.6) and (3.7) frequently because this definition of FT is mathematically simpler to handle for the purpose of proofs.

In general, Fourier transform $\mathcal{G}(\omega)$ is a complex quantity:

$$\mathcal{G}(\omega) = \text{Re}(\omega) + i \text{Im}(\omega) = |\mathcal{G}(\omega)| e^{i\theta(\omega)}, \quad (3.8)$$

where $\text{Re}(\omega)$ is the real part of the FT $\mathcal{G}(\omega)$, $\text{Im}(\omega)$ is the imaginary part of the FT, $|\mathcal{G}(\omega)|$ is the amplitude of a time domain function $g(t)$, and $\theta(\omega)$ is the phase angle of the FT $\mathcal{G}(\omega)$. $|\mathcal{G}(\omega)|$ and $\theta(\omega)$ can be expressed in terms of $\text{Re}(\omega)$ and $\text{Im}(\omega)$ as:

$$|\mathcal{G}(\omega)| = \sqrt{\text{Re}^2(\omega) + \text{Im}^2(\omega)}, \quad (3.9)$$

$$\theta(\omega) = \tan^{-1} \left[\frac{\text{Im}(\omega)}{\text{Re}(\omega)} \right]. \quad (3.10)$$

[3.2] Examples of Fourier Transform

Before discussing important properties of FT, we present representative examples of FT in this section to get the feeling of what FT does.

[3.2.1] Double-Sided Exponential Function

Consider a double-sided exponential function with $A, \alpha \in \mathbb{R}$:

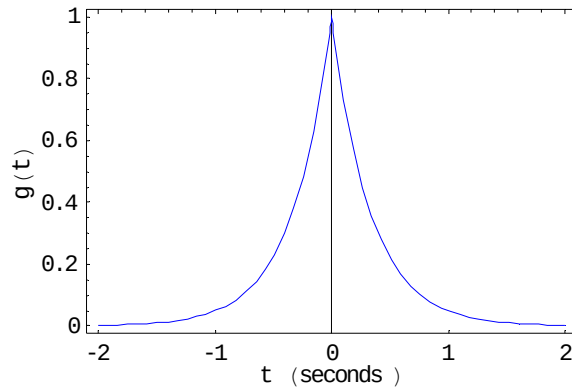
$$g(t) = A e^{-\alpha|t|}.$$

From (3.4):

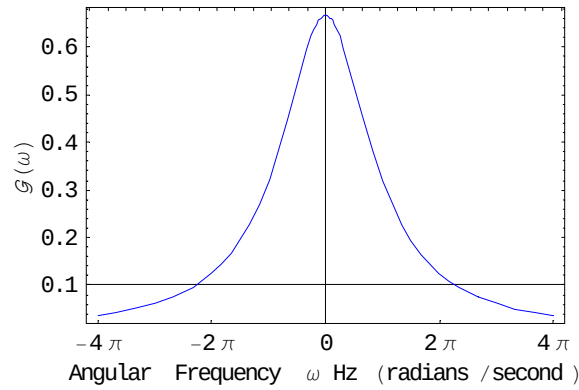
$$\begin{aligned} \mathcal{G}(\omega) &\equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt = \int_{-\infty}^{\infty} e^{i\omega t} A e^{-\alpha|t|} dt = A \left(\int_{-\infty}^0 e^{i\omega t} e^{\alpha t} dt + \int_0^{\infty} e^{i\omega t} e^{-\alpha t} dt \right) \\ \mathcal{G}(\omega) &= A \left(\frac{1}{\alpha + i\omega} + \frac{1}{\alpha - i\omega} \right) = \frac{2A\alpha}{\alpha^2 + \omega^2}. \end{aligned}$$

When $A=1$ and $\alpha=3$, the time domain function $g(t) = e^{-3|t|}$ and its Fourier transform

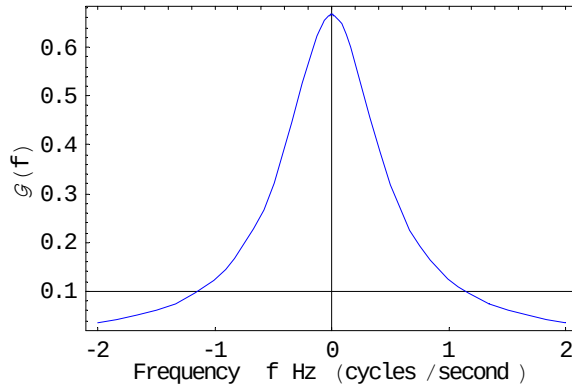
$\mathcal{G}(\omega) = \frac{6}{9 + \omega^2}$ is plotted in Figure 3.1.



A) Plot of a double-sided exponential function $g(t) = e^{-3|t|}$.



B) Plot of FT of $g(t) = e^{-3|t|}$ in Angular Frequency Domain, $\mathcal{G}(\omega)$.



C) Plot of FT of $g(t) = e^{-3|t|}$ in Frequency Domain, $\mathcal{G}(f)$.

Figure 3.1: Plot of Double-Sided Exponential Function $g(t)$ and Its Fourier Transforms $\mathcal{G}(\omega)$ and $\mathcal{G}(f)$.

Using the signal processing definition of FT $(a, b) = (0, -2\pi)$ of the definition (3.6), FT of $g(t) = Ae^{-\alpha|t|}$ is computed as (which is simply obtained by substituting $\omega = 2\pi f$ into $\mathcal{G}(\omega)$):

$$\mathcal{G}(f) = \frac{2A\alpha}{\alpha^2 + 4\pi^2 f^2}.$$

When $A=1$ and $\alpha=3$, $\mathcal{G}(f) = \frac{6}{9+4\pi^2 f^2}$ which is plotted in Panel C of Figure 3.1.

[3.2.2] Rectangular Pulse

Consider a rectangular pulse with $A, T_0 \in \mathbb{R}$:

$$g(t) = \begin{cases} A & -T_0 \leq t \leq T_0 \\ 0 & |t| > T_0 \end{cases},$$

which is an even function of t (symmetric with respect to t).

From (3.4) and use Euler's formula (2.18) $e^{ix} - e^{-ix} = 2i \sin(x)$:

$$\begin{aligned} \mathcal{G}(\omega) &\equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt = A \int_{-T_0}^{T_0} e^{i\omega t} dt = A \left\{ \frac{-(e^{-i\omega T_0} - e^{i\omega T_0})}{i\omega} \right\}, \\ &= A \left\{ \frac{2i \sin(\omega T_0)}{i\omega} \right\} = \frac{2A \sin(\omega T_0)}{\omega}. \end{aligned}$$

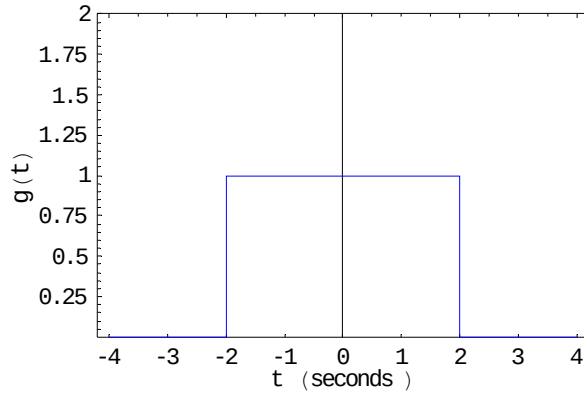
Using the signal processing definition of FT $(a, b) = (0, -2\pi)$ of the definition (3.6), FT of a rectangular pulse is computed as:

$$\mathcal{G}(f) \equiv \int_{-\infty}^{\infty} e^{-2\pi i f t} g(t) dt = A \int_{-T_0}^{T_0} e^{-2\pi i f t} dt = \frac{A \sin(2\pi f T_0)}{\pi f}.$$

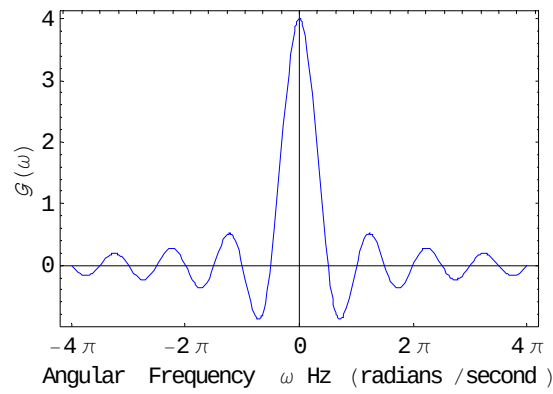
When $A=1$ and $T_0=2$, the time domain function $g(t)$, Fourier transform

$$\mathcal{G}(\omega) = \frac{2 \sin(2\omega)}{\omega} \text{ in angular frequency, and Fourier transform } \mathcal{G}(f) = \frac{\sin(4\pi f)}{\pi f} \text{ in}$$

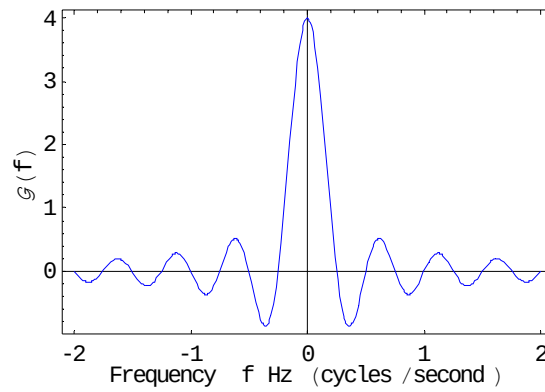
frequency domain are plotted in Figure 3.2.



A) Plot of a rectangular pulse $g(t)$.



B) Plot of the FT of $g(t)$ in Angular Frequency Domain, $\mathcal{G}(\omega)$.



C) Plot of the FT of $g(t)$ in Frequency Domain, $\mathcal{G}(f)$.

Figure 3.2: Plot of Rectangular Pulse $g(t)$ and Its Fourier Transforms $\mathcal{G}(\omega)$ and $\mathcal{G}(f)$.

[3.2.3] Dirac's Delta Function (Impulse Function)

Consider Dirac's delta function scaled by $a \in \mathbb{R}^+$ (Dirac's delta function is discussed in detail in section 3.3.1.):

$$g(t) = a\delta(t) .$$

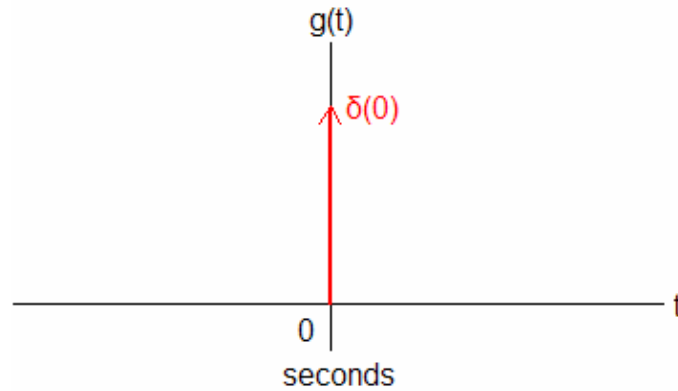
From (3.4):

$$\mathcal{G}(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt = \int_{-\infty}^{\infty} e^{i\omega t} a\delta(t) dt = ae^{i\omega 0} = a .$$

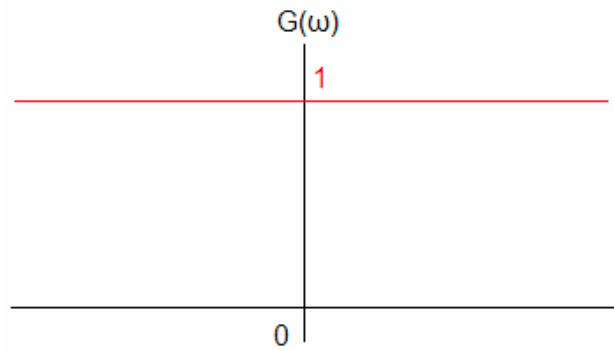
Using the signal processing definition of FT $(a, b) = (0, -2\pi)$ of the definition (3.6), FT of a scaled Dirac's delta is computed as:

$$\mathcal{G}(f) \equiv \int_{-\infty}^{\infty} e^{-2\pi i f t} g(t) dt = \int_{-\infty}^{\infty} e^{-2\pi i f t} a\delta(t) dt = ae^{-2\pi i f 0} = a .$$

When $a = 1$ (i.e. pure Dirac's delta), the time domain function $g(t) = \delta(t)$ and Fourier transforms $\mathcal{G}(\omega) = 1$ and $\mathcal{G}(f) = 1$ are plotted in Figure 3.3.

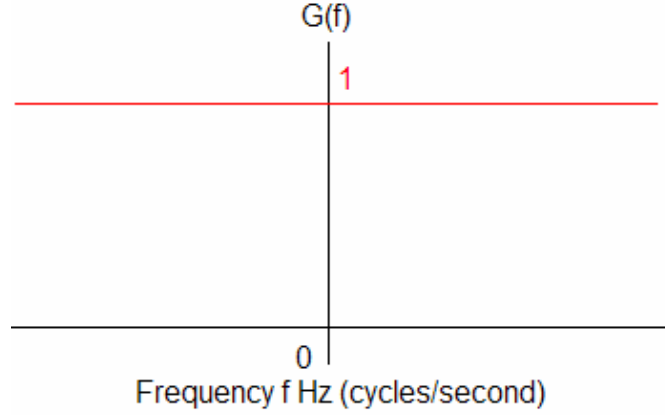


A) Plot of $g(t) = \delta(t)$.



Angular Frequency ω Hz (radians/second)

B) Plot of FT of $g(t)$ in Angular Frequency Domain, $\mathcal{G}(\omega) = 1$.



C) Plot of FT of $g(t)$ in Frequency Domain, $\mathcal{G}(f) = 1$.

Figure 3.3: Plot of Dirac's Delta Function $g(t) = \delta(t)$ and Its Fourier Transforms $\mathcal{G}(\omega)$ and $\mathcal{G}(f)$.

[3.2.4] Gaussian Function

Consider a Gaussian function with $A \in \mathbb{R}^+$:

$$g(t) = e^{-At^2}.$$

From (3.4):

$$\mathcal{G}(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt = \int_{-\infty}^{\infty} e^{i\omega t} e^{-At^2} dt.$$

Use Euler's formula (2.15):

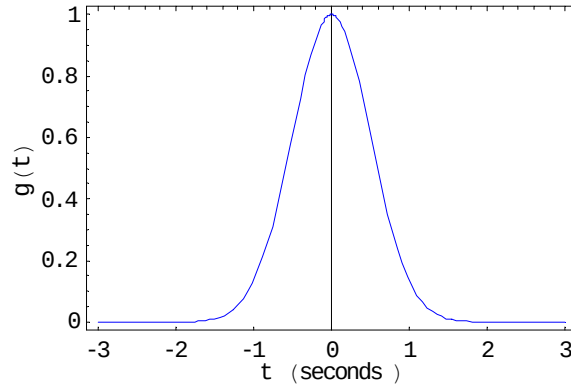
$$\begin{aligned} \mathcal{G}(\omega) &= \int_{-\infty}^{\infty} e^{-At^2} \{\cos(\omega t) + i \sin(\omega t)\} dt \\ &= \int_{-\infty}^{\infty} e^{-At^2} \cos(\omega t) dt + i \int_{-\infty}^{\infty} e^{-At^2} \sin(\omega t) dt \\ &= \sqrt{\frac{\pi}{A}} e^{-\omega^2 / 4A} + i0 = \sqrt{\frac{\pi}{A}} e^{-\omega^2 / 4A}. \end{aligned}$$

Using the signal processing definition of FT $(a, b) = (0, -2\pi)$ of the definition (3.6), FT of a Gaussian function is computed as:

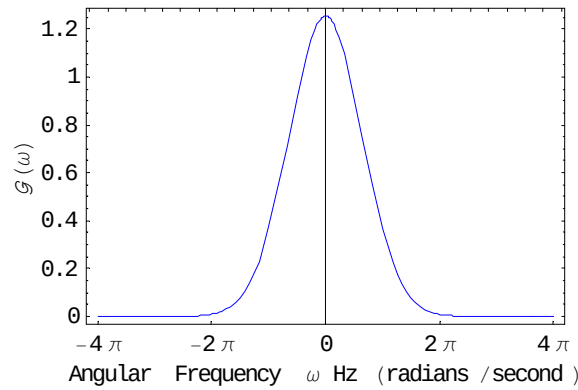
$$\mathcal{G}(f) \equiv \int_{-\infty}^{\infty} e^{-2\pi i f t} g(t) dt = \int_{-\infty}^{\infty} e^{-2\pi i f t} e^{-At^2} dt = \sqrt{\frac{\pi}{A}} e^{-\pi^2 f^2 / A}.$$

When $A = 2$, the time domain function $g(t) = e^{-2t^2}$ and its Fourier transforms

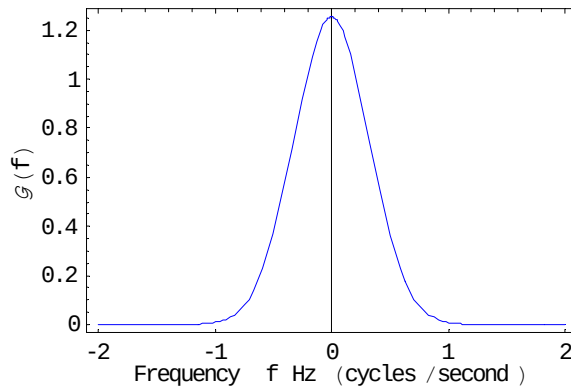
$\mathcal{G}(\omega) = \sqrt{\frac{\pi}{2}} e^{-\omega^2/8}$ and $\mathcal{G}(f) = \sqrt{\frac{\pi}{2}} e^{-\pi^2 f^2/2}$ are plotted in Figure 3.4. Note that FT of a Gaussian Function is another Gaussian function.



A) Plot of Gaussian function $g(t) = e^{-2t^2}$.



B) Plot of FT of $g(t)$ in Angular Frequency Domain, $\mathcal{G}(\omega)$.



C) Plot of FT of $g(t)$ in Frequency Domain, $\mathcal{G}(f)$.

Figure 3.4: Plot of Gaussian Function $g(t) = e^{-2t^2}$ and Fourier Transforms $\mathcal{G}(\omega)$ and $\mathcal{G}(f)$.

[3.2.5] Cosine Wave $g(t) = A\cos(2\pi f_0 t) = A\cos(\omega_0 t)$

Consider a sinusoid $g(t) = A\cos(2\pi f_0 t) = A\cos(\omega_0 t)$. Using the signal processing definition of FT $(a, b) = (0, -2\pi)$ of the definition (3.6), FT of a cosine wave is computed as:

$$\mathcal{G}(f) \equiv \int_{-\infty}^{\infty} e^{-2\pi i f t} g(t) dt = \int_{-\infty}^{\infty} e^{-2\pi i f t} A\cos(2\pi f_0 t) dt.$$

From Euler's formula (2.17):

$$\begin{aligned} \mathcal{G}(f) &= A \int_{-\infty}^{\infty} e^{-2\pi i f t} \frac{1}{2} (e^{i2\pi f_0 t} + e^{-i2\pi f_0 t}) dt \\ &= \frac{1}{2} A \left\{ \int_{-\infty}^{\infty} e^{-2\pi i f t} e^{i2\pi f_0 t} dt + \int_{-\infty}^{\infty} e^{-2\pi i f t} e^{-i2\pi f_0 t} dt \right\} \\ &= \frac{1}{2} A \left\{ \int_{-\infty}^{\infty} e^{-2\pi i (f - f_0) t} dt + \int_{-\infty}^{\infty} e^{-2\pi i (f + f_0) t} dt \right\}. \end{aligned}$$

Use the identity (3.12) of Dirac's delta function:

$$\delta(x - a) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(x-a)} d\omega.$$

Therefore, we obtain:

$$\begin{aligned} \mathcal{G}(f) &= \frac{1}{2} A \{ \delta(f - f_0) + \delta(f + f_0) \} \\ \mathcal{G}(f) &= \frac{A}{2} \delta(f - f_0) + \frac{A}{2} \delta(f + f_0), \end{aligned}$$

which is two impulse functions at $f = \pm f_0$. Thus, FT of a cosine wave (which is an even function) is a real valued even function which means $\mathcal{G}(f)$ is symmetric about $f = 0$.

Next, in terms of angular frequency ω from (3.4):

$$\mathcal{G}(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt = \int_{-\infty}^{\infty} e^{i\omega t} A\cos(\omega_0 t) dt.$$

From Euler's formula (2.17):

$$e^{ix} + e^{-ix} = 2\cos(x).$$

Therefore:

$$\begin{aligned}\mathcal{G}(\omega) &= A \int_{-\infty}^{\infty} e^{i\omega t} \cos(\omega_0 t) dt = A \int_{-\infty}^{\infty} e^{i\omega t} \left\{ \frac{1}{2} (e^{i\omega_0 t} + e^{-i\omega_0 t}) \right\} dt \\ &= \frac{A}{2} \int_{-\infty}^{\infty} e^{i\omega t} e^{i\omega_0 t} dt + \frac{A}{2} \int_{-\infty}^{\infty} e^{i\omega t} e^{-i\omega_0 t} dt = \frac{A}{2} \int_{-\infty}^{\infty} e^{i(\omega+\omega_0)t} dt + \frac{A}{2} \int_{-\infty}^{\infty} e^{i(\omega-\omega_0)t} dt.\end{aligned}$$

Use the identity of Dirac's delta function (3.12):

$$\delta(x-a) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(x-a)} d\omega.$$

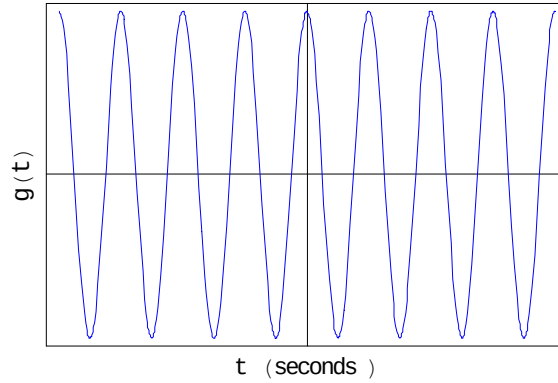
Thus:

$$\begin{aligned}\mathcal{G}(\omega) &= \frac{A}{2} 2\pi \delta(\omega + \omega_0) + \frac{A}{2} 2\pi \delta(\omega - \omega_0) \\ \mathcal{G}(\omega) &= A\pi \delta(\omega + \omega_0) + A\pi \delta(\omega - \omega_0),\end{aligned}$$

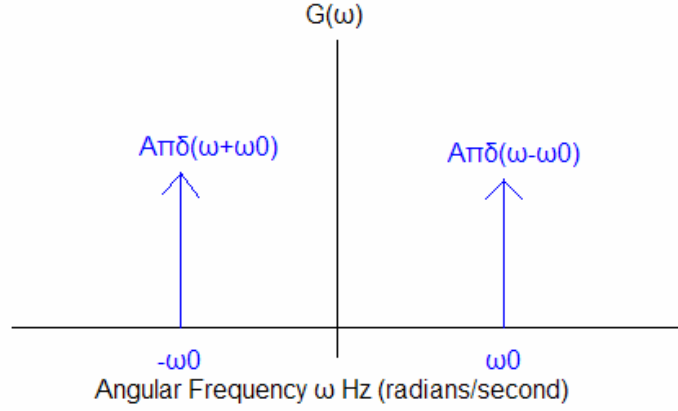
which is two impulse functions at $\omega = \pm\omega_0$. Figure 3.5 illustrates a cosine

wave $g(t) = A\cos(2\pi f_0 t) = A\cos(\omega_0 t)$ and FTs $\mathcal{G}(\omega) = A\pi\delta(\omega + \omega_0) + A\pi\delta(\omega - \omega_0)$ and

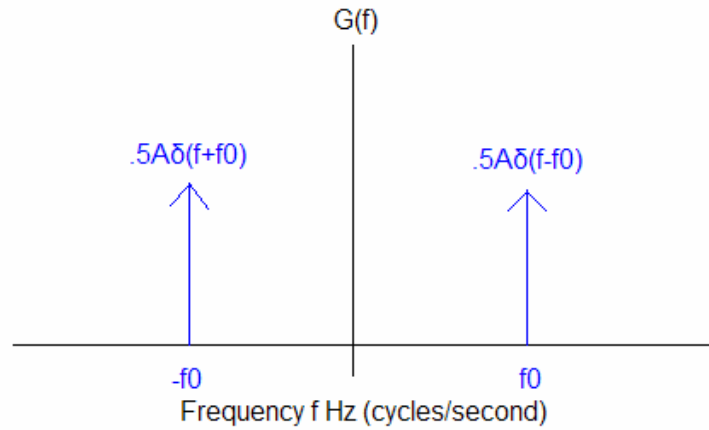
$$\mathcal{G}(f) = \frac{A}{2} \delta(f - f_0) + \frac{A}{2} \delta(f + f_0).$$



A) Plot of a Cosine Wave $g(t) = A\cos(2\pi f_0 t) = A\cos(\omega_0 t)$. Amplitude of the wave is given by A .



B) Plot of FT of $g(t)$ in Angular Frequency Domain, $\mathcal{G}(\omega)$.



C) Plot of FT of $g(t)$ in Frequency Domain, $\mathcal{G}(f)$.

Figure 3.5: Plot of a Cosine Wave $g(t) = A\cos(2\pi f_0 t) = A\cos(\omega_0 t)$ and Fourier Transforms $\mathcal{G}(\omega)$ and $\mathcal{G}(f)$.

[3.2.6] Sine Wave $g(t) = A\sin(2\pi f_0 t) = A\sin(\omega_0 t)$

Consider a sinusoid $g(t) = A\sin(2\pi f_0 t) = A\sin(\omega_0 t)$. Using the signal processing definition of FT $(a, b) = (0, -2\pi)$ of the equation (3.6), FT of a sine wave is computed as:

$$\mathcal{G}(f) \equiv \int_{-\infty}^{\infty} e^{-2\pi ift} g(t) dt = \int_{-\infty}^{\infty} e^{-2\pi ift} A\sin(2\pi f_0 t) dt.$$

From Euler's formula (2.18):

$$\begin{aligned} \mathcal{G}(f) &= A \int_{-\infty}^{\infty} e^{-2\pi ift} \frac{1}{2i} (e^{i2\pi f_0 t} - e^{-i2\pi f_0 t}) dt \\ &= \frac{1}{2i} A \left\{ \int_{-\infty}^{\infty} e^{-2\pi ift} e^{i2\pi f_0 t} dt - \int_{-\infty}^{\infty} e^{-2\pi ift} e^{-i2\pi f_0 t} dt \right\} \end{aligned}$$

$$= \frac{1}{2i} A \left\{ \int_{-\infty}^{\infty} e^{-2\pi i(f-f_0)t} dt - \int_{-\infty}^{\infty} e^{-2\pi i(f+f_0)t} dt \right\}.$$

Multiply $1 = i/i$ to the right hand side:

$$\begin{aligned} \mathcal{G}(f) &= \frac{i}{2i^2} A \left\{ \int_{-\infty}^{\infty} e^{-2\pi i(f-f_0)t} dt - \int_{-\infty}^{\infty} e^{-2\pi i(f+f_0)t} dt \right\} \\ &= \frac{i}{2} A \left\{ \int_{-\infty}^{\infty} e^{-2\pi i(f+f_0)t} dt - \int_{-\infty}^{\infty} e^{-2\pi i(f-f_0)t} dt \right\}. \end{aligned}$$

Using the identity (3.12) of Dirac's delta function, we obtain:

$$\begin{aligned} \mathcal{G}(f) &= \frac{i}{2} A \{ \delta(f+f_0) - \delta(f-f_0) \} \\ &= i \frac{A}{2} \delta(f+f_0) - i \frac{A}{2} \delta(f-f_0), \end{aligned}$$

which is two complex impulse functions at $f = \pm f_0$ which are not symmetric about $f = 0$.

Next, in terms of angular frequency ω from (3.4):

$$\mathcal{G}(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt = A \int_{-\infty}^{\infty} e^{i\omega t} \sin(\omega_0 t) dt.$$

Using Euler's formula (2.18):

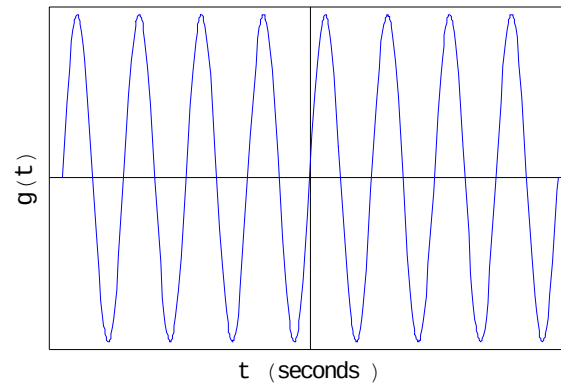
$$\begin{aligned} \mathcal{G}(\omega) &= A \int_{-\infty}^{\infty} e^{i\omega t} \left\{ \frac{1}{2i} (e^{i\omega_0 t} - e^{-i\omega_0 t}) \right\} dt = \frac{A}{2i} \int_{-\infty}^{\infty} e^{i\omega t} e^{i\omega_0 t} dt - \frac{A}{2i} \int_{-\infty}^{\infty} e^{i\omega t} e^{-i\omega_0 t} dt \\ &= \frac{A}{2i} \int_{-\infty}^{\infty} e^{i(\omega+\omega_0)t} dt - \frac{A}{2i} \int_{-\infty}^{\infty} e^{i(\omega-\omega_0)t} dt = \frac{Ai}{2i^2} \int_{-\infty}^{\infty} e^{i(\omega+\omega_0)t} dt - \frac{Ai}{2i^2} \int_{-\infty}^{\infty} e^{i(\omega-\omega_0)t} dt \\ &= \frac{Ai}{2} \int_{-\infty}^{\infty} e^{i(\omega-\omega_0)t} dt - \frac{Ai}{2} \int_{-\infty}^{\infty} e^{i(\omega+\omega_0)t} dt \end{aligned}$$

Use the identity (3.12) of Dirac's delta function:

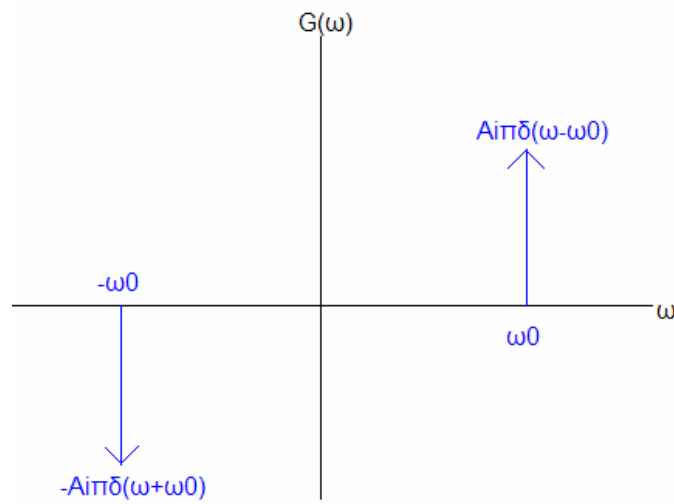
$$\begin{aligned} \mathcal{G}(\omega) &= \frac{Ai}{2} 2\pi \delta(\omega - \omega_0) - \frac{Ai}{2} 2\pi \delta(\omega + \omega_0) \\ &= Ai\pi \delta(\omega - \omega_0) - Ai\pi \delta(\omega + \omega_0). \end{aligned}$$

Figure 3.6 plots a sine wave $g(t) = A \sin(2\pi f_0 t) = A \sin(\omega_0 t)$ and its Fourier transforms

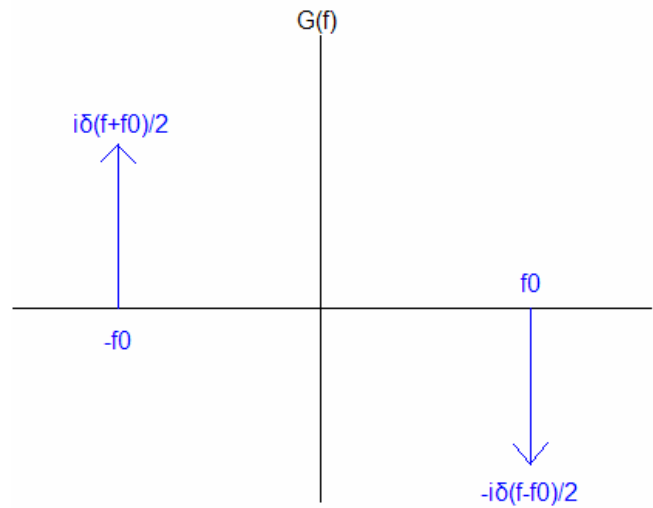
$$\mathcal{G}(\omega) = Ai\pi \delta(\omega - \omega_0) - Ai\pi \delta(\omega + \omega_0) \text{ and } \mathcal{G}(f) = i \frac{A}{2} \delta(f + f_0) - i \frac{A}{2} \delta(f - f_0).$$



A) Plot of a Sine Wave $g(t) = A\sin(2\pi f_0 t) = A\sin(\omega_0 t)$.



B) Plot of FT of $g(t)$ in Angular Frequency Domain, $\mathcal{G}(\omega)$.



C) Plot of FT of $g(t)$ in Frequency Domain, $\mathcal{G}(f)$.

Figure 3.6: Plot of a Sine Wave $g(t) = A\sin(2\pi f_0 t) = A\sin(\omega_0 t)$ and Fourier Transforms $\mathcal{G}(\omega)$ and $\mathcal{G}(f)$.

[3.3] Properties of Fourier Transform

We will discuss important properties of Fourier transform in this section starting from Dirac's delta function which is essential to the understanding of properties of Fourier transform.

[3.3.1] Dirac's Delta Function (Impulse Function)

Consider a function of the form with $n \in \mathbb{R}^+$:

$$h(x) = \frac{n}{\sqrt{\pi}} \exp(-n^2 x^2). \quad (3.11)$$

This function is plotted in Figure 3.7 for three different values for n . The function $h(x)$ becomes more and more concentrated around zero as the value of n increases. The function $h(x)$ has a unit integral:

$$\int_{-\infty}^{\infty} h(x) dx = \int_{-\infty}^{\infty} \frac{n}{\sqrt{\pi}} \exp(-n^2 x^2) dx = 1.$$

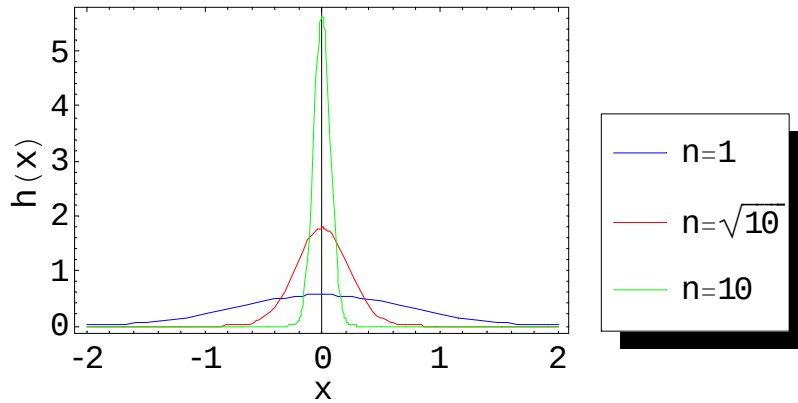


Figure 3.7: Plot of A Function $h(x)$ for $n = 1$, $n = 10^{1/2}$, and $n = 10$.

Dirac's delta function denoted by $\delta(x)$ can be considered as a limit of $h(x)$ when $n \rightarrow \infty$. In other words, $\delta(x)$ is a pulse of unbounded height and zero width with a unit integral:

$$\int_{-\infty}^{\infty} \delta(x) dx = 1.$$

Dirac's delta function $\delta(x)$ evaluates to 0 at all $x \in \mathbb{R}$ other than $x = 0$:

$$\delta(x) = \begin{cases} \delta(0) & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\delta(0)$ is undefined. $\delta(x)$ is called a generalized function not a function because of undefined $\delta(0)$. Therefore, $\delta(x)$ is a distribution with compact support $\{0\}$ meaning that $\delta(x)$ does not occur alone but occurs combined with any continuous functions $f(x)$ and is well defined only when it is integrated.

Dirac's delta function can be defined more generally by its sampling property. Suppose that a function $f(x)$ is defined at $x = 0$. Applying $\delta(x)$ to $f(x)$ yields $f(0)$:

$$\int_{-\infty}^{\infty} f(x)\delta(x)dx = f(0).$$

This is why Dirac's delta function $\delta(x)$ is called a functional because the use of $\delta(x)$ assigns a number $f(0)$ to a function $f(x)$. More generally for $a \in \mathbb{R}$:

$$\delta(x-a) = \begin{cases} \delta(0) & \text{if } x = a \\ 0 & \text{otherwise} \end{cases}$$

and:

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a),$$

or for $\varepsilon > 0$:

$$\int_{a-\varepsilon}^{a+\varepsilon} f(x)\delta(x-a)dx = f(a).$$

$\delta(x)$ has identities such as:

$$\begin{aligned} \delta(ax) &= \frac{1}{|a|} \delta(x), \\ \delta(x^2 - a^2) &= \frac{1}{2|a|} [\delta(x+a) + \delta(x-a)]. \end{aligned}$$

Dirac's delta function $\delta(x)$ can be defined as the limit $n \rightarrow \infty$ of a class of delta sequences:

$$\delta(x) = \lim_{n \rightarrow \infty} \delta_n(x),$$

such that:

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \delta_n(x) f(x) dx = f(0),$$

where $\delta_n(x)$ is a class of delta sequences. Examples of $\delta_n(x)$ other than (3.8) are:

$$\begin{aligned} \delta_n(x) &= \begin{cases} n & \text{if } -1/2n < x < 1/2n \\ 0 & \text{otherwise} \end{cases}, \\ \delta_n(x) &= \frac{1}{2\pi} \int_{-n}^n \exp(iux) du, \\ \delta_n(x) &= \frac{1}{\pi x} \frac{e^{inx} - e^{-inx}}{2i}, \\ \delta_n(x) &= \frac{1}{2\pi} \frac{\sin[(n+1/2)x]}{\sin(x/2)}. \end{aligned}$$

[3.3.2] Useful Identity: Dirac's Delta Function

Dirac's delta function $\delta(x)$ has the following very useful identity which we have used many times before:

$$\delta(x-a) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(x-a)} d\omega. \quad (3.12)$$

PROOF¹

First step is to prove a proposition for all $d = 2, 3, 4, \dots$ and $j = 0, 1, 2, \dots, d-1$ (note that j depends on d):

$$\frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) = 1_{j=0} = \begin{cases} 1 & \text{if } j = 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.13)$$

First, we deal an informal proof of a proposition (3.13).

When $d = 2$ and $j = 0$:

$$\frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) = \frac{1}{2} \sum_{k=0}^{2-1} \exp\left(\frac{2\pi i}{2} 0k\right) = \frac{1}{2} \left\{ \exp\left(\frac{2\pi i}{2} 00\right) + \exp\left(\frac{2\pi i}{2} 01\right) \right\}$$

¹ This is based on "Option Pricing Using Integral Transforms" by Carr, P., Geman, H., Madan, D., and Yor, M.

$$= \frac{1}{2} \{ \exp(0) + \exp(0) \} = 1.$$

When $d = 2$ and $j = 1$:

$$\begin{aligned} \frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) &= \frac{1}{2} \sum_{k=0}^{2-1} \exp\left(\frac{2\pi i}{2} 1k\right) = \frac{1}{2} \left\{ \exp\left(\frac{2\pi i}{2} 10\right) + \exp\left(\frac{2\pi i}{2} 11\right) \right\} \\ &= \frac{1}{2} \{ \exp(0) + \exp(i\pi) \} = \frac{1}{2} \{ 1 - 1 \} = 0. \end{aligned}$$

When $d = 3$ and $j = 0$:

$$\begin{aligned} \frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) &= \frac{1}{3} \sum_{k=0}^{3-1} \exp\left(\frac{2\pi i}{3} 0k\right) \\ &= \frac{1}{3} \left\{ \exp\left(\frac{2\pi i}{3} 00\right) + \exp\left(\frac{2\pi i}{3} 01\right) + \exp\left(\frac{2\pi i}{3} 02\right) \right\} \\ &= \frac{1}{3} \{ \exp(0) + \exp(0) + \exp(0) \} = 1. \end{aligned}$$

When $d = 3$ and $j = 1$:

$$\begin{aligned} \frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) &= \frac{1}{3} \sum_{k=0}^{3-1} \exp\left(\frac{2\pi i}{3} 1k\right) \\ &= \frac{1}{3} \left\{ \exp\left(\frac{2\pi i}{3} 10\right) + \exp\left(\frac{2\pi i}{3} 11\right) + \exp\left(\frac{2\pi i}{3} 12\right) \right\} \\ &= \frac{1}{3} \left\{ \exp(0) + \exp\left(\frac{2\pi i}{3}\right) + \exp\left(\frac{4\pi i}{3}\right) \right\} \\ &= \frac{1}{3} \{ 1 + (0.866025i - 0.5) + (-0.866025i - 0.5) \} \\ &= 0. \end{aligned}$$

When $d = 3$ and $j = 2$:

$$\begin{aligned} \frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) &= \frac{1}{3} \sum_{k=0}^{3-1} \exp\left(\frac{2\pi i}{3} 2k\right) \\ &= \frac{1}{3} \left\{ \exp\left(\frac{2\pi i}{3} 20\right) + \exp\left(\frac{2\pi i}{3} 21\right) + \exp\left(\frac{2\pi i}{3} 22\right) \right\} \\ &= \frac{1}{3} \left\{ \exp(0) + \exp\left(\frac{4\pi i}{3}\right) + \exp\left(\frac{8\pi i}{3}\right) \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{3} \{1 + (-0.866025i - 0.5) + (0.866025i - 0.5)\} \\
&= 0.
\end{aligned}$$

□

(Formal) PROOF of a proposition (3.13)

Rewrite as the below:

$$\frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) = \frac{1}{d} \sum_{k=0}^{d-1} \beta^k, \quad (3.14)$$

where $\beta = \exp\left(\frac{2\pi i}{d} j\right)$. When $j = 0$, $\beta = \exp\left(\frac{2\pi i}{d} 0\right) = 1$. Thus, from (3.14):

$$\frac{1}{d} \sum_{k=0}^{d-1} \beta^k = \frac{1}{d} \sum_{k=0}^{d-1} 1^k = \frac{1}{d} \sum_{k=0}^{d-1} 1 = 1.$$

When $j \neq 0$, consider the term $\sum_{k=0}^{d-1} \beta^k$ which is a geometric series:

$$S_{d-1} = \sum_{k=0}^{d-1} \beta^k = 1 + \beta + \beta^2 + \beta^3 + \dots + \beta^{d-2} + \beta^{d-1}. \quad (3.15)$$

Multiply β to (3.12):

$$\beta S_{d-1} = \beta + \beta^2 + \beta^3 + \dots + \beta^{d-2} + \beta^{d-1} + \beta^d. \quad (3.16)$$

Subtract (3.16) from (3.15):

$$\begin{aligned}
(1 - \beta)S_{d-1} &= 1 - \beta^d \\
S_{d-1} &= \frac{1 - \beta^d}{(1 - \beta)}.
\end{aligned} \quad (3.17)$$

Note that for $j \neq 0$:

$$\beta^d = \exp\left(\frac{2\pi i}{d} j\right)^d = \exp(2\pi i j) = 1.$$

From (3.17):

$$S_{d-1} = \sum_{k=0}^{d-1} \beta^k = \frac{1-\beta^d}{1-\beta} = 0.$$

From (3.14):

$$\frac{1}{d} \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right) = \frac{1}{d} 0 = 0.$$

□

Now we have completed the proof of a proposition (3.13) and we use this now. Multiply d to both sides of a proposition (3.13):

$$d1_{j=0} = \sum_{k=0}^{d-1} \exp\left(\frac{2\pi i}{d} jk\right). \quad (3.18)$$

As the limit $d \rightarrow \infty$ and plug $j = x - a$, (3.18) becomes:

$$\delta(x - a) = \int_{-\infty}^{\infty} e^{i2\pi f(x-a)} df. \quad (3.19)$$

Convert frequency f into angular frequency ω by the equation (3.1) which is

$$\omega = \frac{2\pi}{T} f = 2\pi f. \text{ From (3.19):}$$

$$\delta(x - a) = \int_{-\infty}^{\infty} e^{i\omega(x-a)} d\frac{\omega}{2\pi}.$$

This completes the proof of an identity (3.12).

□

[3.3.3] Linearity of Fourier Transform

Consider time domain functions $f(t)$ and $g(t)$ which have Fourier transforms $\mathcal{F}(\omega)$ and $\mathcal{G}(\omega)$ defined by the equation (3.4). Then:

$$\begin{aligned} \int_{-\infty}^{\infty} \{af(t) + bg(t)\} e^{i\omega t} dt &= a \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt + b \int_{-\infty}^{\infty} g(t) e^{i\omega t} dt \\ &= a\mathcal{F}(\omega) + b\mathcal{G}(\omega). \end{aligned} \quad (3.20)$$

Or, we can write:

$$\mathcal{F}[af(t) + bg(t)](\omega) = a\mathcal{F}[f(t)](\omega) + b\mathcal{F}[g(t)](\omega).$$

Linearity means two things: homogeneity and additivity. Homogeneity of Fourier transform indicates that if the amplitude is changed in one domain by a , the amplitude in the other domain changes by an exactly the same amount a (i.e. scaling property):

$$\mathcal{F}[af(t)](\omega) = a\mathcal{F}[f(t)](\omega).$$

Additivity of Fourier transform indicates that an addition in one domain corresponds to an addition in other domain:

$$\mathcal{F}[f(t) + g(t)](\omega) = \mathcal{F}[f(t)](\omega) + \mathcal{F}[g(t)](\omega).$$

[3.3.4] FT of Even and Odd Functions

A function $g(x)$ is said to be even if for $x \in \mathbb{R}$:

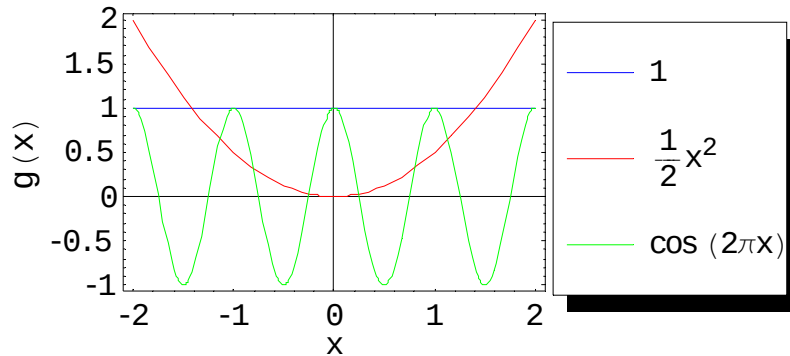
$$g(x) = g(-x), \quad (3.21)$$

which implies that even functions are symmetric with respect to vertical axis. Examples of even functions are illustrated in Panel A of Figure 3.8.

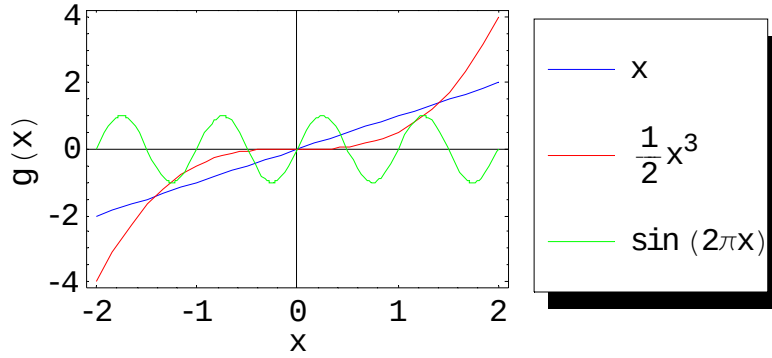
A function $g(x)$ is said to be odd if for $x \in \mathbb{R}$:

$$-g(x) = g(-x), \quad (3.22)$$

which implies that odd functions are symmetric with respect to the origin. Examples of odd functions are illustrated in Panel B of Figure 3.8.



A) Even Functions: $g(x) = 1$, $g(x) = \frac{1}{2}x^2$, and $g(x) = \cos(2\pi x)$.



B) Odd Functions: $g(x) = x$, $g(x) = \frac{1}{2}x^3$, and $g(x) = \sin(2\pi x)$.

Figure 3.8: Plot of Even and Odd Functions

There are several important properties of even and odd functions. The sum of even functions is even and the sum of odd functions is odd. The product of two even functions is even and the product of two odd functions is also even. The product of an even and an odd function is odd.

Let $even(x)$ be an even function and $odd(x)$ be an odd function. Integral properties of even and odd functions are:

$$\int_{-A}^A odd(x)dx = 0, \quad (3.23)$$

$$\int_{-A}^A even(x)dx = 2 \int_0^A even(x)dx. \quad (3.24)$$

Consider FT of an even function $g(t)$. From the definition (3.4) and use Euler's formula:

$$\mathcal{G}(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t)dt = \int_{-\infty}^{\infty} \cos(\omega t)g(t)dt + i \int_{-\infty}^{\infty} \sin(\omega t)g(t)dt. \quad (3.25)$$

Since the imaginary part $\int_{-\infty}^{\infty} \sin(\omega t)g(t)dt$ is zero (because $\sin(\omega t)g(t)$ is odd and use the integral property (3.23)), FT $\mathcal{G}(\omega)$ is real and symmetric with respect to $\omega = 0$. In other words, FT of an even function is also even.

Next, consider FT of an odd function $g(t)$. Since the term $\int_{-\infty}^{\infty} \cos(\omega t)g(t)dt$ in (3.25) becomes zero ($\cos(\omega t)g(t)$ is odd and the integral property (3.23)), FT is given as:

$$\mathcal{G}(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t)dt = i \int_{-\infty}^{\infty} \sin(\omega t)g(t)dt.$$

This means that FT of an odd function $\mathcal{G}(\omega)$ is complex and asymmetric with respect to $\omega = 0$. This property is also illustrated in the section 3.2.5 and 3.2.6.

[3.3.5] Symmetry of Fourier Transform

By the definition of an inverse Fourier transform (3.5):

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{F}(\omega) d\omega.$$

Change the variable in the integration to y :

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyt} \mathcal{F}(y) dy.$$

Consider $2\pi f(-t)$:

$$2\pi f(-t) = \int_{-\infty}^{\infty} e^{iyt} \mathcal{F}(y) dy. \quad (3.26)$$

We can say that the right hand side of (3.26) is by definition the Fourier transform of a function $\mathcal{F}(y)$. Replace t by ω and y by t and (3.26) becomes:

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} e^{it\omega} \mathcal{F}(t) dt. \quad (3.27)$$

The equation (3.27) is called a symmetry property of FT. It means that if a function $f(t)$ has a FT $\mathcal{F}(\omega)$, $\mathcal{F}(t)$ has a FT $2\pi f(-\omega)$. In other words, if $(f(t), \mathcal{F}(\omega))$ is a FT pair, $(\mathcal{F}(t), 2\pi f(-\omega))$ is another FT pair.

This symmetry property of FT can be shown with mathematically simpler form in the frequency domain f Hz (cycles/second). By the definition of an inverse FT (3.7):

$$g(t) \equiv \int_{-\infty}^{\infty} e^{2\pi i f t} \mathcal{G}(f) df.$$

Change the variable in the integration to y :

$$g(t) \equiv \int_{-\infty}^{\infty} e^{2\pi i y t} \mathcal{G}(y) dy.$$

Consider $g(-t)$:

$$g(-t) \equiv \int_{-\infty}^{\infty} e^{-2\pi i y t} \mathcal{G}(y) dy. \quad (3.28)$$

We can say that the right hand side of (3.28) is by definition the FT of a function $\mathcal{G}(y)$. Replace t by f and y by t and (3.28) becomes:

$$g(-f) \equiv \int_{-\infty}^{\infty} e^{-2\pi i t f} \mathcal{G}(t) dt. \quad (3.29)$$

We can state in this case that if $(g(t), \mathcal{G}(f))$ is a FT pair, $(\mathcal{G}(t), g(-f))$ is another FT pair.

[3.3.6] Differentiation of Fourier Transform

By the definition of an inverse Fourier transform (3.5):

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{F}(\omega) d\omega.$$

Differentiate with respect to t :

$$\begin{aligned} \frac{\partial f(t)}{\partial t} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega \mathcal{F}(\omega) \frac{\partial e^{-i\omega t}}{\partial t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega (-i\omega \mathcal{F}(\omega)) e^{-i\omega t} \\ &= -i\omega \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\omega) e^{-i\omega t} d\omega. \end{aligned} \quad (3.30)$$

By the definition of an inverse FT (3.5), the equation (3.30) becomes:

$$\frac{\partial f(t)}{\partial t} = -i\omega f(t). \quad (3.31)$$

Equation (3.31) tells us that FT of $\partial f(t)/\partial t$ is equal to a FT of $f(t)$ multiplied by $-i\omega$:

$$\mathcal{F}[\partial f(t)/\partial t](\omega) = -i\omega \mathcal{F}[f(t)](\omega). \quad (3.32)$$

Next, consider FT in frequency domain f . By the definition of an inverse FT (3.7):

$$g(t) \equiv \int_{-\infty}^{\infty} e^{2\pi i f t} \mathcal{G}(f) df.$$

Differentiate with respect to t :

$$\frac{\partial g(t)}{\partial t} = 2\pi i f \int_{-\infty}^{\infty} e^{2\pi i f t} \mathcal{G}(f) df = 2\pi i f g(t). \quad (3.33)$$

Equation (3.33) tells us that FT of $\partial g(t)/\partial t$ is equal to a Fourier transform of $g(t)$ multiplied by $2\pi if$:

$$\mathcal{F}[\partial g(t)/\partial t](f) = 2\pi if \mathcal{F}[g(t)](f).$$

[3.3.7] Time Scaling of Fourier Transform

Consider a time domain function $f(t)$ and its Fourier transform $\mathcal{F}(\omega)$ by the equation (3.4):

$$\mathcal{F}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} f(t) dt.$$

Then, FT of a function $f(at)$ (i.e. scaled by a real non-zero constant a) can be expressed in terms of $\mathcal{F}(\omega)$ as:

$$\frac{1}{|a|} \mathcal{F}\left(\frac{\omega}{a}\right) = \int_{-\infty}^{\infty} e^{i\omega t} f(at) dt. \quad (3.34)$$

PROOF

Set $at = s$. When $a > 0$:

$$\begin{aligned} \int_{-\infty}^{\infty} e^{i\omega t} f(at) dt &= \int_{-\infty}^{\infty} e^{i\omega s/a} f(s) d\left(\frac{s}{a}\right) = \frac{1}{a} \int_{-\infty}^{\infty} e^{i(\omega/a)s} f(s) ds \\ &= \frac{1}{a} \mathcal{F}\left(\frac{\omega}{a}\right). \end{aligned}$$

When $a < 0$:

$$\begin{aligned} \int_{-\infty}^{\infty} e^{i\omega t} f(at) dt &= \int_{-\infty}^{\infty} e^{i\omega s/a} f(s) d\left(\frac{s}{a}\right) = -\frac{1}{a} \int_{-\infty}^{\infty} e^{i(\omega/a)s} f(s) ds \\ &= -\frac{1}{a} \mathcal{F}\left(\frac{\omega}{a}\right). \end{aligned}$$

□

Similarly, FT of a function $g(at)$ can be expressed in terms of $\mathcal{G}(f)$ in frequency domain f as:

$$\frac{1}{|a|} \mathcal{G}\left(\frac{f}{a}\right) = \int_{-\infty}^{\infty} e^{-2\pi ift} g(at) dt.$$

[3.3.8] Time Shifting of Fourier Transform

Consider a function $f(t)$ and its Fourier transform $\mathcal{F}(\omega)$ by the definition (3.4):

$$\mathcal{F}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} f(t) dt.$$

Then, FT of a function $f(t - t_0)$ (i.e. time t is shifted by $t_0 \in \mathbb{R}$) can be expressed in terms of $\mathcal{F}(\omega)$ as:

$$\int_{-\infty}^{\infty} e^{i\omega t} f(t - t_0) dt = e^{i\omega t_0} \mathcal{F}(\omega). \quad (3.35)$$

PROOF

Set $t - t_0 = t^*$:

$$\begin{aligned} \int_{-\infty}^{\infty} e^{i\omega t} f(t - t_0) dt &= \int_{-\infty}^{\infty} e^{i\omega(t^* + t_0)} f(t^*) d(t^* + t_0) \\ &= e^{i\omega t_0} \int_{-\infty}^{\infty} e^{i\omega t^*} f(t^*) dt^* \\ &= e^{i\omega t_0} \mathcal{F}(\omega). \end{aligned}$$

□

Next, consider FT in frequency domain f . FT of a time domain function $g(t)$ is defined by the definition (3.6) as:

$$\mathcal{G}(f) \equiv \int_{-\infty}^{\infty} e^{-2\pi i f t} g(t) dt.$$

Then, FT of a function $g(t - t_0)$ (i.e. time t is shifted by $t_0 \in \mathbb{R}$) can be expressed in terms of $\mathcal{G}(f)$ as:

$$\int_{-\infty}^{\infty} e^{-2\pi i f t} g(t - t_0) dt = e^{-2\pi i f t_0} \mathcal{G}(f). \quad (3.36)$$

PROOF

Set $t - t_0 = t^*$:

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-2\pi i f t} g(t - t_0) dt &= \int_{-\infty}^{\infty} e^{-2\pi i f(t^* + t_0)} g(t^*) d(t^* + t_0) \\ &= e^{-2\pi i f t_0} \int_{-\infty}^{\infty} e^{-2\pi i f t^*} g(t^*) dt^* = e^{-2\pi i f t_0} \mathcal{G}(f). \end{aligned}$$

□

[3.3.9] Convolution: Time Convolution Theorem

Convolution of time domain functions $f(t)$ and $g(t)$ over a finite interval $[0, t]$ is defined as:

$$f * g \equiv \int_0^t f(\tau)g(t-\tau)d\tau. \quad (3.37)$$

Convolution of time domain functions $f(t)$ and $g(t)$ over an infinite interval $[-\infty, \infty]$ is defined as:

$$f * g \equiv \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau = \int_{-\infty}^{\infty} g(\tau)f(t-\tau)d\tau. \quad (3.38)$$

Convolution can be considered as an integral which measures the amount of overlapping of one function $g(t)$ when $g(t)$ is shifted over another function $f(t)$. Website by mathworld provides an excellent description of convolution with animation. For example, suppose $f(t)$ and $g(t)$ are Gaussian functions:

$$f(t) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(t-\mu_1)^2}{2\sigma_1^2}\right\},$$

$$g(t) = \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(t-\mu_2)^2}{2\sigma_2^2}\right\}.$$

Then, the convolution of two Gaussian functions is calculated as from (3.38):

$$f * g = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_2^2)}} \exp\left\{-\frac{(t-(\mu_1 + \mu_2))^2}{2(\sigma_1^2 + \sigma_2^2)}\right\},$$

which is another Gaussian function. The convolution $f * g$ of two Gaussians for the case $\mu_1 = 0$, $\mu_2 = 0$, $\sigma_1 = 1$, and $\sigma_2 = 2$ is plotted in Figure 3.8.

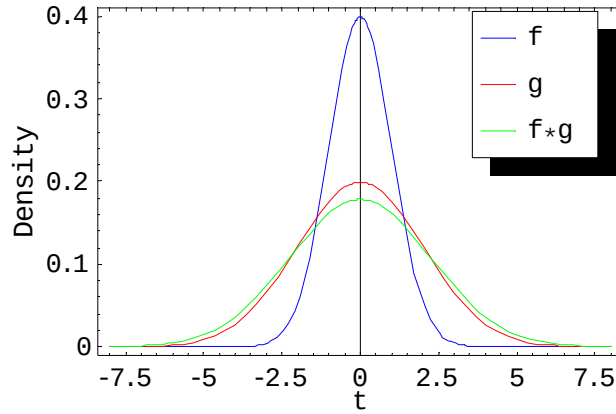


Figure 3.8: Plot of Two Gaussian functions f and g and their convolution $f * g$.

Consider time domain functions $f(t)$ and $g(t)$ with Fourier transforms $\mathcal{F}(\omega)$ and $\mathcal{G}(\omega)$. FT of the convolution of $f(t)$ and $g(t)$ in the time domain is equal to the multiplication in the angular frequency domain (called time-convolution theorem):

$$\mathcal{F}(f * g) \equiv \mathcal{F}\left(\int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau\right) = \mathcal{F}(\omega)\mathcal{G}(\omega). \quad (3.39)$$

PROOF

$$\begin{aligned} \mathcal{F}(f * g) &\equiv \mathcal{F}\left(\int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau\right) = \int_{-\infty}^{\infty} e^{i\omega t} \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau dt \\ &= \int_{-\infty}^{\infty} f(\tau)d\tau \left(\int_{-\infty}^{\infty} g(t-\tau)e^{i\omega t} dt\right). \end{aligned}$$

Use the time shifting property of Fourier transform of the equation (3.35):

$$\begin{aligned} \mathcal{F}(f * g) &= \int_{-\infty}^{\infty} f(\tau)d\tau \left(e^{i\omega\tau} \mathcal{G}(\omega)\right) = \mathcal{G}(\omega) \int_{-\infty}^{\infty} f(\tau)e^{i\omega\tau} d\tau \\ &= \mathcal{F}(\omega)\mathcal{G}(\omega). \end{aligned}$$

□

[3.3.10] Frequency-Convolution Theorem

Consider time domain functions $f(t)$ and $g(t)$ with Fourier transforms $\mathcal{F}(\omega)$ and $\mathcal{G}(\omega)$. Frequency-convolution theorem states that convolution in the angular frequency domain (scaled by $\frac{1}{2\pi}$) is equal to the multiplication in the time domain. In other words, FT of the product $f(t)g(t)$ in the time domain is equal to the convolution $\mathcal{F}(\omega) * \mathcal{G}(\omega)$ (scaled by $\frac{1}{2\pi}$) in the angular frequency domain:

$$\mathcal{F}[f(t)g(t)](\omega) = \frac{1}{2\pi} \mathcal{F}(\omega) * \mathcal{G}(\omega) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\varpi) \mathcal{G}(\omega - \varpi) d\varpi. \quad (3.40)$$

PROOF

There are several different ways to prove the frequency convolution theorem. But we prove this by showing that the inverse FT of the convolution $\mathcal{F} * \mathcal{G}$ in the angular frequency domain is equal to the multiplication $f(t)g(t)$ (scaled by 2π) in the time domain.

Following the definition of inverse FT (3.5):

$$\begin{aligned} \mathcal{F}_\omega^{-1}[\mathcal{F}(\omega) * \mathcal{G}(\omega)](t) &\equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{F}(\omega) * \mathcal{G}(\omega) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \int_{-\infty}^{\infty} \mathcal{F}(\varpi) \mathcal{G}(\omega - \varpi) d\varpi d\omega \\ &= \int_{-\infty}^{\infty} d\varpi \mathcal{F}(\varpi) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{G}(\omega - \varpi) d\omega \right). \end{aligned}$$

Using the frequency shifting (modulation) property of FT (discussed in section 3.3.11):

$$\begin{aligned} \mathcal{F}_\omega^{-1}[\mathcal{F}(\omega) * \mathcal{G}(\omega)](t) &= \int_{-\infty}^{\infty} d\varpi \mathcal{F}(\varpi) e^{-i\varpi t} g(t) \\ &= g(t) \int_{-\infty}^{\infty} d\varpi \mathcal{F}(\varpi) e^{-i\varpi t} = g(t) 2\pi f(t). \end{aligned}$$

□

Next, consider FT in the frequency domain f . Following the definition of inverse FT (3.7):

$$\begin{aligned} \mathcal{F}_f^{-1}[\mathcal{F}(f) * \mathcal{G}(f)](t) &\equiv \int_{-\infty}^{\infty} e^{2\pi i f t} \mathcal{F}(f) * \mathcal{G}(f) df \\ &= \int_{-\infty}^{\infty} e^{2\pi i f t} \int_{-\infty}^{\infty} \mathcal{F}(f^*) \mathcal{G}(f - f^*) df^* df \\ &= \int_{-\infty}^{\infty} df^* \mathcal{F}(f^*) \left(\int_{-\infty}^{\infty} e^{2\pi i f t} \mathcal{G}(f - f^*) df \right). \end{aligned}$$

Using the frequency shifting (modulation) property of FT (discussed in section 3.3.11):

$$\begin{aligned} \mathcal{F}_f^{-1}[\mathcal{F}(f) * \mathcal{G}(f)](t) &= \int_{-\infty}^{\infty} df^* \mathcal{F}(f^*) e^{2\pi i f^* t} g(t) \\ &= g(t) \int_{-\infty}^{\infty} df^* \mathcal{F}(f^*) e^{2\pi i f^* t} = f(t)g(t). \end{aligned}$$

□

[3.3.11] Frequency Shifting: Modulation

Consider time domain function $f(t)$ with Fourier transform $\mathcal{F}(\omega)$. If FT $\mathcal{F}(\omega)$ is shifted by $\omega_0 \in \mathbb{R}$ in the angular frequency domain, then the inverse FT $f(t)$ is multiplied by $e^{i\omega_0 t}$:

$$e^{-i\omega_0 t} f(t) = \mathcal{F}^{-1}(\omega - \omega_0), \quad (3.41)$$

$$\mathcal{F}(e^{-i\omega_0 t} f(t)) = \mathcal{F}(\omega - \omega_0). \quad (3.42)$$

PROOF

Let $s = \omega - \omega_0$. From the definition of an inverse Fourier transform (3.5):

$$\begin{aligned} \mathcal{F}^{-1}(\omega - \omega_0) &\equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{F}(\omega - \omega_0) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i(s+\omega_0)t} \mathcal{F}(s) d(s + \omega_0) \\ &= e^{-i\omega_0 t} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ist} \mathcal{F}(s) ds = e^{-i\omega_0 t} f(t). \end{aligned}$$

□

Similarly, if FT $\mathcal{F}(f)$ is shifted by $f_0 \in \mathbb{R}$ in the frequency domain, then the inverse FT $g(t)$ is multiplied by $e^{2\pi i f_0 t}$:

$$e^{2\pi i f_0 t} g(t) = \mathcal{F}^{-1}[\mathcal{G}(f - f_0)], \quad (3.43)$$

$$\mathcal{F}(e^{2\pi i f_0 t} g(t)) = \mathcal{G}(f - f_0). \quad (3.44)$$

[3.3.12] Parseval's Relation

Let $f(t)$ and $g(t)$ be L^2 -complex functions. A L^2 -function can be informally considered as a square integrable function (i.e. A function $f(t)$ is said to be square-integrable if $\int_{-\infty}^{\infty} |f(t)|^2 dt < \infty$.) Let $\mathcal{F}(\omega)$ and $\mathcal{G}(\omega)$ be the Fourier transforms of $f(t)$ and $g(t)$ defined by (3.4):

$$\mathcal{F}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} f(t) dt,$$

$$\mathcal{G}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt.$$

Let $\bar{g}(t)$ be a complex conjugate of $f(t)$ and $\bar{\mathcal{G}}(\omega)$ be a complex conjugate of $\mathcal{F}(\omega)$ ²:

² A complex conjugate of a complex number $z \equiv a + bi$ is $\bar{z} \equiv a - bi$.

$$\begin{aligned} |f(t)|^2 &= f(t)\bar{g}(t), \\ |\mathcal{F}(\omega)|^2 &= \mathcal{F}(\omega)\bar{\mathcal{G}}(\omega). \end{aligned}$$

Then, Parseval's relation is:

$$\begin{aligned} \int_{-\infty}^{\infty} |f(t)|^2 dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |\mathcal{F}(\omega)|^2 d\omega, \\ \int_{-\infty}^{\infty} f(t)\bar{g}(t)dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\omega)\bar{\mathcal{G}}(\omega)d\omega. \end{aligned} \quad (3.45)$$

Parseval's relation in the case of FT parameters $(a, b) = (1, 1)$ indicates that there is a very simple relationship between the power of a signal function $f(t)$ computed in signal space t or transform space ω of the form (3.45).

Parseval's relation becomes simpler when considered in the frequency domain f instead of angular frequency domain ω . Let $\mathcal{F}(f)$ and $\mathcal{G}(f)$ be the Fourier transforms of $f(t)$ and $g(t)$ defined by (3.6):

$$\begin{aligned} \mathcal{F}(f) &\equiv \int_{-\infty}^{\infty} e^{-2\pi ift} f(t)dt, \\ \mathcal{G}(f) &\equiv \int_{-\infty}^{\infty} e^{-2\pi ift} g(t)dt. \end{aligned}$$

Let $\bar{g}(t)$ be a complex conjugate of $f(t)$ and $\bar{\mathcal{G}}(f)$ be a complex conjugate of $\mathcal{F}(f)$:

$$\begin{aligned} |f(t)|^2 &= f(t)\bar{g}(t), \\ |\mathcal{F}(f)|^2 &= \mathcal{F}(f)\bar{\mathcal{G}}(f). \end{aligned}$$

Then, Parseval's relation is:

$$\begin{aligned} \int_{-\infty}^{\infty} |f(t)|^2 dt &= \int_{-\infty}^{\infty} |\mathcal{F}(f)|^2 df, \\ \int_{-\infty}^{\infty} f(t)\bar{g}(t)dt &= \int_{-\infty}^{\infty} \mathcal{F}(f)\bar{\mathcal{G}}(f)df. \end{aligned} \quad (3.46)$$

This version of Parseval's relation means that the power of a signal function $f(t)$ is same whether it is computed in signal space t or in transform space f .

PROOF

Using inverse Fourier transforms of (3.5):

$$\begin{aligned}\int_{-\infty}^{\infty} |f(t)|^2 dt &= \int_{-\infty}^{\infty} f(t) \bar{g}(t) dt = \int_{-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{F}(\omega) d\omega \right] \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega' t} \bar{\mathcal{G}}(\omega') d\omega' \right] dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\omega) \int_{-\infty}^{\infty} \bar{\mathcal{G}}(\omega') \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{it(\omega' - \omega)} dt \right] d\omega' d\omega.\end{aligned}$$

Use the identity of Dirac's delta function (3.12):

$$\delta(t - a) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(t-a)} d\omega.$$

Thus:

$$\begin{aligned}\int_{-\infty}^{\infty} |f(t)|^2 dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\omega) \int_{-\infty}^{\infty} \bar{\mathcal{G}}(\omega') \delta(\omega' - \omega) d\omega' d\omega \\ \int_{-\infty}^{\infty} |f(t)|^2 dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\omega) \bar{\mathcal{G}}(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\mathcal{F}(\omega)|^2 d\omega.\end{aligned}$$

□

[3.3.13] Summary of Important Properties of Fourier Transform

Table 3.1: Summary of Important Properties of Fourier Transform in Angular Frequency Domain ω Hz (radians/second)

Property	Time Domain Function $y(t)$	Fourier Transform $\mathcal{F}[y(t)](\omega)$
Linearity	$af(t) + bg(t)$	$a\mathcal{F}(\omega) + b\mathcal{G}(\omega)$
Even Function	$f(t)$ is even	$\mathcal{F}(\omega) \in \mathbb{R}$
Odd Function	$f(t)$ is odd	$\mathcal{F}(\omega) \in \mathbb{I}$
Symmetry	$\mathcal{F}(t)$	$2\pi f(-\omega)$
Differentiation	$\frac{df(t)}{dt}$	$-i\omega\mathcal{F}(\omega)$
	$\frac{d^k f(t)}{dt^k}$	$(-i\omega)^k \mathcal{F}(\omega)$
Time Scaling	$f(at)$	$\frac{1}{ a } \mathcal{F}\left(\frac{\omega}{a}\right)$

Time Shifting	$f(t - t_0)$	$e^{i\omega t_0} \mathcal{F}(\omega)$
Convolution	$f * g \equiv \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$	$\mathcal{F}(\omega)\mathcal{G}(\omega)$
Multiplication	$f(t)g(t)$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\varpi)\mathcal{G}(\omega - \varpi)d\varpi$
Modulation (Frequency Shifting)	$e^{-i\omega_0 t} f(t)$	$\mathcal{F}(\omega - \omega_0)$

Table 3.2: Summary of Important Properties of Fourier Transform in Frequency Domain f Hz (cycles/second)

Property	Time Domain Function $y(t)$	Fourier Transform $\mathcal{F}[y(t)](f)$
Linearity	$af(t) + bg(t)$	$a\mathcal{F}(f) + b\mathcal{G}(f)$
Even Function	$g(t)$ is even	$\mathcal{G}(f) \in \mathbb{R}$
Odd Function	$g(t)$ is odd	$\mathcal{G}(f) \in \mathbb{I}$
Symmetry	$\mathcal{G}(t)$	$g(-f)$
Differentiation	$\frac{dg(t)}{dt}$	$2\pi if \mathcal{G}(f)$
	$\frac{d^k g(t)}{dt^k}$	$(2\pi if)^k \mathcal{G}(f)$
Time Scaling	$g(at)$	$\frac{1}{ a } \mathcal{G}(\frac{f}{a})$
Time Shifting	$g(t - t_0)$	$e^{-2\pi if t_0} \mathcal{G}(f)$
Convolution	$f * g \equiv \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$	$\mathcal{F}(f)\mathcal{G}(f)$
Multiplication	$f(t)g(t)$	$\mathcal{F} * \mathcal{G}$

Modulation
(Frequency Shifting)

$$e^{2\pi i f_0 t} g(t)$$

$$\mathcal{G}(f - f_0)$$

[3.4] Existence of the Fourier Integral

Thus far, we've assumed that Fourier integral and its inverse of the definitions (3.4) and (3.5) of a FT pair $(g(t), \mathcal{G}(\omega))$ do exist (i.e. they are well-defined for all functions).

Sufficient (but not necessary) condition for the existence of Fourier transform and its inverse is:

$$\int_{-\infty}^{\infty} |g(t)| dt < \infty, \quad (3.47)$$

which is an integrability condition.

[4] Characteristic Function

[4.1] Definition of a Characteristic Function

Let X be a random variable with its probability density function $\mathbb{P}(x)$. A characteristic function $\phi(\omega)$ with $\omega \in \mathbb{R}$ is defined as the Fourier transform of the probability density function $\mathbb{P}(x)$ using Fourier transform parameters $(a, b) = (1, 1)$. From the definition (3.4):

$$\phi(\omega) \equiv \mathcal{F}[\mathbb{P}(x)] \equiv \int_{-\infty}^{\infty} e^{i\omega x} \mathbb{P}(x) dx = E[e^{i\omega x}]. \quad (4.1)$$

Using the Euler's formula (2.15), $\phi(\omega)$ can be expressed as:

$$\phi(\omega) = E[e^{i\omega x}] = E[\cos(\omega x)] + iE[\sin(\omega x)].$$

Taylor series expansion of a real function $f(x)$ in one dimension about a point $x = b$ is given by:

$$f(x) = f(b) + f'(b)(x-b) + \frac{f''(b)}{2!}(x-b)^2 + \frac{f'''(b)}{3!}(x-b)^3 + \dots \quad (4.2)$$

Taylor series expansion of an exponential function $\exp(i\omega x)$ in one dimension about a point $x = 0$ is given by from (4.2) as:

$$\begin{aligned} \exp(i\omega x) &= \exp(i\omega 0) + \frac{\partial \exp(i\omega x)}{\partial x} \Big|_{x=0} (x-0) + \frac{1}{2!} \frac{\partial^2 \exp(i\omega x)}{\partial x^2} \Big|_{x=0} (x-0)^2 \\ &\quad + \frac{1}{3!} \frac{\partial^3 \exp(i\omega x)}{\partial x^3} \Big|_{x=0} (x-0)^3 + \dots \\ \exp(i\omega x) &= \exp(i\omega 0) + i\omega \exp(i\omega x) \Big|_{x=0} (x-0) + \frac{1}{2!} (i\omega)^2 \exp(i\omega x) \Big|_{x=0} (x-0)^2 \\ &\quad + \frac{1}{3!} (i\omega)^3 \exp(i\omega x) \Big|_{x=0} (x-0)^3 + \dots \\ \exp(i\omega x) &= 1 + i\omega x + \frac{1}{2!} (i\omega x)^2 + \frac{1}{3!} (i\omega x)^3 + \frac{1}{4!} (i\omega x)^4 + \dots \end{aligned} \quad (4.3)$$

Therefore, a characteristic function $\phi(\omega)$ can be rewritten as from equations (4.1) and (4.3) as:

$$\begin{aligned} \phi(\omega) &\equiv \int_{-\infty}^{\infty} e^{i\omega x} \mathbb{P}(x) dx \\ &= \int_{-\infty}^{\infty} \left(1 + i\omega x + \frac{1}{2!} (i\omega x)^2 + \frac{1}{3!} (i\omega x)^3 + \frac{1}{4!} (i\omega x)^4 + \dots \right) \mathbb{P}(x) dx \end{aligned}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} \mathbb{P}(x) dx + i\omega \int_{-\infty}^{\infty} x \mathbb{P}(x) dx + \frac{1}{2!} (i\omega)^2 \int_{-\infty}^{\infty} x^2 \mathbb{P}(x) dx + \frac{1}{3!} (i\omega)^3 \int_{-\infty}^{\infty} x^3 \mathbb{P}(x) dx + \dots \\
&= r_0 + i\omega r_1 - \frac{1}{2!} \omega^2 r_2 - \frac{1}{3!} i\omega^3 r_3 + \frac{1}{4!} \omega^4 r_4 + \dots \\
&= \sum_{n=0}^{\infty} \frac{(i\omega)^n}{n!} r_n,
\end{aligned} \tag{4.4}$$

where r_n is the n -th moment about 0 (called raw moment).

Probability density function $\mathbb{P}(x)$ can be obtained by inverse Fourier transform of the characteristic function using the equation (3.5):

$$\mathbb{P}(x) = \mathcal{F}^{-1}[\phi(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} \phi(\omega) d\omega. \tag{4.5}$$

If X is a discrete random variable with possible values $\{x_k\}_{k=0}^{\infty}$ and $\Pr\{X = x_k\} = a_k$, then $\phi(\omega)$ is obtained by a series expansion instead of an integration as:

$$\phi(\omega) \equiv \sum_{k=0}^{\infty} \exp(i\omega x_k) a_k. \tag{4.6}$$

[4.2] Properties of a Characteristic Function

Let X be a random variable and $\phi_X(\omega)$ be its characteristic function. A characteristic function $\phi_X(\omega)$ is: 1) bounded by 1 (i.e. $|\phi_X(\omega)| \leq 1$, $\omega \in \mathbb{R}$), 2) $\phi_X(0) = 1$, and 3) uniformly continuous in $\mathbb{R}$.

A statistical distribution is uniquely determined by its characteristic function, i.e. one-to-one relationship between distribution functions and characteristic functions. In other words, if two random variables X and Y have the same characteristic functions (i.e. $\phi_X(\omega) = \phi_Y(\omega)$), they have the same distribution.

If $\{X_k, k = 1, \dots, n\}$ are independent random variables, the characteristic function of their sum $X_1 + X_2 + \dots + X_n$ is the product of their characteristic functions:

$$\phi_{X_1 + X_2 + \dots + X_n}(\omega) = \prod_{k=1}^n \phi_{X_k}(\omega). \tag{4.7}$$

A random variable X has a symmetric probability density function $\mathbb{P}(x)$ if and only if its characteristic function $\phi_X(\omega)$ is a real-valued function, i.e.:

$$\phi_X(\omega) \in \mathbb{R} \text{ for } \omega \in \mathbb{R}.$$

For $a, b \in \mathbb{R}$:

$$\phi_{aX+b}(\omega) = e^{i\omega b} \phi_X(a\omega). \quad (4.8)$$

[4.3] Characteristic Exponent: Cumulant-Generating Function

A characteristic exponent of a random variable X , $\Psi_X(\omega)$, is defined as a log of a characteristic function $\phi_X(\omega)$:

$$\Psi_X(\omega) \equiv \ln \phi_X(\omega). \quad (4.9)$$

The n -th cumulant is defined as:

$$cumulant_n = \frac{1}{i^n} \frac{\partial^n \Psi_X(\omega)}{\partial \omega^n} \Big|_{\omega=0}. \quad (4.10)$$

Mean, variance, skewness, and excess kurtosis of the random variable X can be obtained from cumulants as follows:

$$\begin{aligned} \text{Mean of } X &= E[X] = cumulant_1, \\ \text{Variance of } X &= E[(X - E(X))^2] = cumulant_2, \\ \text{Skewness of } X &= \frac{E[(X - E(X))^3]}{\left(\sqrt{E[(X - E(X))^2]}\right)^3} = \frac{cumulant_3}{(cumulant_2)^{3/2}}, \\ \text{Excess kurtosis of } X &= \frac{E[(X - E(X))^4]}{\left(\sqrt{E[(X - E(X))^2]}\right)^4} - 3 = \frac{cumulant_4}{(cumulant_2)^2}. \end{aligned} \quad (4.11)$$

Let's consider one fundamental example. A normal random variable X with mean μ and variance σ^2 has a density:

$$\mathbb{P}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(X - \mu)^2}{2\sigma^2}\right\}.$$

Its characteristic function can be calculated as from the definition (4.1):

$$\phi(\omega) \equiv \mathcal{F}[\mathbb{P}(x)] \equiv \int_{-\infty}^{\infty} e^{i\omega x} \mathbb{P}(x) dx = \exp(i\mu\omega - \frac{\sigma^2 \omega^2}{2}).$$

Its characteristic exponent is:

$$\Psi(\omega) \equiv \ln \phi(\omega) = \ln \left\{ \exp(i\mu\omega - \frac{\sigma^2 \omega^2}{2}) \right\} = i\mu\omega - \frac{\sigma^2 \omega^2}{2}.$$

Cumulants are calculated using $\Psi(\omega)$ from (4.10):

$$\begin{aligned} \text{cumulant}_1 &= \mu, \\ \text{cumulant}_2 &= \sigma^2, \\ \text{cumulant}_3 &= 0, \\ \text{cumulant}_4 &= 0. \end{aligned}$$

This tells us that a normal random variable X has a mean μ and variance σ^2 , zero skewness, and zero excess kurtosis.

[4.4] Laplace Transform

For nonnegative random variables we replace the Fourier transform by Laplace transform in order to obtain characteristic functions. The (unilateral) Laplace transform $\mathcal{L}$ of a function $f(x)$ is defined as:

$$\mathcal{L}[f(x)] \equiv \int_0^{\infty} f(x) e^{-\omega x} dx, \quad (4.12)$$

where $f(x)$ is defined for $x \geq 0$. Thus, the characteristic function of a nonnegative random variable X with its density function $\mathbb{P}(x)$ is given by:

$$\phi_x(\omega) = \mathcal{L}[\mathbb{P}(x)] \equiv \int_0^{\infty} \mathbb{P}(x) e^{-\omega x} dx. \quad (4.13)$$

[4.5] Relationship with Moment Generating Function

Let X be a random variable on $\mathbb{R}$ and $\mathbb{P}(x)$ be its probability density function. A function $M(\omega)$ with $\omega \in \mathbb{R}$ is called a moment generating function if there exists an $h > 0$ for $|\omega| < h$ such that (i.e. if the expectation in (4.14) converges):

$$M(\omega) \equiv \int_{-\infty}^{\infty} e^{\omega x} \mathbb{P}(x) dx \equiv E[\exp(\omega x)]. \quad (4.14)$$

For a continuous random variable X , again using the equation (4.3):

$$\begin{aligned}
 M(\omega) &\equiv \int_{-\infty}^{\infty} e^{\omega x} \mathbb{P}(x) dx \\
 &= \int_{-\infty}^{\infty} \left(1 + \omega x + \frac{1}{2!} (\omega x)^2 + \frac{1}{3!} (\omega x)^3 + \frac{1}{4!} (\omega x)^4 + \dots \right) \mathbb{P}(x) dx \\
 &= 1 + \omega c_1 + \frac{1}{2!} \omega^2 c_2 + \frac{1}{3!} \omega^3 c_3 + \frac{1}{4!} \omega^4 c_4 + \dots,
 \end{aligned}$$

where c_n is the n -th central moment.

If $\{X_k, k=1, \dots, n\}$ are independent random variables, the moment generating function of their sum $X_1 + X_2 + \dots + X_n$ is the product of their moment generating functions:

$$M_{X_1 + X_2 + \dots + X_n}(\omega) = \prod_{k=1}^n M_{X_k}(\omega). \quad (4.15)$$

Its proof is very simple:

$$\begin{aligned}
 M_{X_1 + X_2 + \dots + X_n}(\omega) &= E \left[\exp \left\{ \omega (X_1 + X_2 + \dots + X_n) \right\} \right] \\
 &= E \left[\exp \left\{ \omega X_1 + \omega X_2 + \dots + \omega X_n \right\} \right] \\
 &= E \left[\exp \left\{ \omega X_1 \right\} \exp \left\{ \omega X_2 \right\} \dots \exp \left\{ \omega X_n \right\} \right] \\
 &= E \left[\exp \left\{ \omega X_1 \right\} \right] E \left[\exp \left\{ \omega X_2 \right\} \right] \dots E \left[\exp \left\{ \omega X_n \right\} \right] \\
 &= M_{X_1}(\omega) M_{X_2}(\omega) \dots M_{X_n}(\omega) = \prod_{k=1}^n M_{X_k}(\omega).
 \end{aligned}$$

If the moment generating function $M(\omega)$ is differentiable at zero (defined on a neighborhood $[-\varepsilon, \varepsilon]$ of zero), then the n -th raw moments r_n can be obtained by:

$$r_n = \left. \frac{\partial^n M(\omega)}{\partial \omega^n} \right|_{\omega=0}. \quad (4.16)$$

Thus:

$$\begin{aligned}
 r_1 &= M'(0) = E[X], \\
 r_2 &= M''(0) = E[X^2], \\
 r_3 &= M'''(0) = E[X^3], \\
 r_4 &= M^{(4)}(0) = E[X^4].
 \end{aligned}$$

For example, the mean and variance of the random variable X are computed using raw moments as (discussed in detail in section 4.6):

$$\begin{aligned}\text{Mean of } X &= E[X] = r_1, \\ \text{Variance of } X &= E[X^2] - E[X]^2 = r_2 - r_1^2.\end{aligned}$$

A characteristic function is always well-defined since it is the Fourier transform of a probability measure. But because the integral in (4.14) ($\forall \omega \in \mathbb{R}$) may not converge for some (all) values of ω , a moment generating function is not always well-defined. When $M(\omega)$ is well-defined, the relationship between the moment generating function $M(\omega)$ and the characteristic function $\phi(\omega)$ is given by:

$$M(\omega) = \phi(-i\omega). \quad (4.17)$$

Let's consider one fundamental example. If $X \sim \text{Normal}(\mu, \sigma)$, its moment generating function can be calculated as following the definition (4.14):

$$M(\omega) \equiv \int_{-\infty}^{\infty} e^{\omega x} \mathbb{P}(x) dx \equiv E[\exp(\omega x)] = \exp\left(\mu\omega + \frac{\sigma^2 \omega^2}{2}\right).$$

You can confirm that (4.17) is true for the normal case. Raw moments are calculated as the following:

$$\begin{aligned}r_1 &= M'(0) = \mu, \\ r_2 &= M''(0) = \mu^2 + \sigma^2, \\ r_3 &= M'''(0) = \mu^3 + 3\mu\sigma^2, \\ r_4 &= M^{(4)}(0) = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4.\end{aligned}$$

Using these raw moments, central moments can be calculated as:

$$\begin{aligned}E[X] &= r_1 = \mu, \\ \text{Variance}[X] &= E[X^2] - E[X]^2 = r_2 - r_1^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2, \\ \text{Skewness}[X] &= \frac{E\{X - E[X]\}^3}{(\sqrt{E\{X - E[X]\}^2})^3} = \frac{2r_1^3 - 3r_1r_2 + r_3}{(r_2 - r_1^2)^{3/2}} = 0, \\ \text{Excess Kurtosis}[X] &= \frac{E\{X - E[X]\}^4}{(\sqrt{E\{X - E[X]\}^2})^4} - 3 \\ &= \frac{-6r_1^4 + 12r_1^2r_2 - 3r_2^2 - 4r_1r_3 + r_4}{(r_2 - r_1^2)^2} = 0.\end{aligned}$$

[4.6] Summary: How to Calculate Standardized Moments from Characteristic Function and Moment Generating function

Following is a summary of the relationship between standardized moments, $cumulant_n$, and raw moments r_n . Let X be a random variable. n -th cumulant and n -th raw moment are defined by (5.10) and (5.16):

$$cum_n = \frac{1}{i^n} \frac{\partial^n \Psi_X(\omega)}{\partial \omega^n} \Big|_{\omega=0},$$

$$r_n = \frac{\partial^n M(\omega)}{\partial \omega^n} \Big|_{\omega=0}.$$

Table 4.1: How to Calculate Standardized Moments from Characteristic Function and Moment Generating function

Moments		n -th cumulant	n -th raw moment
Mean	$E[X]$	cum_1	r_1
Variance	$E\{X - E[X]\}^2$	cum_2	$r_2 - r_1^2$
Skewness	$\frac{E\{X - E[X]\}^3}{\left(\sqrt{E\{X - E[X]\}^2}\right)^3}$	$\frac{cum_3}{cum_2^{3/2}}$	$\frac{2r_1^3 - 3r_1r_2 + r_3}{(r_2 - r_1^2)^{3/2}}$
Excess Kurtosis	$\frac{E\{X - E[X]\}^4}{\left(\sqrt{E\{X - E[X]\}^2}\right)^4} - 3$	$\frac{cum_4}{cum_2^2}$	$\frac{-6r_1^4 + 12r_1^2r_2 - 3r_2^2 - 4r_1r_3 + r_4}{(r_2 - r_1^2)^2}$

Note:

- o 2-nd central moment of $X = E\{X - E[X]\}^2 = E[X^2] - E[X]^2 = Var[X]$
 $= cum_2 = r_2 - r_1^2$.
- o 3-rd central moment of $X = E\{X - E[X]\}^3 = cum_3 = 2r_1^3 - 3r_1r_2 + r_3$.
- o Skewness of $X = Skewness[X] = \frac{E\{X - E[X]\}^3}{\left(\sqrt{E\{X - E[X]\}^2}\right)^3} = \frac{cum_3}{cum_2^{3/2}}$
 $= \frac{2r_1^3 - 3r_1r_2 + r_3}{(r_2 - r_1^2)^{3/2}}$.
- o 4-th central moment of $X = E\{X - E[X]\}^4 = cum_4 + 3\{E\{X - E[X]\}^2\}^2$
 $= cum_4 + 3cum_2^2 = (-6r_1^4 + 12r_1^2r_2 - 3r_2^2 - 4r_1r_3 + r_4) + 3(r_2 - r_1^2)^2$.

$$\begin{aligned} \text{Excess Kurtosis of } X &= \frac{E\{X - E[X]\}^4}{\left(\sqrt{E\{X - E[X]\}^2}\right)^4} - 3 = \frac{cum_4 + 3cum_2^2}{cum_2^2} - 3 \\ &= \frac{cum_4}{cum_2^2} = \frac{-6r_1^4 + 12r_1^2r_2 - 3r_2^2 - 4r_1r_3 + r_4}{(r_2 - r_1^2)^2}. \end{aligned}$$

[4.7] Examples of Characteristic Functions

Several examples of characteristic functions using the definition (4.1) for continuous distributions and (4.6) for discrete distributions are given in the Table 4.2.

Table 4.2: Examples of Characteristic Functions

Distribution	$\mathbb{P}(x)$	$\phi(\omega)$
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(X - \mu)^2}{2\sigma^2}\right\}$	$\exp(i\mu\omega - \frac{\sigma^2\omega^2}{2})$
Exponential	ae^{-ax}	$\frac{a}{a - i\omega}$
Gamma	$\frac{b^{-a}e^{-x/b}x^{a-1}}{\Gamma(a)}$	$(1 - ib\omega)^{-a}$
Poisson	$\frac{e^{-\lambda}\lambda^x}{x!}$	$\exp[\lambda(e^{i\omega} - 1)]$

[5] Discrete Fourier Transform (DFT)

DFT is a special case of (continuous) Fourier transform. The results obtained from FT and DFT are identical and the only difference is how we interpret these results.

[5.1] Intuitive Derivation of DFT: Approximation of FT

We first consider DFT of a time domain function $g(t)$ into an angular frequency domain ω Hz (radians/second). This follows the convention in physics. In the field of signal processing which is a major application of FT, frequency f Hz (cycles/second) is used instead of ω . But this difference is not important because ω and f are measuring the same thing (rotation speed/second) in different units and related by:

$$\omega = 2\pi f . \quad (5.1)$$

We saw in section 3 that continuous FT of $g(t)$ and inverse FT of $\mathcal{G}(\omega)$ using FT parameters $(a, b) = (1, 1)$ are defined as:

$$\mathcal{G}(\omega) \equiv \mathcal{F}_t[g(t)](\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t} g(t) dt , \quad (5.2)$$

$$g(t) \equiv \mathcal{F}_{\omega}^{-1}[\mathcal{G}(\omega)](t) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \mathcal{G}(\omega) d\omega . \quad (5.3)$$

The purpose of DFT is to approximate FT as close as possible by sampling a finite number of points N of a continuous time domain function $g(t)$ with time domain sampling interval Δt (seconds) and sampling a finite number of points N of a continuous FT $\mathcal{G}(\omega)$ with angular frequency sampling interval $\Delta\omega$ Hz. In other words, both the original continuous time domain function $g(t)$ and the original continuous FT $\mathcal{G}(\omega)$ are approximated by a sample of N points.

We begin with the notation. Let N be the number of discrete samples taken to approximate $g(t)$ and $\mathcal{G}(\omega)$. Let Δt (seconds/sample) be the time domain sampling interval which is the time increment between samples. Its inverse $f_s \equiv 1/\Delta t$ (samples/second) is called a sampling rate. Let T be the total sampling time:

$$\Delta t \equiv T / N . \quad (5.4)$$

It is extremely important to mention that total sampling time T defined by (5.4) has absolutely nothing to do with period T of oscillation of a wave (i.e. the seconds (time) taken for the wave to complete one wavelength defined by the equation (2.3)). To avoid confusion, period T of oscillation of a wave is called a fundamental period.

If Δt is assumed to be 1, the total sampling time and the number of samples taken are same (i.e. for example, $T = 10$ seconds and $N = 10$ samples). If $\Delta t = 0.01$, 1 sample is

taken in every 0.01 second in a time domain which in turn means that 100 samples are taken every second (i.e. sampling rate $f_s \equiv 1/\Delta t = 100$ Hz).

The first step to DFT is to take N discrete samples of a continuous time domain function $g(t)$ at n -th sampling instant $t_n = n\Delta t$ (seconds) with $n = 0, 1, \dots, N-1$. When $T = 10$ seconds and $N = 10$ samples (i.e. time domain sampling interval $\Delta t = 1$), $g(t)$ is sampled at 0 second, 1 second, 2 seconds,, and 9 seconds. Let $g(t_n = n\Delta t)$ be the sampled values of $g(t)$. In a special case of $\Delta t = 1$:

$$g(t_n) = g(n). \quad (5.5)$$

We call this process as time domain sampling.

Next consider angular frequency domain (ω Hz) sampling. Let $\Delta\omega$ Hz (radians/second) be angular frequency sampling interval:

$$\Delta\omega \equiv \frac{2\pi}{N\Delta t} \equiv \frac{2\pi}{T}. \quad (5.6)$$

The second step to DFT is to take N samples of a continuous FT $\mathcal{G}(\omega)$ at k -th angular frequency sampling instant $\omega_k = k\Delta\omega$ (radians) with $k = 0, 1, \dots, N-1$. When $T = 10$ seconds and $N = 10$ samples (i.e. $\Delta t = 1$), $\mathcal{G}(\omega)$ is sampled at 0 radian, $\pi/5$ radians, $2\pi/5$ radians, $3\pi/5$ radians, ..., and $9\pi/5$ radians. Let $\mathcal{G}(\omega_k)$ be the sampled values of $\mathcal{G}(\omega)$:

$$\mathcal{G}(\omega_k) \equiv \mathcal{G}(k\Delta\omega) \equiv \mathcal{G}(k \frac{2\pi}{N\Delta t}) \equiv \mathcal{G}(k \frac{2\pi}{T}). \quad (5.7)$$

$\mathcal{G}(\omega_k)$ is called a spectrum of $g(t_n)$ at angular frequency ω_k and it is a complex number.

DFT defines the relationship between the sampled wave in time domain $g(t_n)$ and its spectrum at angular frequency ω_k , $\mathcal{G}(\omega_k)$, as:

$$\begin{aligned} \mathcal{G}(\omega_k \equiv k \frac{2\pi}{N\Delta t}) &\equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{i\omega_k t_n\} \\ \mathcal{G}(k \frac{2\pi}{N\Delta t}) &\equiv \sum_{n=0}^{N-1} g(n\Delta t) \exp\{2\pi i k n / N\}, \end{aligned} \quad (5.8)$$

and:

$$\begin{aligned}
g(t_n \equiv n\Delta t) &\equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(\omega_k \equiv k \frac{2\pi}{N\Delta t}) \exp(-i\omega_k t_n) \\
g(n\Delta t) &\equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(k \frac{2\pi}{N\Delta t}) \exp(-2\pi i k n / N).
\end{aligned} \tag{5.9}$$

As you can see, DFT replaces an infinite integral of FT with summation of N points.

[5.2] Definition of DFT

[5.2.1] Physicists' Definition of DFT

Likewise FT, there are several different definitions of DFT depending on the field of study. We first follow physicists' convention. We begin with the most general definition of DFT of a continuous time domain function $g(t)$ into angular frequency domain function $\mathcal{G}(\omega)$ and its inverse using DFT parameters (a, b) as:

$$\mathcal{G}(\omega_k \equiv k \frac{2\pi}{N\Delta t}) \equiv \frac{1}{N^{\frac{1-a}{2}}} \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{ib\omega_k t_n\}, \tag{5.10}$$

$$g(t_n \equiv n\Delta t) \equiv \frac{1}{N^{\frac{1+a}{2}}} \sum_{k=0}^{N-1} \mathcal{G}(\omega_k \equiv k \frac{2\pi}{N\Delta t}) \exp(-ib\omega_k t_n). \tag{5.11}$$

When DFT parameters are $(a, b) = (1, 1)$ for the purpose of calculating characteristic functions, the definitions (5.10) and (5.11) become the definitions (5.8) and (5.9).

[5.2.2] Signal Processing Definition of DFT

In the field of signal processing (and most of the textbooks about FT), frequency f Hz (cycles/second) is used instead of ω Hz (radians/second). Consider frequency domain f sampling. Let Δf Hz be frequency domain sampling interval (also called frequency resolution):

$$\Delta f \equiv \frac{1}{N\Delta t} \equiv \frac{1}{T}. \tag{5.12}$$

Take N samples of a continuous FT $\mathcal{G}(f)$ at k -th frequency sampling instant $f_k = k\Delta f$ Hz with $k = 0, 1, \dots, N-1$. When $T = 10$ seconds and $N = 10$ samples (i.e. $\Delta t = 1$), $\mathcal{G}(f)$ is sampled at 0 Hz, 1/10 Hz, 2/10 Hz, ..., and 9/10 Hz. Let $\mathcal{G}(f_k)$ be the sampled values of $\mathcal{G}(f)$:

$$\mathcal{G}(f_k) \equiv \mathcal{G}(k\Delta f) \equiv \mathcal{G}(k \frac{1}{N\Delta t}) \equiv \mathcal{G}(k \frac{1}{T}). \tag{5.13}$$

$\mathcal{G}(f_k)$ is called a spectrum of $g(t_n)$ at frequency f_k Hz and it is a complex number.

The general definition of DFT of a continuous time domain function $g(t)$ into frequency domain function $\mathcal{G}(f)$ and its inverse using DFT parameters (a, b) become (i.e. equivalent to (5.10) and (5.11)):

$$\mathcal{G}(f_k \equiv k \frac{1}{N\Delta t}) \equiv \frac{1}{N^{\frac{1-a}{2}}} \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{2\pi i b k n / N\}, \quad (5.14)$$

$$g(t_n \equiv n\Delta t) \equiv \frac{1}{N^{\frac{1+a}{2}}} \sum_{k=0}^{N-1} \mathcal{G}(f_k \equiv k \frac{1}{N\Delta t}) \exp(-2\pi i b k n / N). \quad (5.15)$$

In the signal processing, DFT parameters $(1, -1)$ are used:

$$\mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{-2\pi i k n / N\}, \quad (5.16)$$

$$g(t_n \equiv n\Delta t) \equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \exp(2\pi i k n / N). \quad (5.17)$$

[5.2.3] Requirement of DFT

FT and inverse FT defined by the equations (5.10) and (5.11) require both the time domain function $g(t_n \equiv n\Delta t)$ and the angular frequency domain function $\mathcal{G}(\omega_k \equiv k \frac{2\pi}{N\Delta t})$ to be periodic. This means for $\theta = 0, \pm 1, \pm 2, \dots$:

$$g(t_n \equiv n\Delta t) = g[(\theta N + n)\Delta t],$$

$$\mathcal{G}(\omega_k \equiv k \frac{2\pi}{N\Delta t}) = \mathcal{G}\left[(\theta N + k) \frac{2\pi}{N\Delta t}\right].$$

FT and inverse FT defined by the equations (5.16) and (5.17) require both the time domain function $g(t_n \equiv n\Delta t)$ and the frequency domain function $\mathcal{G}(f_k \equiv \frac{k}{N\Delta t})$ to be periodic. This means for $\theta = 0, \pm 1, \pm 2, \dots$:

$$g(t_n \equiv n\Delta t) = g[(\theta N + n)\Delta t],$$

$$\mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) = \mathcal{G}\left[\frac{(\theta N + k)}{N\Delta t}\right].$$

[5.2.4] Sampling Theorem and Nyquist Rule: How to Determine the Time Domain Sampling Interval Δt

At this point we are very familiar with the definition of DFT at least in terms of concept. DFT is simply a discrete approximation of FT by taking a N point sample both in the time domain t (seconds) and frequency (angular frequency) domain f (ω) Hz. When implementing DFT, the following very important question about the frequency of sampling arises. What is the appropriate time domain sampling interval Δt (seconds/sample), 1 second, 0.5 seconds, or 0.01 seconds? In terms of sampling rate $f_s = 1/\Delta t$ (samples/second), question becomes the following. What is the appropriate sampling rate f_s , 1 Hz, 2 Hz, or 100 Hz?

If time domain function $g(t)$ is not sampled at an appropriately high rate (i.e. you chose too large time domain sampling interval Δt or you chose too small sampling rate f_s Hz), it turns out that DFT yields distorted (overlapped) approximation of FT known as aliasing. According to the sampling theorem of Fourier transform, a continuous time domain function $g(t)$ can be uniquely determined by its sampled values $g(t_n \equiv n\Delta t)$ by choosing the time domain sampling interval of $\Delta t = \frac{1}{2f_c}$ (seconds) if the FT $\mathcal{G}(f)$ of $g(t)$ is zero for all frequencies greater than f_c Hz (i.e. if the FT $\mathcal{G}(f)$ of $g(t)$ is band-limited at the frequency f_c Hz):

$$\mathcal{G}(f) = 0 \quad \text{if} \quad |f| > f_c.$$

The time domain sampling interval of $\Delta t = \frac{1}{2f_c}$ (seconds) is the maximum value of the interval without aliasing. In other words, aliasing occurs if $\Delta t > \frac{1}{2f_c}$ and aliasing does not occur if $\Delta t < \frac{1}{2f_c}$.

In terms of a sampling rate $f_s = 1/\Delta t$ Hz, the sampling theorem becomes the following. A continuous time domain function $g(t)$ can be uniquely determined by its sampled values $g(t_n \equiv n\Delta t)$ by choosing the sampling rate of:

$$f_s = 2f_c \text{ Hz}, \quad (5.18)$$

where f_c is the folding frequency of the waveform $g(t)$. The sampling rate of $f_s = 2f_c$ Hz is the minimum value of sampling rate without aliasing. In other words, aliasing occurs if $f_s < 2f_c$ and aliasing does not occur if $f_s \geq 2f_c$.

After simple rearrangement:

$$\Delta t < \frac{1}{2f_c}$$

$$\frac{1}{\Delta t} \equiv f_s < 2f_c$$

$$\frac{1}{2}f_s < f_c.$$

Nyquist rule states that when we sample a time domain function $g(t)$ with sampling rate f_s Hz (samples/second), its FT $\mathcal{G}(f)$ is reliable (i.e. without aliasing) only up to frequency $f_s/2$ Hz. The maximum frequency without aliasing $f_s/2 = f_c$ is called a folding frequency.

Nyquist rule can be stated in terms of k -th frequency sampling instant $f_k = k\Delta f$ Hz with $k = 0, 1, \dots, N-1$. We solve for the value of k such that:

$$k\Delta f = f_s/2.$$

Since $f_s = 1/\Delta t$ and $\Delta f = 1/N\Delta t$:

$$k \frac{1}{N\Delta t} = \frac{1}{2} \frac{1}{\Delta t}$$

$$k = \frac{N}{2}.$$

This means that when we take a N point sample of a time domain function $g(t)$, its FT $\mathcal{G}(f)$ is reliable (i.e. without aliasing) only up to $k = N/2$ -th frequency sampling instant. In other words, only half of N point DFT outputs are reliable.

Consider DFT of a continuous time domain function $g(t)$ with the sampling rate $f_s = 200$ Hz (samples/second) for the total sampling time $T = 3$ (seconds). $g(t)$ is sampled in the time domain with the interval $\Delta t = 1/f_s = 1/200 = 0.005$ (seconds/sample). The number of samples taken is $N = T/\Delta t = 3/0.005 = 600$. Frequency resolution (frequency domain sampling interval) is $\Delta f \equiv 1/N\Delta t \equiv 1/T = 1/3$ Hz.

Following Nyquist rule, FT $\mathcal{G}(f)$ is reliable (i.e. without aliasing) only up to frequency $f_s/2 = 200/2 = 100$ Hz. In other words, out of 600 DFT outputs $\mathcal{G}(f_k \equiv \frac{k}{N\Delta t})$ with $k = 0, 1, 2, \dots, 599$ only the half $N/2 = 600/2 = 300$ are reliable.

Read the chapter 5 of Brigham (1988) which is an excellent book for more rigorous treatment of the sampling theorem.

[5.3] Examples of DFT

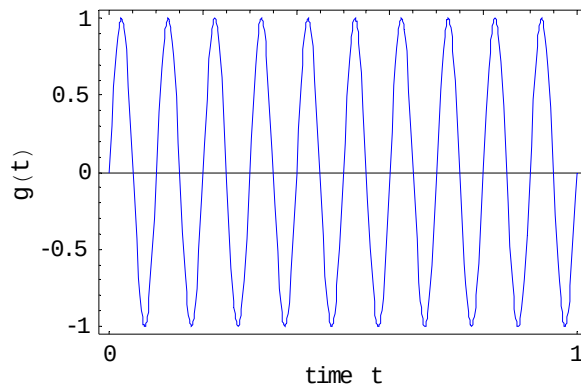
Above all, section 5.3.1 is the most important example. Using an example of a sine wave $g(t) = A\sin(2\pi f_0 t) = A\sin(\omega_0 t)$, following important features of DFT are explained in detail: 1) Approximation error inherent in DFT called leakage, 2) Leakage cannot always be reduced by increasing a sampling rate f_s Hz (cycles/second) with holding the number of samples taken N constant, 3) Guideline of choosing the optimal sampling rate f_s Hz when the folding frequency f_c of the waveform is known (Nyquist rule), and 4) The only remedy for leakage is to sample more points (i.e. higher N).

[5.3.1] Sine Wave $g(t) = A\sin(2\pi f_0 t) = A\sin(\omega_0 t)$

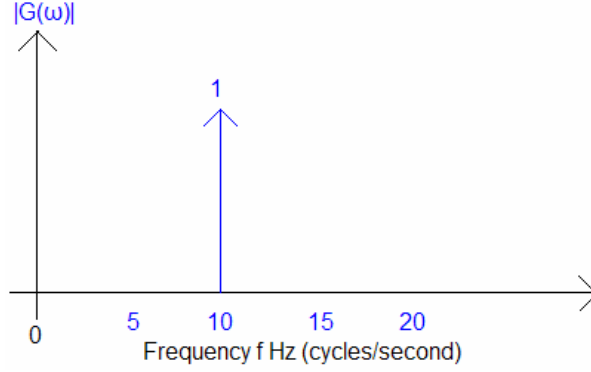
Consider 10 Hz (cycles/second) sine wave (i.e. fundamental frequency $f_0 = 10$) with amplitude $A=1$ volt plotted in Panel A of Figure 5.1:

$$g(t) = \sin(2\pi 10t). \quad (5.19)$$

We saw in section 3.2.6 that the ideal FT of (5.19) is an impulse function of normalized magnitude (frequency spectrum) 1 volt at frequency $f = 10$ Hz plotted in Panel B of Figure 5.1 which implies that the frequency $f = 10$ Hz contains all the energy of the waveform. Normalized magnitude is normalized so that it equals the amplitude A of the waveform $g(t)$.



A) Plot of a 10 Hz sine wave $g(t) = \sin(2\pi 10t)$.



B) Plot of the ideal normalized FT of $g(t) = \sin(2\pi 10t)$.

Figure 5.1: Plot of a 10 Hz Sine Wave and Its Ideal Normalized FT

Next, we consider DFT. Since DFT is only an approximation of FT, the frequency spectrum produced by DFT is not as clean as the Panel B Figure 5.1 meaning that it contains a lot of noise. This degree of cleanness (noise) depends on such factors as the sampling rate f_s Hz (or $\Delta t = 1/f_s$), the point of sample N , and the total sampling time T which are all related by:

$$f_s \equiv N / T . \quad (5.20)$$

First, we sample a time domain function (5.19) by a sampling rate of $f_s = 200$ Hz (samples/second) and the number of samples $N = 256$ (samples). In other words, we take a $N = 256$ point sample of (5.19) with a time domain sampling interval:

$$\Delta t = 1 / f_s = 1 / 200 = 0.005 \text{ (seconds/sample),}$$

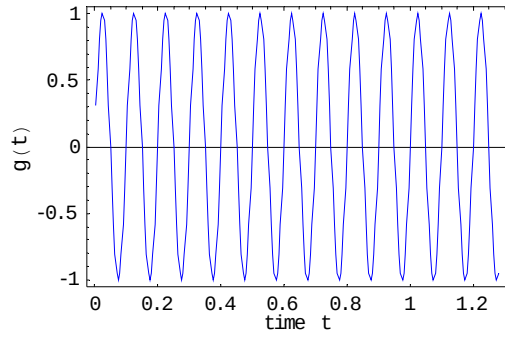
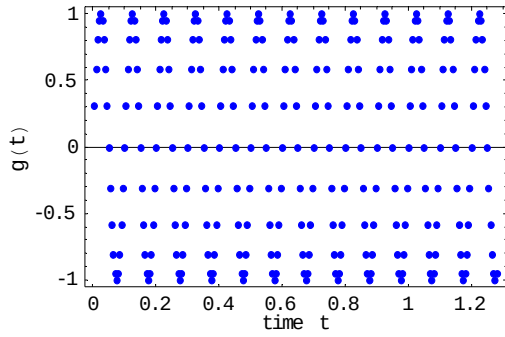
and total sampling time:

$$T = N \Delta t = 256 \times 0.005 = 1.28 \text{ (seconds).}$$

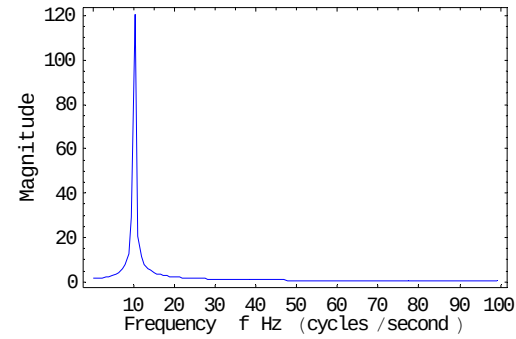
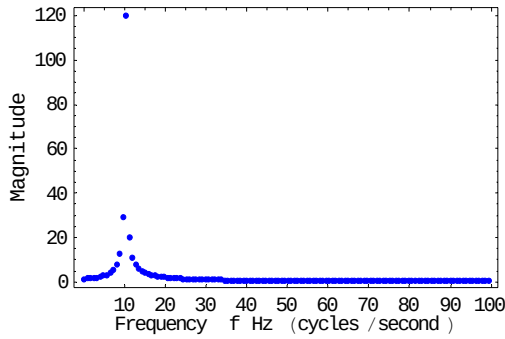
Sampled 10 Hz sine wave (5.19) is plotted in Panel A of Figure 5.2.

Secondly, consider frequency domain f sampling. Frequency resolution (frequency domain sampling interval) Δf Hz (cycles/second) is following (5.12):

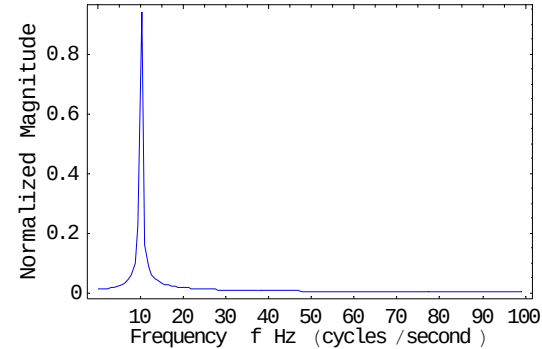
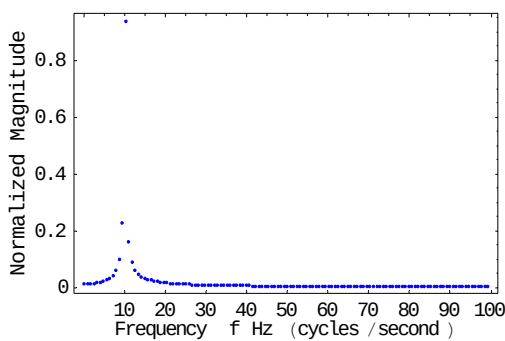
$$\Delta f \equiv \frac{1}{T \equiv N \Delta t} = \frac{1}{1.28} = 0.78125 .$$



A) Plot of sampled 10 Hz sine wave $g(t) = \sin(2\pi 10t)$ on the left. Points are joined on the right.



B) Plot of DFT non-normalized frequency spectrum of sampled 10 Hz sine wave $g(t) = \sin(2\pi 10t)$ on the left. Points are joined on the right.



C) Plot of DFT normalized frequency spectrum of sampled 10 Hz sine wave $g(t) = \sin(2\pi 10t)$ on the left. Points are joined on the right.

Figure 5.2: Plot of Sampled 10 Hz Sine Wave and Its DFT Output

Following the signal processing definition of DFT (5.16), frequency spectrum of sampled 10 Hz sine wave (5.19) is plotted in Panel B of Figure 5.2. The frequency axis goes from 0 to 100 Hz because aliasing does not occur up to the folding frequency $f_s/2 = 200/2 = 100$ Hz. Frequency spectrum in Panel C is normalized so that the normalized magnitude equals the amplitude A of the waveform $g(t)$ (i.e. in this example $A = 1$). This normalization is done by weighting the non-normalized magnitude by:

$$\frac{1}{N/2}.$$

Several important features should be noticed from the Panel C of Figure 5.2. Continuous (normalized) FT $\mathcal{G}(f)$ of a sine wave (5.19) is a unit impulse function at the frequency $f_0 = 10$ Hz (this fundamental frequency is also a folding frequency) as depicted in the Panel B of Figure 5.1. Its approximation by DFT is not an impulse function, but it is rather a spike with significant portion of the energy distributed around 10 Hz (i.e. positive energy in the interval $5 < f < 15$ Hz). In other words, the approximation by DFT leaks the energy around the original impulse. This approximation error inherent in DFT is called a leakage which occurs when the time domain sampling does not end at the phase of the sine wave as it started. As a consequence of leakage, the normalized maximum magnitude of DFT $\mathcal{G}(f)$ is less than 1 compared to the exactly 1 normalized maximum magnitude of an original impulse. Theoretically speaking, the only remedy for leakage is to take infinitely many samples (i.e. $N \rightarrow \infty$). But practically speaking, DFT always contains leakage because it is impossible to take infinitely many samples.

Some readers might think that the degree of leakage can be reduced by taking samples with higher sampling rate f_s Hz with no change in N . The answer is no. To show why this is the case, we redo DFT of a sine wave (5.19) with higher sampling rate $f_s = 1000$ Hz (5 times faster). First, we sample a time domain function (5.19) by a sampling rate of $f_s = 1000$ Hz (samples/second) and the number of samples $N = 256$ (samples). In other words, we take a $N = 256$ point sample of (5.19) with a time domain sampling interval:

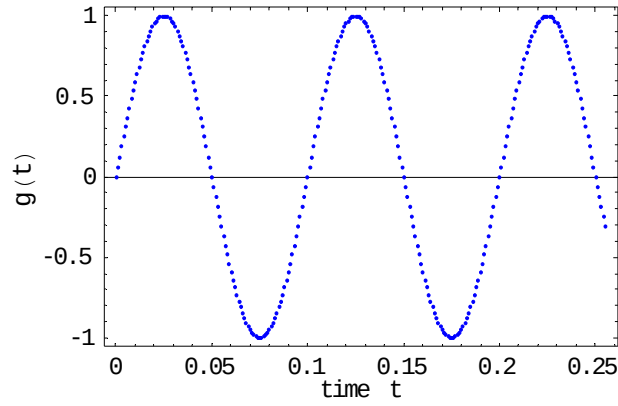
$$\Delta t = 1/f_s = 1/1000 = 0.001 \text{ (seconds/sample),}$$

and total sampling time:

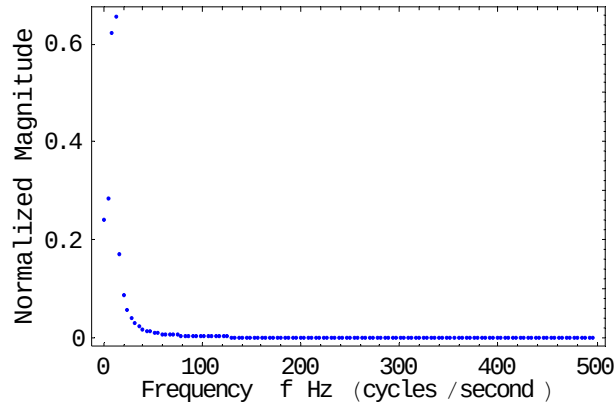
$$T = N\Delta t = 256 \times 0.001 = 0.256 \text{ (seconds).}$$

Sampled 10 Hz sine wave (5.19) is plotted in Panel A of Figure 5.3. Secondly, consider frequency domain f sampling. Frequency resolution (frequency domain sampling interval) Δf Hz (cycles/second) is following (5.12):

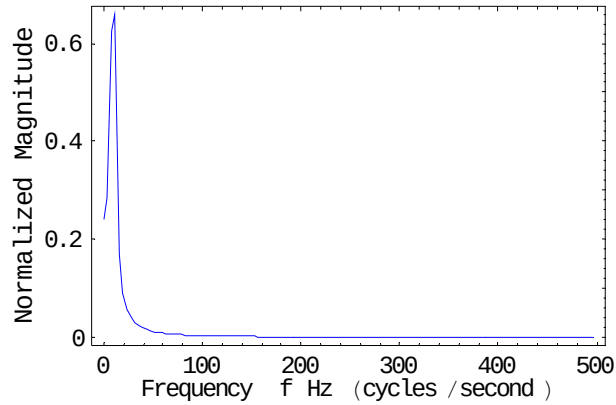
$$\Delta f \equiv \frac{1}{T \equiv N\Delta t} = \frac{1}{0.256} = 3.90625.$$



A) Plot of sampled 10 Hz sine wave with the sampling rate $f_s = 1000$ Hz.



B) Plot of DFT normalized frequency spectrum of sampled 10 Hz sine wave $g(t) = \sin(2\pi 10t)$ with the sampling rate $f_s = 1000$ Hz.



C) Points are joined.

Figure 5.3: Plot of Sampled 10 Hz Sine Wave and Its Normalized DFT Output with Sampling Rate $f_s = 1000$ Hz.

Following the signal processing definition of DFT (5.16), normalized frequency spectrum of sampled 10 Hz sine wave (5.19) with $f_s = 1000$ Hz is plotted in Panel B and C of

Figure 5.3. The frequency axis goes from 0 to 500 Hz because aliasing does not occur up to the folding frequency $f_s/2 = 1000/2 = 500 = 500$ Hz according to Nyquist rule. As you realize now, leakage gets worse as a result of an increase in a sampling rate f_s from 200 to 1000 Hz. We can see that the normalized maximum magnitude of DFT $\mathcal{G}(f)$ is 0.6573267 (significantly less than 1 which is the original normalized magnitude) which in turn indicates that the degree of leakage of energy around 10 Hz is greater. In addition, the normalized maximum magnitude 0.6573267 occurs at 11.7188 Hz, although continuous normalized FT is a unit impulse function at the frequency $f_0 = 10$ Hz. The reason that the accuracy of the approximation by DFT gets poorer is the fact that the frequency resolution gets poorer from:

$$\Delta f \equiv \frac{1}{T \equiv N\Delta t} = \frac{1}{1.28} = 0.78125 \text{ Hz},$$

to:

$$\Delta f \equiv \frac{1}{T \equiv N\Delta t} = \frac{1}{0.256} = 3.90625 \text{ Hz},$$

as a result of an increase in a sampling rate f_s from 200 to 1000 Hz (i.e. because an increase in f_s is an decrease in Δt through the relationship $f_s \equiv \frac{1}{\Delta t}$). The bottom line is that the degree of leakage cannot necessarily be reduced by taking samples with higher sampling rate f_s Hz with no change in N .

The above example indicates that for a 10 Hz sine wave using 1000 Hz sampling rate f_s is too much (i.e. frequency resolution becomes too poor). So the next natural question arises that how can we determine the appropriate sampling rate f_s Hz? Following Nyquist rule of (5.18), the sampling rate of $f_s = 2f_c$ Hz is the minimum value of sampling rate without aliasing. As long as the folding frequency f_c of the waveform $g(t)$ is known (in our example $f_c = 10$ Hz), the minimum value of sampling rate without aliasing can be obtained by doubling f_c . Although this is a nice rule, its practical usefulness is very doubtful because in other than textbook examples the folding frequency f_c of the waveform $g(t)$ is not known a priori. Redo DFT of a sine wave (5.19) using the sampling rate $f_s = 50$ Hz. We take a $N = 256$ point sample of (5.19) with a time domain sampling interval:

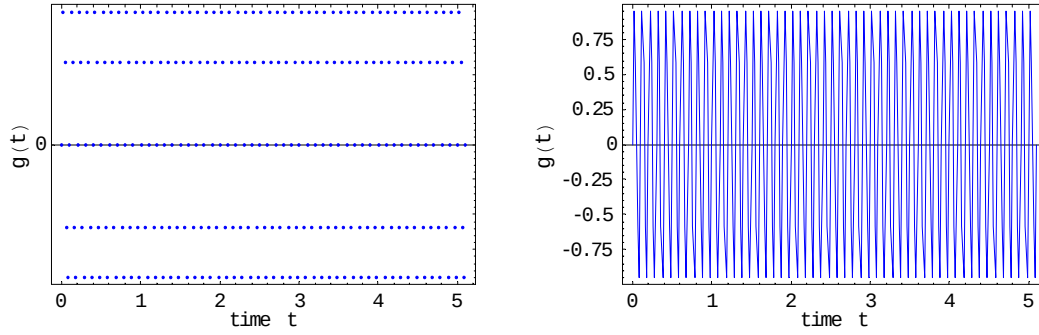
$$\Delta t = 1/f_s = 1/50 = 0.02 \text{ (seconds/sample)},$$

and total sampling time:

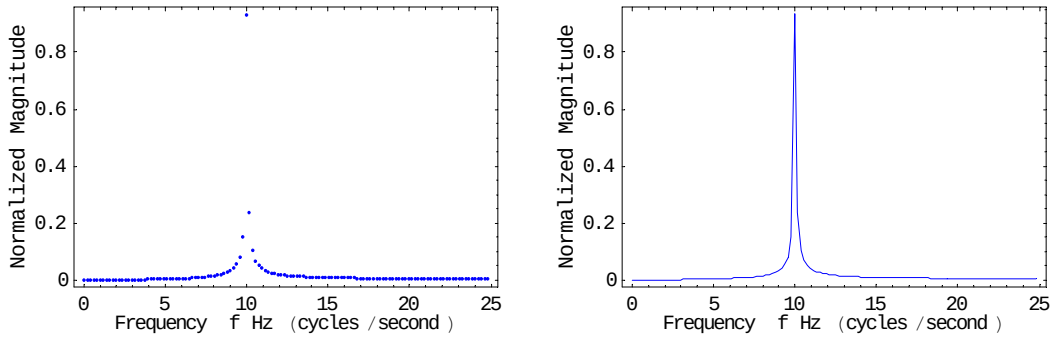
$$T = N\Delta t = 256 \times 0.02 = 5.12 \text{ (seconds)}.$$

Sampled 10 Hz sine wave (5.19) is plotted in Panel A of Figure 5.4. Frequency resolution of DFT (frequency domain sampling interval) Δf Hz (cycles/second) is following (5.12):

$$\Delta f \equiv \frac{1}{T \equiv N\Delta t} = \frac{1}{5.12} = 0.1953125.$$



A) Plot of sampled 10 Hz sine wave $g(t) = \sin(2\pi 10t)$ using the sampling rate $f_s = 50$ Hz on the left. Points are joined on the right.



B) Plot of DFT normalized frequency spectrum of sampled 10 Hz sine wave $g(t) = \sin(2\pi 10t)$ using the sampling rate $f_s = 50$ Hz on the left. Points are joined on the right.

Figure 5.4: Plot of Sampled 10 Hz Sine Wave and Its Normalized DFT Output with Sampling Rate $f_s = 50$ Hz.

By reducing the sampling rate from $f_s = 1000$ Hz to $f_s = 50$ Hz, DFT approximates continuous FT (a unit impulse function at 10 Hz in this example) much better because the frequency resolution Δf increases from 3.90625 Hz to 0.1953125 Hz. Again, the important result is the fact that higher sampling rate f_s does not always reduce leakage (DFT inherent error) depending on the folding frequency f_c of the waveform.

The only way to reduce leakage is to sample more points. To illustrate this, we take a $N = 4 \times 256 = 1024$ point sample of (5.19) using the sampling rate $f_s = 50$ Hz with a time domain sampling interval:

$$\Delta t = 1/f_s = 1/50 = 0.02 \text{ (seconds/sample),}$$

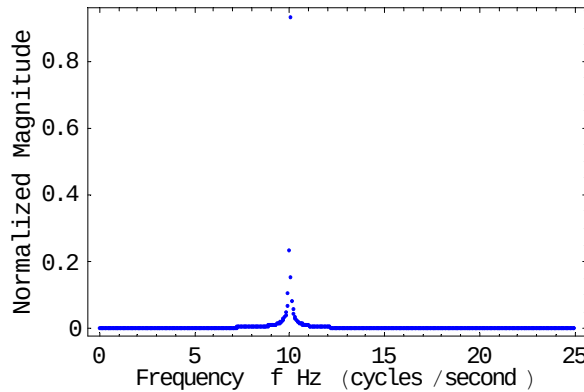
and total sampling time:

$$T = N\Delta t = 1024 \times 0.02 = 20.48 \text{ (seconds).}$$

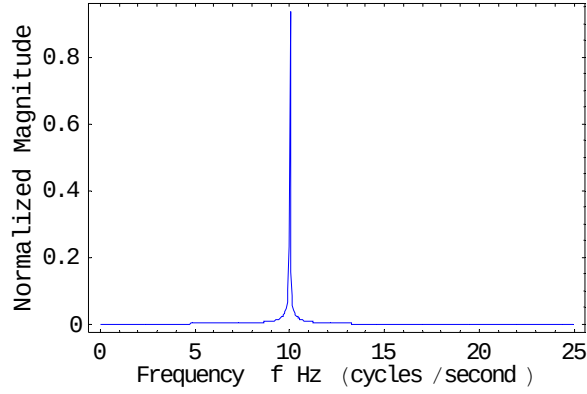
Sampled 10 Hz sine wave (5.19) is plotted in Figure 5.5. Frequency resolution of DFT (frequency domain sampling interval) Δf Hz (cycles/second) is following (5.12):

$$\Delta f \equiv \frac{1}{T \equiv N\Delta t} = \frac{1}{20.48} = 0.048828125.$$

Note that the frequency resolution Δf become 4 times finer because the number of samples taken N increased by four-fold. As a result, leakage has been dramatically reduced. In fact, in Figure 5.5 the peak magnitude is 0.935 at a frequency of 10.01 Hz which is really close (i.e. a good approximation) to the continuous FT of the peak magnitude 1 at 10 Hz.



A) Plot of $N = 1,024$ point-DFT normalized frequency spectrum of 10 Hz sine wave $g(t) = \sin(2\pi 10t)$. Sampling rate $f_s = 50$ Hz is used.



B) Points are joined.

Figure 5.5: Plot of $N = 1,024$ Point-DFT of a 10-Hz Sine Wave and Its Normalized DFT Output with Sampling Rate $f_s = 50$ Hz.

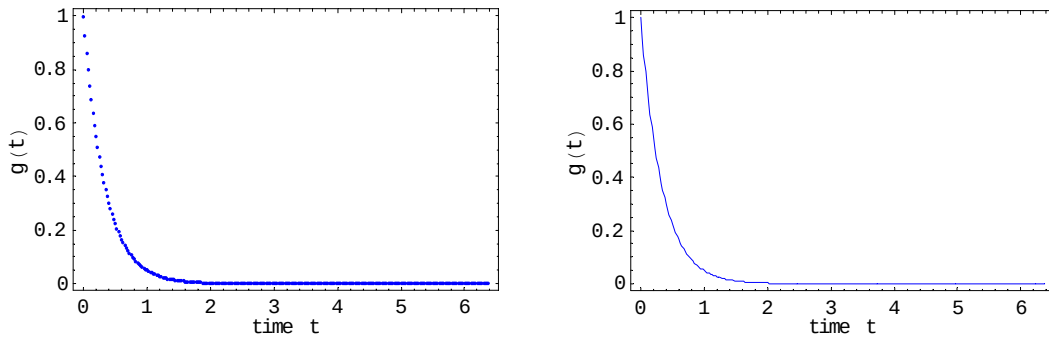
[5.3.2] Double-Sided Exponential

Consider a double-sided exponential function with $A, \alpha \in \mathbb{R}$:

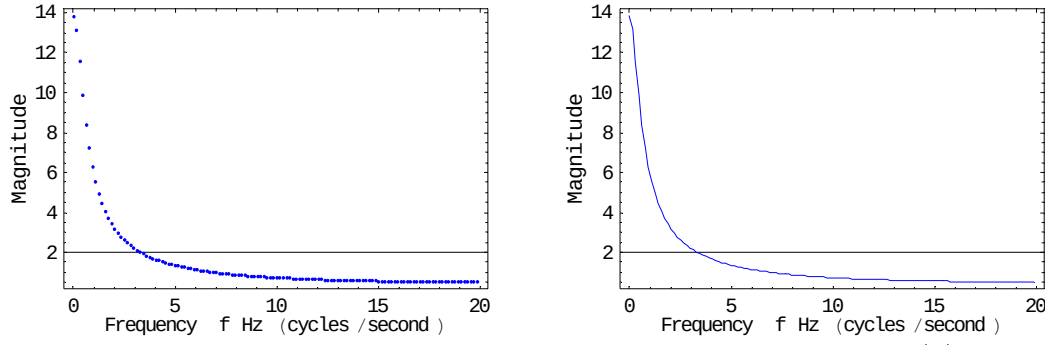
$$g(t) = Ae^{-\alpha|t|}.$$

Set $A=1$ and $\alpha=3$. Following the definition (5.16), $N = 256$ point DFT of a double-sided exponential function with $f_s = 40$ Hz (samples/second) sampling rate is performed on Figure 5.6. Time domain sampling interval Δt (seconds/sample), total sampling time T (seconds), and frequency resolution Δf Hz (cycles/second) are:

$$\begin{aligned}\Delta t &= 1/f_s = 1/40 = 0.025, \\ T &= N\Delta t = 256 \times 0.025 = 6.4, \\ \Delta f &\equiv 1/N\Delta t \equiv 1/T = 1/6.4 = 0.15625.\end{aligned}$$



A) Plot of sampled $g(t) = e^{-3|t|}$ using the sampling rate $f_s = 40$ Hz. Points are joined on the right.



B) Plot of $N = 256$ point DFT frequency spectrum of sampled $g(t) = e^{-3|t|}$ using the sampling rate $f_s = 40$ Hz. Points are joined on the right.

Figure 5.6: Plot of Sampled Double-Sided Exponential Function $g(t) = e^{-3|t|}$ and $N = 256$ Point DFT with Sampling Rate $f_s = 40$ Hz.

[5.3.3] Rectangular Pulse

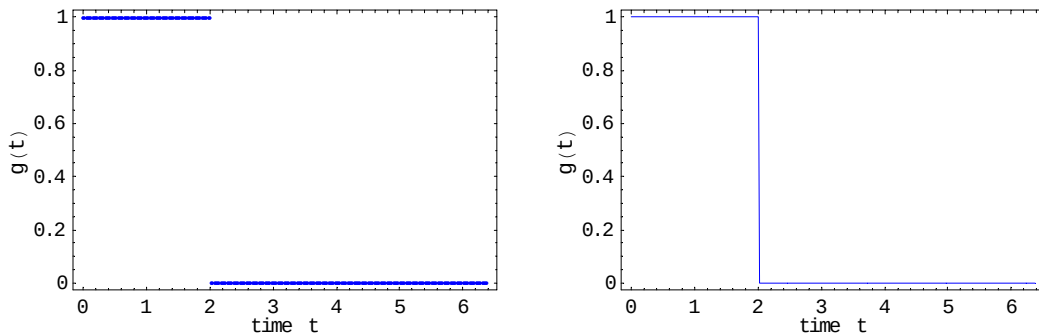
Consider a rectangular pulse with $A, T_0 \in \mathbb{R}$:

$$g(t) = \begin{cases} A & -T_0 \leq t \leq T_0 \\ 0 & |t| > T_0 \end{cases},$$

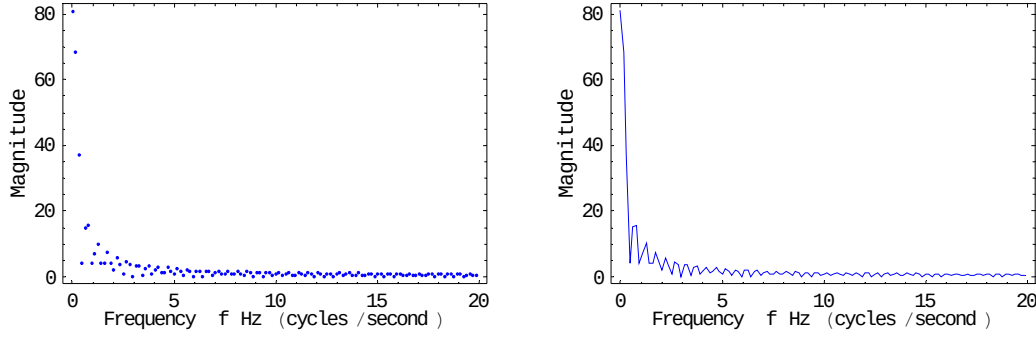
which is an even function of t (symmetric with respect to t).

Set $A=1$ and $T_0 = 2$. Following the definition (5.16), $N = 256$ point DFT of a rectangular pulse with $f_s = 40$ Hz (samples/second) sampling rate is performed on Figure 5.7. Time domain sampling interval Δt (seconds/sample), total sampling time T (seconds), and frequency resolution Δf Hz (cycles/second) are:

$$\begin{aligned} \Delta t &= 1/f_s = 1/40 = 0.025, \\ T &= N\Delta t = 256 \times 0.025 = 6.4, \\ \Delta f &\equiv 1/N\Delta t \equiv 1/T = 1/6.4 = 0.15625. \end{aligned}$$



A) Plot of sampled rectangular pulse using the sampling rate $f_s = 40$ Hz. Points are joined on the right.



B) Plot of $N = 256$ point DFT frequency spectrum of sampled rectangular pulse using the sampling rate $f_s = 40$ Hz. Points are joined on the right.

Figure 5.7: Plot of Sampled Rectangular Pulse and $N = 256$ Point DFT with Sampling Rate $f_s = 40$ Hz.

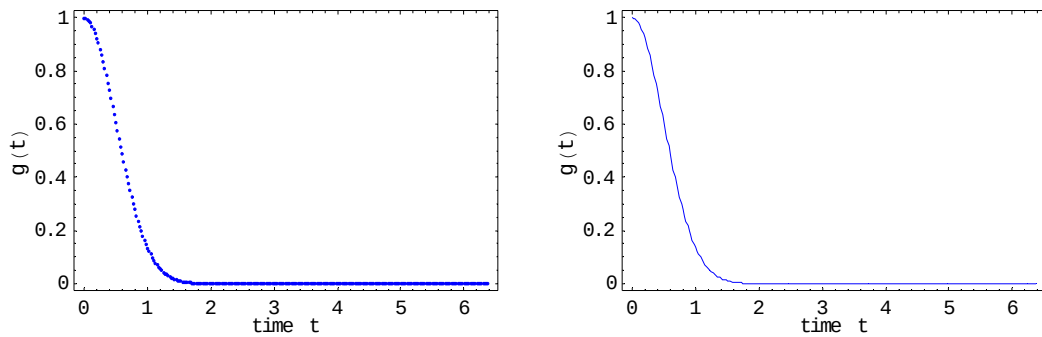
[5.3.4] Gaussian Function

Consider a Gaussian function with $A \in \mathbb{R}^+$:

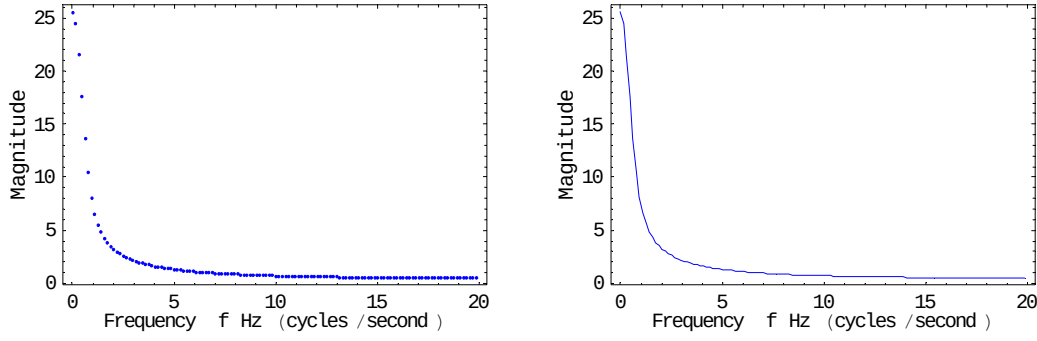
$$g(t) = e^{-At^2}.$$

Set $A = 2$. Following the definition (5.16), $N = 256$ point DFT of a Gaussian function with $f_s = 40$ Hz (samples/second) sampling rate is performed on Figure 5.8. Time domain sampling interval Δt (seconds/sample), total sampling time T (seconds), and frequency resolution Δf Hz (cycles/second) are:

$$\begin{aligned}\Delta t &= 1/f_s = 1/40 = 0.025, \\ T &= N\Delta t = 256 \times 0.025 = 6.4, \\ \Delta f &\equiv 1/N\Delta t \equiv 1/T = 1/6.4 = 0.15625.\end{aligned}$$



A) Plot of sampled $g(t) = e^{-2t^2}$ using the sampling rate $f_s = 40$ Hz. Points are joined on the right.



B) Plot of $N = 256$ point DFT frequency spectrum of sampled $g(t) = e^{-2t^2}$ using the sampling rate $f_s = 40$ Hz. Points are joined on the right.

Figure 5.8: Plot of Sampled Gaussian Function $g(t) = e^{-2t^2}$ and $N = 256$ Point DFT with Sampling Rate $f_s = 40$ Hz.

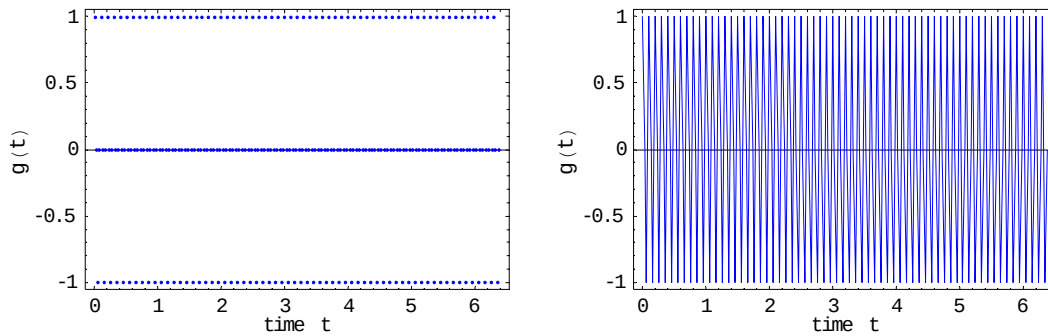
[5.3.5] Cosine Wave $g(t) = A\cos(2\pi f_0 t) = A\cos(\omega_0 t)$

Consider 10 Hz (cycles/second) cosine wave (i.e. fundamental frequency $f_0 = 10$) with amplitude $A = 1$ volt:

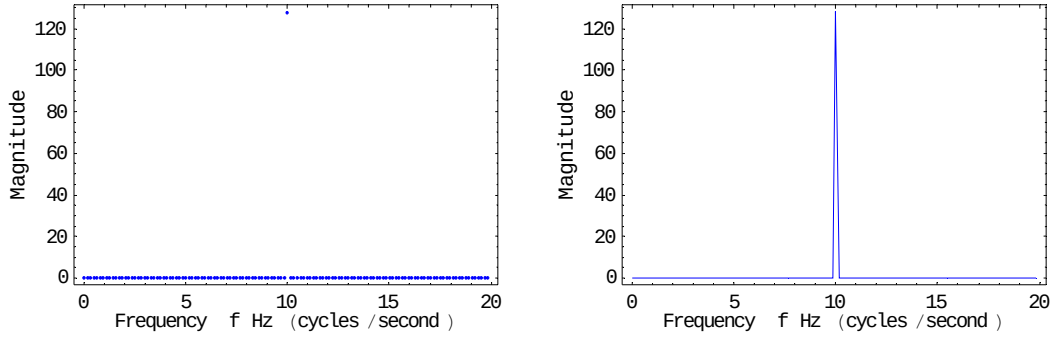
$$g(t) = \cos(2\pi 10t).$$

Following the definition (5.16), $N = 256$ point DFT of a cosine function with $f_s = 40$ Hz (samples/second) sampling rate is performed on Figure 5.9. Time domain sampling interval Δt (seconds/sample), total sampling time T (seconds), and frequency resolution Δf Hz (cycles/second) are:

$$\begin{aligned}\Delta t &= 1/f_s = 1/40 = 0.025, \\ T &= N\Delta t = 256 \times 0.025 = 6.4, \\ \Delta f &\equiv 1/N\Delta t \equiv 1/T = 1/6.4 = 0.15625.\end{aligned}$$



A) Plot of sampled $g(t) = \cos(2\pi 10t)$ using the sampling rate $f_s = 40$ Hz. Points are joined on the right.



B) Plot of $N = 256$ point DFT frequency spectrum of sampled $g(t) = \cos(2\pi 10t)$ using the sampling rate $f_s = 40$ Hz. Points are joined on the right.

Figure 5.9: Plot of Sampled Cosine Wave $g(t) = \cos(2\pi 10t)$ and $N = 256$ Point DFT with Sampling Rate $f_s = 40$ Hz.

[5.4] Properties of DFT

DFT retains all the properties of continuous FT since DFT is a special case of FT.

[5.4.1] Linearity of DFT

Suppose that the time domain sequence $\{f(t_n) : n = 0, 1, \dots, N-1\}$ and $\{g(t_n) : n = 0, 1, \dots, N-1\}$ have discrete Fourier transforms $\{\mathcal{F}(f_k) : k = 0, 1, \dots, N-1\}$ and $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16). Then:

$$a\mathcal{F}(f_k) + b\mathcal{G}(f_k) \equiv \sum_{n=0}^{N-1} (af(t_n) + bg(t_n)) \exp\{-2\pi i kn / N\}. \quad (5.21)$$

PROOF

$$\begin{aligned} & \sum_{n=0}^{N-1} (af(t_n) + bg(t_n)) \exp\{-2\pi i kn / N\} \\ &= a \sum_{n=0}^{N-1} f(t_n) \exp\{-2\pi i kn / N\} + b \sum_{n=0}^{N-1} g(t_n) \exp\{-2\pi i kn / N\} \\ &= a\mathcal{F}(f_k) + b\mathcal{G}(f_k). \end{aligned}$$

[5.4.2] DFT of Even and Odd Functions

Let $even(x)$ be an even function and $odd(x)$ be an odd function. Integral properties of even and odd functions are:

$$\int_{-A}^A \text{odd}(x)dx = 0 ,$$

$$\int_{-A}^A \text{even}(x)dx = 2 \int_0^A \text{even}(x)dx .$$

If the time domain sequence $\{g(t_n) : n = 0, 1, \dots, N-1\}$ is even, i.e. $g(t_n) = g(-t_n)$, then its DFT $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16) is a real-valued even function:

$$\mathcal{G}(f_k) = \sum_{n=0}^{N-1} g(t_n) \cos\left(\frac{2\pi kn}{N}\right). \quad (5.22)$$

PROOF

Start from the DFT definition (5.16):

$$\mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{-2\pi i kn / N\} .$$

Using the Euler's formula (2.16) of $e^{-ix} = \cos(x) - i \sin(x)$:

$$\mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \{\cos(2\pi kn / N) - i \sin(2\pi kn / N)\}$$

$$\mathcal{G}(f_k) = \sum_{n=0}^{N-1} g(t_n) \cos(2\pi kn / N) - i \sum_{n=0}^{N-1} g(t_n) \sin(2\pi kn / N) .$$

Since the product of an even function $g(t_n)$ and an odd function $\sin(2\pi kn / N)$ is odd, from its integral property:

$$\sum_{n=0}^{N-1} g(t_n) \sin(2\pi kn / N) = 0 .$$

Thus:

$$\mathcal{G}(f_k) = \sum_{n=0}^{N-1} g(t_n) \cos(2\pi kn / N) ,$$

which is a real-valued even function since it is a product of two real-valued even functions $g(t_n)$ and $\cos(2\pi kn / N)$.

Next consider an odd case. If the time domain sequence $\{g(t_n) : n = 0, 1, \dots, N-1\}$ is odd, i.e. $g(t_n) = -g(-t_n)$, then its DFT $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16) is a complex-valued odd function:

$$\mathcal{G}(f_k) = -i \sum_{n=0}^{N-1} g(t_n) \sin\left(\frac{2\pi kn}{N}\right). \quad (5.23)$$

PROOF

Start from the DFT definition (5.16):

$$\mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{-2\pi i kn / N\}.$$

Using the Euler's formula (2.16) of $e^{-ix} = \cos(x) - i \sin(x)$:

$$\begin{aligned} \mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) &\equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \{\cos(2\pi kn / N) - i \sin(2\pi kn / N)\} \\ \mathcal{G}(f_k) &= \sum_{n=0}^{N-1} g(t_n) \cos(2\pi kn / N) - i \sum_{n=0}^{N-1} g(t_n) \sin(2\pi kn / N). \end{aligned}$$

Since the product of an odd function $g(t_n)$ and an even function $\cos(2\pi kn / N)$ is odd, from its integral property:

$$\sum_{n=0}^{N-1} g(t_n) \cos(2\pi kn / N) = 0.$$

Thus:

$$\mathcal{G}(f_k) = -i \sum_{n=0}^{N-1} g(t_n) \sin(2\pi kn / N),$$

which is a complex-valued odd function.

[5.4.3] Symmetry of DFT

If the time domain sequence $\{g(t_n) : n = 0, 1, \dots, N-1\}$ has DFT $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16), then the sequence $\left\{\frac{1}{N} \mathcal{G}(t_n) : n = 0, 1, \dots, N-1\right\}$ has DFT

$\{g(-f_k) : k = 0, 1, \dots, N-1\}$. In other words, if $(g(t_n), \mathcal{G}(f_k))$ is a DFT pair, $\left(\frac{1}{N}\mathcal{G}(t_n), g(-f_k)\right)$ is another DFT pair.

PROOF

By the definition of an inverse DFT (5.17):

$$g(t_n \equiv n\Delta t) \equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \exp(2\pi i k n / N).$$

Rewrite it by substituting $-n$ for n :

$$\begin{aligned} g(-t_n \equiv -n\Delta t) &\equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \exp(2\pi i k (-n) / N) \\ g(-t_n) &\equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f_k) \exp(2\pi i k (-n) / N). \end{aligned}$$

Exchange t_n with f_k and n with k , vice versa:

$$g(-f_k) \equiv \frac{1}{N} \sum_{n=0}^{N-1} \mathcal{G}(t_n) \exp(-2\pi i n k / N).$$

From the definition of DFT (5.16), $\left(\frac{1}{N}\mathcal{G}(t_n), g(-f_k)\right)$ is a DFT pair.

[5.4.4] Time Shifting of DFT

Suppose that the time domain sequence $\{g(t_n) : n = 0, 1, \dots, N-1\}$ has DFT $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16). Then, DFT of the time domain sequence $\{g(t_n - t_0\Delta t) : n = 0, 1, \dots, N-1\}$ (i.e. time domain sampling moment t_n is shifted by $t_0\Delta t \in \mathbb{R}$) can be expressed in terms of $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ as:

$$\mathcal{G}(f_k) e^{-2\pi i k t_0 / N} \equiv \sum_{n=0}^{N-1} g(t_n - t_0\Delta t) \exp\{-2\pi i k n / N\}. \quad (5.24)$$

PROOF

By the definition of DFT (5.16):

$$\mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{-i2\pi f_k t_n\} \equiv \sum_{n=0}^{N-1} g(t_n \equiv n\Delta t) \exp\{-2\pi i k n / N\}.$$

By setting $t_n - t_0\Delta t = t^*$:

$$\begin{aligned} \sum_{n=0}^{N-1} g(t_n - t_0\Delta t) \exp\{-i2\pi f_k t_n\} &= \sum_{n=0}^{N-1} g(t^*) \exp\{-i2\pi f_k (t^* + t_0\Delta t)\} \\ &= \exp\{-i2\pi f_k t_0\Delta t\} \sum_{n=0}^{N-1} g(t^*) \exp\{-i2\pi f_k t^*\} \\ &= \exp\{-i2\pi k t_0 / N\} \mathcal{G}(f_k). \end{aligned}$$

□

[5.4.5] Frequency Shifting: Modulation

Suppose that the time domain sequence $\{g(t_n) : n = 0, 1, \dots, N-1\}$ has DFT $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16). If DFT frequency f_k is shifted by $f_0 / N\Delta t \in \mathbb{R}$ Hz (i.e. $\mathcal{G}(f_k - \frac{f_0}{N\Delta t})$), then its inverse DFT $g(t_n)$ is multiplied by $\exp\{2\pi i f_0 t_n / N\}$:

$$e^{2\pi i f_0 t_n / N} g(t_n) = \mathcal{DFT}^{-1} \left[\mathcal{G}(f_k - \frac{f_0}{N\Delta t}) \right], \quad (5.25)$$

$$\mathcal{DFT} \left(e^{2\pi i f_0 t_n / N} g(t_n) \right) = \mathcal{G}(f_k - \frac{f_0}{N\Delta t}). \quad (5.26)$$

PROOF

From the definition of an inverse DFT of (5.17):

$$\begin{aligned} g(t_n \equiv n\Delta t) &\equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \exp(2\pi i k n / N), \\ g(t_n \equiv n\Delta t) &\equiv \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) \exp(i2\pi f_k t_n). \end{aligned}$$

By setting $f_k - \frac{f_0}{N\Delta t} = f^*$:

$$\frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f_k - \frac{f_0}{N\Delta t}) \exp(i2\pi f_k t_n) = \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f^*) \exp\left(i2\pi (f^* + \frac{f_0}{N\Delta t}) t_n\right)$$

$$\begin{aligned}
&= \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f^*) \exp(i2\pi f^* t_n) \exp\left(i2\pi \frac{f_0}{N\Delta t} t_n\right) \\
&= \exp\left(i2\pi \frac{f_0}{N\Delta t} n\Delta t\right) \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}(f^*) \exp(i2\pi f^* t_n) \\
&= \exp(i2\pi n f_0 / N) g(t_n).
\end{aligned}$$

□

[5.4.6] Discrete Convolution: Time Convolution Theorem

Suppose that the time domain sequence $\{f(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ and $\{g(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ have discrete Fourier transforms $\{\mathcal{F}(f_k) : k = 0, 1, \dots, N-1\}$ and $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16). FT and inverse FT defined by the equations (5.16) and (5.17) require both the time domain function $g(t_n \equiv n\Delta t)$ and the frequency domain function $\mathcal{G}(f_k \equiv \frac{k}{N\Delta t})$ to be periodic. This means for $\theta = 0, \pm 1, \pm 2, \dots$:

$$\begin{aligned}
g(t_n \equiv n\Delta t) &= g[(\theta N + n)\Delta t], \\
\mathcal{G}(f_k \equiv \frac{k}{N\Delta t}) &= \mathcal{G}\left[\frac{(\theta N + k)}{N\Delta t}\right],
\end{aligned}$$

where N is the period.

Discrete convolution of time domain sequence $\{f(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ and $\{g(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ for a total of N discrete samples which is denoted as $f(n\Delta t) * g(n\Delta t)$ is defined as:

$$f(n\Delta t) * g(n\Delta t) \equiv \sum_{j=0}^{N-1} f(j\Delta t) h[(n-j)\Delta t]. \quad (5.27)$$

DFT of the discrete convolution of $\{f(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ and $\{g(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ in the time domain is equal to the multiplication in the frequency domain:

$$\mathcal{DFT}(f(n\Delta t) * g(n\Delta t)) \equiv \mathcal{DFT}\left(\sum_{j=0}^{N-1} f(j\Delta t) h[(n-j)\Delta t]\right) = \mathcal{F}(f_k) \mathcal{G}(f_k). \quad (5.28)$$

PROOF

Following the definition of DFT of (5.16):

$$\mathcal{DFT} \left(\sum_{j=0}^{N-1} f(j\Delta t) h[(n-j)\Delta t] \right) = \sum_{n=0}^{N-1} \sum_{j=0}^{N-1} f(j\Delta t) h[(n-j)\Delta t] \exp \{ -2\pi i k n / N \}.$$

Use the time shifting property of DFT of the equation (5.24):

$$\begin{aligned} \mathcal{DFT} \left(\sum_{j=0}^{N-1} f(j\Delta t) h[(n-j)\Delta t] \right) &= \sum_{j=0}^{N-1} f(j\Delta t) \sum_{n=0}^{N-1} h[(n-j)\Delta t] \exp \{ -2\pi i k n / N \} \\ &= \sum_{j=0}^{N-1} f(j\Delta t) \mathcal{G}(f_k) e^{-2\pi i k j / N} = \mathcal{G}(f_k) \sum_{j=0}^{N-1} f(j\Delta t) e^{-2\pi i k j / N} \\ &= \mathcal{F}(f_k) \mathcal{G}(f_k) \end{aligned}$$

□

[5.4.7] Discrete Frequency-Convolution Theorem

Suppose that the time domain sequence $\{f(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ and $\{g(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ have discrete Fourier transforms $\{\mathcal{F}(f_k) : k = 0, 1, \dots, N-1\}$ and $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (6.16). We assume $f(t_n \equiv n\Delta t)$, $g(t_n \equiv n\Delta t)$, $\mathcal{F}(f_k)$, and $\mathcal{G}(f_k)$ are all periodic functions. Discrete convolution of DFTs (frequency convolution) $\{\mathcal{F}(f_k) : k = 0, 1, \dots, N-1\}$ and $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ for a total of N discrete samples which is denoted as $\mathcal{F}(f_k) * \mathcal{G}(f_k)$ is defined as:

$$\mathcal{F}(f_k) * \mathcal{G}(f_k) \equiv \sum_{j=0}^{N-1} \mathcal{F}\left(\frac{j}{N\Delta t}\right) \mathcal{G}\left(\frac{k}{N\Delta t} - \frac{j}{N\Delta t}\right). \quad (5.29)$$

Inverse DFT of the frequency convolution $\mathcal{F}(f_k) * \mathcal{G}(f_k)$ scaled by $1/N$ is equal to the multiplication in the time domain:

$$\mathcal{DFT}^{-1} \left[\frac{1}{N} \mathcal{F}(f_k) * \mathcal{G}(f_k) \right] = f(t_n) g(t_n). \quad (5.30)$$

In other words, DFT of a multiplication in the time domain $f(t_n) g(t_n)$ is equal to a convolution $\mathcal{F}(f_k) * \mathcal{G}(f_k)$ scaled by $1/N$ in the frequency domain:

$$\mathcal{DFT} (f(t_n) g(t_n)) = \frac{1}{N} \mathcal{F}(f_k) * \mathcal{G}(f_k). \quad (5.31)$$

PROOF

There are several different ways to prove the frequency convolution theorem. But we prove this by showing that the inverse DFT of the frequency convolution $\mathcal{F}(f_k) * \mathcal{G}(f_k)$ scaled by $1/N$ is equal to the multiplication in the time domain $f(t_n)g(t_n)$.

Following the definition of an inverse DFT (5.17):

$$\begin{aligned} & \frac{1}{N} \sum_{k=0}^{N-1} \left\{ \frac{1}{N} \mathcal{F}(f_k) * \mathcal{G}(f_k) \right\} \exp(2\pi i k n / N) \\ &= \frac{1}{N} \frac{1}{N} \sum_{k=0}^{N-1} \left\{ \sum_{j=0}^{N-1} \mathcal{F}\left(\frac{j}{N\Delta t}\right) \mathcal{G}\left(\frac{k-j}{N\Delta t}\right) \right\} \exp(2\pi i k n / N) \\ &= \frac{1}{N} \sum_{j=0}^{N-1} \mathcal{F}\left(\frac{j}{N\Delta t}\right) \left\{ \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{G}\left(\frac{k-j}{N\Delta t}\right) \exp(2\pi i k n / N) \right\}. \end{aligned}$$

Using the frequency shifting (modulation) property of DFT of the equation (5.25):

$$\begin{aligned} \frac{1}{N} \sum_{k=0}^{N-1} \left\{ \frac{1}{N} \mathcal{F}(f_k) * \mathcal{G}(f_k) \right\} \exp(2\pi i k n / N) &= \frac{1}{N} \sum_{j=0}^{N-1} \mathcal{F}\left(\frac{j}{N\Delta t}\right) e^{2\pi i j n / N} g(t_n) \\ &= g(t_n) \left[\frac{1}{N} \sum_{j=0}^{N-1} \mathcal{F}\left(\frac{j}{N\Delta t}\right) e^{2\pi i j n / N} \right] \\ &= f(t_n) g(t_n) \end{aligned}$$

□

[5.4.8] Parseval's Relation

Suppose that the time domain sequence $\{f(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ and $\{g(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ have discrete Fourier transforms $\{\mathcal{F}(f_k) : k = 0, 1, \dots, N-1\}$ and $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16). We assume $f(t_n \equiv n\Delta t)$, $g(t_n \equiv n\Delta t)$, $\mathcal{F}(f_k)$, and $\mathcal{G}(f_k)$ are all periodic functions.

Let $\bar{g}(t_n \equiv n\Delta t)$ be a complex conjugate of $f(t_n \equiv n\Delta t)$ and $\bar{\mathcal{G}}(f_k)$ be a complex conjugate of $\mathcal{F}(f_k)$:

$$\begin{aligned} |f(t_n \equiv n\Delta t)|^2 &= f(t_n \equiv n\Delta t) \bar{g}(t_n \equiv n\Delta t), \\ |\mathcal{F}(f_k)|^2 &= \mathcal{F}(f_k) \bar{\mathcal{G}}(f_k). \end{aligned}$$

Parseval's relation states that the power of a signal function $f(t_n \equiv n\Delta t)$ is same whether it is computed in signal (time) space t or transform (frequency) space f after taken care of the weight $1/N$:

$$\begin{aligned}
\sum_{n=0}^{N-1} |f(t_n \equiv n\Delta t)|^2 &= \frac{1}{N} \sum_{k=0}^{N-1} |\mathcal{F}(f_k)|^2, \\
\sum_{n=0}^{N-1} f(t_n \equiv n\Delta t) \bar{g}(t_n \equiv n\Delta t) &= \frac{1}{N} \sum_{k=0}^{N-1} \mathcal{F}(f_k) \bar{\mathcal{G}}(f_k).
\end{aligned} \tag{5.32}$$

[5.4.9] Summary of DFT Properties

Suppose that the time domain sequence $\{f(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ and $\{g(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$ have discrete Fourier transforms $\{\mathcal{F}(f_k) : k = 0, 1, \dots, N-1\}$ and $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$ defined by the equation (5.16). We assume $f(t_n \equiv n\Delta t)$, $g(t_n \equiv n\Delta t)$, $\mathcal{F}(f_k)$, and $\mathcal{G}(f_k)$ are all periodic functions.

Table 5.1: Summary of DFT Properties

Property	$\{g(t_n \equiv n\Delta t) : n = 0, 1, \dots, N-1\}$	DFT $\{\mathcal{G}(f_k) : k = 0, 1, \dots, N-1\}$
Linearity	$af(t_n) + bg(t_n)$	$a\mathcal{F}(f_k) + b\mathcal{G}(f_k)$
Even Function	$g(t_n)$ is even	$\mathcal{G}(f_k) \in \mathbb{R}, \text{ even}$
Odd Function	$g(t_n)$ is odd	$\mathcal{G}(f_k) \in \mathbb{I}, \text{ odd}$
Symmetry	$\frac{1}{N} \mathcal{G}(t_n)$	$g(-f_k)$
Time Shifting	$g(t_n - t_0 \Delta t)$	$\mathcal{G}(f_k) e^{-2\pi i k t_0 / N}$
Convolution	$f(n\Delta t) * g(n\Delta t)$	$\mathcal{F}(f_k) \mathcal{G}(f_k)$
Multiplication	$f(t_n) g(t_n)$	$\frac{1}{N} \mathcal{F}(f_k) * \mathcal{G}(f_k)$
Modulation (Frequency Shifting)	$e^{2\pi i n f_0 / N} g(t_n)$	$\mathcal{G}(f_k - \frac{f_0}{N \Delta t})$

[6] Lévy Processes

In this sequel theorems and propositions are presented without proofs most of the time. This is obviously because we don't need them for our purpose (it is finance) and always inquisitive readers can read Cont and Tankov (2004) and Sato (1999) for more rigorous treatment of the subjects dealt in this section. We recommend for those readers without training in the field of set theory and measure theory to first read Appendix 1 and 2 for the basic knowledge.

[6.1] Definition of Lévy Process

A right-continuous with left limits (cadlag) or an adapted (non-anticipating) stochastic process $\{X_t; 0 \leq t < \infty\}$ on a space $(\Omega, \mathcal{F}, \mathbb{P})$ with values in $\mathbb{R}$ is said to be a Lévy process if it satisfies the following conditions:

- (1) Its increments are independent of the past: $X_u - X_t$ is independent of the filtration $\mathcal{F}_t$ with $0 \leq t < u < \infty$. i.e. $\mathbb{P}(X_u - X_t | \mathcal{F}_t) = \mathbb{P}(X_u - X_t)$.
- (2) Its increments are stationary: $X_{t+h} - X_t$ has the same distribution as X_h . In other words, the distribution of increments does not depend on t (i.e. temporal homogeneity).
- (3) $X_0 = 0$ a.s.
- (4) X_t is continuous in probability: $\forall \varepsilon > 0, \lim_{h \rightarrow 0} \mathbb{P}(|X_{t+h} - X_t| \geq \varepsilon) = 0$.

Processes which satisfy conditions (1), (2), and (3) are said to be processes with stationary independent increments. Condition (4) is satisfied when conditions (1), (2), and (3) are satisfied. Condition (4) does not imply the continuous sample paths. Actually the very opposite is true meaning that most Lévy processes have discontinuous sample paths (i.e. except for one). Condition (4) means that if we are at time t , the probability of a jump at time t is zero because there is no uncertainty about the present. Jumps occur at random times.

From our knowledge, a right-continuous with left limits (cadlag in French) stochastic process, a non-anticipating stochastic process, and an adapted stochastic process define an identical process. A stochastic process $\{X_t; 0 \leq t \leq T\}$ is said to be non-anticipating with respect to the filtration $\{\mathcal{F}_t; 0 \leq t \leq T\}$ or $\mathcal{F}_t$ -adapted if the value of X_t is revealed at time t for each $t \in [0, T]$. In other words, $\{X_t; 0 \leq t \leq T\}$ is said to be non-anticipating if it satisfies for $t \in [0, T]$:

- (1) Left limit of the process $X(t-) \equiv \lim_{s \rightarrow t, s < t} X(s)$ exists.
- (2) Right limit of the process $X(t+) = \lim_{s \rightarrow t, s > t} X(s)$ exists.
- (3) $X(t) = X(t+)$.

Any continuous function is non-anticipating but non-anticipating functions allow discontinuities. Suppose t is a discontinuity point, the jump of X at t is:

$$\Delta X(t) = X(t) - X(t-).$$

A non-anticipating process $\{X_t; 0 \leq t \leq T\}$ can have a finite number of large jumps and countable number (possibly infinite) of small jumps.

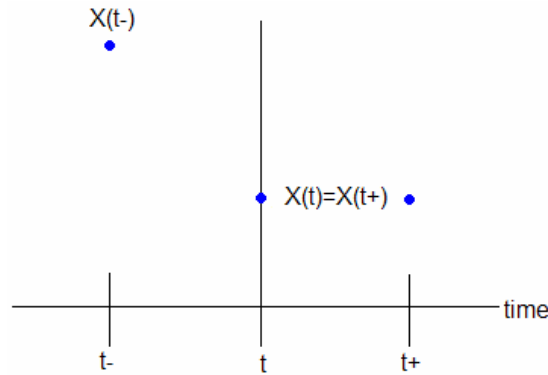


Figure 6.1: Illustration of Non-Anticipating Stochastic Process

Suppose that $X(t)$ is a stock price right now, $X(t-)$ a stock price 1 second ago, and $X(t+)$ a stock price 1 second from now. A stock price process X should be modeled as a non-anticipating process because at time $t-$ we cannot predict $X(t)$ (i.e. it is a future value), but at time $t+$ we already know $X(t)$ (i.e. it is a past value).

We saw the definition of a Lévy process. Next let's discuss infinite divisibility of a distribution. It turns out that we cannot separate Lévy processes from infinitely divisible distributions because Lévy processes are generated by infinitely divisible distributions.

A random variable Y is said to be divisible if it can be represented as the sum of two independent random variables with identical distributions:

$$Y = Y_1 + Y_2.$$

A random variable Y is said to be infinitely divisible if it can be represented as the sum of n independent random variables with identical distributions for any integer $n \geq 2$:

$$Y = Y_1 + Y_2 + \dots + Y_n.$$

Let $\phi(\omega)$ be the characteristic function of the distribution of the infinitely divisible random variable Y and $\phi_n(\omega)$ be the characteristic function of the common distribution of the n summands. Then, the relationship between $\phi(\omega)$ and $\phi_n(\omega)$ is:

$$\phi(\omega) = (\phi_n(\omega))^n \text{ and } \phi_n(\omega) = (\phi(\omega))^{1/n}. \quad (6.1)$$

Examples of infinitely divisible distributions are: the normal distribution, the gamma distribution, α -stable distributions, and the Poisson distribution.

If Y is a normal random variable, i.e. $Y \sim N(\mu, \sigma^2)$, its characteristic function is:

$$\phi(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega Y} \mathbb{P}(Y) dY = \exp(i\mu\omega - \frac{\sigma^2 \omega^2}{2}).$$

Then, the characteristic function for the identically distributed n summands of Y can be computed as using the above relation:

$$\begin{aligned} \phi_n(\omega) &= \{\phi(\omega)\}^{1/n} = \left\{ \exp(i\mu\omega - \frac{\sigma^2 \omega^2}{2}) \right\}^{1/n} \\ &= \exp \left\{ i \left(\frac{\mu}{n} \right) \omega - \frac{(\sigma^2 / n) \omega^2}{2} \right\}. \end{aligned}$$

Thus, the identically distributed n summands of $Y \sim N(\mu, \sigma^2)$ are also normally distributed with the mean μ/n and the variance σ^2/n (because a characteristic function uniquely determines a probability distribution):

$$Y = \sum_{k=0}^{n-1} Y_k, \quad Y_k \sim i.i.d. N(\mu/n, \sigma^2/n).$$

Another example is a Poisson case. If Z is a Poisson random variable (read Appendix 6 for the definition), i.e. $Z \sim \text{Poisson}(\lambda)$, its characteristic function is:

$$\phi(\omega) \equiv \sum_{z=0}^{\infty} \left(\frac{e^{-\lambda} \lambda^z}{z!} \right) e^{i\omega z} = \exp[\lambda(e^{i\omega} - 1)].$$

Its identically distributed n summands follow Poisson law with the parameter λ/n since their characteristic functions take the form:

$$\phi_n(\omega) = \{\phi(\omega)\}^{1/n} = \left[\exp\{\lambda(e^{i\omega} - 1)\} \right]^{1/n} = \exp \left\{ \frac{\lambda}{n} (e^{i\omega} - 1) \right\}.$$

A Lévy process $\{X_t; t \geq 0\}$ possesses this infinite divisibility, i.e. for every t increments of a Lévy process $X_{t+h} - X_t$ has an infinitely divisible law. Conversely, if $\mathbb{P}$ is an infinitely divisible distribution, then there exists a Lévy process $\{X_t; t \geq 0\}$ where the

distribution of increments $X_{t+h} - X_t$ is governed by $\mathbb{P}$. This lemma is extremely important because it means that infinitely divisible distributions (normal, gamma, α -stable, and Poisson distribution) can generate Lévy processes.

$$\text{Lévy process } \{X_t; t \geq 0\} \iff \mathbb{P}(X_{t+h} - X_t) \in \text{Infinitely divisible distributions}$$

[6.2] Standard Brownian Motion Process: The Only Continuous Lévy Process Generated by A Normal Distribution

Some people misinterpret that Lévy processes are discontinuous (i.e. jump) processes. This is not true in a strict sense. It is true to state that most Lévy processes are discontinuous processes. But you can carefully go through the definition of Lévy processes again and you'll notice that no conditions require Lévy processes to possess discontinuous sample paths. A Lévy process can have a continuous sample path. The only example of a continuous Lévy process is a standard Brownian motion process.

A standard Brownian motion process $\{B_t; t \geq 0\}$ is a Lévy process on $\mathbb{R}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that:

$$(1) B_t \sim \text{Normal}(0, t).$$

$$(2) \text{ There is } \Omega_0 \in \mathcal{F} \text{ with } \mathbb{P}(\Omega_0) = 1, \text{ i.e. } X_t(\omega) \text{ is continuous in } t \text{ for every } \omega \in \Omega_0.$$

[6.3] Poisson Process and Compound Poisson Process: Finite Activity Lévy Processes Generated by Poisson Distribution

Appendix 5 gives the in-depth background treatment of the definition and characteristics of an exponential random variable, a Poisson process.

Let $(\tau_i)_{i \geq 1}$ be a sequence of independent exponential random variables with parameter λ and $T_n = \sum_{i=1}^n \tau_i$. A Poisson process with intensity λ is:

$$N_t = \sum_{n \geq 1} 1_{t \geq T_n}.$$

The sample function N_t counts the number of random times (T_n) at which a specified event occurs during the time period between 0 and t where $(T_n - T_{n-1})_{n \geq 1}$ is an *i.i.d.* sequence of exponential variables. Therefore, each possible N_t is represented as a non-decreasing piecewise constant function.

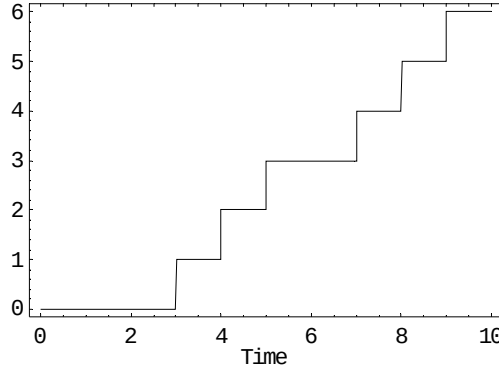


Figure 6.2: Sample Path of a Poisson Process with $\lambda = 1$ and $t = 10$. In this sample, the process has 6 jumps.

A Poisson process (N_t) is a Lévy process because it possesses the following properties:

- (1) Its increments are independent of the past: $N_u - N_t$ is independent of the filtration $\mathcal{F}_t$ with $0 \leq t < u < \infty$. i.e. $\mathbb{P}(N_u - N_t | \mathcal{F}_t) = \mathbb{P}(N_u - N_t)$.
- (2) Its increments are stationary: $N_{t+h} - N_t$ has the same distribution as N_h . In other words, the distribution of increments does not depend on t (i.e. temporal homogeneity).
- (3) $N_0 = 0$ a.s.
- (4) N_t is continuous in probability: $\forall \varepsilon > 0, \lim_{h \rightarrow 0} \mathbb{P}(|N_{t+h} - N_t| \geq \varepsilon) = 0$.
- (5) A sample path of a Poisson process (N_t) is non-anticipating.
- (6) $N_t < \infty$ a.s. for any $t > 0$. A Poisson process has finite number of jumps.

Next, let's take a look at more general version of Poisson process called a compound Poisson process. A compound Poisson process $\{X_t; t \geq 0\}$ with intensity λ is defined as:

$$X_t = \sum_{i=1}^{N_t} Y_i, \quad (6.2)$$

where Y_i are *i.i.d.* jump sizes with the probability density function f . We assume (N_t) and $(Y_i)_{i \geq 1}$ are independent.

Since a compound Poisson process reduces to a Poisson process if $Y_i = 1$ (a Poisson process can be considered as a compound Poisson process with a constant jump size $Y_i = 1$), a compound Poisson process is a Lévy process with properties:

- (1) Its sample paths are piece-wise constant and non-anticipating functions.
- (2) The jump sizes $(Y_i)_{i \geq 1}$ are independent and identically distributed with the probability density function f .

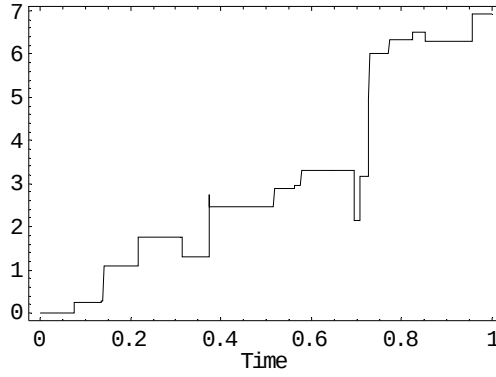


Figure 6.3: Sample Path of a Compound Poisson Process with $\lambda = 20$ and $t = 1$. Jump sizes are drawn from standard normal distribution. In this sample, the process has 19 jumps.

Note that a stochastic process $\{X_t; t \geq 0\}$ is a compound Poisson process, if and only if it is a Lévy process with piecewise constant functions. General Lévy process can be well approximated by a compound Poisson process because any cadlag functions can be approximated by a step function.

The characteristic function of a Poisson process can be obtained by a series expansion:

$$\phi(\omega) \equiv \sum_{z=0}^{\infty} \left\{ \frac{e^{-\lambda t} (\lambda t)^z}{z!} \right\} e^{i\omega z} = \exp \{ \lambda t (e^{i\omega} - 1) \}. \quad (6.3)$$

To obtain the characteristic function of a compound Poisson process $\{X_t; t \geq 0\}$ with intensity λ and the jump size distribution f , we condition the expectation on a Poisson process N_t and let f^* be the characteristic function of f (Cont and Tankov (2004)):

$$\begin{aligned} E[e^{i\omega X_t}] &= E[E[e^{i\omega X_t} | N_t]] = E[f^*(\omega)^{N_t}] = \sum_{n=0}^{\infty} \frac{e^{-\lambda t} (\lambda t)^n f^*(\omega)^n}{n!} \\ &= \exp[\lambda t (f^*(\omega) - 1)] = \exp\{t\lambda \int_{-\infty}^{\infty} (e^{i\omega x} - 1) f(dx)\}, \quad \forall \omega \in \mathbb{R}. \end{aligned} \quad (6.4)$$

We can interpret the characteristic function of a compound Poisson process as a superposition of independent Poisson processes with random jump sizes from distribution f .

Defining a new measure $\ell(dx) = \lambda f(dx)$ which is called a Lévy measure of the Lévy process $\{X_t; 0 \leq t\}$, the above formula can be rewritten as (this is a special case of Lévy-Khinchin representation which will be discussed later):

$$E[e^{i\omega X_t}] = \exp\left\{t \int_{-\infty}^{\infty} (e^{i\omega x} - 1) \ell(dx)\right\}, \quad \forall \omega \in \mathbb{R}.$$

(6.5)

The Lévy measure $\ell(dx)$ represents the arrival rate (i.e. total intensity) of jumps of sizes $[x, x + dx]$. In other words, we can interpret the Lévy measure $\ell(dx)$ of a compound Poisson process as the measure of the average number of jumps per unit of time. There are a couple of extremely important points which should be mentioned about this Lévy measure $\ell(dx)$. Lévy measure is a positive measure on $\mathbb{R}$, but it is not a probability measure since its total mass λ (in the compound Poisson case) does not have to equal 1:

$$\int \ell(dx) = \lambda \in \mathbb{R}^+.$$

Also, Lévy density $\ell(dx)$ of a Lévy process $\{X_t; t \geq 0\}$ is completely different from a probability density of Lévy process $\{X_t; t \geq 0\}$ which will be denoted as $\mathbb{P}(x_t)$ or $vg(x_t)$ in the VG model, for example. We emphasize not to confuse these two densities although they are related (this relation is briefly discussed in Cont and Tankov (2004)).

A Poisson process and a compound Poisson process (i.e. a piecewise constant Lévy process) are called finite activity Lévy processes since their Lévy measures $\ell(dx)$ are finite (i.e. the average number of jumps per unit time is finite):

$$\int_{-\infty}^{\infty} \ell(dx) < \infty.$$

[6.4] Lévy-Itô Decomposition and Infinite Activity Lévy Process

There is a very useful theorem called Lévy-Itô decomposition which basically states that any Lévy process can be represented as the sum of a Brownian motion with drift process (which is a continuous process) and a discontinuous jump process (i.e. compensated (centered) sum of independent jumps). We assume that these two components are independent.

Let $\{X_t; t \geq 0\}$ be a Lévy process on $\mathbb{R}$ with Lévy measure ℓ which is the measure of expected number of jumps per unit of time whose sizes belong to any positive measurable set A . Lévy-Itô decomposition states:

(1) Lévy measure ℓ satisfies:

$$\int_{-\infty}^{\infty} 1(|x| \geq 1) \ell(dx) < \infty \quad \text{and} \quad \int_{\mathbb{R} \setminus \{0\}} x^2 1(|x| < 1) \ell(dx) < \infty.$$

The first condition means that Lévy processes X have finite number of large jumps (large jumps are defined as jumps with absolute values greater than 1).

Second condition states Lévy measure must be square-integrable around the origin.

- (2) There exists a drift b and a Brownian motion process with a diffusion coefficient σ , $(\sigma B_t)_{t \geq 0}$, such that:

$$X_t = bt + \sigma B_t + X_t^L + \lim_{S \downarrow 0} \widetilde{X}_t^S.$$

Following Lévy-Itô decomposition, the distribution of every Lévy process X_t is uniquely determined by its characteristic triplet (Lévy triplet) (b, σ, ℓ) of the process. Condition (2) means that any Lévy process X_t can be decomposed into a continuous (i.e. diffusion) part $bt + \sigma B_t$ and a discontinuous part $X_t^L + \lim_{S \downarrow 0} \widetilde{X}_t^S$. We arbitrarily define large jumps ΔX^L as those with absolute size greater than 1 (i.e. this does not have to be 1):

$$\Delta X^L \equiv |\Delta X| \geq 1,$$

and small jumps as those with absolute size between S and 1:

$$\Delta X^S \equiv S \leq |\Delta X| < 1.$$

X_t^L is the sum of finite number of large jumps during the time interval of $0 \leq r \leq t$:

$$X_t^L = \sum_{0 \leq r \leq t} \Delta X_r^L.$$

X_t^S is the sum of (possibly infinite in the limit $S \downarrow 0$) number of small jumps during the time interval of $0 \leq r \leq t$:

$$X_t^S = \sum_{0 \leq r \leq t} \Delta X_r^S.$$

In the limit case of $S \downarrow 0$, the process can have infinitely many small jumps, therefore X_t^S may not converge. In other words, Lévy measure ℓ can have a singularity at 0 (i.e. infinite arrival rate of small jumps at zero):

$$\int_{-\infty}^{\infty} \ell(dx) = \infty.$$

This type of Lévy process is called an infinite activity Lévy process. Convergence in the condition (2) can be obtained by replacing the jump integral by its compensated version $\widetilde{X}_t^S$. For more rigorous treatment, we recommend Cont and Tankov (2004).

[6.5] Lévy-Khinchin Representation

Lévy-Itô decomposition states that the characteristic function of a Lévy process can be expressed in terms of its characteristic triplet. Let $\{X_t; t \geq 0\}$ be a finite variation Lévy process on $\mathbb{R}$ and (b, σ, ℓ) be its triplet. Then for any $\omega \in \mathbb{R}$, a characteristic function $\phi(\omega)$ and a characteristic exponent $\psi(\omega)$ of a finite variation Lévy process $\{X_t; t \geq 0\}$ can be expressed as:

$$\begin{aligned}\phi(\omega) &\equiv E[\exp(i\omega X_t)] = \exp(t\psi(\omega)), \\ \psi(\omega) &= ib\omega - \frac{\sigma^2 \omega^2}{2} + \int_{-\infty}^{\infty} \{\exp(i\omega x) - 1\} \ell(dx).\end{aligned}\tag{6.6}$$

[6.6] Stable Processes

Consider a real-valued random variable X . Let $\phi(\omega)$ be its characteristic function. X is said to have stable distribution if for any $a > 0$, there exist $b(a) > 0$ and $c(a) \in \mathbb{R}$ such that:

$$\phi_X(\omega)^a = \phi_X(\omega b(a)) \exp(ic(a)), \quad \forall \omega \in \mathbb{R}.\tag{6.7}$$

X is said to have strictly stable distribution if for any $a > 0$, there exist $b(a) > 0$ and $c(a) \in \mathbb{R}$ such that:

$$\phi_X(\omega)^a = \phi_X(\omega b(a)), \quad \forall \omega \in \mathbb{R}.\tag{6.8}$$

At first look these definitions seem complicated, but what they mean is very simple. For example, if Y is a normal random variable, i.e. $Y \sim N(\mu, \sigma^2)$, its characteristic function is $\phi(\omega) = \exp(i\mu\omega - \frac{\sigma^2 \omega^2}{2})$. A normal random variable has stable distribution since it satisfies the equation (6.7) with $b(a) = a^{1/2}$ and $c = (-\sqrt{a} + a)\mu$:

$$\begin{aligned}\phi(\omega)^a &= \{\exp(i\mu\omega - \sigma^2 \omega^2 / 2)\}^a = \exp(i\mu\omega a - \sigma^2 \omega^2 a / 2) \\ \phi(\omega)^a &= \exp(i\mu\omega a - \sigma^2 \omega^2 a / 2) \exp(i\mu\omega\sqrt{a} - i\mu\omega\sqrt{a}) \\ \phi(\omega)^a &= \exp\{i\mu\omega\sqrt{a} - \sigma^2 (\omega\sqrt{a})^2 / 2\} \exp(i\mu\omega a - i\mu\omega\sqrt{a}) \\ \phi(\omega)^a &= \phi_X(\omega\sqrt{a}) \exp\{i(a - \sqrt{a})\mu\omega\}.\end{aligned}$$

A normal random variable with $\mu = 0$ has strictly stable distribution since it satisfies the equation (6.8):

$$\begin{aligned}\phi(\omega)^a &= \{\exp(-\sigma^2 \omega^2 / 2)\}^a = \exp(-\frac{\sigma^2 \omega^2 a}{2}) = \exp\{-\frac{\sigma^2 (\omega \sqrt{a})^2}{2}\} \\ \phi(\omega)^a &= \phi_x(\omega \sqrt{a}).\end{aligned}$$

Let $\{X_t; t \geq 0\}$ be a stochastic process on $\mathbb{R}$. A stochastic process is said to be self-similar if for any $a > 0$, there exists $b > 0$ such that:

$$\{X_{at}; t \geq 0\} \stackrel{d}{=} \{bX_t; t \geq 0\}, \quad (6.9)$$

which we read as that two processes $\{X_{at}; t \geq 0\}$ and $\{bX_t; t \geq 0\}$ are identical in law.

It is said to be broad-sense self-similar if for any $a > 0$, there exists $b > 0$ and a function $c(t)$ such that:

$$\{X_{at}; t \geq 0\} \stackrel{d}{=} \{bX_t + c(t); t \geq 0\}. \quad (6.10)$$

A standard Brownian motion process $\{B_t; t \geq 0\}$ (which is a Lévy process), $B_t \sim \text{Normal}(0, t)$, possesses this self-similarity property since it satisfies the equation (6.9) for $\forall a > 0$:

$$\begin{aligned}\{B_{at}; t \geq 0\} &\sim \text{Normal}(0, at) \\ \{B_{at}; t \geq 0\} &\stackrel{d}{=} \{\sqrt{a}B_t; t \geq 0\}.\end{aligned}$$

A Brownian motion with drift $B_t + \gamma t$ is broad-sense self-similar since it satisfies the equation (6.10) for $\forall a > 0$:

$$\begin{aligned}\{B_{at} + \gamma at; t \geq 0\} &\sim \text{Normal}(\gamma at, at), \\ \{B_{at} + \gamma at; t \geq 0\} &\stackrel{d}{=} \{\sqrt{a}B_t; t \geq 0\} + \gamma at.\end{aligned}$$

Self-similarity means that any change of time scale for the self-similar process has the same effect as some change of spatial scale (also called a scale-invariance property).

Let $\{X_t; t \geq 0\}$ be a Lévy process. A Lévy process is said to be self-similar if for any $a > 0$, there exists $b(a) > 0$ such that:

$$\{X_{at}; t \geq 0\} \stackrel{d}{=} \{b(a)X_t; t \geq 0\}.$$

Following the Lévy-Khinchin representation, the characteristic function of a Lévy process can be expressed as $\phi(\omega) = E[\exp(i\omega X_t)] = \exp(t\psi(\omega))$, $\omega \in \mathbb{R}$. Thus, in terms of

characteristic function a Lévy process is said to be self-similar if for any $a > 0$, there exists $b(a) > 0$ such that:

$$\phi_X(\omega)^a = \phi_X(\omega b(a)),$$

which is the definition of a strictly stable distribution of the equation (6.8). This means that if a Lévy process is self-similar, then it is strictly stable. Formally, a Lévy process $\{X_t; t \geq 0\}$ on $\mathbb{R}$ is self-similar (broad-sense self-similar), if and only if it is strictly stable (stable).

For every stable distribution, there exists a constant $0 < \alpha \leq 2$ such that $b(a) = a^{1/\alpha}$ in the equation (6.7). Let $\phi(\omega)$ be the characteristic function of a stable real-valued random variable X . Then if for any $a > 0$, there exists $0 < \alpha \leq 2$ and $c(a) \in \mathbb{R}$ such that:

$$\phi_X(\omega)^a = \phi_X(\omega a^{1/\alpha}) \exp(ic\omega), \quad \forall \omega \in \mathbb{R}.$$

α is called an index of stability and stable distributions with α are called α -stable distributions. Normal distributions are the only 2-stable distributions.

A strictly α -stable Lévy process (a self-similar Lévy process which is strictly stable with index of stability α) satisfies for any $a > 0$:

$$\{X_{at} : t \geq 0\} \stackrel{d}{=} \{a^{1/\alpha} X_t : t \geq 0\}. \quad (6.11)$$

Brownian motion process is an example of a strictly 2-stable Lévy process.

An α -stable Lévy process satisfies for any $a > 0$:

$$\exists \quad c \in \mathbb{R} \quad \text{such that } \{X_{at} : t \geq 0\} \stackrel{d}{=} \{a^{1/\alpha} X_t + ct : t \geq 0\}.$$

A real-valued random variable is α -stable with $0 < \alpha < 2$ if and only if it is infinitely divisible with characteristic triplet $(b, \sigma = 0, \ell)$ and its Lévy measure is of the form¹:

$$\ell(x) = \frac{A}{x^{\alpha+1}} 1_{x>0} + \frac{B}{|x|^{\alpha+1}} 1_{x<0}, \quad (6.12)$$

where A and B are some positive constants.

¹ For more details, Samorodnitsky, G. and Taqqu, M., 1994, Stable Non-Gaussian Random Processes, Chapman & Hall: New York.

[7] Black-Scholes Model as an Exponential Lévy Model

Even if you were very experienced readers, we recommend you take a look at section 7.6. Other sections may be skipped.

[7.1] Standard Brownian Motion Process: A Lévy Process Generated by a Normal Distribution

As we saw in section 6.2, a standard Brownian motion process $\{B_t; t \geq 0\}$ is a stochastic process on $\mathbb{R}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that:

- (1) $B_{t+u} - B_t \forall u > 0 \sim \text{Normal}(0, u)$.
- (2) There is $\Omega_0 \in \mathcal{F}$ with $\mathbb{P}(\Omega_0) = 1$, i.e. $X_t(\omega)$ is continuous in t for every $\omega \in \Omega_0$.
- (3) $B_{t+u} - B_t \forall u > 0$ are stationary and independent.
- (4) The process $\{B_t; t \geq 0\}$ begins with 0, i.e. $B_0 = 0$.

Stationary increments of the condition (3) mean that the distributions of increments $B_{t+u} - B_t$ do not depend on the time t , but they depend on the time-distance u of two observations (i.e. interval of time). For example, if you model a log stock price $\ln S_t$ as a Brownian motion (with drift) process, the distribution of increment in year 2004 for the next one year $\ln S_{2004+1} - \ln S_{2004}$ is the same as that in year 2050, $\ln S_{2050+1} - \ln S_{2050}$:

$$\ln S_{2004+1} - \ln S_{2004} \stackrel{d}{=} \ln S_{2050+1} - \ln S_{2050}.$$

The conditional probability of the event A given B is assuming $\mathbb{P}(B) > 0$:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

If A and B are independent events:

$$\mathbb{P}(A|B) = \mathbb{P}(A).$$

Independent increments of Brownian motion process of the condition (3) mean that when modeling a log stock price $\ln S_t$ as a Brownian motion (with drift) process, the probability distribution of a log stock price in year 2005 is not affected by whatever happens in year 2004 in the stock price (i.e. such as stock price crash):

$$\mathbb{P}(\ln S_{2005+1} - \ln S_{2005} | \ln S_{2004+1} - \ln S_{2004}) = \mathbb{P}(\ln S_{2005+1} - \ln S_{2005}).$$

These are the two main restrictions imposed by modeling a log stock price process using a Lévy process.

[7.2] Black-Scholes' Distributional Assumptions on a Stock Price

In traditional finance literature almost every financial asset price (stocks, currencies, interest rates) is assumed to follow some variations of Brownian motion with drift process. BS (Black-Scholes) models a stock price increment process in an infinitesimal time interval dt as a log-normal random walk process:

$$dS_t = \mu S_t dt + \sigma S_t dB_t. \quad (7.1)$$

where the drift is μS_t which is a constant expected return on a stock μ proportional to a stock price S_t and the volatility is σS_t which is a constant stock price volatility σ proportional to a stock price S_t . The reason why the process (7.1) is called a log-normal random walk process will be explained very soon. Alternatively, we can state that BS models a percentage change in a stock price process in an infinitesimal time interval dt as a Brownian motion with drift process:

$$\begin{aligned} \frac{dS_t}{S_t} &= \mu dt + \sigma dB_t, \\ \mathbb{P}\left(\frac{dS_t}{S_t}\right) &= \frac{1}{\sqrt{2\pi\sigma^2 dt}} \exp\left[-\frac{(dS_t / S_t - \mu dt)^2}{2\sigma^2 dt}\right]. \end{aligned} \quad (7.2)$$

Let S be a random variable whose dynamics is given by a Ito process:

$$dS = a(S, t)dt + b(S, t)dB,$$

and V be a function dependent on a random variable S and time t . The dynamics of $V(S, t)$ is given by an Ito formula:

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} b^2 \frac{\partial^2 V}{\partial S^2} dt, \quad (7.3)$$

or in terms of a standard Brownian motion process B :

$$\begin{aligned} dV &= \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} (adt + bdB) + \frac{1}{2} b^2 \frac{\partial^2 V}{\partial S^2} dt, \\ dV &= \left(\frac{\partial V}{\partial t} + a \frac{\partial V}{\partial S} + \frac{1}{2} b^2 \frac{\partial^2 V}{\partial S^2} \right) dt + b \frac{\partial V}{\partial S} dB. \end{aligned} \quad (7.4)$$

Dynamics of a log stock price process $\ln S_t$ can be obtained by applying (7.4) to (7.1) as:

$$d \ln S_t = \left(\frac{\partial \ln S_t}{\partial t} + \mu S_t \frac{\partial \ln S_t}{\partial S_t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 \ln S_t}{\partial S_t^2} \right) dt + \sigma S_t \frac{\partial \ln S_t}{\partial S_t} dB_t.$$

Substituting $\frac{\partial \ln S_t}{\partial t} = 0$, $\frac{\partial \ln S_t}{\partial S_t} = \frac{1}{S_t}$, and $\frac{\partial^2 \ln S_t}{\partial S_t^2} = -\frac{1}{S_t^2}$ yields:

$$d \ln S_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dB_t, \quad (7.5)$$

or:

$$\begin{aligned} \ln S_t - \ln S_0 &= \left(\mu - \frac{1}{2} \sigma^2 \right) (t - 0) + \sigma (B_t - B_0) \\ \ln S_t &= \ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t. \end{aligned} \quad (7.6)$$

The equation (7.6) means that BS models a log stock price $\ln S_t$ as a Brownian motion with drift process whose probability density is given by a normal density:

$$\mathbb{P}(\ln S_t) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left[-\frac{\left\{ \ln S_t - \left(\ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t \right) \right\}^2}{2\sigma^2 t}\right]. \quad (7.7)$$

Alternatively, the equation (7.6) means that BS models a log return $\ln(S_t / S_0)$ as a Brownian motion with drift process whose probability density is given by a normal density:

$$\begin{aligned} \ln(S_t / S_0) &= \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t, \\ \mathbb{P}(\ln(S_t / S_0)) &= \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left[-\frac{\left\{ \ln(S_t / S_0) - \left(\mu - \frac{1}{2} \sigma^2 \right) t \right\}^2}{2\sigma^2 t}\right]. \end{aligned} \quad (7.8)$$

An example of BS normal log return $\ln(S_t / S_0)$ density of (7.8) is illustrated in Figure 7.1. Of course, BS log return density is symmetric (i.e. zero skewness) and have zero excess kurtosis because it is a normal density.

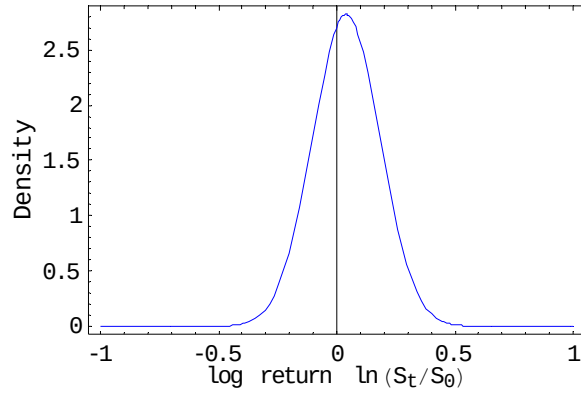


Figure 7.1: An Example of BS normal log return $\ln(S_t/S_0)$ Density. Parameters and variables fixed are $\mu = 0.1$, $\sigma = 0.2$, and $t = 0.5$.

Let y be a random variable. If the log of y is normally distributed with mean a and variance b^2 such that $\ln y \sim N(a, b^2)$, then y is a log-normal random variable whose density is a two parameter family (a, b) :

$$y \sim \text{Lognormal}(e^{a+\frac{1}{2}b^2}, e^{2a+b^2}(e^{b^2}-1)),$$

$$\mathbb{P}(y) = \frac{1}{y\sqrt{2\pi b^2}} \exp\left[-\frac{\{\ln y - a\}^2}{2b^2}\right].$$

From the equation (7.6), we can state that BS models a stock price S_t as a log-normally distributed random variable whose density is given by:

$$\mathbb{P}(S_t) = \frac{1}{S_t \sqrt{2\pi\sigma^2 t}} \exp\left[-\frac{\left\{\ln S_t - \left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right)t\right)\right\}^2}{2\sigma^2 t}\right]. \quad (7.9)$$

Its annualized moments are calculated as:

$$\begin{aligned} \text{Mean}[S_t] &= S_0 e^{\mu}, \\ \text{Variance}[S_t] &= S_0^2 (e^{\sigma^2} - 1) e^{2\mu}, \\ \text{Skewness}[S_t] &= (e^{\sigma^2} + 2) \sqrt{e^{\sigma^2} - 1}, \\ \text{Excess Kurtosis}[S_t] &= -6 + 3e^{2\sigma^2} + 2e^{3\sigma^2} + e^{4\sigma^2}. \end{aligned}$$

An example of BS log-normal stock price density of (7.9) is illustrated in Figure 7.2. Notice that BS log-normal stock price density is positively skewed.

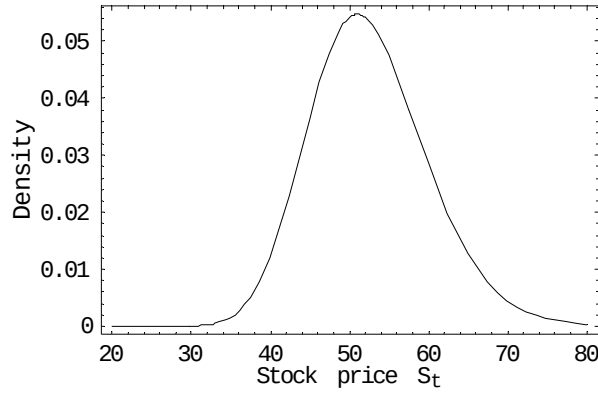


Figure 7.2: An Example of BS Log-Normal Density of a Stock Price. Parameters and variables fixed are $S_0 = 50$, $\mu = 0.1$, $\sigma = 0.2$, and $t = 0.5$.

Table 7.1
Annualized Moments of BS Log-Normal Density of A Stock Price in Figure 2.2

Mean	Standard Deviation	Skewness	Excess Kurtosis
55.2585	11.1632	0.614295	0.678366

From the equation (7.6), we can obtain an integral version equivalent of (7.1):

$$\begin{aligned}
 \exp[\ln S_t] &= \exp[\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t] \\
 S_t &= \exp[\ln S_0] \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t\right] \\
 S_t &= S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t\right].
 \end{aligned} \tag{7.10}$$

Equation (7.10) means that BS models a stock price process S_t as a geometric process with the growth rate given by a Brownian motion with drift process:

$$\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t.$$

[7.3] Traditional Black-Scholes Option Pricing: PDE Approach by Hedging

Consider a portfolio P of the one long option position $V(S, t)$ on the underlying stock S written at time t and a short position of the underlying stock in quantity Δ to derive option pricing function.

$$P_t = V(S_t, t) - \Delta S_t. \tag{7.11}$$

Portfolio value changes in a very short period of time dt by:

$$dP_t = dV(S_t, t) - \Delta dS_t. \quad (7.12)$$

Stock price dynamics is given by a log-normal random walk process of the equation (7.1):

$$dS_t = \mu S_t dt + \sigma S_t dB_t. \quad (7.13)$$

Option price dynamics is given by applying Ito formula of the equation (7.3):

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_t} dS_t + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} dt. \quad (7.14)$$

Now the change in the portfolio value can be expressed as by substituting (7.13) and (7.14) into (7.12):

$$dP_t = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_t} dS_t + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} dt - \Delta dS_t. \quad (7.15)$$

Setting $\Delta = \partial V / \partial S_t$ (i.e. delta hedging) makes the portfolio completely risk-free (i.e. the randomness dS_t has been eliminated) and the portfolio value dynamics of the equation (7.15) simplifies to:

$$dP_t = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} \right) dt. \quad (7.16)$$

Since this portfolio is perfectly risk-free, assuming the absence of arbitrage opportunities the portfolio is expected to grow at the risk-free interest rate r :

$$E[dP_t] = r P_t dt. \quad (7.17)$$

After substitution of (7.11) and (7.16) into (7.17) by setting $\Delta = \partial V / \partial S_t$, we obtain:

$$\left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} \right) dt = r \left(V - \frac{\partial V}{\partial S_t} S_t \right) dt.$$

After rearrangement, Black-Scholes PDE is obtained:

$$\frac{\partial V(S_t, t)}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V(S_t, t)}{\partial S_t^2} + r S_t \frac{\partial V(S_t, t)}{\partial S_t} - r V(S_t, t) = 0. \quad (7.18)$$

BS PDE is categorized as a linear second-order parabolic PDE. The equation (7.18) is a linear PDE because coefficients of the partial derivatives of $V(S_t, t)$ (i.e. $\sigma^2 S_t^2 / 2$ and $r S_t$) are not functions of $V(S_t, t)$ itself. The equation (7.18) is a second-order PDE because it involves the second-order partial derivative $\partial^2 V(S_t, t) / \partial S_t^2$. Generally speaking, a PDE of the form:

$$a + b \frac{\partial V}{\partial t} + c \frac{\partial V}{\partial S} + d \frac{\partial^2 V}{\partial S^2} + e \frac{\partial^2 V}{\partial t^2} + g \frac{\partial^2 V}{\partial t \partial S} = 0,$$

is said to be a parabolic type if:

$$g - 4de = 0. \quad (7.19)$$

The equation (7.18) is a parabolic PDE because it has $g = 0$ and $e = 0$ which satisfies the condition (7.19).

BS solves PDE of (7.18) with boundary conditions:

$$\begin{aligned} &\max(S_T - K, 0) \text{ for a plain vanilla call,} \\ &\max(K - S_T, 0) \text{ for a plain vanilla put,} \end{aligned}$$

and obtains closed-form solutions of call and put pricing functions. Exact derivation of closed-form solutions by solving BS PDE is omitted here (i.e. the original BS approach). Instead we will provide the exact derivation by a martingale asset pricing approach (this is much simpler) in the next section.

[7.4] Traditional Black-Scholes Option Pricing: Martingale Pricing Approach

Let $\{B_t; 0 \leq t \leq T\}$ be a standard Brownian motion process on a space $(\Omega, \mathcal{F}, \mathbb{P})$. Under actual probability measure $\mathbb{P}$, the dynamics of BS stock price process is given by equation (7.9) in the integral form (i.e. which is a geometric Brownian motion process):

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right] \text{ under } \mathbb{P}. \quad (7.20)$$

BS model is an example of a complete model because there is only one equivalent martingale risk-neutral measure $\mathbb{Q} \sim \mathbb{P}$ under which the discounted asset price process $\{e^{-rt} S_t; 0 \leq t \leq T\}$ becomes a martingale. BS finds the equivalent martingale risk-neutral

measure $\mathbb{Q}_{BS} \sim \mathbb{P}$ by changing the drift of the Brownian motion process while keeping the volatility parameter σ unchanged:

$$S_t = S_0 \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma B_t^{\mathbb{Q}_{BS}} \right] \text{ under } \mathbb{Q}_{BS}. \quad (7.21)$$

Note that $B_t^{\mathbb{Q}_{BS}}$ is a standard Brownian motion process on $(\Omega, \mathcal{F}, \mathbb{Q}_{BS})$ and the process $\{e^{-rt} S_t; 0 \leq t \leq T\}$ is a martingale under $\mathbb{Q}_{BS}$. Then, a plain vanilla call option price $C(t, S_t)$ which has a terminal payoff function $\max(S_T - K, 0)$ is calculated as:

$$C(t, S_t) = e^{-r(T-t)} E_t^{\mathbb{Q}_{BS}} [\max(S_T - K, 0)] = e^{-rt} E_t^{\mathbb{Q}_{BS}} [\max(S_T - K, 0)]. \quad (7.22)$$

Note that an expectation operator $E_t^{\mathbb{Q}_{BS}} [\]$ is under a probability measure $\mathbb{Q}_{BS}$ and conditional on time t . Let $\mathbb{Q}(S_T)$ (drop the subscript BS for simplicity) be a probability density function of S_T in a risk-neutral world. From the equation (7.9), a terminal stock price S_T is a log-normal random variable with its density of the form:

$$\mathbb{Q}(S_T) = \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + \left(r - \frac{1}{2} \sigma^2 \right) \tau \right) \right\}^2}{2\sigma^2\tau} \right]. \quad (7.23)$$

Using (7.23), the expectation term in (7.22) can be rewritten as:

$$\begin{aligned} E_t^{\mathbb{Q}} [\max(S_T - K, 0)] &= \int_K^\infty (S_T - K) \mathbb{Q}(S_T) dS_T + \int_0^K (0) \mathbb{Q}(S_T) dS_T \\ E_t^{\mathbb{Q}} [\max(S_T - K, 0)] &= \int_K^\infty (S_T - K) \mathbb{Q}(S_T) dS_T. \end{aligned}$$

Using this, we can rewrite (7.22) as:

$$C(t, S_t) = e^{-rt} \int_K^\infty (S_T - K) \mathbb{Q}(S_T) dS_T. \quad (7.24)$$

Since S_T is a log-normal random variable with its density given by the equation (7.23):

$$\ln S_T \sim \text{Normal} \left(m \equiv \ln S_t + \left(r - \frac{1}{2} \sigma^2 \right) \tau, \sigma^2 \tau \right). \quad (7.25)$$

We use a change of variable technique from a log-normal random variable S_T to a standard normal random variable Z through:

$$Z \equiv \frac{\ln S_T - \left\{ \ln S_t + \left(r - \frac{1}{2}\sigma^2\right)\tau \right\}}{\sigma\sqrt{\tau}} \equiv \frac{\ln S_T - m}{\sigma\sqrt{\tau}} \sim \text{Normal}(0,1), \quad (7.26)$$

with:

$$\mathbb{Z}(Z) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{Z^2}{2}\right].$$

From (7.26):

$$S_T = \exp\left(Z\sigma\sqrt{\tau} + m\right). \quad (7.27)$$

We can rewrite (7.24) as:

$$C(\tau, S_t) = e^{-r\tau} \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \left(\exp\left(Z\sigma\sqrt{\tau} + m\right) - K \right) \mathbb{Z}(Z) dZ,$$

and we express this with more compact form as:

$$C(\tau, S_t) = C_1 - C_2, \quad (7.28)$$

where $C_1 = e^{-r\tau} \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \exp\left(Z\sigma\sqrt{\tau} + m\right) \mathbb{Z}(Z) dZ$ and $C_2 = Ke^{-r\tau} \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \mathbb{Z}(Z) dZ$.

Consider C_1 :

$$\begin{aligned} C_1 &= e^{-r\tau} \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \exp\left(Z\sigma\sqrt{\tau}\right) \exp(m) \mathbb{Z}(Z) dZ \\ C_1 &= \exp(-r\tau) \exp\left(\ln S_t + \left(r - \frac{1}{2}\sigma^2\right)\tau\right) \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \exp\left(Z\sigma\sqrt{\tau}\right) \mathbb{Z}(Z) dZ \\ C_1 &= \exp\left(\ln S_t - \frac{1}{2}\sigma^2\tau\right) \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \exp\left(Z\sigma\sqrt{\tau}\right) \mathbb{Z}(Z) dZ \\ C_1 &= \exp\left(\ln S_t - \frac{1}{2}\sigma^2\tau\right) \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \exp\left(Z\sigma\sqrt{\tau}\right) \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{Z^2}{2}\right] dZ \\ C_1 &= \exp\left(\ln S_t - \frac{1}{2}\sigma^2\tau\right) \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{Z^2 - 2Z\sigma\sqrt{\tau}}{2}\right] dZ \\ C_1 &= \exp\left(\ln S_t - \frac{1}{2}\sigma^2\tau\right) \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z - \sigma\sqrt{\tau})^2 - \sigma^2\tau}{2}\right] dZ \end{aligned}$$

$$\begin{aligned}
C_1 &= \exp\left(\ln S_t - \frac{1}{2}\sigma^2\tau\right) \exp\left(\frac{1}{2}\sigma^2\tau\right) \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z - \sigma\sqrt{\tau})^2}{2}\right] dZ \\
C_1 &= \exp(\ln S_t) \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z - \sigma\sqrt{\tau})^2}{2}\right] dZ \\
C_1 &= S_t \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z - \sigma\sqrt{\tau})^2}{2}\right] dZ .
\end{aligned} \tag{7.29}$$

Use the following relationship:

$$\int_a^b \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z - c)^2}{2}\right] dZ = \int_{a-c}^{b-c} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{Z^2}{2}\right] dZ .$$

Equation (7.29) can be rewritten as:

$$C_1 = S_t \int_{(\ln K - m)/\sigma\sqrt{\tau} - \sigma\sqrt{\tau}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{Z^2}{2}\right] dZ . \tag{7.30}$$

Let $N(\cdot)$ be the standard normal cumulative density function. Using the symmetry of a normal density, (7.30) can be rewritten as:

$$\begin{aligned}
C_1 &= S_t \int_{-\infty}^{-(\ln K - m)/\sigma\sqrt{\tau} + \sigma\sqrt{\tau}} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{Z^2}{2}\right] dZ \\
C_1 &= S_t N\left(\frac{-\ln K + m}{\sigma\sqrt{\tau}} + \sigma\sqrt{\tau}\right) .
\end{aligned} \tag{7.31}$$

From (7.25), substitute for m . The equation (7.31) becomes:

$$\begin{aligned}
C_1 &= S_t N\left(\frac{-\ln K + \ln S_t + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} + \sigma\sqrt{\tau}\right) \\
C_1 &= S_t N\left(\frac{\ln\left(\frac{S_t}{K}\right) + (r - \frac{1}{2}\sigma^2)\tau + \sigma^2\tau}{\sigma\sqrt{\tau}}\right) = S_t N\left(\frac{\ln\left(\frac{S_t}{K}\right) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}\right)
\end{aligned} \tag{7.32}$$

Next, consider C_2 in (7.28):

$$\begin{aligned}
C_2 &= Ke^{-r\tau} \int_{(\ln K - m)/\sigma\sqrt{\tau}}^{\infty} \mathbb{Z}(Z) dZ = Ke^{-r\tau} \int_{-\infty}^{-(\ln K - m)/\sigma\sqrt{\tau}} \mathbb{Z}(Z) dZ \\
C_2 &= Ke^{-r\tau} N\left(\frac{-\ln K + m}{\sigma\sqrt{\tau}}\right).
\end{aligned} \tag{7.33}$$

From (7.25), substitute for m . The equation (7.33) becomes:

$$C_2 = Ke^{-r\tau} N\left(\frac{-\ln K + \ln S_t + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}\right) = Ke^{-r\tau} N\left(\frac{\ln \frac{S_t}{K} + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}\right). \tag{7.34}$$

Substitute (7.32) and (7.34) into (7.28) and we obtain BS plain vanilla call option pricing formula:

$$C(\tau, S_t) = S_t N(d_1) - Ke^{-r\tau} N(d_2), \tag{7.35}$$

$$\text{where } d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} \text{ and } d_2 = \frac{\ln \frac{S_t}{K} + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} = d_1 - \sigma\sqrt{\tau}.$$

Following the similar method, BS plain vanilla put option pricing formula can be obtained as:

$$P(\tau, S_t) = Ke^{-r\tau} N(-d_2) - S_t N(-d_1). \tag{7.36}$$

We can conclude that both PDE approach and martingale approach give the same result. This is because in both approaches we move from a historical probability measure $\mathbb{P}$ to a risk-neutral probability measure $\mathbb{Q}$. This is very obvious for martingale method. But in PDE approach because the source of randomness can be completely eliminated by forming a portfolio of options and underlying stocks, this portfolio grows at a rate equal to the risk-free interest rate. Thus, we switch to a measure $\mathbb{Q}$. For more details, we recommend Neftci (2000) pages 280-282 and 358-365.

[7.5] Alternative Interpretation of Black-Scholes Formula: A Single Integration Problem

Under an equivalent martingale measure $\mathbb{Q} \sim \mathbb{P}$ under which the discounted asset price process $\{e^{-rt}S_t; 0 \leq t \leq T\}$ becomes a martingale, a plain vanilla call and put option price which has a terminal payoff function $\max(S_T - K, 0)$ and $\max(K - S_T, 0)$ are calculated as:

$$C(t, S_t) = e^{-r(T-t)} E_t^{\mathbb{Q}} \left[\max(S_T - K, 0) \right] = e^{-r\tau} E_t^{\mathbb{Q}} \left[\max(S_T - K, 0) \right], \quad (7.37)$$

$$P(t, S_t) = e^{-r(T-t)} E_t^{\mathbb{Q}} \left[\max(K - S_T, 0) \right] = e^{-r\tau} E_t^{\mathbb{Q}} \left[\max(K - S_T, 0) \right]. \quad (7.38)$$

Note that an expectation operator $E_t^{\mathbb{Q}}[\]$ is under a probability measure $\mathbb{Q}$ and conditional on time t . Let $\mathbb{Q}(S_T)$ (drop the subscript BS for simplicity) be a probability density function of S_T under $\mathbb{Q}$. Using $\mathbb{Q}(S_T)$, (7.37) and (7.38) can be rewritten as:

$$\begin{aligned} E_t^{\mathbb{Q}} \left[\max(S_T - K, 0) \right] &= \int_K^{\infty} (S_T - K) \mathbb{Q}(S_T) dS_T + \int_0^K (0) \mathbb{Q}(S_T) dS_T \\ &= \int_K^{\infty} (S_T - K) \mathbb{Q}(S_T) dS_T, \end{aligned}$$

$$\begin{aligned} E_t^{\mathbb{Q}} \left[\max(K - S_T, 0) \right] &= \int_K^{\infty} (0) \mathbb{Q}(S_T) dS_T + \int_0^K (K - S_T) \mathbb{Q}(S_T) dS_T \\ &= \int_0^K (K - S_T) \mathbb{Q}(S_T) dS_T. \end{aligned}$$

Using these, we can rewrite (7.37) and (7.38) as:

$$C(\tau, S_t) = e^{-r\tau} \int_K^{\infty} (S_T - K) \mathbb{Q}(S_T) dS_T, \quad (7.39)$$

$$P(\tau, S_t) = e^{-r\tau} \int_0^K (K - S_T) \mathbb{Q}(S_T) dS_T. \quad (7.40)$$

BS assumes that a terminal stock price S_T is a log-normal random variable with its density of the form:

$$\mathbb{Q}(S_T) = \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right) \right\}^2}{2\sigma^2\tau} \right]. \quad (7.40)$$

Therefore, BS option pricing formula comes down to a very simple single integration problem:

$$C(\tau, S_t) = e^{-r\tau} \int_K^{\infty} (S_T - K) \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right) \right\}^2}{2\sigma^2\tau} \right] dS_T, \quad (7.41)$$

$$P(\tau, S_t) = e^{-r\tau} \int_0^K (K - S_T) \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right) \right\}^2}{2\sigma^2\tau} \right] dS_T. \quad (7.42)$$

This implies that as far as a risk-neutral density of the terminal stock price $\mathbb{Q}(S_T)$ is known, plain vanilla option pricing reduces to a simple integration problem.

[7.6] Black-Scholes Model as an Exponential Lévy Model

The equation (7.10) tells us that BS models a stock price process as a geometric Brownian motion process:

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right].$$

BS model is an exponential Lévy model of the form:

$$S_t = S_0 e^{L_t},$$

where the stock price process $\{S_t : 0 \leq t \leq T\}$ is modeled as an exponential of a Lévy process $\{L_t; 0 \leq t \leq T\}$. Black and Scholes' choice of the Lévy process is a Brownian motion with drift (continuous diffusion process):

$$L_t \equiv \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t. \quad (7.43)$$

The fact that an stock price S_t is modeled as an exponential of Lévy process L_t means that its log-return $\ln(\frac{S_t}{S_0})$ is modeled as a Lévy process such that:

$$\ln\left(\frac{S_t}{S_0}\right) = L_t = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t.$$

BS model can be categorized as a continuous exponential Lévy model apparently because a Brownian motion process is continuous (i.e. no jumps). Later, we will deal with Merton jump-diffusion model (we call it Merton JD model) and variance gamma model by Madan, Carr, and Chang (1998) (we call it VG model). These are all exponential Lévy models of different types. Merton's choice of Lévy process X_t is a Brownian motion with drift process plus a compound Poisson jump process which has a path of continuous path with occasional jumps. Merton JD model can be categorized as a finite activity exponential Lévy model because the expected number of jumps per unit of time (i.e. intensity λ) is finite and small. In other words, the Lévy measure $\ell(dx)$ of Merton JD model is finite:

$$\int \ell(dx) < \infty.$$

VG model can be categorized as an infinite activity exponential Lévy model because the expected number of small jumps per unit of time is infinite:

$$\int \ell(dx) = \infty ,$$

although the expected number of large jumps per unit of time is finite. VG model's Lévy process X_t is a subordinated Brownian motion process by a tempered α -stable subordinator (i.e. a normal tempered α -stable process) which is a pure jump process.

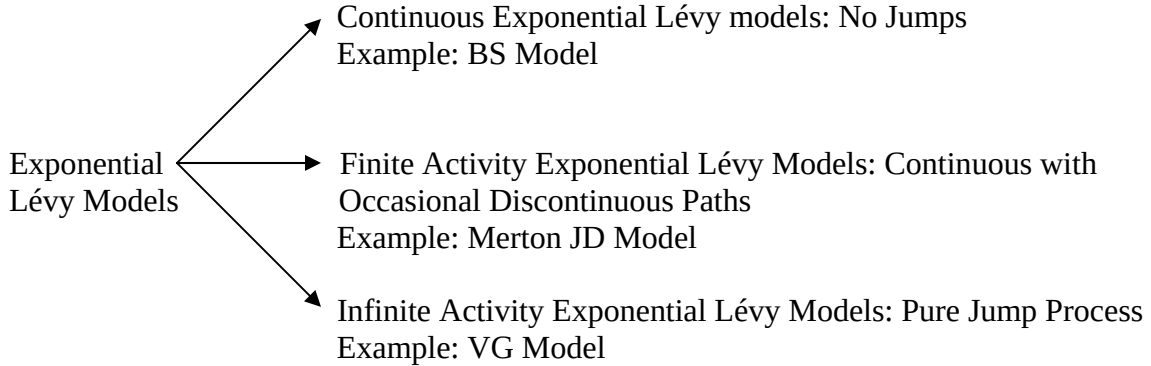


Figure 7.3: Category of Exponential Lévy Models

Recently, there are enormous publications regarding infinite activity exponential Lévy models which we intend to look into in near future.

[8] Option Pricing with Fourier Transform: Black-Scholes Example

In this section we present Fourier transform option pricing approach by Carr and Madan (1999). To illustrate the idea clearly and simply, we present BS model with Fourier transform pricing methodology.

[8.1] Motivation

Let $\mathbb{Q} \sim \mathbb{P}$ be an equivalent martingale measure under which the discounted asset price process $\{e^{-rt}S_t; 0 \leq t \leq T\}$ becomes a martingale and $\{\mathcal{F}_t; 0 \leq t \leq T\}$ be an information flow of the asset price S (i.e. filtration). In an arbitrage-free market, prices of any assets can be calculated as expected terminal payoffs under $\mathbb{Q}$ discounted by a risk-free interest rate r :

$$\begin{aligned} e^{-rt}S_t &= E^{\mathbb{Q}}[e^{-rT}S_T | \mathcal{F}_t], \\ S_t &= e^{-r(T-t)}E^{\mathbb{Q}}[S_T | \mathcal{F}_t], \end{aligned}$$

which are martingale conditions.

Let K be a strike price and T be an expiration of a contingent claim. Plain vanilla call and put option prices are computed as discounted risk-neutral conditional expectations of the terminal payoffs $(S_T - K)^+ \equiv \max(S_T - K, 0)$ and $(K - S_T)^+ \equiv \max(K - S_T, 0)$:

$$C(t, S_t) = e^{-r(T-t)}E^{\mathbb{Q}}[(S_T - K)^+ | \mathcal{F}_t], \quad (8.1)$$

$$P(t, S_t) = e^{-r(T-t)}E^{\mathbb{Q}}[(K - S_T)^+ | \mathcal{F}_t]. \quad (8.2)$$

Intrinsic value of a vanilla call (put) is defined as $(S_t - K)^+$ ($(K - S_t)^+$) which is the value of the call (put) exercised immediately. Obviously, the intrinsic value of out of the money option is zero. Current option price minus its intrinsic value $C(t, S_t) - (S_t - K)^+$ ($P(t, S_t) - (K - S_t)^+$) is called a time value of the option.

Let $\mathbb{Q}(S_T | \mathcal{F}_t)$ be a probability density function of a terminal asset price S_T under $\mathbb{Q}$ conditional on $\mathcal{F}_t$. Using $\mathbb{Q}(S_T | \mathcal{F}_t)$, (8.1) and (8.2) can be rewritten as:

$$\begin{aligned} C(t, S_t) &= e^{-r(T-t)} \left\{ \int_K^\infty (S_T - K) \mathbb{Q}(S_T | \mathcal{F}_t) dS_T + \int_0^K (0) \mathbb{Q}(S_T | \mathcal{F}_t) dS_T \right\} \\ &= e^{-r(T-t)} \int_K^\infty (S_T - K) \mathbb{Q}(S_T | \mathcal{F}_t) dS_T, \end{aligned} \quad (8.3)$$

$$P(t, S_t) = e^{-r(T-t)} \left\{ \int_K^\infty (0) \mathbb{Q}(S_T | \mathcal{F}_t) dS_T + \int_0^K (K - S_T) \mathbb{Q}(S_T | \mathcal{F}_t) dS_T \right\}$$

$$= e^{-r(T-t)} \int_0^K (K - S_T) \mathbb{Q}(S_T | \mathcal{F}_t) dS_T. \quad (8.4)$$

Black-Scholes (BS) assumes that a terminal stock price S_T conditional on $\mathcal{F}_t$ is a log-normal random variable with its density given by:

$$\mathbb{Q}(S_T | \mathcal{F}_t) = \frac{1}{S_T \sqrt{2\pi\sigma^2(T-t)}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + (r - \frac{1}{2}\sigma^2)(T-t) \right) \right\}^2}{2\sigma^2(T-t)} \right].$$

Therefore, BS option pricing formula comes down to single integration problem with respect to S_T since all parameters and variables are known:

$$C_{BS}(t, S_t) = e^{-r\tau} \int_K^\infty (S_T - K) \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + (r - \frac{1}{2}\sigma^2)\tau \right) \right\}^2}{2\sigma^2\tau} \right] dS_T, \quad (8.5)$$

$$P_{BS}(t, S_t) = e^{-r\tau} \int_0^K (K - S_T) \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + (r - \frac{1}{2}\sigma^2)\tau \right) \right\}^2}{2\sigma^2\tau} \right] dS_T. \quad (8.6)$$

This implies that as far as a conditional risk-neutral density of the terminal stock price $\mathbb{Q}(S_T | \mathcal{F}_t)$ is given, plain vanilla option pricing reduces to single integration problem by the equation (8.3) and (8.4).

But for general exponential Lévy models $\mathbb{Q}(S_T | \mathcal{F}_t)$ cannot be expressed using special functions of mathematics or is not known. Therefore, we cannot price plain vanilla options using (8.3) and (8.4). So how do we price options using general exponential Lévy models? The answer is to use a very interesting fact that characteristic functions of general exponential Lévy processes are always known in closed-forms or can be expressed in terms of special functions of mathematics although their probability densities are not. We saw in the section 4 that there is one-to-one relationship between a probability density and a characteristic functions (i.e. through Fourier transform) and both of which uniquely determine a probability distribution. If we can somehow rewrite (8.3) and (8.4) in terms of a characteristic function of $S_T | \mathcal{F}_t$ (i.e. log of $S_T | \mathcal{F}_t$ to be more precise) instead of its probability density $\mathbb{Q}(S_T | \mathcal{F}_t)$, we will be able to price options in general exponential Lévy models.

[8.2] Derivation of Call Price with Fourier Transform: Carr and Madan (1999)

For simplicity, assume $t = 0$ without loss of generality. From the equation (8.3), use a change of variable technique from S_T to $\ln S_T$:

$$C(T, K) = e^{-rT} \int_{\ln K}^{\infty} (e^{\ln S_T} - e^{\ln K}) \mathbb{Q}(\ln S_T | \mathcal{F}_0) d \ln S_T.$$

Let s_T be a log terminal stock price and k be a log strike price, i.e. $s_T \equiv \ln S_T$ and $k \equiv \ln K$. Thus, we have:

$$C(T, k) = e^{-rT} \int_k^{\infty} (e^{s_T} - e^k) \mathbb{Q}(s_T | \mathcal{F}_0) ds_T, \quad (8.7)$$

where $\mathbb{Q}(s_T) \equiv \mathbb{Q}(s_T | \mathcal{F}_0)$ (for simplicity) is a risk-neutral density of a log terminal stock price s_T conditional on filtration $\mathcal{F}_0$. From the equation (4.1), a characteristic function of s_T is a Fourier transform of its density function $\mathbb{Q}(s_T)$:

$$\phi_T(\omega) \equiv \mathcal{F}[\mathbb{Q}(s_T)](\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega s_T} \mathbb{Q}(s_T) ds_T. \quad (8.8)$$

Consider a function $g(t)$. Sufficient (but not necessary) condition for the existence of Fourier transform and its inverse is the equation (3.47):

$$\int_{-\infty}^{\infty} |g(t)| dt < \infty. \quad (8.9)$$

We saw in section 7 that BS models a log terminal stock price s_T as a normal random variable with its normal density given by under $\mathbb{Q}$ (from (7.7)):

$$\mathbb{Q}(s_T) = \frac{1}{\sqrt{2\pi\sigma^2 T}} \exp \left[-\frac{\left\{ s_T - \left(s_0 + \left(r - \frac{1}{2}\sigma^2 \right) T \right) \right\}^2}{2\sigma^2 T} \right]. \quad (8.10)$$

From the equations (8.8) and (8.10), a characteristic function of BS log terminal stock price s_T is easily obtained as:

$$\phi_T(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega s_T} \mathbb{Q}(s_T) ds_T = \exp \left[i \left\{ s_0 + \left(r - \frac{1}{2}\sigma^2 \right) T \right\} \omega - \frac{(\sigma^2 T) \omega^2}{2} \right]. \quad (8.11)$$

When a call price is expressed in terms of a log strike price $k \equiv \ln K$ in the equation (8.7), k approaches $-\infty$ as a strike price K approaches 0 in the limit. Thus, from (7.8):

$$\begin{aligned} C(T, k) &= e^{-rT} \int_{-\infty}^{\infty} (e^{s_T} - e^{-\infty}) \mathbb{Q}(s_T | \mathcal{F}_0) ds_T = e^{-rT} \int_{-\infty}^{\infty} e^{s_T} \mathbb{Q}(s_T | \mathcal{F}_0) ds_T \\ C(T, k) &= e^{-rT} E^{\mathbb{Q}}[e^{s_T} | \mathcal{F}_0]. \end{aligned} \quad (8.12)$$

We know under equivalent martingale measure $\mathbb{Q}$:

$$E^{\mathbb{Q}}[S_T \equiv e^{s_T} | \mathcal{F}_0] = S_0 e^{rT}.$$

Equation (8.12) becomes:

$$C(T, k) = S_0.$$

Therefore, a call price $C(T, k)$ is not integrable (i.e. $C(T, k)$ does not satisfy (8.9)). Therefore, $C(T, k)$ cannot be Fourier transformed. To solve this problem, CM defines a modified call price as:

$$C_{\text{mod}}(T, k) \equiv e^{\alpha k} C(T, k), \quad (8.13)$$

where $C_{\text{mod}}(T, k)$ is expected to satisfy the integrability condition (8.9) by carefully choosing $\alpha > 0$:

$$\int_{-\infty}^{\infty} |C_{\text{mod}}(T, k)| dk < \infty.$$

Consider a FT of a modified call price by the FT definition (3.4):

$$\psi_T(\omega) \equiv \int_{-\infty}^{\infty} e^{i\omega k} C_{\text{mod}}(T, k) dk. \quad (8.14)$$

From (8.14), call price $C(T, k)$ can be obtained by an inverse FT (i.e. the definition (3.5)) of $\psi_T(\omega)$:

$$\begin{aligned} C_{\text{mod}}(T, k) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \psi_T(\omega) d\omega \\ e^{\alpha k} C(T, k) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \psi_T(\omega) d\omega \\ C(T, k) &= \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \psi_T(\omega) d\omega. \end{aligned} \quad (8.15)$$

Now CM derives an analytical expression of $\psi_T(\omega)$ in terms of a characteristic function $\phi_T(\omega)$. Substitute (8.13) into (8.14):

$$\psi_T(\omega) = \int_{-\infty}^{\infty} e^{i\omega k} e^{\alpha k} C(T, k) dk .$$

Substitute (8.7) and interchanging integrals yields:

$$\begin{aligned} \psi_T(\omega) &= \int_{-\infty}^{\infty} e^{i\omega k} e^{\alpha k} e^{-rT} \int_k^{\infty} (e^{s_T} - e^k) \mathbb{Q}(s_T) ds_T dk \\ &= \int_{-\infty}^{\infty} e^{-rT} \mathbb{Q}(s_T) \int_{-\infty}^s e^{i\omega k} (e^{s_T + \alpha k} - e^{(1+\alpha)k}) dk ds_T \\ &= \int_{-\infty}^{\infty} e^{-rT} \mathbb{Q}(s_T) \left(\frac{e^{(\alpha+1+i\omega)s_T}}{\alpha + i\omega} - \frac{e^{(\alpha+1+i\omega)s_T}}{\alpha + 1 + i\omega} \right) ds_T \\ &= \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} . \end{aligned} \quad (8.16)$$

Thus, a call pricing function is obtained by substituting (8.16) into (8.15):

$$C(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} d\omega, \quad (8.17)$$

where $\phi_T(\cdot)$ is a characteristic function of a log terminal stock price s_T conditional on filtration $\mathcal{F}_0$. We can interpret the equation (8.17) which is single numerical integration problem as a characteristic function $\phi_T(\cdot)$ equivalent of the equation (8.3). Table 8.1 illustrates this point.

Table 8.1: Comparison between Traditional and FT Option Pricing Formula

Option Pricing Method	Equation	Formula
Traditional	(8.3)	$C = e^{-r(T-t)} \int_K^{\infty} (S_T - K) \mathbb{Q}(S_T \mathcal{F}_t) dS_T$
Fourier Transform	(8.17)	$C(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} d\omega$ $\phi_T(\cdot) \equiv \int_{-\infty}^{\infty} e^{i\omega \ln(S_T)} \mathbb{Q}(\ln(S_T) \mathcal{F}_0) d \ln(S_T), \quad k \equiv \ln K$

[8.3] How to Choose Decay Rate Parameter α : Carr and Madan (1999)

We mentioned earlier that when a call price is expressed in terms of a log strike price $k \equiv \ln K$ in the equation (8.7), k approaches $-\infty$ as a strike price K approaches 0 in the limit which is illustrated in Figure 8.1.

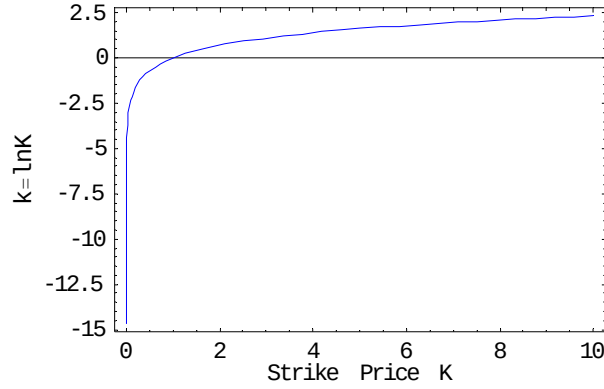


Figure 8.1: Relationship between a Strike Price K and a Log-Strike Price $k \equiv \ln K$.

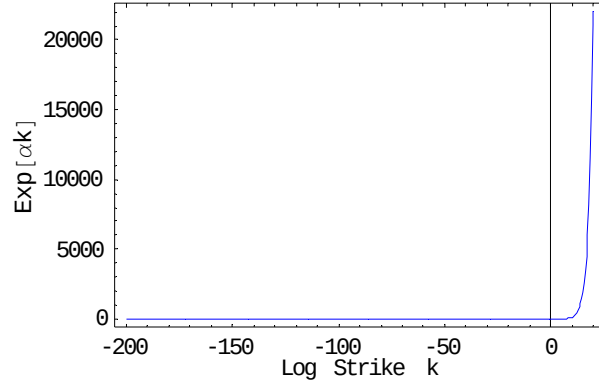


Figure 8.2: Plot of an Exponential Function $e^{0.5k}$ with Respect to Log-Strike Price $k \equiv \ln K$.

Thus, a call price function $C(T, k)$ becomes S_0 as $k \rightarrow -\infty$ discussed before. In order to make a call price integrable, CM multiplies an exponential function $e^{\alpha k}$ with $\alpha \in \mathbb{R}^+$ to $C(T, k)$ and obtains a modified call price $C_{\text{mod}}(T, k)$. As shown by Figure 8.2, the role of $e^{\alpha k}$ with $\alpha \in \mathbb{R}^+$ is to dampen the size of $C(T, k) = S_0$ in the limit $k \rightarrow -\infty$. But this in turn worsens the integrability condition for $k \in \mathbb{R}^+$ (i.e. positive log strike) axis. In order for a modified call price $C_{\text{mod}}(T, k) \equiv e^{\alpha k} C(T, k)$ to be integrable for both positive and negative k axis (i.e. square integrable), CM provides a sufficient condition:

$$\psi_T(0) < \infty.$$

From (8.16):

$$\psi_T(0) = \frac{e^{-rT} \phi_T(-(\alpha + 1)i)}{\alpha^2 + \alpha}.$$

Therefore, the sufficient condition of the square-integrability of $C_{\text{mod}}(T, k)$ is:

$$\phi_T(-(\alpha+1)i) < \infty. \quad (8.18)$$

From the definition of a characteristic function (8.8):

$$\phi_T(\omega) = E[e^{i\omega s_T}] = E[e^{i\omega \ln S_T}] = E[(e^{\ln S_T})^{i\omega}] = E[S_T^{i\omega}]. \quad (8.19)$$

From (8.18) and (8.19):

$$\begin{aligned} \phi_T(-(\alpha+1)i) &< \infty \\ E[S_T^{i\{-(\alpha+1)i\}}] &< \infty \\ E[S_T^{\alpha+1}] &< \infty. \end{aligned} \quad (8.20)$$

CM suggests the use of (8.20) and the analytical expression of the characteristic function to determine an upper bound on α .

[8.4] Black-Scholes Model with Fourier Transform Pricing Method

By substituting a characteristic function of BS log terminal stock price s_T (i.e. the equation (8.11)) into the general FT pricing formula of the equation (8.17), we obtain BS-FT call pricing formula:

$$C_{BS-FT}(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} d\omega, \quad (8.21)$$

with $\phi_T(\omega) = \exp \left[i \left\{ s_0 + \left(r - \frac{1}{2} \sigma^2 \right) T \right\} \omega - \frac{(\sigma^2 T) \omega^2}{2} \right].$

Table 8.2: Comparison between Original BS and BS-FT Option Pricing Formula

Option Pricing Method

Original Black-Scholes

$$C = e^{-r(T-t)} \int_K^{\infty} (S_T - K) \mathbb{Q}(S_T | \mathcal{F}_t) dS_T$$

$$\mathbb{Q}(S_T | \mathcal{F}_t) = \frac{1}{S_T \sqrt{2\pi\sigma^2(T-t)}} \exp \left[-\frac{\left\{ \ln S_T - \left(\ln S_t + (r - \frac{1}{2}\sigma^2)(T-t) \right) \right\}^2}{2\sigma^2(T-t)} \right]$$

Black-Scholes with Fourier Transform Approach

$$C = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} d\omega$$

$$\phi_T(\omega) = \exp \left[i \left\{ s_0 + (r - \frac{1}{2}\sigma^2)T \right\} \omega - \frac{(\sigma^2 T) \omega^2}{2} \right], \quad k \equiv \ln K$$

We implement the BS-FT formula (8.21) with decay rate parameter $\alpha = 1$ and compare the result to the original BS call price. As illustrated by Figure 8.3, as a principle BS-FT call price and the original BS call price are identical. This is no surprise because the original BS formula and BS-FT formula in Table 8.2 are the same person with a different look. BS-FT formula is just frequency representation of the original BS formula.

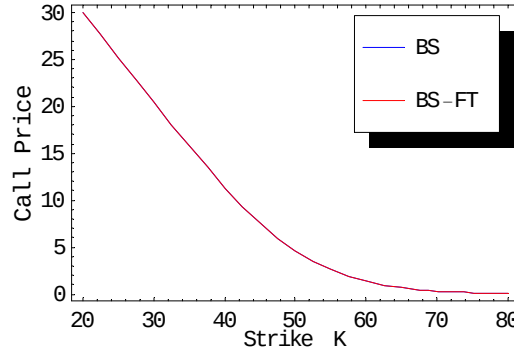


Figure 8.3: Original BS Call Price Vs. BS-FT with $\alpha = 1$ Call Price. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 1$.

Several points should be investigated further now. Firstly, CPU time should be discussed. Consider an ATM call option with $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 1$. Our BS-FT $\alpha = 1$ code needs 0.01 second CPU time while our original BS code needs zero seconds. Although you can state that BS-FT formula is slower than the original BS, speed is not an issue for most of the purposes.

Secondly, let's consider the choice of decay rate parameter α . We saw the selection procedure for α by Carr and Madan (1999) in section 8.3. But we found out that choice of α is not much important as long as $\alpha \in \mathbb{R}^+$. Consider an ATM vanilla call price with $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 0.25$. Its BS price is 2.16794. Figure 8.4 and Table 8.3 indicates that for $\alpha \geq 0.05$ BS-FT price converges to the

original BS price. But it is possible that option prices using FT approach can be sensitive to the choice of α for models other than the BS, so at this point it is safer for us to state that for at least BS model the choice of α is of no importance (i.e. $\alpha = 1$ is fine).

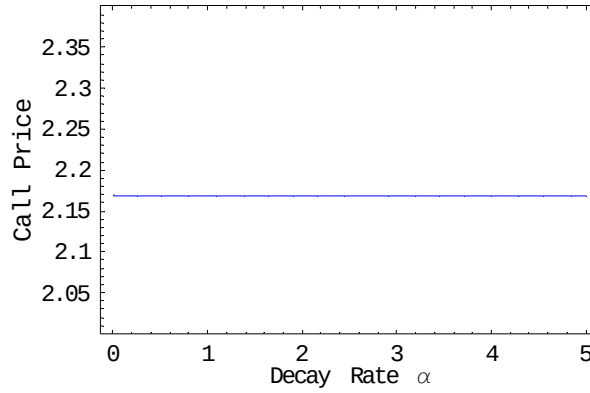


Figure 8.4: Vanilla BS-FT ATM Call Price as a Function of Decay Rate Parameter α . Common parameters and variables fixed are $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 0.25$.

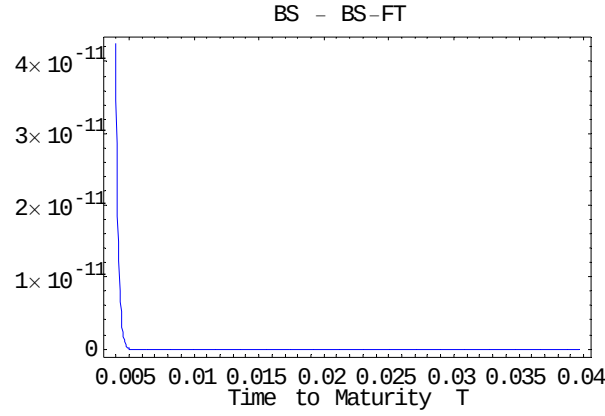
Table 8.3: BS-FT ATM Call Price for Different Values of Decay Rate Parameter α

α	0.01	0.05	0.1	0.5	1	2	5
Call Price	2.16972	2.16794	2.16794	2.16794	2.16794	2.16794	2.16794

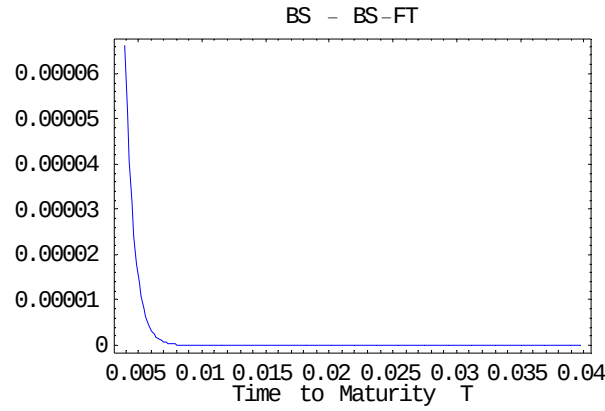
Thirdly, we discuss the FT option pricing with (8.21) for near maturity deep OTM and ITM calls. As CM (1999) points out, for near maturity deep OTM and ITM calls (puts), call (put) prices approach their intrinsic values $(S_t - K)^+$ as $t \rightarrow T$ ($(K - S_t)^+$). Since this makes the inverse Fourier transform integrand highly oscillatory, the numerical integration problem in (8.21) becomes slow and difficult. For example, using our code, in the case of an ATM call with $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$, BS-FT $\alpha = 1$ call price does not face the difficulty in numerical integration at all even for one day to maturity $T = 1/252$. But in the case of a deep OTM call with $S_0 = 50$, $K = 80$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$, BS-FT $\alpha = 1$ call price begins to experience difficulty in numerical integration around 37 trading days to maturity (i.e. $T < 38/252$). For a deep ITM call with $S_0 = 50$, $K = 20$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$, BS-FT $\alpha = 1$ call price begins to experience difficulty around 5 days to maturity (i.e. $T < 5/252$). Of course, this depends on the value of α , your hardware, and your software, etc.

More important question is to investigate the amount of error caused by this numerical integration difficulty for the near maturity deep OTM and ITM options. Consider a call option with $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$. Figure 8.5 plots a series of the difference between the original BS price and BS-FT $\alpha = 1$ price for the range of less than 10 trading days to maturity $1/252 \leq T \leq 10/252$. We find that despite the difficulty in the numerical integration of BS-FT price of (8.21) for the near maturity deep OTM (in Panel

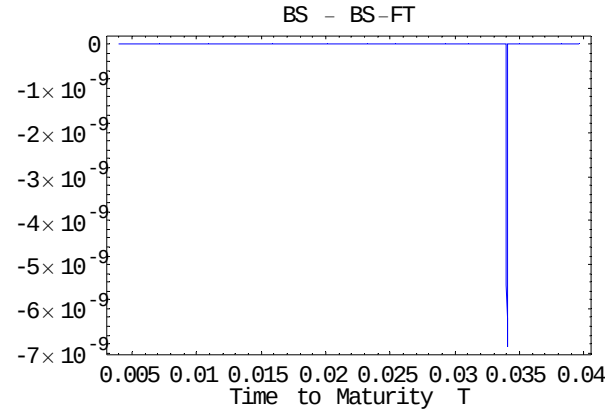
A) and deep ITM (in Panel B) call, our BS-FT code causes no significant errors in terms of pricing. For a near maturity ATM call (in Panel C) which faces no difficulty in numerical integration, BS-FT pricing error is negligible.



A) OTM call with $S_0 = 50$ and $K = 80$.



B) ITM call with $S_0 = 50$ and $K = 20$.



C) ATM Call with $S_0 = 50$ and $K = 50$.

Figure 8.5: Plot of BS Price minus BS-FT $\alpha = 1$ Price for Near-Maturity Vanilla Call with Less Than 10 Trading Days to Maturity $1/252 \leq T \leq 10/252$. Common parameters and variables fixed are $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

We conclude this section by stating the following remarks. In the case of BS model, the FT call price of the equation (8.17) needs equivalent CPU time as the original BS formula and produces negligible pricing errors regardless of the maturity and the moneyness of the call.

[8.5] Derivation of Near-Maturity OTM Vanilla Option Pricing Function with Fourier Transform: Carr and Madan (1999)

Although the numerical integration difficulty of BS-FT call price of (8.21) for the near maturity deep OTM and deep ITM calls causes no significant pricing errors, CM (1999) provides the vanilla option pricing formula specifically designed for them which makes the inverse Fourier transform integrand less oscillatory and facilitates the numerical integration problem.

Let $z_T(k)$ be the time value of an OTM vanilla options with maturity T at the current time $t = 0$ without loss of generality. This implies that when the strike price is less than the spot stock price $K < S_0$ (i.e. $k < s_0$ in log-scale), $z_T(k)$ takes on the OTM put price. When the strike price is greater than the spot stock price $K > S_0$ (i.e. $k > s_0$ in log-scale), $z_T(k)$ takes on the OTM call price:

$$z_T(k) \equiv \begin{cases} P(0, S_0) - (K - S_0)^+ = P(0, S_0) & \text{if } K < S_0 \\ C(0, S_0) - (S_0 - K)^+ = C(0, S_0) & \text{if } K > S_0 \end{cases} .$$

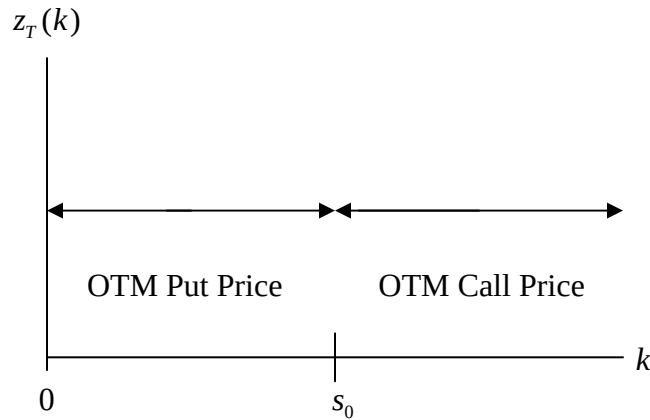


Figure 8.6: Illustration of an OTM Vanilla Option Price $z_T(k)$.

Assume the current stock price $S_0 = 1$ for the simplicity (i.e. taken care in the end) which in turn means $s_0 = 0$.

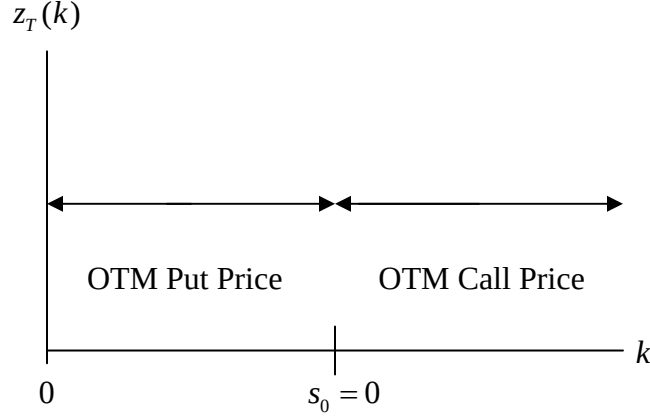


Figure 8.7: Illustration of an OTM Vanilla Option Price $z_T(k)$ Assuming $S_0 = 1$.

Under a risk-neutral measure $\mathbb{Q}$, $z_T(k)$ can be computed as:

$$z_T(k) = e^{-rT} \int_{-\infty}^{\infty} \left[(e^k - e^{s_T}) 1_{s_T < k, k < 0} + (e^{s_T} - e^k) 1_{s_T > k, k > 0} \right] \mathbb{Q}(s_T | \mathcal{F}_0) ds_T, \quad (8.22)$$

where $\mathbb{Q}(s_T) \equiv \mathbb{Q}(s_T | \mathcal{F}_0)$ is a risk-neutral density of a log terminal stock price s_T conditional on filtration $\mathcal{F}_0$. $z_T(k)$ takes on a put payoff $(e^k - e^{s_T})$ if this option is currently OTM put $k < 0$ and if at maturity T this put finishes ITM $s_T < k$. $z_T(k)$ takes on a call payoff $(e^{s_T} - e^k)$ if this option is currently OTM call $k > 0$ and if at maturity T this call finishes ITM $s_T > k$.

Consider a FT of OTM vanilla option price $z_T(k)$ by the FT definition (3.4):

$$\zeta_T(\omega) = \int_{-\infty}^{\infty} e^{i\omega k} z_T(k) dk. \quad (8.23)$$

From (8.23), OTM vanilla option price $z_T(k)$ can be obtained by an inverse FT (i.e. the definition (3.5)) of $\zeta_T(\omega)$:

$$z_T(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \zeta_T(\omega) d\omega. \quad (8.24)$$

By substituting (8.22) into (8.23), CM shows that an analytical expression of $\zeta_T(\omega)$ in terms of a characteristic function $\phi_T(\omega)$ of a log terminal stock price s_T conditional on filtration $\mathcal{F}_0$ can be obtained as:

$$\zeta_T(\omega) = e^{-rT} \left(\frac{1}{1+i\omega} - \frac{e^{rT}}{i\omega} - \frac{\phi_T(\omega-i)}{\omega^2-i\omega} \right). \quad (8.25)$$

To facilitate numerical integration, CM again considers a FT of OTM vanilla option price $z_T(k)$ modified with dampening function $\sinh(\alpha k)$ (compare with (8.23)):

$$\begin{aligned} \gamma_T(\omega) &= \int_{-\infty}^{\infty} e^{i\omega k} \sinh(\alpha k) z_T(k) dk \\ &= \int_{-\infty}^{\infty} e^{i\omega k} \frac{e^{\alpha k} - e^{-\alpha k}}{2} z_T(k) dk \\ &= \frac{\zeta_T(\omega - i\alpha) - \zeta_T(\omega + i\alpha)}{2}. \end{aligned} \quad (8.26)$$

From (8.26), OTM vanilla option price $z_T(k)$ can be obtained by an inverse FT of $\gamma_T(\omega)$:

$$z_T(k) = \frac{1}{\sinh(\alpha k)} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \gamma_T(\omega) d\omega. \quad (8.27)$$

From now on, we call the OTM near maturity option price of the equation (8.27) as FT/TV (i.e. TV indicates the time value approach). Table 8.4 compares the general FT pricing formula using the modified call price (i.e. the equation (8.17)) and FT/TV formula using the time value of OTM options which is specifically designed for near maturity options.

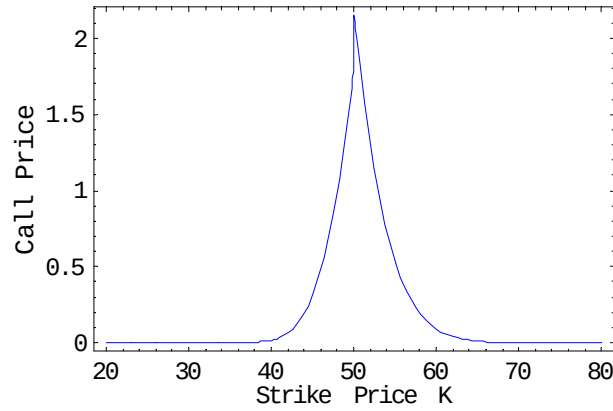
Table 8.4: FT Formula Vs. FT/TV Formula

Approach	Formula
FT	$C(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha + 1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega} d\omega$ $\phi_T(.) \equiv \int_{-\infty}^{\infty} e^{i\omega \ln(S_T)} \mathbb{Q}(\ln(S_T) \mathcal{F}_0) d \ln(S_T), \quad k \equiv \ln K$
FT/TV	$z_T(k) \equiv \begin{cases} P(0, s_0) & \text{if } k < s_0 \\ C(0, s_0) & \text{if } k > s_0 \end{cases}$ $z_T(k) = \frac{1}{\sinh(\alpha k)} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \gamma_T(\omega) d\omega$ $\gamma_T(\omega) = \frac{\zeta_T(\omega - i\alpha) - \zeta_T(\omega + i\alpha)}{2}$

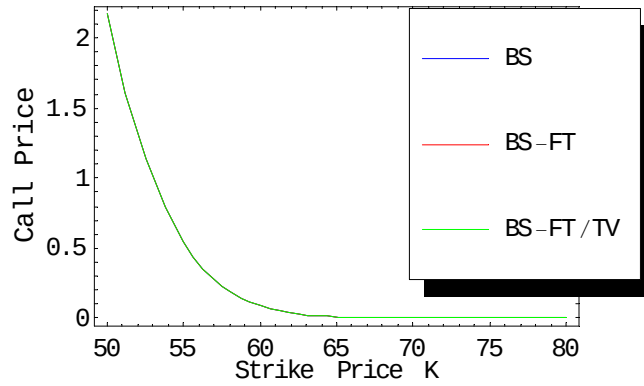
$$\zeta_T(\omega) = e^{-rT} \left(\frac{1}{1+i\omega} - \frac{e^{rT}}{i\omega} - \frac{\phi_T(\omega-i)}{\omega^2-i\omega} \right)$$

$$\phi_T(.) \equiv \int_{-\infty}^{\infty} e^{i\omega \ln(S_T)} \mathbb{Q}(\ln(S_T) | \mathcal{F}_0) d \ln(S_T), \quad k \equiv \ln K$$

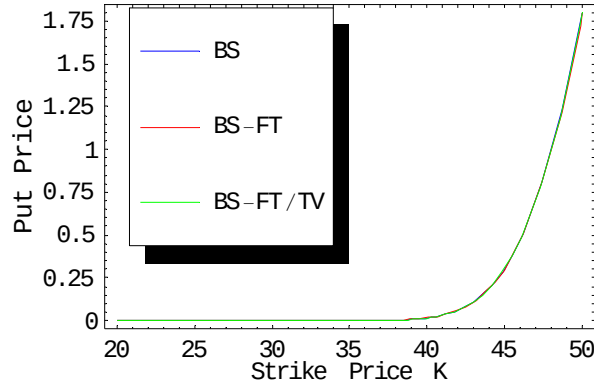
Next the performance of FT-TV formula of the equation (8.27) with decay rate parameter $\alpha = 10$ (we will discuss the choice of α soon) is tested and compared to the original BS price and BS-FT $\alpha = 1$ price for a range of different moneyness in Figure 8.8. Panel A plots $z_T(k)$ which reaches its maximum at ATM and takes the price of OTM put for $K < S_0$ and takes the price of OTM call for $K > S_0$. Panel B and C indicates that as a principle, all three formulae produce identical prices. Again, this is no surprise because all these are same thing with different looks.



A) Plot of $z_T(k)$.



B) For the range $50 \leq K \leq 80$.



C) For the range $20 \leq K \leq 50$.

Figure 8.8: BS Vs. BS-FT $\alpha = 1$ Vs. BS-FT/TV $\alpha = 10$. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 0.25$.

Next, CPU time and the accuracy of BS-FT/TV formula are discussed. Consider vanilla options with common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$. Table 8.5 to 8.7 show CPU time and the accuracy of BS-FT/TV $\alpha=10$ formula compared to other two formulae with three different maturities and with varying moneyness. In terms of CPU time, we notice that BS-FT/TV formula is by far the slowest although most of purposes this will not be an issue. BS-FT/TV $\alpha=10$ formula has marginally larger error than BS-FT $\alpha=1$ formula, but the size of the errors are trivial.

Table 8.5: CPU Time for an OTM Call with $K = 80$

Price is in the bracket.

Method	Time to Maturity		
	$T = 0.1$	$T = 0.5$	$T = 1$
BS	0 seconds (3.77524×10^{-14})	0 seconds (0.00152306)	0 seconds (0.0594469)
BS-FT $\alpha = 1$	0.08 seconds (3.97419×10^{-14})	0.02 seconds (0.00152306)	0.01 seconds (0.0594469)
BS/FT/OTM $\alpha = 10$	0.31 seconds (4.00513×10^{-14})	0.07 seconds (0.00152306)	0.03 seconds (0.0594469)

Table 8.6: CPU Time for an ATM Call with $K = 50$

Price is in the bracket.

Method	Time to Maturity		
	$T = 0.1$	$T = 0.5$	$T = 1$
BS	0 seconds	0 seconds	0 seconds

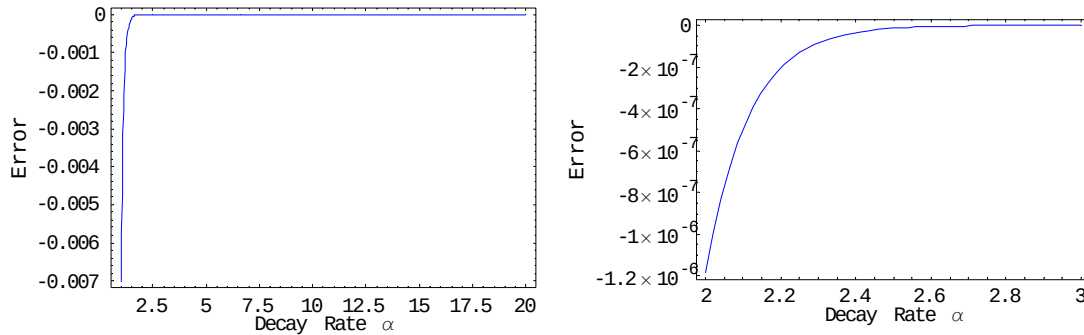
	(1.3331)	(3.15382)	(4.6135)
BS-FT $\alpha = 1$	0.01 seconds (1.3331)	0.01 seconds (3.15382)	0.01 seconds (4.6135)
BS/FT/OTM $\alpha = 10$	0.04 seconds (1.3331)	0.04 seconds (3.15382)	0.07 seconds (4.6135)

Table 8.7: CPU Time for an OTM Put with $K = 20$

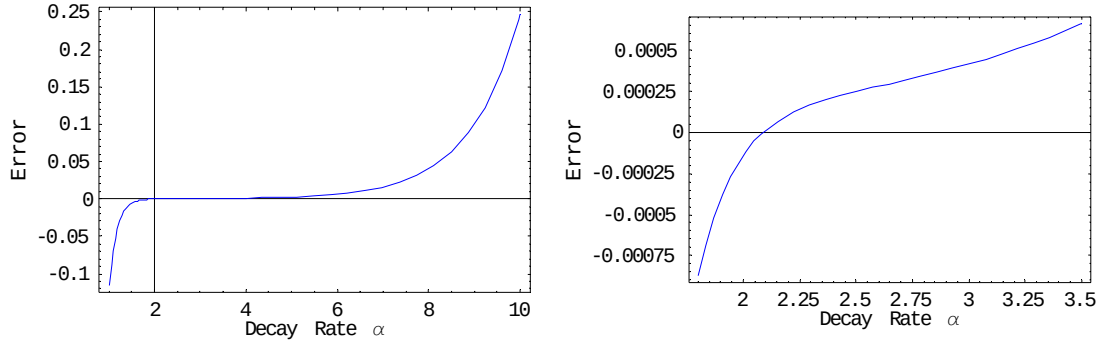
Price is in the bracket.

Method	Time to Maturity		
	$T = 0.1$	$T = 0.5$	$T = 1$
BS	0 seconds (0)	0 seconds (1.4492×10^{-11})	0 seconds (1.32586×10^{-6})
BS-FT $\alpha = 1$	0.04 seconds (-7.10543×10^{-15})	0.02 seconds (1.4481×10^{-11})	0.02 seconds (1.32586×10^{-6})
BS/FT/OTM $\alpha = 10$	0.3 seconds (-3.76588×10^{-13})	0.14 seconds (1.86517×10^{-11})	0.13 seconds (1.32587×10^{-6})

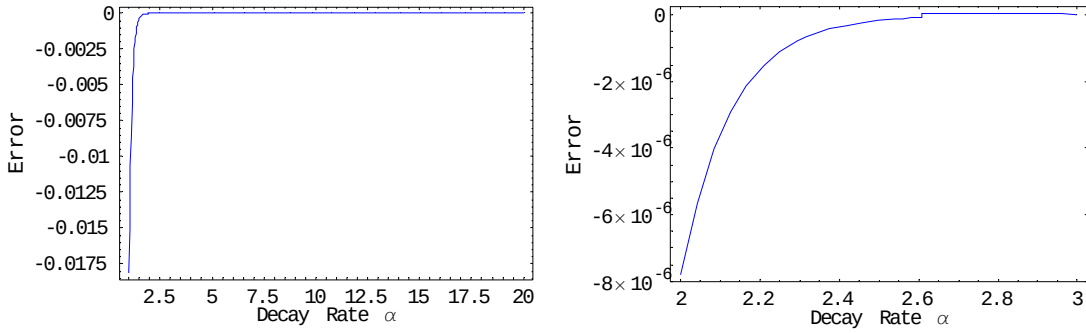
Next, the level of decay rate parameter α for BS-FT/TV formula is dealt. Figure 8.9 illustrates the pricing error of BS-FT/TV formula of a one day to maturity $T = 1/252$ option as a function of varying α . Panel A (for an OTM call) and C (for an ATM call) tells us that for $\alpha \geq 2$, BS-FT/TV formula has effectively zero error relative to BS price. But for an OTM put (Panel B), the error does not monotonically decrease as α rises. It seems that the value of $2 < \alpha < 2.25$ yields the negligible size of the error. Therefore, from now on, we always use $\alpha = 2.1$ when implementing BS-FT/TV formula.



A) For an OTM Call with $K = 80$.



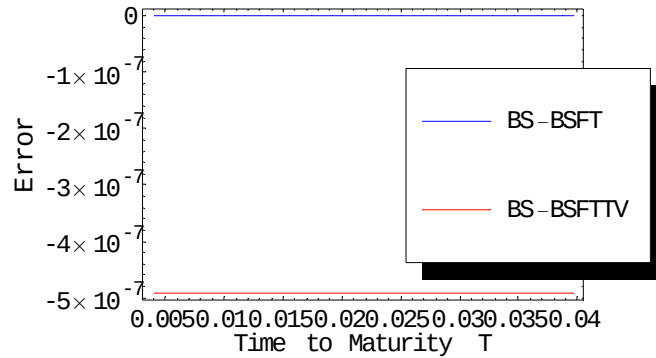
B) For an OTM Put with $K = 20$.



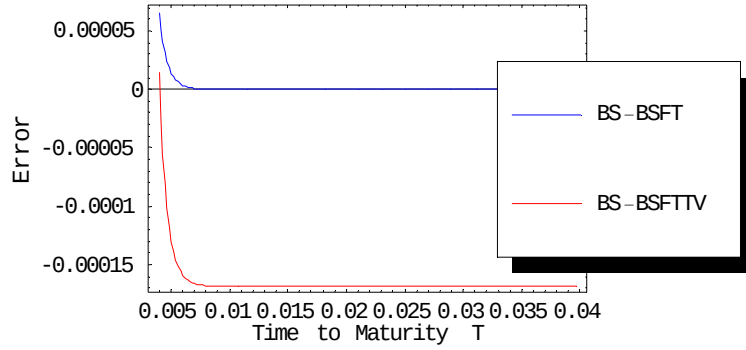
C) For an ATM Call with $K = 50$.

Figure 8.9: BS Price Minus BS-FT/TV Price for One Day to Maturity Option as a Function of Decay Rate Parameter α . Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 1/252$.

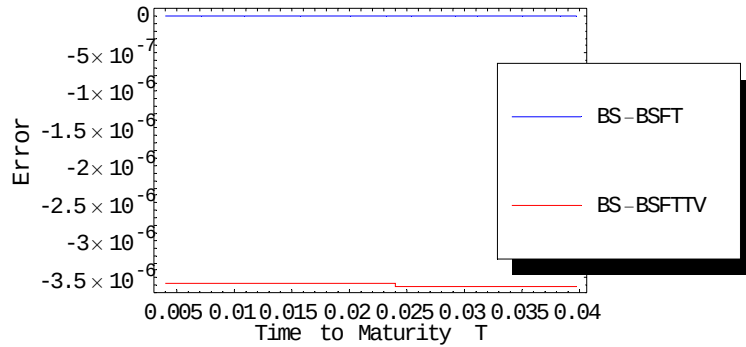
Next, we compare the performance between BS-FT $\alpha = 1$ formula and BS-FT/TV $\alpha = 2.1$ formula and let's see if BS-FT/TV $\alpha = 2.1$ formula improves the pricing accuracy for near maturity options. According to Figure 8.10, BS-FT/TV $\alpha = 2.1$ formula generally has larger error regardless of the moneyness except for one-day to maturity OTM put (in Panel B).



A) OTM call with $K = 80$.



B) OTM put with $K = 20$.



C) ATM Call with $K = 50$.

Figure 8.10: Plot of Error of BS-FT/TV $\alpha = 2.1$ Price Vs. Error of BS-FT $\alpha = 1$ Price with Less Than 10 Trading Days to Maturity $1/252 \leq T \leq 10/252$. Common parameters and variables fixed are $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

We conclude this section by stating the following remarks. Carr and Madan (1999) provides the FT/TV formula of the equation (8.27) in order to improve the pricing accuracy specifically for near maturity options compared to the FT formula of the equation (8.17). Firstly, FT/TV formula is slower than the original BS and FT formula, but this won't be an issue for pricing several options. Secondly, we recommend the use of $2 < \alpha < 2.25$ for the decay rate in FT/TV formula. Thirdly, contrary to the design by CM, FT/TV formula generally has larger error than FT $\alpha = 1$ formula. At this point, we are skeptical about the usefulness of FT/TV formula.

[8.6] Derivation of Call Pricing Function with Discrete Fourier Transform (DFT): Carr and Madan (1999)

There still remains one problem to be solved. Although the numerical integration difficulty of FT call price of (8.17) for the near maturity options causes no significant pricing errors, it makes the evaluation of FT price slow. This speed becomes an issue when calibrating hundreds or thousands of prices (i.e. also in Monte Carlo simulation).

To improve computational time, Carr and Madan (1999) apply discrete Fourier transform to approximate the equation (8.17). Note that our version of DFT call price formula is different from their original formula.

FT call price is (i.e. the equations (8.15) and (8.16)):

$$C_T(k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \psi_T(\omega) d\omega, \quad (8.28)$$

where $\psi_T(\omega) = \frac{e^{-rT} \phi_T(\omega - (\alpha + 1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega}$. The integral is an inverse FT of an angular frequency domain function $\psi_T(\omega)$ into signal space k function. As we saw in section 5, it can be approximated by DFT. Firstly, the infinite integral of (8.28) needs to be truncated as:

$$C_T(k) \approx \frac{e^{-\alpha k}}{2\pi} \int_{-\Omega/2}^{\Omega/2} e^{-i\omega k} \psi_T(\omega) d\omega. \quad (8.29)$$

Secondly, (8.29) can be discretized as:

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n)}{2\pi} \sum_{j=0}^{N-1} w_j \left\{ \exp(-i\omega_j k_n) \psi_T(\omega_j) \Delta\omega \right\}. \quad (8.30)$$

Let us discuss the interpretation of (8.30). The purpose of a DFT (an inverse DFT) is to approximate the transform space function, i.e. in our case FT of a modified call price $\psi_T(\omega)$ (the signal space function, i.e. in our case a call price $C_T(k)$) as close as possible by sampling a finite number of points N of a signal space function $C_T(k)$ with signal space sampling interval Δk and by sampling a finite number of points N of a transform space function $\psi_T(\omega)$ with transform space sampling interval $\Delta\omega$. In other words, both the original continuous signal space function $C_T(k)$ and the original continuous FT $\psi_T(\omega)$ are approximated by a sample of N points.

Let N be the number of discrete samples taken to approximate $C_T(k)$ and $\psi_T(\omega)$. Let Δk be the signal space sampling interval which is the signal increment between signal space samples. Its inverse $f_s \equiv 1/\Delta k$ (samples/1 unit of k) is called a signal space sampling rate. Let K be the total sampling range in the signal space:

$$K \equiv N\Delta k. \quad (8.31)$$

If Δk is 1, the total sampling range in the signal space and the number of samples taken are same (i.e. for example, $K = 10$ dollars and $N = 10$ samples). If $\Delta k = 0.01$, 1 sample

is taken in every 0.01 unit of k in signal space which in turn means that 100 samples are taken per 1 unit of k (i.e. signal space sampling rate $f_s \equiv 1/\Delta k = 100$ Hz).

Consider a sampling in the signal space. Take N discrete samples of a signal space function $C_T(k)$ at n -th sampling instant $k_n \equiv -\frac{N\Delta k}{2} + n\Delta k$ with $n = 0, 1, \dots, N-1$. When $K=10$ and $N=10$ samples (i.e. signal space sampling interval $\Delta k=1$), $k_n \equiv -5 + n$ which means that $C_T(k)$ is sampled at $k = -5, -4, -3, -2, -1, 0, 1, 2, 3$, and 4. Let $C_T(k_n)$ be the sampled values of $C_T(k)$.

Next, consider a sampling in the transform space (i.e. angular frequency ω domain). Let $\Delta\omega$ Hz (radians/1 unit of k) be the angular frequency domain sampling interval:

$$\Delta\omega \equiv \frac{2\pi}{N\Delta k} \equiv \frac{2\pi}{K}. \quad (8.32)$$

The total sampling range in the angular frequency domain Ω is:

$$\Omega \equiv N\Delta\omega. \quad (8.33)$$

Take N samples of a continuous FT $\mathcal{G}(\omega)$ at j -th angular frequency sampling instant:

$$\omega_j \equiv -\frac{N\Delta\omega}{2} + j\Delta\omega \equiv \Delta\omega \left(-\frac{N}{2} + j \right) \equiv \frac{2\pi}{N\Delta k} \left(-\frac{N}{2} + j \right) \equiv -\frac{\pi}{\Delta k} + j\frac{2\pi}{N\Delta k}, \quad (8.34)$$

with $j = 0, 1, \dots, N-1$. For example, when $K=10$ and $N=10$ samples (i.e. $\Delta k=1$), $\mathcal{G}(\omega)$ is sampled at $\omega = -\pi, -4\pi/5, -3\pi/5, -2\pi/5, -\pi/5, 0, \pi/5, 2\pi/5, 3\pi/5, 4\pi/5$ (i.e. $\omega_j = -\pi + j\pi/5$). Let $\mathcal{G}(\omega_j)$ be the sampled values of $\mathcal{G}(\omega)$:

$$\mathcal{G}(\omega_j) \equiv \mathcal{G}\left(-\frac{N\Delta\omega}{2} + j\Delta\omega\right). \quad (8.35)$$

$\mathcal{G}(\omega_j)$ is called a spectrum of $g(k_n)$ at angular frequency ω_j and it is a complex number.

DFT and inverse DFT defines the relationship between the sampled signal space function $g(k_n)$ and its spectrum at angular frequency ω_j , $\mathcal{G}(\omega_j)$, as the following:

$$\mathcal{G}(\omega_j) \equiv \sum_{n=0}^{N-1} g(k_n) \exp\{i\omega_j k_n\}$$

$$\mathcal{G}(\omega_j) \equiv \sum_{n=0}^{N-1} g(k_n) \exp \left\{ i \left(\frac{\pi N}{2} - \pi j - \pi n + \frac{2\pi j n}{N} \right) \right\}$$

$$\mathcal{G}(\omega_j) \equiv \exp(-i\pi j) \exp(i\pi N/2) \sum_{n=0}^{N-1} \left\{ g(k_n) \exp(-i\pi n) \right\} \exp(i2\pi j n / N), \quad (8.36)$$

and:

$$g(k_n) \equiv \frac{1}{N} \sum_{j=0}^{N-1} \mathcal{G}(\omega_j) \exp(-i\omega_j k_n)$$

$$g(k_n) \equiv \frac{1}{N} \sum_{j=0}^{N-1} \mathcal{G}(\omega_j) \exp \left\{ -i \left(\frac{\pi N}{2} - \pi j - \pi n + \frac{2\pi j n}{N} \right) \right\}$$

$$g(k_n) \equiv \exp(-i\pi N/2) \exp(i\pi n) \frac{1}{N} \sum_{j=0}^{N-1} \mathcal{G}(\omega_j) \exp(-i2\pi j n / N) \exp(i\pi j), \quad (8.37)$$

where the following relationships are used:

$$k_n \equiv -\frac{N\Delta k}{2} + n\Delta k$$

$$\omega_j \equiv -\frac{N\Delta\omega}{2} + j\Delta\omega \equiv \Delta\omega \left(-\frac{N}{2} + j \right) \equiv \frac{2\pi}{N\Delta k} \left(-\frac{N}{2} + j \right) \equiv -\frac{\pi}{\Delta k} + j \frac{2\pi}{N\Delta k}$$

$$k_n \omega_j \equiv \frac{\pi N}{2} - \pi j - \pi n + \frac{2\pi j n}{N}.$$

Therefore, from the equation (8.30):

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n)}{2\pi} \sum_{j=0}^{N-1} w_j \left\{ \exp(-i\omega_j k_n) \psi_T(\omega_j) \Delta\omega \right\}$$

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n)}{2\pi} \sum_{j=0}^{N-1} w_j \left[\exp \left\{ -i \left(\frac{\pi N}{2} - \pi j - \pi n + \frac{2\pi j n}{N} \right) \right\} \psi_T(\omega_j) \frac{2\pi}{N\Delta k} \right]$$

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k}$$

$$\times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\} \exp(-i2\pi j n / N), \quad (8.38)$$

where $w_{j \in [0, N-1]}$ are weights for sampled points. For example, when trapezoidal rule is chosen:

$$w_j = \begin{cases} 1/2 & \text{for } j = 0 \text{ and } N-1 \\ 1 & \text{for others} \end{cases}.$$

For near-maturity OTM options, DFT/TV formula which is a DFT approximation of the formula (8.27) is given by simply replacing $\exp(-\alpha k_n)$ by $\frac{1}{\sinh(\alpha k_n)}$ and $\psi_T(\omega_j)$ by $\gamma_T(\omega_j)$ in the DFT call price (8.38) as the following:

$$z_T(k_n) \approx \frac{\exp(i\pi n) \exp(-i\pi N/2)}{\sinh(\alpha k_n) \Delta k} \times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \gamma_T(\omega_j) \right\} \exp(-i2\pi jn/N), \quad (8.39)$$

where $\gamma_T(\omega_j)$ is given by (8.26).

[8.7] Implementation and Performance of DFT Pricing Method with Black-Scholes Model

In this section, performance of DFT call price of the equation (8.38) is tested in the BS case. We will call this as BS-DFT. We implement the formula (8.38) with decay rate parameter $\alpha = 1$ and compare the result to the original BS call price and BS-FT $\alpha = 1$ call price under various settings. We will start from its implementation.

The first step to implement (8.38) is to choose the number of samples taken N and the signal space sampling interval Δk which we will call as log-strike space sampling interval, hereafter. Note that selecting Δk corresponds to selecting the frequency domain sampling interval $\Delta\omega$, because these are related by:

$$\Delta k \Delta\omega \equiv \frac{2\pi}{N}. \quad (8.40)$$

We set $N = 4096$ as suggested by CM. Note that because our code uses DFT not FFT, our choice of N does not need to be a power of 2. We use $\Delta k = 0.005$ which is a half the percentage point in the log-strike space. This corresponds to $\Delta\omega = 0.306796$ radians. The total sampling range in the log-strike space is $K = N\Delta k = 20.48$, its sampling rate is 200 samples per unit of k , and the total sampling range in the angular frequency domain is $\Omega = N\Delta\omega = 1256.64$.

Second step is to construct the $N = 4096$ point-sampling grid in the frequency domain using $\omega_j \equiv -\frac{N\Delta\omega}{2} + j\Delta\omega$. Table 8.8 illustrates this. Using this grid, obtain $N = 4096$ point-samples of DFT integrand in (8.32), i.e. $w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\}$. With common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 0.25$, sampled values of $w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\}$ are shown in Table 8.9.

Table 8.8: Angular Frequency Domain Sampling Grid

Index j	j -th angular frequency point ω_j (Hz)
0	-628.319
1	-628.012
2	-627.705
$\sim$	$\sim$
4095	627.705
4096	628.012

Table 8.9: Angular Frequency Domain Sampled Values

Index j	j -th frequency domain sample $w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\}$
0	$8.6137 \times 10^{-859} + 2.5868 \times 10^{-859} i$
1	$-8.5146 \times 10^{-859} - 1.2340 \times 10^{-857} i$
2	$-7.7265 \times 10^{-857} + 3.5391 \times 10^{-857} i$
$\sim$	$\sim$
4095	$-7.7265 \times 10^{-857} - 3.5391 \times 10^{-857} i$
4096	$-4.2573 \times 10^{-859} + 6.1702 \times 10^{-858} i$

Third step is to perform inverse DFT of Table 8.9 and multiply the result by $\frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k}$ in the equation (8.38). This amounts to the $N = 4096$ point-samples of call price $C_T(k_n)$ in log-strike space. Table 8.10 shows this result and it is plotted in Figure 8.7.

Table 8.10: $N = 4096$ Point-Samples of BS Call Price $C_T(k_n)$ in Log-Strike Space

Index n	n -th sample of call price $C_T(k_n)$
0	49.7525
1	49.752
$\sim$	$\sim$
2500	40.3338
$\sim$	$\sim$
2600	34.2249
$\sim$	$\sim$
2700	24.153
$\sim$	$\sim$
2800	7.6428
$\sim$	$\sim$
2900	0.0006067
$\sim$	$\sim$

4095	8.0931×10^{-17}
4096	8.0929×10^{-17}

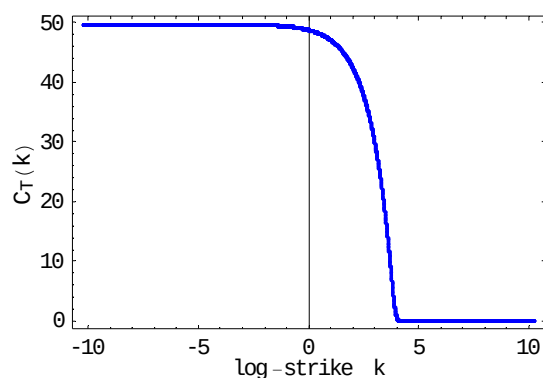


Figure 8.11: N = 4096 Point-Samples of Call Price $C_T(k_n)$ in Log-Strike Space

Fourth step is to match the log-strike price k grid with the strike price grid K as shown in Table 8.11.

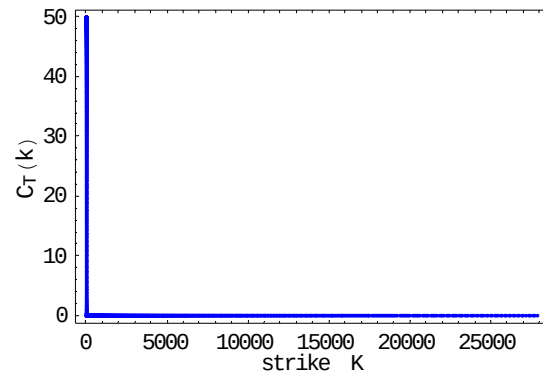
Table 8.11: N = 4096 Point Log-Strike Price (k) Grid and Strike Price (K) Grid

Index n	n -th log-strike price point $n\Delta k$	n -th strike price point $\exp[n\Delta k]$
0	-10.24	0.00003571
~		
1500	-2.745	0.0642
~		
2500	2.255	9.5353
~		
2600	2.755	15.721
~		
2700	3.255	25.9196
~		
2800	3.755	42.7342
~		
2900	4.255	70.4568
~		
4096	10.235	27861.5

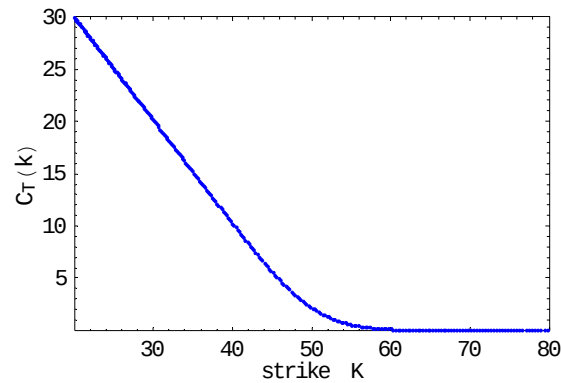
Fifth and final step is to match the strike price K grid with the computed call price as shown in the table 8.12 which is plotted in Figure 8.12. Panel A of Figure 8.12 is for the entire range of the strike price sampled, and Panel B is only for the range of our interest. Remember that we use the spot stock price $S_0 = 50$ and the maturity $T = 0.25$.

Table 8.12: N = 4096 Point Sampled BS-DFT Call Price with respect to Strike Price K_n

Index n	n -th strike price K_n	BS-DFT Call Price
0	0.00003571	49.7525
~		
1500	0.0642	49.6872
~		
2500	9.5353	40.3338
~		
2600	15.721	34.2249
~		
2700	25.9196	24.153
~		
2800	42.7342	7.6428
~		
2900	70.4568	0.0006067
~		
4096	27861.5	8.0929×10^{-17}



A) For the entire strike price range.



B) Our interest.

Figure 8.12: $N = 4096$ Point Sampled BS-DFT Call Price with respect to Strike Price K_n .

This completes the step-by-step procedure of DFT option price computation. Next, we investigate the several important properties of BS-DFT $\alpha = 1$ call price.

Firstly, let's consider the call price computed by the original BS, BS-FT $\alpha = 1$, BS-DFT $\alpha = 1$ with $N = 4096$ and $\Delta k = 0.005$ using common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 0.25$ with respect to a strike price K . The result is plotted in Figure 8.13 which indicates that as a principle these three approaches produce the identical price. Again, this is no surprise because all these are same thing with different looks and different precisions.

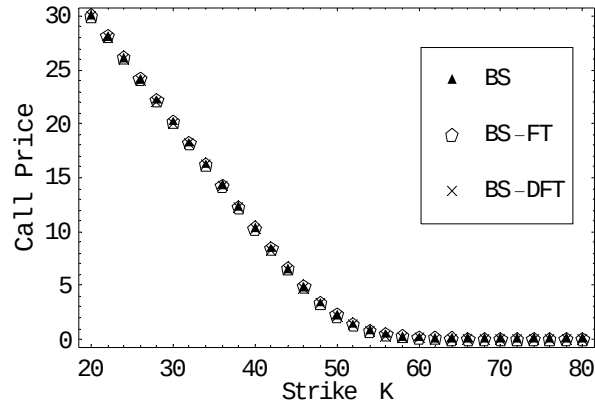


Figure 8.13: BS Vs. BS-FT $\alpha = 1$ Vs. BS-DFT $\alpha = 1$. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 0.25$.

Secondly, let's investigate the difference in CPU time. Consider calculating 100-point call prices for a range of strike price $1 \leq K \leq 100$ with interval 1 with common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 20/252$. Table 8.13 compares CPU time using the original BS, BS-FT $\alpha = 1$, and BS-DFT $\alpha = 1$ with $N = 4096$ and $\Delta k = 0.005$ and Figure 8.14 reports the prices. We notice that there is a significant improvement in the computational time by the use of DFT.

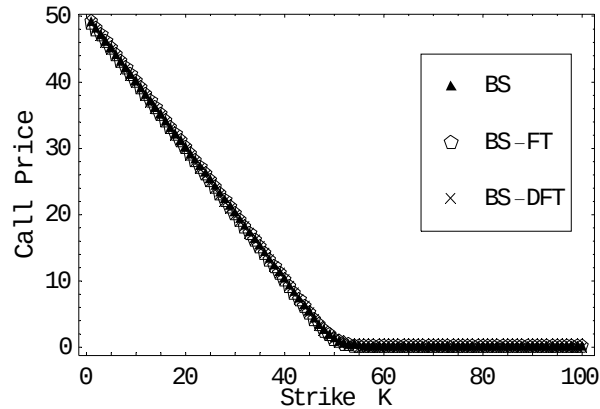


Figure 8.14: BS Vs. BS-FT $\alpha = 1$ Vs. BS-DFT $\alpha = 1$. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 20/252$.

Table 8.13 CPU Time for Calculating 100-point call prices for a range of strike price $1 \leq K \leq 100$ with interval 1 Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 20/252$.

Method	CPU Time
BS	0.03 seconds
BS-FT $\alpha = 1$	5.448 seconds
BS-DFT $\alpha = 1$ $N = 4096$, $\Delta k = 0.005$	0.491 seconds

Thirdly, we pay extra attention to the pricing errors of very near-maturity call prices because these are of Carr and Madan's interest. Consider a call option with common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$. Figures 8.15 to 8.17 plot three series of price differences for vanilla calls computed by the original BS, BS-FT $\alpha = 1$, and BS-DFT $\alpha = 1$ with $N = 4096$ and $\Delta k = 0.005$ as a function of time to maturity of less than a month $1/252 \leq T \leq 20/252$. Figure 8.15 tells us that for deep OTM calls, BS-DFT yields the pricing error around 6×10^{-8} which we interpret negligible. Figure 8.17 tells us that for deep ITM calls, BS-DFT price and BS price are virtually identical except for one day to maturity. Figure 8.16 is the most interesting case among the three. It tells us that as the maturity nears, the error of ATM BS-DFT price monotonically increases and the size of error is large but negligible. In contrast, ATM BS-FT price produces virtually no error. This finding is more clearly illustrated in Figure 8.17 where the $T = 1/252$ ATM pricing error (relative to BS) of BS-DFT $\alpha = 1$ with $N = 4096$ and $\Delta k = 0.005$ is plotted across different moneyness. We again realize that BS-DFT price is virtually identical to BS price for the deep ITM and OTM calls, but its approximation error becomes an issue for ATM call. We have done several experiments trying to reduce the size of this around ATM error by increasing the decay rate parameter α to 10 or by sampling more points $N = 8192$. But these attempts were futile.

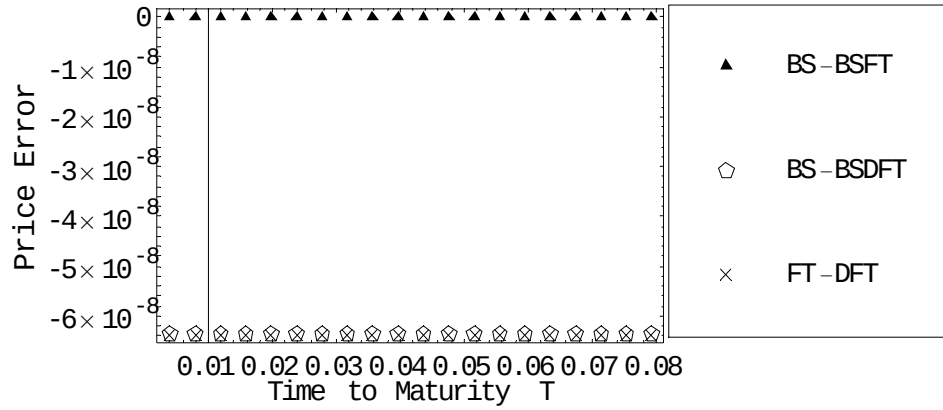


Figure 8.15: Plot of Price Error for Deep-OTM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 80$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

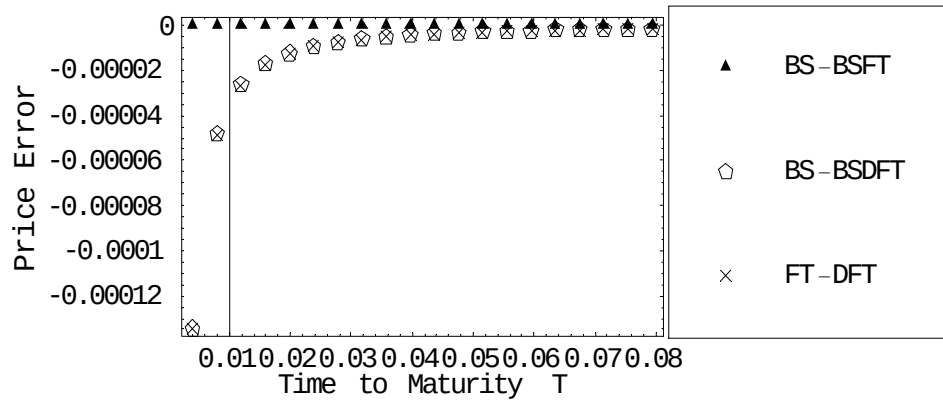


Figure 8.16: Plot of Price Error for ATM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

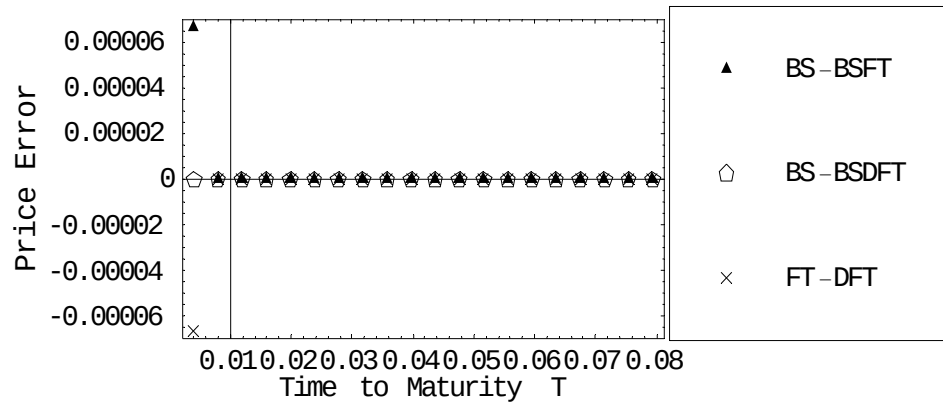


Figure 8.17: Plot of Price Error for Deep-ITM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 20$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

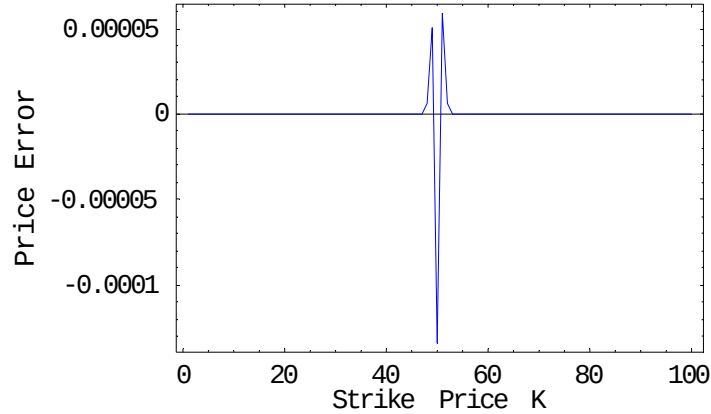


Figure 8.18: Plot of Price Error of BS-DFT Formula for 1-Day-to-Maturity $T = 1/252$ ATM Call as a Function of Strike Price $0 \leq K \leq 100$. Common parameters and variables fixed are $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

We conclude this section by stating the following remarks. Our version of CM (1999) DFT call price formula, the equation (8.38), yields the price virtually identical to the original BS price for OTM and ITM calls even for extreme near-maturity case (i.e. $T = 1/252$) although the size of error is larger than BS-FT formula. But the error of BS-DFT price becomes large around (i.e. ± 3) ATM. In our example used, the maximum error is 0.0001345 which occurs at exactly ATM at $S_0 = K = 50$. Increasing the decay rate parameter α or sampling more points (i.e. larger N) cannot reduce the size of this error. But we can accept this size of error when considering the dramatic improvement in the CPU time to compute hundreds of prices.

[8.8] Summary of Formulae of Option Price with Fourier Transform

Table 8.14: Summary of Formulae of Option Price with Fourier Transform

Method	Formula
Traditional	$C = e^{-r(T-t)} \int_K^\infty (S_T - K) \mathbb{Q}(S_T \mathcal{F}_t) dS_T$
FT	$C(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha + 1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega} d\omega$ $\phi_T(.) \equiv \int_{-\infty}^\infty e^{i\omega \ln(S_T)} \mathbb{Q}(\ln(S_T) \mathcal{F}_0) d \ln(S_T), \quad k \equiv \ln K$

FT/TV

$$z_T(k) = \frac{1}{\sinh(\alpha k)} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \gamma_T(\omega) d\omega$$

$$\gamma_T(\omega) = \frac{\zeta_T(\omega - i\alpha) - \zeta_T(\omega + i\alpha)}{2}$$

$$\zeta_T(\omega) = e^{-rT} \left(\frac{1}{1+i\omega} - \frac{e^{rT}}{i\omega} - \frac{\phi_T(\omega - i)}{\omega^2 - i\omega} \right)$$

$$\phi_T(\cdot) \equiv \int_{-\infty}^{\infty} e^{i\omega \ln(S_T)} \mathbb{Q}(\ln(S_T) | \mathcal{F}_0) d \ln(S_T), \quad k \equiv \ln K$$

DFT

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k}$$

$$\times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\} \exp(-i2\pi jn/N)$$

DFT/TV

$$C_T(k_n) \approx \frac{\exp(i\pi n) \exp(-i\pi N/2)}{\sinh(\alpha k_n) \Delta k}$$

$$\times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \gamma_T(\omega_j) \right\} \exp(-i2\pi jn/N)$$

$$\gamma_T(\omega) = \frac{\zeta_T(\omega - i\alpha) - \zeta_T(\omega + i\alpha)}{2}$$

$$\zeta_T(\omega) = e^{-rT} \left(\frac{1}{1+i\omega} - \frac{e^{rT}}{i\omega} - \frac{\phi_T(\omega - i)}{\omega^2 - i\omega} \right)$$

$$\phi_T(\cdot) \equiv \int_{-\infty}^{\infty} e^{i\omega \ln(S_T)} \mathbb{Q}(\ln(S_T) | \mathcal{F}_0) d \ln(S_T), \quad k \equiv \ln K$$

[9] Merton (1976) Jump-Diffusion Model

[9.1] Model Type

In this section the basic structure of Merton JD model is described without the derivation of the model which will be done in the next section.

Merton JD model is an exponential Lévy model of the form:

$$S_t = S_0 e^{X_t},$$

where the asset price process $\{S_t; 0 \leq t \leq T\}$ is modeled as an exponential of a Lévy process $\{X_t; 0 \leq t \leq T\}$. Merton's choice of the Lévy process is a Brownian motion with drift (continuous diffusion process) plus a compound Poisson process (discontinuous jump process) such that:

$$X_t = \left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma B_t + \sum_{i=1}^{N_t} Y_i,$$

where $\{B_t; 0 \leq t \leq T\}$ is a standard Brownian motion process. The term

$\left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma B_t$ is a Brownian motion with drift process and the term $\sum_{i=1}^{N_t} Y_i$ is a compound Poisson jump process. The only difference between the Black-Scholes and the Merton jump-diffusion is the addition of the term $\sum_{i=1}^{N_t} Y_i$. A compound Poisson jump process $\sum_{i=1}^{N_t} Y_i$ contains two sources of randomness. The first is the Poisson process dN_t with intensity (i.e. average number of jumps per unit of time) λ which causes the asset price to jump randomly (i.e. random timing). Once the asset price jumps, how much it jumps is also modeled random (i.e. random jump size). Merton assumes that log stock price jump size follows normal distribution, $(dx_i) \sim i.i.d. Normal(\mu, \delta^2)$:

$$f(dx_i) = \frac{1}{\sqrt{2\pi\delta^2}} \exp\left\{-\frac{(dx_i - \mu)^2}{2\delta^2}\right\}.$$

It is assumed that these two sources of randomness are independent of each other. By introducing three extra parameters λ , μ , and δ to the original BS model, Merton JD model tries to capture the (negative) skewness and excess kurtosis of the log return density $\mathbb{P}(\ln(S_t / S_0))$ which deviates from the BS normal log return density.

Lévy measure $\ell(dx)$ of a compound Poisson process is given by the multiplication of the intensity and the jump size density $f(dx)$:

$$\ell(dx) = \lambda f(dx) .$$

A compound Poisson process (i.e. a piecewise constant Lévy process) is called finite activity Lévy process since its Lévy measure $\ell(dx)$ is finite (i.e. the average number of jumps per unit time is finite):

$$\int_{-\infty}^{\infty} \ell(dx) = \lambda < \infty .$$

The fact that an asset price S_t is modeled as an exponential of Lévy process X_t means that its log-return $\ln(\frac{S_t}{S_0})$ is modeled as a Lévy process such that:

$$\ln(\frac{S_t}{S_0}) = X_t = (\alpha - \frac{\sigma^2}{2} - \lambda k)t + \sigma B_t + \sum_{i=1}^{N_t} Y_i .$$

Let's derive the model.

[9.2] Model Derivation

In the jump-diffusion model, changes in the asset price consist of normal (continuous diffusion) component that is modeled by a Brownian motion with drift process and abnormal (discontinuous, i.e. jump) component that is modeled by a compound Poisson process. Asset price jumps are assumed to occur independently and identically. The probability that an asset price jumps during a small time interval dt can be written using a Poisson process dN_t as:

$$\Pr \{ \text{an asset price jumps once in } dt \} = \Pr \{ dN_t = 1 \} \cong \lambda dt ,$$

$$\Pr \{ \text{an asset price jumps more than once in } dt \} = \Pr \{ dN_t \geq 2 \} \cong 0 ,$$

$$\Pr \{ \text{an asset price does not jump in } dt \} = \Pr \{ dN_t = 0 \} \cong 1 - \lambda dt ,$$

where the parameter $\lambda \in \mathbb{R}^+$ is the intensity of the jump process (the mean number of jumps per unit of time) which is independent of time t .

Suppose in the small time interval dt the asset price jumps from S_t to $y_t S_t$ (we call y_t as absolute price jump size). So the relative price jump size (i.e. percentage change in the asset price caused by the jump) is:

$$\frac{dS_t}{S_t} = \frac{y_t S_t - S_t}{S_t} = y_t - 1 ,$$

where Merton assumes that the absolute price jump size y_t is a nonnegative random variables drawn from lognormal distribution, i.e. $\ln(y_t) \sim i.i.d.N(\mu, \delta^2)$. This in turn implies that $E[y_t] = e^{\mu + \frac{1}{2}\delta^2}$ and $E[(y_t - E[y_t])^2] = e^{2\mu + \delta^2}(e^{\delta^2} - 1)$. This is because if $\ln x \sim N(a, b)$, then $x \sim Lognormal(e^{a + \frac{1}{2}b^2}, e^{2a + b^2}(e^{b^2} - 1))$.

Merton's jump-diffusion dynamics of asset price which incorporates the above properties takes the SDE of the form:

$$\frac{dS_t}{S_t} = (\alpha - \lambda k)dt + \sigma dB_t + (y_t - 1)dN_t, \quad (9.1)$$

where α is the instantaneous expected return on the asset, σ is the instantaneous volatility of the asset return conditional on that jump does not occur, B_t is a standard Brownian motion process, and N_t is an Poisson process with intensity λ . Standard assumption is that (B_t) , (N_t) , and (y_t) are independent. The relative price jump size of S_t , $y_t - 1$, is lognormally distributed with the mean $E[y_t - 1] = e^{\mu + \frac{1}{2}\delta^2} - 1 \equiv k$ and the variance $E[(y_t - 1 - E[y_t - 1])^2] = e^{2\mu + \delta^2}(e^{\delta^2} - 1)$.¹ This may be confusing to some readers, so we will repeat it again. Merton assumes that the absolute price jump size y_t is a lognormal random variable such that:

$$(y_t) \sim i.i.d.Lognormal(e^{\mu + \frac{1}{2}\delta^2}, e^{2\mu + \delta^2}(e^{\delta^2} - 1)). \quad (9.2)$$

This is equivalent to saying that Merton assumes that the relative price jump size $y_t - 1$ is a lognormal random variable such that:

$$(y_t - 1) \sim i.i.d.Lognormal(k \equiv e^{\mu + \frac{1}{2}\delta^2} - 1, e^{2\mu + \delta^2}(e^{\delta^2} - 1)). \quad (9.3)$$

This is equivalent to saying that Merton assumes that the log price jump size $\ln y_t \equiv Y_t$ is a normal random variable such that:

$$\ln(y_t) \sim i.i.d.Normal(\mu, \delta^2). \quad (9.4)$$

This is equivalent to saying that Merton assumes that the log-return jump size $\ln(\frac{y_t S_t}{S_t})$ is a normal random variable such that:

¹For random variable x , $Variance[x - 1] = Variance[x]$.

$$\ln\left(\frac{y_t S_t}{S_t}\right) = \ln(y_t) \equiv Y_t \sim i.i.d. Normal(\mu, \delta^2). \quad (9.5)$$

It is extremely important to note:

$$E[y_t - 1] = e^{\mu + \frac{1}{2}\delta^2} - 1 \equiv k \neq E[\ln(y_t)] = \mu,$$

because $\ln E[y_t - 1] \neq E[\ln(y_t - 1)] = E[\ln(y_t)]$.

The expected relative price change $E[dS_t / S_t]$ from the jump part dN_t in the time interval dt is $\lambda k dt$ since $E[(y_t - 1)dN_t] = E[y_t - 1]E[dN_t] = k\lambda dt$. This is the predictable part of the jump. This is why the instantaneous expected return on the asset αdt is adjusted by $-\lambda k dt$ in the drift term of the jump-diffusion process to make the jump part an unpredictable innovation:

$$\begin{aligned} E\left[\frac{dS_t}{S_t}\right] &= E[(\alpha - \lambda k)dt] + E[\sigma dB_t] + E[(y_t - 1)dN_t] \\ E\left[\frac{dS_t}{S_t}\right] &= (\alpha - \lambda k)dt + 0 + \lambda k dt = \alpha dt. \end{aligned}$$

Some researchers include this adjustment term for predictable part of the jump $-\lambda k dt$ in the drift term of the Brownian motion process leading to the following simpler (?) specification:

$$\begin{aligned} \frac{dS_t}{S_t} &= \alpha dt + \sigma dB_t + (y_t - 1)dN_t \\ B_t &\sim Normal\left(-\frac{\lambda k t}{\sigma}, t\right) \\ E\left[\frac{dS_t}{S_t}\right] &= \alpha dt + \sigma\left(-\frac{\lambda k dt}{\sigma}\right) + \lambda k dt = \alpha dt. \end{aligned}$$

But we choose to explicitly subtract $\lambda k dt$ from the instantaneous expected return αdt because we prefer to keep B_t as a standard (zero-drift) Brownian motion process. Realize that there are two sources of randomness in the jump-diffusion process. The first source is the Poisson Process dN_t which causes the asset price to jump randomly. Once the asset price jumps, how much it jumps (the jump size) is also random. It is assumed that these two sources of randomness are independent of each other.

If the asset price does not jump in small time interval dt (i.e. $dN_t = 0$), then the jump-diffusion process is simply a Brownian motion with drift process:

$$\frac{dS_t}{S_t} = (\alpha - \lambda k)dt + \sigma dB_t.$$

If the asset price jumps in dt ($dN_t = 1$):

$$\frac{dS_t}{S_t} = (\alpha - \lambda k)dt + \sigma dB_t + (y_t - 1),$$

the relative price jump size is $y_t - 1$. Suppose that the lognormal random drawing y_t is 0.8, the asset price falls by 20%.

Let's solve SDE of (9.1). From (9.1), Merton jump-diffusion dynamics of an asset price is:

$$dS_t = (\alpha - \lambda k)S_t dt + \sigma S_t dB_t + (y_t - 1)S_t dN_t.$$

Appendix 10 gives the Itô formula for the jump-diffusion process as:

$$\begin{aligned} df(X_t, t) = & \frac{\partial f(X_t, t)}{\partial t} dt + b_t \frac{\partial f(X_t, t)}{\partial x} dt + \frac{\sigma_t^2}{2} \frac{\partial^2 f(X_t, t)}{\partial x^2} dt \\ & + \sigma_t \frac{\partial f(X_t, t)}{\partial x} dB_t + [f(X_{t-} + \Delta X_t) - f(X_{t-})], \end{aligned}$$

where b_t corresponds to the drift term and σ_t corresponds to the volatility term of a

jump-diffusion process $X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dB_s + \sum_{i=1}^{N_t} \Delta X_i$. By applying this:

$$\begin{aligned} d \ln S_t = & \frac{\partial \ln S_t}{\partial t} dt + (\alpha - \lambda k)S_t \frac{\partial \ln S_t}{\partial S_t} dt + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 \ln S_t}{\partial S_t^2} dt \\ & + \sigma S_t \frac{\partial \ln S_t}{\partial S_t} dB_t + [\ln y_t S_t - \ln S_t] \\ d \ln S_t = & (\alpha - \lambda k)S_t \frac{1}{S_t} dt + \frac{\sigma^2 S_t^2}{2} \left(-\frac{1}{S_t^2} \right) dt + \sigma S_t \frac{1}{S_t} dB_t + [\ln y_t + \ln S_t - \ln S_t] \\ d \ln S_t = & (\alpha - \lambda k)dt - \frac{\sigma^2}{2} dt + \sigma_t dB_t + \ln y_t \\ \ln S_t - \ln S_0 = & (\alpha - \frac{\sigma^2}{2} - \lambda k)(t - 0) + \sigma_t (B_t - B_0) + \sum_{i=1}^{N_t} \ln y_i \\ \ln S_t = & \ln S_0 + (\alpha - \frac{\sigma^2}{2} - \lambda k)t + \sigma_t B_t + \sum_{i=1}^{N_t} \ln y_i \end{aligned}$$

$$\exp(\ln S_t) = \exp \left\{ \ln S_0 + \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t + \sum_{i=1}^{N_t} \ln y_i \right\}$$

$$S_t = S_0 \exp \left\{ \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t \right\} \exp \left(\sum_{i=1}^{N_t} \ln y_i \right)$$

$$S_t = S_0 \exp \left[\left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t \right] \prod_{i=1}^{N_t} y_i,$$

or alternatively as:

$$S_t = S_0 \exp \left[\left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t + \sum_{i=1}^{N_t} \ln y_i \right].$$

Using the previous definition of the log price (return) jump size $\ln(y_i) \equiv Y_i$:

$$S_t = S_0 \exp \left[\left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t + \sum_{i=1}^{N_t} Y_i \right]. \quad (9.6)$$

This means that the asset price process $\{S_t; 0 \leq t \leq T\}$ is modeled as an exponential Lévy model of the form:

$$S_t = S_0 e^{X_t},$$

where X_t is a Lévy process which is categorized as a Brownian motion with drift (continuous part) plus a compound Poisson process (jump part) such that:

$$X_t = \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t + \sum_{i=1}^{N_t} Y_i.$$

In other words, log-return $\ln\left(\frac{S_t}{S_0}\right)$ is modeled as a Lévy process such that:

$$\ln\left(\frac{S_t}{S_0}\right) = X_t = \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t + \sum_{i=1}^{N_t} Y_i.$$

Note that the compound Poisson jump process $\prod_{i=1}^{N_t} y_i = 1$ (in absolute price scale) or

$\sum_{i=1}^{N_t} \ln y_i = \sum_{i=1}^{N_t} Y_i = 0$ (in log price scale) if $N_t = 0$ (i.e. no jumps between time 0 and t) or positive and negative jumps cancel each other out.

In the Black-Scholes case, log return $\ln(S_t / S_0)$ is normally distributed:

$$S_t = S_0 \exp\left[\left(\alpha - \frac{\sigma^2}{2}\right)t + \sigma B_t\right]$$

$$\ln\left(\frac{S_t}{S_0}\right) \sim \text{Normal}\left[\left(\alpha - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right].$$

But in jump-diffusion case, the existence of compound Poisson jump process $\sum_{i=1}^{N_t} Y_i$ makes log return non-normal. In Merton's case the simple distributional assumption about the log return jump size $(Y_i) \sim N(\mu, \delta^2)$ enables the probability density of log return $x_t = \ln(S_t / S_0)$ to be obtained as a quickly converging series of the following form:

$$\mathbb{P}(x_t \in A) = \sum_{i=0}^{\infty} \mathbb{P}(N_t = i) \mathbb{P}(x_t \in A | N_t = i)$$

$$\mathbb{P}(x_t) = \sum_{i=0}^{\infty} \frac{e^{-\lambda t} (\lambda t)^i}{i!} N(x_t; (\alpha - \frac{\sigma^2}{2} - \lambda k)t + i\mu, \sigma^2 t + i\delta^2) \quad (9.7)$$

where $N(x_t; (\alpha - \frac{\sigma^2}{2} - \lambda k)t + i\mu, \sigma^2 t + i\delta^2)$

$$= \frac{1}{\sqrt{2\pi(\sigma^2 t + i\delta^2)}} \exp\left[-\frac{\left[x_t - \left\{\left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + i\mu\right\}\right]^2}{2(\sigma^2 t + i\delta^2)}\right].$$

The term $\mathbb{P}(N_t = i) = \frac{e^{-\lambda t} (\lambda t)^i}{i!}$ is the probability that the asset price jumps i times during

the time interval of length t . And $\mathbb{P}(x_t \in A | N_t = i) = N(x_t; (\alpha - \frac{\sigma^2}{2} - \lambda k)t + i\mu, \sigma^2 t + i\delta^2)$

is the Black-Scholes normal density of log-return assuming that the asset price jumps i times in the time interval of t . Therefore, the log-return density in the Merton jump-diffusion model can be interpreted as the weighted average of the Black-Scholes normal density by the probability that the asset price jumps i times.

By Fourier transforming the Merton log-return density function with FT parameters $(a, b) = (1, 1)$, its characteristic function is calculated as:

$$\phi(\omega) = \int_{-\infty}^{\infty} \exp(i\omega x_t) \mathbb{P}(x_t) dx_t$$

$$= \exp \left[\lambda t \exp \left\{ \frac{1}{2} \omega (2i\mu - \delta^2 \omega) \right\} - \lambda t (1 + i\omega k) - \frac{1}{2} t \omega \{ -2i\alpha + \sigma^2 (i + \omega) \} \right].$$

After simplification:

$$\phi(\omega) = \exp[t\psi(\omega)]$$

with the characteristic exponent (cumulant generating function):

$$\psi(\omega) = \lambda \left\{ \exp \left(i\omega\mu - \frac{\delta^2 \omega^2}{2} \right) - 1 \right\} + i\omega \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) - \frac{\sigma^2 \omega^2}{2}, \quad (9.8)$$

where $k \equiv e^{\mu + \frac{1}{2}\delta^2} - 1$. The characteristic exponent (9.8) can be alternatively obtained by substituting the Lévy measure of the Meron jump-diffusion model:

$$\begin{aligned} \ell(dx) &= \frac{\lambda}{\sqrt{2\pi\delta^2}} \exp \left\{ -\frac{(dx - \mu)^2}{2\delta^2} \right\} = \lambda f(dx) \\ f(dx) &\sim N(\mu, \delta^2) \end{aligned}$$

into the Lévy-Khinchin representation of the equation (6.6):

$$\begin{aligned} \psi(\omega) &= i\omega\mu - \frac{\sigma^2 \omega^2}{2} + \int_{-\infty}^{\infty} \{ \exp(i\omega x) - 1 \} \ell(dx) \\ \psi(\omega) &= i\omega\mu - \frac{\sigma^2 \omega^2}{2} + \int_{-\infty}^{\infty} \{ \exp(i\omega x) - 1 \} \lambda f(dx) \\ \psi(\omega) &= i\omega\mu - \frac{\sigma^2 \omega^2}{2} + \lambda \int_{-\infty}^{\infty} \{ \exp(i\omega x) - 1 \} f(dx) \\ \psi(\omega) &= i\omega\mu - \frac{\sigma^2 \omega^2}{2} + \lambda \left\{ \int_{-\infty}^{\infty} e^{i\omega x} f(dx) - \int_{-\infty}^{\infty} f(dx) \right\} \end{aligned}$$

Note that $\int_{-\infty}^{\infty} e^{i\omega x} f(dx)$ is the characteristic function of $f(dx)$:

$$\int_{-\infty}^{\infty} e^{i\omega x} f(dx) = \exp \left(i\omega\mu - \frac{\delta^2 \omega^2}{2} \right).$$

Therefore:

$$\psi(\omega) = ib\omega - \frac{\sigma^2 \omega^2}{2} + \lambda \left\{ \exp \left(i\mu\omega - \frac{\delta^2 \omega^2}{2} \right) - 1 \right\},$$

where $b = \alpha - \frac{\sigma^2}{2} - \lambda k$. This corresponds to (9.8). Characteristic exponent (9.8) generates cumulants as follows:

$$\begin{aligned} \text{cumulant}_1 &= \alpha - \frac{\sigma^2}{2} - \lambda k + \lambda \mu, \\ \text{cumulant}_2 &= \sigma^2 + \lambda \delta^2 + \lambda \mu^2, \\ \text{cumulant}_3 &= \lambda(3\delta^2 \mu + \mu^3), \\ \text{cumulant}_4 &= \lambda(3\delta^4 + 6\mu^2 \delta^2 + \mu^4). \end{aligned}$$

Annualized (per unit of time) mean, variance, skewness, and excess kurtosis of the log-return density $\mathbb{P}(x_t)$ are computed from above cumulants as follows:

$$\begin{aligned} E[x_t] &= \text{cumulant}_1 = \alpha - \frac{\sigma^2}{2} - \lambda \left(e^{\mu + \frac{1}{2}\delta^2} - 1 \right) + \lambda \mu \\ \text{Variance}[x_t] &= \text{cumulant}_2 = \sigma^2 + \lambda \delta^2 + \lambda \mu^2 \\ \text{Skewness}[x_t] &= \frac{\text{cumulant}_3}{(\text{cumulant}_2)^{3/2}} = \frac{\lambda(3\delta^2 \mu + \mu^3)}{(\sigma^2 + \lambda \delta^2 + \lambda \mu^2)^{3/2}} \\ \text{Excess Kurtosis}[x_t] &= \frac{\text{cumulant}_4}{(\text{cumulant}_2)^{4/2}} = \frac{\lambda(3\delta^4 + 6\mu^2 \delta^2 + \mu^4)}{(\sigma^2 + \lambda \delta^2 + \lambda \mu^2)^2}. \end{aligned} \quad (9.9)$$

We can observe several interesting properties of Merton's log-return density $\mathbb{P}(x_t)$.

Firstly, the sign of μ which is the expected log-return jump size, $E[Y_t] = \mu$, determines the sign of skewness. The log-return density $\mathbb{P}(x_t)$ is negatively skewed if $\mu < 0$ and it is symmetric if $\mu = 0$ as illustrated in Figure 9.1.

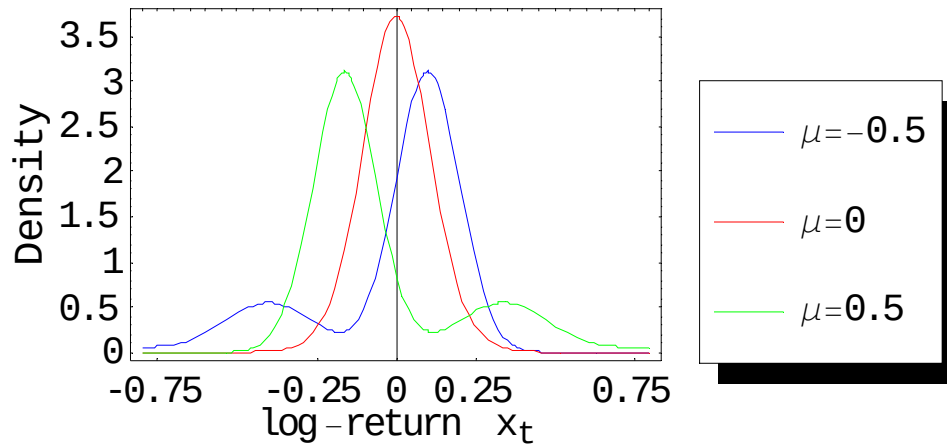


Figure 9.1: Merton's Log-Return Density for Different Values of μ . $\mu = -0.5$ in blue, $\mu = 0$ in red, and $\mu = 0.5$ in green. Parameters fixed are $\tau = 0.25$, $\alpha = 0.03$, $\sigma = 0.2$, $\lambda = 1$, and $\delta = 0.1$.

Table 9.1
Annualized Moments of Merton's Log-Return Density in Figure 9.1

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
$\mu = -0.5$	-0.0996	0.548	-0.852	0.864
$\mu = 0$	0.005	0.3742	0	0.12
$\mu = 0.5$	-0.147	0.5477	0.852	0.864

Secondly, larger value of intensity λ (which means that jumps are expected to occur more frequently) makes the density fatter-tailed as illustrated in Figure 9.2. Note that the excess kurtosis in the case $\lambda = 100$ is much smaller than in the case $\lambda = 1$ or $\lambda = 10$. This is because excess kurtosis is a standardized measure (by standard deviation).

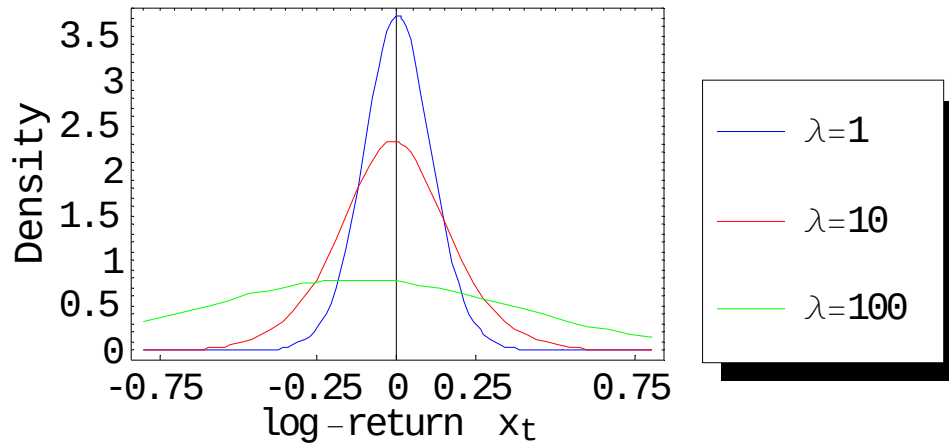


Figure 9.2: Merton's Log-Return Density for Different Values of Intensity λ . $\lambda = 1$ in blue, $\lambda = 10$ in red, and $\lambda = 100$ in green. Parameters fixed are $\tau = 0.25$, $\alpha = 0.03$, $\sigma = 0.2$, $\mu = 0$, and $\delta = 0.1$.

Table 9.2
Annualized Moments of Merton's Log-Return Density in Figure 9.2

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
$\lambda = 1$	0.00499	0.2236	0	0.12
$\lambda = 10$	-0.04012	0.3742	0	0.1531
$\lambda = 100$	-0.49125	1.0198	0	0.0277

Also note that Merton's log-return density has higher peak and fatter tails (more leptokurtic) when matched to the Black-Scholes normal counterpart as illustrated in Figure 9.3.

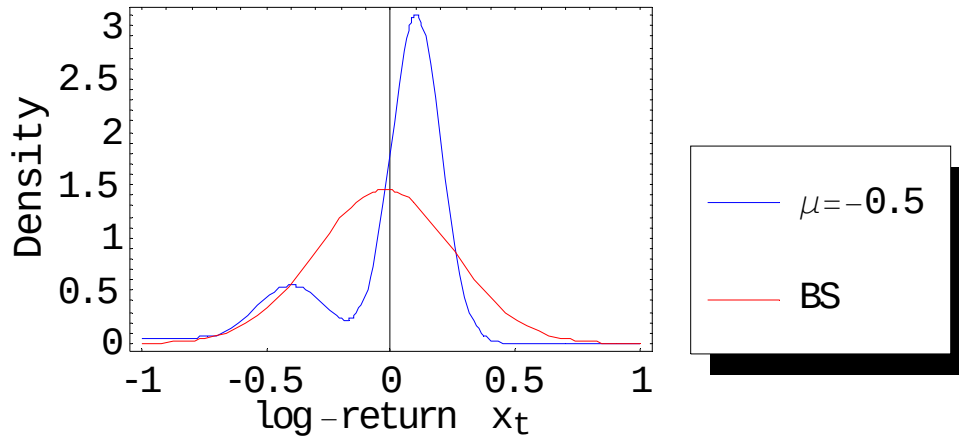


Figure 9.3: Merton Log-Return Density vs. Black-Scholes Log-Return Density (Normal). Parameters fixed for the Merton (in blue) are $\tau = 0.25$, $\alpha = 0.03$, $\sigma = 0.2$, $\lambda = 1$, $\mu = -0.5$, and $\delta = 0.1$. Black-Scholes normal log-return density is plotted (in red) by matching the mean and variance to the Merton.

Table 9.3
Annualized Moments of Merton vs. Black-Scholes Log-Return Density in Figure 9.3

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
Merton with $\mu = -0.5$	-0.0996	0.548	-0.852	0.864
Black-Scholes	-0.0996	0.548	0	0

[9.3] Log Stock Price Process for Merton Jump-Diffusion Model

Log stock price dynamics can be obtained from the equation (9.6) as:

$$\ln S_t = \ln S_0 + \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t + \sum_{i=1}^{N_t} Y_i. \quad (9.10)$$

Probability density of log stock price $\ln S_t$ is obtained as a quickly converging series of the following form (i.e. conditionally normal):

$$\begin{aligned} \mathbb{P}(\ln S_t \in A) &= \sum_{i=0}^{\infty} \mathbb{P}(N_t = i) \mathbb{P}(\ln S_t \in A | N_t = i) \\ \mathbb{P}(\ln S_t) &= \sum_{i=0}^{\infty} \frac{e^{-\lambda t} (\lambda t)^i}{i!} N \left(\ln S_t; \ln S_0 + \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + i\mu, \sigma^2 t + i\delta^2 \right), \end{aligned} \quad (9.11)$$

where:

$$\begin{aligned} &N \left(\ln S_t; \ln S_0 + \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + i\mu, \sigma^2 t + i\delta^2 \right) \\ &= \frac{1}{\sqrt{2\pi(\sigma^2 t + i\delta^2)}} \exp \left[- \frac{\left\{ \ln S_t - \left(\ln S_0 + \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t + i\mu \right) \right\}^2}{2(\sigma^2 t + i\delta^2)} \right]. \end{aligned}$$

By Fourier transforming (9.11) with FT parameters $(a, b,) = (1, 1)$, its characteristic function is calculated as:

$$\begin{aligned} \phi(\omega) &= \int_{-\infty}^{\infty} \exp(i\omega \ln S_t) \mathbb{P}(\ln S_t) d \ln S_t \\ &= \exp \left[\lambda t \left(\exp(i\mu\omega - \frac{\delta^2 \omega^2}{2}) - 1 \right) + i\omega \left(\ln S_0 + \left(\alpha - \frac{\sigma^2}{2} - \lambda k \right) t \right) - \frac{\sigma^2 \omega^2}{2} t \right], \end{aligned} \quad (9.12)$$

where $k = e^{\mu + \frac{\delta^2}{2}} - 1$.

[9.4] Lévy Measure for Merton Jump-Diffusion Model

We saw in section 6.3 that Lévy measure $\ell(dx)$ of a compound Poisson process is given by the multiplication of the intensity and the jump size density $f(dx)$:

$$\ell(dx) = \lambda f(dx).$$

The Lévy measure $\ell(dx)$ represents the arrival rate (i.e. total intensity) of jumps of sizes $[x, x + dx]$. In other words, we can interpret the Lévy measure $\ell(dx)$ of a compound Poisson process as the measure of the average number of jumps per unit of time. Lévy measure is a positive measure on $\mathbb{R}$, but it is not a probability measure since its total mass λ (in the compound Poisson case) does not have to equal 1:

$$\int \ell(dx) = \lambda \in \mathbb{R}^+.$$

A Poisson process and a compound Poisson process (i.e. a piecewise constant Lévy process) are called finite activity Lévy processes since their Lévy measures $\ell(dx)$ are finite (i.e. the average number of jumps per unit time is finite):

$$\int_{-\infty}^{\infty} \ell(dx) < \infty.$$

In Merton jump-diffusion case, the log-return jump size is $(dx_i) \sim i.i.d. Normal(\mu, \delta^2)$:

$$f(dx_i) = \frac{1}{\sqrt{2\pi\delta^2}} \exp\left\{-\frac{(dx_i - \mu)^2}{2\delta^2}\right\}.$$

Therefore, the Lévy measure $\ell(dx)$ for Merton case can be expressed as:

$$\ell(dx) = \lambda f(dx) = \frac{\lambda}{\sqrt{2\pi\delta^2}} \exp\left\{-\frac{(dx - \mu)^2}{2\delta^2}\right\}. \quad (9.13)$$

An example of Lévy measure $\ell(dx)$ for the log-return $x_t = \ln(S_t / S_0)$ in the Merton jump-diffusion model is plotted in Figure 9.5. Each Lévy measure is symmetric (i.e. $\mu = 0$ is used) with total mass 1, 2, and 4 respectively.

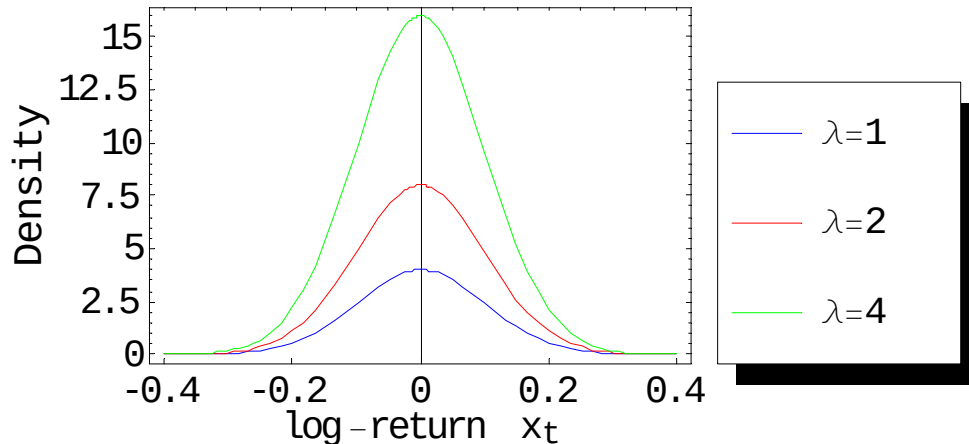


Figure 9.5: Lévy Measures $\ell(dx)$ for the Log-Return $x_t = \ln(S_t / S_0)$ in the Merton Jump-Diffusion Model for Different Values of Intensity λ . Parameters used are $\mu = 0$ and $\delta = 0.1$.

[9.5] Option (Contingent Claim) Pricing: PDE Approach by Hedging

Consider a portfolio P of the one long option position $V(S, t)$ on the underlying asset S written at time t and a short position of the underlying asset in quantity Δ to derive option pricing functions in the presence of jumps:

$$P_t = V(S_t, t) - \Delta S_t. \quad (9.14)$$

Portfolio value changes by in a very short period of time:

$$dP_t = dV(S_t, t) - \Delta dS_t. \quad (9.15)$$

Merton jump-diffusion dynamics of an asset price is given by equation (9.1) in the differential form as:

$$\begin{aligned} \frac{dS_t}{S_t} &= (\alpha - \lambda k)dt + \sigma dB_t + (y_t - 1)dN_t, \\ dS_t &= (\alpha - \lambda k)S_t dt + \sigma S_t dB_t + (y_t - 1)S_t dN_t. \end{aligned} \quad (9.16)$$

Appendix 8 gives the Itô formula for the jump-diffusion process as:

$$\begin{aligned} df(X_t, t) &= \frac{\partial f(X_t, t)}{\partial t} dt + b_t \frac{\partial f(X_t, t)}{\partial x} dt + \frac{\sigma_t^2}{2} \frac{\partial^2 f(X_t, t)}{\partial x^2} dt \\ &\quad + \sigma_t \frac{\partial f(X_t, t)}{\partial x} dB_t + [f(X_{t-} + \Delta X_t) - f(X_{t-})], \end{aligned}$$

where b_t corresponds to the drift term and σ_t corresponds to the volatility term of a jump-diffusion process $X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dB_s + \sum_{i=1}^{N_t} \Delta X_i$. Apply this to our case of option price function $V(S, t)$:

$$\begin{aligned} dV(S_t, t) &= \frac{\partial V}{\partial t} dt + (\alpha - \lambda k)S_t \frac{\partial V}{\partial S_t} dt + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} dt + \sigma S_t \frac{\partial V}{\partial S_t} dB_t \\ &\quad + [V(y_t S_t, t) - V(S_t, t)]dN_t. \end{aligned} \quad (9.17)$$

The term $[V(y_t S_t, t) - V(S_t, t)]dN_t$ describes the difference in the option value when a jump occurs. Now the change in the portfolio value can be expressed as by substituting (9.16) and (9.17) into (9.15):

$$\begin{aligned}
dP_t &= dV(S_t, t) - \Delta dS_t \\
dP_t &= \frac{\partial V}{\partial t} dt + (\alpha - \lambda k) S_t \frac{\partial V}{\partial S_t} dt + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} dt + \sigma S_t \frac{\partial V}{\partial S_t} dB_t + [V(y_t S_t, t) - V(S_t, t)] dN_t \\
&\quad - \Delta \{ (\alpha - \lambda k) S_t dt + \sigma S_t dB_t + (y_t - 1) S_t dN_t \} \\
dP_t &= \left\{ \frac{\partial V}{\partial t} + (\alpha - \lambda k) S_t \frac{\partial V}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} - \Delta (\alpha - \lambda k) S_t \right\} dt + \left(\sigma S_t \frac{\partial V}{\partial S_t} - \Delta \sigma S_t \right) dB_t \\
&\quad + \{ V(y_t S_t, t) - V(S_t, t) - \Delta (y_t - 1) S_t \} dN_t. \tag{9.18}
\end{aligned}$$

If there is no jump between time 0 and t (i.e. $dN_t = 0$), the problem reduces to Black-Scholes case in which setting $\Delta = \partial V / \partial S_t$ makes the portfolio risk-free leading to the following (i.e. the randomness dB_t has been eliminated):

$$\begin{aligned}
dP_t &= \left\{ \frac{\partial V}{\partial t} + (\alpha - \lambda k) S_t \frac{\partial V}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} - \frac{\partial V}{\partial S_t} (\alpha - \lambda k) S_t \right\} dt + \left(\sigma S_t \frac{\partial V}{\partial S_t} - \frac{\partial V}{\partial S_t} \sigma S_t \right) dB_t \\
dP_t &= \left\{ \frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} \right\} dt.
\end{aligned}$$

This in turn means that if there is a jump between time 0 and t (i.e. $dN_t \neq 0$), setting $\Delta = \partial V / \partial S_t$ does not eliminate the risk. Suppose we decided to hedge the randomness caused by diffusion part dB_t in the underlying asset price (which are always present) and not to hedge the randomness caused by jumps dN_t (which occur infrequently) by setting $\Delta = \partial V / \partial S_t$. Then, the change in the value of the portfolio is given by from equation (9.18):

$$\begin{aligned}
dP_t &= \left\{ \frac{\partial V}{\partial t} + (\alpha - \lambda k) S_t \frac{\partial V}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} - \frac{\partial V}{\partial S_t} (\alpha - \lambda k) S_t \right\} dt + \left(\sigma S_t \frac{\partial V}{\partial S_t} - \frac{\partial V}{\partial S_t} \sigma S_t \right) dB_t \\
&\quad + \left\{ V(y_t S_t, t) - V(S_t, t) - \frac{\partial V}{\partial S_t} (y_t - 1) S_t \right\} dN_t, \\
dP_t &= \left\{ \frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} \right\} dt + \left\{ V(y_t S_t, t) - V(S_t, t) - \frac{\partial V}{\partial S_t} (y_t - 1) S_t \right\} dN_t. \tag{9.19}
\end{aligned}$$

Merton argues that the jump component (dN_t) of the asset price process S_t is uncorrelated with the market as a whole. Then, the risk of jump is diversifiable (non-systematic) and it should earn no risk premium. Therefore, the portfolio is expected to grow at the risk-free interest rate r :

$$E[dP_t] = rP_t dt. \quad (9.20)$$

After substitution of (9.14) and (9.19) into (9.20) by setting $\Delta = \partial V / \partial S_t$:

$$\begin{aligned} E\left[\left\{\frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2}\right\} dt + \left\{V(y_t S_t, t) - V(S_t, t) - \frac{\partial V}{\partial S_t}(y_t - 1)S_t\right\} dN_t\right] &= r\{V(S_t, t) - \Delta S_t\} dt \\ \left(\frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2}\right) dt + E[V(y_t S_t, t) - V(S_t, t) - \frac{\partial V}{\partial S_t}(y_t - 1)S_t] E[dN_t] &= r\{V(S_t, t) - \frac{\partial V}{\partial S_t} S_t\} dt \\ \left(\frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2}\right) dt + E[V(y_t S_t, t) - V(S_t, t) - \frac{\partial V}{\partial S_t}(y_t - 1)S_t] \lambda dt &= r\{V(S_t, t) - \frac{\partial V}{\partial S_t} S_t\} dt \\ \frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} + \lambda E[V(y_t S_t, t) - V(S_t, t) - \frac{\partial V}{\partial S_t}(y_t - 1)S_t] &= r\{V(S_t, t) - \frac{\partial V}{\partial S_t} S_t\} \end{aligned}$$

Thus, the Merton jump-diffusion counterpart of Black-Scholes PDE is:

$$\frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} + rS_t \frac{\partial V}{\partial S_t} - rV + \lambda E[V(y_t S_t, t) - V(S_t, t)] - \lambda S_t \frac{\partial V}{\partial S_t} E[y_t - 1] = 0. \quad (9.21)$$

where the term $E[V(y_t S_t, t) - V(S_t, t)]$ involves the expectation operator and

$E[y_t - 1] = e^{\mu + \frac{1}{2}\delta^2} - 1 \equiv k$ (which is the mean of relative asset price jump size). Obviously, if jump is not expected to occur (i.e. $\lambda = 0$), this reduces to Black-Scholes PDE:²

$$\frac{\partial V}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} + rS_t \frac{\partial V}{\partial S_t} - rV = 0.$$

Merton's simple assumption that the absolute price jump size is lognormally distributed (i.e. the log-return jump size is normally distributed, $Y_t \equiv \ln(y_t) \sim N(\mu, \delta^2)$) makes it possible to solve the jump-diffusion PDE to obtain the following price function of European vanilla options as a quickly converging series of the form:

$$\sum_{i=0}^{\infty} \frac{e^{-\bar{\lambda}\tau} (\bar{\lambda}\tau)^i}{i!} V_{BS}(S_t, \tau = T - t, \sigma_i, r_i), \quad (9.22)$$

where $\bar{\lambda} = \lambda(1 + k) = \lambda e^{\mu + \frac{1}{2}\delta^2}$,

$$\sigma_i^2 = \sigma^2 + \frac{i\delta^2}{\tau},$$

² This equation not only contains local derivatives but also links together option values at discontinuous values in S . This is called non-local nature.

$$r_i = r - \lambda k + \frac{i \ln(1+k)}{\tau} = r - \lambda(e^{\mu + \frac{1}{2}\delta^2} - 1) + \frac{i(\mu + \frac{1}{2}\delta^2)}{\tau},$$

and V_{BS} is the Black-Scholes price without jumps.

Thus, Merton's jump-diffusion option price can be interpreted as the weighted average of the Black-Scholes price conditional on that the underlying asset price jumps i times to the expiry with weights being the probability that the underlying jumps i times to the expiry.

[9.6] Option (Contingent Claim) Pricing: Martingale Approach

Let $\{B_t; 0 \leq t \leq T\}$ be a standard Brownian motion process on a space $(\Omega, \mathcal{F}, \mathbb{P})$. Under actual probability measure $\mathbb{P}$, the dynamics of Merton jump-diffusion asset price process is given by equation (9.6) in the integral form:

$$S_t = S_0 \exp[(\alpha - \frac{\sigma^2}{2} - \lambda k)t + \sigma B_t + \sum_{k=1}^{N_t} Y_k] \text{ under } \mathbb{P}.$$

We changed the index from $\sum_{i=1}^{N_t} Y_i$ to $\sum_{k=1}^{N_t} Y_k$. This is trivial but readers will find the reason soon. Merton jump-diffusion model is an example of an incomplete model because there are many equivalent martingale risk-neutral measures $\mathbb{Q} \sim \mathbb{P}$ under which the discounted asset price process $\{e^{-rt} S_t; 0 \leq t \leq T\}$ becomes a martingale. Merton finds his equivalent martingale risk-neutral measure $\mathbb{Q}_M \sim \mathbb{P}$ by changing the drift of the Brownian motion process while keeping the other parts (most important is the jump measure, i.e. the distribution of jump times and jump sizes) unchanged:

$$S_t = S_0 \exp[(r - \frac{\sigma^2}{2} - \lambda k)t + \sigma B_t^{\mathbb{Q}_M} + \sum_{k=1}^{N_t} Y_k] \text{ under } \mathbb{Q}_M. \quad (9.23)$$

Note that $B_t^{\mathbb{Q}_M}$ is a standard Brownian motion process on $(\Omega, \mathcal{F}, \mathbb{Q}_M)$ and the process $\{e^{-rt} S_t; 0 \leq t \leq T\}$ is a martingale under $\mathbb{Q}_M$. Then, a European option price $V^{Merton}(t, S_t)$ with payoff function $H(S_T)$ is calculated as:

$$V^{Merton}(t, S_t) = e^{-r(T-t)} E^{\mathbb{Q}_M}[H(S_T) | \mathcal{F}_t]. \quad (9.24)$$

Standard assumption is $\mathcal{F}_t = S_t$, thus:

$$V^{Merton}(t, S_t) = e^{-r(T-t)} E^{\mathbb{Q}_M}[H(S_t \exp[(r - \frac{\sigma^2}{2} - \lambda k)(T-t) + \sigma B_{T-t}^{\mathbb{Q}_M} + \sum_{k=1}^{N_{T-t}} Y_k]) | S_t]$$

$$V^{Merton}(t, S_t) = e^{-r(T-t)} E^{\mathbb{Q}_M} [H(S_t \exp[(r - \frac{\sigma^2}{2} - \lambda k)(T-t) + \sigma B_{T-t}^{\mathbb{Q}_M} + \sum_{k=1}^{N_{T-t}} Y_k])]. \quad (9.25)$$

Poisson counter is (we would like to use index i for the number of jumps):

$$N_{T-t} = 0, 1, 2, \dots \equiv i.$$

And the compound Poisson process is distributed as:

$$\sum_{k=1}^{N_{T-t}} Y_k \sim \text{Normal}(i\mu, i\delta^2).$$

Thus, $V^{Merton}(t, S_t)$ can be expressed as from equation (9.25) (i.e. by conditioning on i):

$$\begin{aligned} V^{Merton}(t, S_t) \\ = e^{-r(T-t)} \sum_{i \geq 0} \mathbb{Q}_M(N_{T-t} = i) E^{\mathbb{Q}_M} [H(S_t \exp[(r - \frac{\sigma^2}{2} - \lambda k)(T-t) + \sigma B_{T-t}^{\mathbb{Q}_M} + \sum_{k=1}^i Y_k])]. \end{aligned}$$

Use $\tau = T - t$:

$$\begin{aligned} V^{Merton}(t, S_t) \\ = e^{-r\tau} \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} E^{\mathbb{Q}_M} \left[H \left(S_t \exp \left[\left\{ r - \frac{\sigma^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + \sigma B_{\tau}^{\mathbb{Q}_M} + \sum_{k=1}^i Y_k \right] \right) \right]. \quad (9.26) \end{aligned}$$

Inside the exponential function is normally distributed:

$$\begin{aligned} \left\{ r - \frac{\sigma^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + \sigma B_{\tau}^{\mathbb{Q}_M} + \sum_{k=1}^i Y_k \\ \sim \text{Normal} \left(\left\{ r - \frac{\sigma^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + i\mu, \sigma^2 \tau + i\delta^2 \right). \end{aligned}$$

Rewrite it so that its distribution remains the same:

$$\begin{aligned} \left\{ r - \frac{\sigma^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + i\mu + \sqrt{\frac{\sigma^2 \tau + i\delta^2}{\tau}} B_{\tau}^{\mathbb{Q}_M} \\ \sim \text{Normal} \left(\left\{ r - \frac{\sigma^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + i\mu, \sigma^2 \tau + i\delta^2 \right). \end{aligned}$$

Now we can rewrite equation (9.24) as (we can do this operation because a normal density is uniquely determined by only two parameters: its mean and variance):

$$V^{Merton}(t, S_t) = e^{-r\tau} \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} E^{\mathbb{Q}_M} \left[H \left(S_t \exp \left[\left\{ r - \frac{\sigma^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + i\mu + \sqrt{\frac{\sigma^2\tau + i\delta^2}{\tau}} B_\tau^{\mathbb{Q}_M} \right] \right) \right]$$

We can always add $(\frac{i\delta^2}{2\tau} - \frac{i\delta^2}{2\tau}) = 0$ inside the exponential function:

$$\begin{aligned} V^{Merton}(t, S_t) &= e^{-r\tau} \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} \times \\ &E^{\mathbb{Q}_M} \left[H \left(S_t \exp \left[\left\{ r - \frac{\sigma^2}{2} + \left(\frac{i\delta^2}{2\tau} - \frac{i\delta^2}{2\tau} \right) - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + i\mu + \sqrt{\sigma^2 + \frac{i\delta^2}{\tau}} B_\tau^{\mathbb{Q}_M} \right] \right) \right] \\ V^{Merton}(t, S_t) &= e^{-r\tau} \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} \times \\ &E^{\mathbb{Q}_M} \left[H \left(S_t \exp \left[\left\{ r - \frac{1}{2}(\sigma^2 + \frac{i\delta^2}{\tau}) + \frac{i\delta^2}{2\tau} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + i\mu + \sqrt{\sigma^2 + \frac{i\delta^2}{\tau}} B_\tau^{\mathbb{Q}_M} \right] \right) \right] \end{aligned}$$

Set $\sigma_i^2 = \sigma^2 + \frac{i\delta^2}{\tau}$ and rearrange:

$$\begin{aligned} V^{Merton}(t, S_t) &= e^{-r\tau} \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} E^{\mathbb{Q}_M} \left[H \left(S_t \exp \left[\left\{ r - \frac{1}{2}\sigma_i^2 + \frac{i\delta^2}{2\tau} - \lambda(e^{\mu+\delta^2/2} - 1) \right\} \tau + i\mu + \sigma_i B_\tau^{\mathbb{Q}_M} \right] \right) \right] \\ &= e^{-r\tau} \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} E^{\mathbb{Q}_M} \left[H \left(S_t \exp \left\{ i\mu + \frac{i\delta^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1)\tau \right\} \exp \left\{ \left(r - \frac{1}{2}\sigma_i^2 \right) \tau + \sigma_i B_\tau^{\mathbb{Q}_M} \right\} \right) \right]. \end{aligned}$$

Black-Scholes price can be expressed as:

$$V^{BS}(\tau = T - t, S_t; \sigma) = e^{-r\tau} E^{\mathbb{Q}_{BS}} [H\{S_t \exp(r - \frac{1}{2}\sigma^2)\tau + \sigma B_\tau^{\mathbb{Q}_{BS}}\}].$$

Finally, Merton jump-diffusion pricing formula can be obtained as a weighted average of Black-Scholes price conditioned on the number of jumps i :

$$\begin{aligned} V^{Merton}(t, S_t) &= \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} V^{BS}(\tau, S_i \equiv S_t \exp\{i\mu + \frac{i\delta^2}{2} - \lambda(e^{\mu+\delta^2/2} - 1)\tau\}; \sigma_i \equiv \sqrt{\sigma^2 + \frac{i\delta^2}{\tau}}). \quad (9.27) \end{aligned}$$

Alternatively:

$$\begin{aligned}
V^{Merton}(t, S_t) &= e^{-r\tau} \sum_{i \geq 0} \frac{e^{-\lambda\tau} (\lambda\tau)^i}{i!} E^{\mathbb{Q}_M} \left[H \left\{ S_t \exp \left\{ \left(r - \lambda(e^{\mu+\delta^2/2} - 1) + \frac{i\mu + i\delta^2/2}{\tau} - \frac{1}{2} \sigma_i^2 \right) \tau + \sigma_i B_\tau^{\mathbb{Q}_M} \right\} \right\} \right] \\
&= \sum_{i \geq 0} \frac{e^{-\bar{\lambda}\tau} (\bar{\lambda}\tau)^i}{i!} V^{BS}(\tau, S_t; \sigma_i \equiv \sqrt{\sigma^2 + \frac{i\delta^2}{\tau}}, r_i \equiv r - \lambda(e^{\mu+\delta^2/2} - 1) + \frac{i\mu + i\delta^2/2}{\tau}). \quad (9.28)
\end{aligned}$$

where $\bar{\lambda} = \lambda(1+k) = \lambda e^{\mu + \frac{1}{2}\delta^2}$. As you might notice, this is the same result as the option pricing formula derived from solving a PDE by forming a risk-free portfolio in equation (9.22). PDE approach and Martingale approach are different approaches but they are related and give the same result.

[9.7] Option Pricing Example of Merton Jump-Diffusion Model

In this section the equation (9.22) is used to price hypothetical plain vanilla options: current stock price $S_t = 50$, risk-free interest rate $r = 0.05$, continuously compounded dividend yield $q = 0.02$, time to maturity $\tau = 0.25$ years.

We need to be careful about volatility σ . In the Black-Scholes case, the t -period standard deviation of log-return x_t is:

$$\text{Standard Deviation}_{BS}(x_t) = \sigma_{BS} \sqrt{t}. \quad (9.29)$$

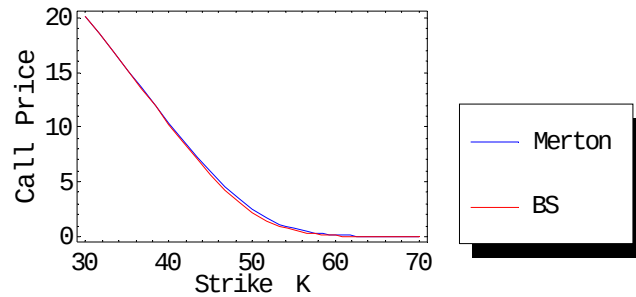
Equation (9.10) tells that the t -period standard deviation of log-return x_t in the Merton model is given as:

$$\text{Standard Deviation}_{Merton}(x_t) = \sqrt{(\sigma_{Merton}^2 + \lambda\delta^2 + \lambda\mu^2)t}. \quad (9.30)$$

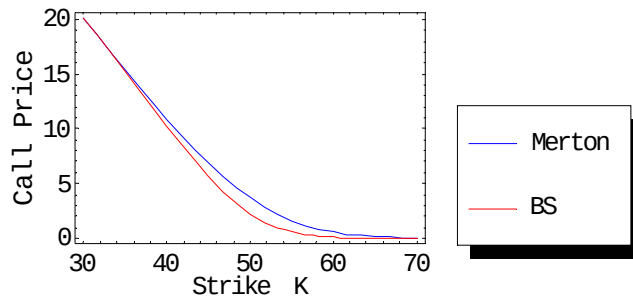
This means that if we set $\sigma_{BS} = \sigma_{Merton}$, Merton jump-diffusion prices are always greater (or equal to) than Black-Scholes prices because of the extra source of volatility λ (intensity), μ (mean log-return jump size), δ (standard deviation of log-return jump) (i.e. larger volatility is translated to larger option price):

$$\begin{aligned}
\text{Standard Deviation}_{BS}(x_t) &\leq \text{Standard Deviation}_{Merton}(x_t) \\
\sigma_{BS} \sqrt{t} &\leq \sqrt{(\sigma_{Merton}^2 + \lambda\delta^2 + \lambda\mu^2)t}.
\end{aligned}$$

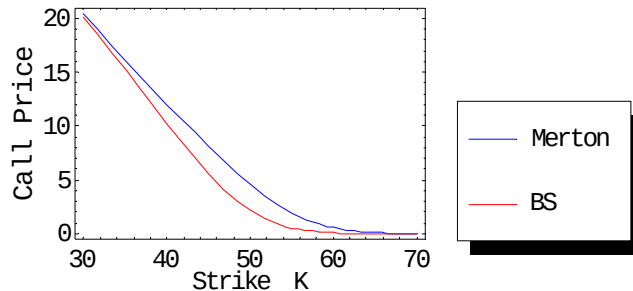
This very obvious point is illustrated in Figure 9.6 where diffusion volatility is set $\sigma_{BS} = \sigma_{Merton} = 0.2$. Note the followings: (1) In all four panels Merton jump-diffusion price is always greater (or equal to) than Black-Scholes price. (2) When Merton parameters (λ , μ , and δ) are small in Panel A, the difference between these two prices is small. (3) As intensity λ increases (i.e. increased expected number of jumps per unit of time), the t -period Merton standard deviation of log-return x_t increases (equation (9.30)) leading to the larger difference between Merton price and Black-Scholes price as illustrated in Panel B. (4) As Merton mean log-return jump size μ increases, the t -period Merton standard deviation of log-return x_t increases (equation (9.30)) leading to the larger difference between Merton price and Black-Scholes price as illustrated in Panel C. (5) As Merton standard deviation of log-return jump size δ increases, the t -period Merton standard deviation of log-return x_t increases (equation (9.30)) leading to the larger difference between Merton price and Black-Scholes price as illustrated in Panel D.



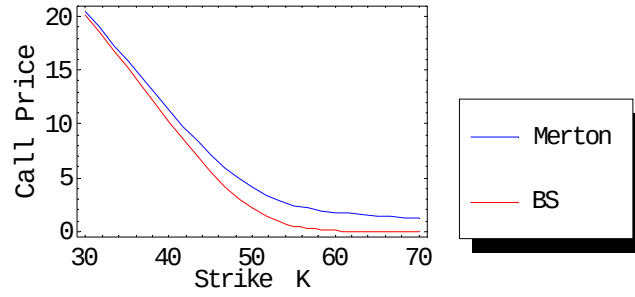
A) Merton parameters: $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$.



B) Merton parameters: $\lambda = 5$, $\mu = -0.1$, and $\delta = 0.1$.



C) Merton parameters: $\lambda = 1$, $\mu = -0.5$, and $\delta = 0.1$.



D) Merton parameters: $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.5$.

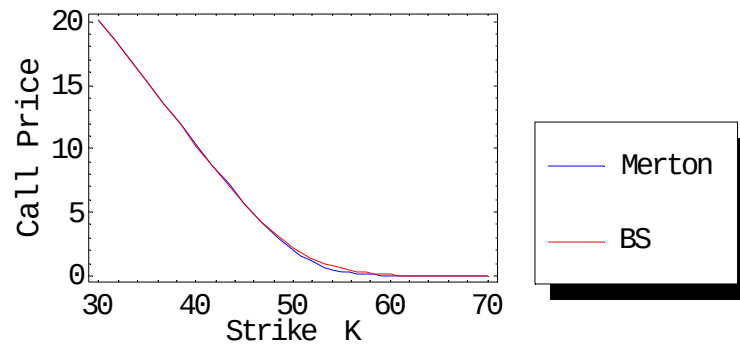
Figure 9.6: Merton Jump-Diffusion Call Price vs. Black-Scholes Call Price When Diffusion Volatility σ is same. Parameters and variables used are $S_t = 50$, $r = 0.05$, $q = 0.02$, $\tau = 0.25$, and $\sigma_{BS} = \sigma_{Merton} = 0.2$.

Next we consider a more delicate case where we restrict diffusion volatilities σ_{BS} and σ_{Merton} such that standard deviations of Merton jump-diffusion and Black-Scholes log-return densities are the same:

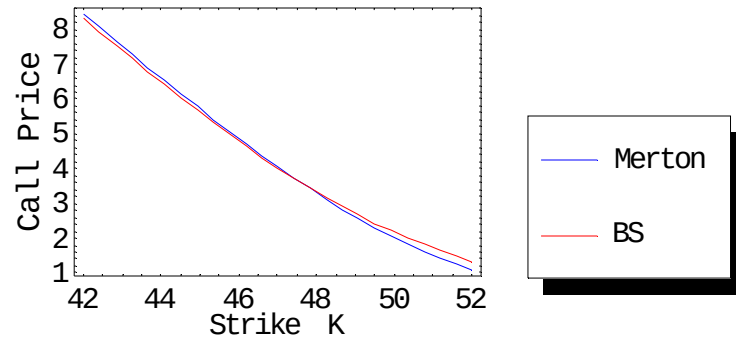
$$\text{Standard Deviation}_{BS}(x_t) = \text{Standard Deviation}_{Merton}(x_t),$$

$$\sigma_{BS} \sqrt{t} = \sqrt{(\sigma_{Merton}^2 + \lambda \delta^2 + \lambda \mu^2) t}.$$

Using the Merton parameters $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$ and Black-Scholes volatility $\sigma_{BS} = 0.2$, Merton diffusion volatility is calculated as $\sigma_{Merton} = 0.141421$. In this same standard deviation case, call price function is plotted in Figure 9.7 and put price function is plotted in Figure 9.8. It seems that Merton jump diffusion model overestimates in-the-money call and underestimates out-of-the-money call when compared to Black-Scholes model. And Merton jump diffusion model overestimates out-of-the-money put and underestimates in-the-money put when compared to Black-Scholes model.

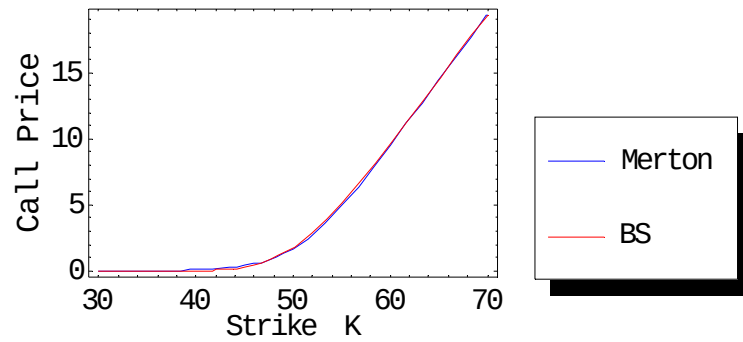


A) Range 30 to 70.

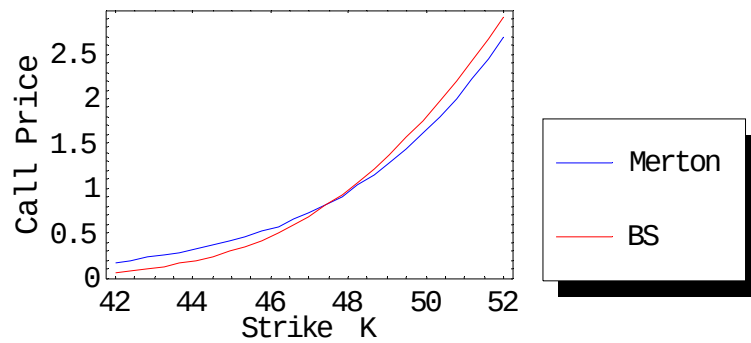


B) Range 42 to 52.

Figure 9.7: Merton Jump-Diffusion Call Price vs. Black-Scholes Call Price When Restricting Merton Diffusion Volatility σ_{Merton} . We set $\sigma_{BS} = 0.2$ and $\sigma_{Merton} = 0.141421$. Parameters and variables used are $S_t = 50$, $r = 0.05$, $q = 0.02$, $\tau = 0.25$.



A) Range 30 to 70.



B) Range 42 to 52.

Figure 9.8: Merton Jump-Diffusion Put Price vs. Black-Scholes Put Price When Restricting Merton Diffusion Volatility σ_{Merton} . We set $\sigma_{BS} = 0.2$ and $\sigma_{Merton} = 0.141421$. Parameters and variables used are $S_t = 50$, $r = 0.05$, $q = 0.02$, $\tau = 0.25$.

[10] Merton (1976) Jump-Diffusion Model with Fourier Transform Pricing

We saw in section 9 that the most important character of Merton JD model is its conditional Black-Scholes property. Because Merton JD model is a simple addition of compound Poisson jump process to the original BS model (i.e. geometric Brownian motion with drift), everything in the Merton JD model such as log return density $\ln(S_t / S_0)$ or even call price can be expressed as the weighted average of the BS (log return density or price) conditional on that the underlying asset price jumps i times to the expiry with weights being the probability that the underlying jumps i times to the expiry. Traditional pricing approach with Merton JD model is discussed in the section 9.5 (PDE approach by hedging) and 9.6 (martingale approach).

In this section we present Fourier transform option pricing approach with Merton JD model. You will see FT method is much more general and simpler.

[10.1] Merton JD Model with Fourier Transform Pricing Method

The first step to FT option pricing is to obtain the characteristic function of the log stock price $\ln S_t$. Similar arguments can be seen in section 9.3. Risk-neutral log stock price dynamics can be obtained from the equation (9.6) as:

$$\ln S_t = \ln S_0 + \left(r - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma B_t + \sum_{i=1}^{N_t} Y_i, \quad (10.1)$$

where $k \equiv e^{\mu + \frac{1}{2}\delta^2} - 1$. Probability density of log stock price $\ln S_t$ is obtained as a quickly converging series of the following form (i.e. conditionally normal):

$$\begin{aligned} \mathbb{P}(\ln S_t \in A) &= \sum_{i=0}^{\infty} \mathbb{P}(N_t = i) \mathbb{P}(\ln S_t \in A | N_t = i) \\ \mathbb{P}(\ln S_t) &= \sum_{i=0}^{\infty} \frac{e^{-\lambda t} (\lambda t)^i}{i!} N \left(\ln S_t; \ln S_0 + \left(r - \frac{\sigma^2}{2} - \lambda k \right) t + i\mu, \sigma^2 t + i\delta^2 \right), \end{aligned} \quad (10.2)$$

where:

$$\begin{aligned} &N \left(\ln S_t; \ln S_0 + \left(r - \frac{\sigma^2}{2} - \lambda k \right) t + i\mu, \sigma^2 t + i\delta^2 \right) \\ &= \frac{1}{\sqrt{2\pi(\sigma^2 t + i\delta^2)}} \exp \left[-\frac{\left\{ \ln S_t - \left(\ln S_0 + \left(r - \frac{\sigma^2}{2} - \lambda k \right) t + i\mu \right) \right\}^2}{2(\sigma^2 t + i\delta^2)} \right]. \end{aligned}$$

Let $s_t \equiv \ln S_t$ as before. By Fourier transforming (10.2) with FT parameters $(a, b) = (1, 1)$, its characteristic function is calculated as:

$$\begin{aligned}\phi(\omega) &= \int_{-\infty}^{\infty} e^{i\omega s_t} \mathbb{P}(s_t) ds_t \\ &= \exp \left[\lambda t \left(\exp(i\mu\omega - \frac{\delta^2 \omega^2}{2}) - 1 \right) + i\omega \left(s_0 + (r - \frac{\sigma^2}{2} - \lambda k)t \right) - \frac{\sigma^2 \omega^2}{2} t \right].\end{aligned}\quad (10.3)$$

By substituting a characteristic function of Merton log terminal stock price s_T , the equation (10.3), into the general FT pricing formula of the equation (8.17), we obtain Merton-FT call pricing formula:

$$C_{\text{Merton-FT}}(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha + 1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega} d\omega, \quad (10.4)$$

where:

$$\phi_T(\omega) = \exp \left[\lambda T \left(\exp(i\mu\omega - \frac{\delta^2 \omega^2}{2}) - 1 \right) + i\omega \left(s_0 + (r - \frac{\sigma^2}{2} - \lambda k)T \right) - \frac{\sigma^2 \omega^2}{2} T \right].$$

We implement the Merton-FT formula (10.4) with decay rate parameter $\alpha = 1$ and compare the result to the original Merton call price of the equation (9.22) using common parameters and variables fixed $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$. As illustrated by Figure 10.1, as a principle Merton-FT call price and the original Merton call price are identical. This is no surprise because the original Merton formula (9.22) and Merton-FT formula (10.4) are the same person with a different look. Merton-FT formula is just frequency representation of the original Merton formula.

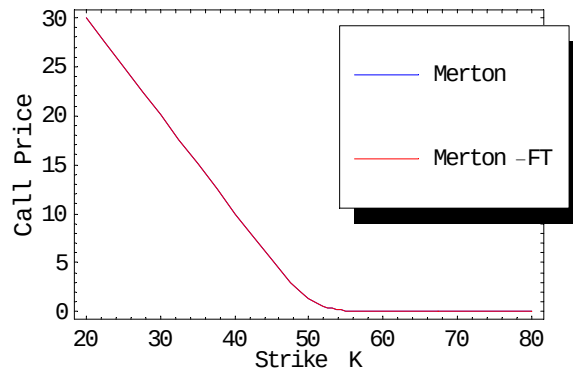


Figure 10.1: Original Merton Call Price Vs. Merton-FT with $\alpha = 1$ Call Price. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$.

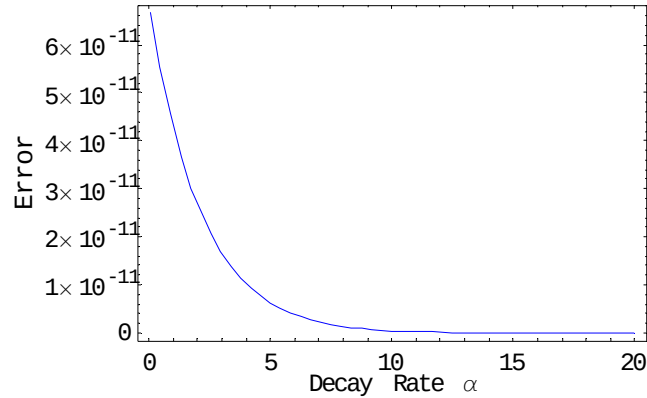
Next, CPU time should be discussed. Consider call options with $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 20/252$ as a function of varying strike price K . We use Merton JD parameters $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$. Table 10.1 reveals that although Merton-FT formula is slower than the original Merton, speed is not an issue for most of the purposes.

Table 10.1: CPU Time for Calls with Different Moneyness

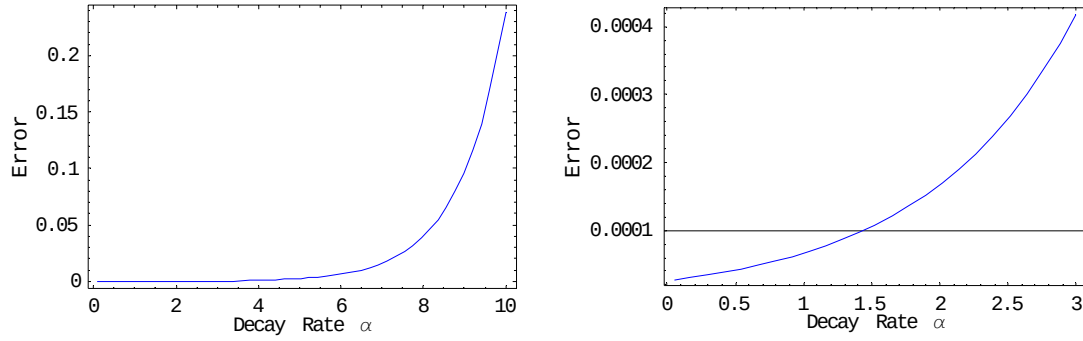
Price is in the bracket.

Method	$K = 20$	Strike $K = 50$	$K = 80$
Merton	0.01 seconds (29.9999)	0.01 seconds (1.32941)	0.01 seconds (1.19634×10^{-7})
Merton-FT $\alpha = 1$	0.06 seconds (29.9999)	0.02 seconds (1.32941)	0.12 seconds (1.19634×10^{-7})

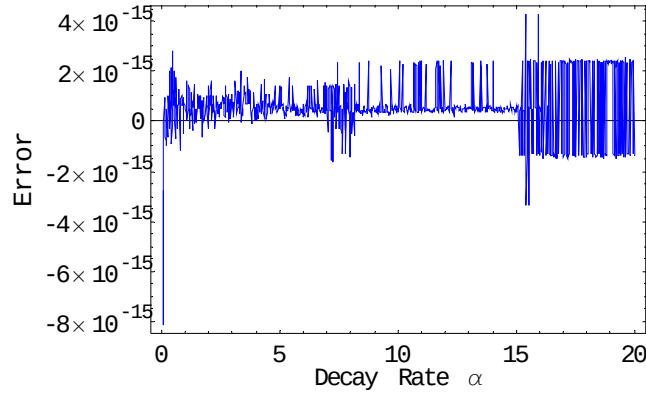
Secondly, we investigate the level of decay rate parameter α for Merton-FT formula. Figure 10.2 illustrates the pricing error (i.e. relative to the original Merton) of Merton-FT formula of a one day to maturity $T = 1/252$ call as a function of varying α . Panel A (for an OTM call) and C (for an ATM call) tells us that for $0.05 \leq \alpha \leq 20$, Merton-FT formula has effectively zero error relative to the original Merton price. But it seems for an ITM call (Panel B), the error monotonically increases as α rises. Therefore, from now on, we always use $\alpha = 1$ when implementing Merton-FT formula.



A) For an OTM Call with $K = 80$.



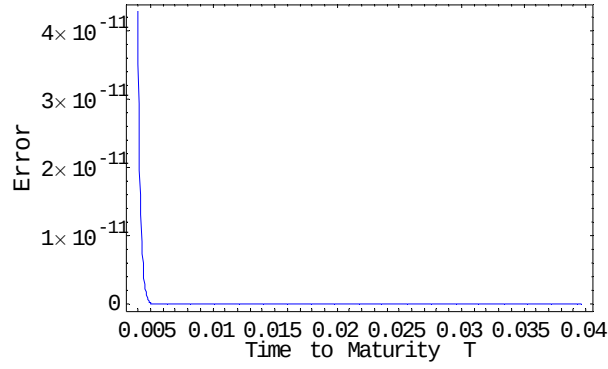
B) For an ITM Call with $K = 20$.



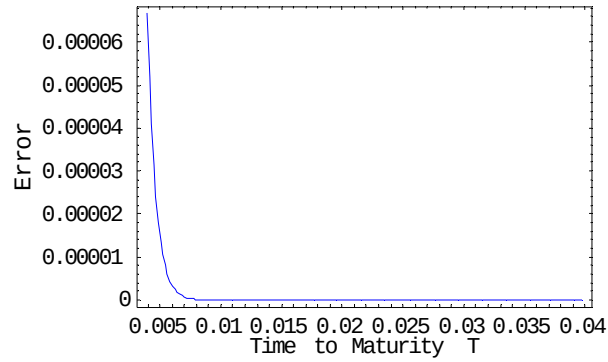
C) For an ATM Call with $K = 50$.

Figure 10.2: Original Merton Price Minus Merton-FT Price for One Day to Maturity Call as a Function of Decay Rate Parameter α . Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 1/252$, $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$.

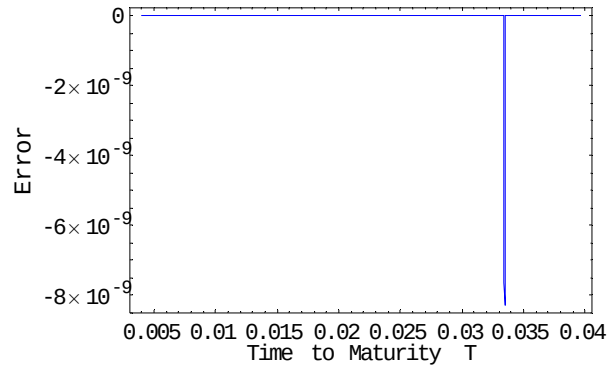
Thirdly, we investigate the amount of error caused by the difficulty in numerical integration in (10.4) for the near maturity calls. Consider a call option with $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$. Figure 10.3 plots a series of the difference between the original Merton price and Merton-FT $\alpha = 1$ price for the range of less than 10 trading days to maturity $1/252 \leq T \leq 10/252$. We find that despite the difficulty in the numerical integration of Merton-FT price of (10.4) for the near maturity deep OTM (in Panel A) and deep ITM (in Panel B) call, our Merton-FT code yields effectively zero error in terms of pricing.



A) OTM call with $K = 80$.



B) ITM call with $S_0 = 50$ and $K = 20$.



C) ATM Call with $S_0 = 50$ and $K = 50$.

Figure 10.3: Plot of Merton Price minus Merton-FT $\alpha = 1$ Price for Near-Maturity Vanilla Call with Less Than 10 Trading Days to Maturity $1/252 \leq T \leq 10/252$.
Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$.

We summarize this section. Although Merton-FT call price of the equation (10.4) is slower than the original Merton price formula, it produces negligible pricing errors

regardless of the maturity and the moneyness of the call. We recommend the use of $\alpha = 1$ for its decay rate.

[10.2] Discrete Fourier Transform (DFT) Call Pricing Formula with Merton Jump-Diffusion Model

Although the numerical integration difficulty of Merton-FT call price of (10.4) for the near maturity options yields virtually zero pricing error, it makes the evaluation slow. This speed becomes an issue when calibrating hundreds or thousands of prices (i.e. also in Monte Carlo simulation). To improve computational time, we apply our version of DFT call price formula in Merton case.

From the equation (8.38), Merton-DFT call price is given as:

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k} \times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\} \exp(-i2\pi jn/N), \quad (10.5)$$

where $w_{j \in [0, N-1]}$ are trapezoidal rule weights:

$$w_j = \begin{cases} 1/2 & \text{for } j = 0 \text{ and } N-1 \\ 1 & \text{for others} \end{cases},$$

and:

$$\psi_T(\omega) = \frac{e^{-rT} \phi_T(\omega - (\alpha + 1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega},$$

with $\phi_T(\cdot)$ given by (10.3).

[10.3] Implementation and Performance of DFT Pricing Method with Merton Jump-Diffusion Model

In this section, performance of Merton-DFT $\alpha = 1$ call price of the equation (10.5) is tested by comparing results to the original Merton call price and Merton-FT $\alpha = 1$ call price of the equation (10.4) under various settings. Merton-DFT $\alpha = 1$ call price is implemented using $N = 4096$ samples and log strike space sampling interval $\Delta k = 0.005$. This corresponds to angular frequency domain sampling interval of $\Delta\omega = 0.306796$ radians, the total sampling range in the log strike space is $K = N\Delta k = 20.48$, its sampling rate is 200 samples per unit of k , and the total sampling range in the angular frequency domain is $\Omega = N\Delta\omega = 1256.64$.

Firstly, let's investigate the difference in price and CPU time. Consider calculating 100-point call prices for a range of strike price $1 \leq K \leq 100$ with interval 1 with common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$. Figure 10.4 reports the price and Table 10.2 compares CPU time. We notice that call prices are virtually identical (i.e. they are supposed to be identical) and the use of DFT significantly improves the computational time although it is slower than the original Merton price.

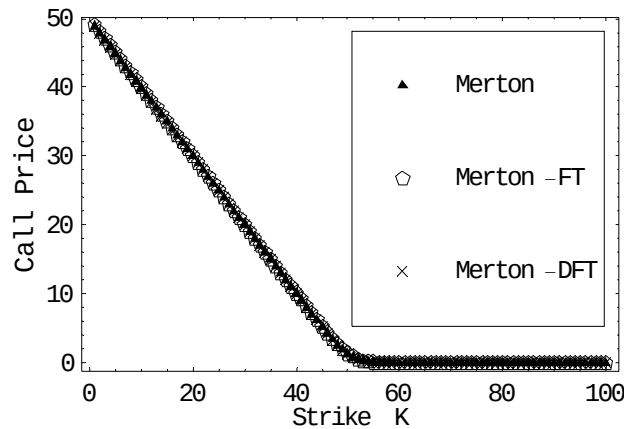


Figure 10.4: Merton Vs. Merton-FT $\alpha = 1$ Vs. Merton-DFT $\alpha = 1$. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$.

Table 10.2 CPU Time for Calculating 100-point call prices for a range of strike price $1 \leq K \leq 100$ with interval 1 Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$.

Method	CPU Time
Merton	0.14 seconds
Merton-FT $\alpha = 1$	7.22 seconds
Merton-DFT $\alpha = 1$ $N = 4096$, $\Delta k = 0.005$	1.31 seconds

Secondly, we pay extra attention to the pricing errors of very near-maturity calls. Consider a call option with common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$. Merton jump-diffusion parameters are set as $\lambda = 1$, $\mu = -0.1$, and $\delta = 0.1$. Figures 10.5 to 10.7 plot three series of price differences for vanilla calls computed by the original Merton, Merton-FT $\alpha = 1$, and Merton-DFT $\alpha = 1$ with $N = 4096$ and $\Delta k = 0.005$ as a function of time to maturity of less than a month $1/252 \leq T \leq 20/252$. Figure 10.5 tells us that for deep OTM calls, Merton-DFT yields

the pricing error around 6×10^{-8} which we interpret negligible. Figure 10.7 tells us that for deep ITM calls, Merton-DFT price is virtually identical to the original Merton except for one day to maturity. Very similar to the BS case, Figure 10.6 tells us that as the maturity nears, the error of ATM Merton-DFT price monotonically increases and the size of error is large (compared to ITM and OTM errors) but still negligible. In contrast, ATM Merton-FT price produces virtually no error. This finding is more clearly illustrated in Figure 10.7 where the $T = 1/252$ ATM pricing error of Merton-DFT $\alpha = 1$ with $N = 4096$ and $\Delta k = 0.005$ is plotted across different moneyness. We again realize that Merton-DFT price is virtually identical to BS price for the deep ITM and OTM calls, but its approximation error becomes an issue for ATM call. We have done several experiments trying to reduce the size of this around ATM error by increasing the decay rate parameter α to 10 or by sampling more points $N = 8192$. But these attempts were futile.

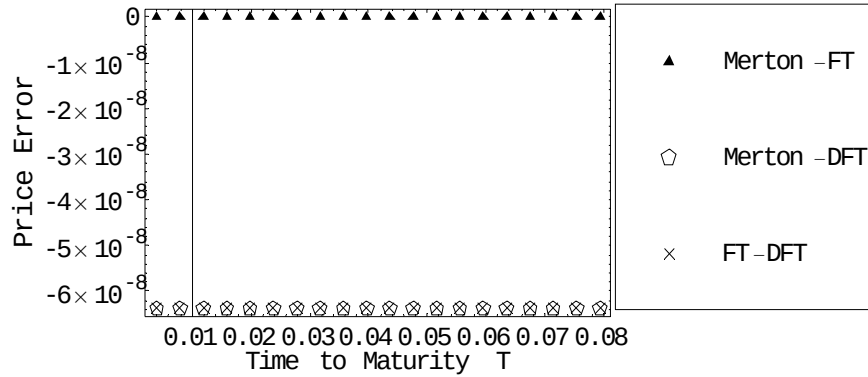


Figure 10.5: Plot of Price Error for Deep-OTM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 80$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

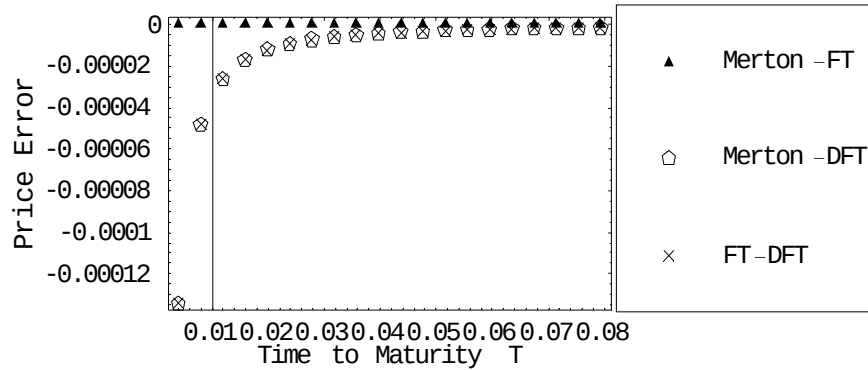


Figure 10.6: Plot of Price Error for ATM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

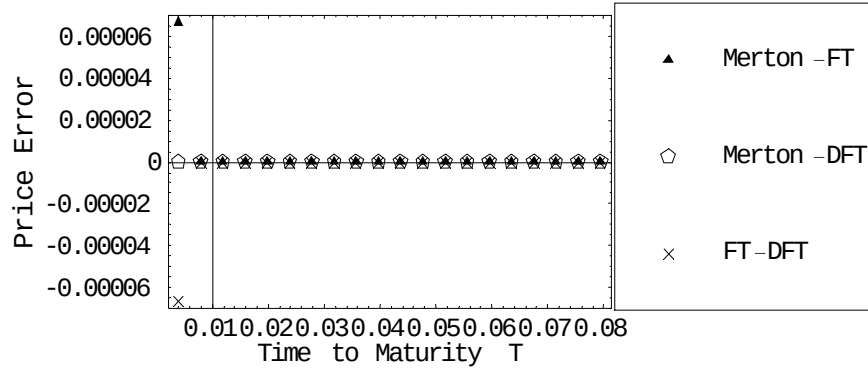


Figure 10.7: Plot of Price Error for Deep-ITM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 20$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

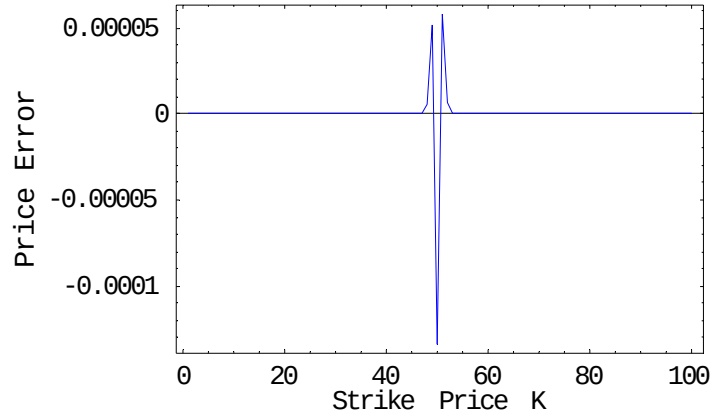


Figure 10.8: Plot of Price Error of Merton-DFT Formula for 1-Day-to-Maturity $T = 1/252$ ATM Call as a Function of Strike Price $0 \leq K \leq 100$. Common parameters and variables fixed are $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$.

We conclude this section by stating the following remarks. Our Merton-DFT $\alpha = 1$ call price formula, the equation (10.5), yields the price virtually identical to the original Merton price for OTM and ITM calls even for extreme near-maturity case (i.e. $T = 1/252$) although the size of error is larger than Merton-FT formula. But the error of Merton-DFT price becomes large around (i.e. ± 3) ATM. In our example used, the maximum error is 0.0001343 which occurs at exactly ATM at $S_0 = K = 50$. Increasing the decay rate parameter α or sampling more points (i.e. larger N) cannot reduce the size of this error. But we can accept this size of error when considering the dramatic improvement in the CPU time to compute hundreds of prices.

[10.4] Summary of Formulae of Option Price with Fourier Transform in Merton Jump-Diffusion Model

Table 10.10: Summary of Formulae for Merton Jump-Diffusion Model

Method	Formula
Original Merton	$C_{Merton} = \sum_{j=0}^{\infty} \frac{e^{-\bar{\lambda}T} (\bar{\lambda}T)^j}{j!} C_{BS}(S_0, T, \sigma_j, r_j)$ $\bar{\lambda} = \lambda(1+k) = \lambda \exp\left(\mu + \frac{1}{2}\delta^2\right), \sigma_j^2 = \sigma^2 + \frac{j\delta^2}{T}$ $r_j = r - \lambda k + \frac{i \ln(1+k)}{T} = r - \lambda \left(e^{\mu + \frac{1}{2}\delta^2} - 1\right) + \frac{i\left(\mu + \frac{1}{2}\delta^2\right)}{T}$
FT	$C_{Merton-FT}(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} d\omega$ $\phi_T(\omega) = \exp\left[\lambda T \left(\exp(i\mu\omega - \frac{\delta^2\omega^2}{2}) - 1\right) + i\omega \left(s_0 + \left(r - \frac{\sigma^2}{2} - \lambda k\right)T\right) - \frac{\sigma^2\omega^2}{2}T\right]$ $k \equiv e^{\mu + \frac{1}{2}\delta^2} - 1$
DFT	$C_T(k_n) \approx \frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k}$ $\times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\} \exp(-i2\pi jn/N)$ $\psi_T(\omega_j) = \frac{e^{-rT} \phi_T(\omega_j - (\alpha+1)i)}{\alpha^2 + \alpha - \omega_j^2 + i(2\alpha+1)\omega_j}$

[11] Variance Gamma (VG) Model by Madan, Carr, and Chang (1998)

[11.1] Model Type

In this section the basic structure of VG model is described without the derivation of the model which will be done in the next section.

VG model is an exponential Lévy model of the form:

$$S_t = S_0 e^{L_t} ,$$

where the asset price process $\{S_t; 0 \leq t \leq T\}$ is modeled as an exponential of a Lévy process $\{L_t; 0 \leq t \leq T\}$. Choice of the Lévy process by Madan, Carr, and Chang (1998) is a VG process, $VG(x_t; \theta_{\mathbb{P}}, \sigma_{\mathbb{P}}, \kappa_{\mathbb{P}})$, plus a drift:

$$L_t \equiv \left\{ m + \frac{1}{\kappa_{\mathbb{P}}} \ln \left(1 - \theta_{\mathbb{P}} \kappa_{\mathbb{P}} - \frac{\sigma_{\mathbb{P}}^2 \kappa_{\mathbb{P}}}{2} \right) \right\} t + VG(x_t; \theta_{\mathbb{P}}, \sigma_{\mathbb{P}}, \kappa_{\mathbb{P}}) .$$

A VG process $VG(x_t; \theta_{\mathbb{P}}, \sigma_{\mathbb{P}}, \kappa_{\mathbb{P}})$ is defined as a stochastic process $\{X_t; 0 \leq t \leq T\}$ created by random time changing (i.e. subordinating) a Brownian motion with drift process $\theta t + \sigma B_t$ by a tempered 0-stable subordinator (i.e. a gamma subordinator) $\{S_t; 0 \leq t \leq T\}$ with unit mean rate:

$$X_t \equiv \theta(S_t) + \sigma B_{S_t} .$$

The CF of a VG process can be easily obtained by the use of the subordination theorem described in the section 11.2.1 and its probability density function is obtained in closed form.

A VG process $VG(x_t; \theta_{\mathbb{P}}, \sigma_{\mathbb{P}}, \kappa_{\mathbb{P}})$ is characterized as a pure jump Lévy process with infinite arrival rate of jumps. In other words, the Lévy measure of a VG process has an infinite integral:

$$\int_{-\infty}^{\infty} \ell(x) dx = \infty .$$

This means that a VG process has infinitely many small jumps but a finite number of large jumps.

Introduction of two extra parameters by the VG model captures the (negative) skewness and excess kurtosis of the log return density $\mathbb{P}(\ln(S_t / S_0))$ which deviates from the BS normal log return density. One is variance rate parameter κ which controls the degree of

the randomness of the subordination. Larger κ implies the fatter tails of the log return density $\mathbb{P}(\ln(S_t/S_0))$. The other is the drift parameter of the subordinated Brownian motion process θ which captures the skewness of the log return density.

Let's derive the model.

[11.2] Model Derivation

[11.2.1] Subordination Theorem of Lévy Processes

A transformation of a stochastic process to a new stochastic process through random time change by an increasing Lévy process (subordinator) independent of the original process is called subordination. Time is an increasing sequence of numbers: 0, 1, 2, 3, ... or 1, 10, 23, 30, 50, ... So is subordinator because subordinator is an increasing (i.e. non-decreasing) Lévy process. This non-decreasing property of a subordinator makes it a possible candidate for a time indicator.

Let $\{S_t; 0 \leq t \leq T\}$ be a subordinator with its characteristic triplet $(b_s, \sigma_s = 0, \ell_s)$. Any increasing function is of finite variation in one dimension and following the Lévy-Khinchin representation in section 6.5, the MGF (moment generating function) of S_t is:

$$M_s(\omega) \equiv E[\exp(\omega S_t)] = \exp\{t\mathcal{L}_s(\omega)\},$$

where the Laplace exponent is given by:

$$\mathcal{L}_s(\omega) = b_s \omega + \int_0^\infty (e^{\omega x} - 1) \ell_s(dx). \quad (11.1)$$

We used a relationship between the MGF M_x and the CF (characteristic function) given by $M_x(t) = \phi_x(-it)$ which holds when M_x is well-defined. Laplace transform is used instead of Fourier transform because Lévy measure of an increasing Lévy process is concentrated on the positive real-axis (i.e. by definition). $\mathcal{L}_s(\omega)$ satisfies the following:

$$\begin{aligned} \int_0^\infty 1_{dx \geq 1} \rho(dx) &< \infty \text{ (i.e. a finite variation condition),} \\ b_s &\geq 0 \text{ (i.e. a positive drift),} \end{aligned}$$

which are the conditions of being an increasing Lévy process.

Let $\{X_t; 0 \leq t \leq T\}$ be a Lévy process on $\mathbb{R}$ with its characteristic triplet (b, σ, ℓ) and its CF given by the Lévy-Khinchin representation:

$$\phi_x(\omega) \equiv E[\exp(i\omega X_t)] = \exp\{t\psi_x(\omega)\},$$

where the CE (characteristic exponent) is given by:

$$\psi_X(\omega) = ib\omega - \frac{\sigma^2 \omega^2}{2} + \int_{-\infty}^{\infty} \{\exp(i\omega x) - 1\} \ell(dx). \quad (11.2)$$

We assume $\{X_t; 0 \leq t \leq T\}$ and $\{S_t; 0 \leq t \leq T\}$ are independent.

Define a new stochastic process $\{Z_t; 0 \leq t \leq T\}$ by random time changing (i.e. subordinating) the original Lévy process $\{X_t; 0 \leq t \leq T\}$:

$$Z_t \equiv X_{S_t}. \quad (11.3)$$

Then, the process $\{Z_t; 0 \leq t \leq T\}$ is a Lévy process on $\mathbb{R}$ with its CF given by:

$$\phi_Z(\omega) \equiv E \{ \exp(i\omega Z_t) \} = \exp \{ t \mathcal{L}_S(\psi_X(\omega)) \}. \quad (11.4)$$

This is an extremely powerful and useful theorem since we can obtain the CF of a random time-changed Lévy process Z_t by simply substituting the CE of an original Lévy process $\psi_X(\omega)$ into the Laplace exponent of the subordinator $\mathcal{L}_S(\omega)$. The characteristic triplet of a random time-changed Lévy process Z_t , (b_Z, σ_Z, ℓ_Z) , is given by:

$$\begin{aligned} b_Z &= b_S b + \int_0^\infty \ell_S(ds) \int_{|x| \leq 1} x \mathbb{P}_X(dx), \\ \sigma_Z^2 &= b_S^2 \sigma^2, \\ \ell_Z(dx) &= b_S \ell(dx) + \int_0^\infty \mathbb{P}_X(dx) \ell_S(ds), \end{aligned} \quad (11.5)$$

where $\mathbb{P}_X()$ is the probability density function of X_t . Transformation of a Lévy process $\{X_t; 0 \leq t \leq T\}$ into another Lévy process $\{Z_t; 0 \leq t \leq T\}$ is called subordination by the subordinator $\{S_t; 0 \leq t \leq T\}$.

[11.2.2] Tempered α -Stable Subordinator: General Case

For the reason of pure mathematical tractability of the process, we restrict our attention to the tempered α -stable subordinator. Let $\{S_t; 0 \leq t \leq T\}$ be a tempered α -stable subordinator which is Esscher transformed stable subordinator (i.e. whose tails of both probability density and Lévy density are damped symmetrically) and $(b_S, \sigma_S = 0, \ell_S)$ be its characteristic triplet, then:

- (1) $\{S_t; 0 \leq t \leq T\}$ is a α -stable process with the index of stability $0 < \alpha < 1$.

- (2) $\int_{-\infty}^0 \ell_s(dx) = 0$: $\{S_t; 0 \leq t \leq T\}$ has no negative jump. In other words, Lévy measure concentrated on the positive real-axis.
- (3) $b_s \geq 0$: Positive drift.
- (4) Lévy measure is three-parameter family and is simply a tempered version of the Lévy measure of a stable subordinator of the equation (6.12):

$$\ell_s(x) = \frac{c \exp(-\lambda x)}{x^{\alpha+1}} 1_{x>0}, \quad (11.6)$$

where $c > 0$ and $\lambda > 0$. The intensity of jumps of all sizes is controlled by the parameter c and the decay rate of large jumps is controlled by λ . And the relative importance of small jumps in the trajectory is determined by α .

Let $\{S_t; 0 \leq t \leq T\}$ be a tempered α -stable subordinator with zero drift for simplicity (i.e. $b_s = 0$) in the general case $0 < \alpha < 1$. Following the Lévy-Khinchin representation in section 6.5, the MGF of S_t is given by (11.1) as:

$$M_s(\omega) \equiv E[\exp(\omega S_t)] = \exp\{t \mathcal{L}_s(\omega)\},$$

where the Laplace exponent is given by:

$$\begin{aligned} \mathcal{L}_s(\omega) &= b_s \omega + \int_0^\infty (e^{\omega x} - 1) \ell_s(dx) \\ &= \int_0^\infty (e^{\omega x} - 1) \frac{c e^{-\lambda x}}{x^{\alpha+1}} 1_{x>0} dx \\ &= c\{(\lambda - \omega)^\alpha - \lambda^\alpha\} \Gamma(-\alpha). \end{aligned} \quad (11.7)$$

Using the MGF, the mean, variance, skewness, kurtosis of the tempered α -stable subordinator with zero drift $b_s = 0$ in the general case are obtained as:

$$\begin{aligned} E[S_t] &= m_1 = -ct\alpha\lambda^{\alpha-1}\Gamma[-\alpha], \\ \text{Var}[S_t] &= m_2 - m_1^2 = ct(\alpha-1)\alpha\lambda^{\alpha-2}\Gamma[-\alpha], \\ \text{Skewness}[S_t] &= \frac{2m_1^3 - 3m_1m_2 + m_3}{(m_2 - m_1^2)^{3/2}} = \frac{2 - \alpha}{\lambda\sqrt{ct(\alpha-1)\alpha\lambda^{\alpha-2}\Gamma[-\alpha]}}, \\ \text{Excess Kurtosis}[S_t] &= \frac{-6m_1^4 + 12m_1^2m_2 - 3m_2^2 - 4m_1m_3 + m_4}{(m_2 - m_1^2)^2} = \frac{\lambda^{-\alpha}(6 - 5\alpha + \alpha^2)}{ct(\alpha-1)\alpha\Gamma[-\alpha]}. \end{aligned} \quad (11.8)$$

Note that the probability density of the tempered α -stable subordinator with zero drift $b_s = 0$ in the general case $0 < \alpha < 1$ is not known.

[11.2.3] Gamma Subordinator (Process): Special Case of Tempered α -Stable Subordinator When $\alpha = 0$

Consider a special case when the stability index takes zero, i.e. $\alpha = 0$. A tempered 0-stable subordinator with zero drift $b_s = 0$ is called a gamma subordinator. Lévy density of the gamma subordinator S_t is obtained by setting $\alpha = 0$ in the equation (11.6) as:

$$\ell_s(x) = \frac{ce^{-\lambda x}}{x} \mathbf{1}_{x>0}. \quad (11.9)$$

Following the Lévy-Khinchin representation of the equation (11.1), the MGF of the gamma subordinator S_t is:

$$M_s(\omega) \equiv E[\exp(\omega S_t)] = \exp\{t\mathcal{L}_s(\omega)\},$$

where the Laplace exponent is given by substituting (11.9):

$$\begin{aligned} \mathcal{L}_s(\omega) &= b_s\omega + \int_0^\infty (e^{\omega x} - 1)\ell_s(dx) \\ &= \int_0^\infty (e^{\omega x} - 1) \frac{ce^{-\lambda x}}{x} dx \\ &= c \ln\left(\frac{\lambda}{\lambda - \omega}\right). \end{aligned}$$

Therefore, the MGF of the gamma subordinator S_t is explicitly calculated as the following:

$$\begin{aligned} M_s(\omega) &\equiv \exp\{t\mathcal{L}(\omega)\} = \exp\left\{tc \ln\left(\frac{\lambda}{\lambda - \omega}\right)\right\} \\ &= \exp\left\{\ln\left(\frac{\lambda}{\lambda - \omega}\right)^{tc}\right\} = \left(\frac{\lambda}{\lambda - \omega}\right)^{tc} \\ &= \left(\frac{1}{1 - \omega\lambda^{-1}}\right)^{tc} = \frac{1}{(1 - \omega\lambda^{-1})^{ct}}. \end{aligned} \quad (11.10)$$

Note that the equation (11.10) cannot be obtained by setting $\alpha = 0$ in (11.7).

The probability density of the gamma subordinator S_t follows a gamma distribution:

$$Gamma_t(g) = \frac{\lambda^{ct}}{\Gamma(ct)} g^{ct-1} e^{-\lambda g} \mathbf{1}_{g>0}. \quad (11.11)$$

Using moment generating function (11.10) the mean, variance, skewness, and excess kurtosis of the gamma subordinator are calculated as:

$$\begin{aligned} E[S_t] &= \frac{ct}{\lambda}, \\ \text{Var}[S_t] &= \frac{ct}{\lambda^2}, \\ \text{Skewness}[S_t] &= \frac{2}{\sqrt{ct}}, \\ \text{Excess Kurtosis}[S_t] &= \frac{6}{ct}. \end{aligned} \tag{11.12}$$

Note that MCC (1998) uses the different parameterization such that:

$$\begin{aligned} \mu = \text{mean rate} &= \frac{c}{\lambda} \quad \text{and} \quad \kappa = \text{variance rate} = \frac{c}{\lambda^2}, \\ (c = \frac{\mu^2}{\kappa} \quad \text{and} \quad \lambda = \frac{\mu}{\kappa}). \end{aligned} \tag{11.13}$$

Don't be confused by this different parameterization. We and MCC (1998) are the same. Alternatively, (11.12) can be obtained by setting $\alpha = 0$ in the equation (11.8).

A gamma process is an infinite activity Lévy process because a gamma distribution possesses the infinite divisibility and its Lévy measure (11.9) has an infinite integral. This means that a gamma process has infinitely many small jumps (i.e. infinite arrival rate of small jumps) and a finite number of large jumps (i.e. $\ell_S(dx)$ is concentrated at the origin). A gamma process is a pure jump process since it has no continuous (Brownian motion) component.

[11.2.4] Re-Parameterizing Tempered α -Stable Subordinator Using Its Scale-Invariance (Self-Similarity) Property

A tempered α -stable subordinator $S_t(\alpha, \lambda, c)$ possesses a very important property called the scale-invariance since the process S at scale $w^\alpha t$ has the same law as the process S at scale t after appropriate rescaling. In other words, if $\{S_t(\alpha, \lambda, c); 0 \leq t \leq T\}$ is a tempered α -stable subordinator with zero drift $b_S = 0$ and with parameters α , λ , and c , then for every positive constant w :

$$wS_t(\alpha, \lambda, c) \stackrel{d}{=} S_{w^\alpha t}(\alpha, \lambda/w, c),$$

which is equivalent to stating that the process S is self-similar (read section 6.6 for the definition). Remember that if a Lévy process (i.e. includes subordinator) is self-similar,

then it is strictly stable. This indicates by definition that a tempered α -stable subordinator possesses this scale-invariance property. We saw in section 6.6 that a standard Brownian motion process B_t is a strictly 2-stable Lévy process:

$$wB_t \stackrel{d}{=} B_{w^2 t}.$$

Because of this scale invariance property of a tempered α -stable subordinator $\{S_t(\alpha, \lambda, c); t \geq 0\}$ and a standard Brownian motion process B_t , we consider only tempered α -stable subordinators with $E[S_t] = t$ in subordinated models. In other words, we re-parameterize a tempered α -stable subordinator so that it has the mean t (i.e. called unit mean rate). Remember that in the general case $0 < \alpha < 1$ the mean of a tempered α -stable subordinator is given by the equation (11.8). The re-parameterization is done by setting:

$$c = \frac{1}{\Gamma(1-\alpha)} \left(\frac{1-\alpha}{\kappa} \right)^{1-\alpha} \quad \text{and} \quad \lambda = \frac{1-\alpha}{\kappa}, \quad (11.14)$$

such that:

$$E[S_t] = -\frac{1}{\Gamma(1-\alpha)} \left(\frac{1-\alpha}{\kappa} \right)^{1-\alpha} t \alpha \left(\frac{1-\alpha}{\kappa} \right)^{\alpha-1} \Gamma[-\alpha] = t,$$

where use $\frac{\Gamma(-c)}{\Gamma(1-c)} = -\frac{1}{c}$. And the Lévy measure is re-parameterized from three-parameter measure (11.6) to two-parameter measure:

$$\ell_s(x) = \frac{1}{\Gamma(1-\alpha)} \left(\frac{1-\alpha}{\kappa} \right)^{1-\alpha} \frac{\exp\left\{-(1-\alpha)x/\kappa\right\}}{x^{1+\alpha}}, \quad (11.15)$$

where α is the stability index and $\kappa > 0$ is a variance rate which is equal to the variance of these subordinators at time 1, i.e. $\text{Var}[S_1] = \kappa$. Thus, κ determines the degree of the randomness of the time change (i.e. subordination). When $\kappa = 0$, the process is deterministic.

Following the Lévy-Khinchin representation of the equation (11.1), the MGF of the re-parameterized tempered α -stable subordinator S_t is:

$$M_s(\omega) \equiv E[\exp(\omega S_t)] = \exp\{t \mathcal{L}_s(\omega)\},$$

where the Laplace exponent is given by substituting (11.15):

$$\begin{aligned}
\mathcal{L}_s(\omega) &= b_s \omega + \int_0^\infty (e^{\omega x} - 1) \ell_s(dx) \\
&= \int_0^\infty (e^{\omega x} - 1) \frac{1}{\Gamma(1-\alpha)} \left(\frac{1-\alpha}{\kappa}\right)^{1-\alpha} \frac{\exp\{-(1-\alpha)x/\kappa\}}{x^{1+\alpha}} dx \\
&= \frac{1-\alpha}{\kappa\alpha} \left\{ 1 - \left(1 - \frac{\kappa\omega}{1-\alpha}\right)^\alpha \right\}.
\end{aligned} \tag{11.16}$$

From the MGF, the variance, skewness, and excess kurtosis of the tempered α -stable subordinator with unit mean rate are calculated as:

$$\begin{aligned}
\text{Var}[S_t] &= \kappa t, \\
\text{Skewness}[S_t] &= \frac{2-\alpha}{1-\alpha} \sqrt{\frac{\kappa}{t}}, \\
\text{Excess Kurtosis}[S_t] &= \frac{\kappa}{t} \frac{\alpha^2 - 5\alpha + 6}{(1-\alpha)^2}.
\end{aligned} \tag{11.17}$$

Note that the probability density of the tempered α -stable subordinator with unit mean rate and zero drift $b_s = 0$ in the general case $0 < \alpha < 1$ is not known.

[11.2.5] Gamma Subordinator (Process): Special Case of Tempered α -Stable Subordinator with Unit Mean Rate When $\alpha = 0$

Again, consider a special case when $\alpha = 0$, the tempered α -stable subordinator with unit mean rate becomes a gamma subordinator with unit mean rate $E[S_t] = t$. Its Lévy density can be expressed by setting $\alpha = 0$ in the equation (11.15) as:

$$\ell_s(x) = \frac{1}{\kappa} \frac{\exp(-x/\kappa)}{x} 1_{x>0}. \tag{11.18}$$

Because the Laplace exponent of the gamma subordinator with unit mean rate cannot be obtained by simply substituting $\alpha = 0$ in (11.16), we need to derive it step-by-step. The MGF of the gamma subordinator S_t with unit mean rate can be expressed as using Lévy-Khinchin representation:

$$M_s(\omega) \equiv E[\exp(\omega S_t)] = \exp\{t \mathcal{L}_s(\omega)\},$$

where the Laplace exponent is given as using (11.18):

$$\mathcal{L}_s(\omega) = \int_0^\infty (e^{\omega x} - 1) \ell_s(x) dx = \int_0^\infty (e^{\omega x} - 1) \frac{1}{\kappa} \frac{\exp(-x/\kappa)}{x} dx$$

$$= \frac{-\log \kappa - \log\left(\frac{1}{\kappa} - \omega\right)}{\kappa}. \quad (11.19)$$

The mean, variance, skewness, and excess kurtosis of the gamma subordinator with unit mean rate are calculated as using MGF or by setting $\alpha = 0$ in (11.17):

$$\begin{aligned} E[S_t] &= t, \\ \text{Var}[S_t] &= \kappa t, \\ \text{Skewness}[S_t] &= 2\sqrt{\frac{\kappa}{t}}, \\ \text{Excess Kurtosis}[S_t] &= \frac{6\kappa}{t}. \end{aligned} \quad (11.20)$$

Consider obtaining the probability density of the gamma subordinator S_t with unit mean rate. The equation (11.11) is the probability density of the gamma subordinator S_t with the mean rate c/λ (check the equation (11.12)):

$$\text{Gamma}_t(g) = \frac{\lambda^{ct}}{\Gamma(ct)} g^{ct-1} e^{-\lambda g} \mathbf{1}_{g>0}.$$

Re-parameterization terms for the purpose of having unit mean rate for the gamma subordinator case are obtained by setting $\alpha = 0$ in the equation (11.14):

$$c = \frac{1}{\kappa} \text{ and } \lambda = \frac{1}{\kappa}.$$

Obviously, the mean rate after this re-parameterization is:

$$\frac{c}{\lambda} = \frac{1/\kappa}{1/\kappa} = 1.$$

Thus, the probability density of a gamma subordinator S_t with unit mean rate $E[S_t] = t$ has a gamma density of the following form:

$$\begin{aligned} \text{Gamma}_t(g) &= \frac{(1/\kappa)^{(1/\kappa)t}}{\Gamma((1/\kappa)t)} g^{(1/\kappa)t-1} e^{-(1/\kappa)g} \mathbf{1}_{g>0} \\ &= \frac{\left(\frac{1}{\kappa}\right)^{t/\kappa}}{\Gamma(t/\kappa)} g^{\frac{t}{\kappa}-1} e^{-g/\kappa} \mathbf{1}_{g>0}. \end{aligned} \quad (11.21)$$

[11.2.6] Subordinated Brownian Motion Process with Tempered α -Stable Subordinator with Unit Mean Rate: Normal Tempered α -Stable Process

Firstly, we deal with a general case. Let $\{B_t; 0 \leq t \leq T\}$ be a standard Brownian motion process and $\theta t + \sigma B_t \sim N(\theta t, \sigma^2 t)$ be a Brownian motion with drift process. Define a new stochastic process $\{X_t; 0 \leq t \leq T\}$ by random time changing (i.e. subordinating) a Brownian motion with drift process $\theta t + \sigma B_t$ by a tempered α -stable subordinator with unit mean rate $\{S_t; 0 \leq t \leq T\}$:

$$X_t \equiv \theta(S_t) + \sigma B_{S_t}.$$

Then, the stochastic process $\{X_t; t \geq 0\}$ is said to be a normal tempered α -stable process. The CF of a normal tempered α -stable process can be obtained by the use of the subordination theorem of the equation (11.4). As the first step, obtain the CF of the original Lévy process which is a Brownian motion with drift process $\theta t + \sigma B_t$:

$$\phi_B(\omega) = E[e^{i\omega(\theta t + \sigma B_t)}] = \exp(i\theta\omega t - \frac{\sigma^2 \omega^2}{2} t) = \exp\{t\psi_B(\omega)\},$$

where the characteristic exponent is:

$$\psi_B(\omega) = i\theta\omega - \frac{\sigma^2 \omega^2}{2}. \quad (11.22)$$

We saw that the Laplace exponent of the tempered α -stable subordinator S_t with $E[S_t] = t$ can be expressed as in the general case $0 < \alpha < 1$ by (11.16). Following the subordination theorem of the equation (11.2), the CF of the normal tempered α -stable process $X_t \equiv \theta(S_t) + \sigma B_{S_t}$ can be obtained by a substitution of (11.22) into (11.16):

$$\phi_X(\omega) \equiv E\{\exp(i\omega X_t)\} = \exp\{t\mathcal{L}_S(\psi_B(\omega))\},$$

with its characteristic exponent:

$$\begin{aligned} \psi_X(\omega) &= \mathcal{L}_S(\psi_B(\omega)) = \mathcal{L}_S\left(i\theta\omega - \frac{\sigma^2 \omega^2}{2}\right) \\ &= \frac{1-\alpha}{\kappa\alpha} \left[1 - \left(1 + \frac{\kappa\left(\frac{\sigma^2 \omega^2}{2} - i\theta\omega\right)}{1-\alpha} \right)^\alpha \right]. \end{aligned} \quad (11.23)$$

Using the characteristic exponent (11.23), the mean, variance, skewness, and excess kurtosis of the normal tempered α -stable process with unit mean rate are calculated as in the general case $0 < \alpha < 1$:

$$\begin{aligned} E[X_t] &= \theta t, \\ \text{Var}[X_t] &= (\kappa\theta^2 + \sigma^2)t, \\ \text{Skewness}[X_t] &= \frac{\kappa\theta(\kappa\theta^2(\alpha - 2) + 3\sigma^2(\alpha - 1))}{\alpha - 1}t, \\ \text{Excess Kurtosis}[X_t] &= \frac{\kappa(\kappa^2\theta^4(\alpha^2 - 5\alpha + 6) + 6\kappa\theta^2\sigma^2(\alpha^2 - 3\alpha + 2) + 3\sigma^4(\alpha - 1)^2)}{(\alpha - 1)^2(\kappa\theta^2 + \sigma^2)^2}t. \end{aligned} \quad (11.24)$$

Cont and Tankov (2004) shows that the Lévy density of the normal tempered α -stable process is given by (again we assume the drift of the subordinator is zero for simplicity, i.e. $b_s = 0$):

$$\ell(x) = \exp\left(\frac{\theta}{\sigma^2}x\right) \frac{F(\alpha, \kappa, \sigma, \theta)}{|x|^{\alpha+\frac{1}{2}}} K_{\alpha+\frac{1}{2}}\left(\frac{|x|\sqrt{\theta^2 + \frac{2}{\kappa}\sigma^2(1-\alpha)}}{\sigma^2}\right), \quad (11.25)$$

where $F(\alpha, \kappa, \sigma, \theta) = \left\{\theta^2 + \frac{2}{\kappa}\sigma^2(1-\alpha)\right\}^{\frac{\alpha+1}{2}+\frac{1}{4}} \frac{2}{\Gamma(1-\alpha)\sqrt{2\pi\sigma^2}} \left(\frac{1-\alpha}{\kappa}\right)^{1-\alpha}$ and K is a modified Bessel function of the second kind (Appendix 7 gives its details).

Note that because the probability density of tempered α -stable subordinator is not known in the general case (i.e. known only for $\alpha = 0$ and $1/2$), the probability density of the normal tempered α -stable process is not known in the general case $0 < \alpha < 1$, either.

[11.2.7] Variance Gamma (VG) Process: Subordinated Brownian Motion Process with Tempered 0-Stable Subordinator with Unit Mean Rate: Normal Tempered 0-Stable Process

Consider a special case when $\alpha = 0$. Define a new stochastic process $\{X_t; 0 \leq t \leq T\}$ by random time changing (i.e. subordinating) a Brownian motion with drift process $\theta t + \sigma B_t$ by a tempered $\alpha = 0$ -stable subordinator $\{S_t; 0 \leq t \leq T\}$ with unit mean rate:

$$X_t \equiv \theta(S_t) + \sigma B_{S_t}.$$

Then, the stochastic process $\{X_t; 0 \leq t \leq T\}$ is a normal tempered 0-stable process which is called a variance gamma (VG) process. Because the CF of VG process cannot be obtained by simply substituting $\alpha = 0$ in (11.23), we need to do this step-by-step.

Following the subordination theorem of the equation (11.2), the characteristic function of the VG (normal tempered 0-stable) process $X_t \equiv \theta(S_t) + \sigma B_{S_t}$ can be obtained by substituting the CE of the Brownian motion with drift process (11.22) into the Laplace exponent of the gamma subordinator $\{S_t; 0 \leq t \leq T\}$ with unit mean rate (11.19):

$$\phi_X(\omega) \equiv E\{\exp(i\omega X_t)\} = \exp\{t\mathcal{L}_S(\psi_B(\omega))\},$$

with its characteristic exponent:

$$\psi_X(\omega) = \mathcal{L}_S(\psi_B(\omega)) = \mathcal{L}\left(i\theta\omega - \frac{\sigma^2\omega^2}{2}\right) = -\frac{1}{\kappa} \ln\left(1 + \frac{\sigma^2\omega^2\kappa}{2} - i\theta\kappa\omega\right). \quad (11.26)$$

The characteristic function of the VG process $\phi_X(\omega)$ can be explicitly calculated as follows (corresponding to MCC (1998) equation 7):

$$\begin{aligned} \phi_X(\omega) &= \exp\{t\mathcal{L}_S(\psi_B(\omega))\} = \exp\left[t\left\{-\frac{1}{\kappa} \ln\left(1 + \frac{\sigma^2\omega^2\kappa}{2} - i\theta\kappa\omega\right)\right\}\right] \\ \ln \phi_X(\omega) &= -\frac{t}{\kappa} \ln\left(1 + \frac{\sigma^2\omega^2\kappa}{2} - i\theta\kappa\omega\right) = \ln\left(1 + \frac{\sigma^2\omega^2\kappa}{2} - i\theta\kappa\omega\right)^{-\frac{t}{\kappa}} \\ \phi_X(\omega) &= \left(1 + \frac{\sigma^2\omega^2\kappa}{2} - i\theta\kappa\omega\right)^{-\frac{t}{\kappa}}. \end{aligned} \quad (11.27)$$

Consider obtaining a probability density function of a VG process $\{X_t; 0 \leq t \leq T\}$. Start from the fact that a Brownian motion with drift $W_t \equiv \theta t + \sigma B_t$ has a normal density:

$$\mathbb{P}(W_t) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left\{-\frac{(W_t - \theta t)^2}{2\sigma^2 t}\right\}.$$

Because of the subordination structure $X_t \equiv \theta(S_t) + \sigma B_{S_t}$, by conditioning on the fact that the realized value of the random time change by a gamma subordinator S_t with unit mean rate is $S_t = g$, the conditional probability density of VG process can be expressed as:

$$VG(x_t | t = S_t = g) = \frac{1}{\sqrt{2\pi\sigma^2 g}} \exp \left\{ -\frac{(x_t - \theta g)^2}{2\sigma^2 g} \right\}.$$

Therefore, the unconditional density of the VG process is calculated by multiplying out the gamma probability density of the equation (11.21):

$$VG(x_t) = \int_0^\infty \frac{1}{\sqrt{2\pi\sigma^2 g}} \exp \left\{ -\frac{(x_t - \theta g)^2}{2\sigma^2 g} \right\} \frac{(1/\kappa)^{t/\kappa}}{\Gamma(t/\kappa)} g^{\frac{t}{\kappa}-1} e^{-g/\kappa} dg, \quad (11.28)$$

which corresponds to MCC (1998) equation 6. After tedious algebra:

$$VG(x_t) = \frac{\sqrt{2} \exp\left(\frac{\theta}{\sigma^2} x_t\right)}{\kappa^{t/\kappa} \sigma \sqrt{\pi} \Gamma(t/\kappa)} \left(\frac{x_t^2}{\frac{2\sigma^2}{\kappa} + \theta^2} \right)^{\frac{t}{2\kappa} - \frac{1}{4}} K_{\frac{t}{\kappa} - \frac{1}{2}} \left(\frac{\sqrt{x_t^2 \left(\frac{2\sigma^2}{\kappa} + \theta^2 \right)}}{\sigma^2} \right), \quad (11.29)$$

which is MCC (1998) equation 23 and K is a modified Bessel function of the second kind.

Using the CF (11.27), cumulants of the VG process are calculated as:

$$\begin{aligned} \text{cumulant}_1 &= \theta t, \\ \text{cumulant}_2 &= (\kappa\theta^2 + \sigma^2)t, \\ \text{cumulant}_3 &= (2\theta^3\kappa^2 + 3\sigma^2\theta\kappa)t, \\ \text{cumulant}_4 &= (3\sigma^4\kappa + 6\theta^4\kappa^3 + 12\sigma^2\theta^2\kappa^2)t. \end{aligned}$$

From these cumulants (or by setting $\alpha = 0$ in the equation (11.24)), annualized (per unit of time) mean, variance, skewness, and excess kurtosis of the variance gamma process are calculated as (corresponding to MCC (1998) equations 18, 19, and 20):

$$\begin{aligned} E[X_t] &= \theta, \\ \text{Var}[X_t] &= \kappa\theta^2 + \sigma^2, \\ \text{Skewness}[X_t] &= 2\theta^3\kappa^2 + 3\sigma^2\theta\kappa, \\ \text{Excess Kurtosis}[X_t] &= \frac{3\sigma^4\kappa + 6\theta^4\kappa^3 + 12\sigma^2\theta^2\kappa^2}{(\kappa\theta^2 + \sigma^2)^2}. \end{aligned} \quad (11.30)$$

Note that these standardized moments are not equivalent to those of log-return density of VG model. These are simply the moments of VG density. This point will be explained in section 11.4. Several very interesting properties of VG density should be worth

mentioning. If $\theta = 0$ (i.e. subordination of zero-drift Brownian motion process with gamma subordinator), then:

$$\begin{aligned} E[X_t] &= 0, \\ \text{Var}[X_t] &= \sigma^2, \\ \text{Skewness}[X_t] &= 0, \\ \text{Excess Kurtosis}[X_t] &= 3\kappa, \end{aligned}$$

which indicates that VG density is symmetric (zero-skewness) and its excess kurtosis (i.e. tail behavior) is determined by the variance rate κ of the gamma subordinator S_t .

Generally, the sign of the drift parameter of the Brownian motion with drift process θ determines the skewness of VG density as illustrated in Figure 11.1. The case $\theta = -0.2$ is a mirror image of the case $\theta = 0.2$.

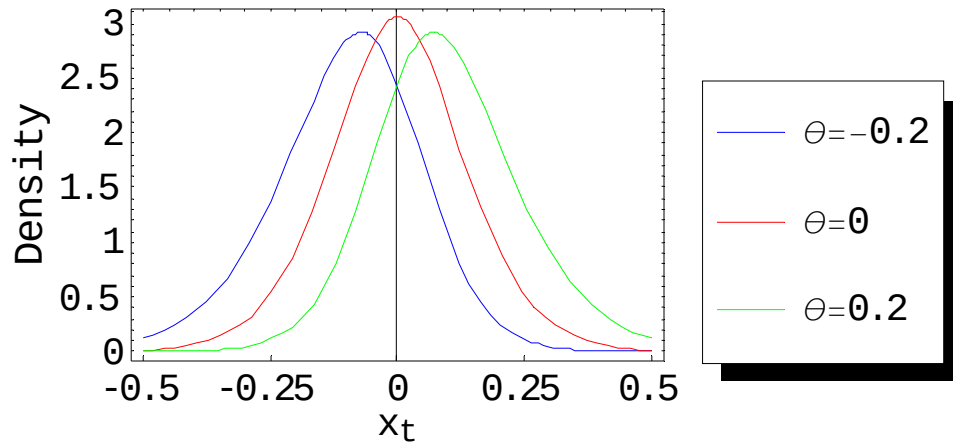


Figure 11.1: VG Density for Different Values of Drift Parameter of the Brownian Motion Process θ . Parameters fixed are $t = 0.5$, $\sigma = 0.2$, and $\kappa = 0.1$.

Table 11.1
Annualized Moments of VG Density in Figure 11.1

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
$\theta = -0.2$	-0.2	0.209762	-0.00256	0.352066
$\theta = 0$	0	0.2	0	0.3
$\theta = 0.2$	0.2	0.209762	0.00256	0.352066

Secondly, larger value of variance rate κ (i.e. the variance rate of the gamma subordinator S_t with unit mean rate and determines the degree of randomness of the subordination) makes the density fatter-tailed and higher-peaked as illustrated in Figure 11.2.

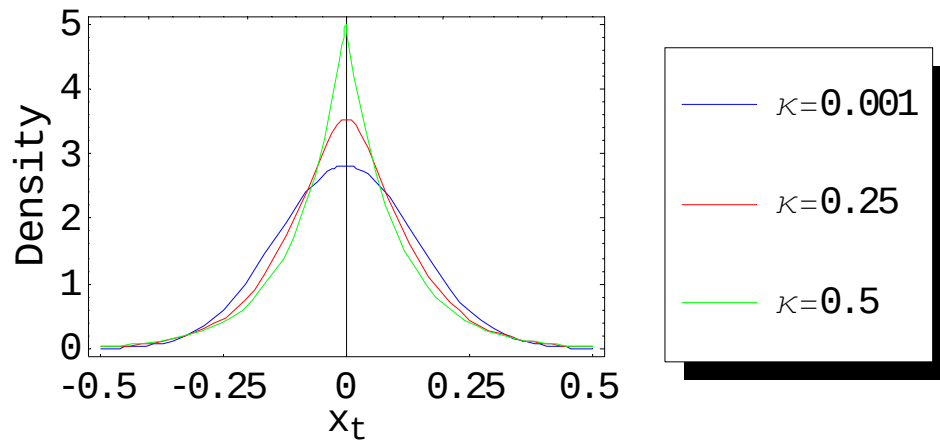


Figure 11.2: VG Density for Different Values of Variance Rate Parameter κ . Parameters fixed are $t = 0.5$, $\sigma = 0.2$, and $\theta = 0$.

Table 11.2
Annualized Moments of VG Density in Figure 11.2

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
$\kappa = 0.001$	0	0.2	0	0.003
$\kappa = 0.25$	0	0.2	0	0.75
$\kappa = 0.5$	0	0.2	0	1.5

Also note that VG density has higher peak and fatter tails (more leptokurtic) when matched to the Normal density as illustrated in Figure 11.3.

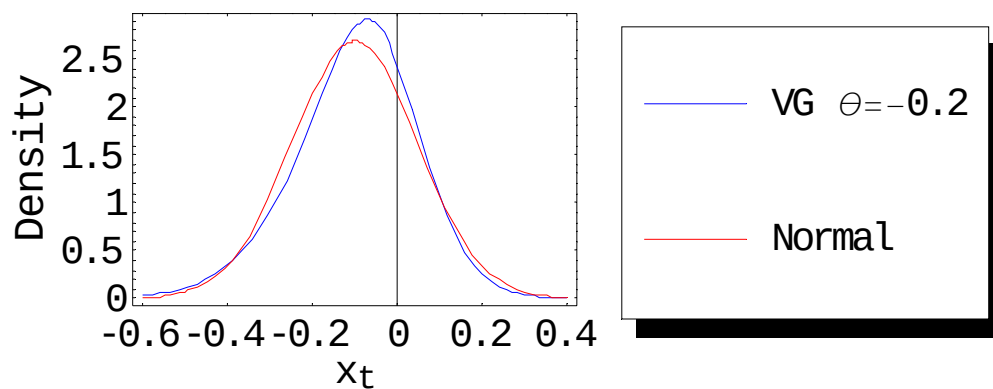


Figure 11.3: VG Density vs. Normal Density. Parameters fixed for the VG density are $t = 0.5$, $\theta = -0.2$, $\sigma = 0.2$, and $\kappa = 0.1$. Normal density is plotted by matching the mean and variance to the VG.

Table 11.3
Annualized Moments of VG Density in Figure 11.3

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
VG with $\theta = -0.2$	-0.2	0.209762	-0.00256	0.352066
Normal	-0.2	0.209762	0	0

[11.3] Lévy Measure for Variance Gamma (VG) Process

We saw that the Lévy measure for the normal tempered α -stable process in the general case $0 < \alpha < 1$ is given by the equation (11.25). The Lévy density of the VG process can be obtained by setting $\alpha = 0$ in (11.25) as:

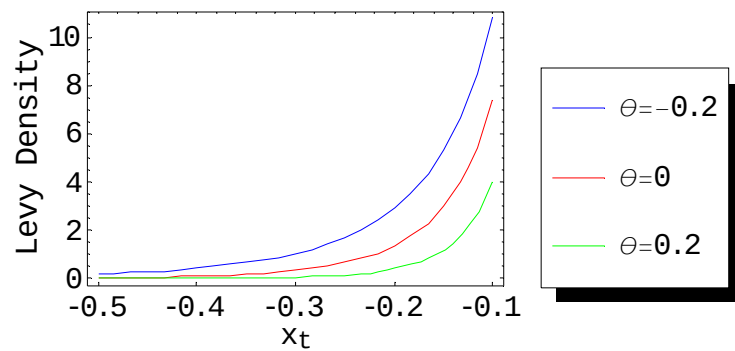
$$\ell(x) = \frac{1}{\kappa|x|} \exp\{Ax - B|x|\}, \quad (11.31)$$

where $A = \frac{\theta}{\sigma^2}$ and $B = \frac{\sqrt{\frac{2\sigma^2}{\kappa} + \theta^2}}{\sigma^2}$ and (11.31) corresponds to MCC (1998) equation 14.

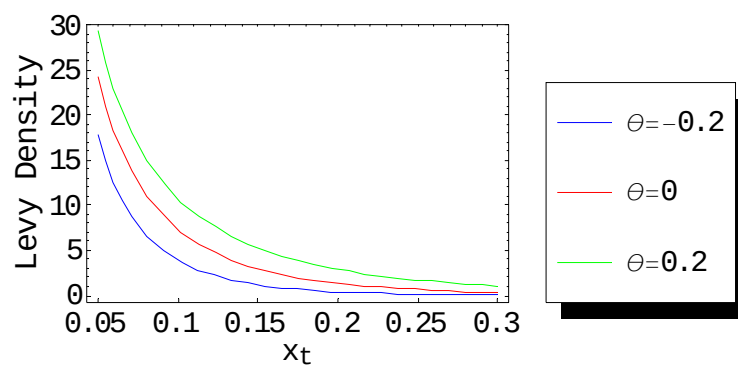
Tails of both Lévy density of the equation (11.31) and probability density of the equation (11.29) of the VG process are exponentially decayed with rates $\lambda_+ = B - A$ (for the upper tail) and $\lambda_- = B + A$ (for the lower tail).

A VG process is an infinite activity Lévy process since the integral of Lévy measure (11.31), $\int_{-\infty}^{\infty} \ell(x)dx$, does not converge in $[0, \infty]$ (i.e. an infinite integral). This means that a VG process has infinitely many small jumps but a finite number of large jumps inheriting the infinite arrival rate of jumps from the gamma subordinator. In other words, the Lévy measure $\ell(x)$ of the VG process is concentrated around zero which will be illustrated soon.

An example of Lévy measure $\ell(x)$ for the VG process is plotted in Figure 11.4 where panel A plots the lower tail and panel B plots the upper tail because $\ell(x)$ becomes complex infinity when $x = 0$ (i.e. infinite arrival rate). Generally, the sign of the drift parameter of the Brownian motion with drift process θ determines the skewness of VG Lévy density as illustrated and the case $\theta = -0.2$ is a mirror image of the case $\theta = 0.2$. This means that if $\theta = 0$, we have a symmetric VG Lévy density. Larger values of the variance rate parameter κ for the gamma subordinator with unit variance rate (i.e. higher degree of randomness of the subordination) leads to lower exponential decay rate of the Lévy measure $\ell(x)$ (i.e. lower B in the above), thus the tails of Lévy measure becomes fatter indicating higher probability of large jumps. This point is illustrated in Figure 11.5.

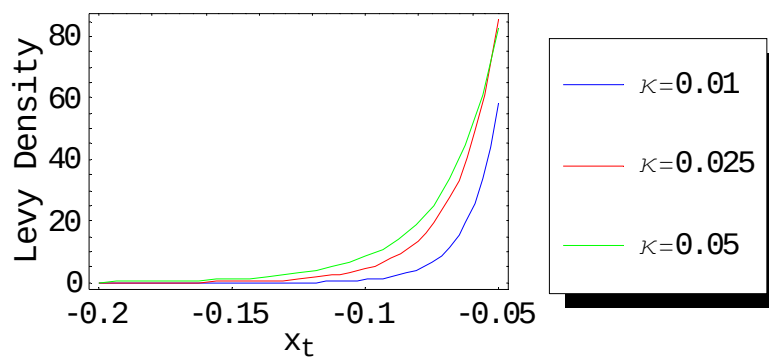


A) Lower tail.

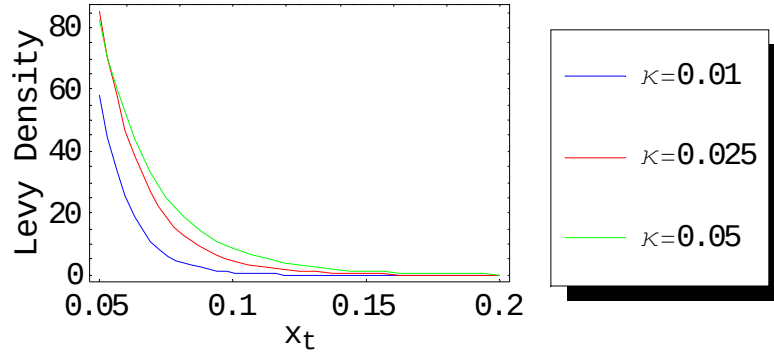


B) Upper tail.

Figure 11.4: Lévy Measure $\ell(x)$ for VG Process with Different Values for the Drift Parameter of the Brownian Motion Process θ . Parameters fixed are $t = 0.5$, $\sigma = 0.2$, and $\kappa = 0.5$.



A) Lower tail.



B) Upper tail.

Figure 11.5: Lévy Measure $\ell(x)$ for VG Process with Different Values for the Variance Rate Parameter κ . Parameters fixed are $t = 0.5$, $\theta = 0$, and $\sigma = 0.2$.

[11.4] Modeling Stock Price Dynamics with VG Process and VG Log-Return Density

We saw in section 7.6 that in the BS case, a stock price process $\{S_t; 0 \leq t \leq T\}$ is modeled as an exponential Lévy process of the form:

$$S_t = S_0 e^{X_t},$$

where $\{X_t; 0 \leq t \leq T\}$ is a Lévy process. BS choose a Brownian motion with drift (continuous diffusion process) as the Lévy process:

$$X_t = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t.$$

The fact that a stock price S_t is modeled as an exponential of Lévy process X_t means that its log-return $\ln(\frac{S_t}{S_0})$ is modeled as a Lévy process such that:

$$\ln\left(\frac{S_t}{S_0}\right) = X_t = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t.$$

BS model can be categorized as the only continuous exponential Lévy model apparently because a Brownian motion process is the only continuous (i.e. no jumps) Lévy process.

Now by replacing a standard Brownian motion process B_t with a VG process $VG(x_t; \theta, \sigma, \kappa)$ and κ being the variance rate of the gamma subordinator, stock price

dynamics under historical probability measure $\mathbb{P}$ can be expressed as (MCC (1998) equation 21):

$$S_t = S_0 \exp \left[\left\{ m + \frac{1}{\kappa_{\mathbb{P}}} \ln \left(1 - \theta_{\mathbb{P}} \kappa_{\mathbb{P}} - \frac{\sigma_{\mathbb{P}}^2 \kappa_{\mathbb{P}}}{2} \right) \right\} t + VG(x_t; \theta_{\mathbb{P}}, \sigma_{\mathbb{P}}, \kappa_{\mathbb{P}}) \right], \quad (11.32)$$

where m is the instantaneous rate of return on the stock under $\mathbb{P}$.

A stock price process $\{S_t; 0 \leq t \leq T\}$ is not a martingale under $\mathbb{P}$ because risk-averse investors expect S_t to grow at a rate greater than the constant risk-free interest rate r :

$$E_t^{\mathbb{P}}[e^{-r\Delta} S_{t+\Delta}] > S_t.$$

Thus, for the purpose of option pricing we convert non-martingales into martingales by changing the probability measure. In other words, we will try to find an equivalent probability measure $\mathbb{Q}$ (called risk-neutral measure) under which the stock price discounted by the risk-free interest rate becomes martingale:

$$E_t^{\mathbb{Q}}[e^{-r\Delta} S_{t+\Delta}] = S_t.$$

This risk-neutral dynamics is given by:

$$S_t = S_0 \exp \left[\left\{ r + \frac{1}{\kappa_{\mathbb{Q}}} \ln \left(1 - \theta_{\mathbb{Q}} \kappa_{\mathbb{Q}} - \frac{\sigma_{\mathbb{Q}}^2 \kappa_{\mathbb{Q}}}{2} \right) \right\} t + VG(x_t; \theta_{\mathbb{Q}}, \sigma_{\mathbb{Q}}, \kappa_{\mathbb{Q}}) \right]. \quad (11.33)$$

In VG model, log-return $z_t \equiv \ln(S_t / S_0)$ is modeled as under historical probability measure $\mathbb{P}$:

$$z_t = \left\{ m + \frac{1}{\kappa_{\mathbb{P}}} \ln \left(1 - \theta_{\mathbb{P}} \kappa_{\mathbb{P}} - \frac{\sigma_{\mathbb{P}}^2 \kappa_{\mathbb{P}}}{2} \right) \right\} t + VG(x_t; \theta_{\mathbb{P}}, \sigma_{\mathbb{P}}, \kappa_{\mathbb{P}}). \quad (11.34)$$

VG process x_t has VG density of the form of equation (11.29). Use the relationship (11.34):

$$x_t = z_t - \left\{ m + \frac{1}{\kappa_{\mathbb{P}}} \ln \left(1 - \theta_{\mathbb{P}} \kappa_{\mathbb{P}} - \frac{\sigma_{\mathbb{P}}^2 \kappa_{\mathbb{P}}}{2} \right) \right\} t, \quad (11.35)$$

where the term inside the curly bracket is deterministic. Therefore, log-return $z_t \equiv \ln(S_t / S_0)$ density under historical probability measure $\mathbb{P}$ in VG model can be written as (MCC (1998) equation 23):

$$VG(z_t) = \frac{\sqrt{2} \exp\left(\frac{\theta}{\sigma^2} x_t\right)}{\kappa^{t/\kappa} \sigma \sqrt{\pi} \Gamma(t/\kappa)} \left(\frac{x_t^2}{\frac{2\sigma^2}{\kappa} + \theta^2} \right)^{\frac{t}{2\kappa} - \frac{1}{4}} K_{\frac{t}{\kappa} - \frac{1}{2}} \left(\frac{\sqrt{x_t^2 \left(\frac{2\sigma^2}{\kappa} + \theta^2 \right)}}{\sigma^2} \right), \quad (11.36)$$

where $x_t = z_t - \left\{ m + \frac{1}{\kappa} \ln \left(1 - \theta \kappa - \frac{\sigma^2 \kappa}{2} \right) \right\} t$. This answers to the statement we made in section 11.2 that those standardized moments of a VG process x_t in the equation (11.30) are not equivalent to those of log-return $z_t \equiv \ln(S_t / S_0)$ density of VG model because of a correction term $-\left\{ m + \frac{1}{\kappa} \ln \left(1 - \theta \kappa - \frac{\sigma^2 \kappa}{2} \right) \right\} t$. VG log-return density $VG(z_t)$ does not have a nice closed form expression for the standardized moments like those for a VG process. Thus, we will simply numerically calculate those when necessary. By taking care of this correction term we plot several figures here for the purpose of illustration. Several very interesting properties of VG log-return density $VG(z_t)$ should be worth mentioning. If $\theta = 0$ (i.e. subordination of zero-drift Brownian motion process with gamma subordinator), VG log-return density is symmetric (zero-skewness). Generally, the sign of the drift parameter of the Brownian motion with drift process θ determines the skewness of VG log-return density as illustrated in Figure 11.6. The case $\theta = -0.2$ is a mirror image of the case $\theta = 0.2$.

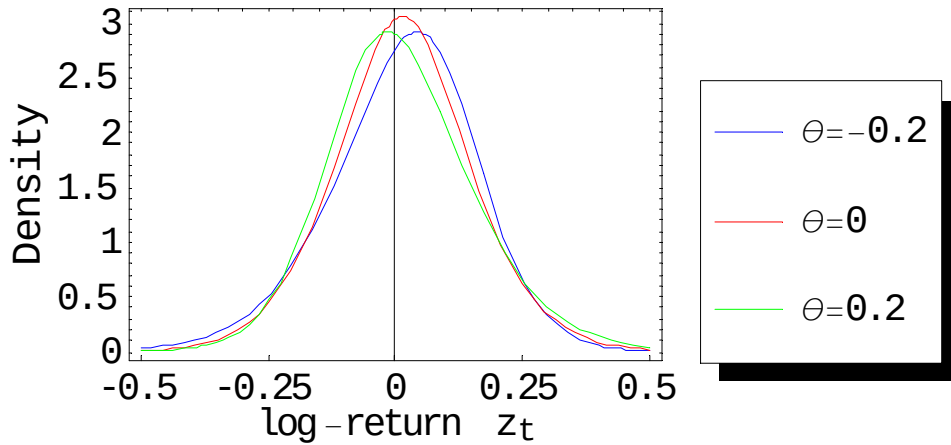


Figure 11.6: VG Log-Return Density for Different Values of Drift Parameter of the Brownian Motion Process θ . Parameters fixed are $t = 0.5$, $\sigma = 0.2$, $\kappa = 0.1$, and $m = 0.05$.

Table 11.4
Annualized Moments of VG Log-Return Density in Figure 11.6

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
$\theta = -0.2$	0.028399	0.209762	-0.392262	0.704132
$\theta = 0$	0.02998	0.2	0	0.6
$\theta = 0.2$	0.275439	0.209762	0.392262	0.704132

Secondly, larger value of variance rate κ (which is the variance rate of the gamma subordinator with $E[S_t] = t$ and determines how random the subordination is) makes the density fatter-tailed and higher-peaked as illustrated in Figure 11.7.

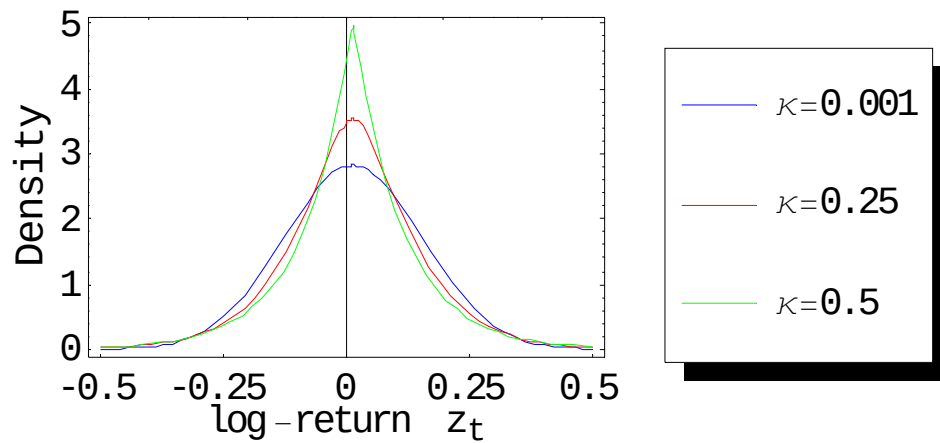


Figure 11.7: VG Log-Return Density for Different Values of Variance Rate Parameter κ . Parameters fixed are $t = 0.5$, $\theta = 0$, $\sigma = 0.2$, and $m = 0.05$.

Table 11.5
Annualized Moments of VG Log-Return Density in Figure 11.7

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
$\kappa = 0.001$	0.0299998	0.2	0	0.006
$\kappa = 0.25$	0.0299498	0.2	0	1.5
$\kappa = 0.5$	0.0298993	0.2	0	3

Also note that VG log return density has higher peak and fatter tails (more leptokurtic) when matched to the BS normal log return density as illustrated in Figure 11.8.

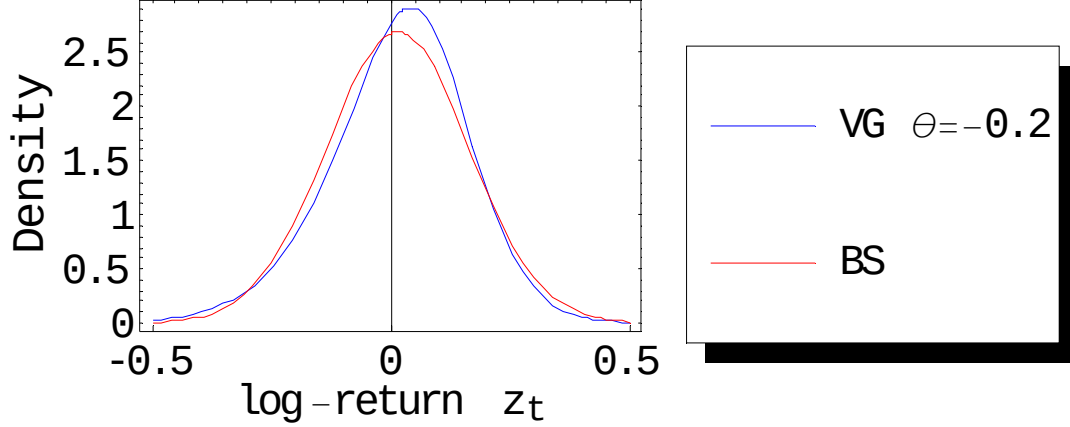


Figure 11.8: VG Log-Return Density vs. Black-Scholes Normal Log-Return Density. Parameters fixed for the VG log-return density are $t = 0.5$, $\theta = -0.2$, $\sigma = 0.2$, $\kappa = 0.1$, and $m = 0.05$. Black-Scholes normal density is plotted by matching the mean and variance to the VG.

Table 11.6
Annualized Moments of VG Log-Return Density in Figure 11.8

Model	Mean	Standard Deviation	Skewness	Excess Kurtosis
VG with $\theta = -0.2$	0.0283992	0.209762	-0.392262	0.704132
BS	0.0283992	0.209762	0	0

[11.5] Option Pricing with VG Model

We saw that the risk-neutral dynamics of a stock price is given by the equation (11.33):

$$S_t = S_0 \exp \left[\left\{ r + \frac{1}{\kappa_{\mathbb{Q}}} \ln \left(1 - \theta_{\mathbb{Q}} \kappa_{\mathbb{Q}} - \frac{\sigma_{\mathbb{Q}}^2 \kappa_{\mathbb{Q}}}{2} \right) \right\} t + VG(x_t; \theta_{\mathbb{Q}}, \sigma_{\mathbb{Q}}, \kappa_{\mathbb{Q}}) \right],$$

where $\{x_t; 0 \leq t \leq T\}$ is a VG process on a space $(\Omega, \mathcal{F}, \mathbb{Q})$ and r is the constant risk-free interest rate. Then, a European call option price $C_{VG}(S_t; \tau = T - t)$ is calculated as:

$$C_{VG}(S_t; \tau) = e^{-r\tau} E^{\mathbb{Q}} [\max(S_T - K, 0) | \mathcal{F}_t].$$

We saw in section 11.2.7 that because of the subordination structure $X_t \equiv \theta(S_t) + \sigma B_{S_t}$ of the VG process $\{X_t; 0 \leq t \leq T\}$, the probability density of VG process can be expressed as

the conditionally normal by conditioning on the fact that the realized value of the random time change by a gamma subordinator S_t with unit mean rate is $S_t = g$:

$$VG(x_t | t = S_t = g) = \frac{1}{\sqrt{2\pi\sigma^2 g}} \exp\left\{-\frac{(x_t - \theta g)^2}{2\sigma^2 g}\right\}.$$

Using this conditional normality of VG process, the call price conditioned on the fact that the realized value of the random time change by a gamma subordinator S_t with unit mean rate is $S_t = g$ can be obtained as a BS type formula (equation 6.5 in Madan and Milne (1991)):

$$\begin{aligned} C_{VG} | g = S_t & \left(1 - \frac{\kappa(\alpha + s)^2}{2}\right)^{\frac{\tau}{\kappa}} \exp\left(\frac{(\alpha + s)^2 g}{2}\right) N\left(\frac{d}{\sqrt{g}} + (\alpha + s)\sqrt{g}\right) \\ & - K \exp(-r\tau) \left(1 - \frac{\kappa\alpha^2}{2}\right)^{\frac{\tau}{\kappa}} \exp\left(\frac{\alpha^2 g}{2}\right) N\left(\frac{d}{\sqrt{g}} + \alpha\sqrt{g}\right), \end{aligned} \quad (11.37)$$

where N is a cumulative distribution function for a standard normal variable. And the parameterizations are follows:

$$\begin{aligned} \zeta &= -\frac{\theta}{\sigma^2}, \quad s = \frac{\sigma}{\sqrt{1 + \frac{\theta^2 \kappa}{2\sigma^2}}}, \quad \alpha = \zeta s = -\frac{\theta}{\sqrt{\sigma^2 + \frac{\theta^2 \kappa}{2}}}, \quad c_1 = \frac{\kappa}{2}(\alpha + s)^2, \quad c_2 = \frac{\kappa}{2}\alpha^2, \\ \frac{1-c_1}{1-c_2} &= 1 + \kappa\left(\theta - \frac{\sigma^2}{2}\right), \quad \text{and } d = \frac{1}{s} \left[\ln\left(\frac{S_t}{K}\right) + r(T-t) + \frac{T-t}{\kappa} \ln\left(\frac{1-c_1}{1-c_2}\right) \right]. \end{aligned}$$

Unconditional call price is then obtained by integrating the conditional price $C_{VG} | g$ with respect to gamma density of the equation (11.21):

$$C_{VG}(S_t; \tau) = \int_0^\infty (C_{VG} | g) \frac{(1/\kappa)^{t/\kappa}}{\Gamma(t/\kappa)} g^{\frac{t}{\kappa}-1} e^{-g/\kappa} dg. \quad (11.38)$$

MCC (1998) defines a user specified function $\Psi(a, b, \gamma)$:

$$\Psi(a, b, \gamma) = \int_0^\infty N\left(\frac{a}{\sqrt{u}} + b\sqrt{u}\right) \frac{u^{\gamma-1} \exp(-u)}{\Gamma(\gamma)} du. \quad (11.39)$$

After numerous changes of variables, a European call price for VG model is obtained as (MCC (1998) theorem 2):

$$C_{VG}(S_t; \tau) = S_t \Psi \left(d \sqrt{\frac{1-c_1}{\kappa}}, (\alpha + s) \sqrt{\frac{\kappa}{1-c_1}}, \frac{\tau}{\kappa} \right) - K \exp(-rt) \Psi \left(d \sqrt{\frac{1-c_2}{\kappa}}, \alpha s \sqrt{\frac{\kappa}{1-c_2}}, \frac{\tau}{\kappa} \right). \quad (11.40)$$

Because the equation (11.39) is not a closed-form function, the call price (11.40) is not a closed-form option pricing function. In order to obtain a closed-form option pricing function, MCC (1998) rewrites (11.39) in terms of the modified Bessel function of the second kind K and the confluent hypergeometric function of two variables $\Phi(\alpha, \beta, \gamma; x, y)$ as follows:

$$\begin{aligned} & \Psi(a, b, \gamma) \\ &= \frac{c^{\gamma+1/2} \exp(\text{sign}(a)c)(1+u)^\gamma}{\sqrt{2\pi}\Gamma(\gamma)\gamma} K_{\gamma+1/2}(c) \Phi(\gamma, 1-\gamma, 1+\gamma; \frac{1+u}{2}, -\text{sign}(a)c(1+u)) \\ & - \text{sign}(a) \frac{c^{\gamma+1/2} \exp(\text{sign}(a)c)(1+u)^{1+\gamma}}{\sqrt{2\pi}\Gamma(\gamma)(1+\gamma)} K_{\gamma-1/2}(c) \Phi(1+\gamma, 1-\gamma, 2+\gamma; \frac{1+u}{2}, -\text{sign}(a)c(1+u)) \\ & + \text{sign}(a) \frac{c^{\gamma+1/2} \exp(\text{sign}(a)c)(1+u)^\gamma}{\sqrt{2\pi}\Gamma(\gamma)\gamma} K_{\gamma-1/2}(c) \Phi(\gamma, 1-\gamma, 1+\gamma; \frac{1+u}{2}, -\text{sign}(a)c(1+u)), \end{aligned}$$

where:

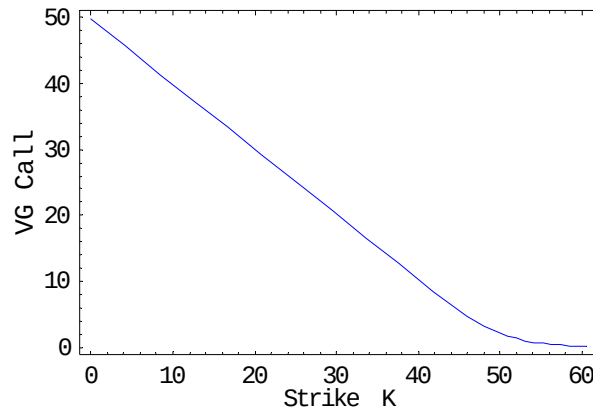
$$\begin{aligned} \Phi(\alpha, \beta, \gamma; x, y) &= \frac{\Gamma(\gamma)}{\Gamma(\alpha)\Gamma(\gamma-\alpha)} \int_0^1 u^{\alpha-1} (1-u)^{\gamma-\alpha-1} (1-ux)^{-\beta} \exp(uy) du, \\ c &= |a| \sqrt{2+b^2}, \\ u &= \frac{b}{\sqrt{2+b^2}}. \end{aligned} \quad (11.41)$$

We were unsuccessful implementing the, so called, closed form VG call price of the equation (11.40) with (11.41). We believe this is due to the confluent hypergeometric function of two variables $\Phi(\alpha, \beta, \gamma; x, y)$ having the singularity at $u = 1$. Therefore, we use the numerical VG call price of the equation (11.40) with (11.39).

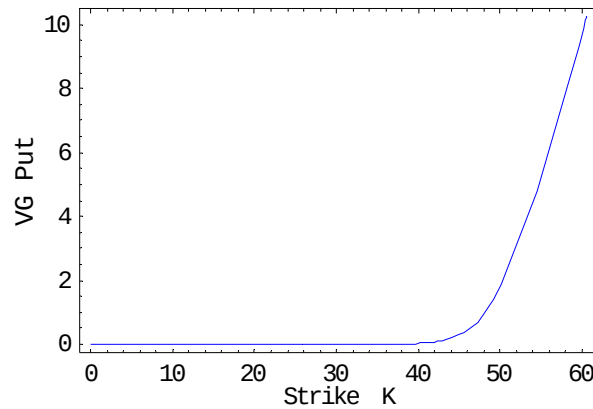
[11.6] Option Pricing Example of VG Model

In this section we calculate the numerical VG call prices of the equation (11.40) with (11.39) and compare them with BS counterparts. Consider pricing a plain vanilla call with common parameters and variables fixed $S_t = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $\tau = 0.25$. VG parameters are set at $\theta = -0.1$ and $\kappa = 0.1$, unless mentioned otherwise.

The first thing we noticed when implementing the numerical VG call price is that its failure to achieve convergence since the integrand (11.39) has a non-integrable singularity at $u = 0$. Practical problem with this singularity is that the numerical VG call price cannot be calculated for deep OTM calls with $60 < K$ as illustrated in Figure 11.9 where the numerical VG call prices are computed for the range of the strike price $0 \leq K \leq 100$. The level of out of moneyness of calls whose prices can be computed with the numerical VG formula depends on the VG parameters θ and κ . This is a big problem for the purpose of calibration because we are mainly interested in OTM options due to their high liquidity. For puts, the numerical VG price cannot be calculated for deep ITM puts. But this won't be a huge problem.



A) Call.

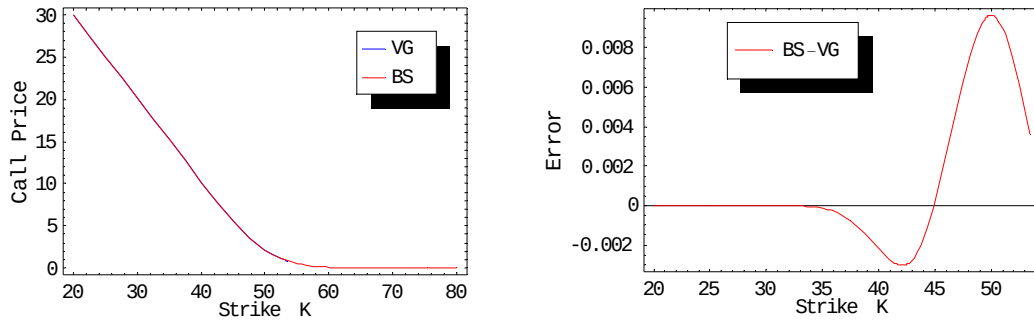


B) Put.

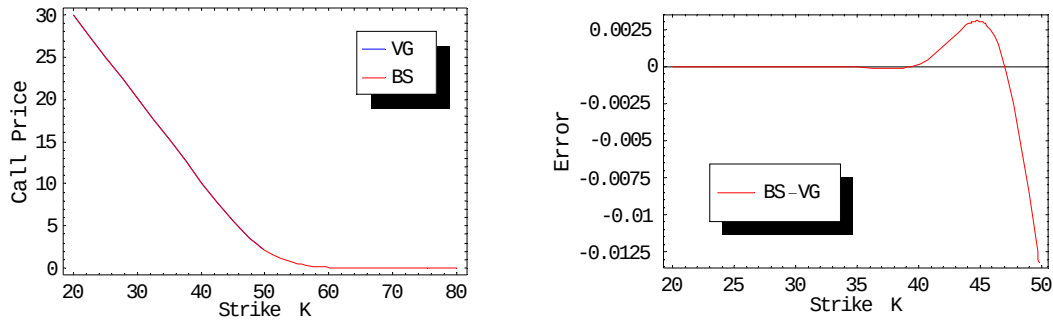
Figure 11.9: Numerical VG Prices. $S_t = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $\tau = 0.25$. VG parameters are set at $\theta = -0.1$ and $\kappa = 0.1$.

Figure 11.10 compares numerical VG call prices with BS counterparts. Note the followings: (1) When VG parameters (θ and κ) are small, these two prices cannot be distinguished by naked eyes in Panel A left. When the difference is plotted on the right, the numerical VG price underprices ATM calls by approximately 0.008. (2) The size of the drift parameter of the subordinated Brownian motion process θ has very little impact on the numerical VG call price. As the size of θ increases from -0.01 to -0.3 (i.e.

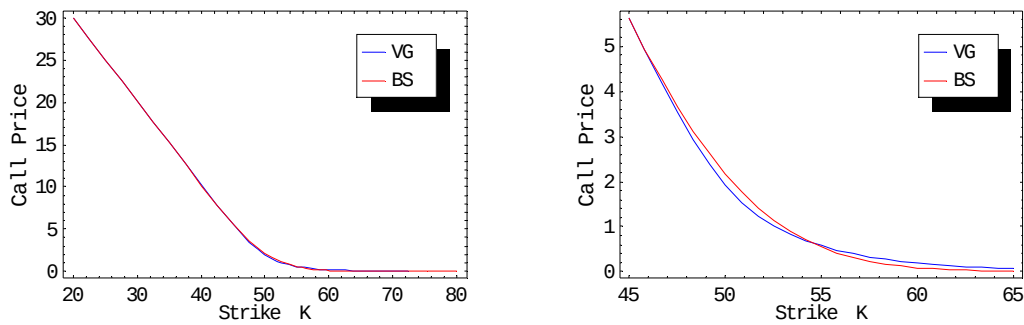
increased skewness of log-return density), the numerical VG price and the BS price are virtually identical as illustrated in Panel B where the numerical VG price overprices ATM call by approximately 0.0125. (3) As the variance rate parameter of the gamma subordinator κ increases (i.e. increased randomness associated with time change), VG model underprices ATM call and overprices OTM call as illustrated in Panel C. Panel D shows the distinct character of the numerical VG call price is that the overvaluation of OTM call.

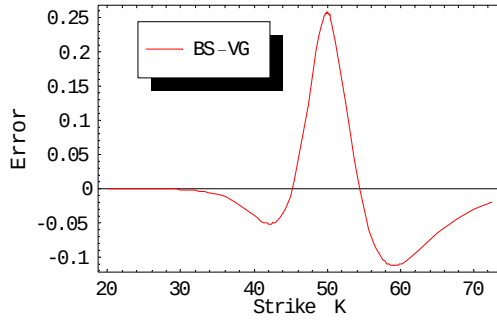


A) VG parameters: $\theta = -0.01$ and $\kappa = 0.01$.

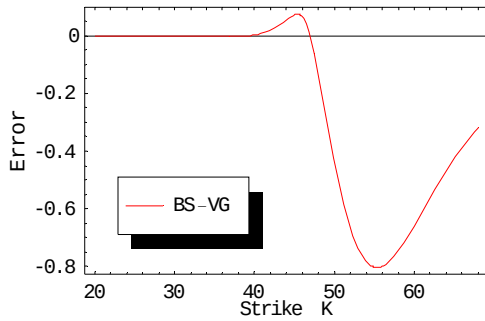
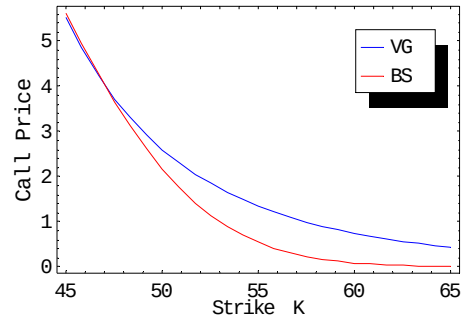
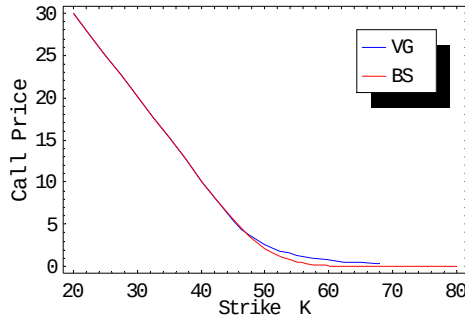


B) VG parameters: $\theta = -0.3$ and $\kappa = 0.01$.





C) VG parameters: $\theta = -0.01$ and $\kappa = 0.3$.

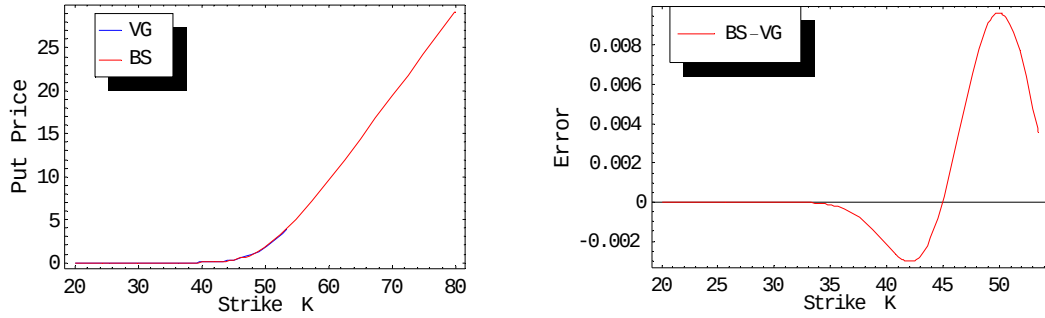


D) VG parameters: $\theta = -0.3$ and $\kappa = 0.3$.

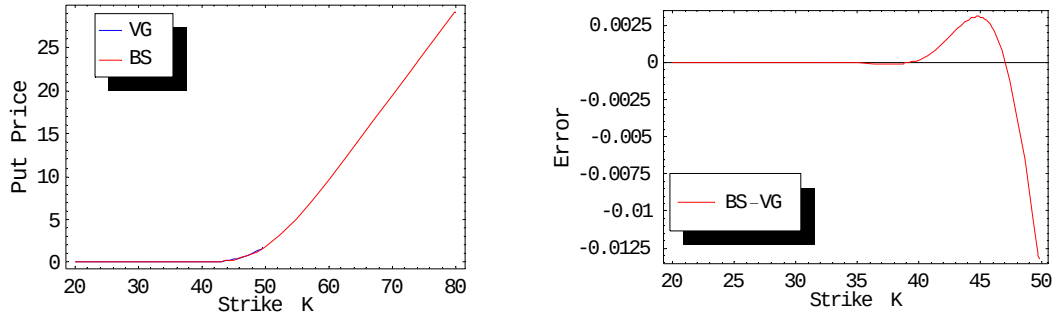
Figure 11.10: VG Call Price vs. Black-Scholes Call Price. Parameters and variables used are $S_t = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $\tau = 0.25$.

Figure 11.11 compares numerical VG put prices with BS counterparts. Note the followings: (1) When VG parameters (θ and κ) are small, these two prices cannot be distinguished by naked eyes in Panel A left. When the difference is plotted on the right, the numerical VG price slightly underprices ATM puts by approximately 0.008. (2) The size of the drift parameter of the subordinated Brownian motion process θ has very little impact on the numerical VG put price. As the size of θ increases from -0.01 to -0.3 (i.e. increased skewness of log-return density), the numerical VG price and the BS price are virtually identical as illustrated in Panel B where the numerical VG price overprices ATM put slightly by approximately 0.0125. (3) As the variance rate parameter of the gamma subordinator κ increases (i.e. increased randomness associated with time

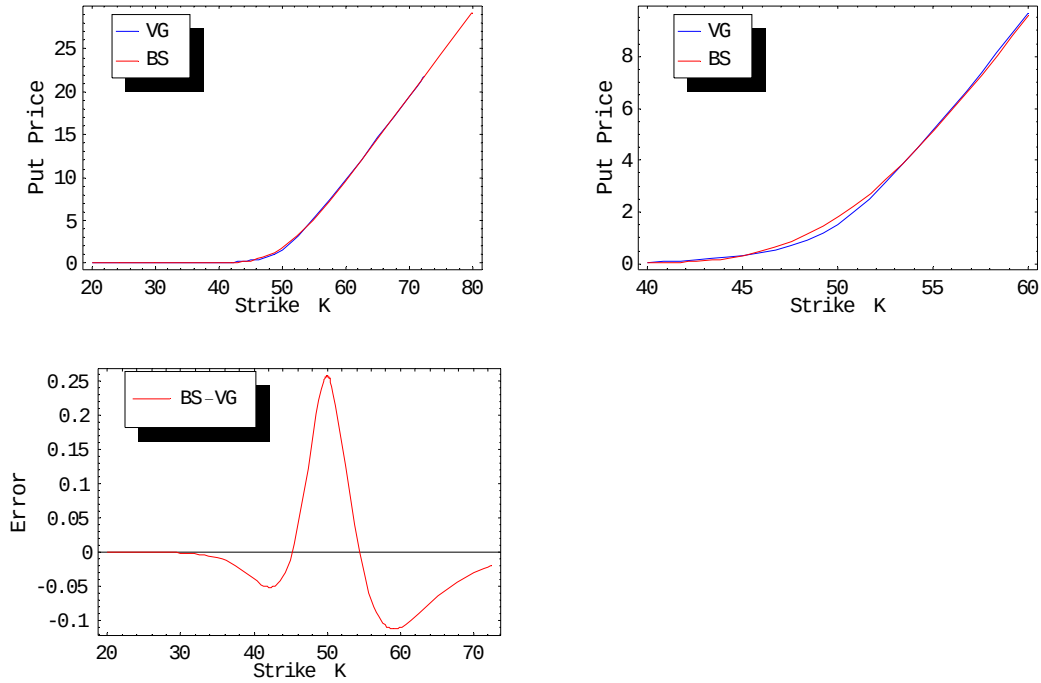
change), VG model underprices ATM puts and overprices ITM calls as illustrated in Panel C. Panel D shows the distinct character of the numerical VG put price is that the overvaluation of ITM put.



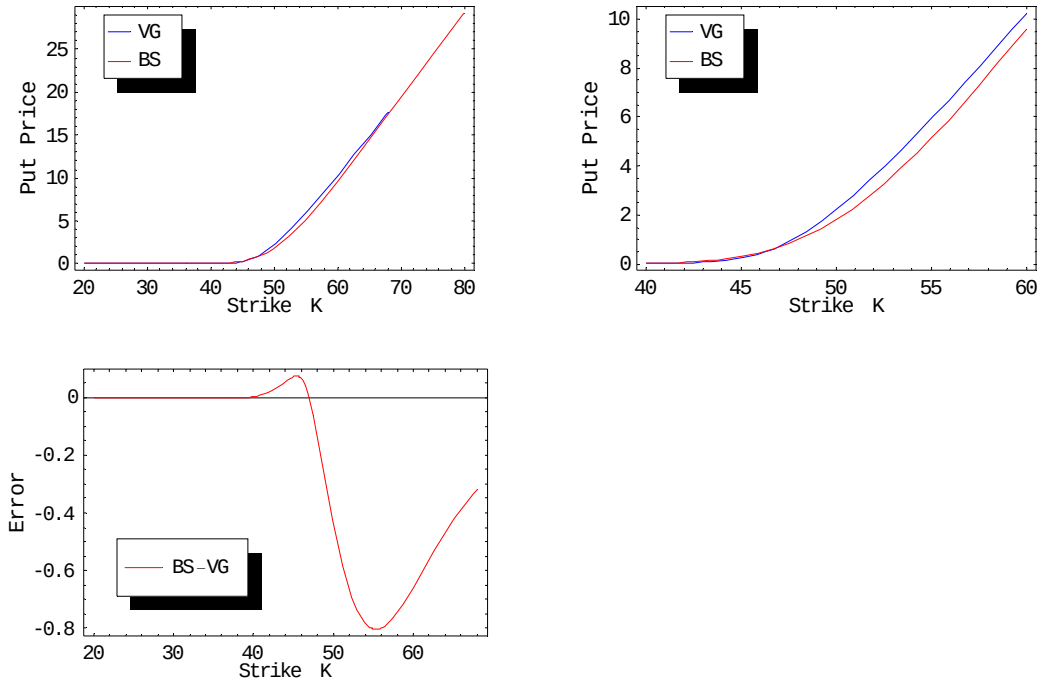
A) VG parameters: $\theta = -0.01$ and $\kappa = 0.01$.



B) VG parameters: $\theta = -0.3$ and $\kappa = 0.01$.



C) VG parameters: $\theta = -0.01$ and $\kappa = 0.3$.



D) VG parameters: $\theta = -0.3$ and $\kappa = 0.3$.

Figure 11.11: VG Put Price vs. Black-Scholes Put Price. Parameters and variables used are $S_t = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $\tau = 0.25$.

[11.7] Lévy Measure of VG Log-Return z_t

We saw in section 11.3 that the Lévy measure of the VG process is given by the equation (11.31). We saw in section 11.4 that the log-return $z_t \equiv \ln(S_t / S_0)$ and a VG process x_t are related by the equation (11.35). Therefore, the Lévy measure of the VG log-return z_t can be expressed as:

$$\ell(z) = \frac{1}{\kappa |x|} \exp \{ Ax - B |x| \}, \quad (11.43)$$

where:

$$x = z - \left\{ m + \frac{1}{\kappa_{\mathbb{P}}} \ln \left(1 - \theta_{\mathbb{P}} \kappa_{\mathbb{P}} - \frac{\sigma_{\mathbb{P}}^2 \kappa_{\mathbb{P}}}{2} \right) \right\},$$

$$A = \frac{\theta_{\mathbb{P}}}{\sigma_{\mathbb{P}}^2},$$

$$B = \frac{\sqrt{\frac{2\sigma_{\mathbb{P}}^2}{\kappa_{\mathbb{P}}} + \theta_{\mathbb{P}}^2}}{\sigma_{\mathbb{P}}^2}.$$

An example of Lévy measure of the VG log-return z_t is plotted in Figure 11.12. Generally, the sign of the drift parameter of the subordinated Brownian motion with drift process θ determines the skewness of Lévy measure $\ell(z)$ of the VG log-return z_t as illustrated and the case $\theta = -0.2$ is a mirror image of the case $\theta = 0.2$. This means that if $\theta = 0$, we have a symmetric Lévy measure $\ell(z)$. Larger values of the variance rate parameter κ for the gamma subordinator with unit variance rate (i.e. higher degree of randomness of the subordination) leads to lower exponential decay rate of the Lévy measure $\nu(dx)$, thus the tails of Lévy measure becomes fatter indicating higher probability of large jumps. This point is illustrated in Figure 11.13.

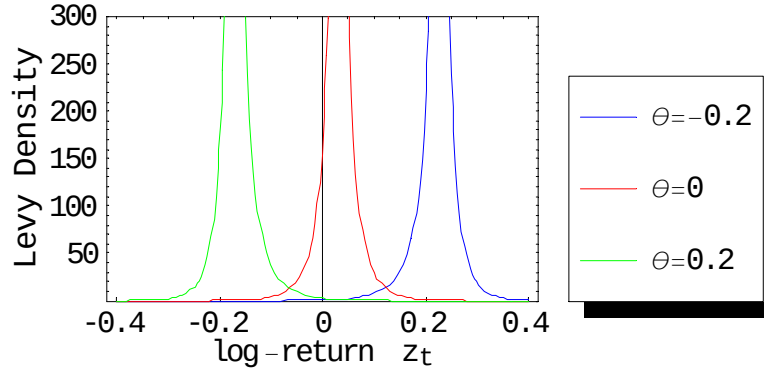


Figure 11.12: Lévy Measures of VG Log-Return z_t with Different Values for the Drift Parameter of the Subordinated Brownian Motion Process θ . Parameters fixed are $\sigma = 0.2$, $\kappa = 0.1$, and $m = 0.05$.

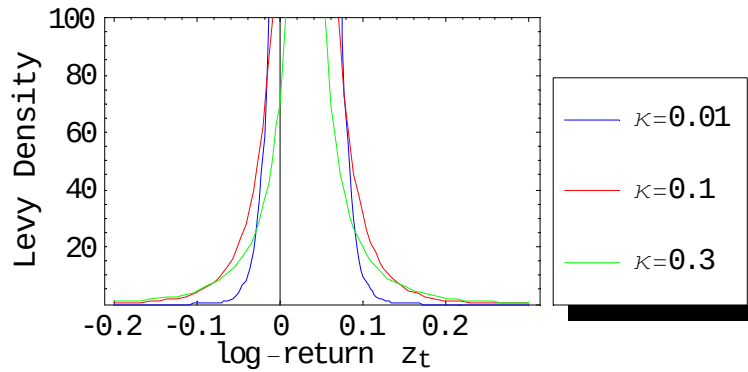


Figure 11.13: Lévy Measures of VG Log-Return z_t with Different Values for the Variance Rate Parameter of the Gamma Subordinator κ . Parameters fixed are $\sigma = 0.2$, $\theta = 0$, and $m = 0.05$.

[12] VG (Variance Gamma) Model with Fourier Transform Pricing

We saw in section 11 that the most important character of VG model is its conditional normal property. Because of the subordination structure $X_t \equiv \theta(S_t) + \sigma B_{S_t}$ of the VG process $\{X_t; 0 \leq t \leq T\}$, the probability density of VG process can be expressed as the conditionally normal by conditioning on the fact that the realized value of the random time change by a gamma subordinator S_t with unit mean rate is $S_t = g$. Similarly, the conditional call price can be obtained as a BS type formula of (11.37) and the unconditional call price is then obtained by integrating the conditional price with respect to gamma density. As a result, a call price can be expressed as single numerical integration problem of the equation (11.40) with (11.39).

But as we realize MCC (1998) use very complicated process of numerous changes of variables in order to obtain a closed form call price of the equation (11.40) with (11.41) and to express the call price in terms of special functions of mathematics. This procedure is not only cumbersome, but also deficient of generality.

As far as the numerical VG call price is concerned, it is simpler to implement, but it has a critical problem of failing to achieve convergence for deep OTM calls since the integrand (11.39) has a non-integrable singularity at $u = 0$.

Thus, we are in desperate need of more general and simpler method of pricing for VG model and other exponential Lévy models. In this section we present our version of Fourier transform option pricing with VG model. We would like readers to appreciate its generality and simplicity.

[12.1] VG Model with Fourier Transform Pricing Method

The first step to FT option pricing is to obtain the CF of the log stock price $\ln S_t$. Risk-neutral log stock price dynamics can be obtained from the equation (11.33) as:

$$\ln S_t = \ln S_0 + \left\{ r + \frac{1}{\kappa_Q} \ln \left(1 - \theta_Q \kappa_Q - \frac{\sigma_Q^2 \kappa_Q}{2} \right) \right\} t + VG(x_t; \theta_Q, \sigma_Q, \kappa_Q). \quad (12.1)$$

VG process x_t has VG density of the form of equation (11.29). After rearrangement:

$$x_t = \ln S_t - \left[\ln S_0 + \left\{ r + \frac{1}{\kappa_Q} \ln \left(1 - \theta_Q \kappa_Q - \frac{\sigma_Q^2 \kappa_Q}{2} \right) \right\} t \right], \quad (12.2)$$

where the term inside the square bracket is deterministic. Therefore, risk-neutral log stock price density in VG model can be written as:

$$VG(\ln S_t) = \frac{\sqrt{2} \exp\left(\frac{\theta}{\sigma^2} x_t\right)}{\kappa^{t/\kappa} \sigma \sqrt{\pi} \Gamma(t/\kappa)} \left(\frac{x_t^2}{\frac{2\sigma^2}{\kappa} + \theta^2} \right)^{\frac{t}{2\kappa} - \frac{1}{4}} K_{\frac{t}{\kappa} - \frac{1}{2}} \left(\frac{\sqrt{x_t^2 \left(\frac{2\sigma^2}{\kappa} + \theta^2 \right)}}{\sigma^2} \right), \quad (12.3)$$

where $x_t = \ln S_t - \left[\ln S_0 + \left\{ r + \frac{1}{\kappa_{\mathbb{Q}}} \ln \left(1 - \theta_{\mathbb{Q}} \kappa_{\mathbb{Q}} - \frac{\sigma_{\mathbb{Q}}^2 \kappa_{\mathbb{Q}}}{2} \right) \right\} t \right]$. By Fourier transforming

(12.3) with FT parameters $(a, b, \kappa) = (1, 1, \kappa)$, its CF can be calculated. But this direct computation of CF is difficult because it involves the Bessel function.

Consider a following example. The CF function of a multiplicative standard Brownian motion process $\{\sigma B_t; 0 \leq t \leq T\}$ is:

$$\begin{aligned} \phi_B(\omega) &\equiv \int_{-\infty}^{\infty} e^{i\omega B_t} \mathbb{P}(B_t) dB_t = \int_{-\infty}^{\infty} e^{i\omega B_t} \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{B_t^2}{2\sigma^2 t}\right) dB_t \\ &= \exp\left(-\frac{\sigma^2 t \omega^2}{2}\right). \end{aligned}$$

When the drift term At is added, the CF function of a Brownian motion with drift process $\{W_t \equiv At + \sigma B_t; 0 \leq t \leq T\}$ is:

$$\begin{aligned} \phi_W(\omega) &\equiv \int_{-\infty}^{\infty} e^{i\omega W_t} \mathbb{P}(W_t) dW_t = \int_{-\infty}^{\infty} e^{i\omega W_t} \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left\{-\frac{(W_t - At)^2}{2\sigma^2 t}\right\} dW_t \\ &= \exp\left(iAt\omega - \frac{\sigma^2 t \omega^2}{2}\right) = \exp(iAt\omega) \exp\left(-\frac{\sigma^2 t \omega^2}{2}\right). \end{aligned}$$

This implies that the CF of a Brownian motion with drift process can be obtained by simple multiplication of the CF function of a multiplicative standard Brownian motion process and $\exp(iAt\omega)$. Let's use this logic in our context. We derived the CF of a VG process x_t which is given by the equation (11.27):

$$\phi_X(\omega) = \left(1 + \frac{\sigma^2 \omega^2 \kappa}{2} - i\theta \kappa \omega \right)^{-\frac{t}{\kappa}}. \quad (11.27)$$

From the VG risk-neutral log stock price dynamics of the equation (12.1), by taking care of the drift term $At \equiv \ln S_0 + \left\{ r + \frac{1}{\kappa_Q} \ln \left(1 - \theta_Q \kappa_Q - \frac{\sigma_Q^2 \kappa_Q}{2} \right) \right\} t$, the CF of log stock price can be expressed as:

$$\phi_{\ln S_t}(\omega) = \exp \left[i\omega \left\{ \ln S_0 + \left(r + \frac{1}{\kappa_Q} \ln \left(1 - \theta_Q \kappa_Q - \frac{\sigma_Q^2 \kappa_Q}{2} \right) \right) t \right\} \right] \left(1 + \frac{\sigma^2 \omega^2 \kappa}{2} - i\theta \kappa \omega \right)^{-\frac{t}{\kappa}}.$$

Set $t = T$ and $\ln S_T \equiv s_T$. And drop Q :

$$\phi_T(\omega) = \exp \left[i\omega \left\{ s_0 + \left(r + \frac{1}{\kappa} \ln \left(1 - \theta \kappa - \frac{\sigma^2 \kappa}{2} \right) \right) T \right\} \right] \left(1 + \frac{\sigma^2 \omega^2 \kappa}{2} - i\theta \kappa \omega \right)^{-\frac{T}{\kappa}}. \quad (12.4)$$

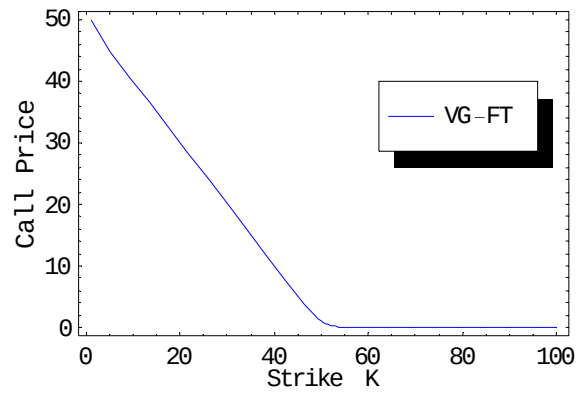
By substituting the above CF into the general FT call pricing formula of the equation (8.17), we obtain VG-FT call pricing formula:

$$C_{VG-FT}(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha + 1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega} d\omega, \quad (12.5)$$

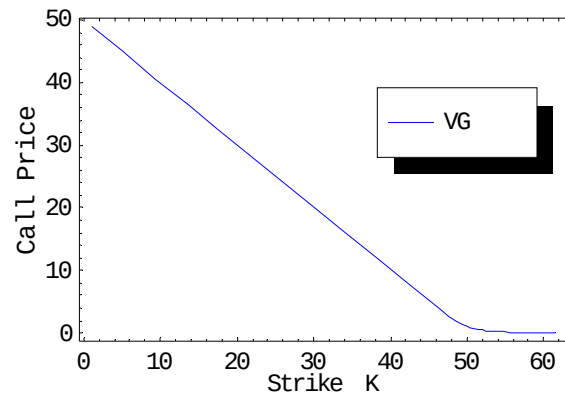
where:

$$\phi_T(\omega) = \exp \left[i\omega \left\{ s_0 + \left(r + \frac{1}{\kappa} \ln \left(1 - \theta \kappa - \frac{\sigma^2 \kappa}{2} \right) \right) T \right\} \right] \left(1 + \frac{\sigma^2 \omega^2 \kappa}{2} - i\theta \kappa \omega \right)^{-\frac{T}{\kappa}}.$$

We implement the VG-FT formula (12.5) with decay rate parameter $\alpha = 1.5$ and compare the result to the numerical VG call price using common parameters and variables fixed $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$. Figure 12.1 demonstrates two important points. One is that as previously mentioned, the numerical VG price cannot be calculated for deep OTM calls because they fail to converge which is shown in Panel B. The other is that although the VG-FT price can be calculated for deep OTM calls but they also fail to achieve convergence at the acceptable level of accuracy. On top of this, the additional issue with VG-FT price is that it oscillates severely for the deep ITM and OTM calls as illustrated by Figure 12.2. Especially for the deep OTM calls, this oscillation yields negative call prices. This interesting oscillation pattern for deep OTM calls occur for the BS-FT call price with much smaller degree. Merton-FT call price, in contrast, has no oscillation.

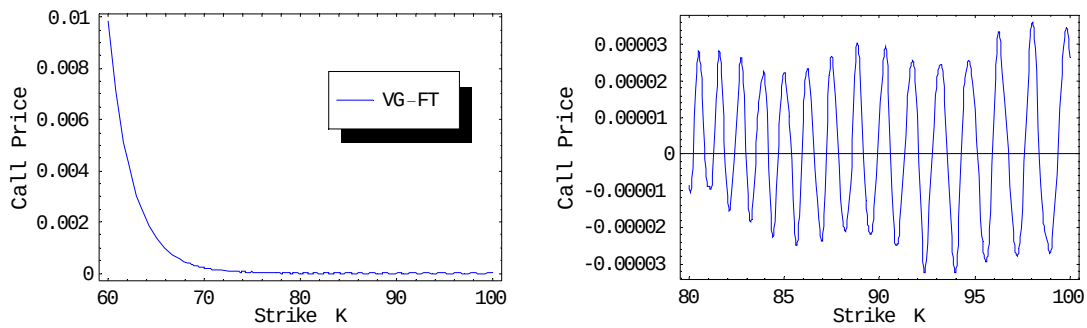


A) VG-FT $\alpha = 1.5$ call price.

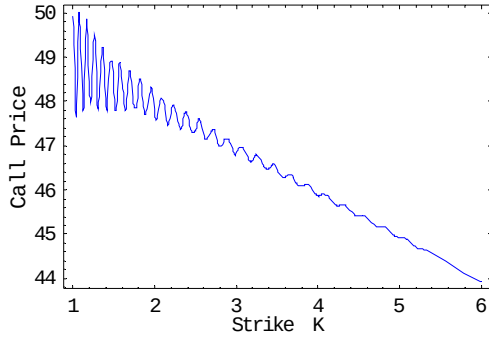


B) Numerical VG call price.

Figure 12.1: VG-FT $\alpha = 1.5$ Call Price Vs. Numerical VG Call Price. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$.



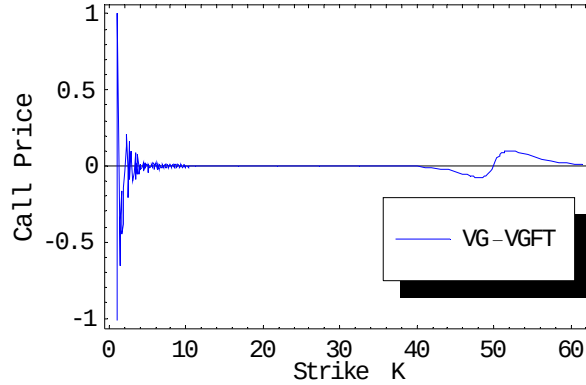
A) Deep OTM call price for $60 \leq K \leq 100$ on the left and for $80 \leq K \leq 100$ on the right.



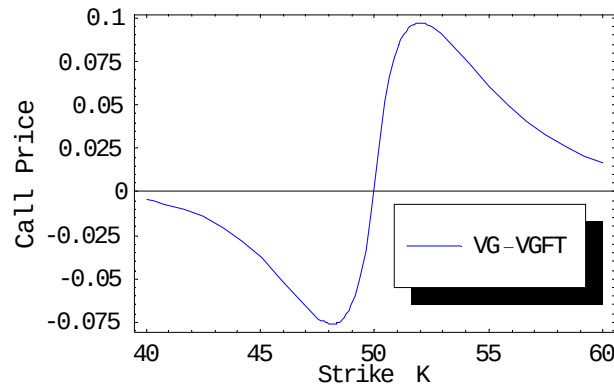
B) Deep ITM call price for $1 \leq K \leq 6$.

Figure 12.2: Oscillation of VG-FT $\alpha = 1.5$ Price for Deep ITM and Deep OTM Calls. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$.

As a principle, VG-FT $\alpha = 1.5$ call price and the numerical VG call price are same (i.e. they are supposed to be). Figure 12.3 plots the difference between the numerical VG and VG-FT call price for the range of strikes with which the numerical VG price can be computed. Panel A shows that the above mentioned oscillation of VG-FT price causes the price difference to oscillate for deep ITM calls of $1 \leq K \leq 5$. For example, for $K = 1$ call the numerical VG price is 48.9247 while VG-FT price is 49.9419 leading to an error of 1.0173. But deep ITM calls are not most researchers' interests. More important error occurs around ATM calls of the strike $45 \leq K \leq 55$ which is illustrated in Panel B. The maximum size of the error is about 0.1.



A) For the range of strike price $1 \leq K \leq 100$.



B) For the range of strike price $40 \leq K \leq 60$

Figure 12.3: Numerical VG Call Price Minus VG-FT $\alpha = 1.5$ Call Price. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$.

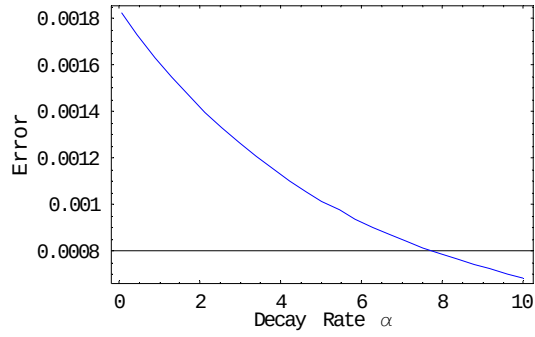
Next, we discuss CPU time considering calls with $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, and $T = 20/252$ as a function of varying strike price K . VG parameters are set as $\theta = -0.1$ and $\kappa = 0.1$. Table 12.1 reveals that VG-FT formula is slower than the numerical VG formula, but speed is not an issue for most of the pricing purposes.

Table 12.1: CPU Time for Calls with Different Moneyness

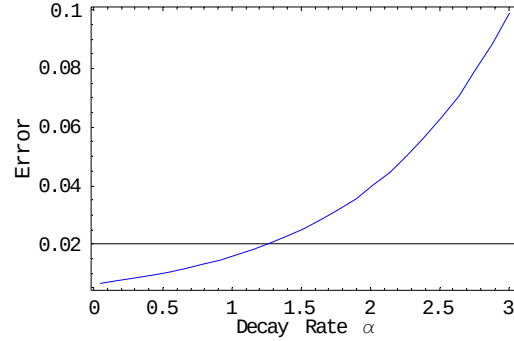
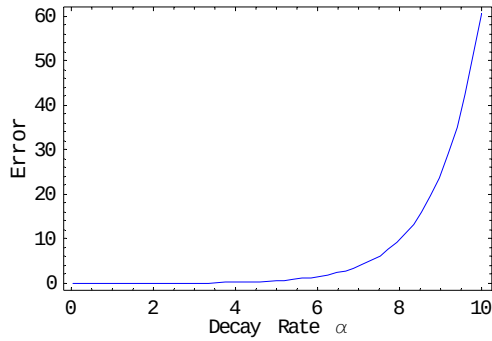
Price is in the bracket.

Method	Strike $K = 20$	Strike $K = 50$	Strike $K = 80$
VG (numerical)	0.08 seconds (29.9999)	0.05 seconds (1.04663)	0.11 seconds (NA)
VG-FT $\alpha = 1$	0.17 seconds (30)	0.09 seconds (1.04107)	0.16 seconds (-8.41091×10^{-6})

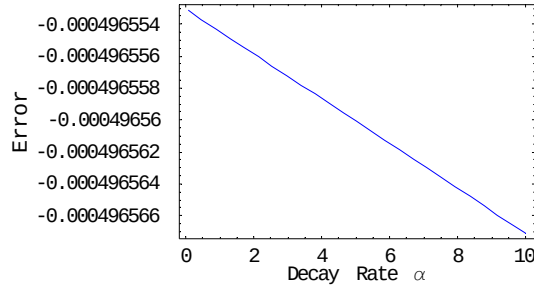
We investigate the level of decay rate parameter α for VG-FT formula. Figure 12.4 illustrates the pricing error of VG-FT price relative to the numerical VG price for a one day to maturity $T = 1/252$ call as a function of varying α and Figure 12.5 is for one month to maturity $T = 20/252$ call. First of all, Panel A and C of Figure 12.4 and 12.5 tell us that the size of the error does not vary significantly for $0.05 \leq \alpha \leq 10$ in the case of OTM and ATM call. Small exception is for OTM call with $T = 1/252$ in Panel A of Figure 12.4 where larger α makes the error smaller. While Panel B in Figure 12.4 and 5 shows that smaller α is better for deep ITM calls. Judging from these findings, we recommend the choice of $1 \leq \alpha \leq 2$ for the purpose of general pricing (i.e. ATM, OTM, and ITM calls) which includes the suggested value by MCC (1998) of $\alpha = 1.5$.



A) For an OTM Call with $K = 60$. Numerical VG price cannot be calculated for deep OTM calls with $K = 70$ or 80 .

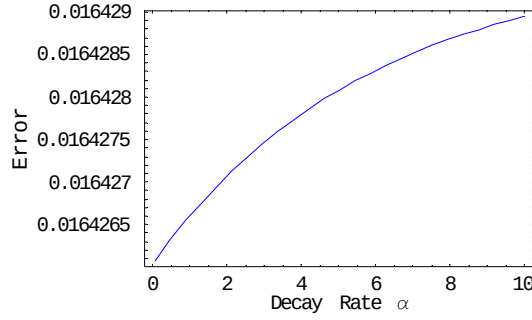


B) For an ITM Call with $K = 20$.

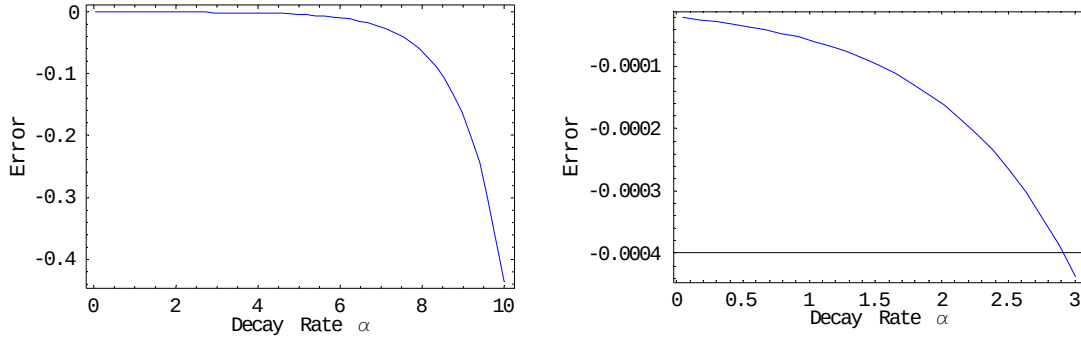


C) For an ATM Call with $K = 50$.

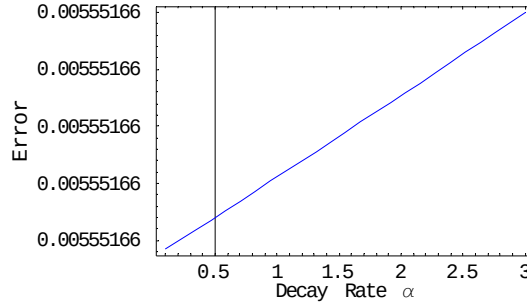
Figure 12.4: Numerical VG Price Minus VG-FT Price for One Day to Maturity Call as a Function of Decay Rate Parameter $0.05 \leq \alpha \leq 10$. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 1/252$, $\theta = -0.1$, and $\kappa = 0.1$.



A) For an OTM Call with $K = 60$. Numerical VG price cannot be calculated for deep OTM calls with $K = 70$ or 80 .



B) For an ITM Call with $K = 20$.



C) For an ATM Call with $K = 50$.

Figure 12.5: Numerical VG Price Minus VG-FT Price for Call with $T = 20/252$ as a Function of Decay Rate Parameter $0.05 \leq \alpha \leq 10$. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $\theta = -0.1$, and $\kappa = 0.1$.

Which is better, VG-FT price of the equation (12.5) or the numerical VG price (11.40) with (11.39)? It is a tough call. Numerical VG call price has a critical problem of failing to achieve convergence for deep OTM calls since the integrand (11.39) has a non-integrable singularity at $u = 0$. VG-FT formula calculates the prices for calls regardless of the moneyness, but it also has convergence problem and oscillation problem for deep ITM and OTM calls which yields negative prices. Can DFT solve these problems

associated with FT price? Let's see in the next section. The value of the decay rate parameter $\alpha = 1.5$ suggested by MCC (1998) seems to be appropriate.

[12.2] Discrete Fourier Transform (DFT) Call Pricing Formula with VG Model

To improve computational time of the VG-FT call price, we apply our version of DFT call price formula.

VG-DFT call price can be simply obtained by substituting the CF of VG log stock price of the equation (12.5) into the general DFT call price of the equation (8.38):

$$C_T(k_n) \approx \frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k} \times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\} \exp(-i2\pi jn/N), \quad (12.6)$$

where $w_{j \in [0, N-1]}$ are trapezoidal rule weights:

$$w_j = \begin{cases} 1/2 & \text{for } j = 0 \text{ and } N-1 \\ 1 & \text{for others} \end{cases},$$

and:

$$\psi_T(\omega) = \frac{e^{-rT} \phi_T(\omega - (\alpha + 1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega},$$

with $\phi_T(\cdot)$ given by (12.4).

[12.3] Implementation and Performance of DFT Pricing Method with VG Model

In this section, performance of VG-DFT $\alpha = 1.5$ call price of the equation (12.6) is tested by comparing results to the VG-FT $\alpha = 1.5$ call price of the equation (12.5) under various settings. VG-DFT $\alpha = 1.5$ call price is implemented using $N = 4096$ samples and log strike space sampling interval $\Delta k = 0.005$. This corresponds to angular frequency domain sampling interval of $\Delta \omega = 0.306796$ radians, the total sampling range in the log strike space is $K = N\Delta k = 20.48$, its sampling rate is 200 samples per unit of k , and the total sampling range in the angular frequency domain is $\Omega = N\Delta \omega = 1256.64$.

Firstly, let's investigate the difference in price and CPU time. Consider calculating 100-point call prices for a range of strike price $1 \leq K \leq 100$ with interval 1 with common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$. Figure 12.6 reports the price and Table 12.2 compares CPU time. We notice

that prices are basically same (i.e. they are supposed to be same because DFT is an approximation of FT) and the use of DFT significantly improves the computational time.

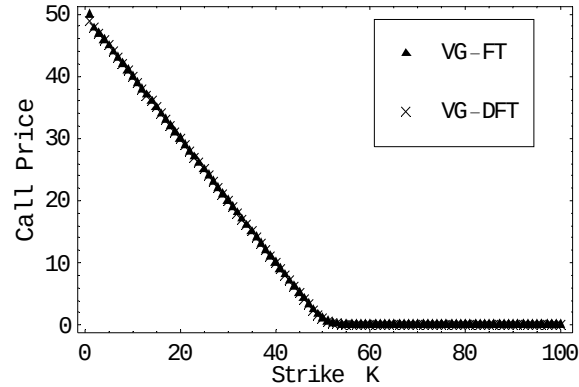
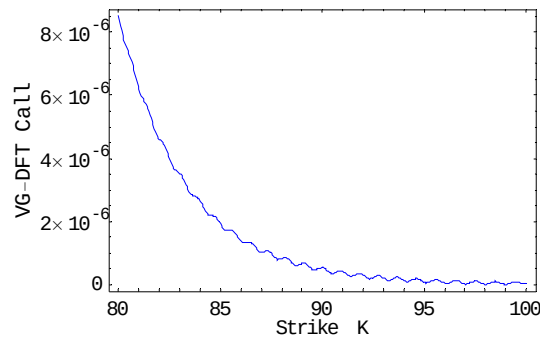


Figure 12.6: VG-FT $\alpha = 1.5$ Vs. VG-DFT $\alpha = 1.5$. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$.

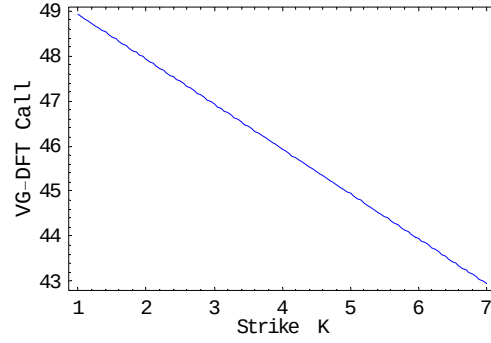
Table 12.2 CPU Time for Calculating 100-point call prices for a range of strike price $1 \leq K \leq 100$ with interval 1 Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$.

Method	CPU Time
VG-FT $\alpha = 1.5$	15.512 seconds
VG-DFT $\alpha = 1.5$ $N = 4096$, $\Delta k = 0.005$	0.571 seconds

Secondly, we saw in section 12.1 that VG-FT price has an oscillation pattern for deep ITM and OTM calls. We find that VG-DFT price has almost zero oscillation for deep ITM calls and very small (i.e. negligible) oscillation for deep OTM calls demonstrated by Figure 12.7.



A) Deep OTM call price for $80 \leq K \leq 100$.



B) Deep ITM call price for $1 \leq K \leq 7$.

Figure 12.7: Oscillation of VG-DFT $\alpha = 1.5$ Price for Deep ITM and Deep OTM Calls. Common parameters and variables fixed are $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, $q = 0.02$, $T = 20/252$, $\theta = -0.1$, and $\kappa = 0.1$.

Thirdly, we pay attention to the price difference between VG-DFT and VG-FT especially for very near-maturity calls. Consider a call option with common parameters and variables $S_0 = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$. VG parameters are set as $\theta = -0.1$, and $\kappa = 0.1$. Figures 12.8 to 12.10 plot three series of price differences computed by the VG-FT $\alpha = 1.5$ and VG-DFT $\alpha = 1.5$ with $N = 4096$ and $\Delta k = 0.005$ as a function of time to maturity of less than a month $1/252 \leq T \leq 20/252$. These figures demonstrate that, as we expected, the price difference tends to increase as the maturity nears regardless of the moneyness of the call. We consider the maximum price difference of the size 0.025 negligible.

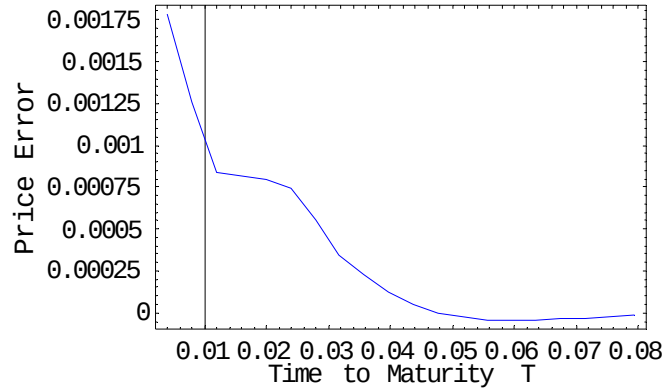


Figure 12.8: Plot of VG-FT $\alpha = 1.5$ Price Minus VG-DFT $\alpha = 1.5$ Price for Deep-OTM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 80$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$, $\theta = -0.1$, and $\kappa = 0.1$.

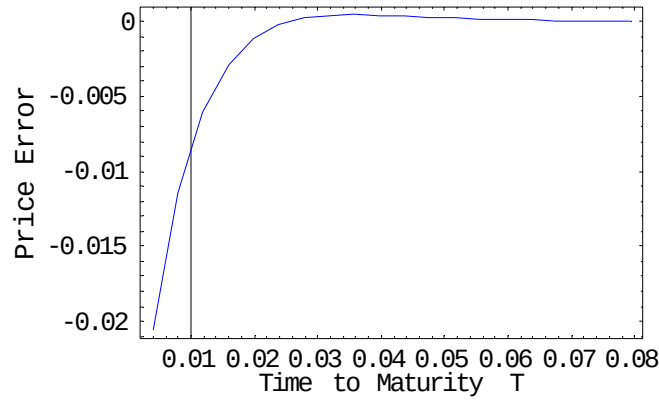


Figure 12.9: Plot of VG-FT $\alpha = 1.5$ Price Minus VG-DFT $\alpha = 1.5$ Price for ATM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$ $\theta = -0.1$, and $\kappa = 0.1$.

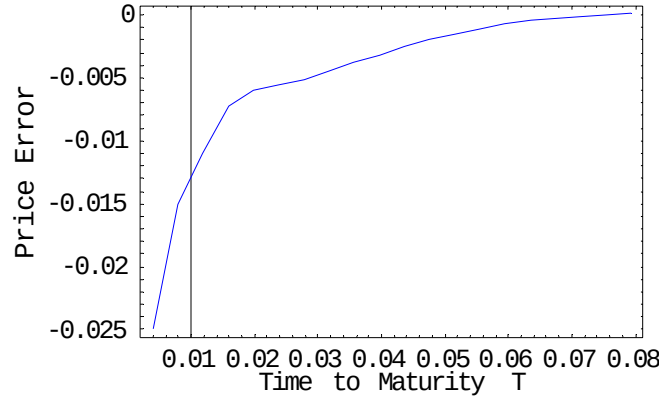
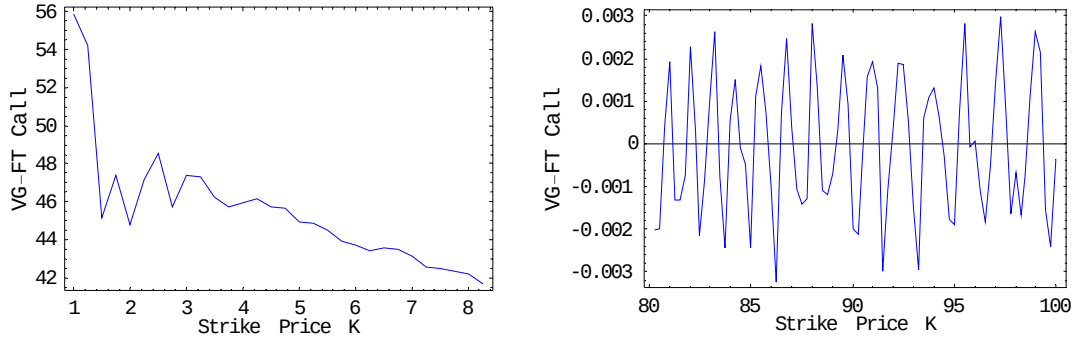
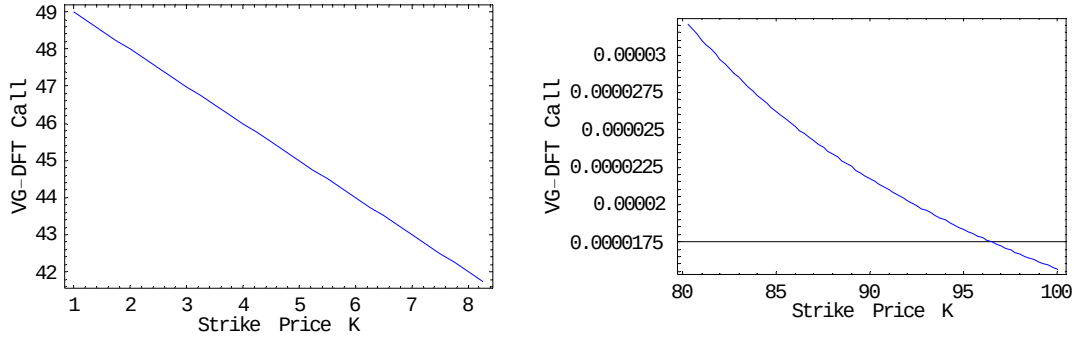


Figure 12.10: Plot of VG-FT $\alpha = 1.5$ Price Minus VG-DFT $\alpha = 1.5$ Price for Deep-ITM Call as a Function of Time to Maturity $1/252 \leq T \leq 20/252$. Common parameters and variables fixed are $S_0 = 50$, $K = 20$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$ $\theta = -0.1$, and $\kappa = 0.1$.

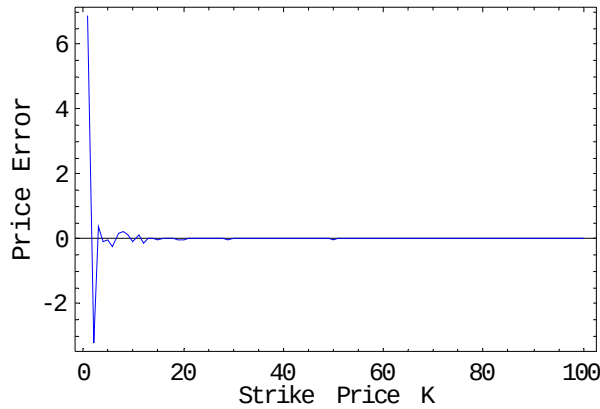
Figure 12.11 clarifies the superiority of VG-DFT price over VG-FT price for one day to maturity call. Panel A exhibits the oscillation of VG-FT price for deep ITM and OTM calls. In contrast, VG-DFT price has almost zero oscillation shown in Panel B. Panel C reminds us that as a principle these two prices are same despite the large deviation for deep ITM calls which is not so important.



A) Oscillation of VG-FT prices for Deep ITM and OTM calls.



B) Almost zero oscillation of VG-DFT prices for Deep ITM and OTM calls.



C) VG-FT Call Price – VG-DFT Call Price.

Figure 12.11: VG-DFT $\alpha = 1.5$ Price Vs. VG-FT $\alpha = 1.5$ Price for 1-Day-to-Maturity ($T = 1/252$). Common parameters and variables fixed are $S_0 = 50$, $K = 50$, $\sigma = 0.2$, $r = 0.05$, and $q = 0.02$ $\theta = -0.1$, and $\kappa = 0.1$.

We conclude this section by stating the following remarks. As a principle, VG-DFT $\alpha = 1.5$ price and VG-FT $\alpha = 1.5$ price are same. Which is better? The clear-cut answer is VG-DFT price. One reason is that VG-DFT needs significantly smaller CPU time to compute hundreds of option prices which is very important for the calibration. Another reason is that VG-DFT price does not have the oscillation pattern observed with VG-FT price for deep ITM and OTM options.

[12.4] Summary of Formulae of Option Price with Fourier Transform in VG Model

Table 12.3: Summary of Formulae for VG Model

Method	Formula
Numerical VG	$C_{VG}(S_t; \tau) = S_t \Psi \left(d \sqrt{\frac{1-c_1}{\kappa}}, (\alpha + s) \sqrt{\frac{\kappa}{1-c_1}}, \frac{\tau}{\kappa} \right) - K \exp(-rt) \Psi \left(d \sqrt{\frac{1-c_2}{\kappa}}, \alpha s \sqrt{\frac{\kappa}{1-c_2}}, \frac{\tau}{\kappa} \right)$ $\Psi(a, b, \gamma) = \int_0^\infty N \left(\frac{a}{\sqrt{u}} + b\sqrt{u} \right) \frac{u^{\gamma-1} \exp(-u)}{\Gamma(\gamma)} du$
FT	$C_{VG-FT}(T, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega k} \frac{e^{-rT} \phi_T(\omega - (\alpha+1)i)}{\alpha^2 + \alpha - \omega^2 + i(2\alpha+1)\omega} d\omega$ $\phi_T(\omega) = \exp \left[i\omega \left\{ s_0 + \left(r + \frac{1}{\kappa} \ln(1 - \theta\kappa - \frac{\sigma^2 \kappa}{2}) \right) T \right\} \right] \left(1 + \frac{\sigma^2 \omega^2 \kappa}{2} - i\theta\kappa\omega \right)^{-\frac{T}{\kappa}}$
DFT	$C_T(k_n) \approx \frac{\exp(-\alpha k_n) \exp(i\pi n) \exp(-i\pi N/2)}{\Delta k} \times \frac{1}{N} \sum_{j=0}^{N-1} w_j \left\{ \exp(i\pi j) \psi_T(\omega_j) \right\} \exp(-i2\pi jn/N)$ $\psi_T(\omega_j) = \frac{e^{-rT} \phi_T(\omega_j - (\alpha+1)i)}{\alpha^2 + \alpha - \omega_j^2 + i(2\alpha+1)\omega_j}$

[13] Conclusion

We implemented the general FT call price (8.17) and our version of the general DFT call price (8.38) for three different types of exponential Lévy models. DFT call price is implemented using $N = 4096$ samples and log strike space sampling interval $\Delta k = 0.005$. This corresponds to angular frequency domain sampling interval of $\Delta\omega = 0.306796$ radians, the total sampling range in the log strike space is $K = N\Delta k = 20.48$, its sampling rate is 200 samples per unit of k , and the total sampling range in the angular frequency domain is $\Omega = N\Delta\omega = 1256.64$.

In the classical Black-Scholes model, our version of DFT $\alpha = 1$ call price yields the price virtually identical to the original BS price for OTM and ITM calls even for extreme near-maturity case (i.e. $T = 1/252$) although the size of error is larger than BS-FT formula. But the error of BS-DFT price becomes relatively large around (i.e. ± 3) ATM. In our example used, the maximum error is 0.0001345 which occurs at exactly ATM at $S_0 = K = 50$. Increasing the decay rate parameter α or sampling more points (i.e. larger N) cannot reduce the size of this error. But we can accept this size of error when considering the dramatic improvement in the CPU time to compute hundreds of prices.

The result of Merton JD model is the same as that of the BS. Our Merton-DFT $\alpha = 1$ call price yields the price virtually identical to the original Merton price for OTM and ITM calls even for extreme near-maturity case (i.e. $T = 1/252$) although the size of error is larger than Merton-FT formula. But the error of Merton-DFT price becomes relatively large around (i.e. ± 3) ATM. In our example used, the maximum error is 0.0001343 which occurs at exactly ATM at $S_0 = K = 50$.

In the VG model as a principle, VG-DFT $\alpha = 1.5$ price and VG-FT $\alpha = 1.5$ price are same. As we expected, the price difference tends to increase as the maturity nears regardless of the moneyness of the call. We consider the maximum price difference of the size approximately 0.025 negligible. Which is better? The clear-cut answer is VG-DFT price. One reason is that VG-DFT needs significantly smaller CPU time to compute hundreds of option prices which is very important for the calibration. Another reason is that VG-DFT price does not have the oscillation pattern observed with VG-FT price for deep ITM and OTM options.

We hope readers appreciate the excellence of FT and DFT option pricing in the sense of its simplicity and generality and this sequel would be a good starting point.

Appendix

[A.1] Set Theory: Notation and Basics¹

- o $\mathbb{R}$: Collection of all real numbers. Intervals in $\mathbb{R}$ are denoted via each endpoint. A square bracket $[$ indicates its inclusion and an open bracket $($ indicates its exclusion. For example, $[c, d) = \{x \in \mathbb{R} : c \leq x < d\}$. Unbounded intervals are described using ∞ and $-\infty$. Examples are $(-\infty, a) = \{x \in \mathbb{R} : x < a\}$ and $[0, \infty) = \{x \in \mathbb{R} : x \geq 0\} = \mathbb{R}^+$.
- o $\mathbb{R}^+$: Collection of nonnegative elements of $\mathbb{R}$.
- o $\mathbb{R}^d$: d -dimensional Euclidian space. Its elements $x = (x_k)_{k=1,\dots,d}$ and $y = (y_k)_{k=1,\dots,d}$ are column vectors with d real components.
- o $\mathbb{N}$: Collection of all positive integers.
- o $\mathbb{Z}$: Collection of all integers.
- o $\mathbb{Q}$: Collection of all rational numbers.
- o Ω : Universal set (a set of all scenarios).
- o $\emptyset$: Empty set. It has no members.
- o $x \in A$: Membership. The element x is a member of the set A .
- o $A \subset B$: Set inclusion. Every member of A is a member of B .
- o $\{x \in A : P(x)\}$: The set of elements of A with property P .
- o $\mathcal{P}(A)$: The set of all subsets (power set) of A .
- o $A \cap B = \{x : x \in A \text{ and } x \in B\}$. Intersection.
- o $A \cup B = \{x : x \in A \text{ or } x \in B\}$. Union.
- o A^C : The complement of A . The elements of Ω which are not members of A .
 $A^C = \Omega \setminus A$
- o $B \setminus A$: The difference between the set A and B . $B \setminus A = \{x \in B : x \notin A\} = B \cap A^C$
- o $\exists$: There exists.
- o $\forall$: For all.
- o $\bigcap_{\alpha \in \Lambda} A_\alpha = \{x : x \in A_\alpha \text{ for all } \alpha \in \Lambda\} = \{x : \forall \alpha \in \Lambda, x \in A_\alpha\}$
- o $\bigcup_{\alpha \in \Lambda} A_\alpha = \{x : x \in A_\alpha \text{ for some } \alpha \in \Lambda\} = \{x : \exists \alpha \in \Lambda, x \in A_\alpha\}$
- o de Morgan's law: $(\bigcup_{\alpha} A_\alpha)^C = \bigcap_{\alpha} A_\alpha^C$, $(\bigcap_{\alpha} A_\alpha)^C = \bigcup_{\alpha} A_\alpha^C$
- o Disjoint sets: A and B are disjoint if $A \cap B = \emptyset$.

¹ Based on Capinski, Kopp, and Kopp (2004).

- o Pairwise disjoint sets: A family of sets $(A_\alpha)_{\alpha \in \Lambda}$ is pairwise disjoint if $A_\alpha \cap A_\beta = \emptyset$ whenever $\alpha \neq \beta$ ($\alpha, \beta \in \Lambda$).
- o Cartesian product: Cartesian product $C \times D$ of sets C and D is the set of ordered pairs $C \times D = \{(c, d) : c \in C, d \in D\}$.
- o A function $f : C \rightarrow D$: A subset of $C \times D$ where each first coordinate determines the second. Its scope is described by its domain $\mathcal{D}_f = \{c \in C : \exists d \in D, (c, d) \in f\}$ and range $\mathcal{R}_f = \{d \in D : \exists c \in C, (c, d) \in f\}$. A function f associates elements of D with those of C , such that each $c \in C$ has at most one image $d \in D$. This is written as $d = f(c)$.
- o Indicator function : Indicator function 1_C of the set C is the function
$$1_C(x) = \begin{cases} 1 & \text{for } x \in C \\ 0 & \text{for } x \notin C. \end{cases}$$
- o Upper bound: u is an upper bound for a set $A \subset \mathbb{R}$ if $a \leq u$ for all $a \in A$.
- o Lower bound: l is a lower bound for a set $A \subset \mathbb{R}$ if $l \leq a$ for all $a \in A$.
- o $\sup A$: Supremum (least upper bound) of A . Minimum of all upper bounds.
- o $\inf A$: Infimum (greatest lower bound) of A . Maximum of all lower bounds.

[A.2] Measure²

[A.2.1] Null Sets

Consider an interval to define the length of a set. Let H be a bounded interval $H = [a, b]$, $H = [a, b)$, $H = (a, b]$, or $H = (a, b)$. In each case the length of H is $l(H) = b - a$. A one element set is null since $l(\{b\}) = l([b, b]) = 0$ (The length of a single point is 0). Next, consider the length of a finite set $B = \{1, 5, 20\}$. The length of a finite set is 0 since $l(B) = l(1) + l(5) + l(20) = 0$. Thus the length of a set can be calculated by adding the lengths of its pieces.

Definition: Null set

A null set $A \subseteq \mathbb{R}$ is a set that may be covered by a sequence of intervals of arbitrarily small total length, that is, a sequence $\{H_n : n \geq 1\}$ of intervals given any $\varepsilon > 0$ such that

$$A \subseteq \bigcup_{k=1}^{\infty} H_n$$

$$\sum_{n=1}^{\infty} l(H_n) < \varepsilon.$$

Any one-element set $\{x\}$ is null. For example, when $H_1 = (x - \frac{\varepsilon}{8}, x + \frac{\varepsilon}{8})$ and $H_n = [0, 0]$ for $n \geq 2$,

² Based on Capinski, Kopp, and Kopp (2004).

$$\sum_{n=1}^{\infty} l(H_n) = l(H_1) = \frac{\varepsilon}{4} < \varepsilon.$$

More generally, any countable set $A = \{x_1, x_2, \dots\}$ is null.

Theorem: Union of a Sequence of Null Sets

Let $(N_k)_{k \geq 1}$ be a sequence of null sets, then their union

$$N = \bigcup_{k=1}^{\infty} N_k$$

is also null.

Any countable set is null since it is quite sparse when compared with an interval and it does not contribute to its length.

[A.2.2] Outer Measure

Definition: (Lebesgue) Outer Measure

The (Lebesgue) outer measure of any set $A \subseteq \mathbb{R}$ is the non-negative real number

$$m^*(A) = \inf Z_A$$

where

$$Z_A = \left\{ \sum_{n=1}^{\infty} l(H_n) : H_n \text{ are intervals, } A \subseteq \sum_{n=1}^{\infty} H_n \right\}.$$

The outer measure is the infimum (greatest lower bound) of length of all possible covers of A .

Theorem

$A \subseteq \mathbb{R}$ is a null set if and only if $m^*(A) = 0$.

Note that $m^*(\emptyset) = 0$, $m^*({x}) = 0$ for any $x \in \mathbb{R}$, $m^*(\mathbb{Q}) = 0$, and $m^*(Y) = 0$ for any countable Y .

Proposition

If $A \subset B$, $m^*(A) \leq m^*(B)$.

This means that m^* is monotone: the larger the set, the greater its outer measure.

Theorem

The outer measure of an interval is equal to its length.

Theorem

Outer measure is countably subadditive, i.e. for any sequence of sets $\{B_n\}$

$$m^*\left(\bigcup_{n=1}^{\infty} B_n\right) \leq \sum_{n=1}^{\infty} m^*(B_n).$$

When an interval is shifted along the real line, there is no change in its length:

$$l([c, d]) = l([c + u, d + u]) = d - c.$$

Proposition

Outer measure is translation invariant,

$$m^*(A) = m^*(A + u)$$

where $A \subset \mathbb{R}$, $u \in \mathbb{R}$, and $A + u = \{a + u : a \in A\}$.

[A.2.3] Lebesgue Measurable Sets and Lebesgue Measure

Consider the class of good sets for which outer measure is countably additive:

$$m^*\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} m^*(B_n)$$

for pairwise disjoint (B_n) .

Definition: Lebesgue Measurability

A set $B \subseteq \mathbb{R}$ is (Lebesgue) measurable if for every set $A \subseteq \mathbb{R}$ we have

$$m^*(A) = m^*(A \cap B) + m^*(A \cap B^c)$$

and we write $B \in \mathcal{M}$.

Theorem

Any null set is measurable. Any interval is measurable.

Theorem: Fundamental Properties of the Class $\mathcal{M}$ of All Lebesgue-Measurable Subsets

- (i) $\mathbb{R} \in \mathcal{M}$.
- (ii) If $B \in \mathcal{M}$, then $B^c \in \mathcal{M}$.
- (iii) If $B_n \in \mathcal{M}$ for all $n = 1, 2, \dots$, then $\bigcup_{n=1}^{\infty} B_n \in \mathcal{M}$.

If $B_n \in \mathcal{M}$ for all $n = 1, 2, \dots$ and $B_j \cap B_k = \emptyset$ for $j \neq k$, then

$$m^*\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} m^*(B_n).$$

A family of sets is a σ -field if it contains the base set and is closed under complements and countable unions. Thus, theorem above means that $\mathcal{M}$ is a σ -field. A $[0, \infty]$ -valued

function defined on a σ -field is called a measure if it satisfies the theorem above for pairwise disjoint sets.

More general approach to measure theory is to begin with the above theorem as axioms. Start with a measure space $(\Omega, \mathcal{F}, \mu)$ where Ω is an abstractly given set (a set of given scenarios), $\mathcal{F}$ is a σ -field of subsets of Ω , and $\mu: \mathcal{F} \mapsto [0, \infty]$ is a function satisfying the above theorem.

Proposition

If $B_k \in \mathcal{M}$, $k = 1, 2, \dots$, then

$$B = \bigcap_{k=1}^{\infty} B_k \in \mathcal{M}.$$

$\mathcal{M}$ is closed under countable unions, countable intersections, and complements. $\mathcal{M}$ contains intervals and all null sets.

Summary: Lebesgue Measure

Lebesgue measure $m: \mathcal{M} \mapsto [0, \infty]$ is a countably additive set function defined on the σ -field $\mathcal{M}$ of measurable sets. Lebesgue measure of an interval equals its length. Lebesgue measure of a null set is zero.

[A.2.4] σ -field³

Let Ω be an arbitrary set and $\mathcal{F}$ be a family of subsets of Ω , and A be a set of $\mathbb{R}^d$. $\mathcal{F}$ is said to be a σ -field on Ω , if it satisfies:

- (1) $\Omega \in \mathcal{F}$, $\emptyset \in \mathcal{F}$ (It contains empty set.)
- (2) If $A_n \in \mathcal{F}$ for $n = 1, 2, \dots$ (disjoint A_n), then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ and $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$.
Stability under unions and intersections.
- (3) If $A \in \mathcal{F}$, then $A^C \in \mathcal{F}$ (It contains the complements of every element.).

$(\Omega, \mathcal{F})$ is called a measurable space.

Let B be a subset of A , i.e. $B \subset A$, and the measure of a subset B be a positive finite (possibly infinite) number, $\mu(B) \in [0, \infty]$. B is said to be a measurable set. Properties of measurable sets include:

1. An empty set $\emptyset$ has measure 0, i.e. $\mu(\emptyset) = 0$.
2. Additive property: If B and C are disjoint measurable sets, the measure of the union $B \cup C$ is $\mu(B \cup C) = \mu(B) + \mu(C)$.

³ Based on Cont and Tankov (2004).

3. σ -additivity: Let $(B_n)_{n \in \mathbb{N}}$ be a infinite sequence of disjoint measurable subsets. Then:

$$\mu\left(\bigcup_{n \geq 1} B_n\right) = \sum_{n \geq 1} \mu(B_n).$$

4. It is possible that the measure of a set or subset is infinite, i.e. $\mu(A) = \infty$ and $\mu(B) = \infty$.
5. If a set A is finite (i.e. $\mu(A) < \infty$) for any measurable set B , the measure of the complement B^c ($B \cup B^c = A$) is $\mu(B^c) = \mu(A) - \mu(B)$.

[A.2.5] Borel σ -Field

Definition: Borel σ -field

$\mathcal{B} = \bigcap \{ \mathcal{F} : \mathcal{F} \text{ is a } \sigma\text{-field containing all intervals} \}.$

Borel σ -field is the collection of all Borel sets on $\mathbb{R}^d$ which is denoted by $\mathcal{B}(\mathbb{R}^d)$. Borel σ -field is the σ -field generated by the open sets in $\mathbb{R}^d$. In other words, it is the smallest σ -field containing all open sets in $\mathbb{R}^d$. A real-valued function $f(x)$ on $\mathbb{R}^d$ is said to be measurable if it is $\mathcal{B}(\mathbb{R}^d)$ -measurable.

[A.2.6] Probability

Lebesgue measure restricted to $[0,1]$ is a probability measure. We select a number from $[0,1]$ at random, restrict Lebesgue measure m to the interval $B = [0,1]$, and consider the σ -field of $\mathcal{M}_{[0,1]}$ of measurable subsets of $[0,1]$. $m_{[0,1]}$ is a probability measure because it is a measure on $\mathcal{M}_{[0,1]}$ with total mass 1.

Definition: Probability Space and Probability Measure $(\Omega, \mathcal{F}, \mathbb{P})$

Let Ω be an arbitrary set (a set of scenarios), $\mathcal{F}$ be a σ -field of subsets of Ω , and $\mathbb{P}$ be a mapping from $\mathcal{F}$ into $\mathbb{R}$. Let A be a set of $\mathbb{R}^d$. A triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is said to be a probability space and $\mathbb{P}$ is said to be a probability measure if it satisfies the conditions:

- (1) $\mathbb{P}(\Omega) = 1$. (A probability space is a measure space with total mass 1.)
- (2) $0 \leq \mathbb{P}(A) \leq 1$ and $\mathbb{P}(\emptyset) = 0$.
- (3) If $A_n \in \mathcal{F}$ for $n = 1, 2, \dots$ and they are disjoint (i.e. $A_n \cap A_m = \emptyset$ for $n \neq m$), then $\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

Let B be an event which is a measurable set $B \in \mathcal{F}$. A probability measure $\mathbb{P}$ assigns a probability between 0 and 1 to each event:

$$\begin{aligned}\mathbb{P}: \mathcal{F} &\rightarrow [0,1] \\ B &\mapsto \mathbb{P}(B).\end{aligned}$$

If two probability measures $\mathbb{P}$ and $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ define the same impossible events, they are said to be equivalent:

$$\mathbb{P} \sim \mathbb{Q} \Leftrightarrow [\forall B \in \mathcal{F}, \mathbb{P}(B) = 0 \Leftrightarrow \mathbb{Q}(B) = 0].$$

Definition: Random Variable

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. A mapping X from Ω into $\mathbb{R}^d$, i.e.

$X: \Omega \rightarrow \mathbb{R}^d$, is said to be an $\mathbb{R}^d$ -valued random variable if it is $\mathcal{F}$ -measurable:

$$\{\omega: X(\omega) \in B\} \in \mathcal{F} \text{ for each } B \in \mathcal{B}(\mathbb{R}^d).$$

We can interpret that $X(\omega)$ is the outcome of the random variable when the scenario ω happens. Probability measures on $\mathcal{B}(\mathbb{R}^d)$ are said to be distributions (laws) on $\mathbb{R}^d$.

Definition: Property A Almost Surely (a.s.)

A random variable X is said to have a property A almost surely (or with probability 1), if there is $\Omega_0 \in \mathcal{F}$ with $\mathbb{P}(\Omega_0) = 1$ such that $X(\omega)$ has the property A for every $\omega \in \Omega_0$.

Definition: Conditional Probability

Let $\mathbb{P}(B) > 0$. The conditional probability of the event A given B is

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Given some disjoint hypotheses, the probability of an event can be calculated by means of conditional probabilities:

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A|O_i) \mathbb{P}(O_i)$$

where O_i are pairwise disjoint events such that $\bigcup_{i=1}^{\infty} O_i \in \Omega$ and $\mathbb{P}(O_i) \neq 0$.

Definition: Independence

Let X_j be an $\mathbb{R}^{d_j}$ -valued random variable for $j = 1, \dots, n$. The family $\{X_1, \dots, X_n\}$ is said to be independent if, for every $B_j \in \mathcal{B}(\mathbb{R}^{d_j})$:

$$\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) = \mathbb{P}(X_1 \in B_1) \dots \mathbb{P}(X_n \in B_n).$$

We often say that $X_1, \dots, X_n$ are independent rather than saying that the family $\{X_1, \dots, X_n\}$ is independent.

[A.3] Stochastic Process

A stochastic process $\{X_t\}$ is a family $\{X_t\}_{t \in [0, \infty)}$ of random variables on $\mathbb{R}^d$ defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For any increasing sequence of time $0 \leq t_1 < t_2 < \dots < t_n$, $\mathbb{P}(X(t_1) \in B_1, \dots, X(t_n) \in B_n)$ determines a probability measure on $\mathcal{B}((\mathbb{R}^d)^n)$. Sample function (path) of $\{X_t\}$ is $X(t, \omega)$.

Let $\{X_t\}$ and $\{Y_t\}$ be two stochastic processes. $\{X_t\}$ and $\{Y_t\}$ are said to be identical in law:

$$\{X_t\} \stackrel{d}{=} \{Y_t\},$$

if the systems of their finite-dimensional distributions are identical.

A stochastic process $\{X_t\}$ on $\mathbb{R}^d$ is said to be continuous in probability (stochastically continuous) if, for every $t \geq 0$ and $\varepsilon > 0$:

$$\lim_{s \rightarrow t} \mathbb{P}(|X_s - X_t| > \varepsilon) = 0.$$

[A.3.1] Filtration (Information Flow)

An increasing family of σ -fields $(\mathcal{F}_t)_{t \in [0, T]} : \forall t \geq s \geq 0, \mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ is called a filtration or information flow on $(\Omega, \mathcal{F}, \mathbb{P})$. We can interpret $\mathcal{F}_t$ as the information known at time t . $\mathcal{F}_t$ increases as time progresses.

[A.3.2] Non-anticipating (Adapted) Process

A stochastic process $(X_t)_{t \in [0, T]}$ is said to be non-anticipating with respect to the filtration $(\mathcal{F}_t)_{t \in [0, T]}$ or $\mathcal{F}_t$ -adapted if the value of X_t is revealed at time t for each $t \in [0, T]$.

[A.4] Martingales

[A.4.1] General Concept

Consider a trend of a time series of a stochastic process. A stochastic process is said to be a martingale if its time series have no trend. A process with increasing trend is called a submartingale and a process with decreasing trend is called a supermartingale.

Definition: Martingale

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with an filtration (information flow) $\mathcal{F}_t$. A cadlag process $(X_t)_{t \in [0, T]}$ is said to be a martingale with respect to its filtration $\mathcal{F}_t$ and the probability measure $\mathbb{P}$ if X is nonanticipating (adapted to $\mathcal{F}_t$), $E[|X_t|]$ is finite for any $t \in [0, T]$ and:

$$\forall s > t, \quad E[X_s | \mathcal{F}_t] = X_t.$$

The best prediction of a martingale's future value is its present value. The definition of martingale makes sense only when the underlying probability measure P and the information flow $(\mathcal{F}_t)_{t \in [0, T]}$ have been specified.

The fundamental property of a martingale process is that its future variations are completely unpredictable with the information flow $\mathcal{F}_t$:

$$\forall u > 0, \quad E[x_{t+u} - x_t | \mathcal{F}_t] = E[x_{t+u} | \mathcal{F}_t] - E[x_t | \mathcal{F}_t] = x_t - x_t = 0.$$

Example of Continuous-Time Martingale 1)⁴

Let $(X_t)_{t \in [0, T]}$ be a continuous stochastic process whose increments are normally distributed. Let Δ be a small time interval. Then:

$$\begin{aligned} \Delta X_t &\sim N(\mu\Delta, \sigma^2\Delta) \\ \forall s \neq t, \quad E[(\Delta X_s - \mu\Delta)(\Delta X_t - \mu\Delta)] &= 0. \end{aligned}$$

Consider an expectation with the filtration $\mathcal{F}_t$ and with respect to a probability distribution described above:

$$\begin{aligned} \forall u > 0, \quad X_{t+u} &= X_t + \int_t^{t+u} dX_v \\ E[X_{t+u} | \mathcal{F}_t] &= E[X_t + \int_t^{t+u} dX_v | \mathcal{F}_t] = E[X_t | \mathcal{F}_t] + E[\int_t^{t+u} dX_v | \mathcal{F}_t] = X_t + \mu u \neq X_t. \end{aligned}$$

Obviously, $(X_t)_{t \in [0, T]}$ is not a martingale. But the process $(Y_t)_{t \in [0, T]}$:

$$Y_t = X_t - \mu u,$$

is a martingale since:

$$E[Y_{t+u} | \mathcal{F}_t] = E[X_{t+u} - \mu u | \mathcal{F}_t] = E[X_t + (X_{t+u} - X_t) - \mu u | \mathcal{F}_t]$$

⁴ Based on Neftci (2000).

$$= E[X_t | \mathcal{F}_t] + E[X_{t+u} - X_t | \mathcal{F}_t] + E[-\mu u | \mathcal{F}_t] = X_t + \mu u - \mu u = X_t .$$

Example of Continuous-Time Martingale 2) Standard Brownian Motion Process

Consider a standard Brownian motion process $(B_t)_{t \in [0, T]}$ defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Standard Brownian motion is a continuous adapted process with properties that $B_0 = 0$, increments $B_t - B_s = 0$ for $0 \leq s < t$ are independent of $\mathcal{F}_s$ and are normally distributed with mean 0 and variance $t - s$. Obviously, $(B_t)_{t \in [0, T]}$ is a martingale:

$$\begin{aligned} \forall u > 0, \quad B_{t+u} &= B_t + \int_t^{t+u} dB_v \\ E[B_{t+u} | \mathcal{F}_t] &= E[B_t + \int_t^{t+u} dB_v | \mathcal{F}_t] = E[B_t | \mathcal{F}_t] + E[\int_t^{t+u} dB_v | \mathcal{F}_t] = B_t + 0 = B_t . \end{aligned}$$

[A.4.2] Martingale Asset Pricing

Most of financial asset prices are not martingales (i.e. are not completely unpredictable). Consider a risky stock price S_t at time t and let r be the risk-free interest rate. In a small time interval Δ risk-averse investors expect S_t to grow at some positive rate. This can be written as under actual probability measure $\mathbb{P}$:

$$E_t^{\mathbb{P}}[S_{t+\Delta}] > S_t .$$

Obviously, $S_{t+\Delta}$ is not a martingale. To be more precise risk-averse investors expect S_t to grow at a rate greater than r :

$$E_t^{\mathbb{P}}[e^{-r\Delta} S_{t+\Delta}] > S_t .$$

The stock price discounted by the risk-free interest rate $e^{-r\Delta} S_{t+\Delta}$ is not martingale under $\mathbb{P}$.

But interestingly, non-martingales can be converted to martingales by changing the probability measure. We will try to find an equivalent probability measure $\mathbb{Q}$ (called risk-neutral measure) under which the stock price discounted by the risk-free interest rate becomes martingale:

$$E_t^{\mathbb{Q}}[e^{-r\Delta} S_{t+\Delta}] = S_t .$$

[A.4.3] Continuous Martingales, Right Continuous Martingales, Square-Integrable Martingales

Let $\{X_t; 0 \leq t\}$ be a continuous martingale. Continuous martingales have continuous trajectories, when $\Delta \rightarrow 0$:

$$P(\Delta S_t > \varepsilon) \rightarrow 0 \quad \text{for all } \varepsilon > 0.$$

It tells us when the time step becomes extremely small, the probability that the stock price changes by some amount approaches zero. Example is a standard Brownian motion process.

In contrast to continuous martingales, right continuous martingales basically change by jumps. Compensated Poisson process is an example of right continuous martingales.

A continuous martingale $\{X_t; 0 \leq t\}$ is said to be a continuous square integrable martingale if X has a finite second moment:

$$E[X_t^2] < \infty.$$

[A.5] Poisson Process⁵

[A.5.1] Exponential Distribution

Suppose a positive random variable X follows an exponential distribution with parameter $\lambda > 0$. Its probability density function is:

$$f_X(x) = \lambda e^{-\lambda x} \quad \text{for } x > 0.$$

The distribution function is:

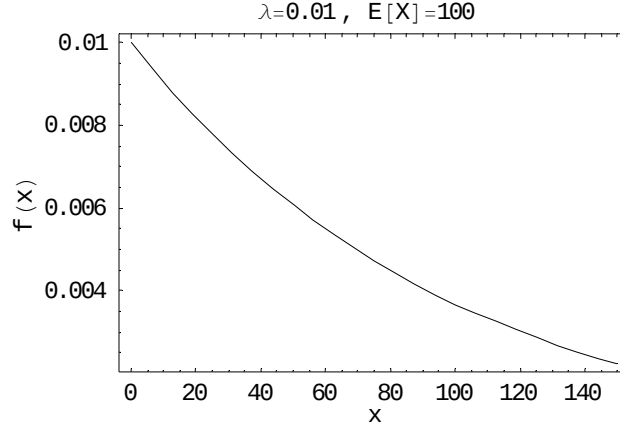
$$F_X(x) = \Pr(X \leq x) = 1 - e^{-\lambda x} \quad \text{for } x > 0.$$

Its mean and variance are:

$$E[X] = \frac{1}{\lambda} \quad \text{and} \quad \text{Var}[X] = \frac{1}{\lambda^2}.$$

For example, the probability density function of an exponential random variable X with $\lambda = 0.01$ is plotted below.

⁵ Based on Cont and Tankov (2004).



The exponential distribution has an important feature called memory-less property. Suppose that the arrival time of a large earthquake, X , follows an exponential distribution with mean 100 years. Consider a situation where $s=10$ years have passed since the last large earthquake and let $\{X-s|X>s\}$ be the remaining arrival time. Since x and s are both positive,

$$\begin{aligned} \Pr\{X-s>x|X>s\} &= \frac{\Pr\{X-s>x \text{ and } X>s\}}{\Pr\{X>s\}} = \frac{\Pr\{X-s>x\}}{\Pr\{X>s\}} = \frac{\Pr\{X>x+s\}}{\Pr\{X>s\}} \\ &= \frac{1-F_X(x+s)}{1-F_X(s)} = \frac{1-(1-e^{-\lambda(x+s)})}{1-F_X(s)} = \frac{e^{-\lambda x}e^{-\lambda s}}{1-F_X(s)} = \frac{(1-F_X(x))(1-F_X(s))}{1-F_X(s)} = 1-F_X(x) \\ &= \Pr\{X>x\}. \end{aligned}$$

So the expected remaining arrival time of a next large earthquake is 100 years, i.e. the exponential random variable X does not remember that 10 years have passed since the last large earthquake.

Memory-less property of exponential distribution

If a random variable X follows an exponential distribution,

$$\forall x, s > 0, \quad \Pr\{X > x+s|X > s\} = \Pr\{X > x\}.$$

If X is a random time, the distribution of $X-s$ given $X>s$ is the same as the distribution of X itself.

[A.5.2] Poisson Distribution

If $(\tau_i)_{i \geq 1}$ are independent exponential random variables with parameter λ , the random variable for any $t > 0$:

$$N_t = \inf\{n \geq 1, \sum_{i=1}^n \tau_i > t\},$$

follows a Poisson distribution with parameter λt :

$$\forall n \in \mathbb{N}, \mathbb{P}(N_t = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}.$$

If X_1 and X_2 are independent Poisson variables with parameters λ_1 and λ_2 , then $X_1 + X_2$ follows a Poisson distribution with parameter $\lambda_1 + \lambda_2$. This indicates that a Poisson random variable X is infinitely divisible, i.e. X can be divided into an arbitrary number of *i.i.d.* random variables:

$$\begin{aligned} &\text{if } X \sim \text{Poisson}(\lambda), \\ &\forall n \in \mathbb{N}, \quad X = \sum_{i=1}^n \frac{X}{n}, \quad \frac{X}{n} \sim \text{i.i.d. Poisson}\left(\frac{\lambda}{n}\right). \end{aligned}$$

[A.5.3] Compensated Poisson Process

A compensated Poisson process $(\widetilde{N}_t)_{t \geq 0}$ is a centered version of a Poisson process $(N_t)_{t \geq 0}$:

$$\widetilde{N}_t = N_t - \lambda t.$$

The mean and variance of a compensated Poisson process are:

$$\begin{aligned} E[\widetilde{N}_t] &= E[N_t - \lambda t] = \lambda t - \lambda t = 0, \\ \text{Var}[\widetilde{N}_t] &= E[\widetilde{N}_t^2] = E[(N_t - \lambda t)^2] = \text{Var}[N_t] = \lambda t. \end{aligned}$$

Increments of $\widetilde{N}$ are independent and $\widetilde{N}$ is a martingale:

$$\begin{aligned} E[\widetilde{N}_t | \widetilde{N}_s, s \leq t] &= E[\widetilde{N}_t - \widetilde{N}_s + \widetilde{N}_s | \widetilde{N}_s] = E[\widetilde{N}_t - \widetilde{N}_s] + \widetilde{N}_s \\ &= E[\widetilde{N}_t] - E[\widetilde{N}_s] + \widetilde{N}_s = 0 - 0 + \widetilde{N}_s = \widetilde{N}_s. \end{aligned}$$

$(\lambda t)_{t \geq 0}$ called the compensator of a Poisson process $(N_t)_{t \geq 0}$ is the quantity which needs to be subtracted in order to make the process a martingale.

Unlike a Poisson process, a compensated Poisson process is not integer-valued and not a counting process.

A compensated Poisson process $(\widetilde{N}_t)_{t \geq 0}$ behaves like a Brownian motion after rescaling it by $1/\lambda$ because:

$$E\left[\frac{\widetilde{N}_t}{\lambda}\right] = 0 \quad \text{and} \quad \text{Var}\left[\frac{\widetilde{N}_t}{\lambda}\right] = t.$$

Formally, when the intensity λ of its jumps becomes larger, the compensated Poisson process converges in distribution to a Brownian motion:

$$\left(\frac{\widetilde{N}_t}{\lambda}\right)_{t \in [0, T]} \Rightarrow (B_t)_{t \in [0, T]} \quad \text{when} \quad \lambda \rightarrow \infty.$$

[A.6] Other Distributions Used

[A.6.1] Gamma Function

Let c be a complex number. The integral called gamma function:

$$\Gamma(c) = \int_0^\infty t^{c-1} e^{-t} dt$$

converges absolutely, if the real part of c is positive. By integration by parts:

$$\Gamma(c+1) = c\Gamma(c).$$

Since $\Gamma(1) = 1$, for all natural numbers n :

$$\Gamma(n+1) = n!.$$

Properties of a gamma function which are often used in the process of calculation are:

$$\begin{aligned} \Gamma(1-c) &= -c\Gamma(-c), \\ \frac{\Gamma(-c)}{\Gamma(1-c)} &= -\frac{1}{c}, \\ \frac{\Gamma(1+c)}{\Gamma(c)} &= c, \\ \frac{\Gamma(2+c)}{\Gamma(c)} &= \frac{\Gamma(2+c)}{\Gamma(1+c)} \frac{\Gamma(1+c)}{\Gamma(c)} = (1+c)c. \end{aligned}$$

[A.6.2] Incomplete Gamma Function

The incomplete gamma function is defined by an indefinite integral of the same integrand $t^{c-1}e^{-t}$. For $x \in \mathbb{R}^+$ and complex variable c , if the real part of c is positive:

$$\Gamma(c, x) = \int_x^\infty t^{c-1} e^{-t} dt,$$

$$\gamma(c, x) = \int_0^x t^{c-1} e^{-t} dt,$$

where $\Gamma(c, x)$ is when the lower limit of integration is variable and $\gamma(c, x)$ is when upper limit of the integration is variable. Obviously, we have following relationship:

$$\Gamma(c) = \gamma(c, x) + \Gamma(c, x)$$

$$\Gamma(c) = \Gamma(c, 0)$$

$$\gamma(c, x) \rightarrow \Gamma(c) \quad \text{as } x \rightarrow \infty.$$

[A.6.3] Gamma Distribution

Let X be a gamma distributed random variable with the shape parameter $c > 0$ and the scale parameter (controls the tail behavior) $\lambda > 0$. Its probability density function can be expressed as:

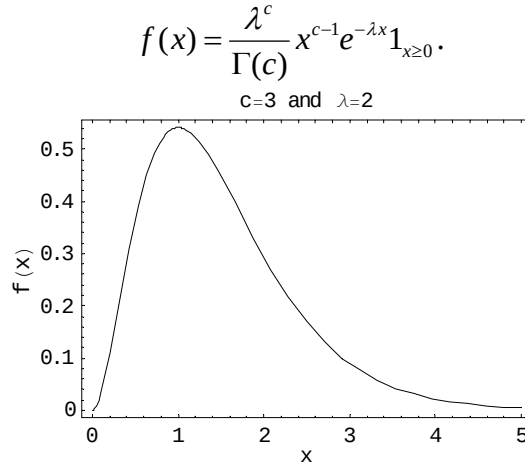


Figure A.6.1: The Gamma Density with $c = 3$ and $\lambda = 2$

Its distribution function in terms of the incomplete gamma function is

$$F(x) = \int_0^x f(z) dz = \frac{\gamma(c, \lambda x)}{\Gamma(c)} 1_{x \geq 0}.$$

Its characteristic function $\phi(\omega)$ for any $\omega \in \mathbb{R}$ is calculated as

$$\phi(\omega) = \mathcal{L}_x[f(x)](\omega) = \int_0^\infty e^{i\omega x} f(x) dx = E[e^{i\omega x}] = \frac{1}{(1 - i\omega\lambda^{-1})^c}.$$

The mean, variance, skewness, and excess kurtosis of a gamma random variable are follows:

$$E[X] = \frac{c}{\lambda},$$

$$\text{Var}[X] = \frac{c}{\lambda^2},$$

$$\text{Skewness}[X] = \frac{2}{\sqrt{c}},$$

$$\text{Excess Kurtosis}[X] = \frac{6}{c}.$$

A gamma distributed random variable X with the shape parameter $c > 0$ and the scale parameter $\lambda > 0$ possesses the following properties:

- o Suppose $X_1, X_2, \dots, X_n$ are independent gamma distributed random variables with parameters $(c_1, \lambda), (c_2, \lambda), \dots, (c_n, \lambda)$, then $\sum_{i=1}^n X_i \sim \text{Gamma}(\sum_{i=1}^n c_i, \lambda)$.
- o If $c = 1$, the gamma distribution reduces to an exponential distribution with parameter λ .

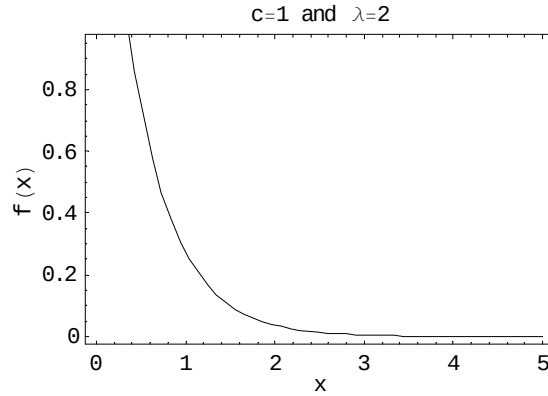


Figure A.6.2: The Gamma Density with $c = 1$ and $\lambda = 2$

[A.7] Modified Bessel Functions: Modified Bessel Function of the First Kind $I_\nu(z)$ and Second Kind $K_\nu(z)$

Modified Bessel differential equation is the second-order ordinary differential equation of the form

$$z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} - (z^2 + \nu^2)y = 0.$$

Its solutions can be expressed for $n = 0, 1, 2, \dots$,

$$\begin{aligned} y &= a_1 J_n(-iz) + a_2 Y_n(-iz) \\ &= b_1 I_n(z) + b_2 K_n(z), \end{aligned}$$

where $J_\nu(z)$ is called a Bessel function of the first kind of order ν , $Y_\nu(z)$ a Bessel function of the second kind of order ν , $I_\nu(z)$ a modified Bessel function of the first kind of order ν , $K_\nu(z)$ a modified Bessel function of the second kind of order ν , and $\nu \in \mathbb{R}^+$. The modified Bessel function of the first kind $I_\nu(z)$ admits the series expansion

$$I_\nu(z) = \left(\frac{1}{2}z\right)^\nu \sum_{k=0}^{\infty} \frac{\left(\frac{1}{4}z^2\right)^k}{k!\Gamma(k+\nu+1)} = \sum_{k=0}^{\infty} \frac{(z/2)^{\nu+k}}{k!\Gamma(k+\nu+1)} \quad \text{for } z \geq 0 \text{ and } \nu \geq 0,$$

which is the solution of the modified Bessel differential equation which is bounded when $z \rightarrow 0$. For $n = 0, 1, 2, \dots$

$$I_{-n}(z) = I_n(z).$$

$I_\nu(z)$ has asymptotic behavior

$$I_\nu(z) = \frac{e^z}{\sqrt{2\pi z}} \left[1 + o\left(\frac{1}{z}\right)\right] \quad \text{when } z \rightarrow +\infty,$$

$$I_\nu(z) \sim \frac{(z/2)^\nu}{\Gamma(\nu+1)} \quad \text{when } z \rightarrow 0.$$

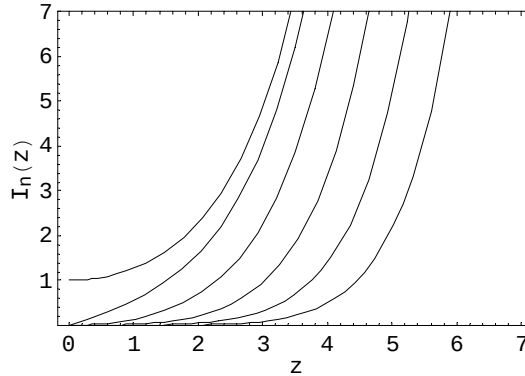


Figure A.7.1: Plot of the modified Bessel functions of the first kind. $I_\nu(z)$ for $\nu = 1, 2, 3, 4, 5$ from left to right.

The modified Bessel function of the second kind $K_\nu(z)$ admits the series expansion

$$K_\nu(z) = \frac{\pi}{2} \frac{I_{-\nu}(z) - I_\nu(z)}{\sin(\pi\nu)} \quad \text{for } z \geq 0 \text{ and } \nu \geq 0,$$

which is the solution of the modified Bessel differential equation which is bounded when $z \rightarrow +\infty$. For all orders ν :

$$K_{-v}(z) = K_v(z).$$

$I_v(z)$ has asymptotic behavior:

$$\begin{aligned} K_v(z) &= e^{-z} \sqrt{\frac{\pi}{2z}} [1 + o(\frac{1}{z})] \quad \text{when } z \rightarrow +\infty, \\ K_0(z) &\sim -\log z \quad \text{for } v=0, \\ K_v(z) &\sim \frac{1}{2} \Gamma(v) (z/2)^{-v} \quad \text{for } v > 0. \end{aligned}$$

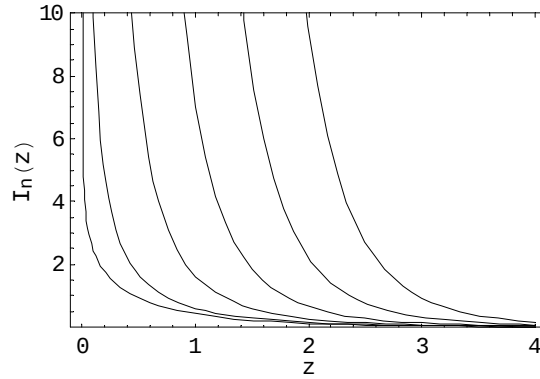


Figure A.7.2: Plot of the modified Bessel functions of the second kind. $K_v(z)$ for $v = 1, 2, 3, 4, 5$ from left to right.

[A.8] Itô Formula

[A.8.1] Itô Formula for Brownian Motion

Let $B_t \sim \text{Normal}(0, t)$ be a standard Brownian motion and $f(B)$ be an arbitrary function of a Brownian motion. Consider a very small time interval h . A Taylor series expansion of a function $f(B_{t+h})$ about a point B_t is given by:

$$\begin{aligned} f(B_{t+h}) - f(B_t) &= \frac{df(B_t)}{dB} (B_{t+h} - B_t) + \frac{d^2 f(B_t)}{dB^2} \frac{1}{2!} (B_{t+h} - B_t)^2 \\ &\quad + \frac{d^3 f(B_t)}{dB^3} \frac{1}{3!} (B_{t+h} - B_t)^3 + \dots \end{aligned}$$

Thus:

$$\begin{aligned} &\sum_{j=1}^n (f(B_{t+jh}) - f(B_{t+(j-1)h})) \\ &= (f(B_{t+h}) - f(B_t)) + (f(B_{t+2h}) - f(B_{t+h})) + \dots + (f(B_{t+nh}) - f(B_{t+(n-1)h})) \end{aligned}$$

$$\begin{aligned}
&= f(B_{t+nh}) - f(B_t) \\
&= \frac{df(B_t)}{dB} (B_{t+h} - B_t) + \frac{d^2 f(B_t)}{dB^2} \frac{1}{2!} (B_{t+h} - B_t)^2 + \frac{d^3 f(B_t)}{dB^3} \frac{1}{3!} (B_{t+h} - B_t)^3 + \dots \\
&+ \frac{df(B_{t+h})}{dB} (B_{t+2h} - B_{t+h}) + \frac{d^2 f(B_{t+h})}{dB^2} \frac{1}{2!} (B_{t+2h} - B_{t+h})^2 + \frac{d^3 f(B_{t+h})}{dB^3} \frac{1}{3!} (B_{t+2h} - B_{t+h})^3 + \dots \\
&+ \frac{df(B_{t+(n-1)h})}{dB} (B_{t+nh} - B_{t+(n-1)h}) + \frac{d^2 f(B_{t+(n-1)h})}{dB^2} \frac{1}{2!} (B_{t+nh} - B_{t+(n-1)h})^2 \\
&\quad + \frac{d^3 f(B_{t+(n-1)h})}{dB^3} \frac{1}{3!} (B_{t+nh} - B_{t+(n-1)h})^3 + \dots \\
&= \sum_{j=1}^n \frac{df(B_{t+(j-1)h})}{dB} (B_{t+jh} - B_{t+(j-1)h}) + \frac{1}{2!} \sum_{j=1}^n \frac{d^2 f(B_{t+(j-1)h})}{dB^2} (B_{t+jh} - B_{t+(j-1)h})^2 \\
&\quad + \frac{1}{3!} \sum_{j=1}^n \frac{d^3 f(B_{t+(j-1)h})}{dB^3} (B_{t+jh} - B_{t+(j-1)h})^3 + \dots
\end{aligned}$$

Use the approximation:

$$\frac{d^2 f(B_{t+(j-1)h})}{dB^2} = \frac{d^2 f(B_t)}{dB^2} \quad \text{and} \quad \frac{d^3 f(B_{t+(j-1)h})}{dB^3} = \frac{d^3 f(B_t)}{dB^3}.$$

Thus,

$$\begin{aligned}
f(B_{t+nh}) - f(B_t) &= \sum_{j=1}^n \frac{df(B_{t+(j-1)h})}{dB} (B_{t+jh} - B_{t+(j-1)h}) \\
&+ \frac{1}{2!} \frac{d^2 f(B_t)}{dB^2} \sum_{j=1}^n (B_{t+jh} - B_{t+(j-1)h})^2 + \frac{1}{3!} \frac{d^3 f(B_t)}{dB^3} \sum_{j=1}^n (B_{t+jh} - B_{t+(j-1)h})^3 + \dots
\end{aligned}$$

Note the following:

$$\begin{aligned}
\sum_{j=1}^n \frac{df(B_{t+(j-1)h})}{dB} (B_{t+jh} - B_{t+(j-1)h}) &\equiv \int_t^{t+\Delta t} \frac{df}{dB} dB, \\
\frac{1}{2!} \frac{d^2 f(B_t)}{dB^2} \sum_{j=1}^n (B_{t+jh} - B_{t+(j-1)h})^2 &= \frac{1}{2} \frac{d^2 f}{dB^2} \int_t^{t+\Delta t} (dB)^2 = \frac{1}{2} \frac{d^2 f}{dB^2} \Delta t,^6 \\
\frac{1}{3!} \frac{d^3 f(B_t)}{dB^3} \sum_{j=1}^n (B_{t+jh} - B_{t+(j-1)h})^3 &+ \dots \approx 0.
\end{aligned}$$

Therefore, we have the integral version of Itô formula for the Brownian motion:

⁶ In the mean square limit $\int_0^t (dX)^2 = t$.

$$f(B_{t+\Delta t}) - f(B_t) = \int_t^{t+\Delta t} \frac{df(B_\tau)}{dB} dB_\tau + \frac{1}{2} \int_t^{t+\Delta t} \frac{df^2(B_\tau)}{dB^2} d\tau,$$

or in terms of time interval between 0 and t :

$$f(B_t) = f(B_0) + \int_0^t \frac{df(B_s)}{dB} dB_s + \frac{1}{2} \int_0^t \frac{df^2(B_s)}{dB^2} ds.$$

Its differential version is:

$$df = \frac{df}{dB} dB + \frac{1}{2} \frac{d^2 f}{dB^2} dt.$$

Consider a simple example of $f(B) = B^2$ where:

$$\frac{df}{dB} = 2B \quad \text{and} \quad \frac{d^2 f}{dB^2} = 2.$$

Thus the function $f(B)$ satisfies the stochastic differential equation (SDE):

$$df = 2BdB + dt.^7$$

[A.8.2] Wimott's (1998) Rule of Thumb of Itô Formula for Brownian Motion

A Taylor series expansion of a function $f(B + dB)$ about a point B is given by⁸:

$$\begin{aligned} f(B + dB) - f(B) &= \frac{df(B)}{dB} dB + \frac{1}{2} \frac{d^2 f(B)}{dB^2} dB^2, \\ df &= \frac{df(B)}{dB} dB + \frac{1}{2} \frac{d^2 f(B)}{dB^2} dB^2. \end{aligned}$$

Since in the mean square limit $\int_0^t (dX)^2 = t$, setting $dB^2 = dt$ yields the Itô formula:

$$df = \frac{df(B)}{dB} dB + \frac{1}{2} \frac{d^2 f(B)}{dB^2} dt.$$

This is of course technically wrong, but it is very useful.

[A.8.3] Itô Formula for Brownian Motion with Drift

⁷ In ordinary calculus (if B were deterministic variable), $df = 2BdB$.

⁸ Ignore higher order terms.

Consider a function of a random variable $V(S)$ and S is a Brownian motion with drift:

$$dS = a(S)dt + b(S)dB,$$

where the drift parameter a and the volatility parameter b depends on S . Using Wilmott's rule of thumb, the function $V(S)$ satisfies the SDE:

$$dV = \frac{dV}{dS} dS + \frac{1}{2} b^2 \frac{d^2V}{dS^2} dt.$$

In terms of a Brownian motion B , the function $V(S)$ satisfies the SDE:

$$\begin{aligned} dV &= \frac{dV}{dS} (a(S)dt + b(S)dB) + \frac{1}{2} b(S)^2 \frac{d^2V}{dS^2} dt \\ dV &= \left(a(S) \frac{dV}{dS} + \frac{1}{2} b(S)^2 \frac{d^2V}{dS^2} \right) dt + b(S) \frac{dV}{dS} dB. \end{aligned}$$

[A.8.4] Itô Formula for Brownian Motion with Drift in Higher Dimensions

Consider a function of a random variable $V(S,t)$ and S is a Brownian motion with drift

$$dS = a(S,t)dt + b(S,t)dB,$$

where the drift parameter a and the volatility parameter b depends on S and time t . The function $V(S,t)$ satisfies the SDE:

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} b^2 \frac{\partial^2 V}{\partial S^2} dt.$$

An example is that if S is a Brownian motion with drift of the form⁹:

$$dS = \mu S dt + \sigma S dB,$$

the function $V(S,t)$ satisfies the SDE:

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} dt.$$

[A.8.5] Itô Formula for Jump-Diffusion (Finite Activity Lévy) Processes

⁹ Geometric Brownian motion.

Let X be a jump-diffusion process of the form:

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dB_s + \sum_{i=1}^{N_t} \Delta X_i,$$

where $\int_0^t b_s ds$ is a sum of a drift term, $\int_0^t \sigma_s dB_s$ is a stochastic integral of a

(multiplicative) Brownian motion process with $E[\int_0^T \sigma_t^2 dt] < \infty$, and $\sum_{i=1}^{N_t} \Delta X_i$ is a

compound Poisson Process. Drift and volatility processes b and σ are continuous and nonanticipating.

Let $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ be any $C^{1,2}$ function. Then, the integral version of Itô formula for jump-diffusion processes is:

$$\begin{aligned} f(X_t, t) - f(X_0, 0) = & \int_0^t \left[\frac{\partial f(X_s, s)}{\partial s} + b_s \frac{\partial f(X_s, s)}{\partial x} \right] ds \\ & + \frac{1}{2} \int_0^t \sigma_s^2 \frac{\partial^2 f(X_s, s)}{\partial x^2} ds + \int_0^t \frac{\partial f(X_s, s)}{\partial x} \sigma_s dB_s \\ & + \sum_{\{i \geq 1, T_i \leq t\}} [f(X_{T_i-} + \Delta X_i) - f(X_{T_i-})]. \end{aligned}$$

Its differential version becomes:

$$\begin{aligned} df(X_t, t) = & \frac{\partial f(X_t, t)}{\partial t} dt + b_t \frac{\partial f(X_t, t)}{\partial x} dt + \frac{\sigma_t^2}{2} \frac{\partial^2 f(X_t, t)}{\partial x^2} dt \\ & + \sigma_t \frac{\partial f(X_t, t)}{\partial x} dB_t + [f(X_{t-} + \Delta X_t) - f(X_{t-})]. \end{aligned}$$

For more details, see Cont and Tankov (2004).

[A.8.6] Itô Formula for General (Finite and Infinite Activity) Scalar Lévy Processes

Let $\{X_t; t \geq 0\}$ be a general scalar Lévy process with its characteristic triplet (σ^2, ν, γ) and $f : \mathbb{R} \rightarrow \mathbb{R}$ be any C^2 function. Then, the integral version of Itô formula is:

$$\begin{aligned} f(X_t) = & f(0) + \int_0^t \frac{\sigma^2}{2} \frac{\partial^2 f(X_s)}{\partial x^2} ds + \int_0^t \frac{\partial f(X_{s-})}{\partial x} dX_s \\ & + \sum_{\substack{0 \leq s \leq t \\ \Delta X_s \neq 0}} [f(X_{s-} + \Delta X_s) - f(X_{s-}) - \Delta X_s \frac{\partial f(X_{s-})}{\partial x}]. \end{aligned}$$

Its differential version is:

$$df(X_t) = \frac{\sigma^2}{2} \frac{\partial^2 f(X_t)}{\partial x^2} dt + \frac{\partial f(X_{t-})}{\partial x} dX_t + f(X_t) - f(X_{t-}) - \Delta X_t \frac{\partial f(X_{t-})}{\partial x}.$$

The integral version of Itô formula in higher dimension is:

$$\begin{aligned} f(X_t, t) - f(X_0, 0) = & \int_0^t \frac{\partial f(X_{s-}, s)}{\partial x} dX_s + \int_0^t \left[\frac{\partial f(X_s, s)}{\partial s} + \frac{\sigma^2}{2} \frac{\partial^2 f(X_s, s)}{\partial x^2} \right] ds \\ & + \sum_{\substack{0 \leq s \leq t \\ \Delta X_s \neq 0}} [f(X_{s-} + \Delta X_s, s) - f(X_{s-}, s) - \Delta X_s \frac{\partial f(X_{s-}, s)}{\partial x}]. \end{aligned}$$

For more details, see Cont and Tankov (2004).

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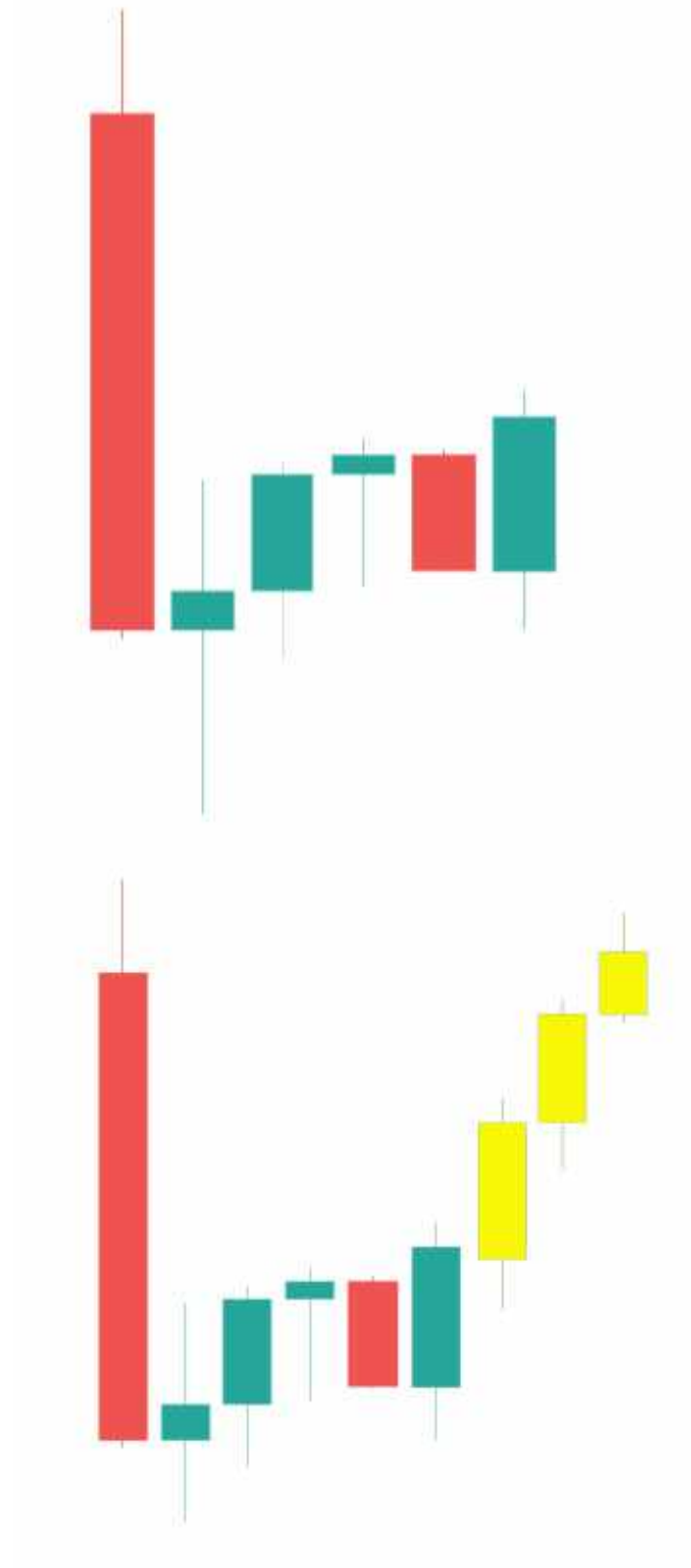
Libro dei Frattali



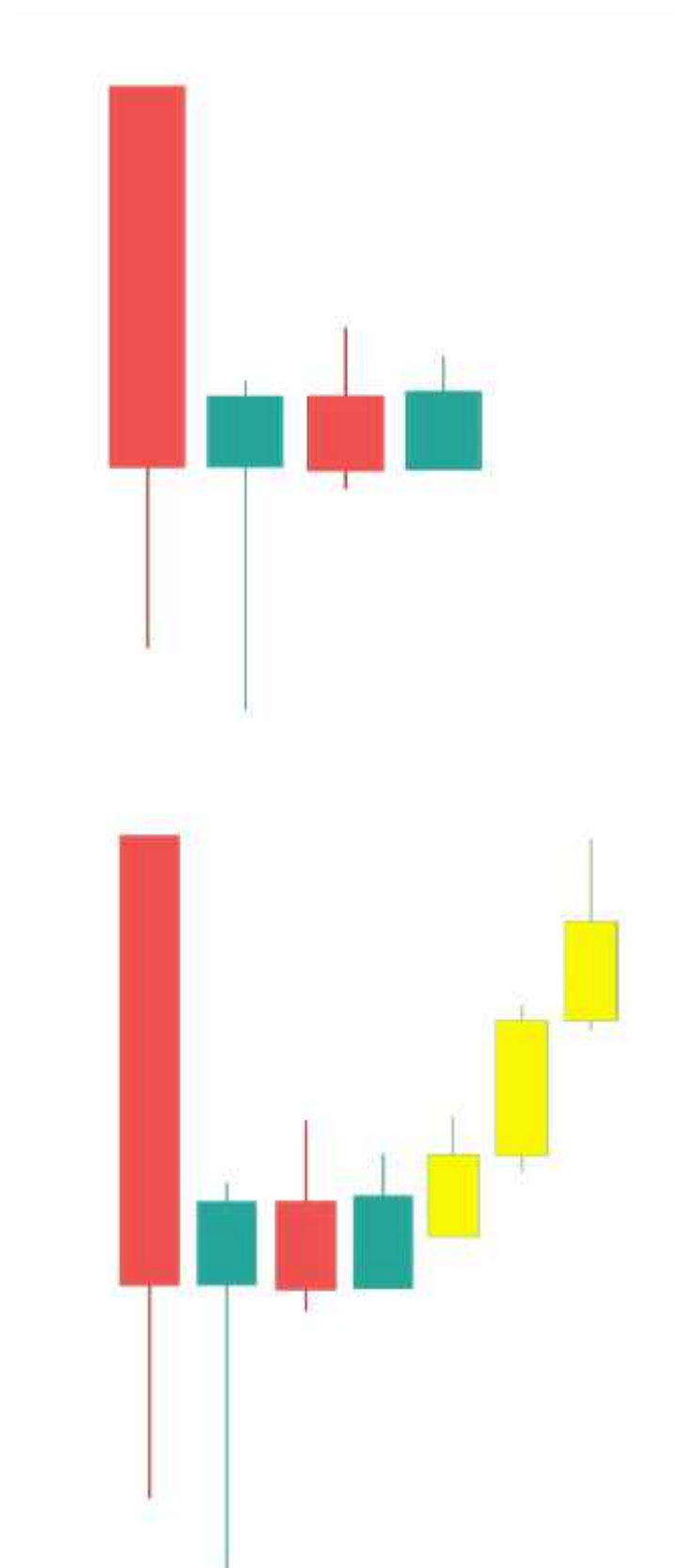
Luca Serleto
Corrado Malanga

Pattern da 3 candele a massimo 6 candele (LONG)

Pattern 1



Pattern 2



Pattern 3



Pattern 4



Pattern 5



Pattern 6



Pattern 7



Pattern 8



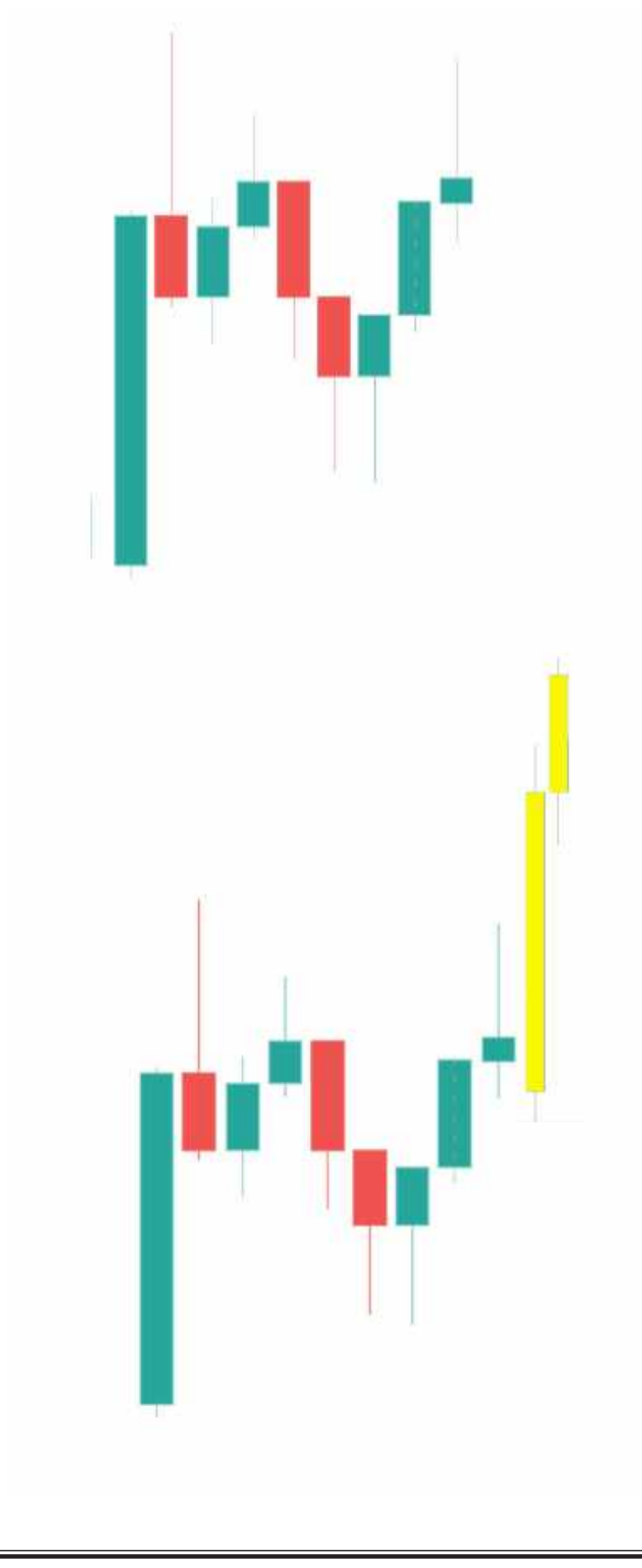
Pattern 9



Pattern 10



Pattern 11



Pattern 12



Pattern 13



Pattern 14



Pattern 15



Pattern 16



Pattern 17



Pattern 18



Pattern 19



Pattern 20



Pattern 21



Pattern 22



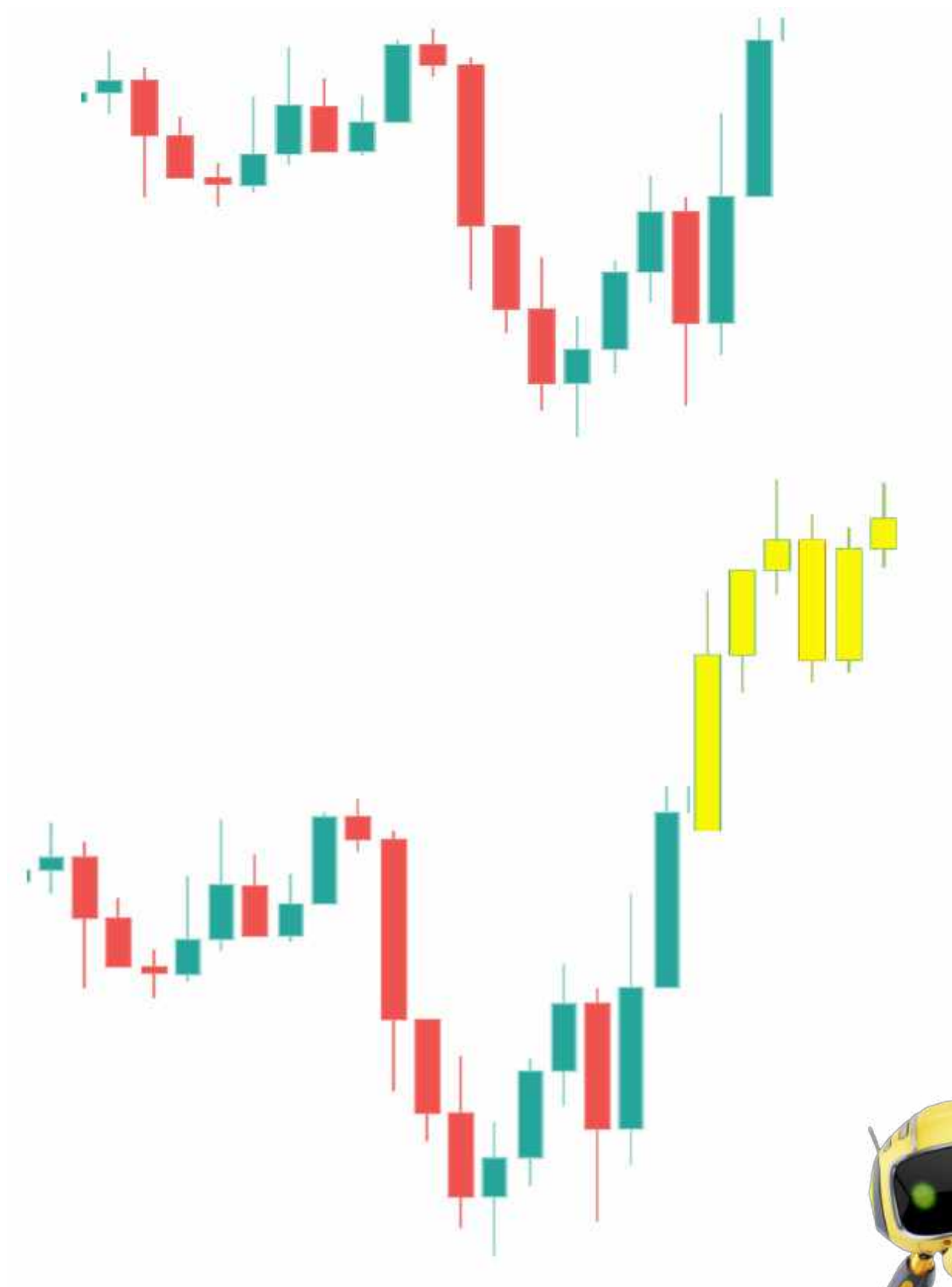
Pattern 23



Pattern 24



Pattern 25



Pattern 26



Pattern 27



Pattern 28



Pattern 29



Pattern 30



Pattern 31



Pattern 32



Pattern 33



Pattern 34



Pattern 35



Pattern 36



Pattern 37



Pattern 38



Pattern 39



Pattern 40



Pattern 41



Pattern 42



Pattern 43



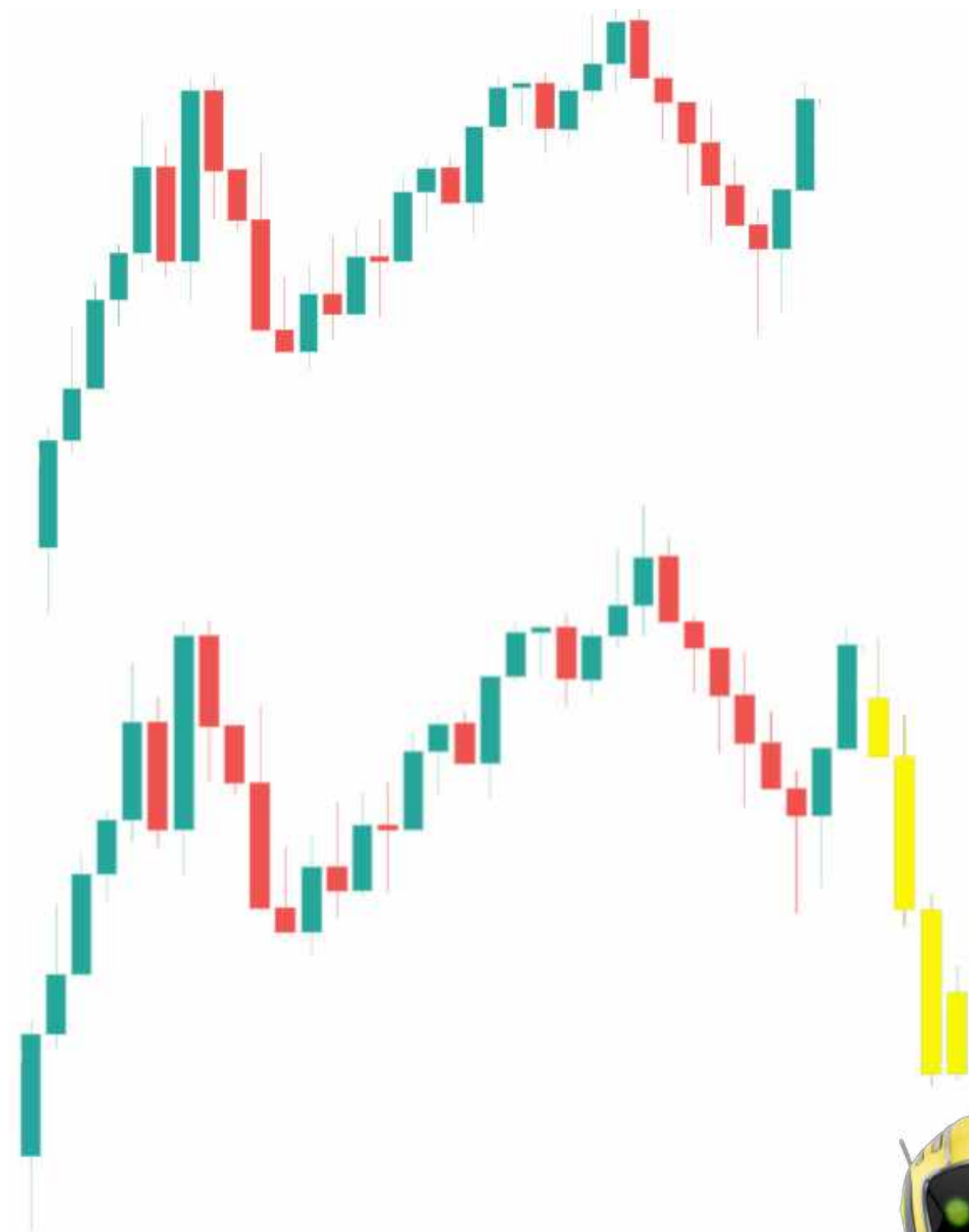
Pattern 44



Pattern 45



Pattern 46



Pattern 47



Pattern 48



Pattern 49



Pattern 50



Pattern 51



Pattern 52



Pattern 53



Pattern 54



Pattern 55



Pattern 56



Pattern 57





Luca Serleto
Corrado Malanga

Introduzione :

Questa ricerca nasce dall'unione dello studio sui frattali del Professor Corrado Malanga, traslando la parte teorica in un sistema applicativo dei frattali nei mercati finanziari. Unendo l'esperienza del trader Luca Serleto, è stato creato uno strumento che sfrutta i frattali come base della creazione, con l'obiettivo di portare una nuova consapevolezza individuale attraverso l'esperienza di non lasciarsi dominare dalla ricerca del guadagno a tutti i costi o dalla paura della sua mancanza.

Il denaro dovrebbe essere uno strumento al servizio delle persone, non il contrario. Senza ipocrisie, oggi il denaro rappresenta il condizionamento del 90% delle nostre scelte; per questo, sviluppare un rapporto sano con il denaro permette di evitare di cadere nell'ansia dell'accumulo o nella frenesia del consumo.

Essere liberi dal dominio del denaro consente di prendere decisioni più consapevoli e in linea con i propri valori, mantenendo sempre il controllo senza oscillare nella dualità e comprendendo infine la vera natura del denaro, ovvero essere un'estensione di noi stessi, non uno strumento di divisione.



Il Sottile Confine tra Magia e Scienza:

La Predizione del Futuro tra Arti Magiche e Probabilità Quantistica

Predire il futuro è stato da sempre un obiettivo di molte persone che, per vari scopi, hanno basato la loro esistenza su maldestri tentativi di sopravvivenza, affidando ad ipotetiche facoltà paranormali ad altrettante improbabili leggi della fisica, la possibilità di prevedere eventi futuri. La ragione che spingeva l'uomo a cercare di indovinare il futuro era legata all'incapacità di prendere su di sé le responsabilità degli accadimenti del presente che, se malamente gestiti, avrebbero condotto l'uomo ad un invivibile futuro. Conoscere ciò che accadrà sarebbe stato fondamentale per evitare di commettere errori decisionali nel presente. Se da un lato i sostenitori delle arti magiche si affidavano alle divinazioni di qualche mago, la scienza non faceva di meglio o affidandosi alla possibilità che la statistica metteva a disposizione. Da una parte avevamo un approccio totalmente animico attraverso il quale, alcuni soggetti producevano visioni del futuro collegandosi a loro dire con le forze del cosmo che essi stessi sarebbero stati in grado di leggere. Tal e approccio, da un punto di vista totalmente femminile per la psicoanalisi, metteva in atto un processo di sensibilizzazione della propria parte animica che "leggeva" gli eventi del futuro, non avendo essa stessa nozione del tempo. Infatti se la parte animica rappresentava l'archetipo del femminile e se essa non aveva consapevolezza del tempo che leggeva in modo o "non locale", essa sarebbe stata in grado di vedere l'asse del tempo come un solo evento costruito da tanti eventi, tutti copresenti nel presente. Facendo focalizzare l'attenzione del soggetto solo su uno di quegli eventi, sia che esso appartenesse al passato od al presente, esso si sarebbe materializzato nella zona del cervello che produce le immagini (Kosslin) e si sarebbe reso manifesto cioè leggibile ed interpretabile dal cervello umano. Tale tipo di approccio sarebbe stato possibile perché il tempo, per la fisica quantistica moderna, non esisterebbe e si manifesterebbe come una proprietà non locale dell'Universo Virtuale (Alan Aspect).

Sostenere a priori che questo tipo di approccio fosse impraticabile si è rivelato profondamente errato ed, a dimostrazione di ciò, basta analizzare alcuni rapporti scientifici segreti oggi declassificati con il FOIA della CIA che portano alla luce studi sulle ricerche segrete delle attività cerebrali effettuati dall'Istituto Monroe. In tali studi si evidenziava la capacità di alcune zone del cervello di leggere il tempo in modo olografico (Karl Pribram) in accordo con la visione olografica della fisica moderna (Bohm) che sosteneva che il cervello stesso altro non fosse che un lettore di ologrammi.



Del resto la storia dell'umanità è piena di racconti nei quali si narra di previsioni di eventi poi realmente verificatisi, prodotte da stati di consapevolezza "allargati" sovente prodotti da pratiche meditative, da stati di ipnosi, da sostanze a llucinogene eccetera . Andava però considerato il fatto che in molte altre situazioni, le previsioni fatte non si fossero verificate o si fossero "quasi verificate", mettendo in luce l'ipotesi che in realtà, quando esse si verificavano si trattasse di un caso fortunato. Un errore di un improbabile metodologia scientifica è quello di pensare che se su dieci eventi predetti, nove non si verificano ed il decimo accade, allora il sistema predittivo non funziona potrebbe non essere totalmente corretto. Infatti la fisica quantistica ha portato ad una rivoluzione culturale importante che comporta l'idea che non esista la certezza che un evento possa accadere ma che esista la probabilità che un evento possa accadere.

Dalla Probabilità alla Certezza:

Un Nuovo Approccio per la Predizione del Futuro

Se applichiamo questo paradigma ai risultati del nostro esperimento dovremmo dedurre non che il sistema per predire il futuro sia errato ma che esso ha la probabilità del 10% di accadere.

Ciò cambia tutto perché se il sistema predittivo è errato esso non funzionerà mai ma se il sistema predittivo è corretto al 10% possiamo trovare il parametro che trasformi tale probabilità nella certezza che l'evento sia correttamente sempre previsto con una prevedibilità del 100%.

Ciò significa che ci possiamo mettere a cercare quelle condizioni per cui il risultato finale prima o poi verrà raggiunto.

Come si può capire da tutto ciò, in questo approccio al problema, non sono importanti le formule matematiche ma il modo di porsi di fronte al problema da risolvere, il che sarebbe come dire: Se si può fare, in qualche modo lo faremo!

Se in questa ipotesi iniziale, il nostro approccio fosse corretto, ciò porterebbe alla conclusione che cercare le condizioni perché la prevedibilità di un evento diventi certezza, non sarebbe tempo perso. Questo approccio sarebbe identificato da un punto di vista psicoanalitico, un approccio femminile al problema cioè un approccio che lasciamo sviluppare dall'emisfero destro del nostro cervello, dedicato alla creatività, alla fantasia, alla mancanza di comportamenti convenzionali, all'inutilità della esistenza di regole, all'imprevedibilità delle azioni, ad una visione atemporale della Realtà Virtuale.



La seconda tipologia di approccio è una visione psicoanalitica maschile che si sviluppa nel cervello sinistro e che prevede la necessità che esista una certezza di descrivere la Virtualità attraverso osservazione continua di essa e la susseguente descrizione, attraverso un opportuno algoritmo, della stessa. La scienza chiama impropriamente questo modello di approccio, modello galileiano.

In questo contesto ciò che appare è misurabile e se misurabile, può essere trattato come un algoritmo. In realtà tale approccio scientifico risulta anch'esso imperfetto perchè per tale approccio si potrebbe dimostrare che l'Universo non esiste in quanto non ripetibile come evento quindi non studiabile poichè non descrivibile attraverso algoritmi.

La Scienza come Religione: Limiti della Matematica e la Deresponsabilizzazione dell'Uomo

Questo approccio ha dunque la necessità di essere assistito da formule ed il suo punto debole è che le formule sono matematiche e la matematica è un linguaggio; come tutti i linguaggi anche esso è imperfetto.

Inoltre da un punto di vista psicologico, l'affidarsi a regole di matematica cioè del linguaggio rappresenta l'incapacità di assumersi la responsabilità di poter osservare l'Universo in modo personale, al di là di quello che dice la scienza ma percependo la falsità di certe formule che agli occhi del moderno sperimentatore diventano solo esercizi ideologici senza capo né coda. Tale aspetto dell'approccio scientifico mette in atto un processo di deresponsabilizzazione della parte maschile dell'essere umano che demanda tutta la responsabilità di ciò che nel mondo accade all'esistenza o meno di un algoritmo matematico. Sarebbe come dire che: Se le mele cascano per terra è perché esiste la formula di Newton che lo dice e se essa no esistesse, le mele cascherebbero dove gli pare.

In realtà le mele se ne fregano della esistenza della formula di Newton.

Esse fanno ciò che la Realtà Virtuale gli impone di fare non ciò che gli dicono le formule che invece vengono dopo la Realtà Virtuale e non la comandano ma tentano di descriverla. Se uno scienziato perde di vista questo paradigma ecco che trasforma la scienza in religione dove prima ci sono le regole e poi il comportamento del Cosmo. Dio ha fatto le regole ed il cosmo le segue. Per la scienza prima nasce il cosmo e poi vengono le regole che lo descrivono.

Dunque quale può essere l'approccio scientifico messo in atto dal cervello maschile per prevedere il futuro se non l'uso della statistica?



Oltre l'Algoritmo:

L'Errore del Buon Senso e il Ruolo della Coscienza

La statistica è una scienza esatta, si osa dire, quando tratta dei grandi numeri, cioè quando ci sono tanti dati sperimentali che descrivano un fenomeno, si dice che esiste un algoritmo che può descrivere il fenomeno e questo algoritmo, pur essendo una approssimazione della Virtualità è abbastanza preciso da poterla descrivere.

Tale approccio prevede di osservare un fenomeno per molto tempo e cercare di capire se parti di questo fenomeno tendano a riprodursi nel tempo e se sì, con quale algoritmo lo fanno. Ma quali sono i presupposti che mi diranno che le mie osservazioni sono state condotte in modo corretto? Ammettiamo che esista un gruppo di scienziati che osserva e studia la vita delle pulci ed abbia notato che esse hanno la tendenza a saltare sulle loro zampe e credendo che ci sia una relazione tra l'azione del saltare e la morfologia delle zampette delle pulci, cominciano a studiarle.

Gli scienziati scoprono che togliendo una zampetta alle pulci esse saltano su tre zampette ma mostrano qualche difficoltà. Tolgono un'altra zampetta alle pulci ed esse pur saltando ancora, saltano con maggiore difficoltà e di meno. Tolgono la terza zampetta e le pulci pur con una sola zampa continuano a volte a saltare. Togliendo l'ultima zampa le pulci non saltano più.

Il risultato di tale studio scientifico ha portato gli scienziati a scoprire che se toglie tutte le zampe alle pulci esse non hanno più voglia di saltare. Questo esempio mostra né più né meno quello che è accaduto con il Covid-19 durante gli ultimi tre anni e sta tutt'ora accadendo.

A questo punto possiamo introdurre quel parametro che fa la differenza tra un osservatore e l'altro osservatore. A tale parametro alcuni scienziati hanno cercato di dare un nome per definirlo ma la cosa appare decisamente sfuggitiva.

I fisici moderni hanno chiamato questo parametro coll'appellativo di "nascosto", intendendo con questo termine un parametro sicuramente tangibile in futuro ma per ora non identificato nel mondo della Virtualità, che sicuramente in futuro verrà compreso.

Per ora tale parametro non appare e non può apparire mai, secondo noi, perché il fisico commette un errore ontologico di partenza sostenendo che questo sia un parametro fisico, quando non ne ha alcuna certezza. In parole povere la scienza vende la pelle dell'orso prima di averlo catturato.



Alcuni identificano questo parametro misterioso con il “ buon senso”, intendendo con tale appellativo che, se non si ha buon senso nell’esaminare la realtà della Virtualità, ecco che essa ci può apparire completamente falsata. Se un problema matematicamente mi dà due soluzioni ma una delle due è improponibile ecco che il buon senso mi fa scegliere solo quella attuabile e mi fa scartare l’altra. Ma ci dobbiamo chiedere perché la matematica mi avrebbe fornito due soluzioni mentre la realtà ne mostra solo una? I casi sono due: o l’algoritmo matematico è imperfetto, cioè descrive in modo errato la Realtà Virtuale, o io faccio un errore nel non considerare una seconda soluzione anch’essa plausibile e del tutto esistente. Così è successo ad Einstein che nella sua celebre formula $E=mc^2$ ci si dimentica di considerare i valori di energia negativi, dicendo a quei tempi, che non aveva senso e prendendo una grossa cantonata. Anche il buon senso a volte fa acqua da tutte le parti. In questo lavoro proponiamo che il parametro nascosto sia la Coscienza Umana.

Il Ritorno degli Eventi:

Circolarità del Tempo e Consapevolezza della Coscienza

La nostra analisi dunque si differenzierà da tutte le altre analisi per l’introduzione di questo parametro, che non è descrivibile matematicamente se non con qualche trucco, ma di questo parleremo in seguito. Riprendiamo ora il percorso dell’emisfero sinistro del nostro cervello.

L’analisi statistica di un fenomeno segue un comportamento incredibile. Essa mostra come tutti gli eventi dell’universo abbiano un modo di comportarsi che alla lunga diviene ripetibile. In meccanica hamiltoniana il teorema di ricorrenza di Henri Poincaré stabilisce che, nell’evoluzione di un sistema dinamico che ha uno spazio delle fasi limitato, il sistema può trovarsi in uno stato arbitrariamente vicino a quello di partenza dopo un tempo sufficientemente lungo (<https://youtube.com/shorts/OnM09uMCS50?si=wkwnlOAWpBWB0e1>).

Questa osservazione ci permette di capire che gli eventi della fisica nel tempo, stabiliscono che il tempo, se esso è una misura degli spazi percorsi appare circolare e non lineare. Per il momento dunque sappiamo che gli eventi sono ricorrenti anche se per ora non sappiamo come i diversi percorsi con cui l’evento ricorre, siano relazionati fra loro. Una cosa accadrà qualche tempo fa tornerà ad accadere e questo ci spinge a determinare le modalità con cui ciò accadrà, visto che sicuramente ciò avverrà. Dunque non stiamo sprecando tempo nel cercare di verificare se il futuro possa essere veramente predetto con algoritmi matematici.



L'approccio coscenziale alla Virtualità, prevede di conoscere esattamente come l'Universo sia stato creato. Rimandiamo in questa sede tale descrizione alla consultazione di alcune nostre opere precedentemente pubblicate che tendono a mostrare come l'Uomo è il Creatore di tutto ma, allo stato attuale della sua consapevolezza, egli non se ne può accorgere.

(Corrado Malanga in "io e Dio" e "Trilogia della Coscienza" Ed. Spazio Interiore, Roma).

Sebbene la Coscienza non sia misurabile perchè appartenente al mondo della Realtà Reale, essa proietta nella virtualità la sua "ombra" per così dire, che abbiamo chiamato Consapevolezza di sé. La Coscienza è rappresentabile con l'entropia "S", parametro proprio del secondo principio della termodinamica, che rappresenta l'ordine, la simmetria, l'informazione attraverso il teorema di Shannon. Così il valore totale dell'entropia non può essere calcolato perché all'inizio dei tempi non eravamo lì a calcolarlo.

Ammettendo che all'inizio dell'Universo regnasse il totale Caos e che alla fine dell'universo tutto sarà tornato al suo posto, possiamo dire che all'inizio il valore dell'entropia sia meno infinito ed alla fine il suo valore, espresso dalla equazione di Boltzmann ($S = -\lg W$) sarà 0, (dove W sono i microstati in cui si trova la materia). Ciò porta a concludere che l'entropia totale dell'universo vale infinito.

La consapevolezza, in questa ottica, viene definita come la differenza di entropia tra un istante e quello successivo ed essendo che essa aumenta sempre, ciò significa che essa è misurabile e dalla fine dell'universo essa sarà pari ad infinito.

Dunque quello che aumenta in realtà per il secondo principio della termodinamica non è S ma il ΔS cioè la consapevolezza della Coscienza

L'approccio al trading coscenziale

Il trading è l'attività pratica che segue l'analisi. L'analisi è la fase preliminare in cui un trader o un investitore raccoglie informazioni, studia i dati e valuta la situazione del mercato per fare previsioni sui futuri movimenti dei prezzi. Nel nostro caso, l'indicatore frattale, come vedremo in seguito, ci dà dei riferimenti precisi sul presente e sul futuro, ma questi riferimenti non hanno un piano applicativo poiché lo scopo dell'analisi non è quello di avere un piano esecutivo, ma di avere dei riferimenti sul futuro.

Il trader che vuole esercitare il suo ruolo deve essere in grado di percepire in qualche modo, attraverso la propria consapevolezza, cosa accadrà in futuro usando un approccio che metta d'accordo il



modus operandi della parte animica e quello della parte spirituale.

In altre parole deve integrare due metodi di lettura della virtualità in modo aspiariale ed atemporale. L'unico sistema a nostra disposizione per ottenere questo risultato è integrare le due parti del Sé con la consapevolezza dell'essere.

Per fare ciò bisogna prima dimostrare che questo percorso può essere effettuato sia su base animica che spirituale, cioè sia con il percepire l'universo che nel calcolarne gli algoritmi. La prima osservazione che facemmo risale ormai a diversi anni addietro, quando ci accorgemmo che esisteva un pattern in grado di rappresentare una unità costruttiva dell'Universo.

Lo spunto ci venne fornito dagli studi di Hartman e Curry, che lievemente modificati hanno dato vita al indicatore frattale, questo indicatore sembrava descrivere l'Universo intero. Praticamente eravamo di fronte ad una struttura tridimensionale costruita sui numeri Universali che sembrava costruire tutti gli oggetti della virtualità.

In altre parole potevamo sostenere che l'Universo fosse sì un ologramma ma di conseguenza avesse struttura frattalica. Se ciò era vero, con questa struttura che si ripeteva all'infinito avremmo potuto costruire un grafico del trading con caratteristiche di ripetibilità, che poteva descrivere l'andamento di alcuni asset finanziari partendo dall'osservazione del passato, verificando che il modello si produceva nel presente e d'estrappolando il modello al futuro. In questa fase va sottolineato come la matematica del nostro pianeta si basa sui numeri.

La nostra numerologia prevede che i numeri servano per contare gli oggetti: e questo approccio è corretto nel senso che è applicabile a tutti i punti dell'Universo. Quando però il numero identifica una misura ecco che sorge il problema di sottolineare come le nostre unità di misura non siano valide in tutto il sistema.

Inoltre se l'Universo viene considerato un ologramma frattalico va da sé che non esistono unità di misura a causa della adimensionalità dell'ologramma. Posso infatti dire che l'ologramma è composto da N pezzi ma non ha senso dare delle dimensioni ad un unico pezzo.

L'Universo in questo contesto sarebbe a dimensionale anche se al nostro cervello appare dimensionabile. Ciò accade perché il cervello sarebbe un lettore di ologrammi, dove le dimensioni ce le metterebbe lui.



La Mappa non è il Territorio:

Percezione, Consapevolezza e il Potenziale del Trading

La percezione non è la realtà così come per la Programmazione Neuro Linguistica, la mappa non è il territorio. Noi infatti percepiamo l'esterno come, ignari creatori del Tutto, crediamo che esso debba essere creato. E nel crearlo in quel modo, lo percepiremo in quello stesso modo. Cambiare consapevolezza del Tutto significava crearlo in modo diverso e conseguentemente percepirlo in modo diverso. Ciò mette in evidenza come il trader fino ad oggi andava solamente dietro gli asset finanziari in modo passivo.

Quando abbiamo a disposizione dei dati per vedere il futuro, ciò non è garanzia di successo, poichè il futuro si costruisce un momento presente dopo l'altro, e in termini pratici ciò vuol dire: se oggi l'indicatore frattale mi dice che domani mattina il prezzo dell'oro arriverà a 2700 e io compro l'oro, m e n t r e in una prima fase il prezzo scende, per evitare ulteriori p e r d i t e , decido d i chiudere la posizione. Successivamente, nelle ore seguenti, il prezzo arriva a 2700. In questo caso, il prezzo è arrivato al target, m a io non ho guadagnato nonostante sapessi che nel futuro il prezzo sarebbe andato a 2700. Quando si fa trading, il potenziale futuro che intercettiamo va contestualizzato in base al time frame, alla consapevolezza dell'asset, alla leva finanziaria (se la sto utilizzando) e ad altri fattori. Il nostro punto di vista è che in una certa qual misura, noi stessi creiamo il futuro ma ciò accade solo se ne abbiamo consapevolezza. Questo si chiama "essere", questo vuol dire diventare un'estensione d i noi stessi e questo è ciò che intendiamo.

La Dualità della Coscienza:

Oltre l'Indeterminazione di Heisenberg

Tale approccio è in realtà sotto gli occhi di tutti i fisici che per altro, non considerando il famoso parametro nascosto coscenziale, non riescono a comprendere il vero significato delle formule da loro stessi trovate. Un esempio per tutti è rappresentato dalla indeterminazione di Heisenberg che non è un principio come erroneamente detto sulla maggior parte dei libri di testo di fisica moderna ma una formula ricavata da altri principi irriducibili.

Per Heisenberg una misura che riguardi qualcosa che ha a che fare con lo spazio-tempo non può essere effettuata contemporaneamente ad una misura sull'asse delle energie di qualcosa che riguarda questa grandezza. Ciò porta erroneamente i fisici a sostenere che esiste una incertezza nelle misurazioni d i questo tipo per cui ne scaturisce u n piccolo errore determinabile in multipli o sottomultipli della costante di Planck.



In questo contesto non si può prevedere neanche il presente perché ci sarà un errore nella nostra previsione inversamente proporzionale alla accuratezza della previsione. In realtà se noi siamo i costruttori del Tutto, non avrebbe senso che costruissimo un Universo che poi non sappiamo gestire, a meno che non introducessimo il paramentro coscenziale. In questo contesto il discorso da farsi è un altro. La formula è corretta ma bisogna tener conto del suo intrinseco significato. Noi siamo venuti in questo universo per comprendere le cose che non abbiamo compreso e l'unica cosa che la Coscienza primordiale, di cui siamo emanazione, non capisce è la dualità, essendo Essa una cosa sola. Studiare la dualità porta a capirla facendone esperienza: e l'esperienza si può fare in un laboratorio che va creato. Questo laboratorio si chiama Universo. In questo contesto la Coscienza crea l'uomo che non può avere la capacità di vedere tutto assieme, altrimenti non potrebbe sperimentare la dualità. Egli ha a disposizione un cervello, quale lettore di ologrammi, che osserva il piano spazio-temporale e ne fa una misura. Subito dopo e non contemporaneamente effettua una misura della energia di quel punto dello spazio-tempo. Avendo a disposizione un solo strumento non è in grado di fare le due misure assieme. Dunque non si tratterebbe di una deficienza dell'uomo ma di una necessità della Coscienza non permettersi, in questa fase di apprendimento, di vedere tutto assieme. Se così fosse infatti, non esisterebbe la separazione e non potremmo viverla nella dualità.

Dualità e Consapevolezza:

Superare i Limiti della Percezione nella Creazione dell'Universo

Consci ora di questo, cioè avendone preso consapevolezza ecco che si comprendono molti più aspetti della Virtualità. Si comprende infatti come mai la teoria dello ZPE (zero point energy) risulta non rispettare l'indeterminazione di Heisenberg che infatti non è una indeterminazione ma una limitazione imposta dalle condizioni al contorno, che l'uomo stesso si impone, creando l'Universo. Nella ZPE infatti si dice che l'Universo non è vuoto nei punti dove non c'è nulla ma è pieno di tutto ed anti-tutto. Le particelle così si annichilirebbero in un mare di particelle virtuali ed il nulla ed il tutto, alla fine dei giochi, sarebbero la stessa cosa. Ciò, se da una parte mostra l'evidenza che la Dualità che noi siamo venuti a sperimentare, in realtà è solo prodotta artificialmente, dalla Coscienza pur non essendo mai esistita.



Ma d'altro canto si dimostra come in quel punto in cui non c'è niente perché il tutto ed il suo contrario si annichiliscono, si conoscono perfettamente le coordinate spazio-temporali ed il valore nullo della energia. Dunque non esiste nessuna incertezza sulla doppia misura.

Un altro aspetto che limita l'osservazione da parte dell'uomo creatore dell'Universo da lui creato è espresso dal secondo teorema di Shannon che dimostra di come, per poter descrivere un fenomeno oscillante nel tempo, ad una certa frequenza, bisogna applicare alle capacità osservative dell'osservatore, una frequenza doppia di quella che caratterizza l'osservabile.

In parole semplici quando osserviamo i cerchioni di una auto che parte noteremo che ad un certo punto della accelerazione, il cerchione sembra fermarsi per poi iniziare a ruotare al contrario. Ciò accade perché quando la frequenza di rotazione del cerchione è eguale a quella dell'occhio umano, il nostro cervello percepirà erroneamente il cerchione fermo. Per evitare tutto ciò bisogna

aumentare il numero di fotogrammi che il nostro cervello produce nel fenomeno della visualizzazione. Dunque se tutto può essere descritto come qualcosa che vibra (Louis de Broglie) la nostra Coscienza crea l'Universo con una certa frequenza vibratoria che può essere al massimo la stessa che la nostra Coscienza mette in atto in un processo osservativo del suo Creato.

In realtà per poter osservare il Creato da noi stessi costruito ci vorrebbe una frequenza doppia. Tutto ciò non è però da considerarsi una limitazione ma una necessità che ci permette di comprendere la separazione all'interno dell'Universo, in questo caso tra creato e creatore. Una volta acquisita la consapevolezza che le cose stanno così, attraverso il parametro nascosto della Coscienza, l'uomo sa cosa sta osservando conoscendo i limiti della sua percezione e sa anche come modificare la propria percezione per vedere oltre.



Il Trading come Specchio:

Comprendere il Mercato per Comprendere Sè Stessi

Quindi nella osservazione di un fenomeno, si devono mettere in conto due punti di vista: quello coscenziale animico, che percepisce un universo atemporale e quello spirituale che concepisce un universo senza spazio. Tutti e due questi punti di vista possiedono una idea della energia completamente differente e ciò produce l'assenza di una teoria del Tutto che i fisici vanno bramando da sempre ma che le loro limitazioni coscenziali non permettono di approcciare correttamente. Ricordiamo che il metodo scientifico galileiano prevede che se una cosa non si vede e non si può misurare, essa è come se non esistesse perché si suppone non interagisca con il sistema che stiamo studiando. Dunque siccome la Coscienza per definizione, non è misurabile, non solo non esiste ma non interagisce nella creazione e non conterebbe nulla. Così pensando l'uomo si taglia di netto la possibilità di capire sè stesso applicando il paradigma che se la Coscienza esiste essa è fuori di lui e quindi senza possibilità di interazione attiva. Il nostro approccio al trading per la prima volta, vuole il trader al centro dell'universo e non il mercato finanziario. Nessuno fino ad oggi ha costruito un approccio consapevole al trading, il che non vuol essere dire consapevole delle leggi del mercato ma consapevole del trader stesso. Ciò porta l'uomo al centro del problema e ci ricorda come per la fisica quantistica moderna, nella interpretazione di Von Neumann, non esiste separazione tra lo sperimentatore e la sperimentazione.

Oltre che a sottolineare ancora una volta che comprendere che la separazione in verità non esiste o meglio appare solo fino al momento in cui si capisce che essa non esiste, si comprende come capire le regole del mercato è come fare un viaggio dentro se stessi poiché quello che studiamo altro non è che uno specchio che creiamo al di fuori di noi per comprendere la parte di noi stessi che non abbiamo ancora compreso.

Appare evidente che il trader consapevole si deve subito chiedere perché sta facendo quel lavoro. Capire che il target finale non è fare soldi ma capire sè stessi vuol dire crescere in consapevolezza e siccome l'assenza di consapevolezza garantirà il fallimento del raggiungimento del target, è necessario che il trader, prima di affrontare qualsiasi azione sul mercato abbia capito la sua vera essenza. Il problema del target si manifesta nell'uomo moderno in molti campi della sua esistenza e la consapevolezza può mettere a nudo alcuni errori che si fanno nel credere che i nostri obiettivi siano quelli corretti.



Per esempio se un uomo è malato può erroneamente credere che il suo obiettivo sia guarire e pur avendo consapevolezza che lui stesso, per qualche motivo inconscio è stato l'artefice della sua malattia, pensa che il suo scopo sia guarire. Quello che accade in questi casi è che il soggetto, per quanti sforzi faccia non guarirà mai. La cosa accade perché il soggetto ha sbagliato bersaglio, che non era guarire ma capire perché si è ammalato. Il verbo ammalarsi è riflessivo. Non si dice: tu mi hai ammalato ma io mi sono ammalato, sottolineando che il processo è personale.

Nell'istante in cui il soggetto comprende che si è ammalato e che lui stesso comprendendo che la malattia è una opportunità che lui stesso si è inconsapevolmente dato per comprendere, ecco che capirà anche che il target è comprendere non guarire perché la guarigione arriverà da sola, automaticamente dopo la comprensione. Da questo punto di vista il percorso del trader non sarebbe diventare ricco ma conoscere sè stesso ed attraverso le regole del trading, farsi domande sulla propria natura.

La comprensione dei meccanismi universali

Per comprendere sè stessi si può cercare di analizzare come abbiamo costruito l'Universo che, essendo una nostra creazione rappresenterà di noi, anche uno specchio. A metà del 1600 il matematico Fourier trova un modo per descrivere un evento che abbia una sua funzione con un abile trucco. Egli sosteneva che ogni funzione continua poteva essere scomposta in molte funzioni continue caratterizzate dalla possibilità di essere descritte come seni o coseni di qualcosa.

Il magico mondo della trasformata di Fourier

Then CTFT becomes

- Fourier Transform of a 1D continuous signal

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$

"Euler's formula" $e^{-j2\pi ux} = \cos(2\pi ux) - j \sin(2\pi ux)$

- Inverse Fourier Transform

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$

2D Fourier Transform

Fourier Transform:

$$F(u, v) = \iint_{-\infty}^{\infty} f(x, y) e^{-i2\pi(ux+vy)} dx dy$$

u and v are frequencies along x and y , respectively

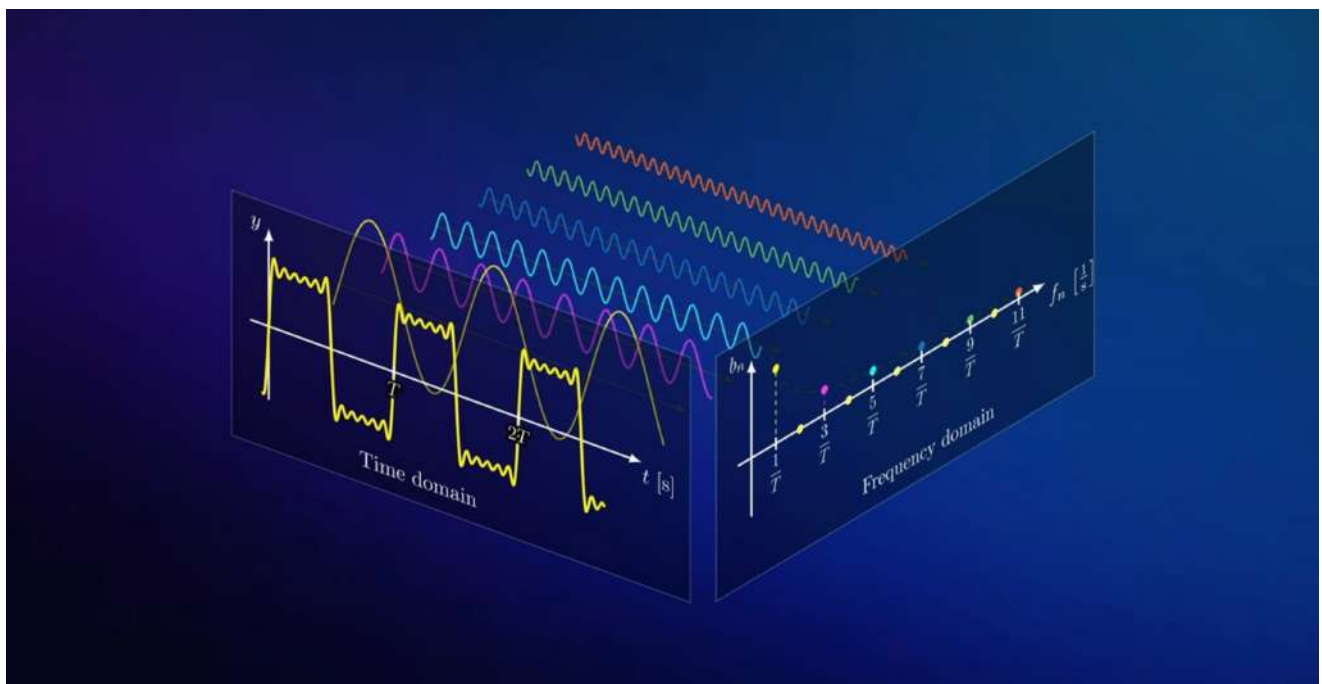
Inverse Fourier Transform:

$$f(x, y) = \iint_{-\infty}^{\infty} F(u, v) e^{i2\pi(xu+yv)} du dv$$



In parole povere qualsiasi tratto di penna su un foglio, per quanto complicato possibile, sarebbe stato considerato come somma di sinusoidi semplici. Era evidente che con questo sistema, qualsiasi andamento di qualsiasi funzione continua, poteva essere scomposto nelle sue componenti sinusoidali. Tali componenti potevano essere analizzate con semplicità ed essere considerate quali semplici componenti della intera funzione che aveva un grado di complicazione apparentemente intrattabile. Le diverse funzioni sinusoidali potevano essere dunque descritte nel tempo a seconda della frequenza che possedevano ed in ampiezza.

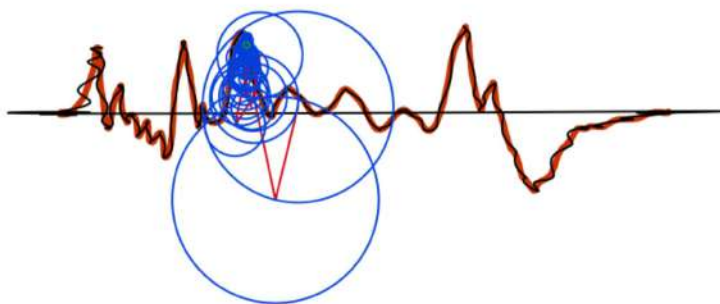
L' algoritmo che descrive la trasformata in una dimensione dipende solo da una variabile che chiameremo "x". la somma di tutte le funzioni che va analizzata da più infinito a meno infinito prende forma di integrale, cioè di somma delle aree sottese dalle singole funzioni. Il termine esponenziale in base "e" è caratterizzato dalla informazione matematica che dice: tutte le volte che incontri una funzione che si ripete ogni 2π volte, tienine conto.



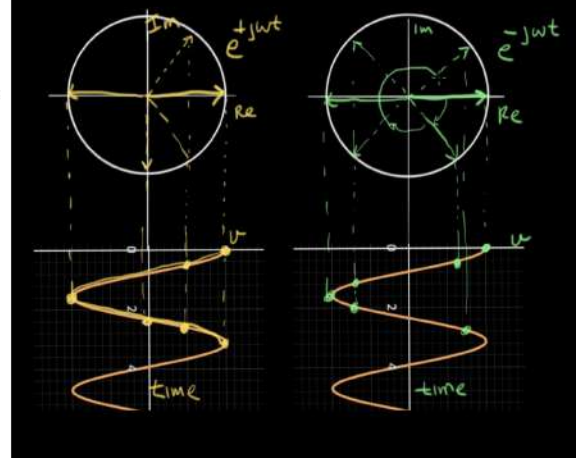
In questo contesto l'algoritmo vede all'interno di una qualsiasi funzione continua i possibili componenti che la costituiscono che si ripetono nel tempo, se essi esistono e li intercetta, separandoli. Così per esempio un'onda quadra, analizzata con la trasformata, viene scomposta in sinusoidi di cui si conoscerà tutto. Appare evidente che se l'andamento di un asset finanziario sembra complesso, la trasformata di Fourier lo scompone in tutti i suoi pezzi e permette l'analisi di ogni componente, che, essendo ripetitiva nel tempo, potrebbe essere facilmente intercettata nel tempo futuro.



Da un altro punto di vista prettamente geometrico, ogni funzione sinusoidale è descrivibile come una circonferenza il cui raggio di lunghezza “ r ” viene ad essere caratterizzato da una sua lunghezza e da una frequenza di rotazione attorno al suo centro, come una rotellina che gira ad una opportuna velocità. Tutte queste rotelline sono però fra loro legate e fanno parte di un ingranaggio più complicato. Bisogna notare come tale approccio è valido matematicamente non solo per valori di frequenza positivi (rotazioni dei raggi dei cerchietti in senso orario) ma anche per frequenze negative (rotazioni antiorarie). La trasformata di Fourier può essere utilizzata per movimenti indietro nel tempo e non solo in avanti in quanto i risultati sarebbero indistinguibili tra loro e le funzioni sinusoidali ottenute, sarebbero indistinguibili, come si può notare dal grafico riportato qui sopra. Le applicazioni di questo algoritmo sono praticamente infinite.



Ogni funzione seno può essere graficata come una proiezione sul piano del raggio di un cerchio mentre ruota con una frequenza omega



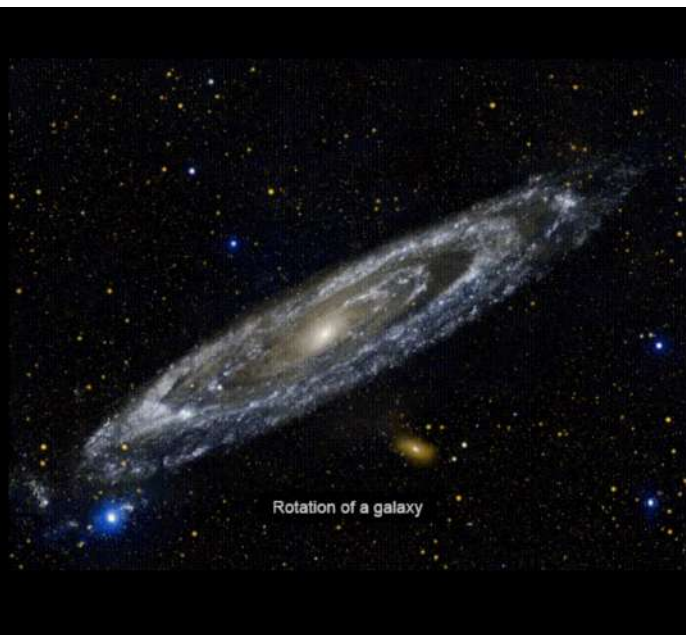
Per esempio l'analisi fotografica per il miglioramento delle foto o l'analisi di particolari che nelle foto devono essere messi in evidenza si ottiene proprio con la trasformata di Fourier, così come l'analisi dei suoni.

In parole povere tutto ciò che è onda può essere analizzato e siccome de Broglie sostiene che in questo universo tutto è onda ecco che la trasformata ne diviene un descrittore. In altre parole potremmo dire che se faccio una foto dell'universo e sottopongo il risultato all'algoritmo di Fourier, esso scompone l'universo in tutte le sue parti che considererà come variabili che si ripetono nel tempo con opportune frequenze.

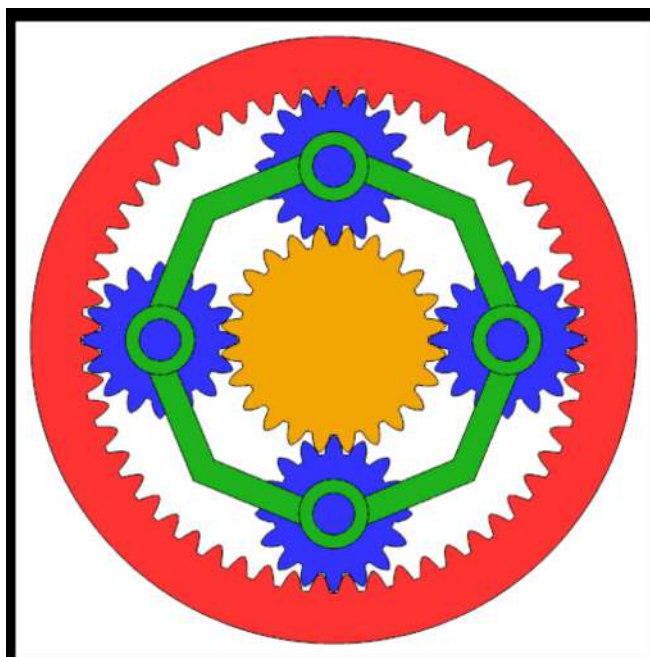


Da un punto di vista coscenziale ciò ci riporta alle antiche credenze induiste. Gli antichi infatti sostenevano che l'universo fosse una gigantesca ruota chiamata Chakra.

Ciò vuol dire che un evento complesso è costituito da molte componenti irriducibili che rappresentano vibrazioni, tutte collegate da multipli e sottomultipli di una stessa fase. Ciò vuol dire che l'Universo in movimento può essere descritto da una trasformata di Fourier dove tutti gli eventi che la compongono risultano essere in fase tra loro. Si tratterebbe di una immensa ruota con infiniti ingranaggi (chakra) tutti però collegati fra loro.



Questa ruota era la parte di un immenso ingranaggio fatto di tanti Chakra più piccoli. Ad ogni giro della grande ruota, le altre ruotine più o meno piccole componenti di e l'gigantesco meccanismo, facevano diversi giri con valori di frequenza multipli di quella della grande ruota. Dunque quest'ultima osservazione, contiene un importante aspetto che caratterizza l'universo intero. Attraverso la trasformata di Fourier dell'universo potremmo dire che qualsiasi evento accade all'interno di esso è per forza di cose collegato al tutto e che prima o poi tenderà a ripresentarsi come era all'inizio dei tempi come sostiene il teorema di ricorrenza di Poincaré citato in precedenza.



Nel totale apparente chaos di una funzione matematica, ci sarebbe uno stretto ordine delle sue molteplici componenti. Ma ciò significherebbe che tutto è collegato a Tutto, facendo dell'Universo una sola creatura vivente



Ciò, per un trader ha un solo significato: il mercato attuale si ripeterà in futuro: o per meglio dire: alcuni parametri del mercato o diurno si stanno già ripetendo e possiamo prevedere il futuro.

Questa visione però è ancora semplicistica poiché descrive un universo armonico che in realtà non esiste. Infatti se è vero che tutti gli eventi che descrivono l'universo costituiscono un gigantesco ingranaggio di cui se si conosce la frequenza di una rotella, si può ricavare la frequenza di qualsiasi altra parte dell'ingranaggio, almeno da un punto di vista teorico, bisogna supporre che i diversi ingranaggi siano fatti di gomma. In altre parole essi subiscono alterazioni durante il movimento della grande ruota primaria, alterando con la alterazione della propria geometria, anche il movimento frequenziale di tutte le altre ruote.

Questa alterazione di forma dipende dalla consapevolezza che noi abbiamo di tutti questi processi e di conseguenza della armonia dell'intero sistema. Ma come entra il parametro, ora evidenziato, della consapevolezza nella formula di Fourier? La cosa è veramente semplice. Tale parametro infatti è dato dal numero N di cerchietti che compongono la trasformata stessa. Infatti se io ricostruisco la mia funzione complessa a piacere con pochi sub-elementi, per esempio una decina, avrò una ricostruzione approssimativa della mia funzione finale; ma se io sono in grado di distinguere duecento componenti sinusoidali che compongono la funzione che sto studiando, ecco che il disegno di tale funzione, cioè il grafico che definisce le variabili del sistema sarà più accurato.

In altre parole è la nostra consapevolezza del fenomeno che sto osservando che stabilisce il numero N di sottofunzioni. Il nostro cervello, nel mettere a fuoco la visione della Virtualità mette in opera una trasformata di Fourier della Virtualità, scomponendola in tutti i suoi componenti.

C'è chi è capace di osservare la Virtualità in modo da ottenere un numero N di componenti elevato, rispetto ad altri che vedono la realtà con un numero N non sufficiente ad avere una mappa del territorio che assomigli abbastanza alla Realtà delle cose. Ebbene, il parametro N è una misura della consapevolezza. Nel caso specifico il trader che non è in grado di osservare con consapevolezza l'andamento del suo asset è destinato ad avere una previsione del futuro poco attinente a quella che in futuro si mostrerà. Da un punto di vista teorico e solamente speculativo, è probabile che l'Universo abbia un numero di subcomponenti non infinito che sia eguale al numero di Avogadro ($N = 6.022 \times 10^{23}$). Tale osservazione condurrebbe anche ad ammettere che l'Universo è finito e frattalico, dove il numero degli oggetti contenuti è un numero preciso e dove ogni oggetto è



costituito da un numero di oggetti più piccoli che è sempre lo stesso numero, come il numero di molecole di un gas ideale contenute in una mole di sostanza in condizioni standard od il numero di elettroni che passa in un filo elettrico in un secondo e che sviluppa un Farad di corrente elettrica o il numero di galassie nell'Universo eccetera.

La Trasformata di Fourier nel Trading: Il Ruolo della Consapevolezza oltre l'Algoritmo

Impiegare la trasformata in campo finanziario era già stato tentato in passato con scarsi risultati a causa della mancanza della applicazione del famoso parametro nascosto che noi indichiamo con il termine di "consapevolezza". Tale parametro ci permette di non lasciare la valutazione delle proiezioni dell'andamento futuro del mercato semplicemente ad una macchina che calcola un algoritmo ma ci permette di interagire con il software in modo diretto per compensare la mancata consapevolezza coscienziale che un computer non potrà mai replicare, nemmeno con i sofisticati moderni sistemi di A.I.

La trasformata di Fourier per esempio è una fotografia del presente e per poterla usare come proiezione per il futuro ha bisogno della consapevolezza che il presente è la somma di due equazioni d'onda che rappresentano il passato ed il futuro. La termodinamica quantistica moderna prevede che, esistendo solo il presente, esso conterrà tutte le informazioni del passato e del futuro che costruiscono il continuo presente.

Una volta si diceva che esisteva un principio di causa-effetto secondo il quale se nel passato faccio una azione, ecco che nel presente ci sarà una reazione a quella azione. Oggi si dice che le azioni del passato e del futuro costruiscono l'unica realtà esistente che è il presente. Dunque per poter costruire il tracciato di un asset come proiezione futura bisogna analizzare l'immediato passato e confrontarlo con il presente per estrarre il futuro. Questa operazione prevede consapevolezza del trading che va acquisita attraverso l'esperienza. Se da un lato ciò significa che non si deve lasciare al computer la decisione di come andrà il mercato, perché questo atteggiamento poco coscienziale prevedrebbe lo scarico di responsabilità di un eventuale fallimento al computer stesso; si dovrà fare in modo che il computer tenga in qualche modo conto dell'immediato passato perché lì ci saranno le informazioni per costruire il frattale del futuro. Inoltre, nell'esaminare i primi risultati ottenuti come proiezioni del futuro, ci siamo accorti che ancora qualcosa era perfezionabile.



Infatti ogni tanto la trasformata di Fourier segnalava correttamente l'andamento dell'asset preso in considerazione ma ad esso si sommava un segnale imprevisto che non avrebbe dovuto esserci. Tale segnale rappresenta un "disturbo" le cui cause non possono che essere volute appositamente da chi gestisce il mercato e vengono messe in atto probabilmente per evitare che il trader non consapevole non riesca a trovare un algoritmo proiettivo corretto.

La nostra idea, basata sulla nostra esperienza, ci faceva credere in quella fase della ricerca, che chi comanda il mondo, sappia perfettamente le leggi che lo governano e siccome il denaro viene utilizzato solo come strumento divisivo, che abbatta l'entropia, cioè la coscienza dell'umanità, era evidente che da parte della Elite, qualunque essa fosse, c'era la necessità di creare disturbi perturbativi che avessero lo scopo di fuorviare anche Fourier.

La trasformata però segnalava queste perturbazioni e l'idea che anche esse fossero prevedibili, essendo parte comunque del meccanismo generale che regola l'universo, ci ha permesso di constatare la prevedibilità e la corrispondente forma frattale su tutti gli asset da noi analizzati.

La Coscienza crea la virtualità

In questa carrellata di osservazioni coscenziali e geometrico-matematiche bisogna sottolineare come si parta dall'idea che siamo noi i creatori del Tutto ed anche quindi del presente. Se ciò fosse vero, noi comunque, nel mettere in atto un tentativo di previsione dei nostri asset, non faremmo altro che inconsapevolmente crearli come crediamo che essi debbano essere creati. Ancora una volta la causa e l'effetto verrebbero sovrapposti costruendo un unico evento. Sono io che con la mia coscienza prevedo l'evento futuro oppure mentre cerco di prevederlo lo costruisco? La risposta esatta è che accadono contemporaneamente entrambe le cose. Il trader cosciente deve conoscere questo aspetto della realtà e per questo esso deve qui essere dimostrato e per farlo dobbiamo analizzare l'equazione di Erwin Shroedinger, premio Nobel per la fisica nel 1933. La sua equazione calcolava l'energia di una particella subatomica, per esempio un elettrone, all'interno di un orbitale molecolare prendendo in esame la variazione della equazione d'onda attribuita alla particella elementare nello spazio e nel tempo.





KINETIC ENERGY

POTENTIAL ENERGY

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \cdot \psi$$

OVER TIME

Notare l'analogia con l'equazione termodinamica dell'energia libera di Gibbs che per processi spontanei con entalpia nulla è $\Delta G = \Delta H - T\Delta S$ con $\Delta H = 0$

<https://youtu.be/qOYra2Yxemk?si=eW1mR940i1-eXiNE>

Come si può notare, l'equazione generica è fatta di due parti: la prima che riguarda l'energia cinetica della particella ed una seconda che riguarda l'energia potenziale che per processi spontanei è nulla (vedi l'esempio della pallina che spontaneamente cade a valle partendo da una posizione più in alto).

Notare che spontaneamente non succede il contrario). Notare inoltre come la parte seconda della formula, che indica l'energia cinetica ha un segno "meno" proprio come disegnato nella equazione della energia libera di Gibbs per quanto riguarda il termine $-T\Delta S$. Notare infine che l'energia totale dipende dal tempo ma la cinetica assume la forma della derivata seconda della funzione d'onda. Tale derivata seconda ha il significato geometrico di piegatura della stessa funzione d'onda ed è l'equivalente einsteiniano di una accelerazione ed in questa sede, senza approfondire il discorso, dobbiamo ricordare che in un'altro studio da noi fatto, abbiamo dimostrato come la costante della forza di gravità universale "G" è un aspetto della Coscienza. (<https://youtu.be/hHig-PP4Mcs?si=IKVKLqtn6Krl69UW>). Ma che cosa rappresenta la funzione d'onda?



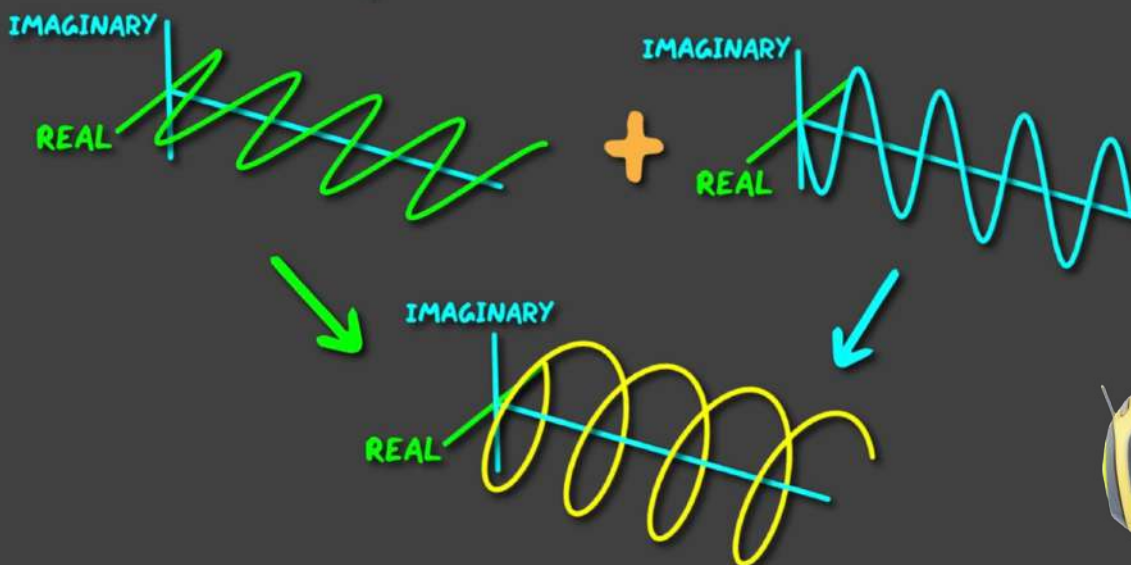
WHAT IS A WAVEFUNCTION?

$$\Psi = A \cos(kx - \omega t) - A i \sin(kx - \omega t)$$


$$\Psi = A e^{-i(kx - \omega t)}$$

Essa è la rappresentazione formale di una sola delle componenti di una ipotetica trasformata di Fourier che descrive un sistema complesso di oggetti che potrebbero essere le componenti di un atomo.

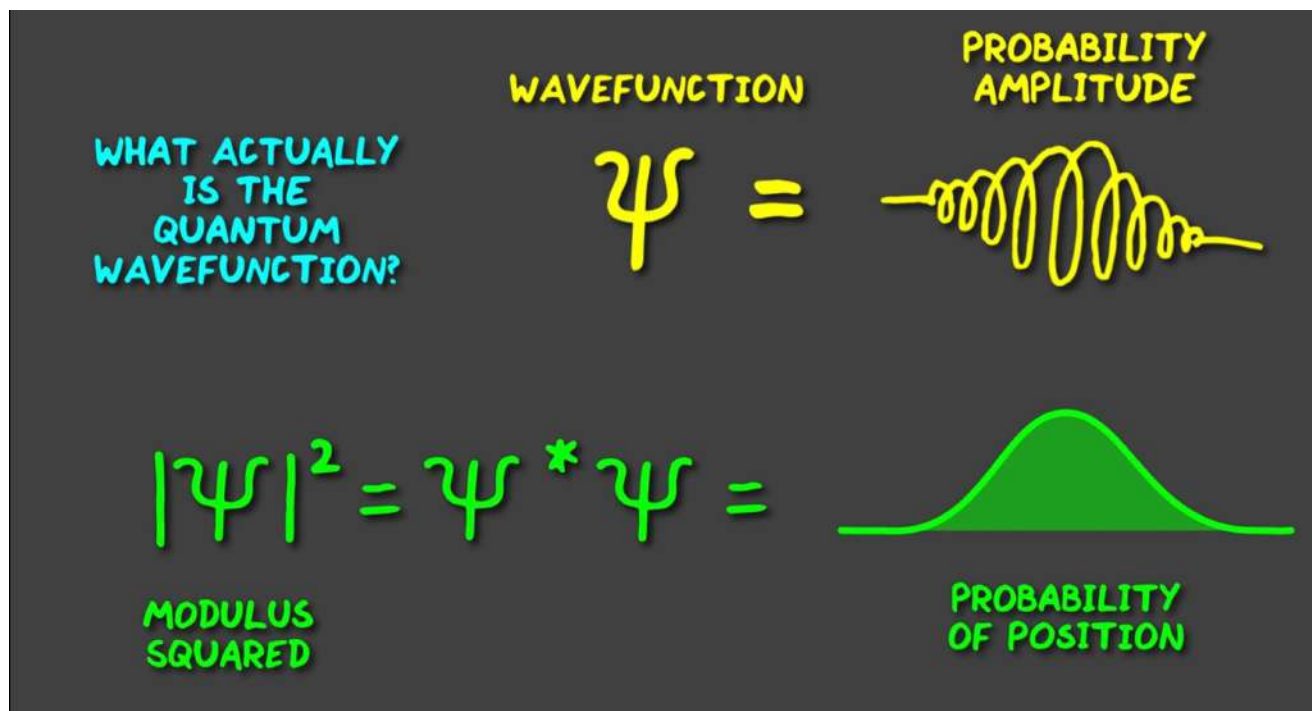
$$\Psi = A \cos(kx - \omega t) - A i \sin(kx - \omega t)$$



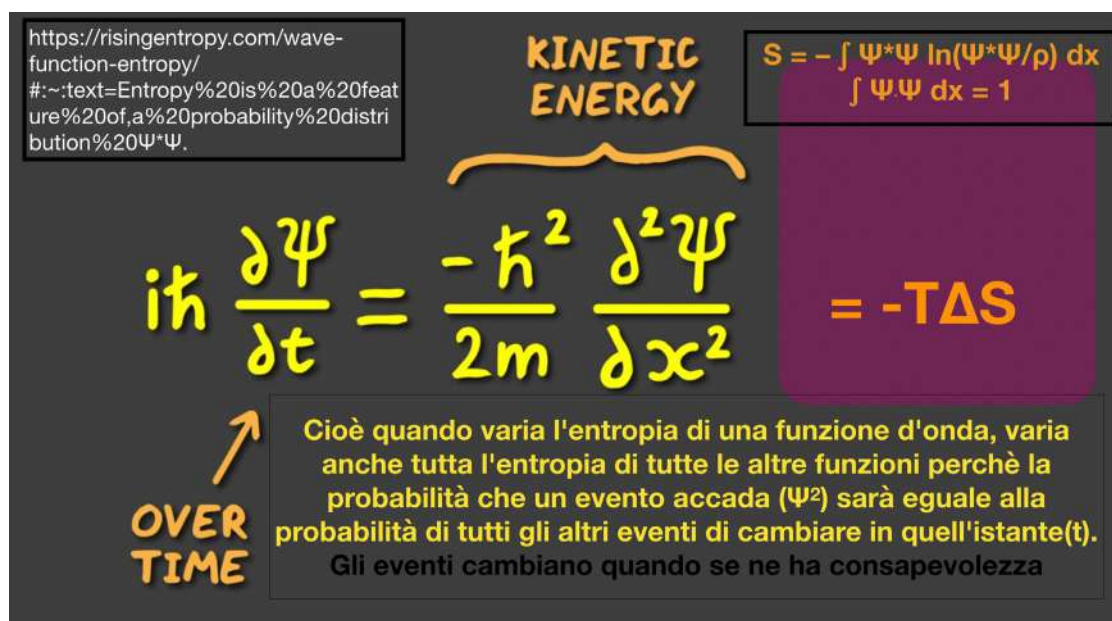
Si noterà che esistono due componenti una caratterizzata dal coseno, componente reale e l'altra descritta come seno nel campo immaginario.

Da qui la necessità di Shroedinger di moltiplicare l'energia totale per $-i$, dove i è la radice di -1 , per rendere reale il risultato della sua equazione.

La funzione Ψ rappresenta l'ampiezza della funzione d'onda mentre Ψ^2 rappresenta la probabilità nello spazio di trovare l'onda stessa.



Dunque appare evidente che il termine entropico della equazione di Gibbs corrisponda esattamente al termine che identifica l'energia cinetica della equazione di Heisenberg.



Ma tutto ciò ha un solo significato; essendo infatti che l'entropia è la rappresentazione della consapevolezza umana e che il ΔS della equazione di Gibbs rappresenta la consapevolezza, è chiaro che la consapevolezza è la variabile che stabilisce la forma e la probabilità che l'equazione d'onda dell'elettrone esista.

In altre parole siamo noi i creatori di tutto, siamo noi che diamo forma alle cose e ne stabiliamo l'energia e la variazione nello spazio. In questo contesto, il significato di probabilità di un evento è strettamente legato alla misura della propria consapevolezza. Infatti se lanciamo una moneta per aria senza alcuna consapevolezza, essa mostrerà una probabilità del 50% di cadere come testa o come croce sul pavimento. Ma se noi fossimo coscienti di essere i creatori di tutto, come descritto dalla equazione di Shroedinger, lanciando la moneta per aria potremmo sempre decidere al 100% prima di conoscere il risultato del lancio, cosa accadrà. Dunque la probabilità ancora una volta è solo una misura della nostra consapevolezza che dipende dall'entropia del sistema.

Tutto questo conduce ad una sola ed inevitabile conclusione. Il futuro non solo può essere da noi previsto ma può essere da noi creato. Mai prima d'ora era stato tentato un approccio simile a riguardo del trading.

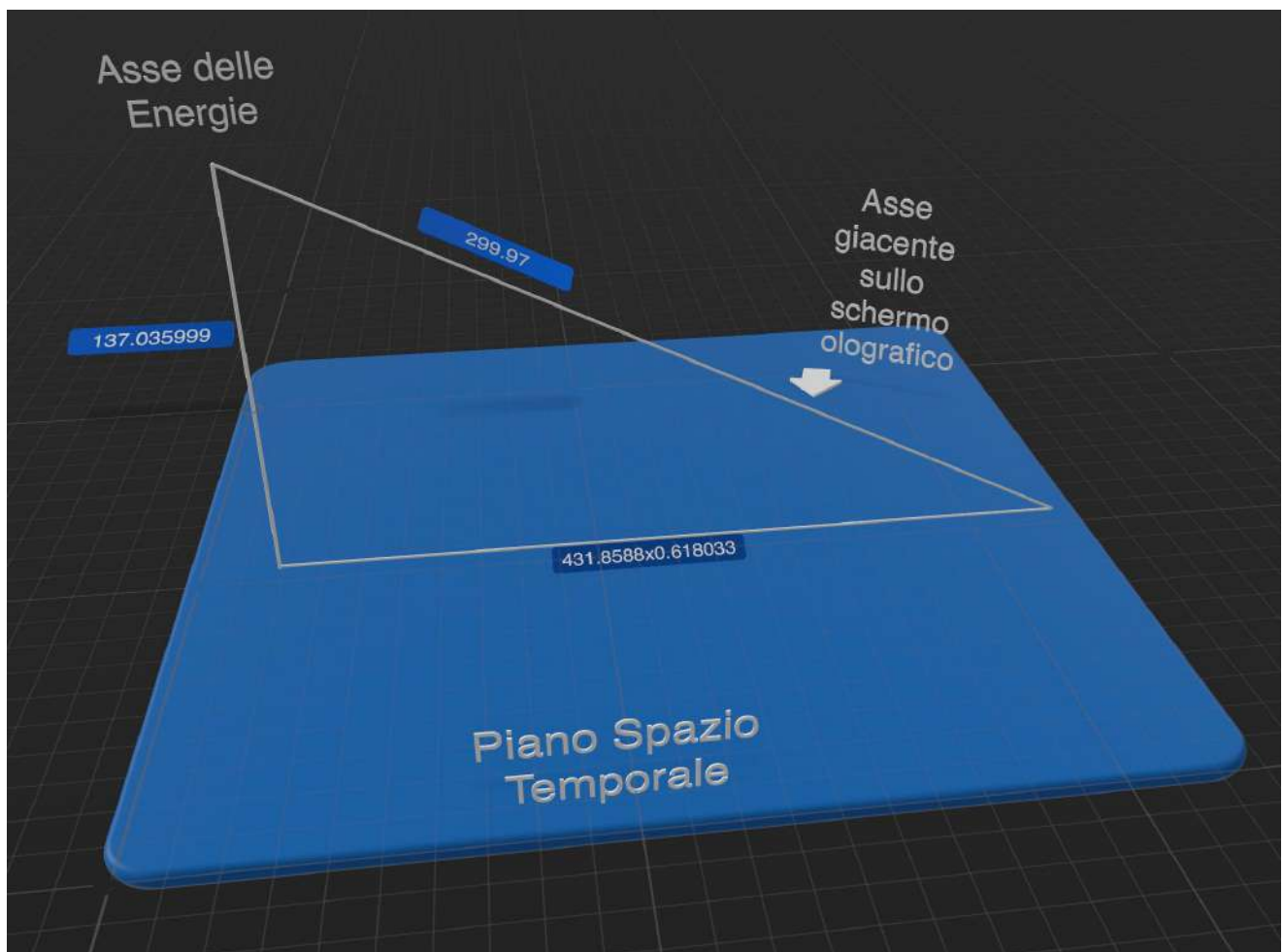
Il problema della inversione del pattern

Durante la costruzione della proiezione del pattern del presente nel futuro ci siamo sperimentalmente accorti che tale unità di realtà andava ulteriormente manipolata con tre operazioni geometriche.

Infatti sovente parti di pattern dovevano essere "rovesciate" specularmente rispetto all'asse verticale e rispetto all'asse orizzontale e ridimensionate. La teoria che stava dietro queste operazioni geometriche dipende dalla struttura stessa dell'universo evideonico.

L'universo evideonico si definisce come un luogo di punti caratterizzati da tre assi fondamentali che rappresentano lo spazio, il tempo e l'energia di tutto il sistema.

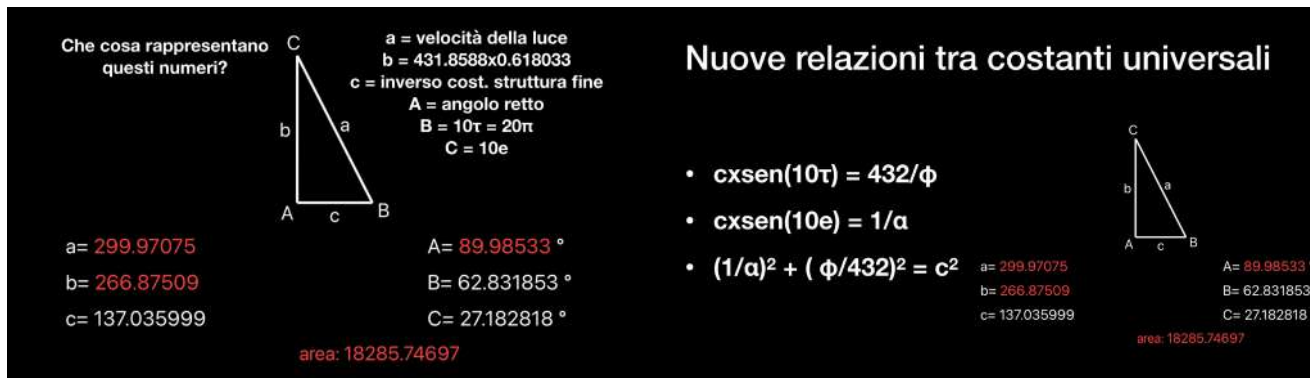




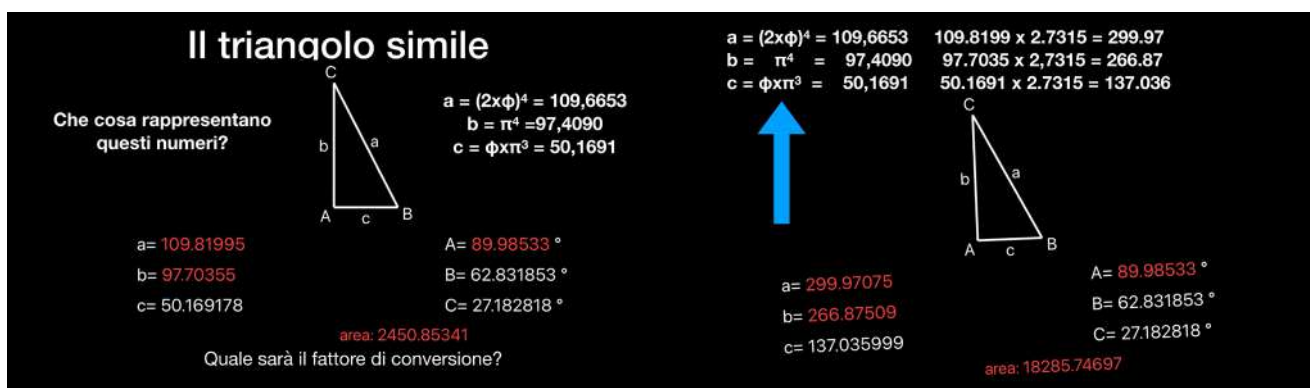
Questi assi hanno valori precisi che ricalcano i numeri dell'universo e non quelli della nostra matematica. Tali numeri sono: la sezione aurea, Pigreco, il numero di Eulero, la radice di 2 eccetera. Tali numeri sono infatti irrazionali e trascendenti nel senso che trascendono la comprensione umana ma hanno dentro di sé tutte le informazioni dell'universo intero. Tali numeri rappresentano tutte le costanti universali che noi comunemente utilizziamo nella nostra matematica.

La relazione tra le nostre costanti universali e i veri numeri dell'universo è espressa da un triangolo rettangolo che contiene tutte queste informazioni, i cui lati contengono dati su espressioni di Velocità.





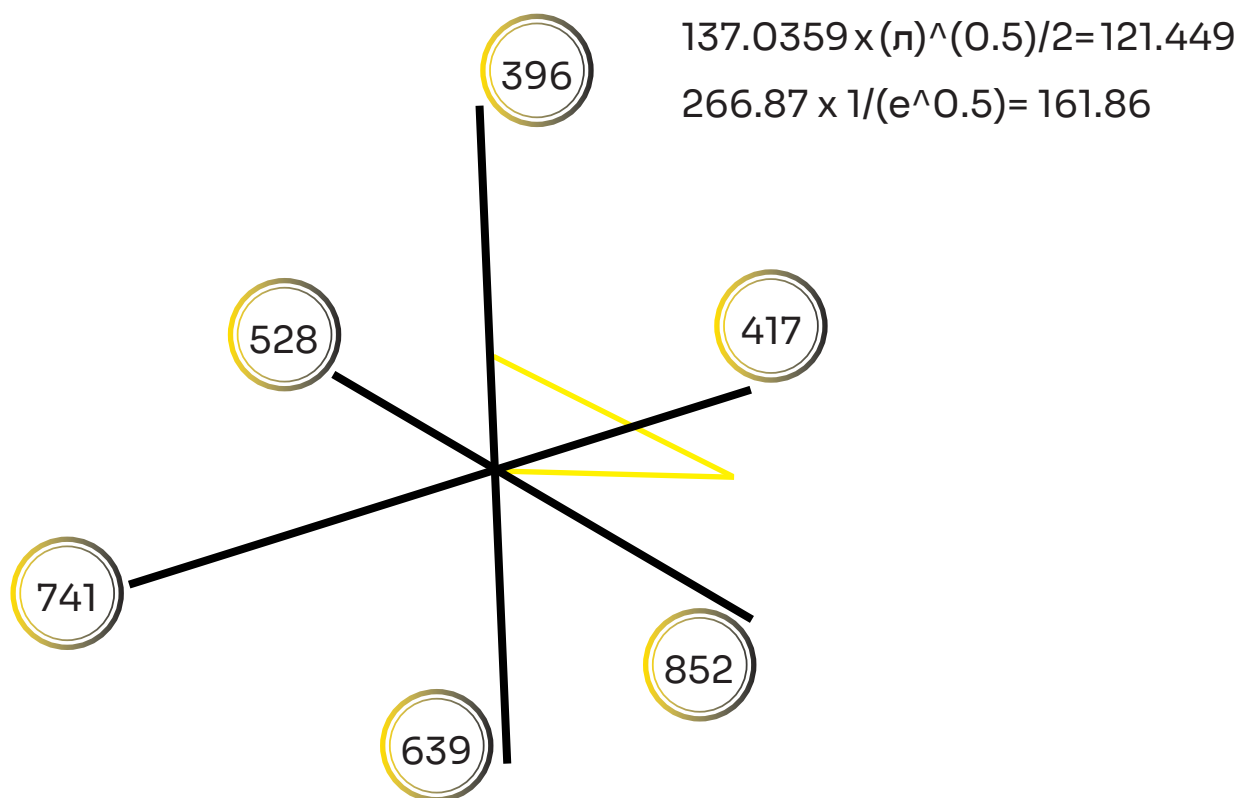
Tali dati opportunamente rielaborati mostrano come tutte le costanti che noi utilizziamo per i calcoli di fisica sono in realtà multipli di quei numeri irrazionali e trascendenti elencati prima.



Come si nota basta infatti moltiplicare i valori dei lati del triangolo simile creato con valori di π e ϕ per il valore della temperatura assoluta di Kelvin/100 per ottenere il triangolo che contiene tutte le nostre costanti universali. Tale triangolo va introdotto nel modello di universo videonico ed adattato agli assi dello spazio-tempo ed energia, moltiplicando i cateti per opportuni coefficienti sempre basati sul numero di Nepero e pigreco. Per una più profonda ed accurata descrizione dell'Evideon si può vedere la seguente conferenza esplicativa della sua struttura (https://youtu.be/HTIYZBrzZbM?si=PHf7HWGOASm-8s_Z).

Nella descrizione dell'universo e videonico bisogna tener presente che, essendo l'universo un frattale, l'oggetto più piccolo dell'universo stesso è il fotone che avrà la stessa struttura del Tutto.





Se il fotone rappresenta il Tutto, in questa sede va sottolineato come esso risulti caratterizzato da un piano spazio-temporale rigido e da un asse delle energie mobile che oscilla in continuazione facendo in modi di convertire il fotone nella sua immagine speculare così velocemente da non poter essere detectato.

Quello che si vedrà in queste condizioni è solo un fotone virtuale senza massa e carica che sarà l'ibrido di risonanza tra le due forme limiti di risonanza, cioè il fotone e l'anti fotone. Se si identifica il fotone come un unico tensore, esso avrà un dominio di esistenza a forma di doppio cono, nel dominio di spazio, tempo, energia. Tale oggetto possiede dunque una frequenza oscillatoria nel tempo e deve rispettare la simmetria CPT (carica, parità e tempo).

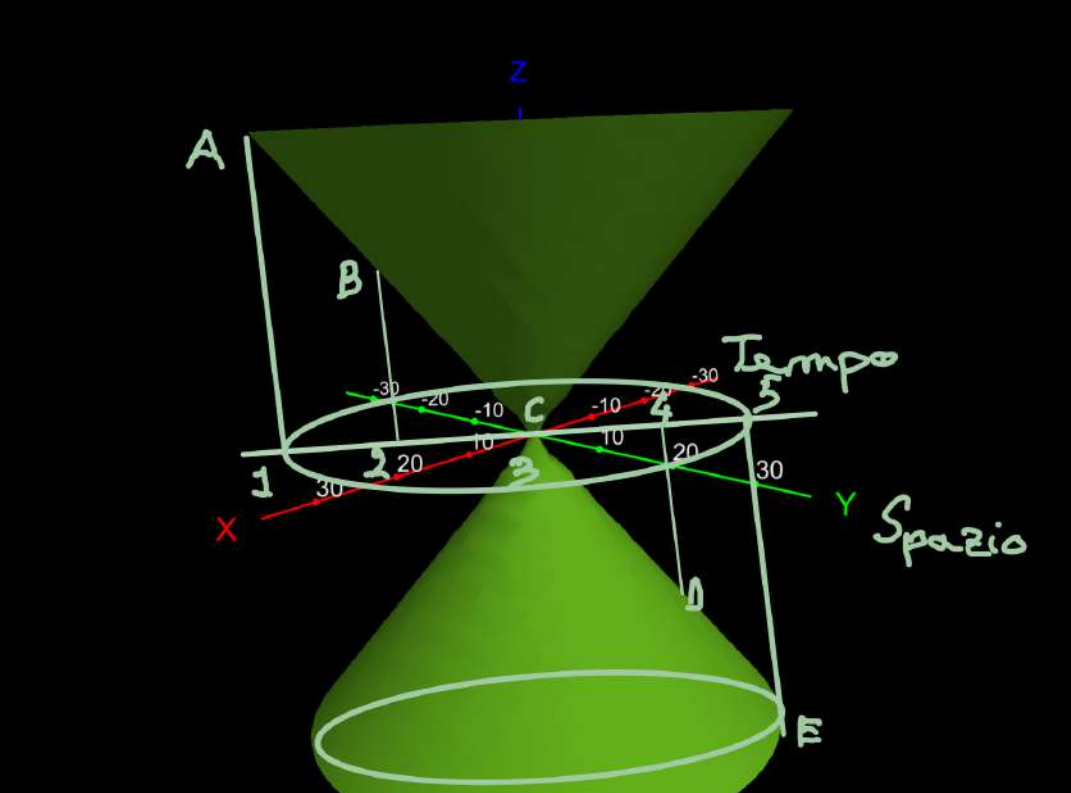
Il comportamento del fotone nel tempo sarà lo stesso del grafico di un asset perché tutti e due gli oggetti sono collocati all'interno dello stesso universo e frattalmente rispondono alle stesse regole geometriche.

Nella analisi dei frattali del trading vedremo, quando passeremo alla parte pratica contenente i dati che permettono di operare sul frattale che, una volta individuato un frattale, esso verrà proiettato nel futuro



manipolazione deriva dalla conoscenza della struttura dell'Evideon.

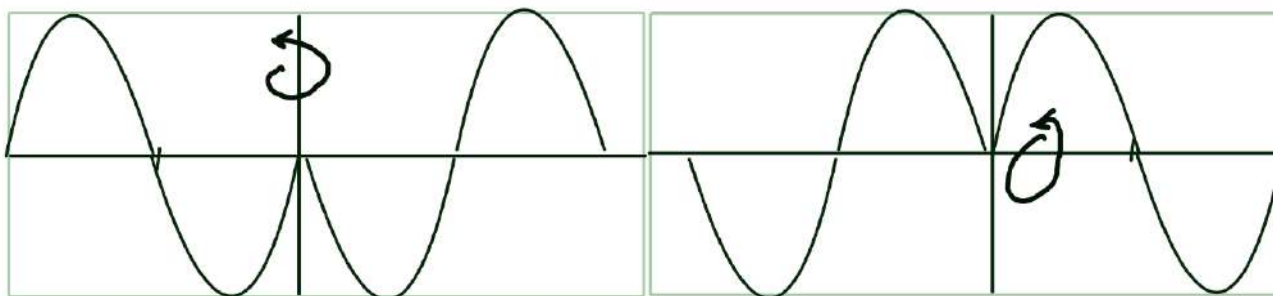
$F(x,y,z): x^2+y^2-(2/3)z^2=1.62$



A 3D plot of a hyperboloid of two sheets. The upper sheet is light blue and the lower sheet is dark blue. A horizontal white ellipse is drawn on the upper sheet, representing the intersection of the surface with a plane. The axes are labeled: 'x' in red, 'y' in green, and 'z' in blue. The 'z' axis is vertical. The 'x' and 'y' axes are horizontal. The ellipse has points labeled 1 through 5. The 'x' axis has points labeled 1, 2, 3, 4, 5. The 'y' axis has points labeled 1, 2, 3, 4, 5. The 'z' axis has points labeled 1, 2, 3, 4, 5. The intersection of the surface with the plane is labeled 'C'.



Riportiamo ora tale operazione su un ipotetico asset semplice, costituito da una sola funzione seno di x .



Come si può notare, due rotazioni del pattern equivalgono ad una traslazione temporale del pattern stesso che, a prima vista sembra che nel riprodursi, cioè nel ripresentarsi nel futuro, si sia semplicemente riprodotto traslando. In realtà l'operazione geometrica che sta dietro questa proiezione dell'asset è più complessa e nella realtà evideonica, essa si rappresenta con una unica operazione geometrica, identificabile come centro di inversione. Ma quando si prende in considerazione il pattern in un tempo sottomultiplo del tempo di partenza, il punto A di partenza si trasforma nel punto D su l cono evideonico e questo punto D ha una proiezione sul piano spaziotemporale, che non corrisponde al punto 5, come nel caso del punto E, ma da origine al punto 4. Nello spaziotempo, il vettore C-4 è più corto del corrispondente vettore C-5 e questo corrisponde a una contrattura del pattern che subirà deformazioni, sia sull'asse del tempo che sull'asse delle energie, cioè l'asse verticale dell'asset, analogamente con lo stesso meccanismo opposto si potranno generare dilatazioni sui due assi; ma tutto ciò sarà ben visibile nella parte pratica di questo trattato.

Bisogna sottolineare come nel semplice esempio di una senoide riportato sopra, siamo stati in grado di vedere coscientemente l'inizio del pattern e la fine del ciclo, perché tali punti rappresentano i nodi della funzione seno, cioè i punti reali di inizio e fine temporale del pattern.

In una situazione più complessa l'operatore trader, nell'individuare l'inizio di un pattern potrebbe individuare quei nodi delle componenti del pattern complesso che in quel punto sono effettivamente a zero ma magari, ci sono alcune componenti che in quel punto non si azzerano. Questo porta alla considerazione che non è sempre facile trovare i punti dove tutte indistintamente le componenti della trasformata di Fourier si azzerino.



Sarà frequente non riuscire, se non operiamo con consapevolezza, a localizzare quei punti in cui tutte le frequenze delle componenti della trasformata, si azzerino ma l'importante è sapere, e d'ora lo sappiamo, perché ciò accade.

Infatti utilizzando una grande dose di consapevolezza/esperienza, sarà possibile riuscire in questa impresa soprattutto calcolando pattern corti nel tempo, perché in tempi brevi non tutte le componenti della trasformata fanno in tempo ad azzerarsi. In quel caso potremmo verificare che la proiezione temporale del pattern si dilata su uno solo dei due assi e si restringe sull'altro. L'importante in questa sede è però sapere, cioè essere coscienti, del perché ciò accade e noi ora qui forniamo una chiave di lettura in accordo con la teoria che permette di sostenere il nostro modello geometrico.

Una ulteriore osservazione sulla equazione di Shroedinger

L'equazione ha sostanzialmente tre termini. Volendo ammettere che il trading si esprima come la somma di tanti parametri da considerarsi, come prima approssimazione, spontanei, in questo contesto il termine "V" di energia potenziale che rappresenta l'Entalpia di un qualsiasi processo, va considerato nullo. Si nota così come il termine che contiene una derivata seconda, cioè la variazione della derivata prima della funzione d'onda, rappresenti una "piegatura" che, nella accezione comune sarebbe la piegatura dello spazio-tempo, considerato come un unico componente, che nella equazione abbiamo indicato con "x".

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = -T\Delta S$$



In realtà la piegatura corrisponderebbe a quella dello schermo olografico sul quale si proiettano le immagini della realtà virtuale (Verlinde, Maldacena). Tale formula descrive l'andamento nello spazio-tempo del trading indicando che la probabilità che un evento accada, appare legata alla consapevolezza stessa che l'operatore ha dell'evento. Praticamente q u e s t o termine descrive l'andamento del trading lungo l'asse delle ascisse. I n questo contesto “ m ” rappresenterebbe una massa fittizia del “volume d'affari”, una sorta di grandezza che rappresenterebbe l a quantità di operazioni effettuate nell'unità di tempo e di spazio. L'identificazione di questo numero non è importante ai nostri fini perché può essere considerato una costante essendo sempre molto grande. L'al t ro termine invece prevede che ci sia la derivata prima della funzione d'onda sull'asse delle energie che è un asse immaginario all'interno dell'Evideon.

$$i\hbar \frac{\partial \psi}{\partial t} = -T\Delta S$$

Da qui si nota come l'entropia, c i o è la consapevolezza sia la responsabile anche della altezza delle “candele” all'interno del grafico sull'andamento del trading. In parole povere, la nostra consapevolezza influenza anche le variazioni in altezza sull'asse delle ordinate. Il segno “meno” è legato alle convenzioni utilizzate dalla fisica termodinamica, dove l'energia e l'entropia, per ragioni storiche e religiose hanno segno contrario.

L'ampiezza della funzione d'onda rappresenterebbe l'altezza delle “candele” e l'espansione in verticale del grafico di trading.

Dietro questo concetto appare l'evidenza che, se da un lato una grandezza immaginaria non può essere direttamente identificata nel campo reale, è pur vero che essa modifica la parte reale alterando la rotazione del vettore rotazione di Pointing, di cui noi però possiamo evidenziarne la proiezione finale sul piano dello spazio-tempo (parte reale).



Una ulteriore finale osservazione v è effettuata sulla finestra di Overton che determina l'osservazione del presente. Abbiamo infatti detto che per comprendere il futuro bisogna avere informazioni sul passato e manipolarle nel presente per estrapolare il futuro, tenendo conto che nel presente esiste sia il passato che il futuro a livello di contributo entropico. In realtà l'osservazione del presente come tale nel nostro universo è resa impossibile dalla limitazione della velocità della luce che, per quanto elevata, ci permette solo di vedere un passato molto vicino al presente ma non il vero presente.

Tenendo conto di questa insignificante approssimazione, possiamo dire di osservare il presente anche se realmente ciò non sia affatto vero. In questo contesto ci dobbiamo chiedere quanto dura effettivamente il presente di cui abbiamo consapevolezza.

Infatti sapendo che il nostro programma che calcola le trasformate di Fourier come proiezioni del presente e passato nel futuro, dobbiamo chiarire che nella scelta che si fa, esiste un parametro, detto finestra di Overton, che rappresenta la lunghezza sull'asse delle ascisse del parametro "x" cioè l'apertura dello spazio-tempo che noi "arbitrariamente" consideriamo presente. Una volta identificata la frequenza di Overton impiegata, è evidente che la stessa finestra verrà utilizzata per prendere informazioni dal passato per produrre una identica finestra del futuro. "L'apertura sintetica" di questa finestra è proporzionale alla qualità delle previsioni effettuate; infatti più l'apertura considerata è larga, più dati vengono computati dalla trasformata di Fourier e più preciso sarà il dato proiettivo nel futuro. In pratica tale apertura corrisponde alla lunghezza di una antenna radar di un satellite che osserva oggetti sulla superficie terrestre.

Più grande è l'antenna più piccola sarà la dimensione dell'oggetto visibile nella foto satellitare. Ricordiamo in questa sede che il Radar ad Apertura Sintetica SAR produce una trasformata di Fourier dei dati acquisiti, dove la stessa trasformata rappresenta la capacità di mettere a fuoco un determinato target.

La nostra Coscienza con lo stesso meccanismo osserva le curve di trading ed è evidente che alcuni operatori hanno più consapevolezza

di altri e riusciranno ad ottenere immagini messe a fuoco con maggior definizione di altri, provvedendo a previsioni più reali. Infine il termine di energia potenziale $V(\Psi)$ che rappresenta l'energia potenziale e che appare sempre essere una funzione d'onda cioè appare sempre essere sotto controllo dell'Entropia, rappresenta quel fattore non spontaneo che produce alterazioni nelle previsioni di spontaneità che la funzione finale di trading dovrebbe avere.



È evidente che questo fattore rappresenta una perturbazione provocata artificialmente nel trading da componenti esterne che possono produrre alterazioni sulle previsioni di spontaneità dei grafici di trading. È altresì evidente che per quanto possa sembrare assurdo, queste alternazioni dalla spontaneità rientrano sempre in una prevedibilità dettata dalla trasformata di Fourier. In altre parole, chiunque tenti disperatamente di modificare il percorso degli eventi in qualsiasi campo o asset, sarà costretto a usare sempre le stesse leggi di ripetibilità dell'Universo e ciò significa che le sue azioni "disturbanti" appariranno solo ad un primo esame di tipo randomizzate ma seguiranno obbligatoriamente un percorso descrivibile da una o più funzioni trigonometriche. Da un punto di vista matematico formale, si dovrebbe obiettare che non è possibile, in linea di principio, eguagliare una derivata ad un differenziale o ad una differenza.

Non è comunque vero che le notazioni sono le medesime: il differenziale si indica con df , la derivata (o derivata parziale) si indica con una frazione della forma $\frac{d}{d\text{variabile}}f$, $\frac{\partial}{\partial\text{variabile}}f$, o in alternativa con $D(f)$, $D_{\text{variabile}}(f)$.

$$\frac{dv}{dx} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{v(t + \Delta t) - v(t)}{\Delta t}$$

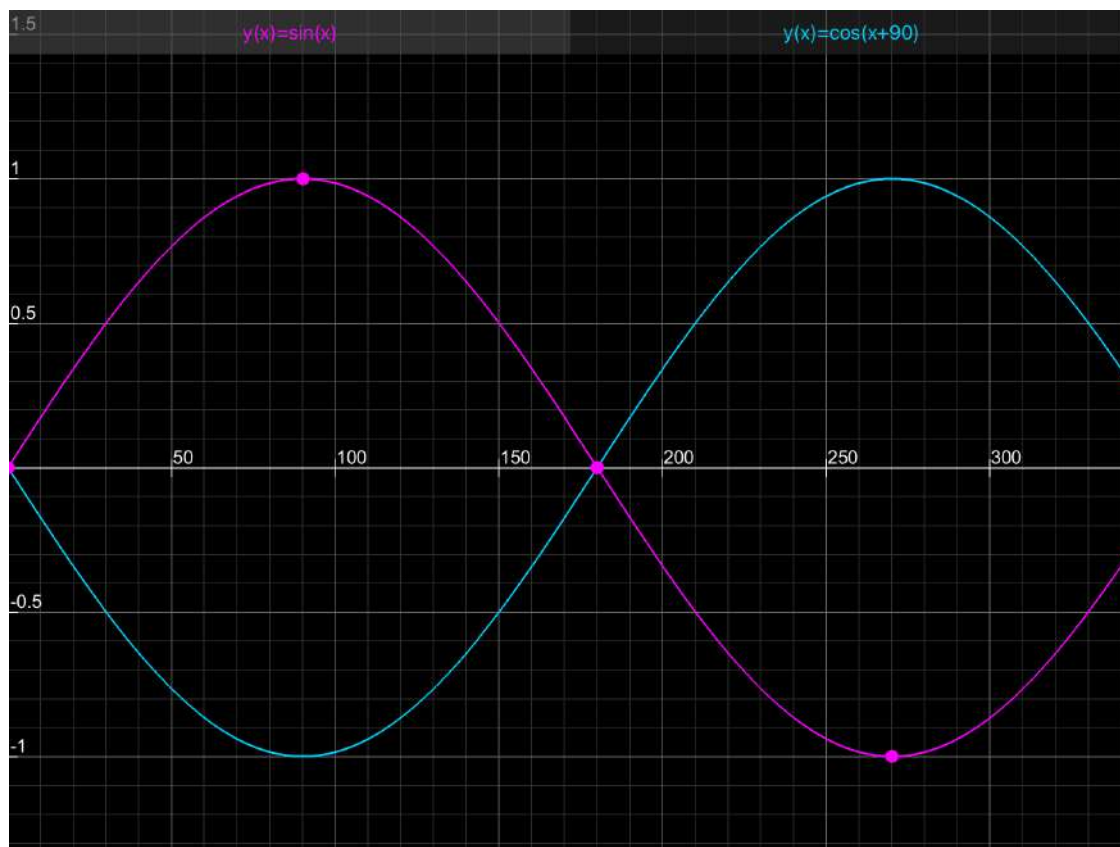
Dalla parte sinistra dell'equazione di Shroedinger paragonata alla equazione di Gibbs, abbiamo termini che indicano come varia una equazione d'onda mentre dalla parte destra abbiamo una funzione di stato che è indipendente dalla sua variazione o nel tempo o nello spazio-tempo ma dipende solo dai suoi valori iniziali e finali, cioè dallo stato iniziale e finale della entropia ma non da come essa varia durante la sua variazione. In realtà l'equazione così scritta ci dice che la derivata seconda della funzione d'onda nello spazio-tempo alla fine ed all'inizio dell'analisi è la rappresentazione della probabilità che l'evento sia accaduto e tale valore, solo alla fine dell'evento, è sempre eguale all'1, indipendentemente dallo svolgimento degli accadimenti; e tutto ciò rende i due termini dell'equazione compatibili tra loro, solo se essi calcolano l'inizio e la fine dell'evento nella sua completezza, rendendo il valore di ΔS sostanzialmente dipendente solo dal valore di "m".

Nel trading ciò significa che una previsione a lungo termine viene considerata da noi come un evento iniziato e finito dove siamo sicuri del valore finale della funzione su ascisse e coordinate. Noi cioè sappiamo che, a lungo termine, il mercato porterà a quella variabile con quel valore finale ma non siamo sicuri di come quel valore sarà raggiunto, perché, nel tempo, la consapevolezza dell'evento e di noi che lo stiamo creando, cambia.



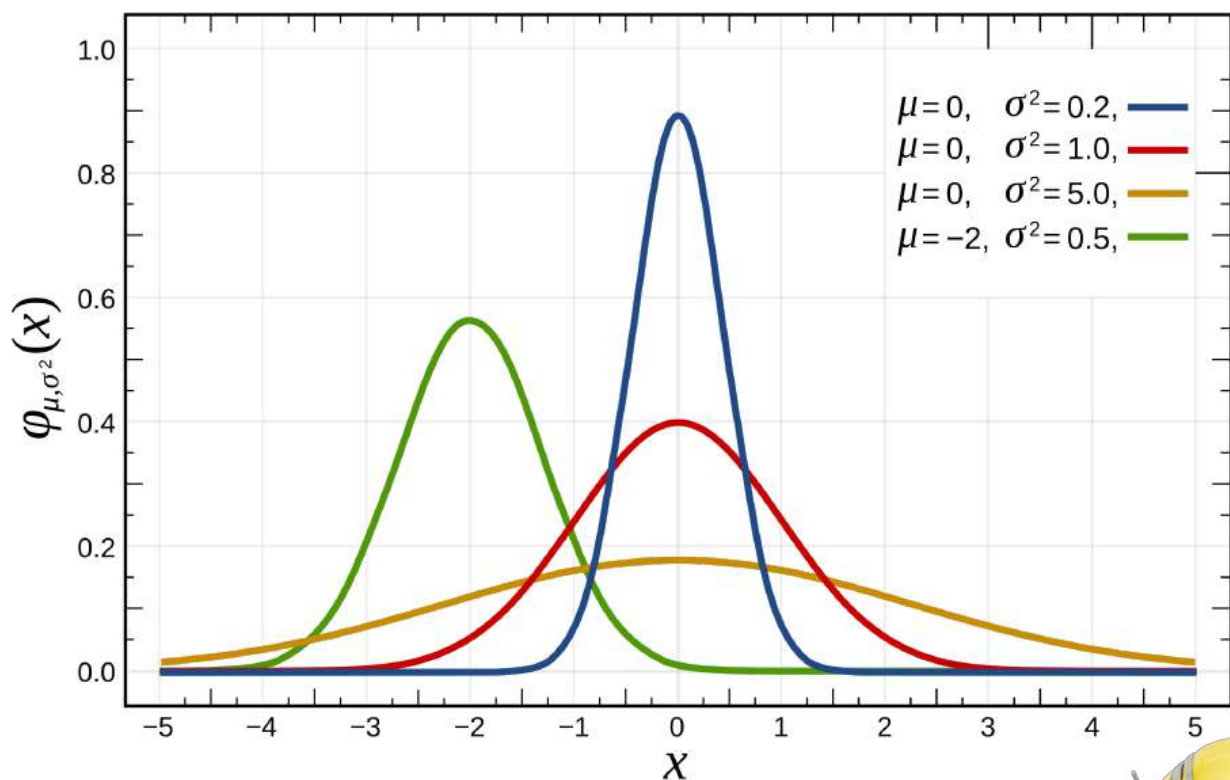
Dunque l'operatore, avendo a disposizione la sicurezza della fine dell'evento o postulandola tale, nelle condizioni al contorno del problema, potrà ricalcoare la trasformata di Fourier in sotto campi spazio-temporali più corti, dove la trasformata verrà via via modificata, per adattarsi alle condizioni cioè alle informazioni del passato e del presente per proiettarle in un futuro il cui punto finale è obbligatoriamente fissato dalle condizioni iniziali del problema.

Giunti alla fine della previsione a lungo termine, integrata dalle sotto previsioni a breve termine, si deve considerare l'evento terminato e si ripartirà con un altro evento. Considerando così il trading una serie di eventi compiuti, si potrà sostenere che la derivata seconda della funzione d'onda, che caratterizza questo percorso, abbia sempre probabilità fissa ed identificabile, di accadere. In termini matematici ciò significa che: qualsiasi percorso abbia compiuto la funzione d'onda per arrivare al suo punto finale dell'evento preso in considerazione, l'area descritta dalla sua probabilità finale sarà sempre la stessa, il che significa che l'integrale di tale funzione sarà, alla fine, indipendente dalla forma della funzione ma dipenderà solo dal punto di partenza e da quello di arrivo, come nell'esempio seguente:



rendendo compatibile il formale segno δ con il segno Δ , nella nostra eguaglianza. Analogamente il significato dell'eguaglianza tra il primo termine a sinistra della equazione di Shroedinger, che indica l'energia totale del sistema, quale entità immaginaria eguagliata al termine entropico $-T\Delta S$, va a sottolineare come, anche in questo caso, all'inizio ed alla fine di un evento, la quantità di energia messa in gioco, sia sempre la stessa in un sistema isolato (vedi teoria dello Zero Point Energy).

Per chiarire questo ulteriore aspetto coscienziale della realtà virtuale, facciamo l'esempio della vita. Un soggetto viene inconsapevolmente al mondo avendo lo scopo (inconsapevole) di capire la dualità che lo porterà, partendo da zero energia, al tempo zero, ad arrivare a zero energia al tempo finale t . Il suo percorso sarà nella ipotesi più semplice una gaussiana più o meno accentuata a seconda della capacità di mirare il target finale da raggiungere nel minor sforzo possibile, sempre avendo a disposizione lo stesso tempo.



In questo grafico, a seconda del percorso che uno decide di fare (il più diretto è il più consapevole lungo le ascisse che rappresentano l'asse del tempo, mentre il più tortuoso prevede innalzamenti e cambi di direzione), si vede che la derivata prima della funzione d'onda rappresenta la pendenza delle curve.

Questo parametro indica quanto noi si sia in grado di centrare l'obiettivo finale, divagando il meno possibile.

Dunque, ancora una volta, siamo di fronte ad un parametro di consapevolezza che ci dice come il valore della derivata prima della funzione d'onda rappresenta la pendenza delle curve.

Questo parametro indica quanto noi si sia in grado di centrare l'obiettivo finale, divagando il meno possibile. Dunque, ancora una volta, siamo di fronte ad un parametro di consapevolezza che ci dice come il valore della derivata prima della funzione d'onda, all'inizio ed alla fine del percorso sia, in modulo, sempre lo stesso, dove il segno $+$ o $-$, indica invece se ci stiamo allontanando od avvicinando al valore finale/iniziale dell'energia. Praticamente l'ordinata di ogni punto del nostro percorso di consapevolezza, indica di quanto siamo distanti dal percorso più breve, energeticamente parlando, per ottenere la fine dell'esperienza. Ma ora che la teoria ci dice che il nostro modello è in grado di prevedere qualitativamente e quantitativamente il futuro, possiamo applicare tutto ciò con consapevolezza alla pratica, che verrà qui di seguito esposta con esempi pratici.



Frattali



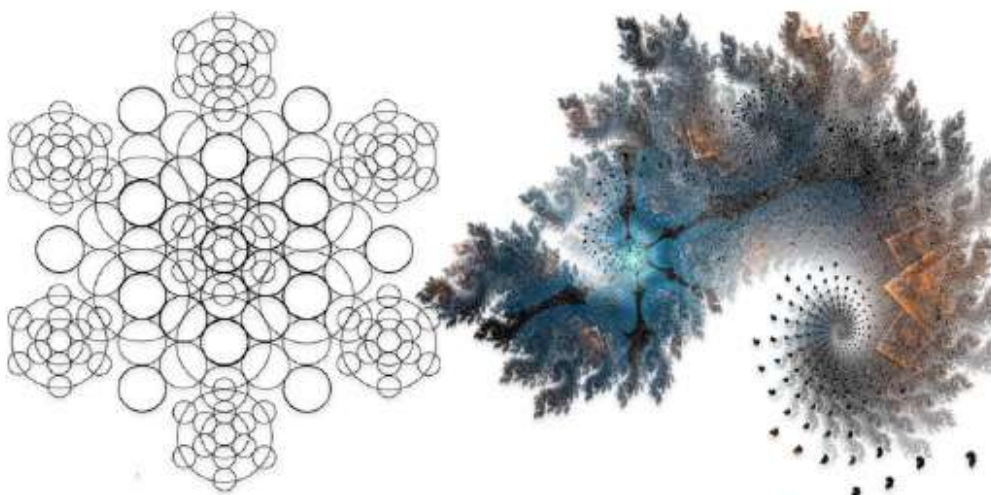
Cosa sono i frattali

I frattali sono strutture geometriche complesse che presentano un pattern ripetitivo a diverse scale. Questa proprietà è nota come autosimilarità.

I frattali sono interessanti sia per la loro bellezza naturale sia per le loro applicazioni pratiche in vari campi della scienza e della tecnologia e nella finanza.

Origine e Sviluppo

Il termine “frattale” fu coniato dal matematico Benoît Mandelbrot nel 1975. Mandelbrot ha introdotto questo concetto mentre cercava di descrivere forme geometriche irregolari e frammentate che non potevano essere rappresentate dalle tradizionali figure della geometria euclidea.



Caratteristiche dei Frattali

- **Autosimilarità:** I frattali mostrano dettagli simili a se stessi su scale diverse. Ad esempio, ingrandendo una parte di un frattale, si troverà una struttura simile a quella più grande.

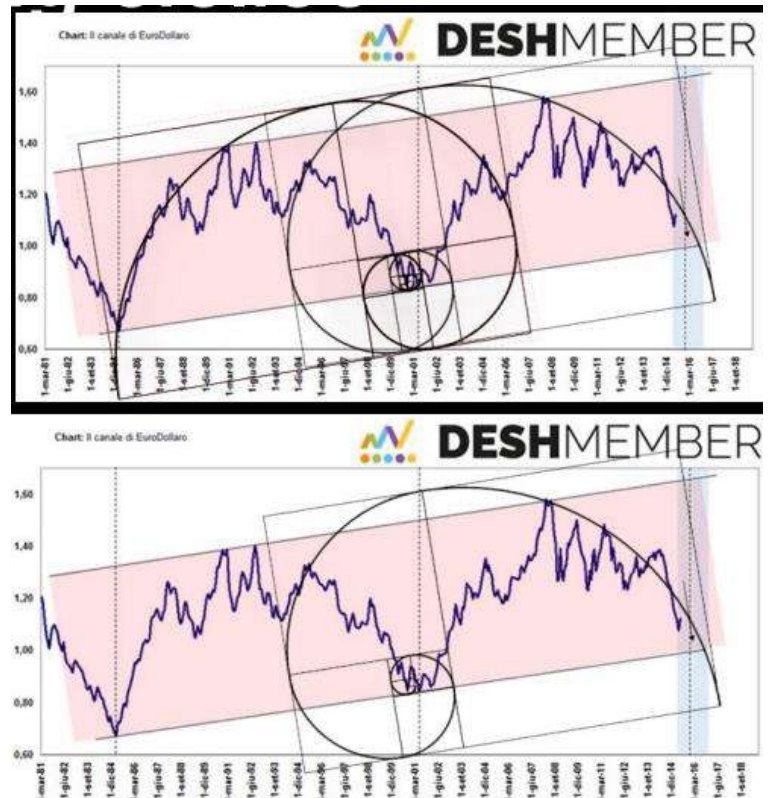
- **Struttura a Scale Multiple:** Un frattale contiene dettagli su scale che variano enormemente. Questa proprietà li rende utili per modellare fenomeni naturali che presentano forme complesse a diverse dimensioni, come le coste, le nuvole o i sistemi di vasi sanguigni.

- **Dimensione Frattale :** I frattali sono spesso caratterizzati da una “dimensione frattale”, che è una misura statistica della loro complessità. La dimensione frattale può essere un numero non intero, a differenza delle dimensioni convenzionali.

Questa spiegazione generica ha l'obiettivo di introdurre il concetto di base dei frattali applicati ai mercati finanziari, dove il denaro, con tutti i suoi aspetti duali di avidità e paura, altera e complica i concetti alla base del frattale. Questo non avviene tanto per la complessità operativa dell'applicazione, ma perché pone le persone di fronte a se stesse, alle limitazioni della personalità e ai condizionamenti che l'essere umano subisce costantemente. È importante considerare che i frattali influenzano tutti gli ambiti della nostra vita; così come in altri settori, anche il mercato finanziario ha origini frattale. Tuttavia, l'impatto dei frattali nella finanza è immediato e ci porta a rapidi aumenti di entropia che richiedono di vivere intensamente il momento presente e di sentire il peso dell'apprendimento mentre se ne fa esperienza.

Nei frattali che incontriamo nella vita quotidiana, spesso non abbiamo modo di comprendere appieno queste esperienze, poiché il tempo di sviluppo del frattale è rallentato e le distrazioni e i condizionamenti impediscono di percepire questo cambiamento, finché lo stesso cambiamento non è già avvenuto nella nostra vita.





Ci siamo chiesti il perché, e la risposta è risultata abbastanza evidente: nessuno ha mai adottato un'altra prospettiva, ovvero quella di vedere il mercato finanziario con un approccio coscienziale. Questo approccio non pone il guadagno come obiettivo principale, ma lo considera una conseguenza del risultato ottenuto. Combinando l'esperienza da trader di Serleto con questa nuova visione, ci siamo subito accorti della limitazione imposta dal tempo: più il tempo è dilatato, meno è probabile fare l'esperienza di comprensione e di predittività del frattale futuro. Al contrario, se il tempo è ridotto, la predizione del frattale futuro è meno affidabile poiché l'informazione non è completa. Possiamo pensare al tempo come a una serie di ingranaggi (esempio). In letteratura finanziaria si trovano diversi lavori che hanno tentato di applicare questi metodi, ma il tempo troppo ridotto o troppo dilatato non ha portato a risultati replicabili.

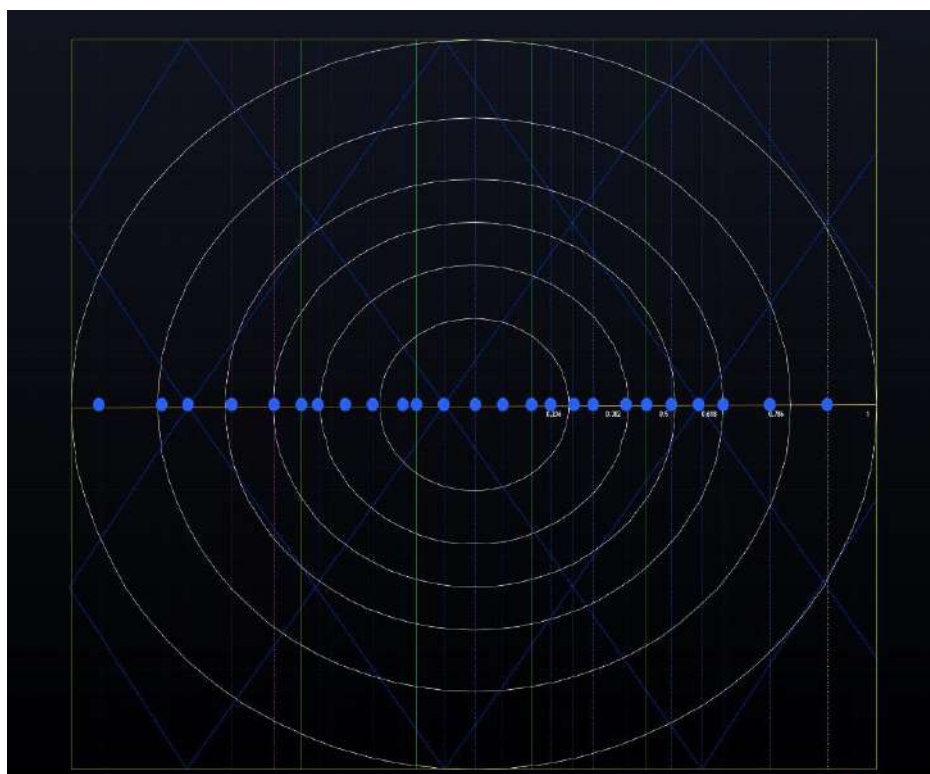


Da pattern HC a indicatore frattale

La prima parte del lavoro è stata quella di trasformare il disegno del pattern H-C in un algoritmo che, in base alla consapevolezza di chi lo utilizza, permette di individuare la scala di applicazione corretta rispetto al contesto. Questo include l'identificazione della consapevolezza dell'asset, il tempo di movimento dell'asset all'interno del pattern, e la fusione di livelli di massimi e minimi con linee temporali di accelerazione.

Scala di applicazione:

Grazie all'algoritmo che incorpora i dati del frattale, identificando un massimo e un minimo importanti, è possibile estendere il frattale nel futuro, risolvendo il problema della discrezionalità sul trend, ovvero su come adattare l'inclinazione del pattern su un grafico dinamico. In ogni caso, il risultato dell'estensione dell'indicatore è determinato dalla consapevolezza individuale, che consente di identificare i massimi e i minimi più appropriati rispetto agli obiettivi che si vogliono raggiungere.



Consapevolezza dell'asset:

La consapevolezza dell'asset è ciò che nei mercati finanziari viene chiamata volatilità, ma identificata con una percentuale definita e diversa per ogni asset. Infatti, sebbene sia possibile prevedere le strutture future, queste si formano con un disegno simile ma con un'estensione diversa. È essenziale che il pattern sia parametrato correttamente per ogni asset in base alla sua percentuale di consapevolezza.

Tempo: Il tempo è identificato come il time frame scelto per utilizzare il pattern, poiché ogni candela è determinata dal time frame. Ad esempio, su un grafico a 15 minuti, in un'ora avremo 4 candele, mentre se scegliamo il time frame di 1 ora, avremo solo una candela. Questa differenza influisce sul risultato dell'estensione dell'indicatore frattale. Per un'analisi di breve termine, è funzionale il time frame da 1 ora, mentre per un'analisi di più lungo termine è più efficace il time frame giornaliero. Bisogna tenere in considerazione un fattore essenziale: l'analisi non è il trading, e comprendere che, anche se abbiamo un'indicazione di un prezzo futuro di un asset, esso non è garanzia di guadagno. L'indicatore frattale è uno strumento per analizzare e prevedere il comportamento futuro in termini di accelerazioni al rialzo o al ribasso e prevedere i punti di contatto di massimi e minimi delle proiezioni dei frattali.

Analisi: L'analisi è la fase preliminare in cui un trader o un investitore raccoglie informazioni, studia i dati e valuta la situazione del mercato per fare previsioni sui futuri movimenti dei prezzi. Nel nostro caso, l'indicatore frattale, come vedremo in seguito, ci dà dei riferimenti precisi sul presente e sul futuro, ma questi riferimenti non hanno un piano applicativo poiché lo scopo dell'analisi non è quello di avere un piano esecutivo, ma di avere dei riferimenti sul futuro.

Trading: Il trading è l'attività pratica che segue l'analisi. Ci sono varie filosofie e stili di trading, e una delle grandi lacune è che non si può fare trading se questo non è una nostra estensione. Avere a disposizione dei dati per vedere il futuro, non è garanzia di successo, poiché il futuro si costruisce un momento presente dopo l'altro, e in termini pratici vuol dire: se oggi l'indicatore frattale mi dice che domani mattina il prezzo dell'oro arriverà a 2500 e io compro l'oro mentre, in una prima fase, il prezzo scende, e, per evitare ulteriori perdite, io decido di chiudere la posizione. Successivamente, nelle ore seguenti, il prezzo arriva a 2500. In questo caso, il prezzo è arrivato al target, ma io non ho guadagnato nonostante sapessi che nel futuro il prezzo sarebbe andato a 2500.



Quando si fa trading, il potenziale futuro che intercettiamo va contestualizzato in base al time frame, alla consapevolezza dell'asset, alla leva finanziaria (se la sto utilizzando) e ad altri fattori. La padronanza di questi e altri concetti si chiama Consapevolezza, e l'applicazione di questa consapevolezza nel momento presente ti permette di sapere quando e quanto comprare. Se la posizione che ho aperto è in passivo, so come gestirla in attesa del target futuro. Questo si chiama "essere", questo vuol dire diventare un'estensione di noi stessi e questo è ciò che intendiamo quando utilizziamo il termine "trading" in questa ricerca.

Indicatore frattale

L'algoritmo dell'indicatore frattale è disponibile sulla piattaforma TradingView tramite una licenza dedicata. L'indicatore frattale viene tracciato da un minimo a un massimo e, in questa sequenza, raccoglie i dati del frattale passato ed estende 5 sezioni in cui sono racchiusi tutti i dati del frattale.

Questi dati vengono espressi in:

- **Linee verticali:** Le linee verticali indicano cambiamenti ciclici all'interno del frattale. Ci sono 25 linee verticali all'interno di 588 candele, che, in base al time frame, possono essere 1H (588 ore), 1D (588 giorni), ecc. Il 90% di queste linee segnala un'accelerazione, quindi un aumento di volatilità, mentre il restante 10% indica un'eventuale lateralizzazione. È importante considerare che, se si usa un time frame di 1H o 4H, dalla data e ora precisa in cui si verifica la linea verticale, bisogna considerare una tolleranza di 3 ore prima e dopo, durante le quali l'accelerazione può anticipare o ritardare.
- **Linee inclinate (blu):** Queste linee indicano aree di supporto e resistenze condizionate dal tempo. Ogni linea indica che, quando il prezzo interagirà con ognuna di esse, ci sarà un arresto (se il prezzo viene da una salita) o una ripartenza (se il prezzo viene da una discesa).
- **Cerchi bianchi:** I cerchi seguono l'andamento dei prezzi sui mercati in forma circolare, in base al frattale. Quando il prezzo entra in contatto con un cerchio e proviene da un rialzo, il prezzo può arrestarsi momentaneamente; al rialzo come prossimo obiettivo. Se invece veniamo da una discesa, il prezzo può arrestarsi momentaneamente e possiamo considerare il cerchio successivo al ribasso come prossimo obiettivo. Utilizzando singolarmente o simultaneamente questi riferimenti, è possibile prevedere con precisione aumenti di volatilità, massimi e minimi futuri.



L'accuratezza di questi riferimenti può variare in base a chi li utilizza. Ricordiamo che il frattale si estende tra due punti, minimo e massimo, e questa azione è influenzata dalla consapevolezza personale. Quindi, possiamo affermare che il risultato dell'indicatore frattale cambia a seconda della consapevolezza di chi lo utilizza.



Trasformata di Fourier



Parte Didattica

Successivamente all'indicatore frattale, il professor Corrado Malanga ha introdotto nel lavoro che avevamo svolto fino a quel momento la trasformata di Fourier (spiegazione). Luca Serleto l'ha poi adattata con parametri di mercato specifici, calibrando le scale corrette e considerando le varie consapevolezza e dell'asset. Da qui è nato qualcosa di straordinario: ci siamo accorti che potevamo identificare pattern nascosti che non erano immediatamente visibili.

Isolando i fattori sia emotivi che applicativi che contraddistinguono gli operatori di mercato, si riesce a risolvere, una dopo l'altra, tutte le incertezze, arrivando a un risultato senza precedenti.

Il Tempo: Come prima cosa, abbiamo dovuto risolvere il problema del tempo. Inserendo la trasformata di Fourier e identificando diverse sequenze temporali in cui avviene la costruzione dei frattali, abbiamo isolato le varie componenti di frequenza. In una fase successiva, abbiamo cercato di rimuovere il rumore di fondo per isolare i pattern, ma i risultati non erano corretti. Lavorandoci sopra, ci siamo resi conto che quando si seleziona un range temporale in una costruzione frattale in un momento specifico, i dati del passato sembrano scomparire, senza più influenzare la creazione di pattern frattali futuri. È come dire che l'esperienza fatta a un anno di vita è registrata in noi, ma oggi quel passato non ha alcuna influenza sulla creazione del nostro futuro nel momento presente.

Risolto questo problema e identificato il tempo corretto, aggiungendo delle variabili di dati alla trasformata di Fourier, siamo riusciti non solo a isolare il tempo che non aveva influenza sulla costruzione del frattale nel futuro, ma anche a identificare diversi pattern che si ripetono su tutti i mercati del mondo.

In questa applicazione useremo i grafici di Bitcoin, Ethereum, Oro, Argento, contro il dollaro ma potremo usare qualsiasi asset del mondo. Inizieremo a prendere come riferimento il pattern che nel book dei frattali estrapolato dalla trasformata di Fourier in allegato è il numero 1:



Pattern 1

Bitcoin: h 00:00 del 28 agosto 24 (time frame 15m)



Ethereum: h 00:00 del 28 agosto 24 (time frame 15m)



Oro: h14:30 del 29 agosto 24 (time frame 15m)



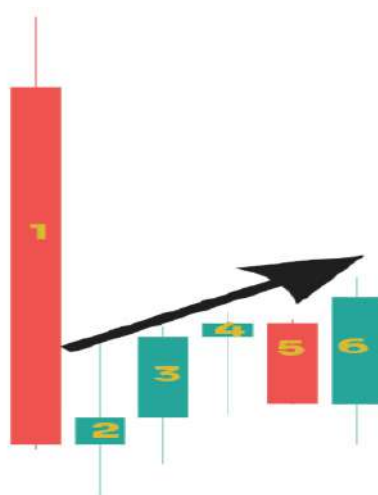
Argento : h18:15 del 17 luglio 24 (time frame 15m)



Pattern frattalici: Come possiamo vedere in questo esempio, su asset di natura diversa, abbiamo lo stesso pattern, tenendo in considerazione che possono esistere diversi pattern simili che possono variare per inclinazione, ampiezza o numero di candele.

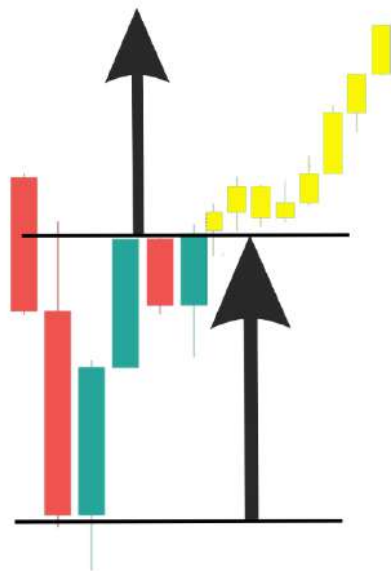
Il pattern che abbiamo utilizzato nel primo esempio, identificato nel libro come “Pattern 1”, non è così chiamato perché sia il primo in assoluto, ma perché i pattern possono essere riconosciuti in base al tempo, che in questo caso è rappresentato dalle candele al suo interno.

Questa tipologia di pattern può estendersi fino a 6 candele, per questo motivo nel libro i pattern sono identificati con range da 3 a 6, da 6 a 12, da 12 a 24, da 24 a 39 e da 39 a 56. Oltre al tempo, il pattern può variare anche nell’inclinazione, pur mantenendo la stessa forma. Queste caratteristiche non alterano la proiezione futura del frattale, poiché la figura rimane la stessa, ma cambiano i tempi in cui avviene la proiezione.



Se prendiamo ora in considerazione i Pattern 2 e 3, abbiamo la stessa costruzione frattale ma con un numero diverso di candele, sempre appartenenti allo stesso range da 3 a 6, con inclinazioni e curvature differenti.





Proiezioni dei pattern: Dopo aver compreso la natura del frattale e come le forme, sebbene uguali in termini di costruzione, possano essere differenti in termini di variabili, è bene comprendere che queste variabili non condizionano il risultato in termini di proiezione, ma possono influenzare l'operatività in termini applicativi, che approfondiremo in seguito. La proiezione del frattale è già ben definita nella natura della sua costruzione e viene determinata dall'ampiezza tra il punto più basso della struttura e la linearità più alta della costruzione della struttura. L'estensione della proiezione avviene sempre con una costruzione frattale che, anche se utilizziamo il pattern dello stesso range, il target sarà sempre quello definito all'interno del frattale, ma il modo in cui si raggiunge può essere diverso. La piattaforma di Pythagoras, che integra la trasformata di Fourier, può analizzare questo con i dati del presente e fornire tempi e proiezioni più dettagliate.

Tuttavia, nel contesto attuale, stiamo applicando i dati forniti dalla trasformata di Fourier, ma non sono adattati ai dati del presente in cui verranno utilizzati.

Pertanto, possiamo vedere le strutture dei frattali grazie al libro dei frattali, possiamo conoscere il potenziale futuro, ma non siamo in grado di determinare un range di tempo definito per applicarlo, a meno che non si abbia una consapevolezza tale da poterlo determinare.

Questo, nell'ambito della ricerca e dell'analisi, non ha alcun impatto, poiché non si ha lo scopo di trarre profitto da uno scenario di mercato. Tuttavia, se questo strumento verrà utilizzato per il trading e per drenare liquidità dal sistema, è essenziale unire questi dati con quelli presenti al momento dell'utilizzo. Per questo, Pythagoras diventa importante per chi ha tali obiettivi.

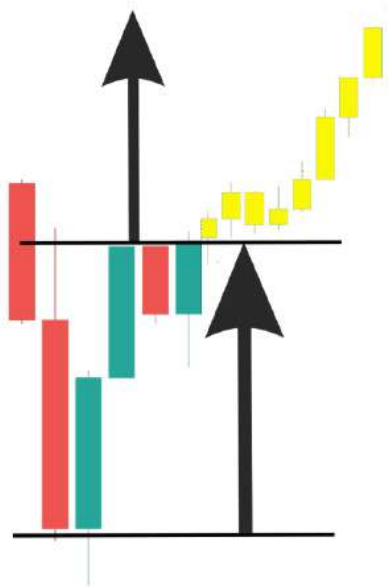


Esempi:

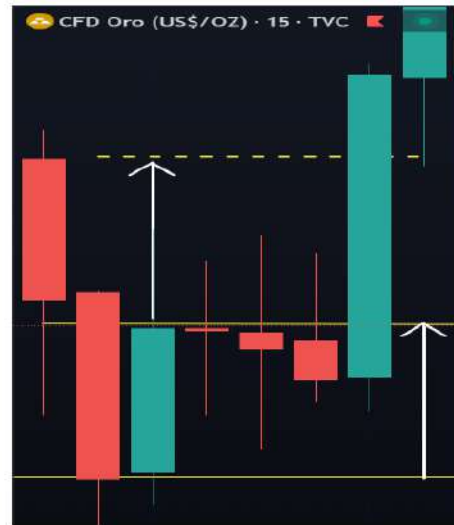
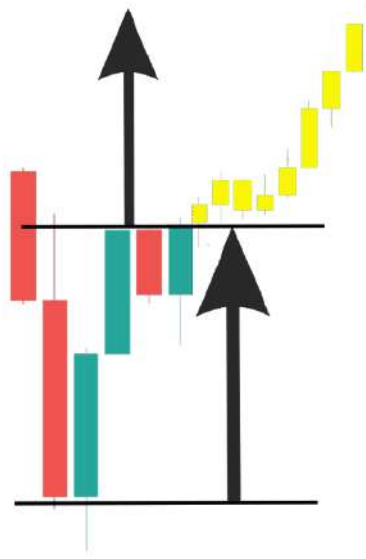
Bitcoin: h 18:30 del 27 agosto 24 (time frame 15m)



Ethereum: h 18:30 del 27 agosto 24 (time frame 15m)



Oro: h 08:45 del 30 agosto 24 (time frame15m)



Argento: h 18:15 del 17 Luglio 24 (time frame15m)



Come si evince dagli esempi, abbiamo identificato su asset diversi le stesse strutture, che si caratterizzano per un numero minimo di candele che varia da 3 a 6 in questo caso, ma si differenziano per variabili di inclinazione, ampiezza e curvatura. Tuttavia, in ogni caso, il target si ripete sempre allo stesso modo. Come si può vedere dalle immagini con le proiezioni in giallo sulla sinistra, il pattern appare nitido, mentre sul grafico, solo disponendo dei dati della struttura, inizia a visualizzarsi.

Questo accade perché, quando si osserva sul grafico, il pattern è collegato ad altre strutture frattale che hanno già completato la loro proiezione e rende complessa la sua individuazione.

Frattali complessi: Nel Book dei frattali troviamo diverse figure e diversi range. Tutti i pattern superiori a 6 candele iniziano a diventare frattali complessi, ovvero agganciano al frattale altri pezzi di frattale, portando alla creazione di figure complesse. La complessità della struttura, che richiede un tempo più dilatato per attivare la sua proiezione, è irrilevante a scopo di studio, ricerca e analisi di mercato, mentre a scopo operativo può avere un impatto significativo. Prendendo ad esempio il pattern che nel Book è indicato come 13, scomponendolo, troviamo al suo interno il pattern 3.

Pattern 3



Pattern 11



Pattern 13



L'innovazione di queste conoscenze, portata dall'introduzione della trasformata di Fourier nei mercati finanziari, cambia completamente il paradigma. Non solo possiamo individuare il prezzo futuro di un asset identificando le singole strutture, ma unendo i frattali possiamo creare strutture complesse che permettono una predizione del prezzo con una percentuale di accuratezza molto più elevata, in alcuni casi anticipando anche notizie di geopolitica e molto altro.

Attentato a Donald Trump, 14 luglio 2024:

Molti commentatori si sono divisi sulla veridicità di questo evento, ma con questa ricerca possiamo dimostrare scientificamente che il mercato, nello specifico il grafico della DJT Trump Media & Technology Group Corp quotata al NASDAQ, già in data 12 luglio 2024, un venerdì prima della chiusura dei mercati, aveva creato un frattale complesso con indicazione di rialzo. Questo frattale escludeva la possibilità di un evento negativo, e solo dopo 2 giorni siamo venuti a conoscenza dell'attentato a Trump, dal quale è uscito illeso.

Grafico Trump Media & Technology Group Corp



Pattern 15



Come si evince chiaramente dalle foto, sovrapponendo il pattern 15, abbiamo lo stesso identico pattern.

In questo caso, cambia leggermente la visualizzazione delle candele, poiché il pattern 15 è graficamente estrapolato da un time frame di 15 minuti nel mercato crypto, che è sempre aperto, mentre quello di Trump Media & Technology Group Corp è un grafico giornaliero, con ore di chiusura giornaliera delle contrattazioni. Trovando strutture complesse nei mercati finanziari, dove passa tutto ciò che accade e si muove nel mondo, e portando avanti questa nuova conoscenza e consapevolezza, è possibile che un giorno, quando le persone riusciranno a vedere il “chiodo nel muro”, potranno evitare di essere vittime di manipolazioni ed eventi, comprendendo il significato delle manipolazioni e dei condizionamenti che, direttamente o indirettamente, ci influenzano.

Tornando al frattale che abbiamo preso in esame precedentemente il Pattern 13 con la sua proiezione e andiamo a fare alcune dimostrazioni:

Pattern 13

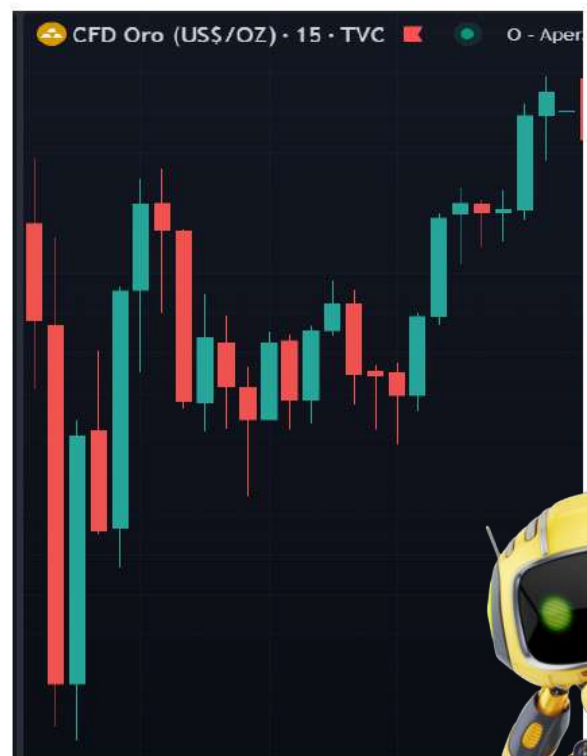
Bitcoin: h 23:15 del 24 agosto 24 (time frame 15m)



Ethereum: h 23:45 del 24 agosto 24 (time frame15m)



Oro: h 16:45 del 2 agosto 24 (time frame15m)



Argento: h 16:45 del 2 agosto 24 (time frame 15m)



Come si evince dai dati forniti e verificabili, prendendo i frattali dal libro e applicandoli alla data e ora specifica, troviamo il frattale con le sue proiezioni. Questo dovrebbe far sorgere una domanda: sei trader, gli operatori di mercato e tutti gli investitori comprano in maniera indipendente in mercati apparentemente differenti e con fondamentali diversi, come è possibile che si creino sempre le stesse strutture con le stesse proiezioni?

La risposta che ci siamo dati è che gli operatori di mercato non compiono azioni consapevoli, e quindi i loro acquisti e vendite sono condizionati da fattori esterni in maniera organizzata. Partendo da questi presupposti e applicando un aspetto consapevole a queste informazioni, si può velocizzare l'aumento dell'entropia attraverso l'esperienza di utilizzare il denaro come strumento per acquisire una conoscenza di noi stessi che non abbiamo ancora compreso.

Questi dati sono interamente estrapolati dalla trasformata di Fourier e hanno uno scopo didattico; richiedono applicazione e studio per associare correttamente tutte le strutture e applicarle in modo ripetitivo e automatico per prevedere i prezzi futuri. Questo modello didattico, se non si ha una conoscenza approfondita, permette sì di prevedere il prezzo futuro ed eventualmente di comprendere eventi, come ad esempio quello di Trump, ma è difficile trarne profitto finanziario poiché, come abbiamo visto, anche se partiamo da un frattale semplice, la visione della costruzione del frattale complesso sarà sempre in ritardo.



Trasformata di Fourier: La soluzione Pythagoras



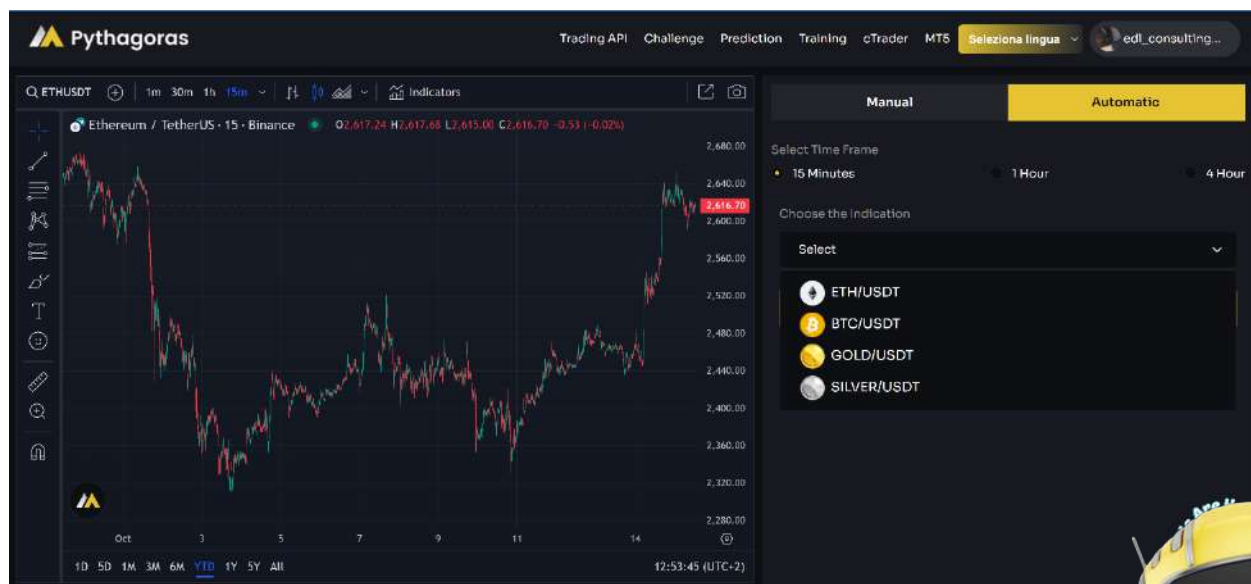
Pythagoras è la piattaforma che incorpora al suo interno, attraverso algoritmi proprietari, la Trasformata di Fourier. La sua peculiarità, applicata su diversi asset, è che può essere utilizzata nel presente prendendo istantaneamente i dati del passato e del presente, proiettando i potenziali prezzi futuri attraverso le strutture frattali.

Oltre a questo potentissimo strumento, la piattaforma offre la possibilità di studiare e aumentare la propria consapevolezza attraverso corsi dedicati alla comprensione dei frattali e all'utilizzo di soluzioni innovative che possono rendere il trading un'estensione del nostro essere.

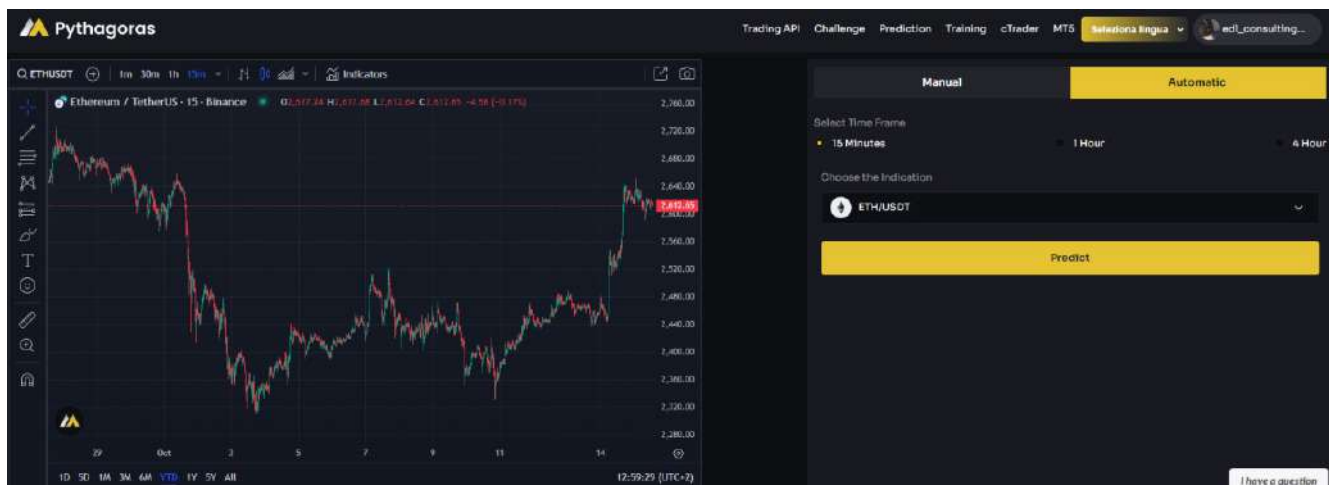
Pythagoras Automatic MODE:

La piattaforma dispone di una funzione automatica che permette ai neofiti, che non hanno ancora la capacità di identificare i frattali, di selezionare un asset a propria scelta con il time frame specifico (15 minuti, 1 ora, o 4 ore) e attivare la Trasformata di Fourier, la quale identificherà automaticamente il frattale e proietterà il potenziale futuro.

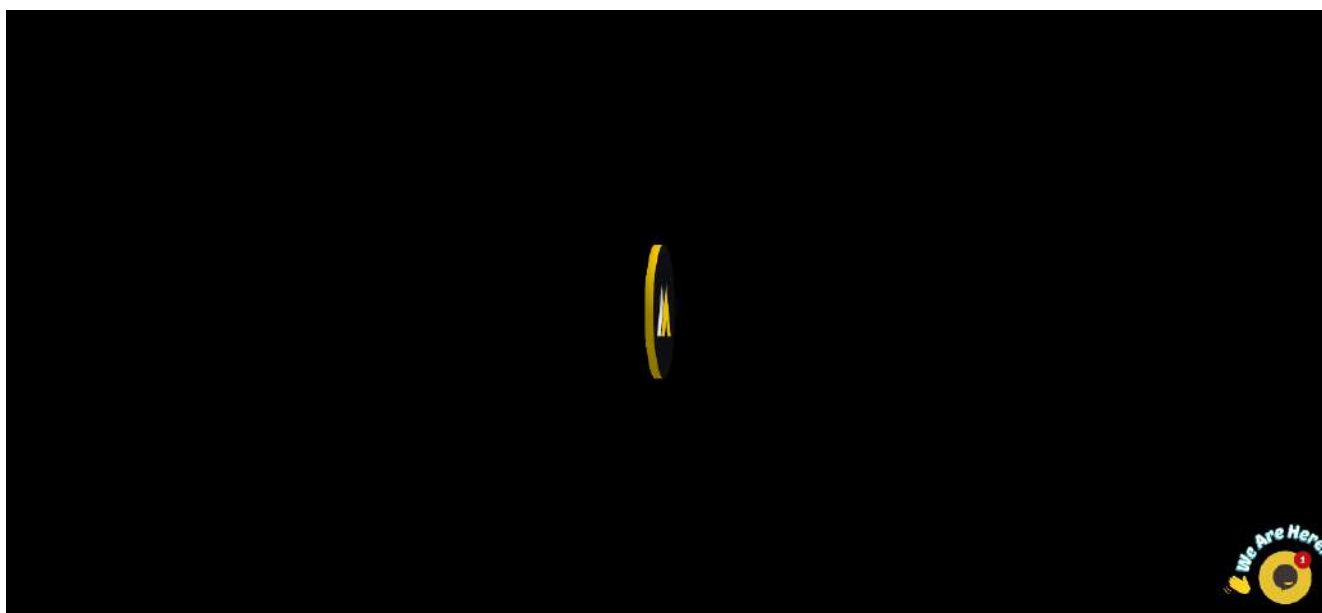
Step 1: selezionare la modalità automatica è uno tra gli asset disponibili.



Step 2: Attivare la trasformata di Fourier cliccando sul tasto “Predict”.



Step 3: Attendere il caricamento.



Step 4: In giallo si avrà il potenziale futuro a disposizione con i potenziali futuri



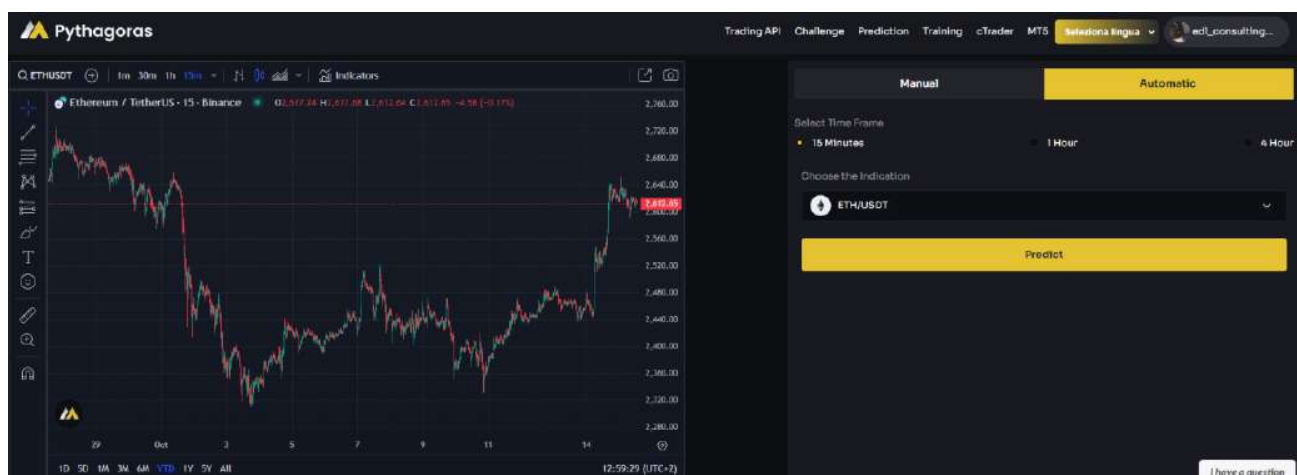
La modalità automatica è molto efficace e offre un grande supporto a chi inizia ad approcciarsi al mondo del trading con un aspetto consapevole. È bene sottolineare che, anche se si conosce un potenziale futuro, questo non è garanzia di guadagno. Solo la consapevolezza del trader, basata sull'esperienza, potrà trasformare la predizione di un potenziale futuro in un profitto concreto.



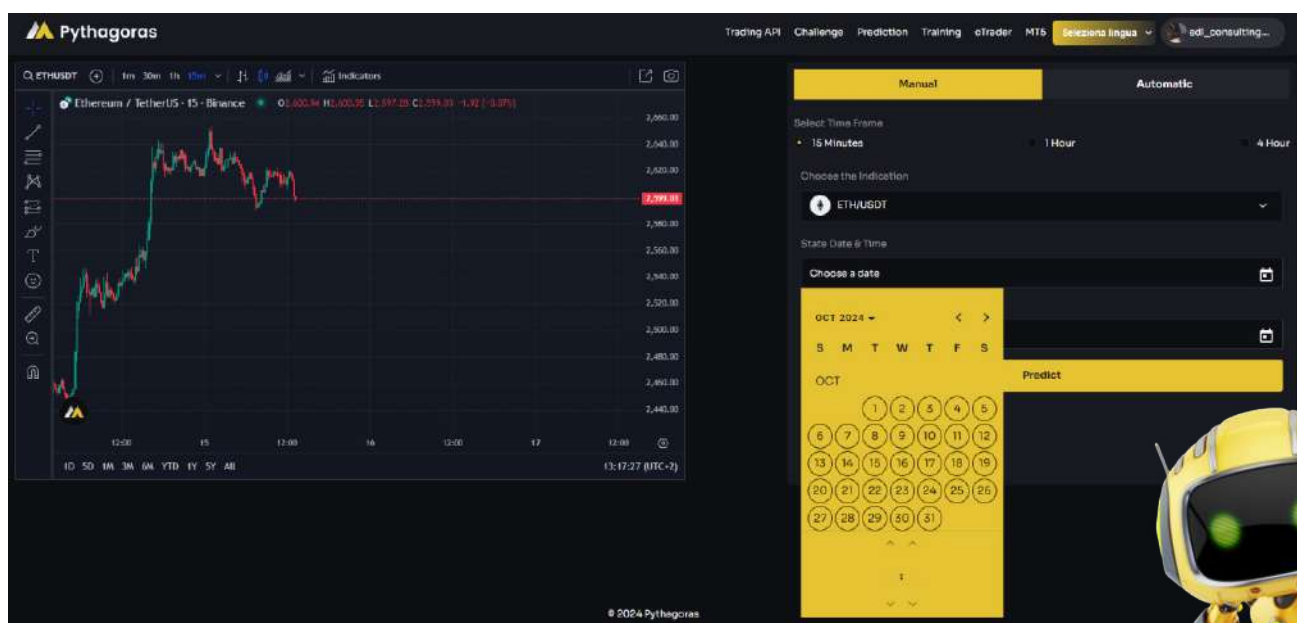
Pythagoras Manual MODE:

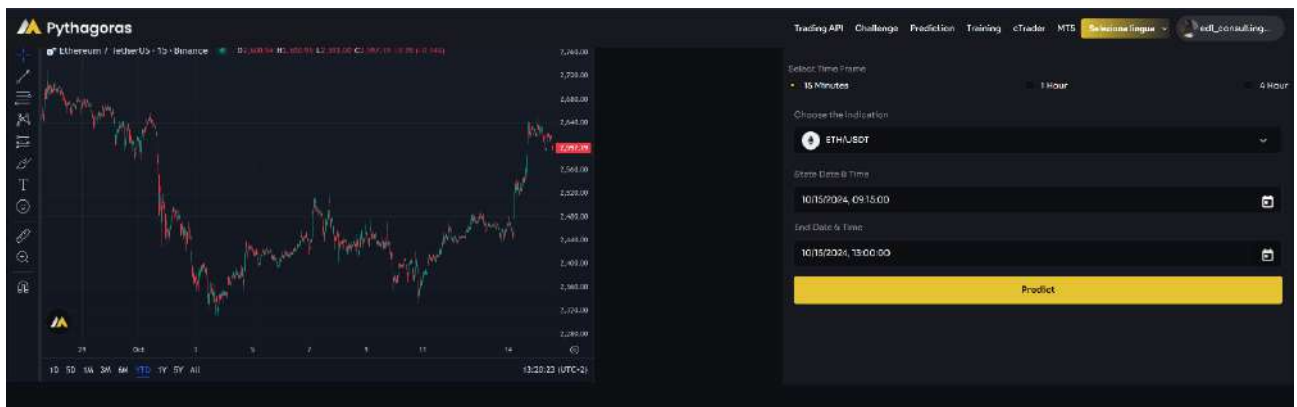
La piattaforma dispone anche di una Funzione Manuale che permette al trader che ha già acquisito una certa consapevolezza e un proprio modello operativo di selezionare il frattale, che può trovare nel book dedicato, individuandolo con il tempo specifico e attivare la trasformata di Fourier. Sostanzialmente, la modalità manuale differisce da quella automatica poiché il frattale viene individuato in base alla consapevolezza del trader.

Step 1: selezionare la modalità manuale è uno tra gli asset disponibili.

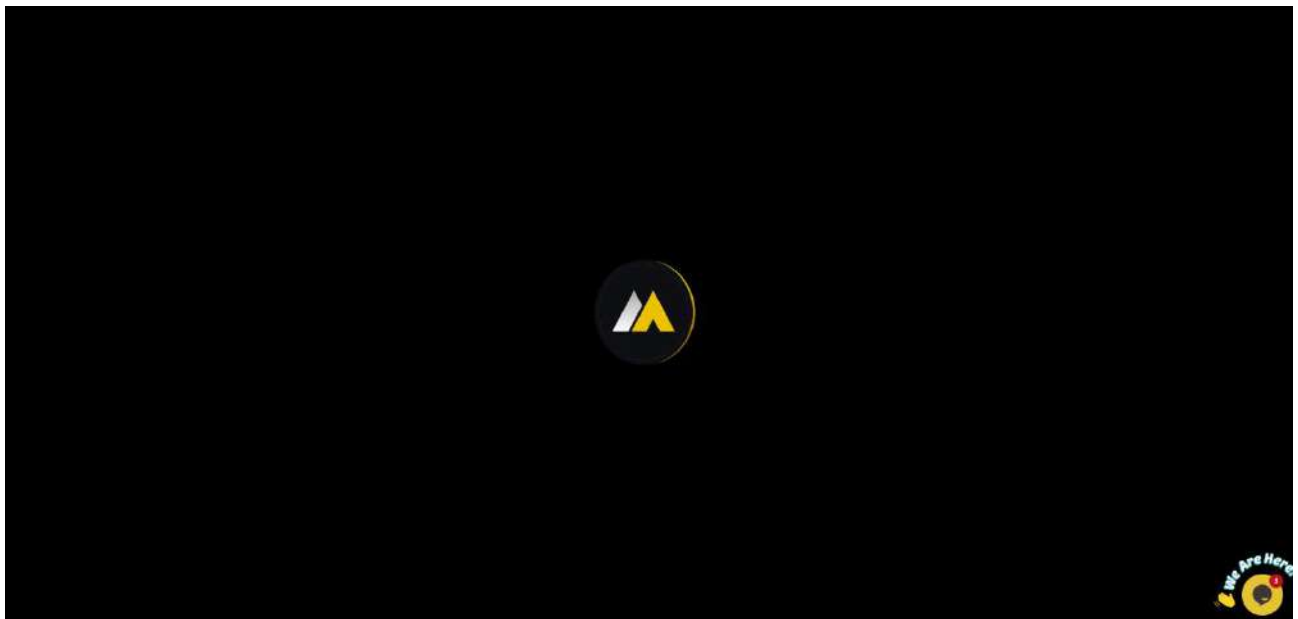


Step 2: Selezionare il frattale attraverso le date e attivare la trasformata di Fourier cliccando sul tasto “Predict”.

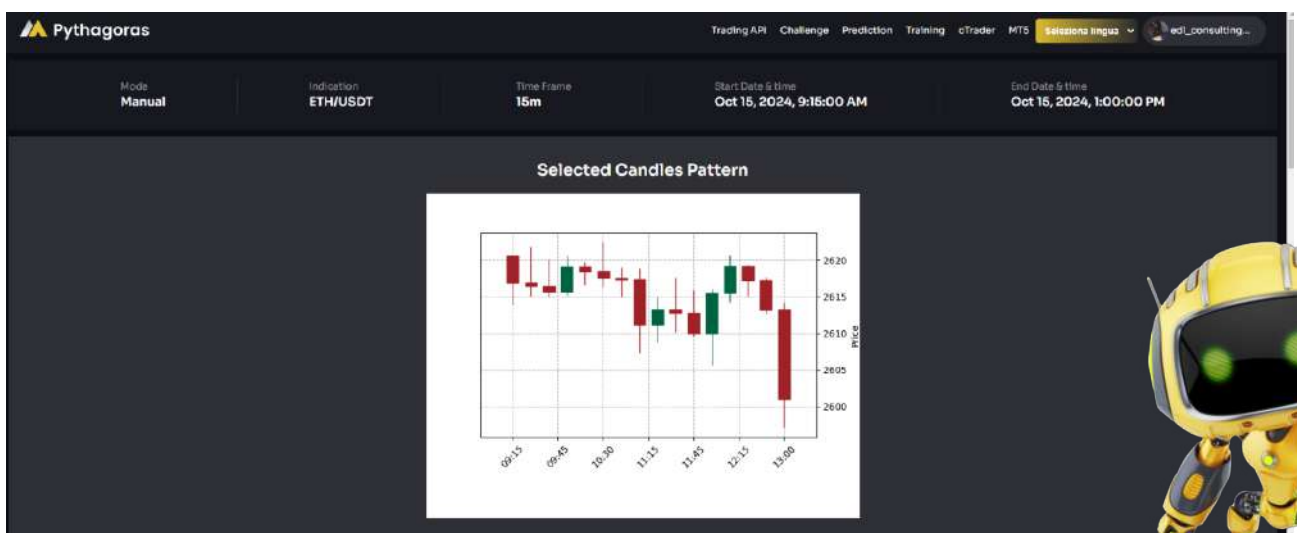


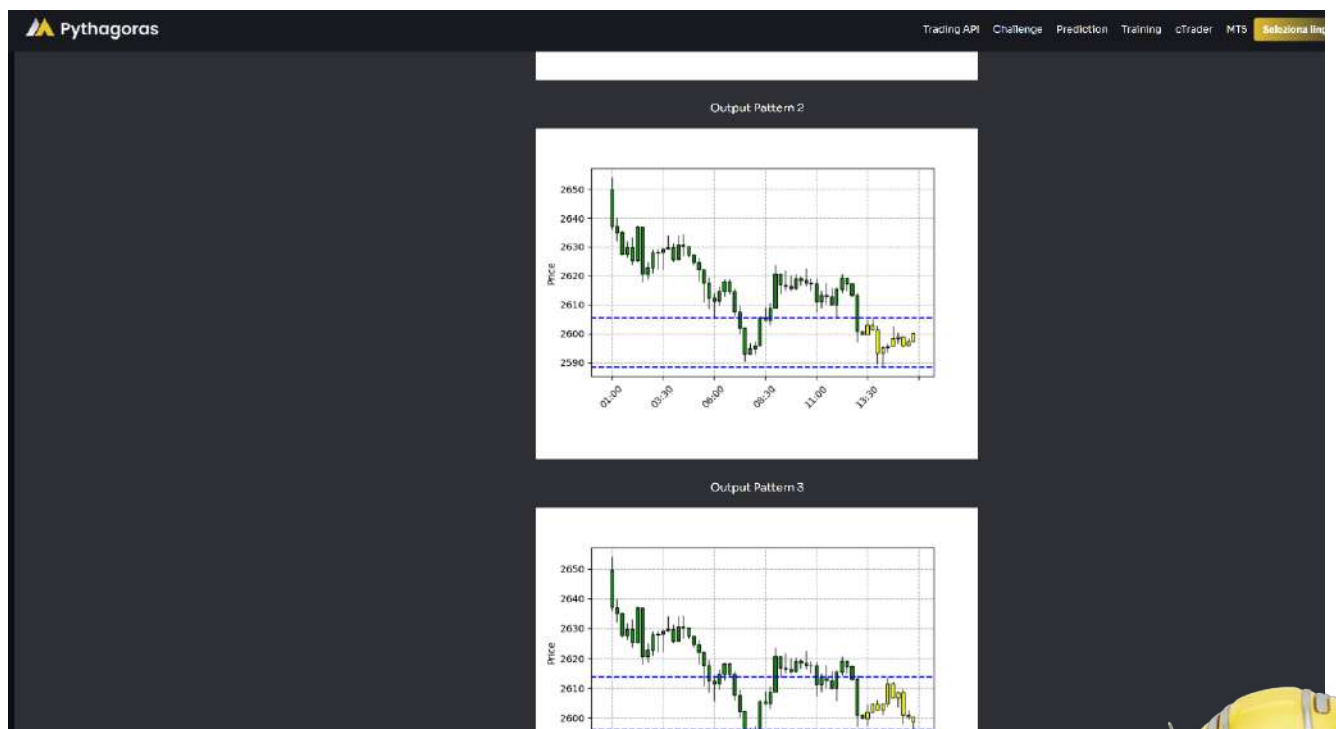


Step 3: Attendere il caricamento



Step 4: La trasformata identica il pattern e proietta i potenziali futuri





Output Pattern 4



Output Pattern 5



Il primo output rappresenta il futuro senza alterazioni, evidenziato in giallo, con la sua proiezione accompagnata da linee blu tratteggiate.

Poiché il futuro può essere alterato dal condizionamento attraverso diversi fattori che impattano sull'essere umano, la Trasformata di Fourier, utilizzando i dati del passato e del presente, tiene già conto di queste alterazioni e fornisce ulteriori potenziali futuri basati su possibili variazioni. Questo consente al trader non solo di visualizzare un frattale complesso, ma anche di essere in grado di gestire tali alterazioni.

Questi due strumenti permettono di accompagnare il trader nel suo percorso di apprendimento e conoscenza, nella scoperta di sé stesso, utilizzando il denaro come strumento di evoluzione.

Oltre a questi strumenti, è possibile utilizzare il **book dei frattali**, che contiene tutte le strutture necessarie per intercettare i frattali semplici e complessi. L'obiettivo di questo lavoro non è necessariamente quello di guadagnare più soldi, ma di utilizzare il denaro per fare esperienza personale e aumentare la propria consapevolezza. Forse un giorno, dopo aver fatto questa esperienza, il denaro non avrà più alcuna importanza.

